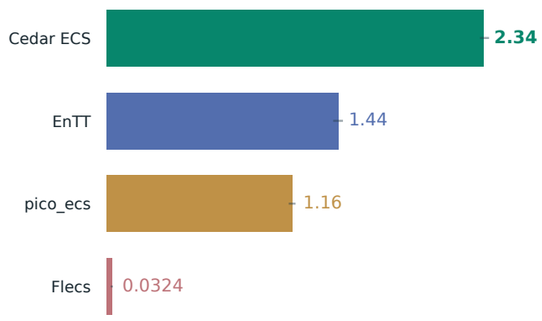
Entity comparison

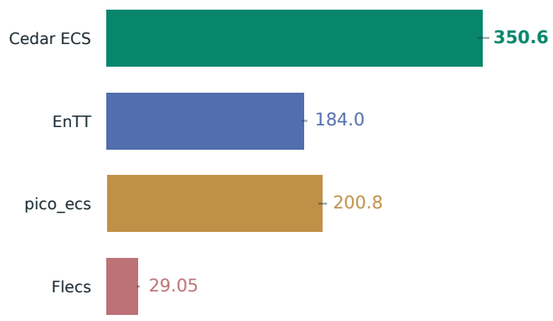
Mop/s ↑

Cedar ECS EnTT pico_ecs Flecs

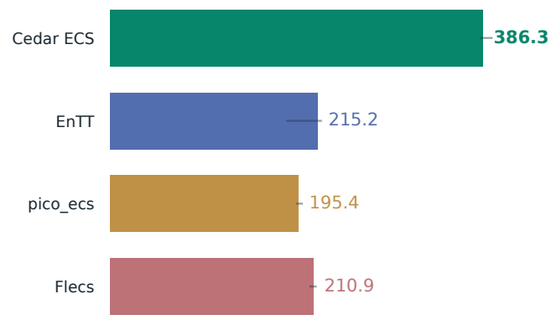
create_empty
N=1



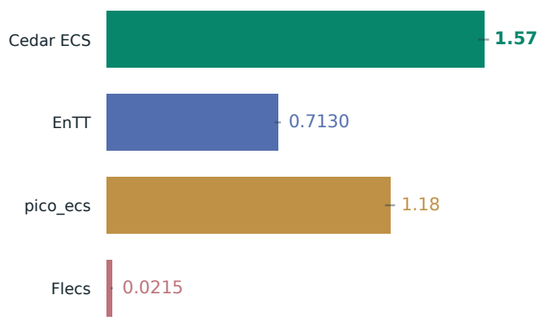
create_empty
N=1,024



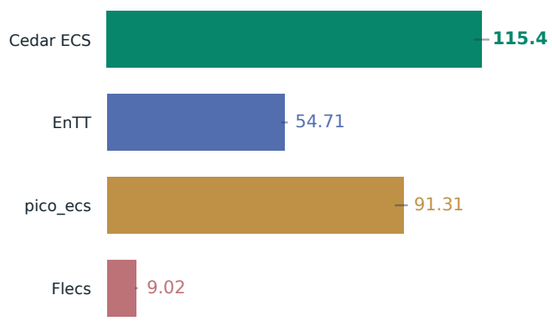
create_empty
N=65,536



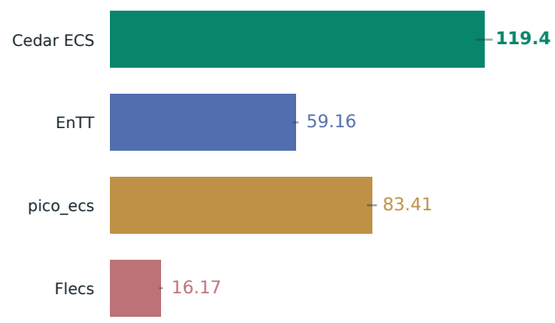
create_components
N=1



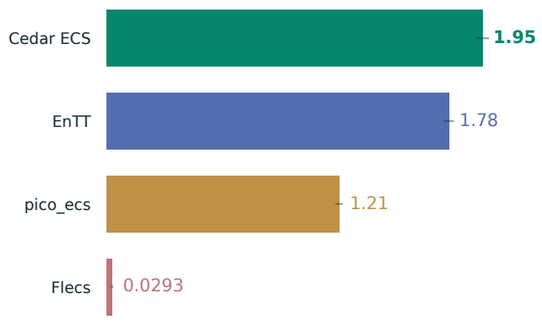
create_components
N=1,024



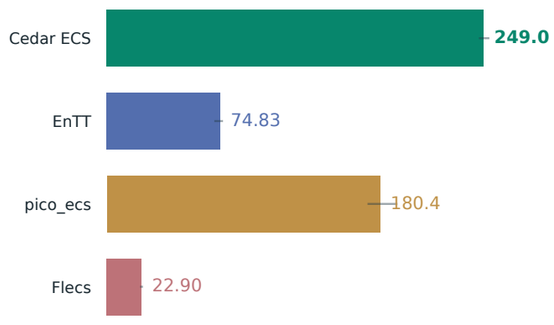
create_components
N=65,536



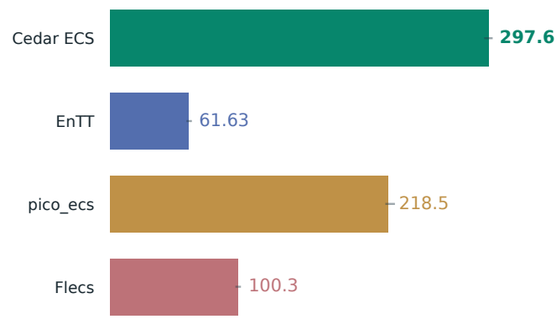
destroy_entities
N=1



destroy_entities
N=1,024



destroy_entities
N=65,536



add_component
N=1



add_component
N=1,024



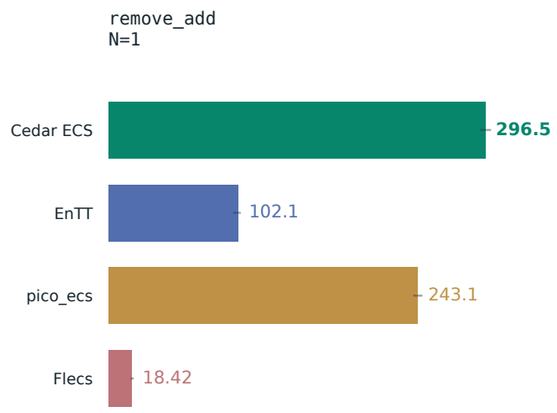
add_component
N=65,536

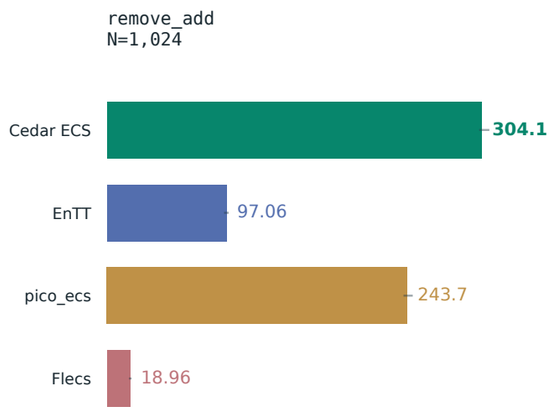


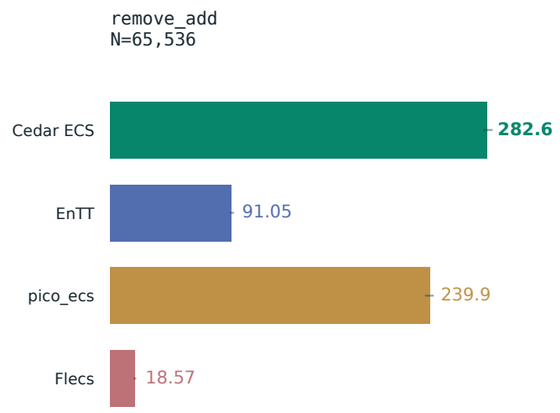
Entity comparison

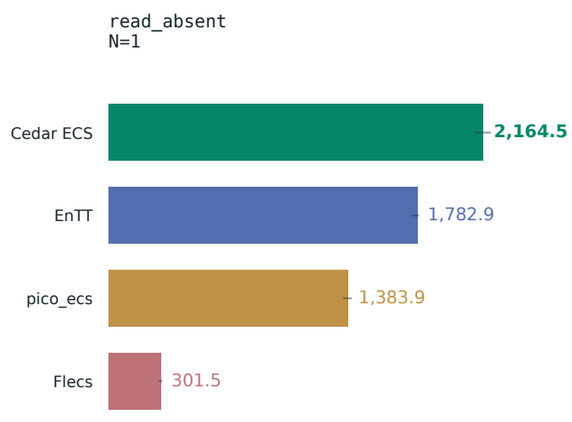
Mop/s ↑

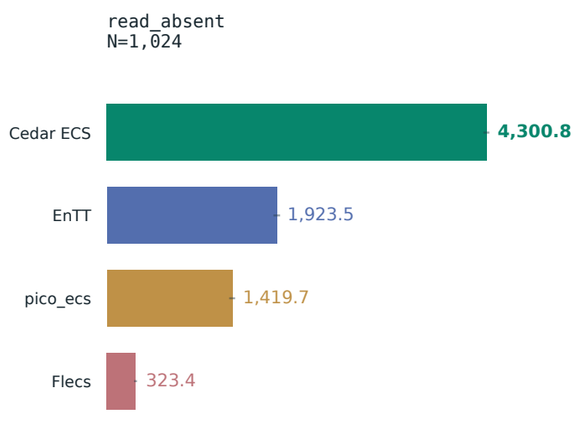
Cedar ECS EnTT pico_ecs Flecs

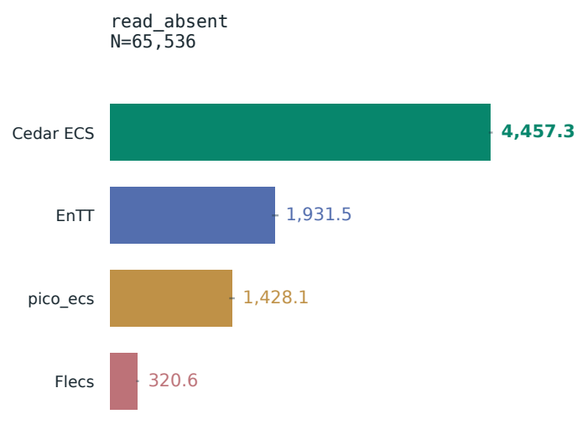


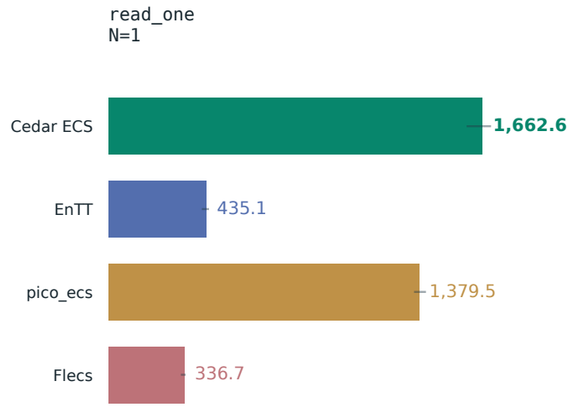


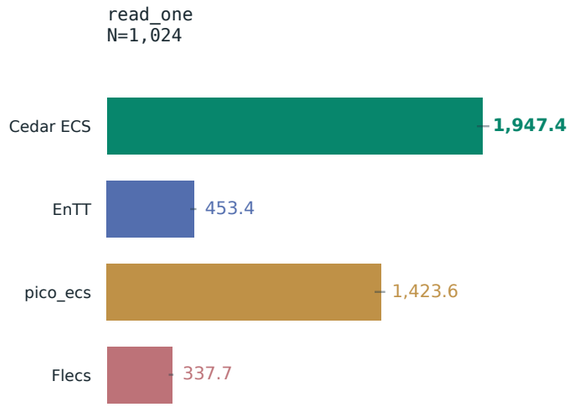


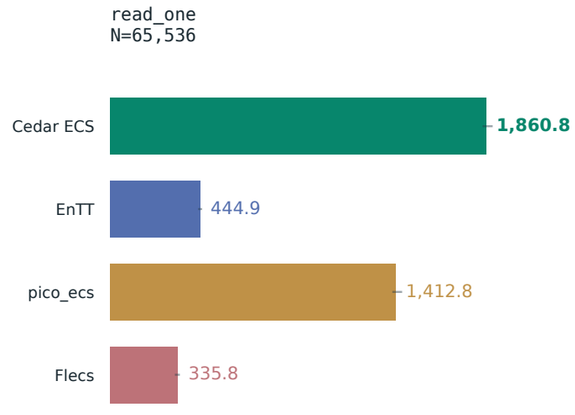


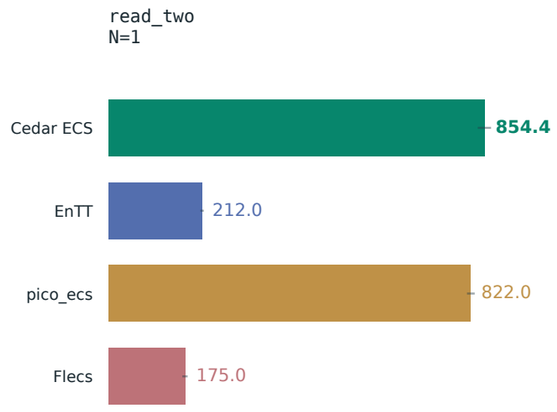


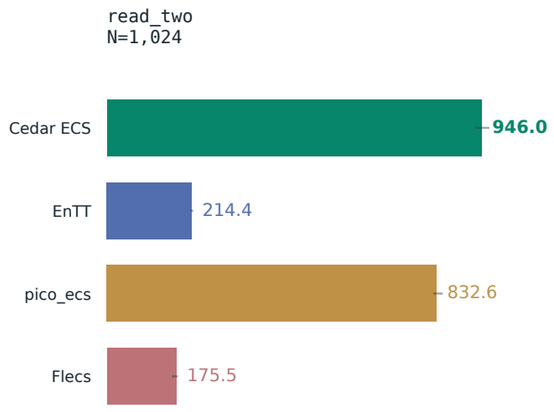


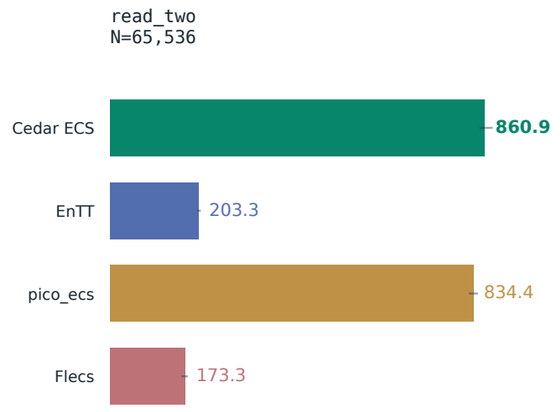








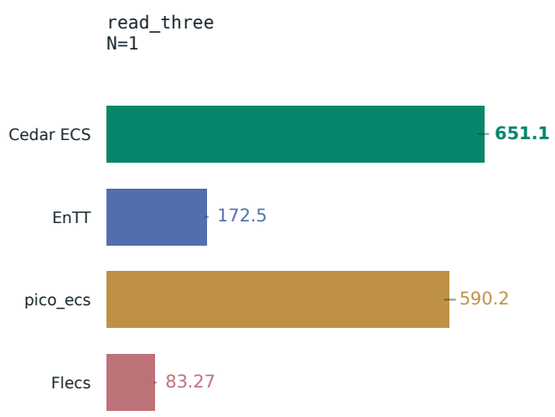


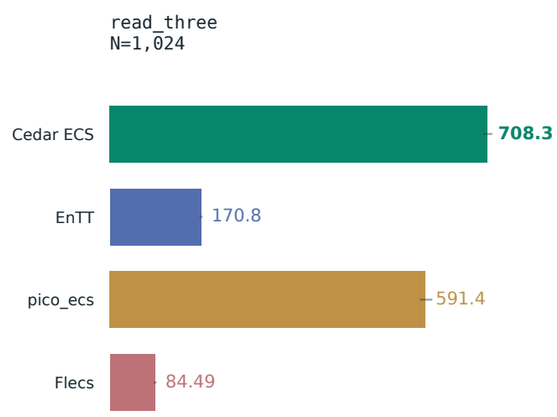


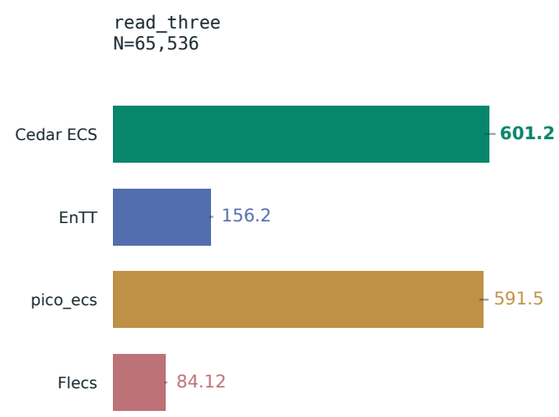
Entity comparison

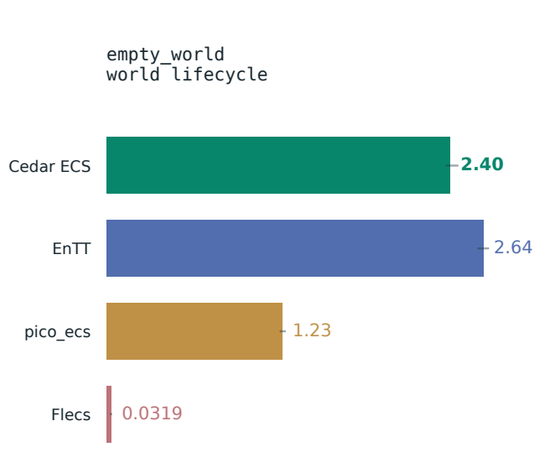
Mop/s ↑

Cedar ECS EnTT pico_ecs Flecs







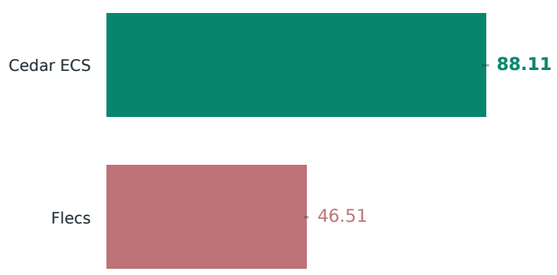


Original · Cedar vs Flecs

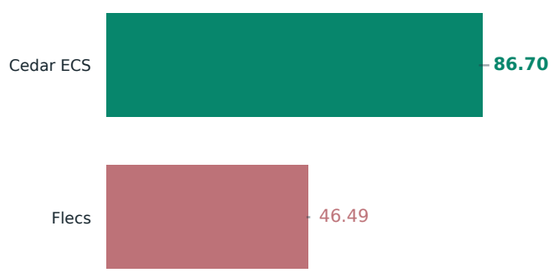
Mop/s ↑

Cedar ECS Flecs

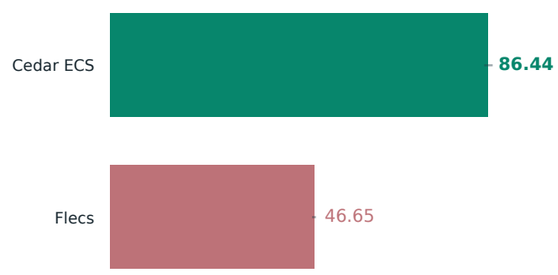
add_remove
N=10k, T=1



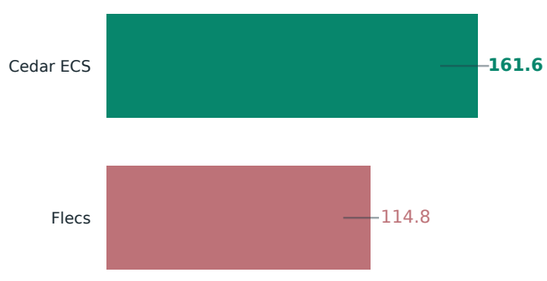
add_remove
N=100k, T=1



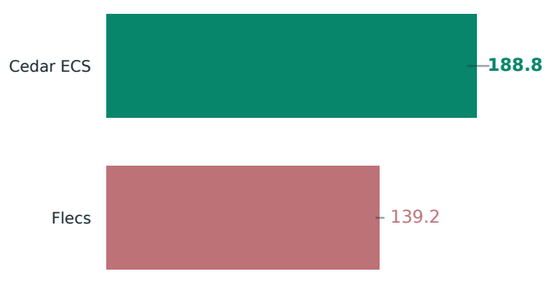
add_remove
N=1M, T=1



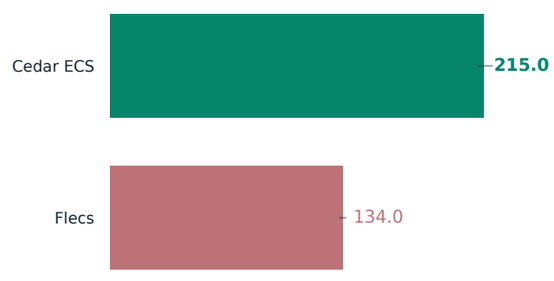
create_empty
N=10k, T=1



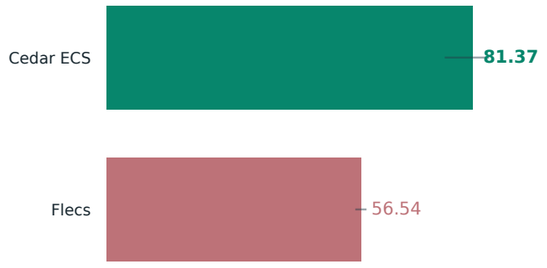
create_empty
N=100k, T=1



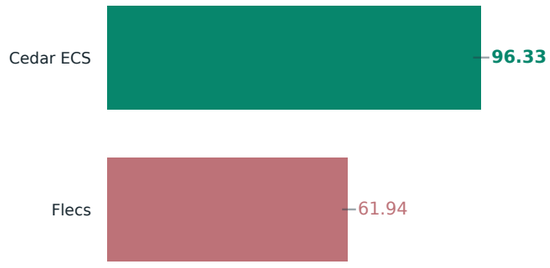
create_empty
N=1M, T=1



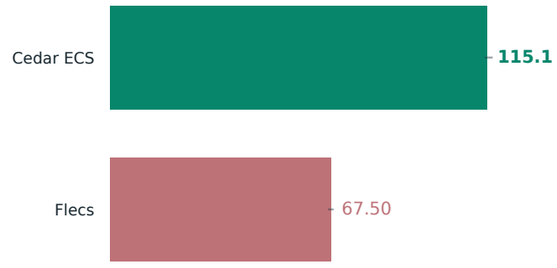
create_pv
N=10k, T=1



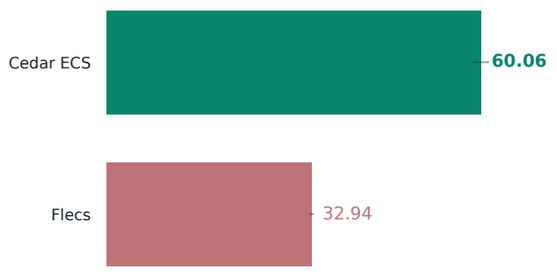
create_pv
N=100k, T=1



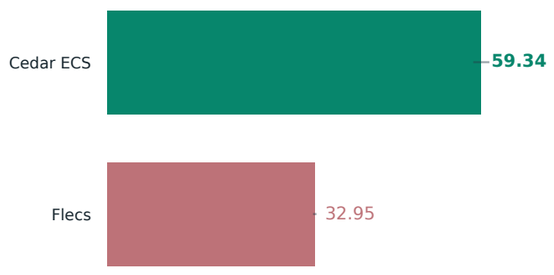
create_pv
N=1M, T=1



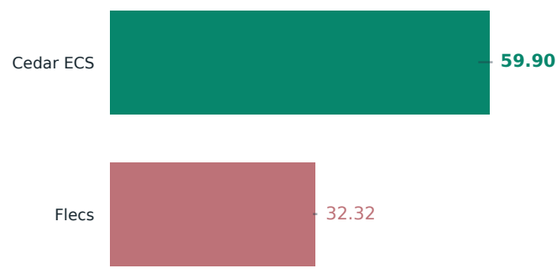
defer_add_remove
N=10k, T=1



defer_add_remove
N=100k, T=1



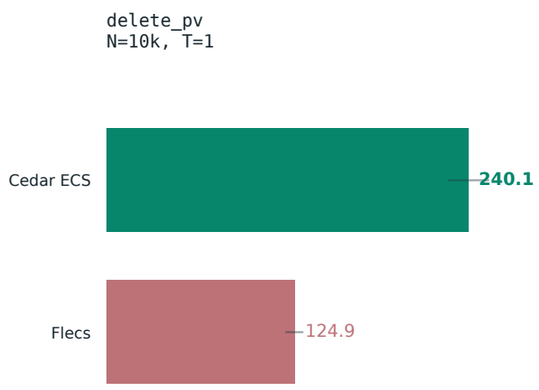
defer_add_remove
N=1M, T=1

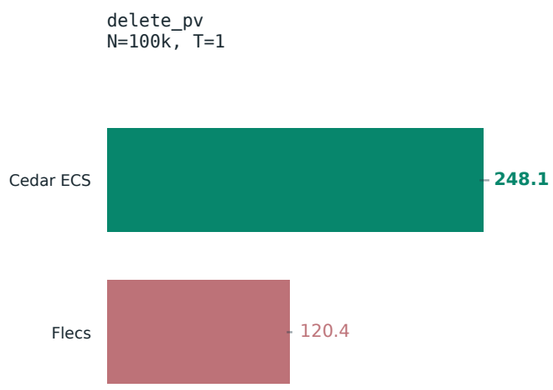


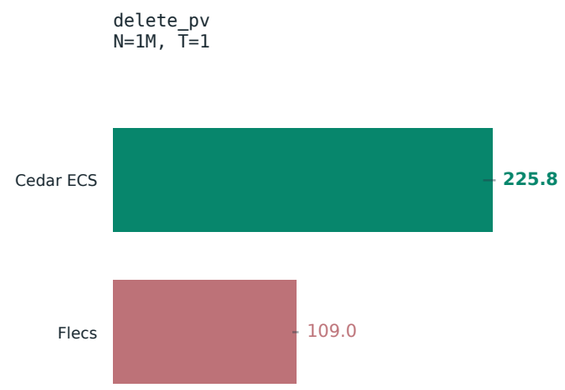
Original · Cedar vs Flecs

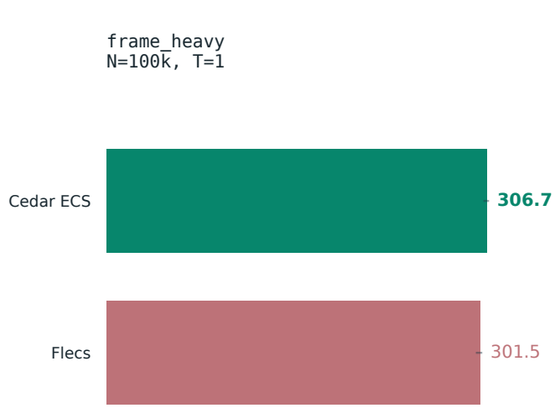
Mop/s ↑

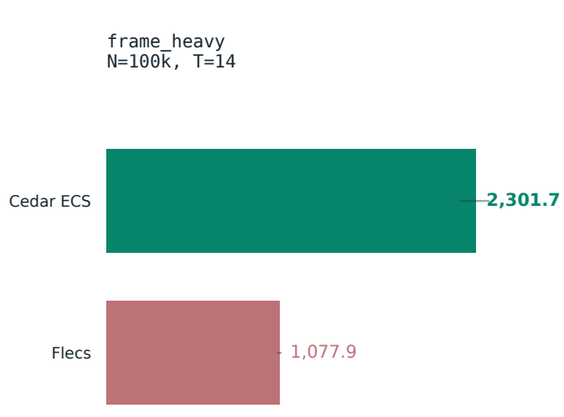
Cedar ECS Flecs

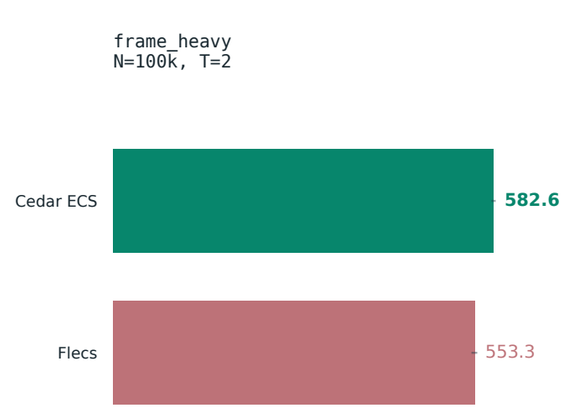


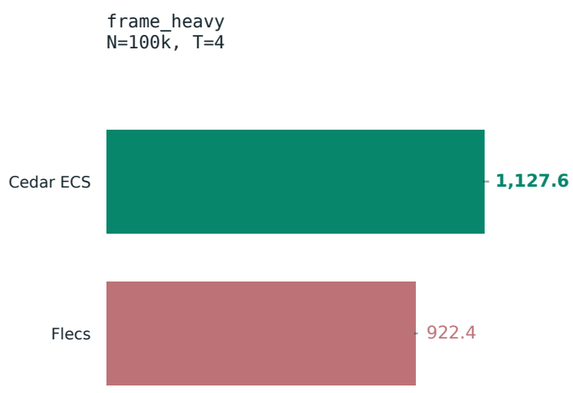


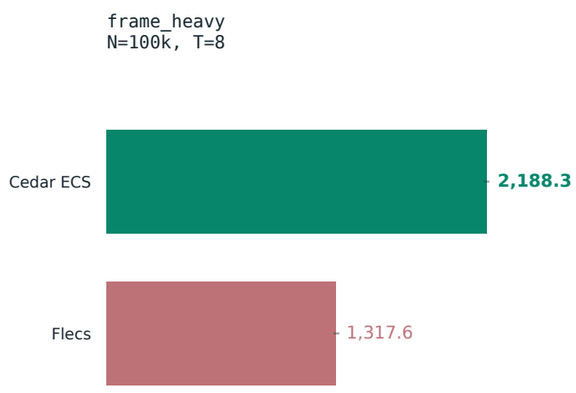


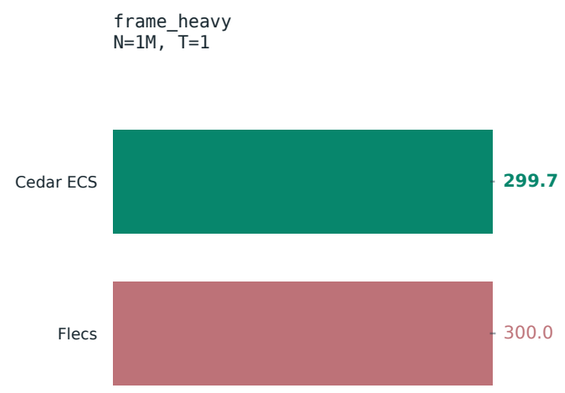


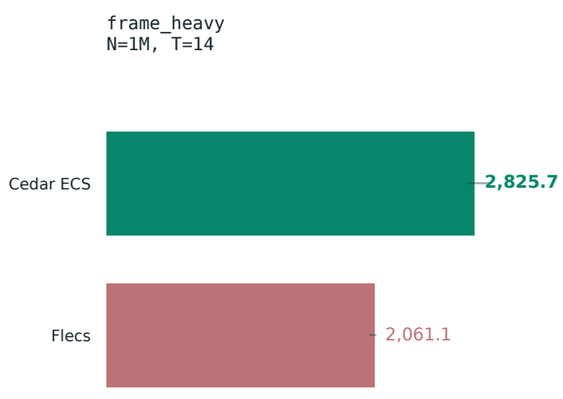


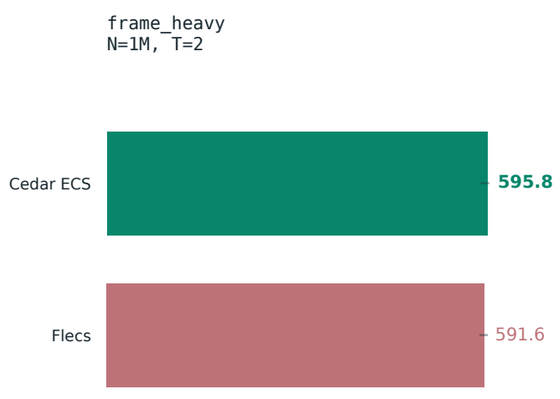


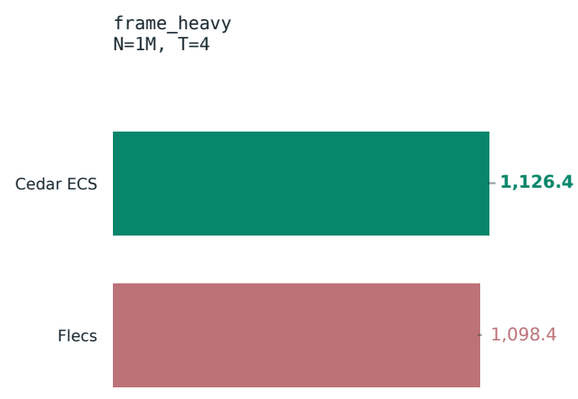












Original · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

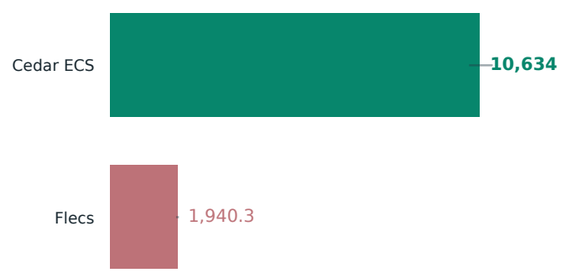
frame_heavy
N=1M, T=8



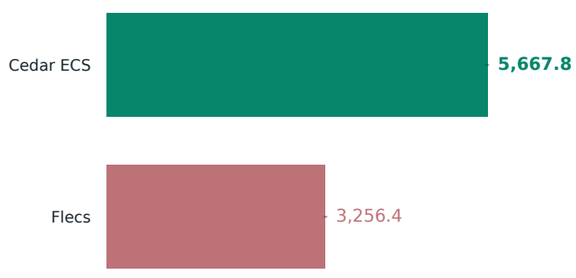
frame_light
N=100k, T=1



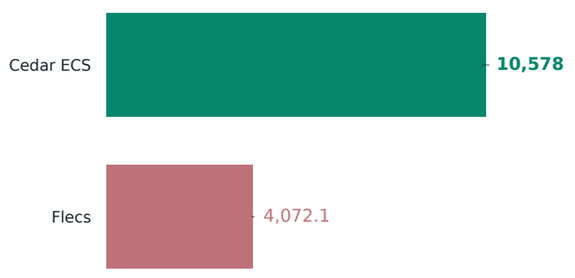
frame_light
N=100k, T=14



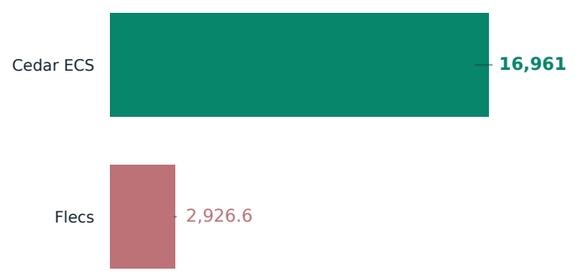
frame_light
N=100k, T=2



frame_light
N=100k, T=4



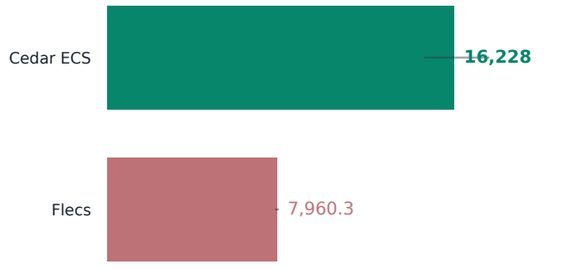
frame_light
N=100k, T=8



frame_light
N=1M, T=1



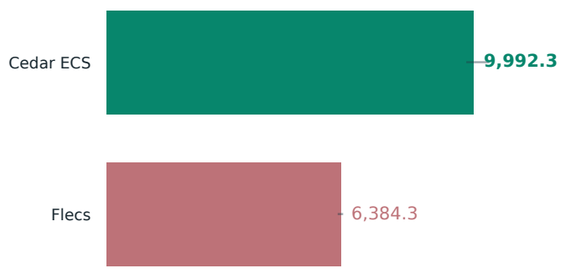
frame_light
N=1M, T=14



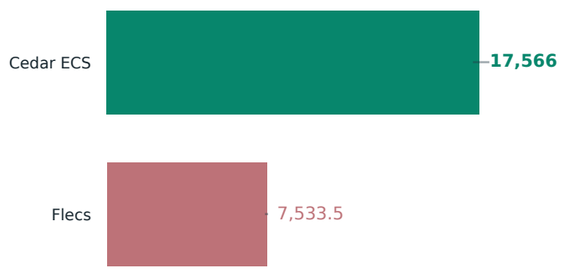
frame_light
N=1M, T=2



frame_light
N=1M, T=4



frame_light
N=1M, T=8



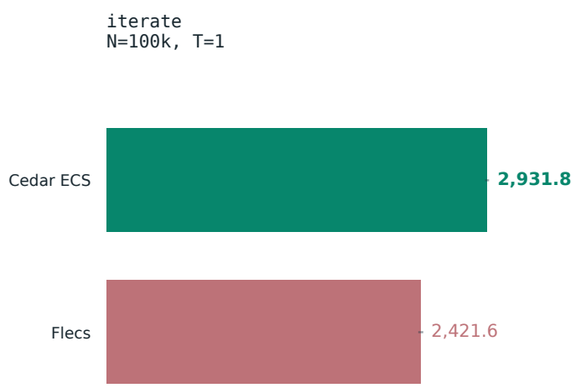
iterate
N=10k, T=1

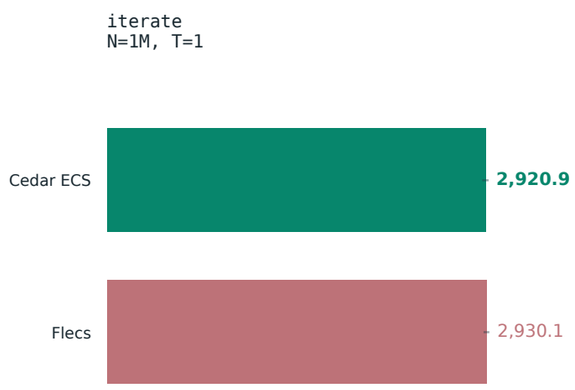


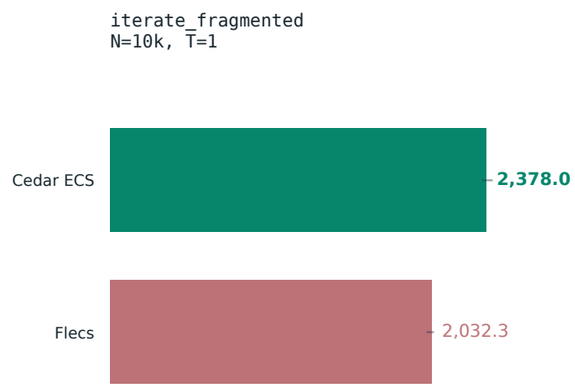
Original · Cedar vs Flecs

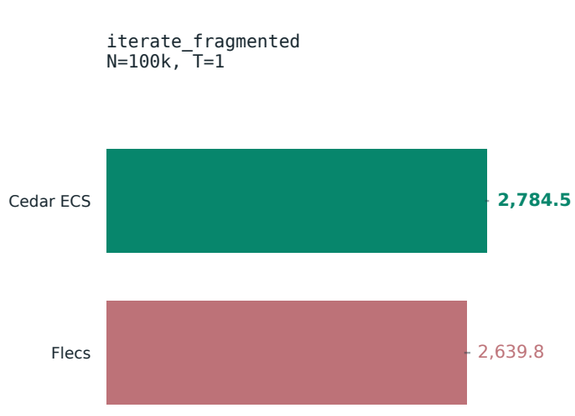
Mop/s ↑

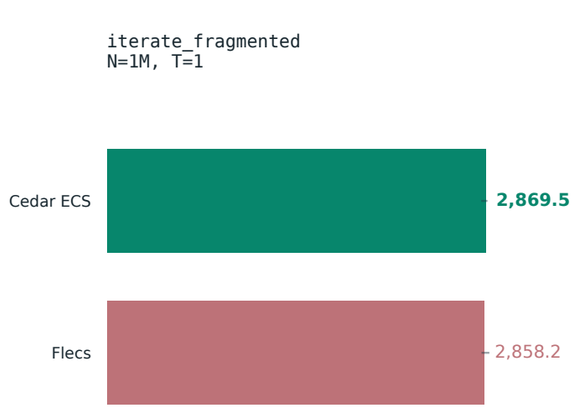
Cedar ECS Flecs

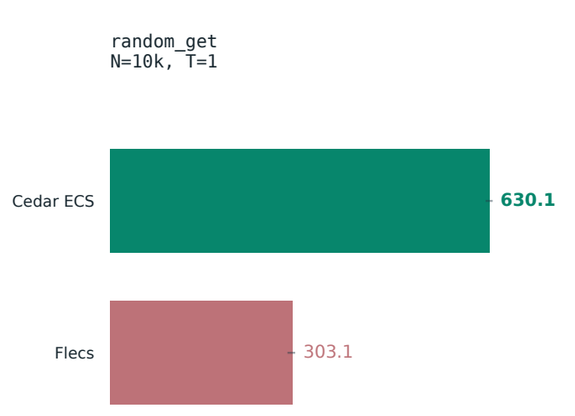


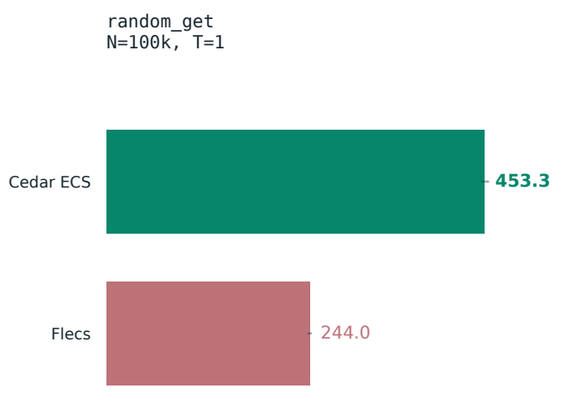


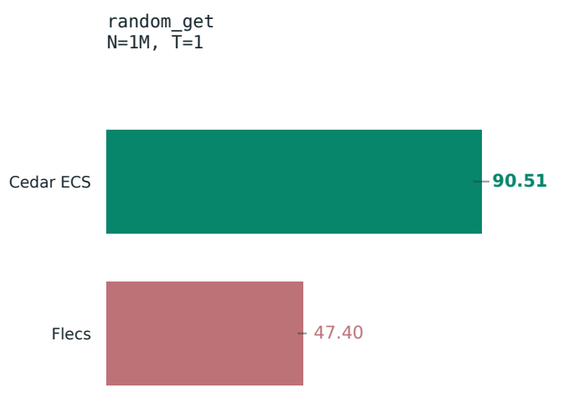


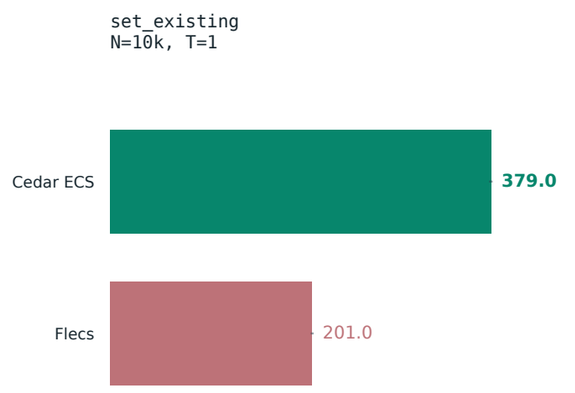


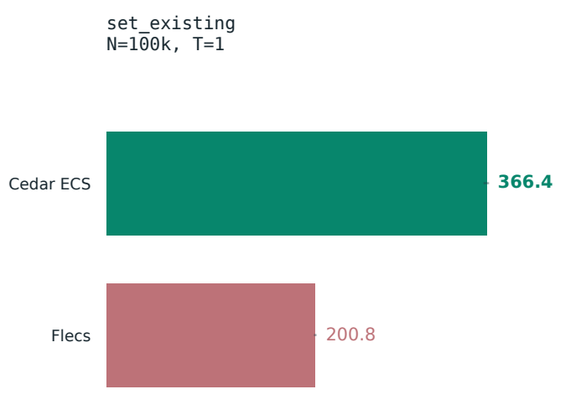


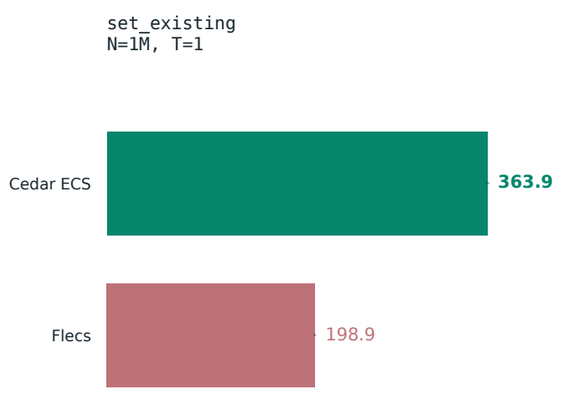


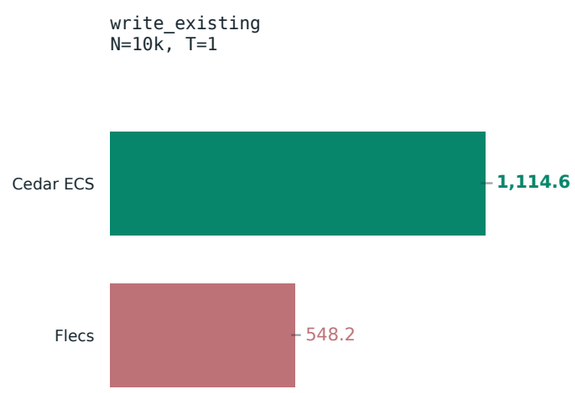








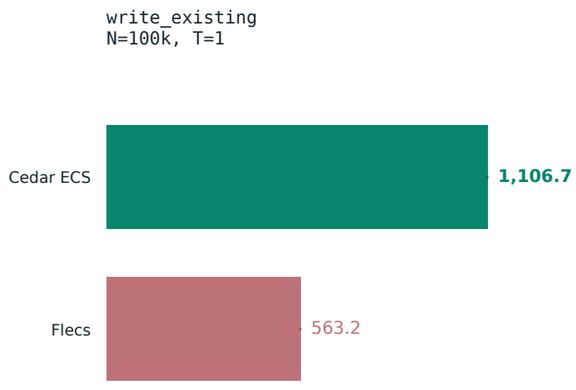


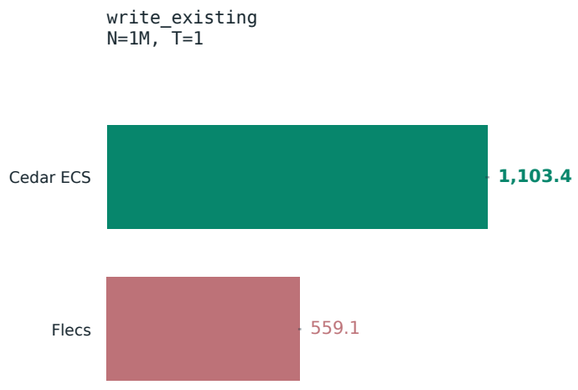


Original · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

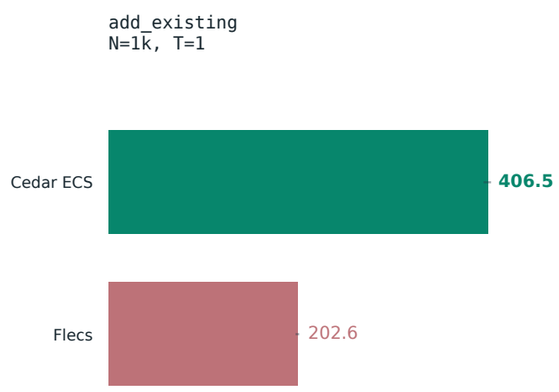


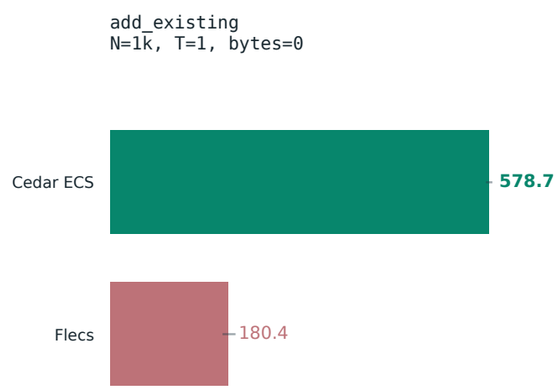


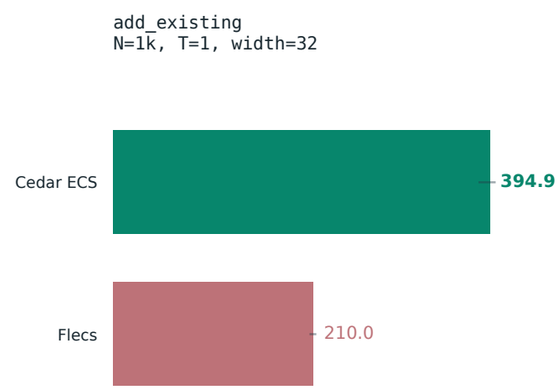
Packed · Cedar vs Flecs

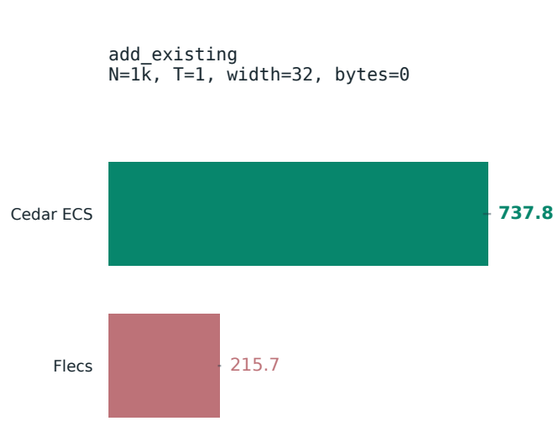
Mop/s ↑

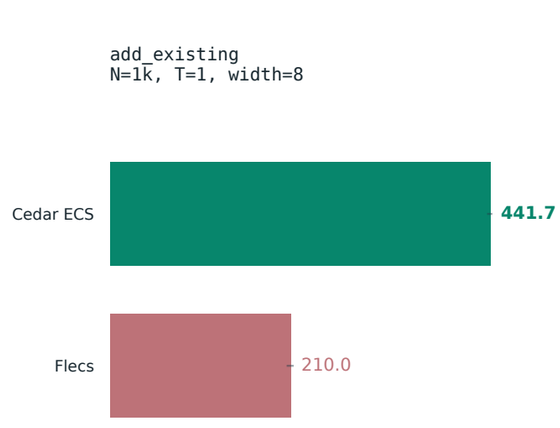
Cedar ECS Flecs

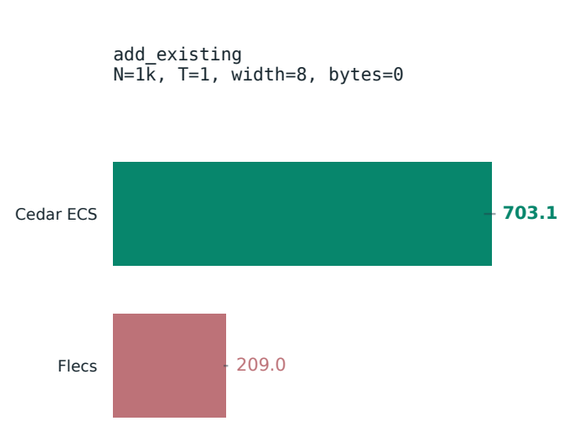


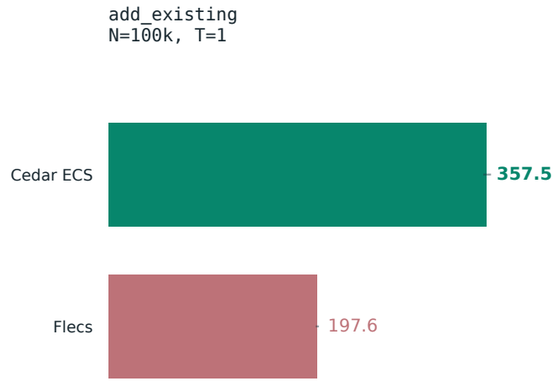


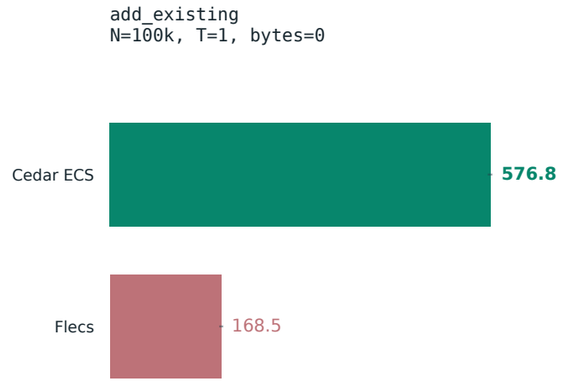


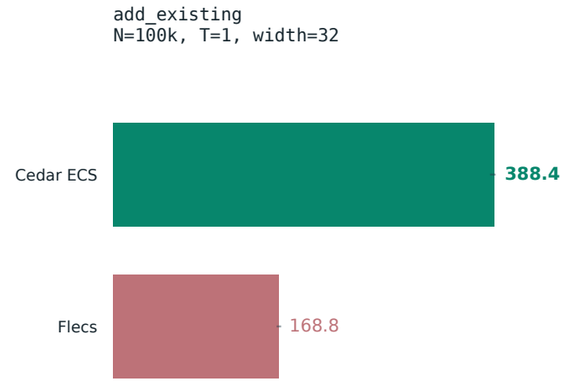


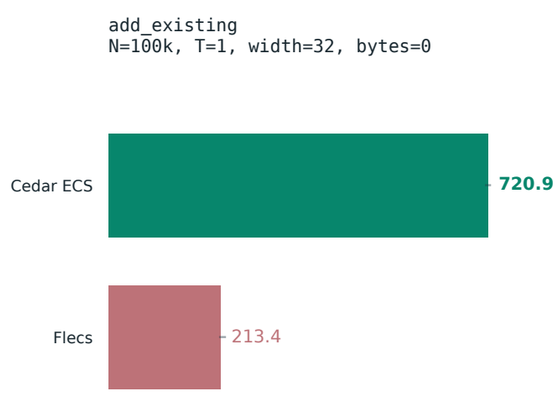


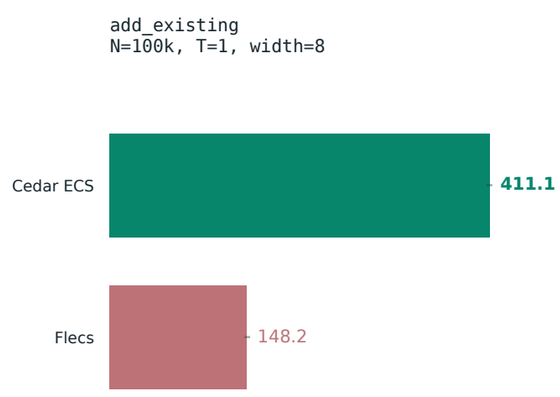


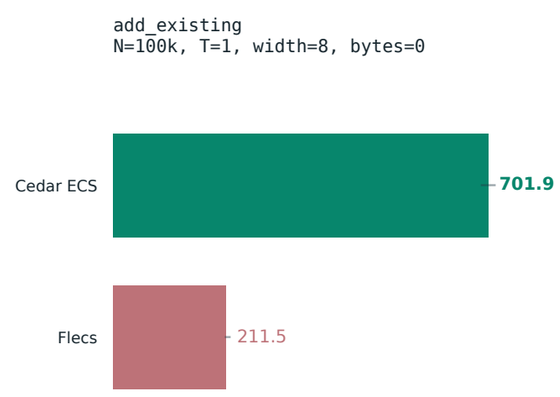












Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

churn
N=10k, T=1, width=2, percent=1, shuffle=1



churn
N=10k, T=1, width=2, percent=50, shuffle=1



churn
N=10k, T=1, width=2, shuffle=1



churn
N=100k, T=1, width=2, percent=1, shuffle=1



churn
N=100k, T=1, width=2, percent=50, shuffle=1



churn
N=100k, T=1, width=2, shuffle=1



create
N=1k, T=1



create
N=1k, T=1, reserve=1



create
N=1k, T=1, width=32



create
N=1k, T=1, width=32, reserve=1



create
N=1k, T=1, width=8



create
N=1k, T=1, width=8, reserve=1



Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

create
N=10k, T=1, width=8, bytes=64



create
N=10k, T=1, width=8, bytes=64, reserve=1



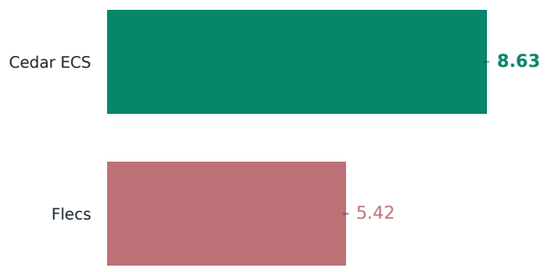
create
N=100k, T=1



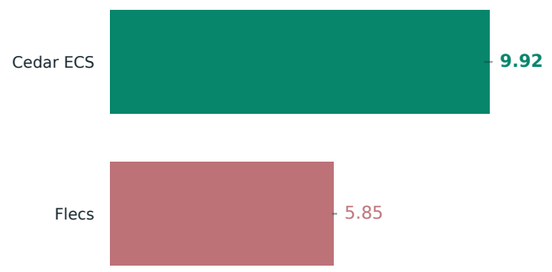
create
N=100k, T=1, reserve=1



create
N=100k, T=1, width=32



create
N=100k, T=1, width=32, reserve=1



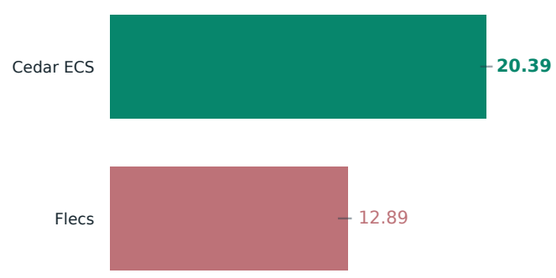
create
N=100k, T=1, width=8



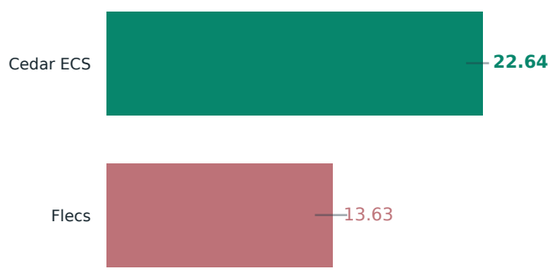
create
N=100k, T=1, width=8, reserve=1



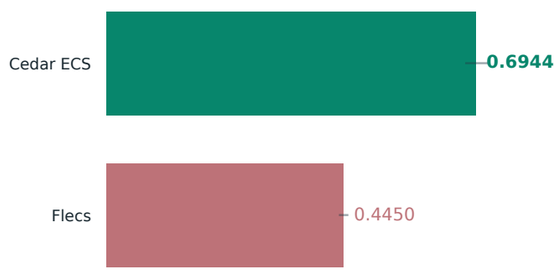
create_incremental
N=1k, T=1



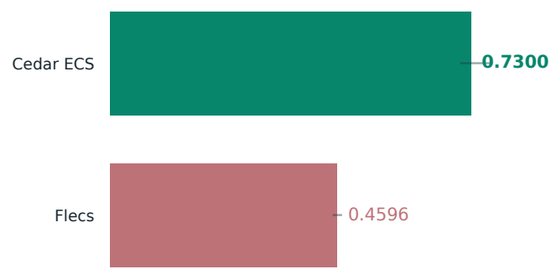
create_incremental
N=1k, T=1, reserve=1



create_incremental
N=1k, T=1, width=32



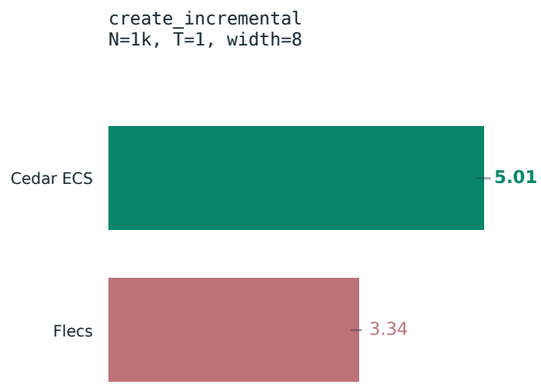
create_incremental
N=1k, T=1, width=32, reserve=1

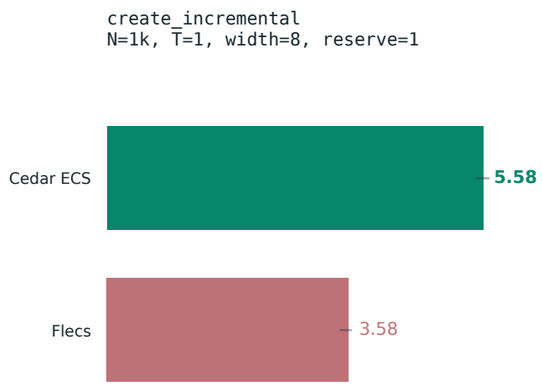


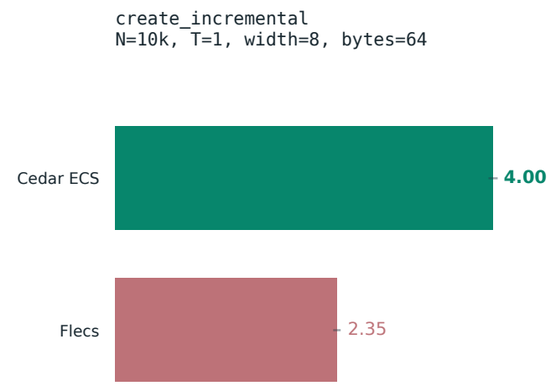
Packed · Cedar vs Flecs

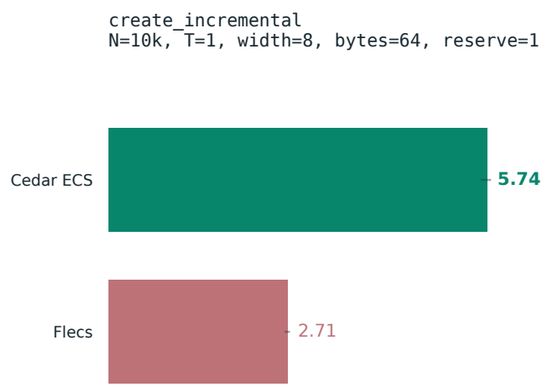
Mop/s ↑

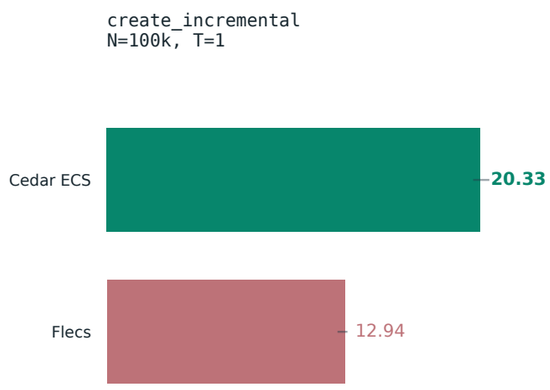
Cedar ECS Flecs

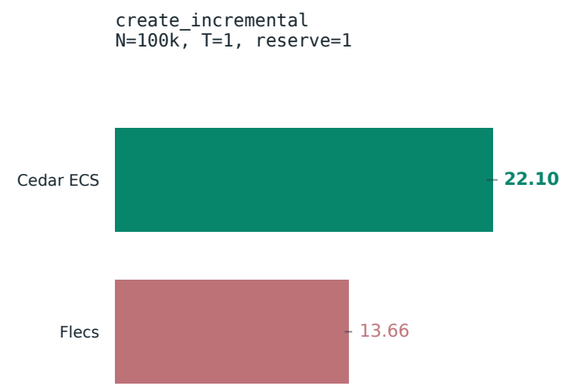


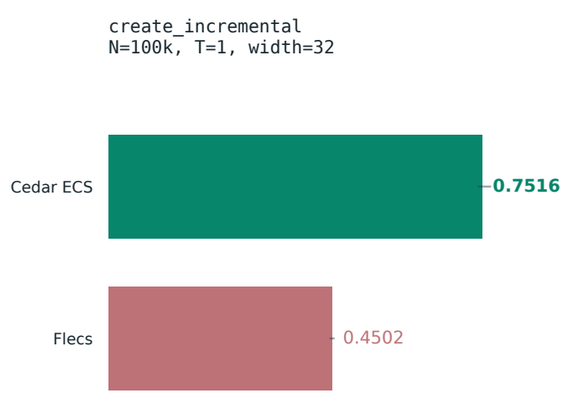


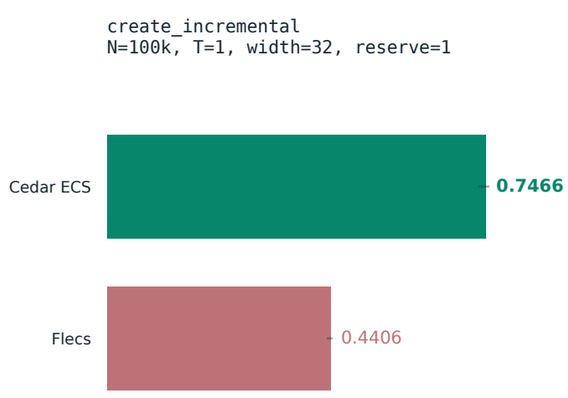


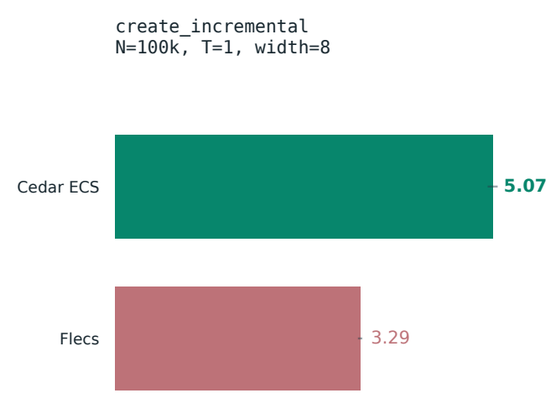


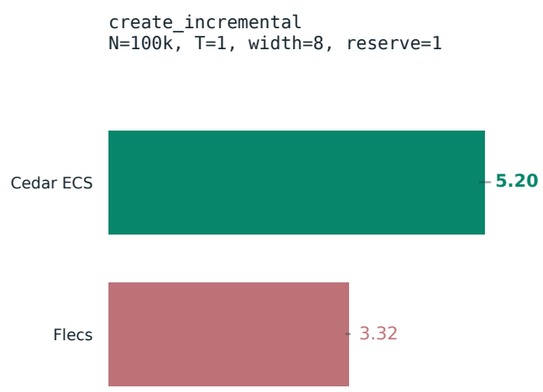


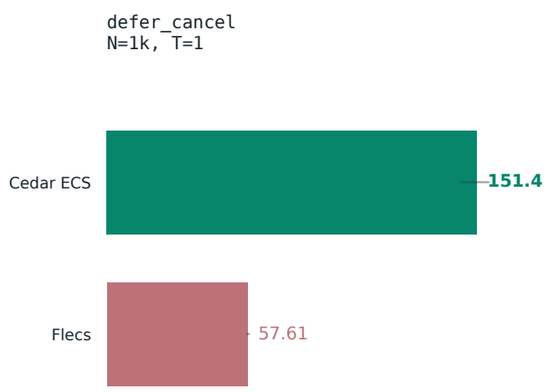


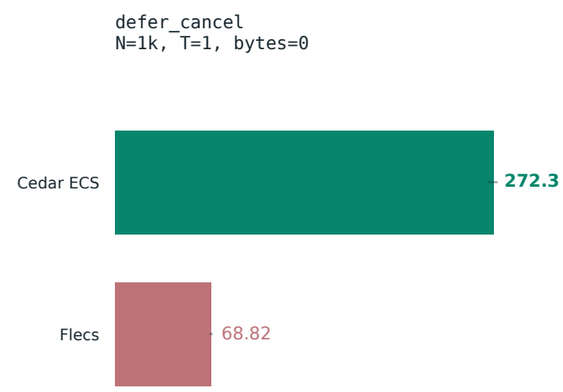








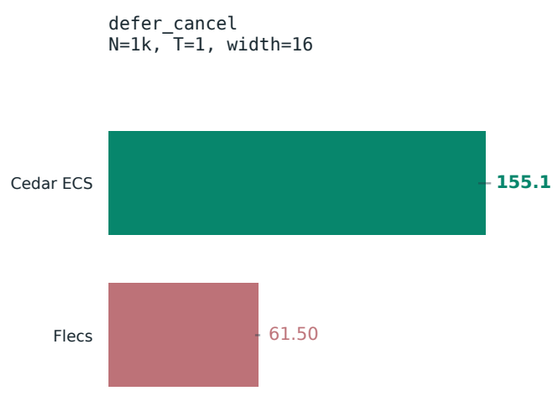


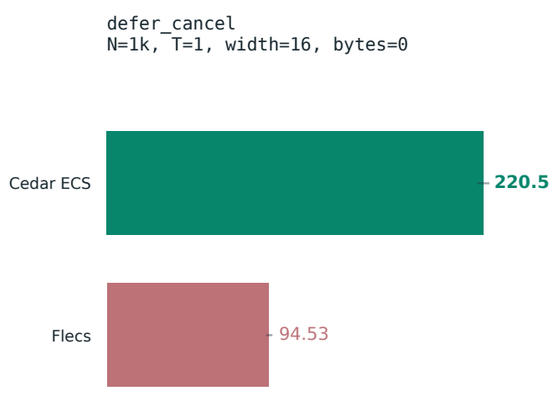


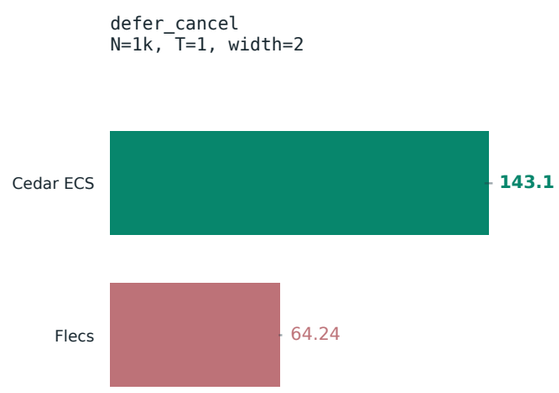
Packed · Cedar vs Flecs

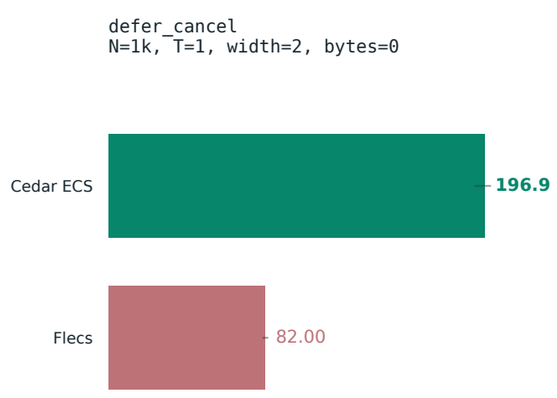
Mop/s ↑

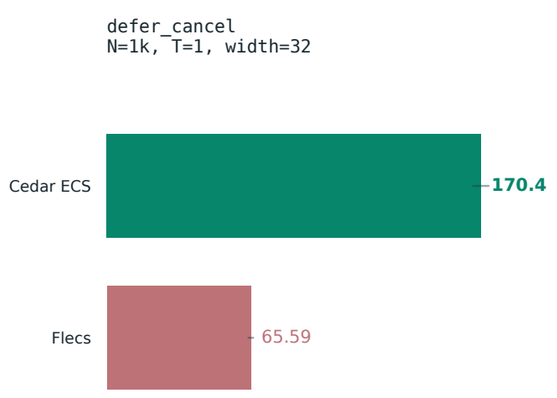
Cedar ECS Flecs

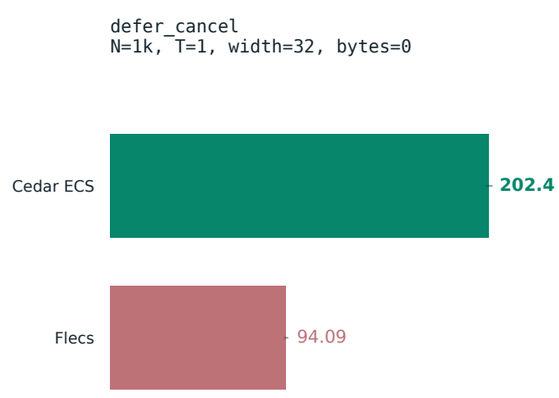


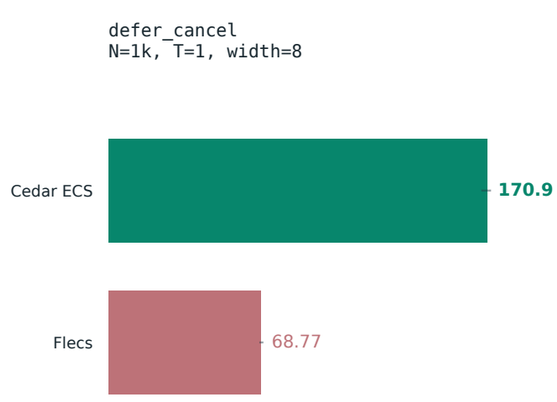


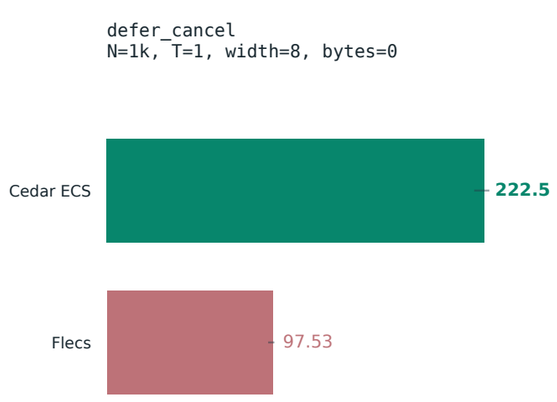


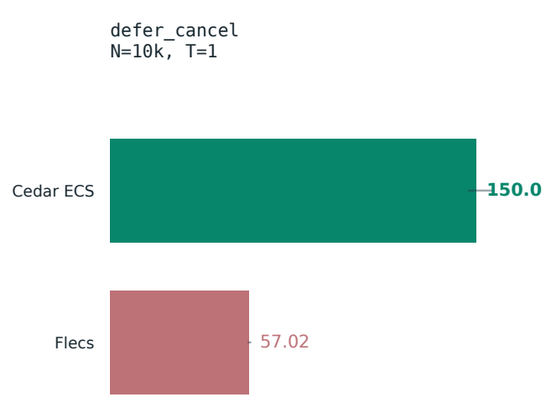


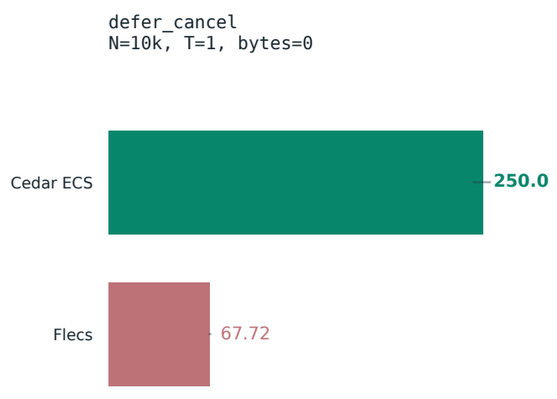


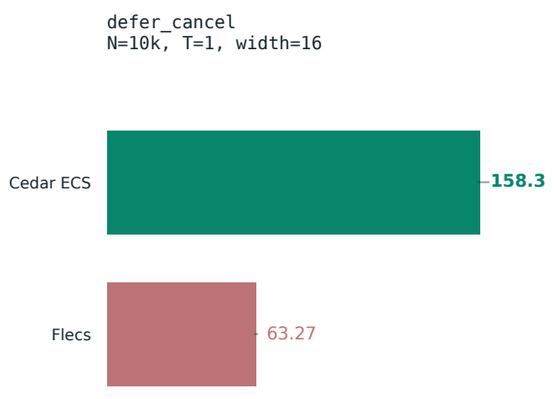


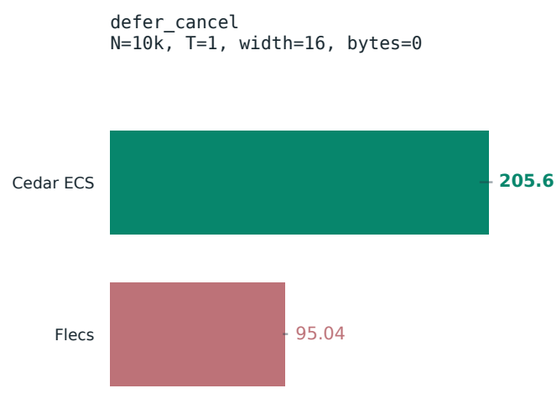








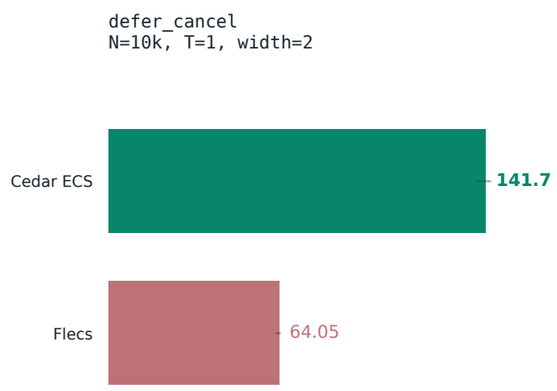


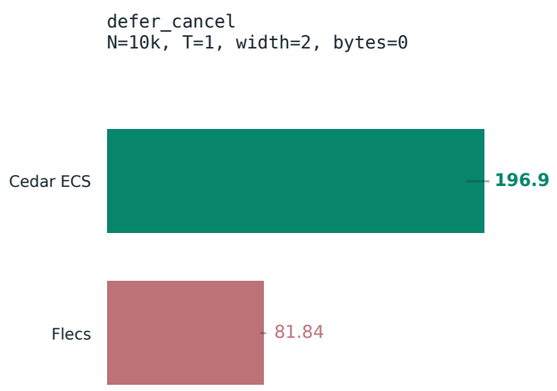


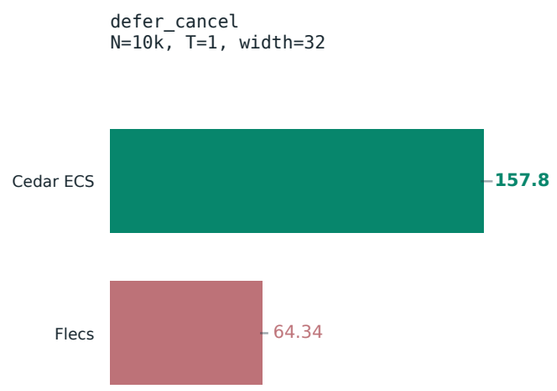
Packed · Cedar vs Flecs

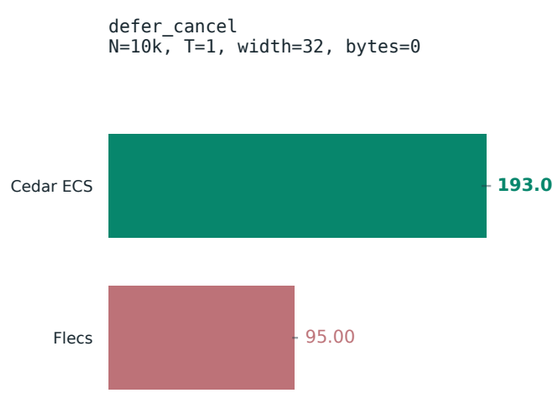
Mop/s ↑

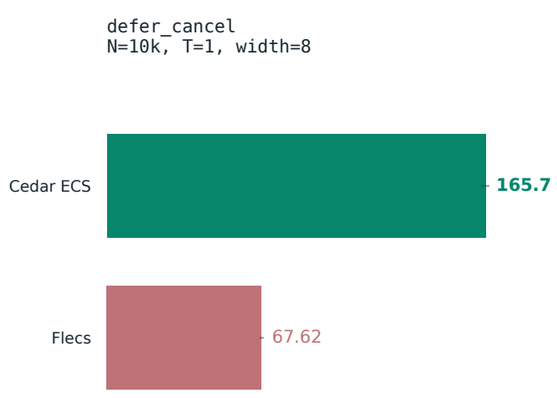
Cedar ECS Flecs

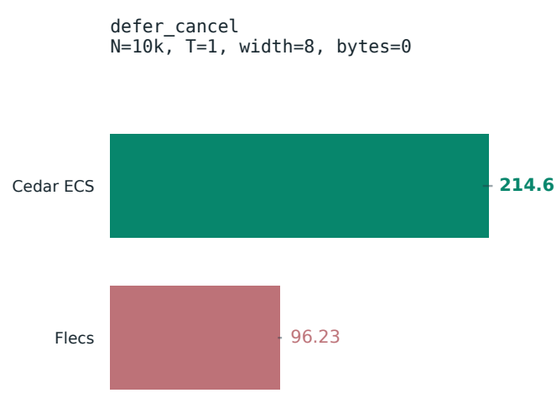


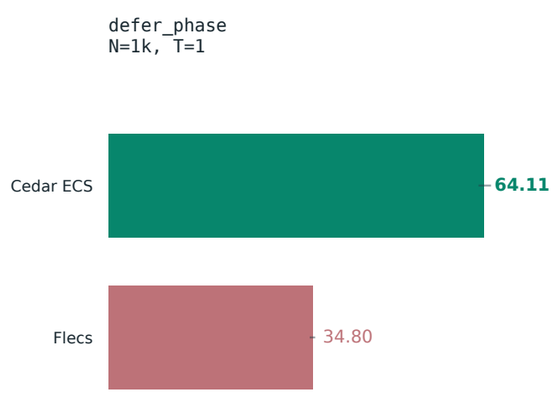


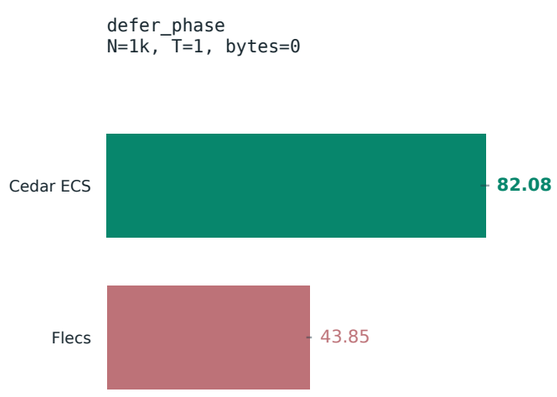


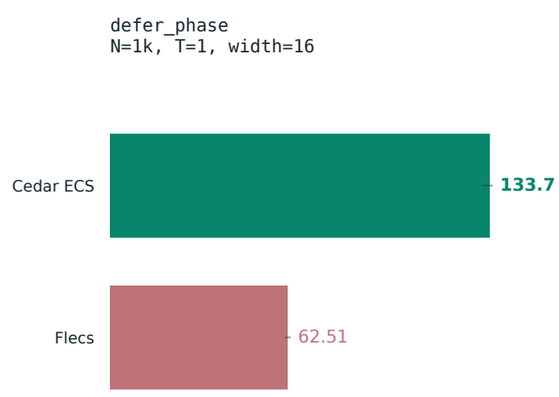


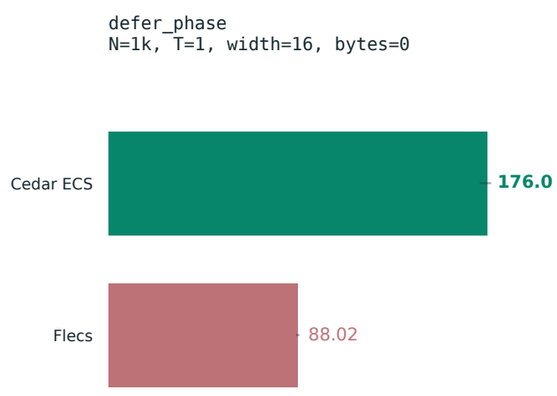


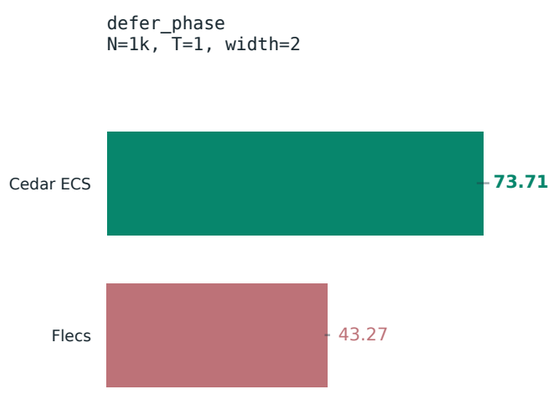


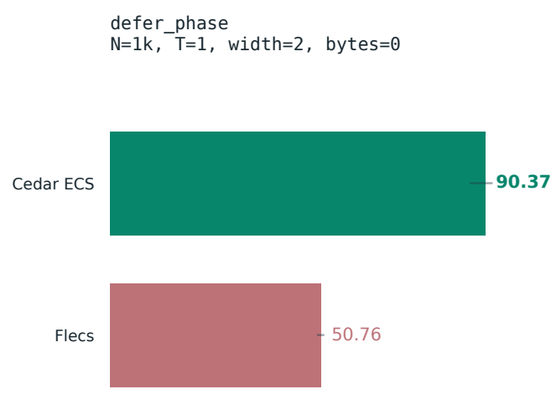








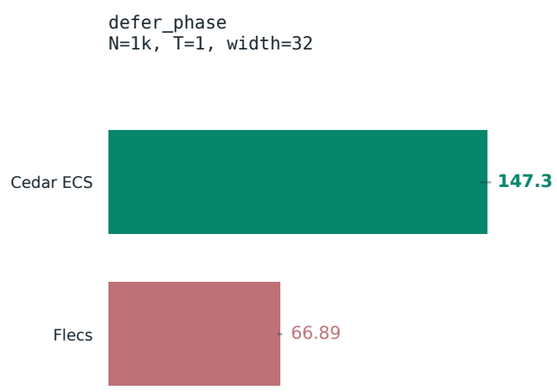


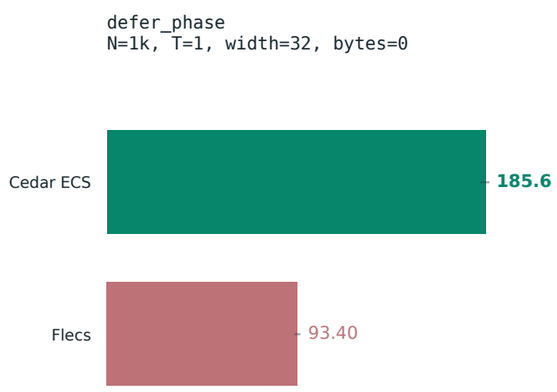


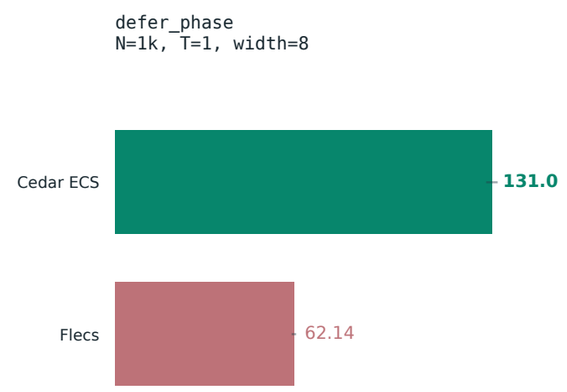
Packed · Cedar vs Flecs

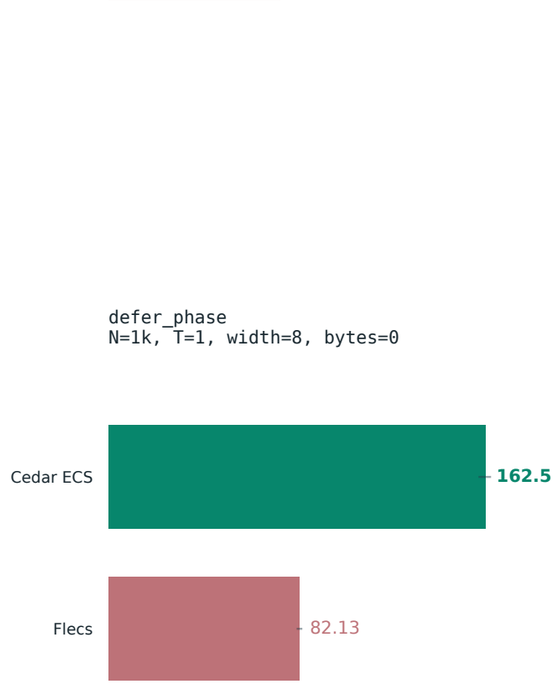
Mop/s ↑

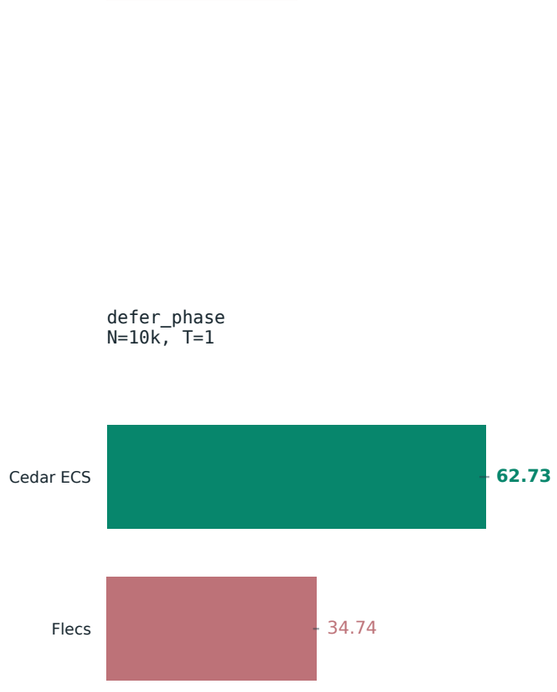
Cedar ECS Flecs

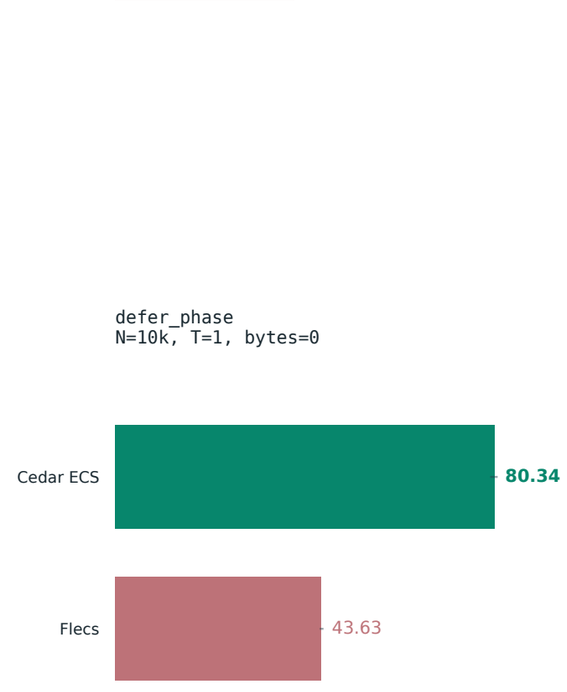


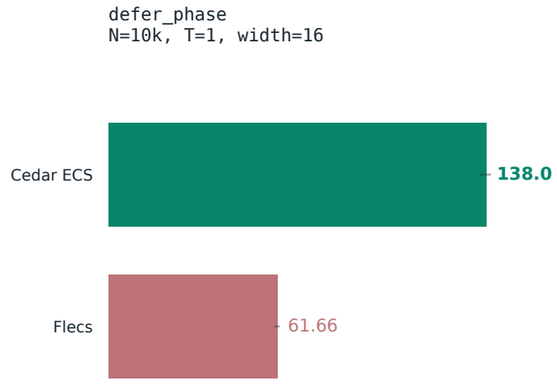


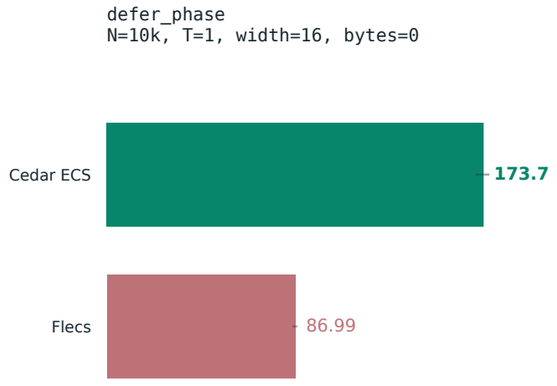






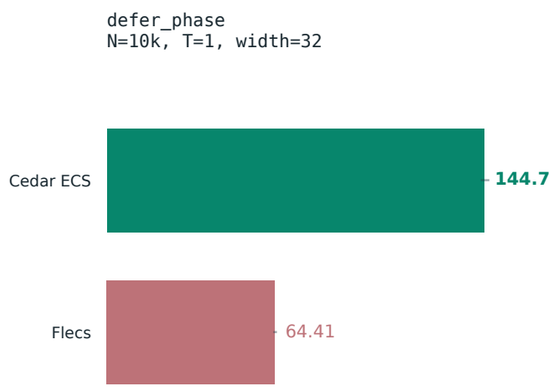


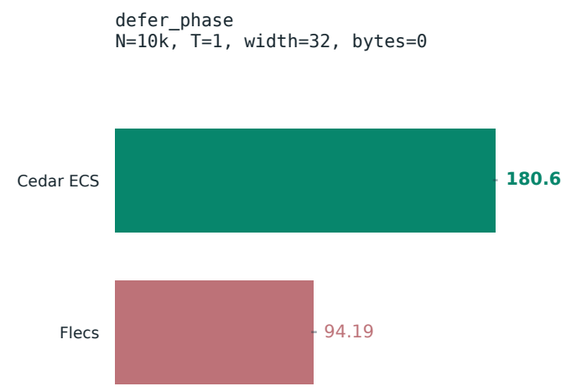








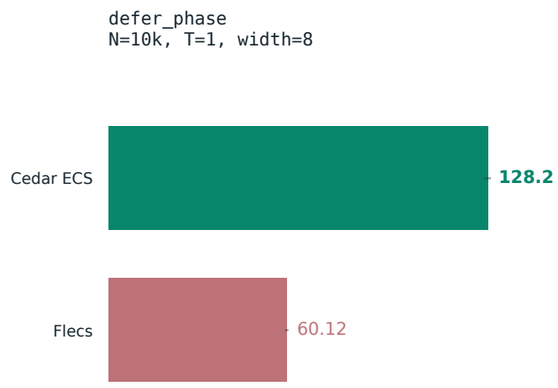


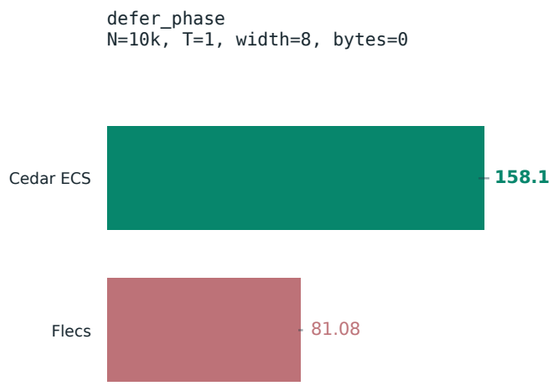


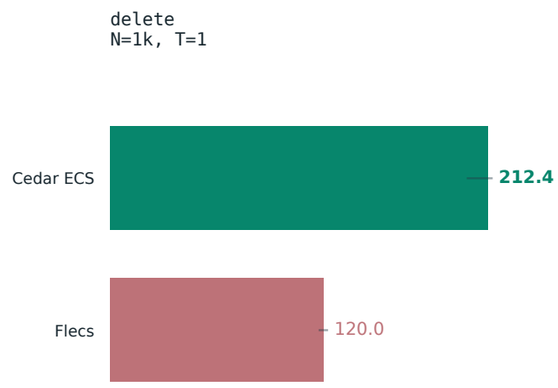
Packed · Cedar vs Flecs

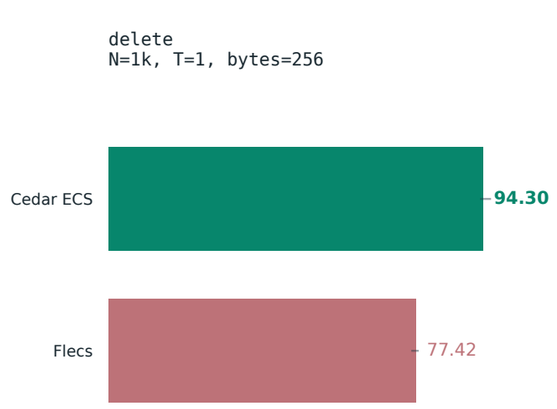
Mop/s ↑

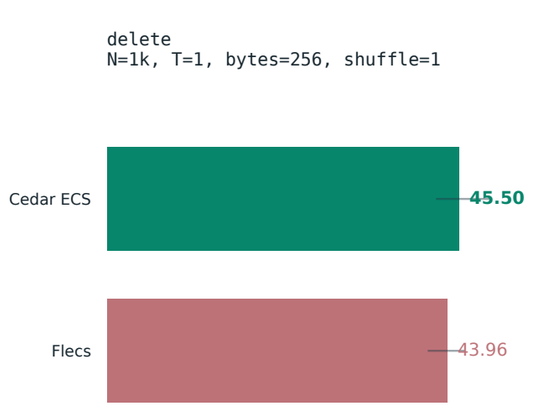
Cedar ECS Flecs

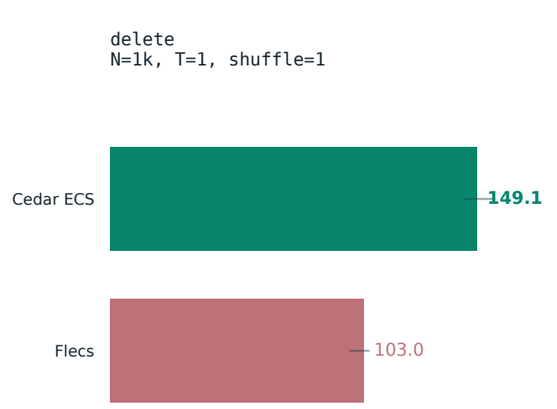


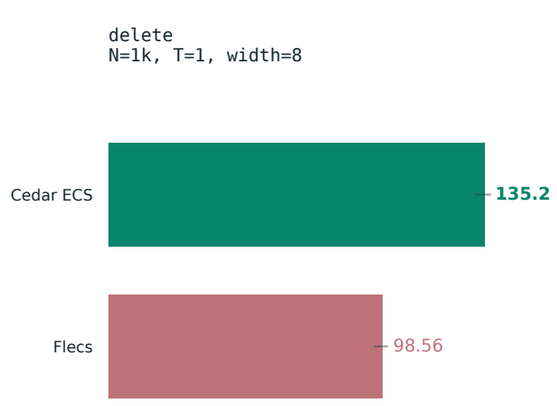


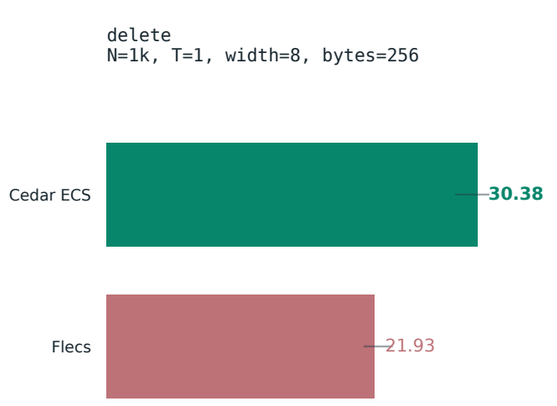


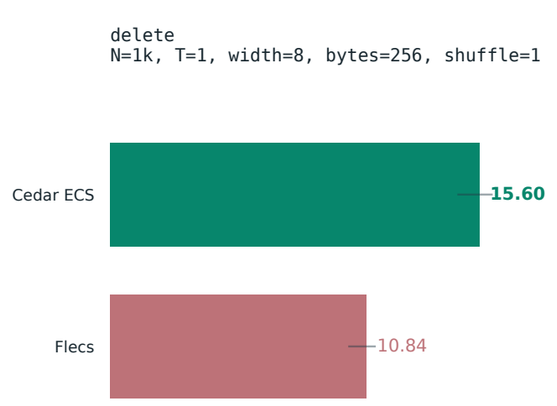




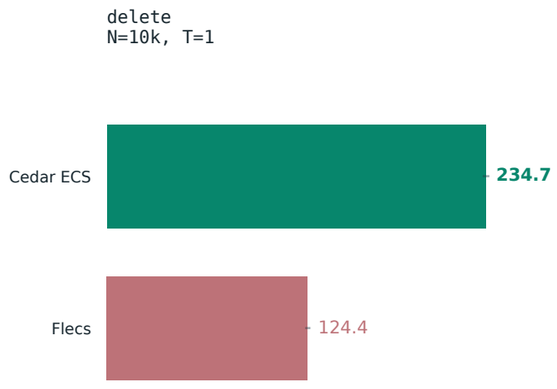


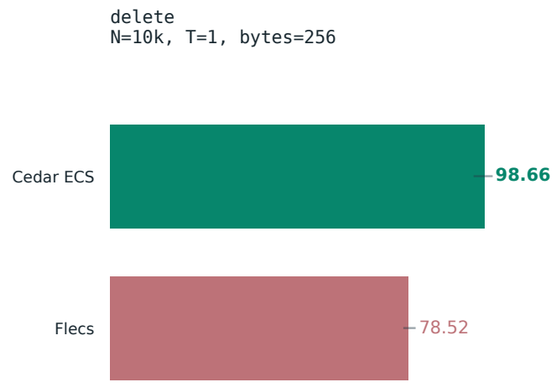








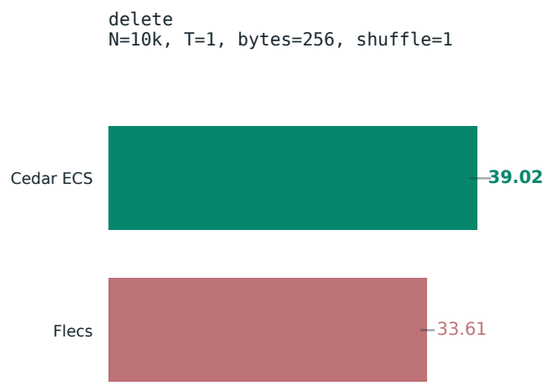


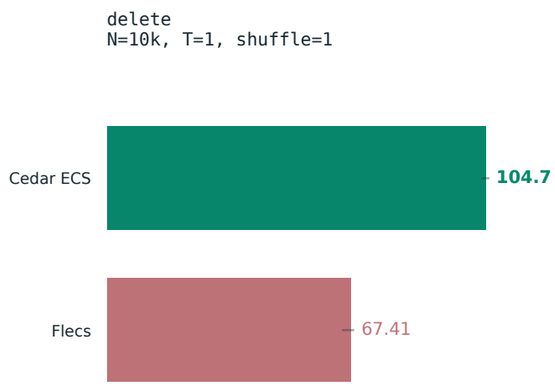


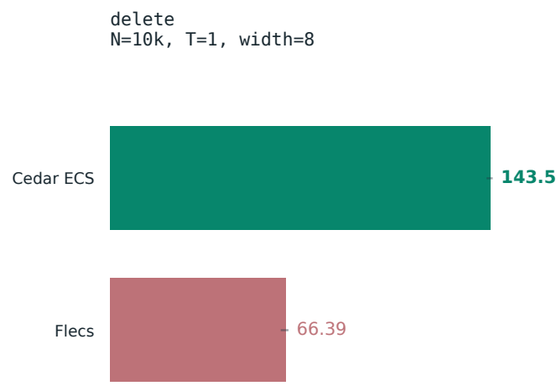
Packed · Cedar vs Flecs

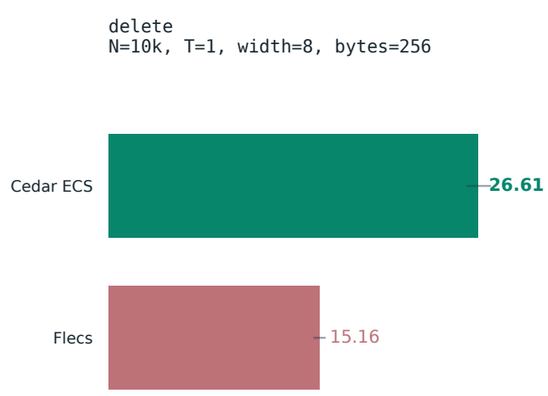
Mop/s ↑

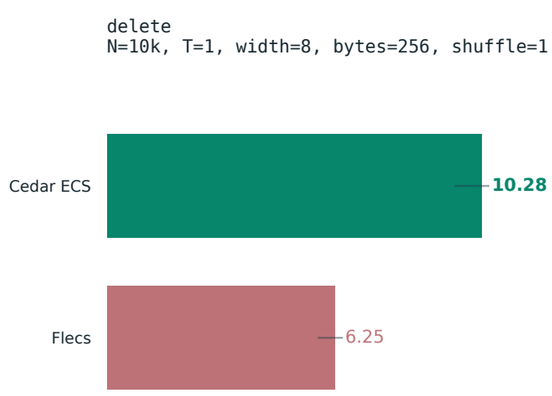
Cedar ECS Flecs



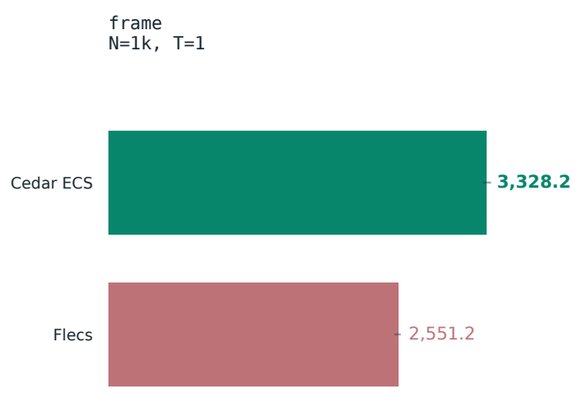




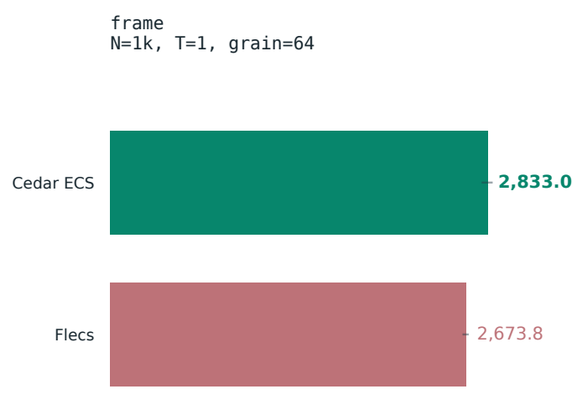


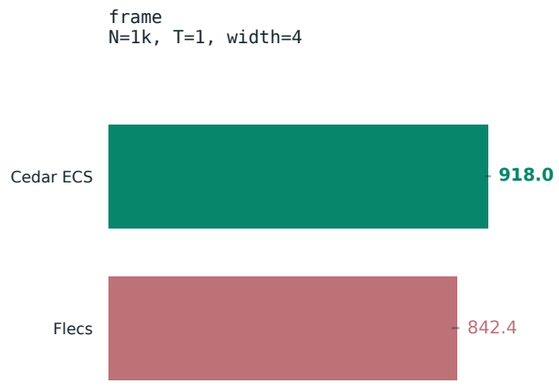


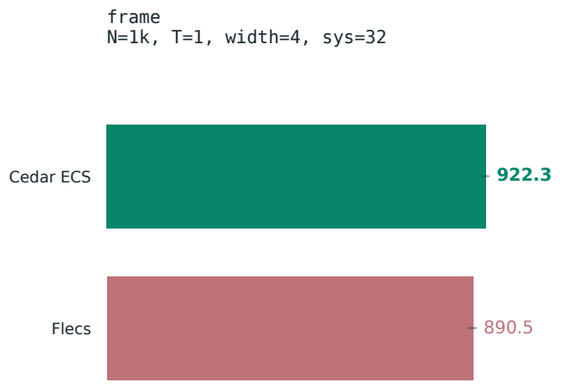


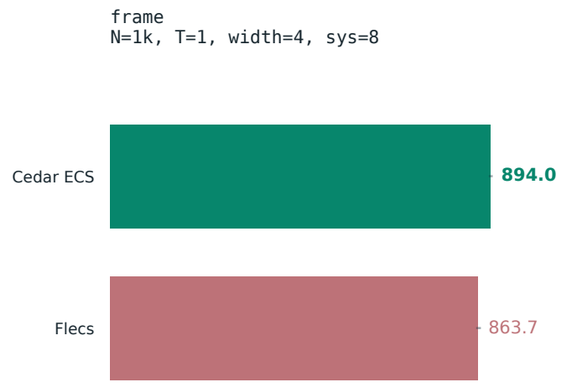








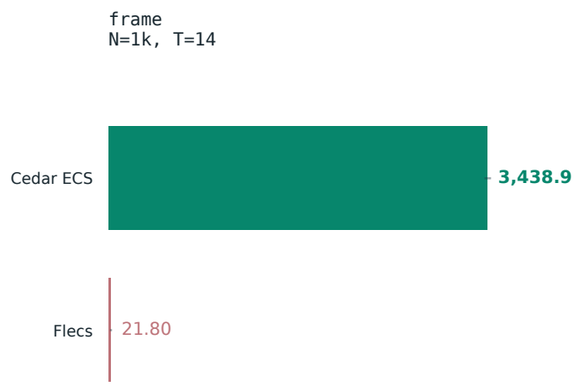


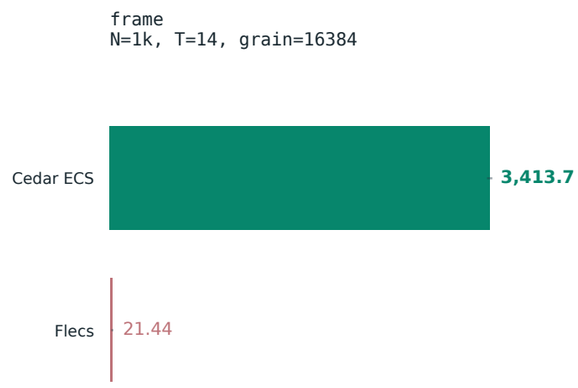


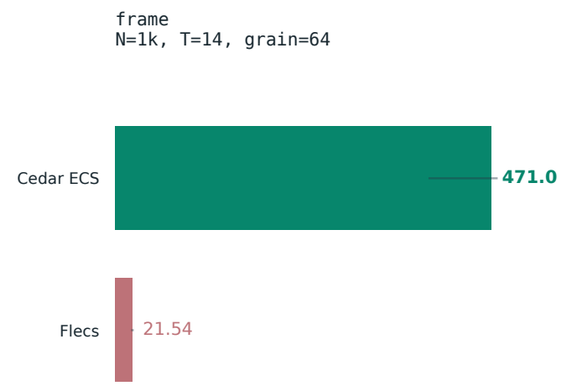
Packed · Cedar vs Flecs

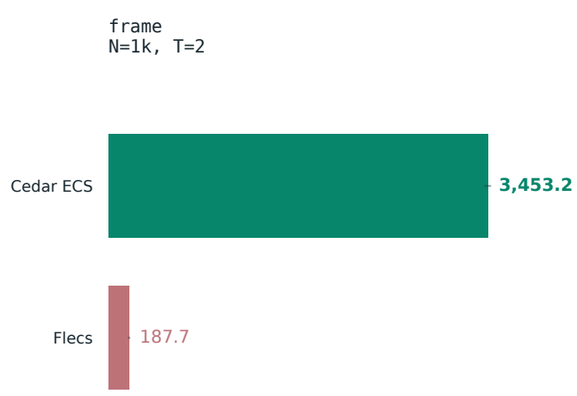
Mop/s ↑

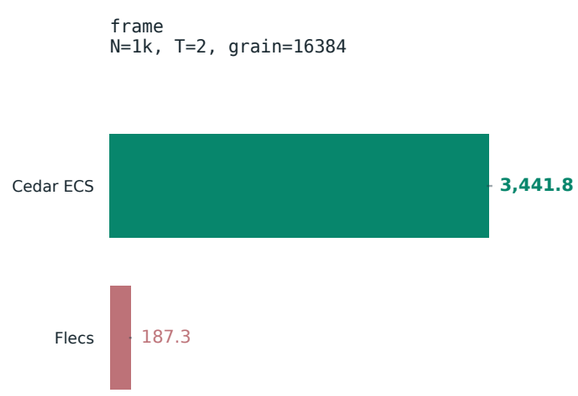
Cedar ECS Flecs

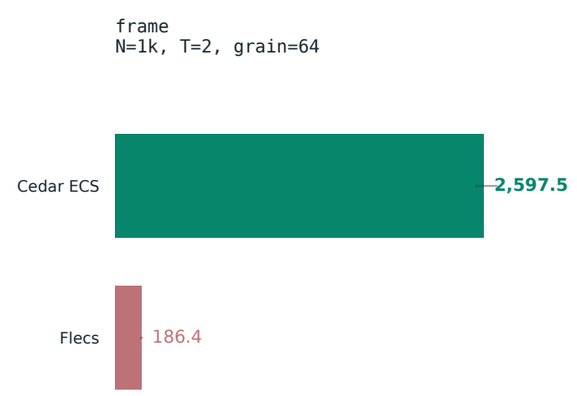


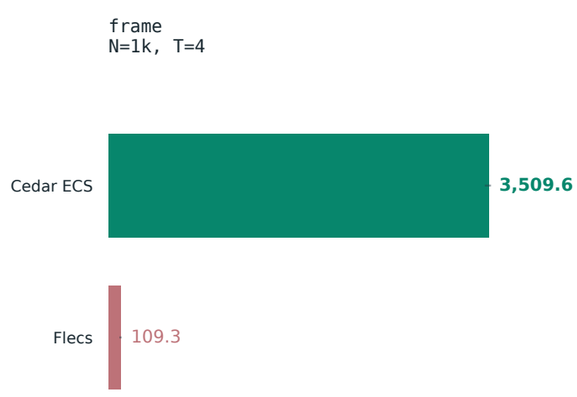




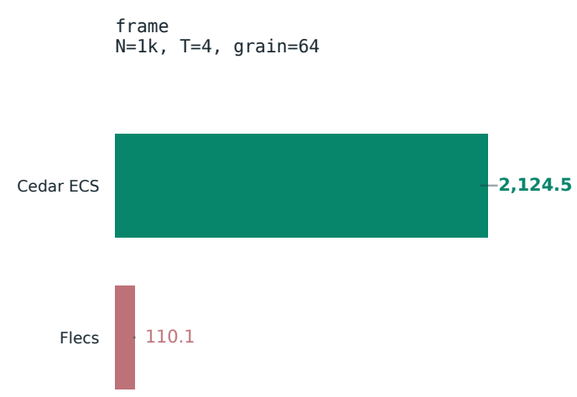


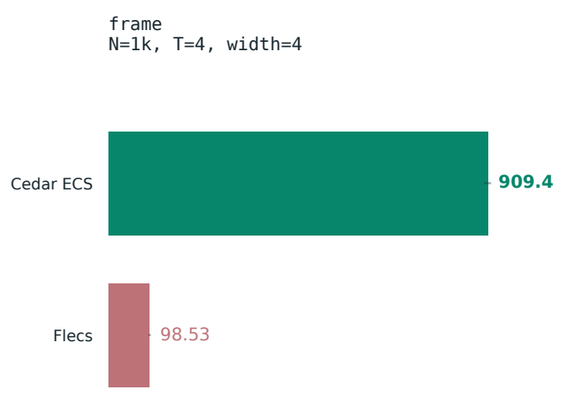


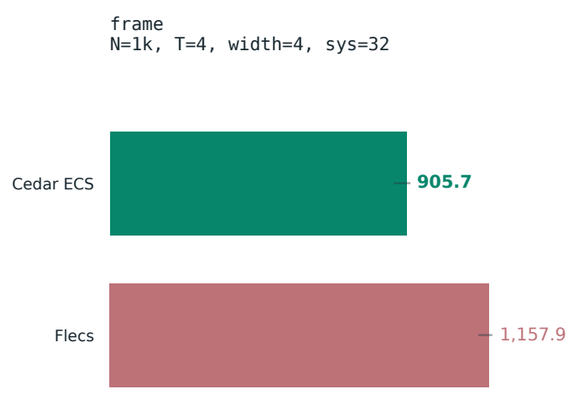










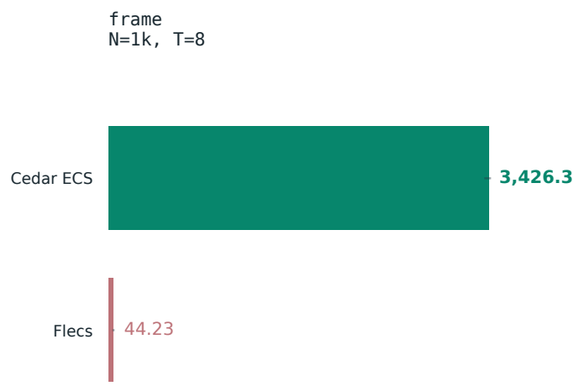




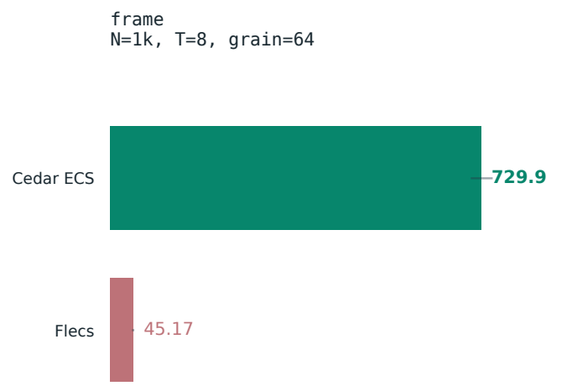
Packed · Cedar vs Flecs

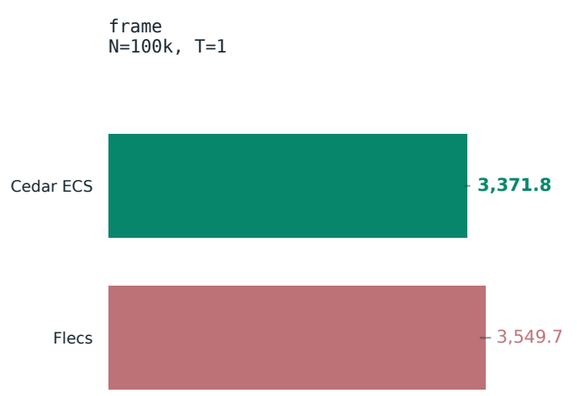
Mop/s ↑

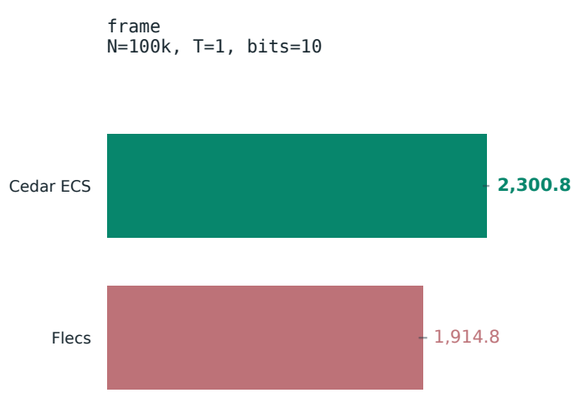
Cedar ECS Flecs

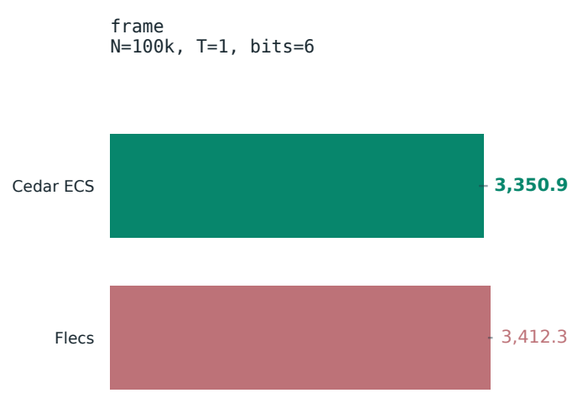


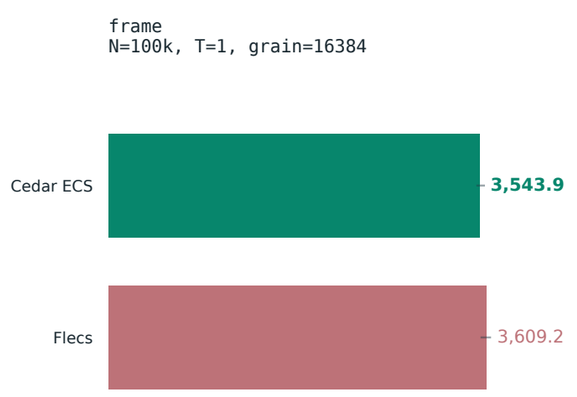


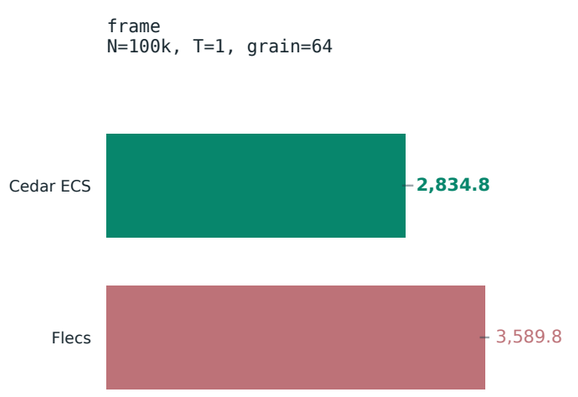


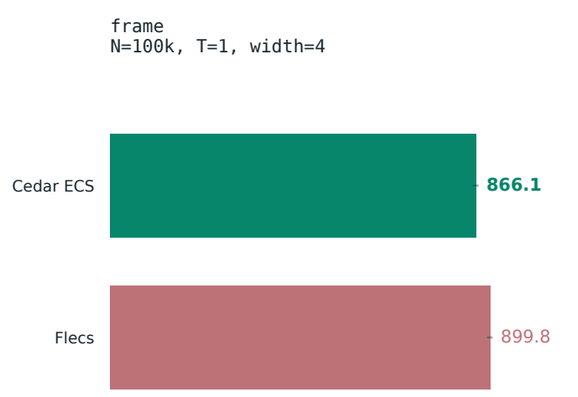


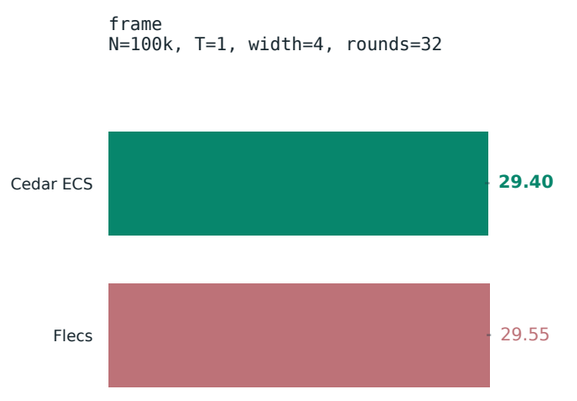


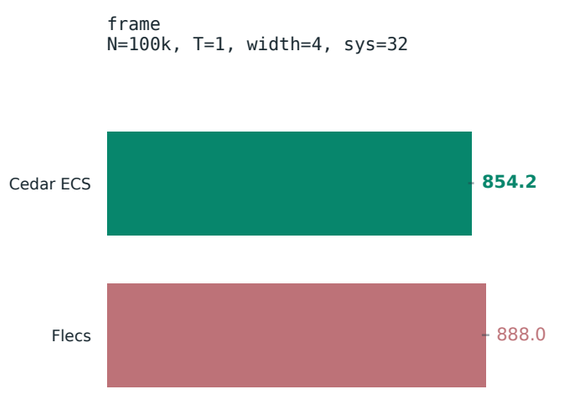


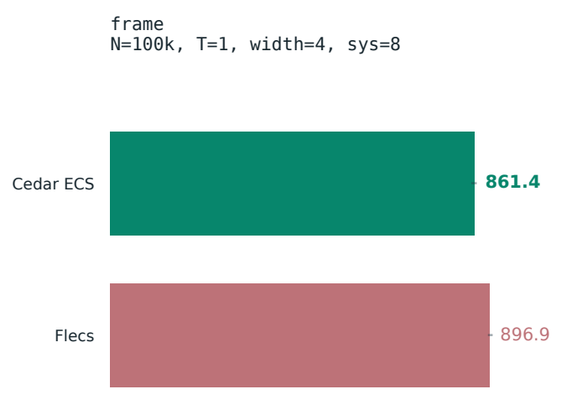








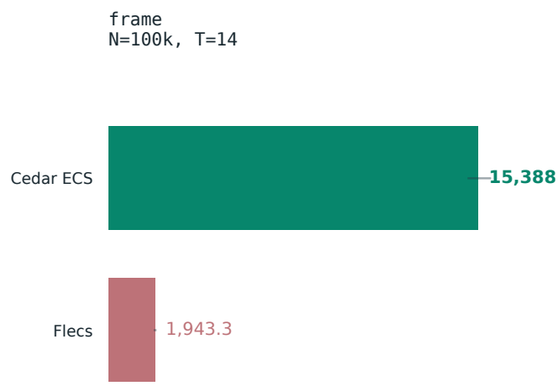


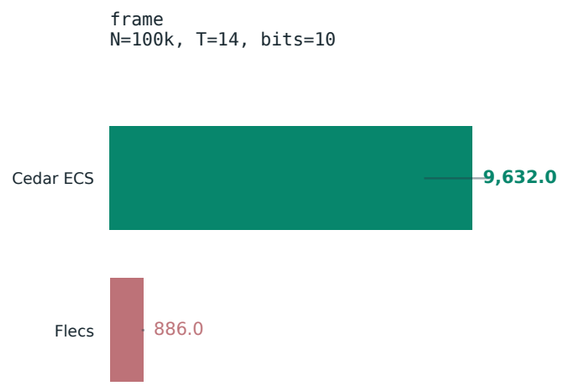


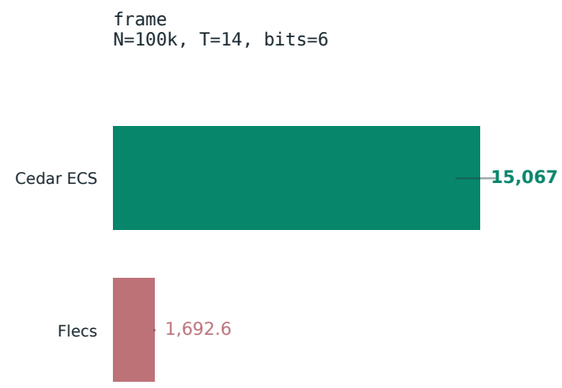
Packed · Cedar vs Flecs

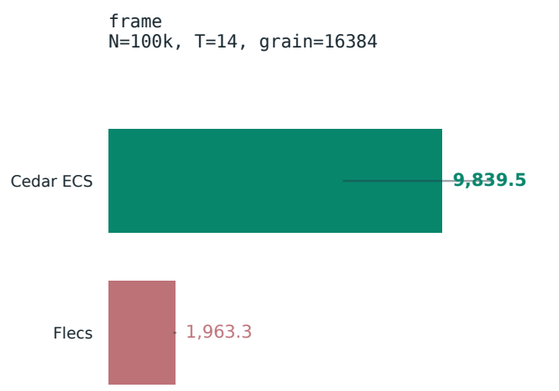
Mop/s ↑

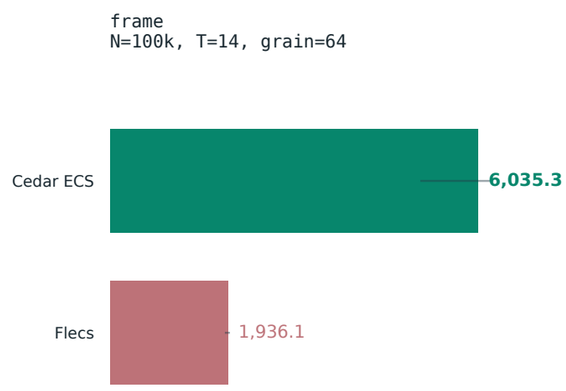
Cedar ECS Flecs

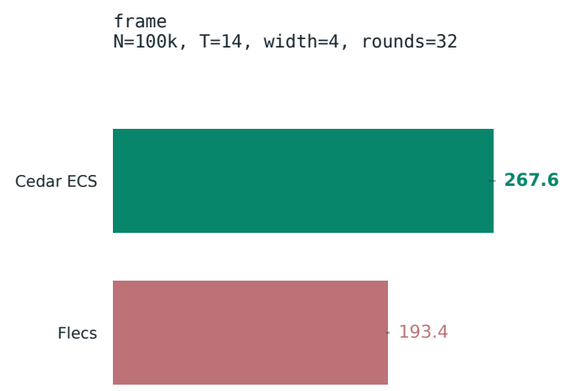


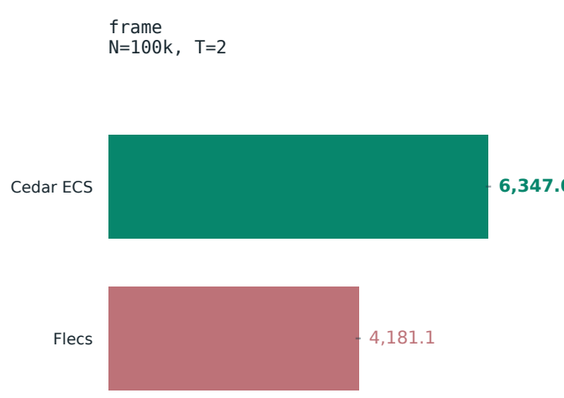


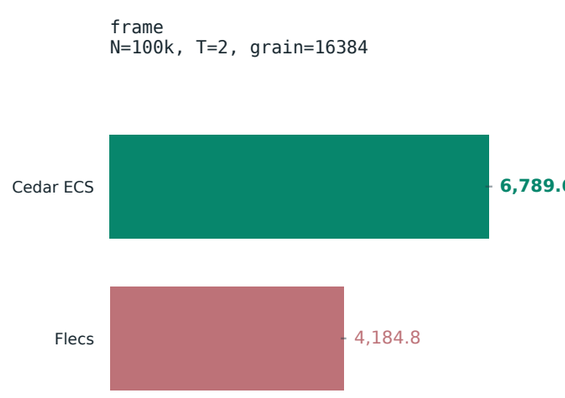


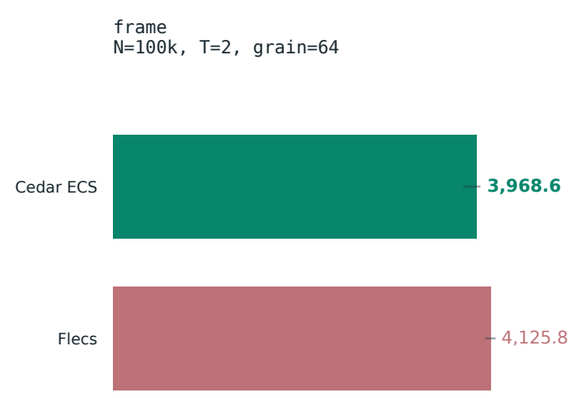


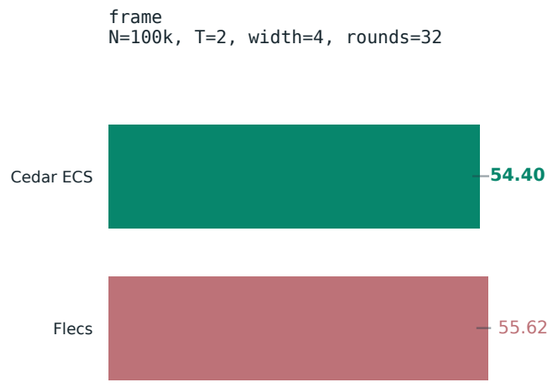


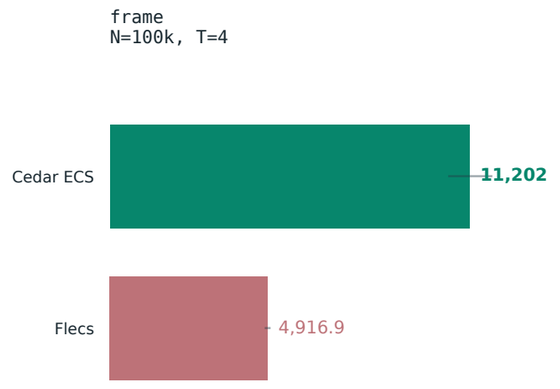


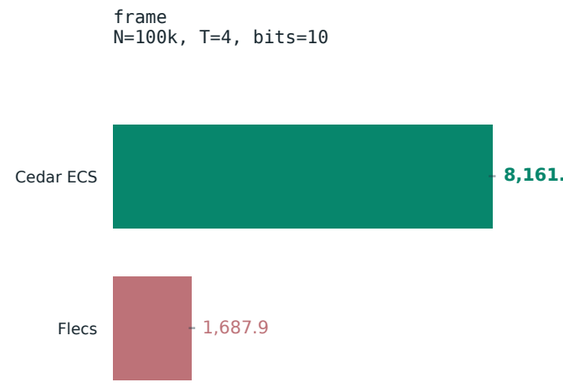








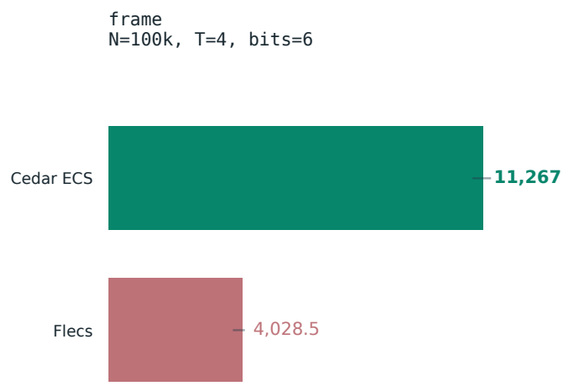


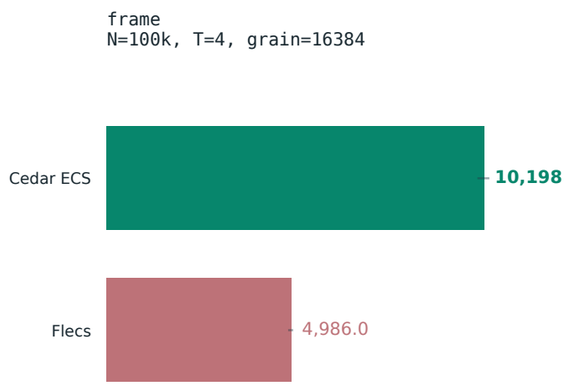


Packed · Cedar vs Flecs

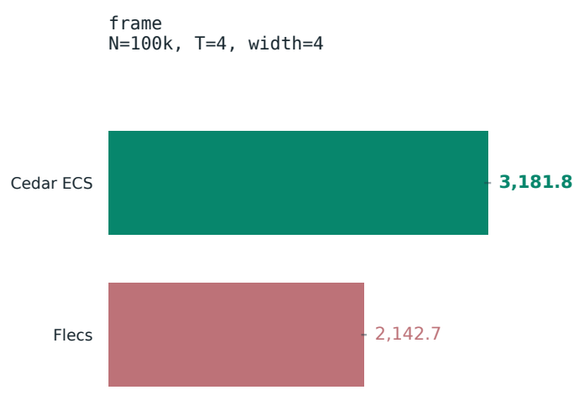
Mop/s ↑

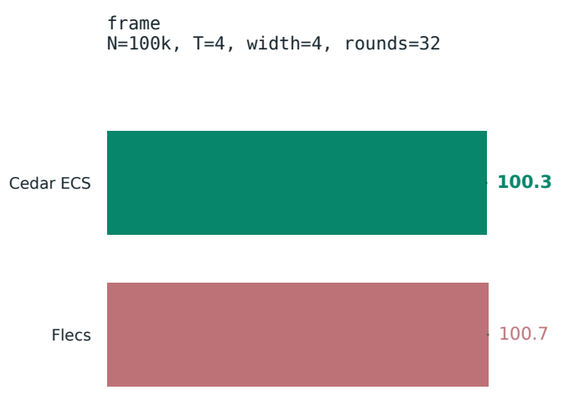
Cedar ECS Flecs

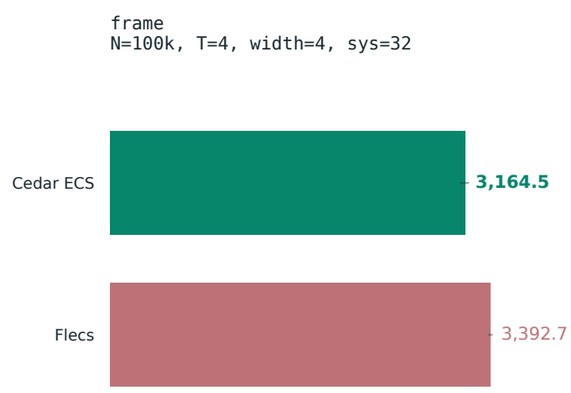


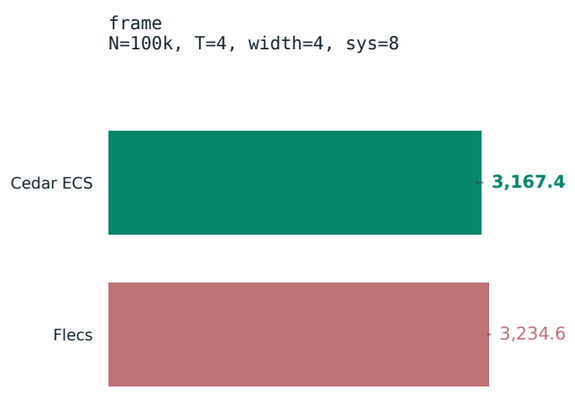


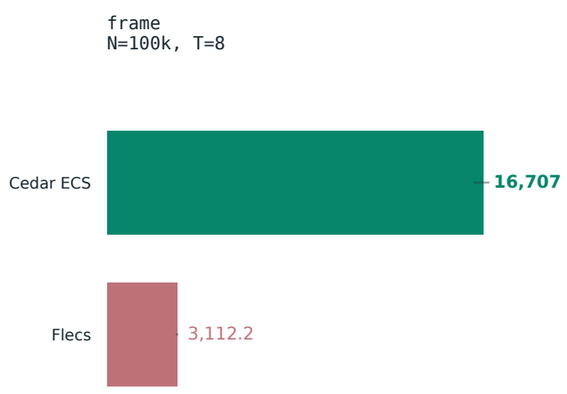


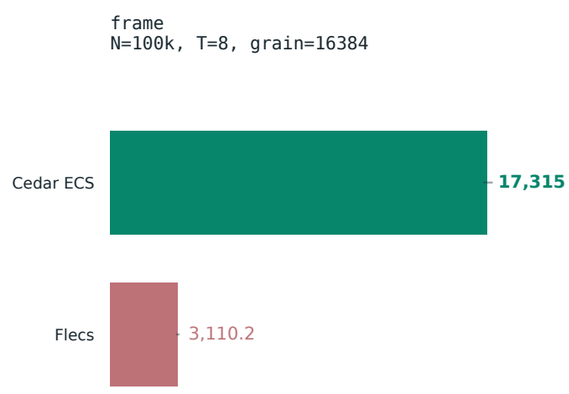


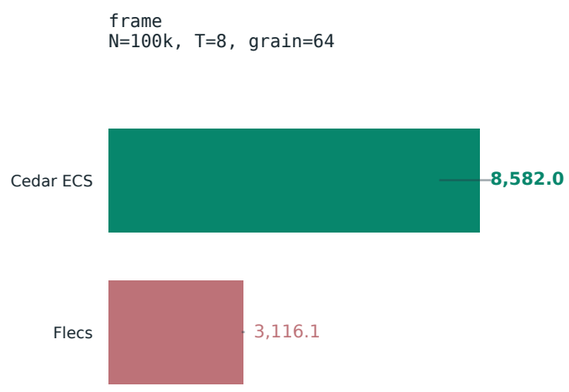


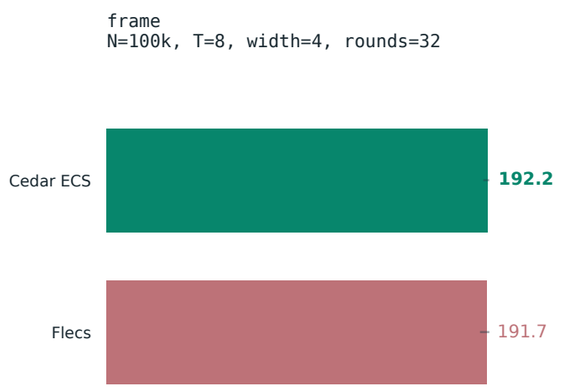


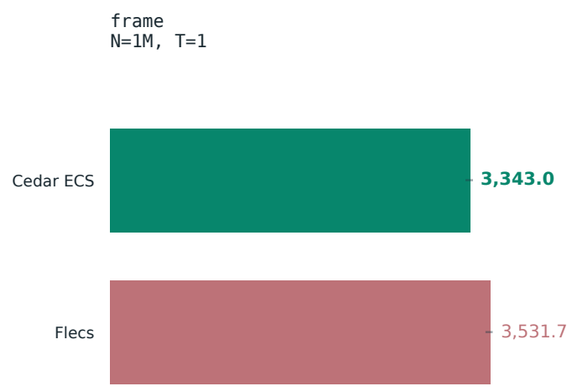








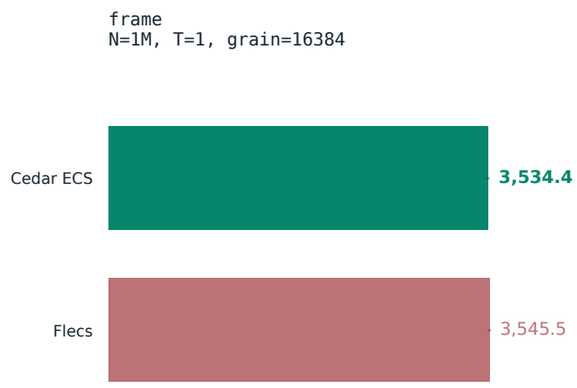


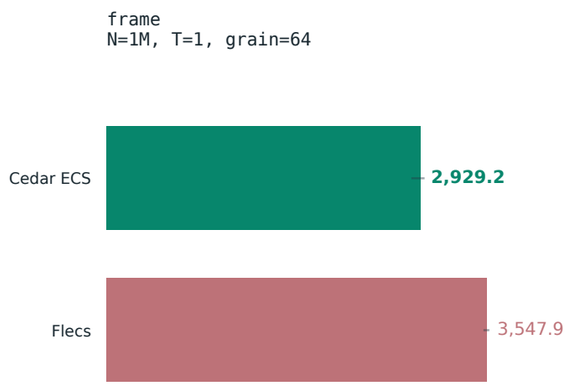


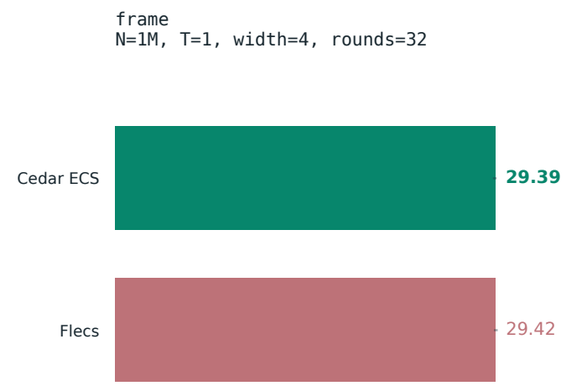
Packed · Cedar vs Flecs

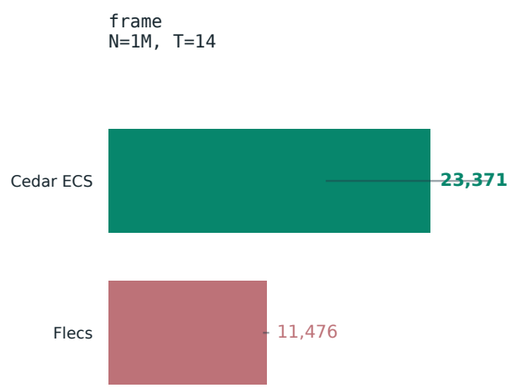
Mop/s ↑

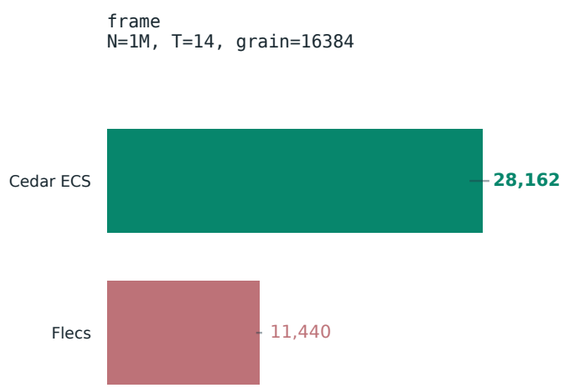
Cedar ECS Flecs

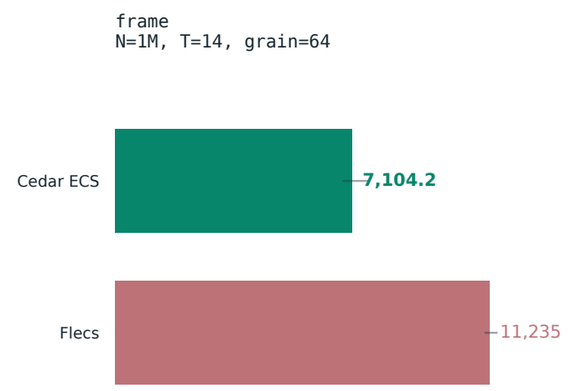


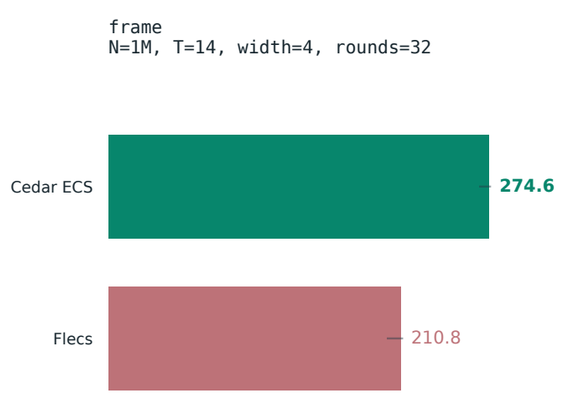




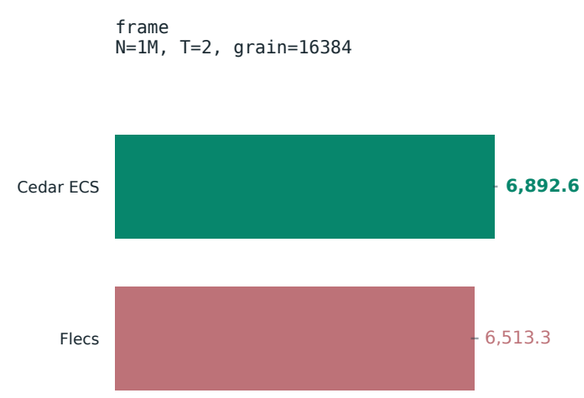


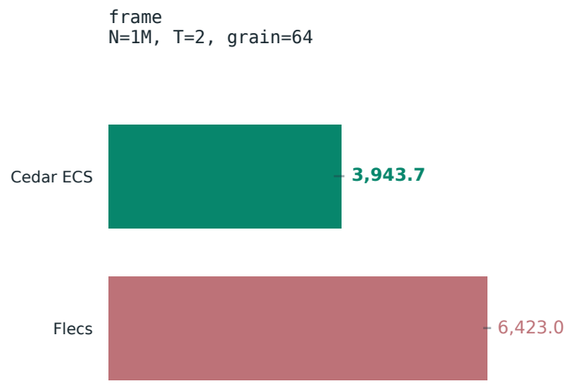


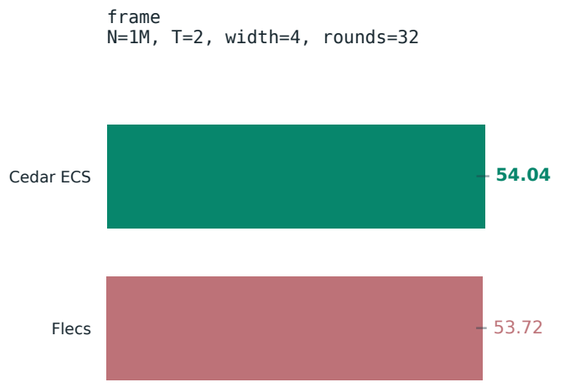


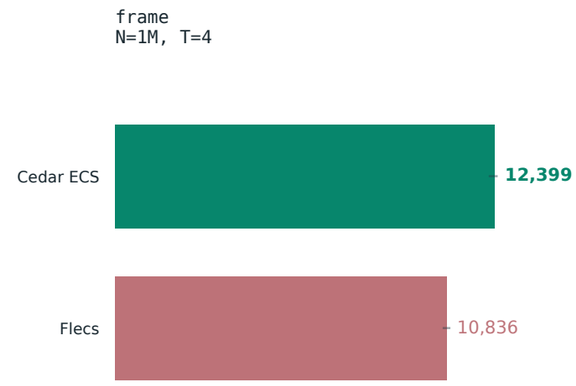








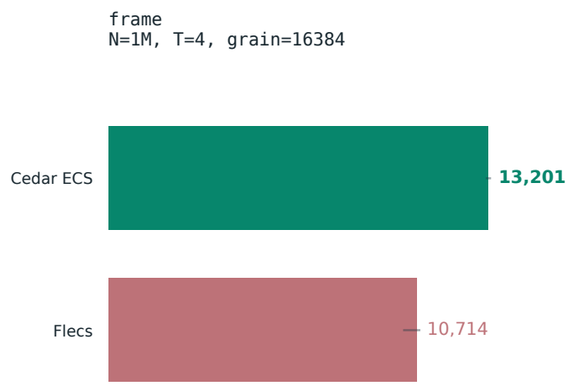


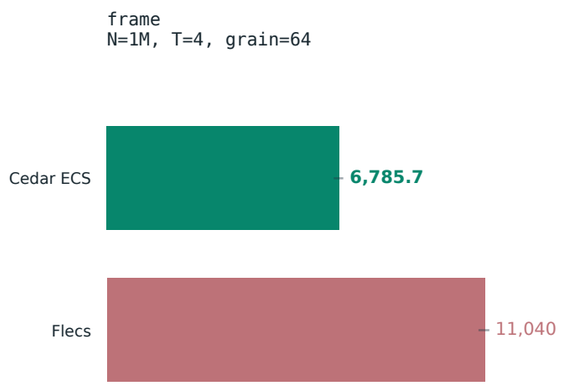


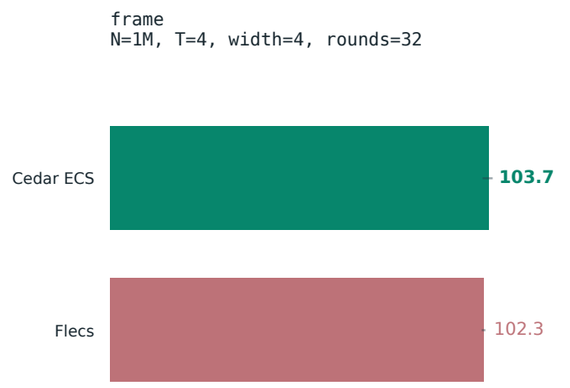
Packed · Cedar vs Flecs

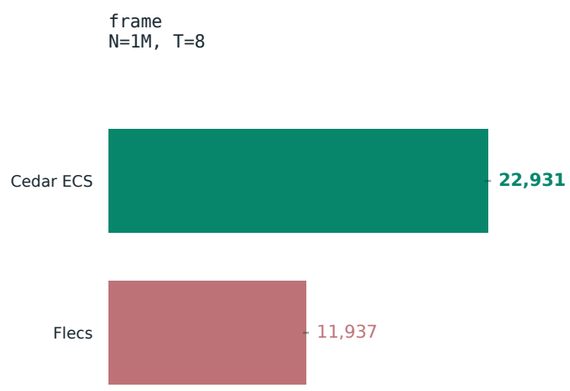
Mop/s ↑

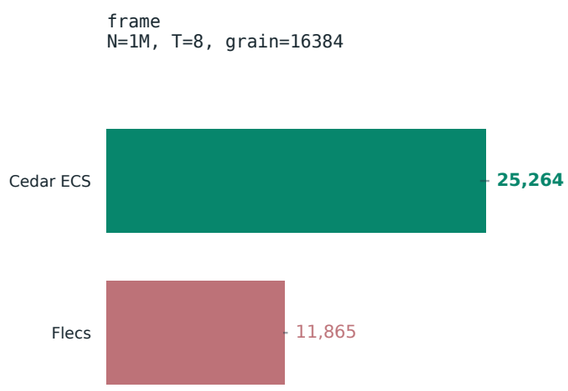
Cedar ECS Flecs

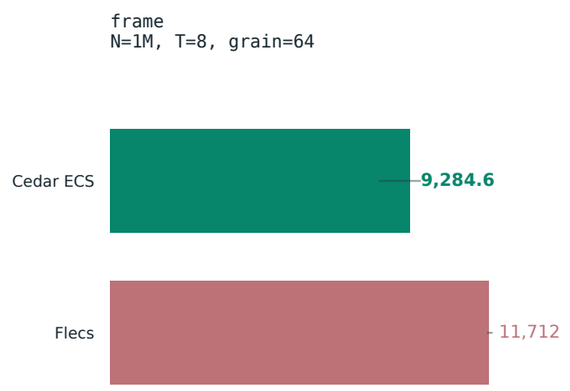


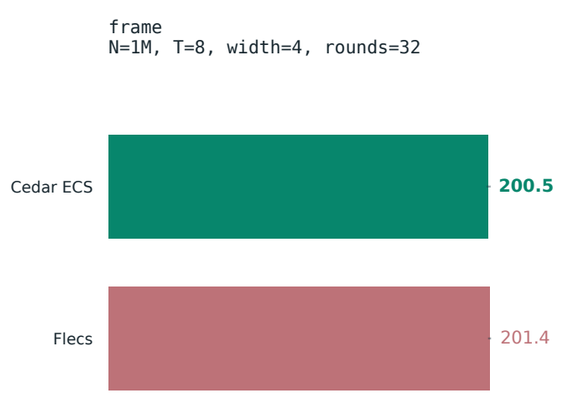


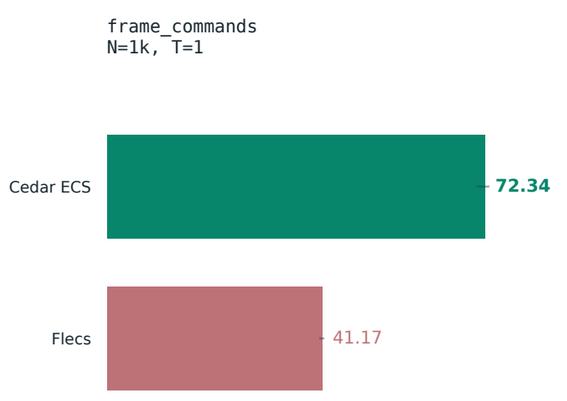


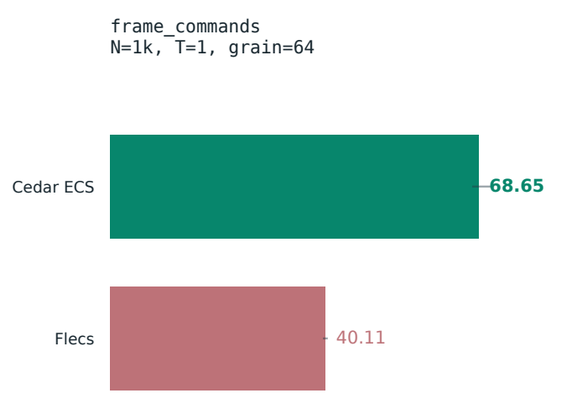


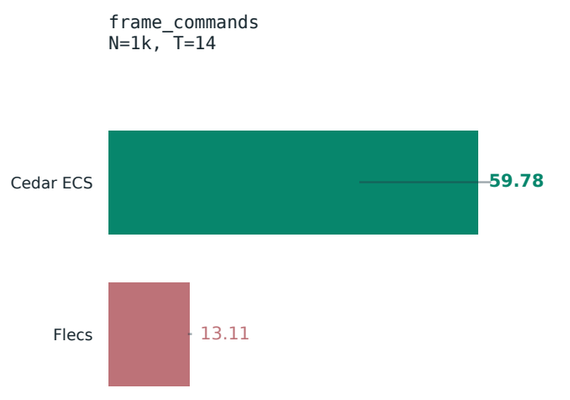


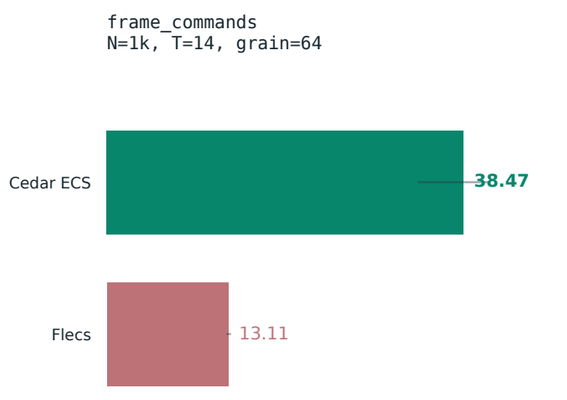


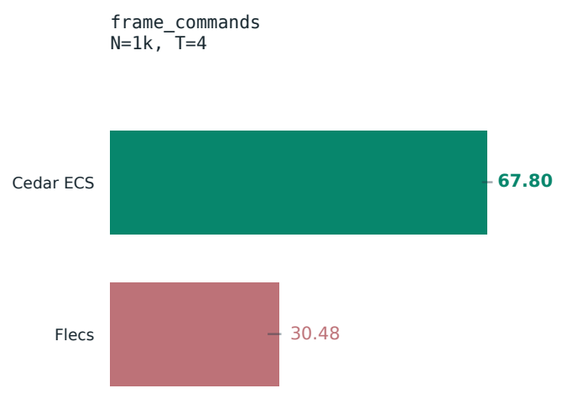








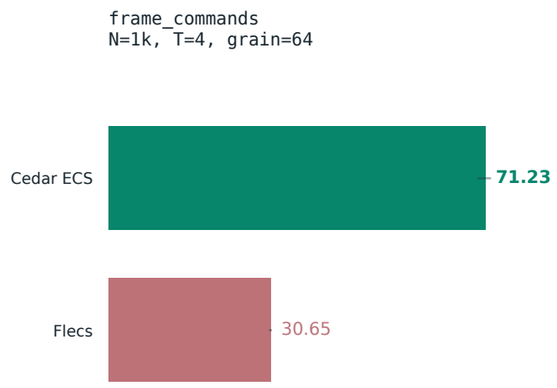


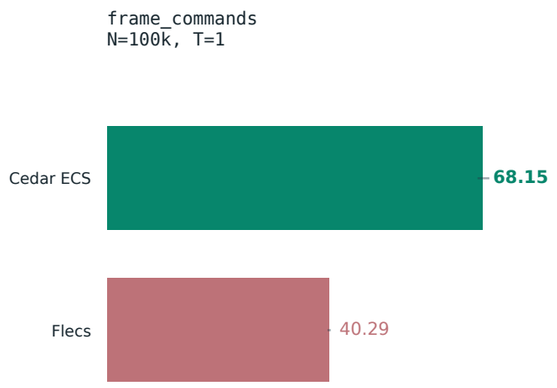


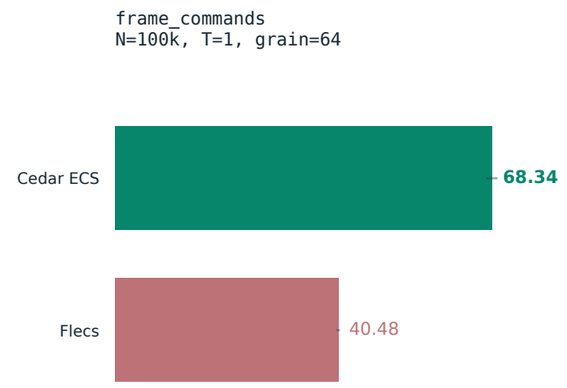
Packed · Cedar vs Flecs

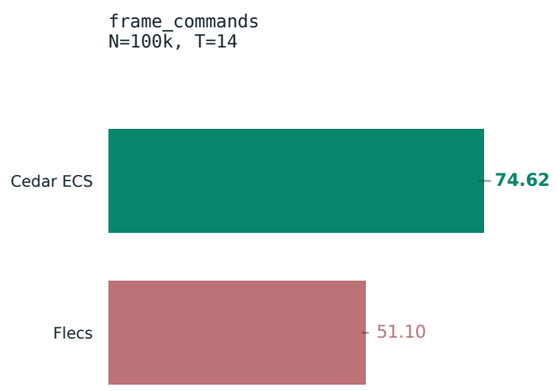
Mop/s ↑

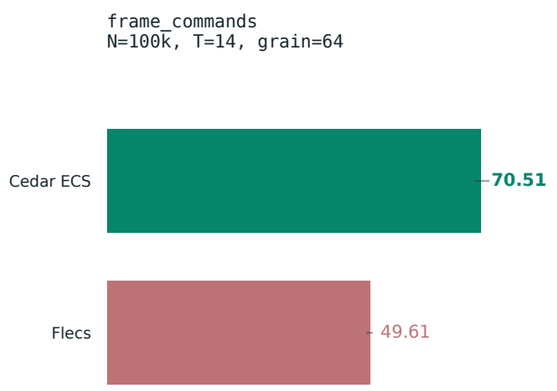
Cedar ECS Flecs

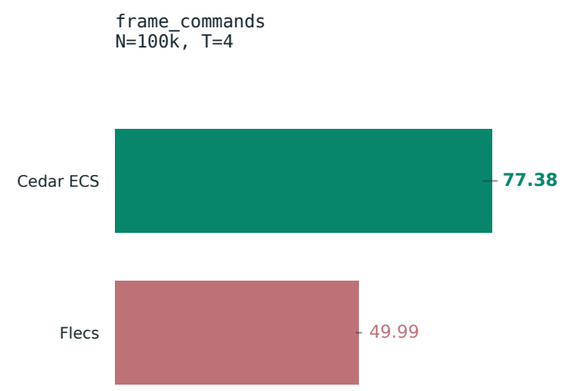


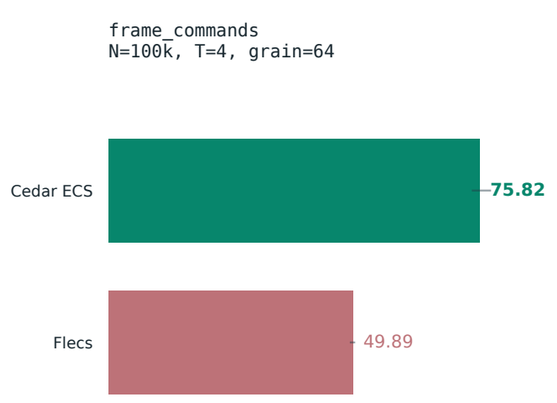


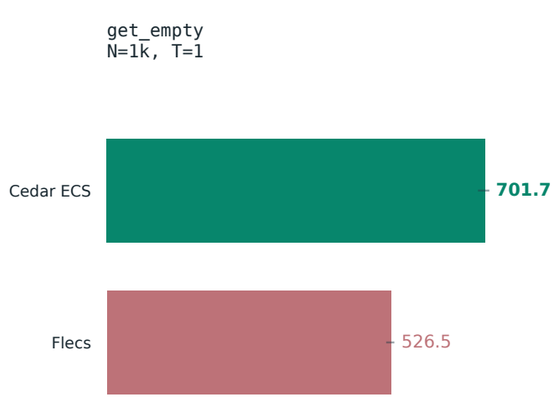


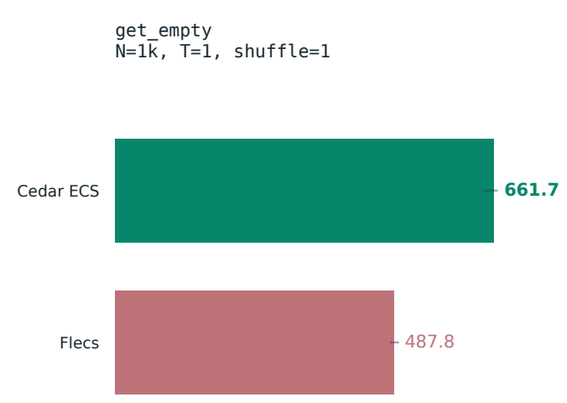


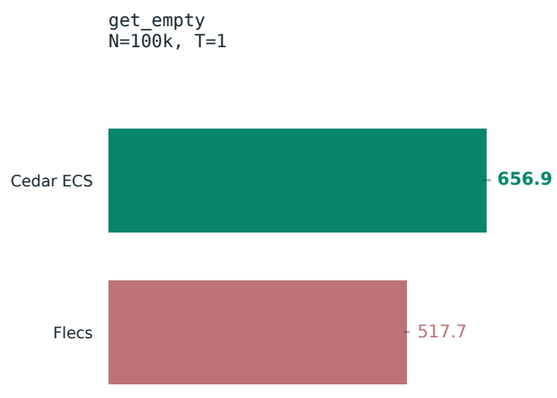


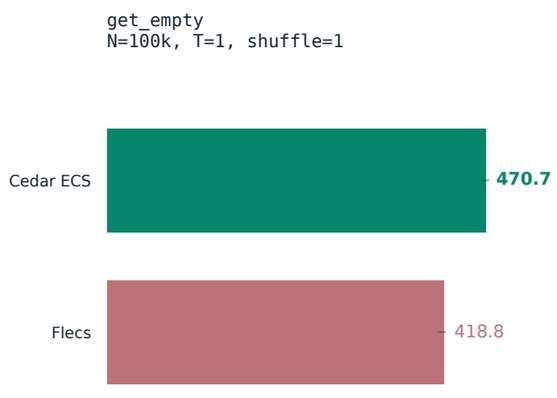


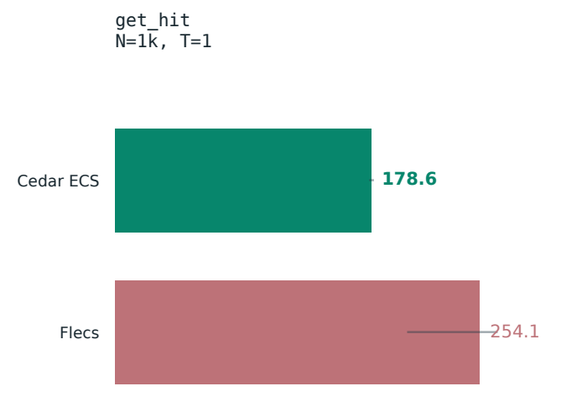








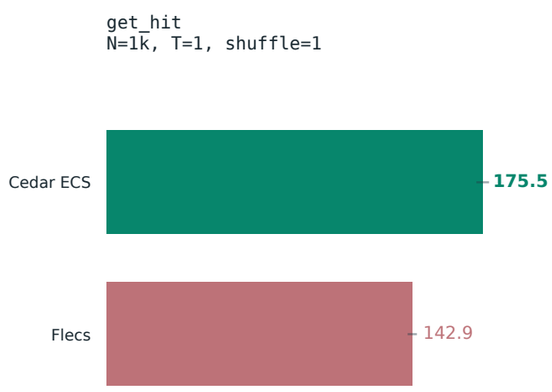




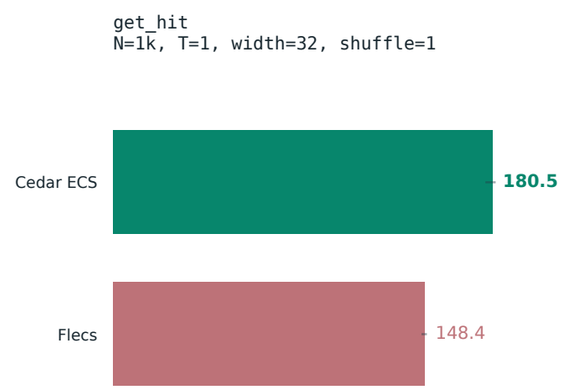
Packed · Cedar vs Flecs

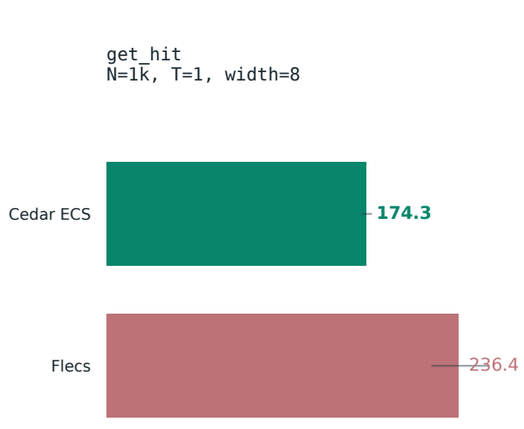
Mop/s ↑

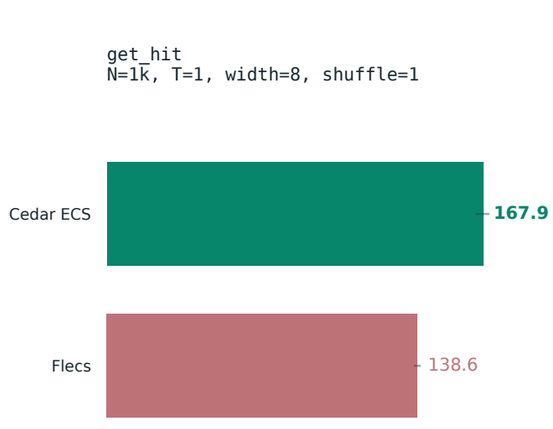
Cedar ECS Flecs

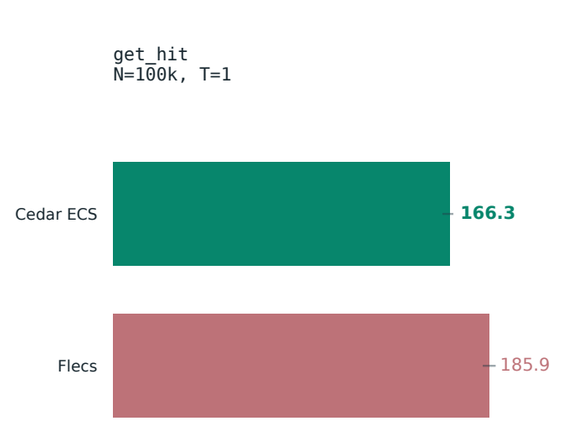


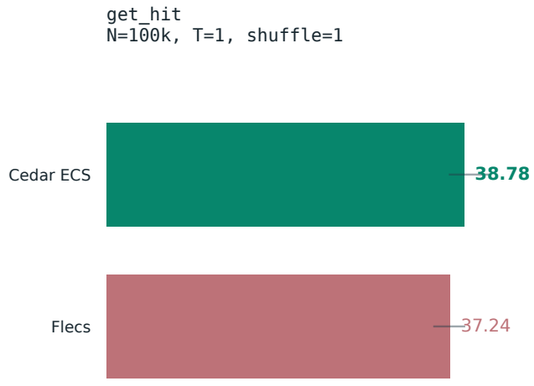




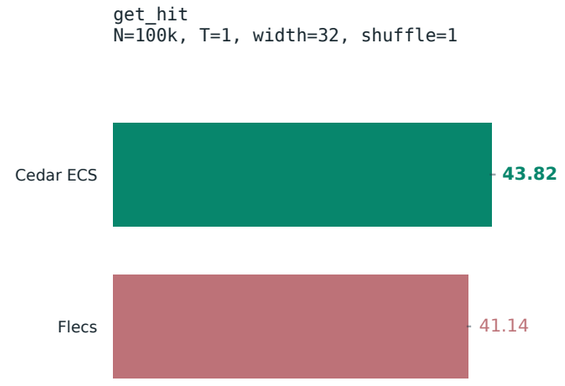


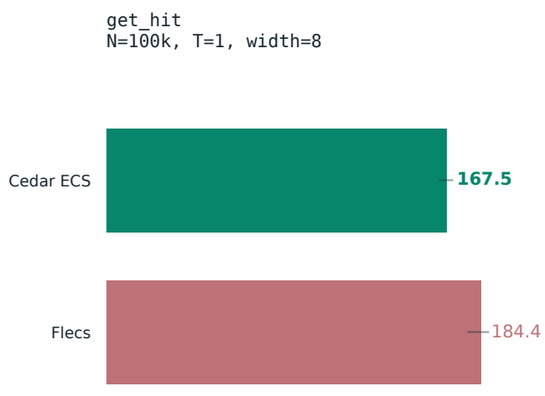


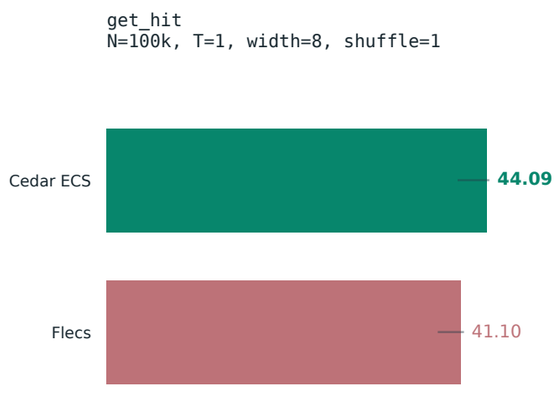


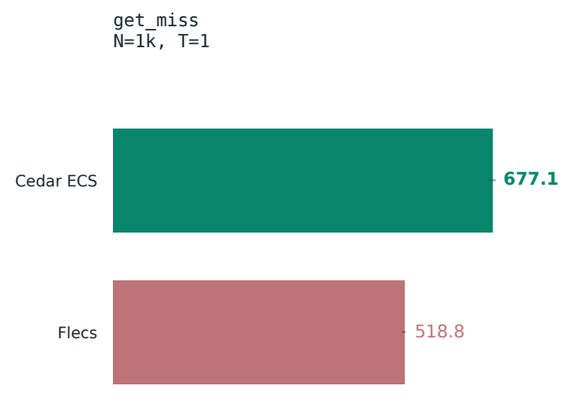










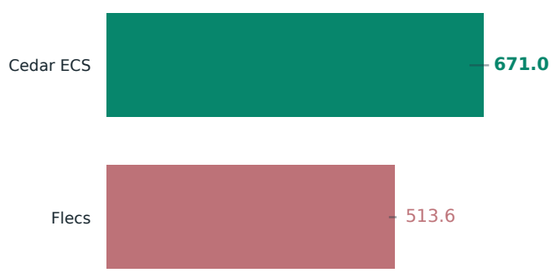


Packed · Cedar vs Flecs

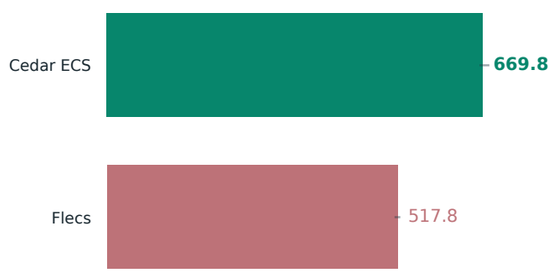
Mop/s ↑

Cedar ECS Flecs

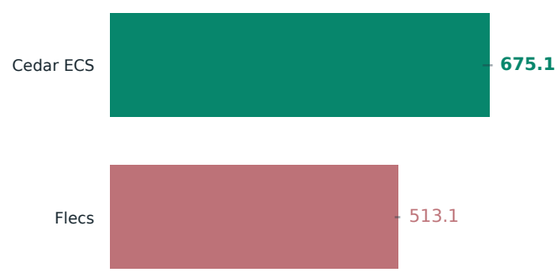
get_miss
N=1k, T=1, shuffle=1



get_miss
N=1k, T=1, width=32



get_miss
N=1k, T=1, width=32, shuffle=1



get_miss
N=100k, T=1



get_miss
N=100k, T=1, shuffle=1



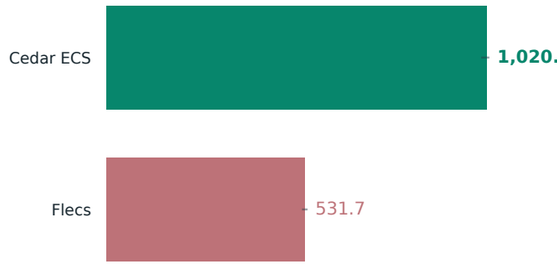
get_miss
N=100k, T=1, width=32



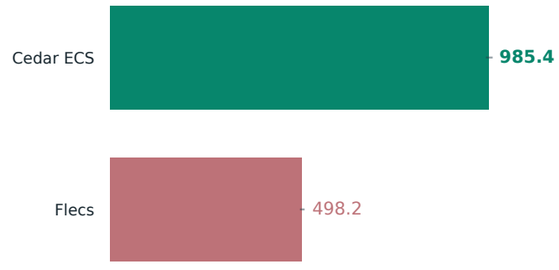
get_miss
N=100k, T=1, width=32, shuffle=1



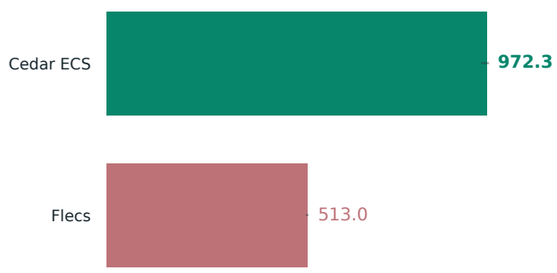
get_stale
N=1k, T=1



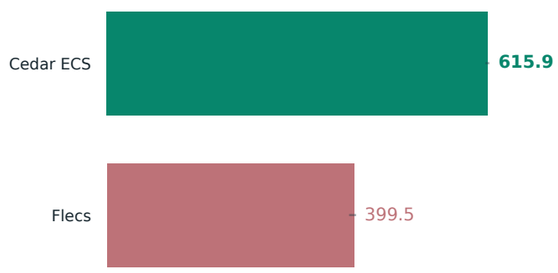
get_stale
N=1k, T=1, shuffle=1



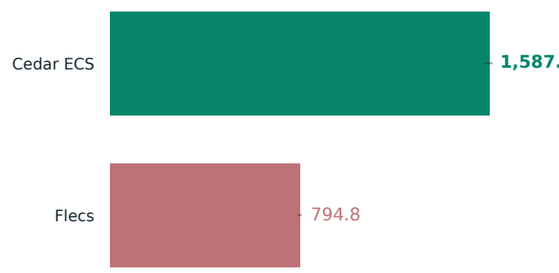
get_stale
N=100k, T=1



get_stale
N=100k, T=1, shuffle=1



has_mixed
N=1k, T=1

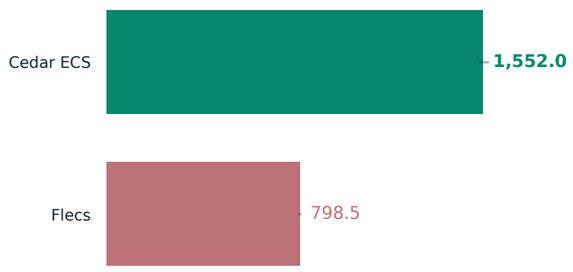


Packed · Cedar vs Flecs

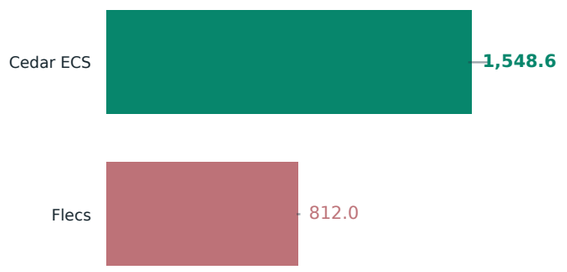
Mop/s ↑

Cedar ECS Flecs

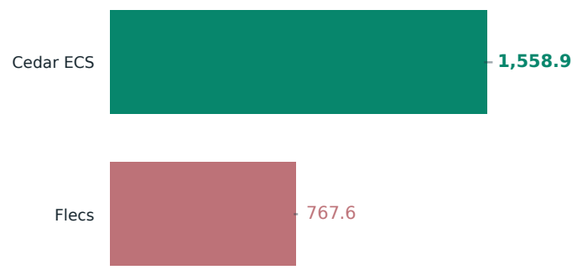
has_mixed
N=1k, T=1, percent=50



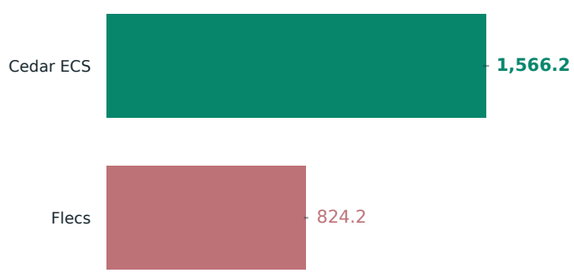
has_mixed
N=1k, T=1, percent=50, shuffle=1



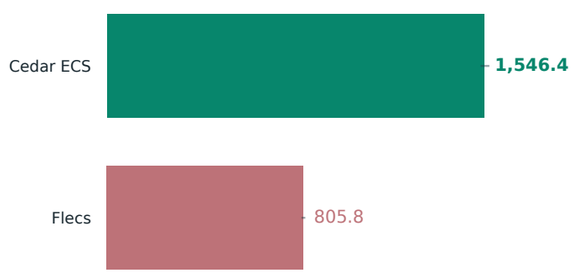
has_mixed
N=1k, T=1, percent=90



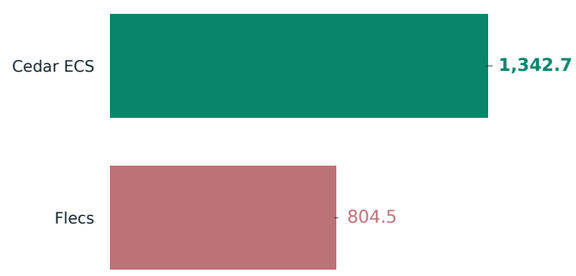
has_mixed
N=1k, T=1, percent=90, shuffle=1



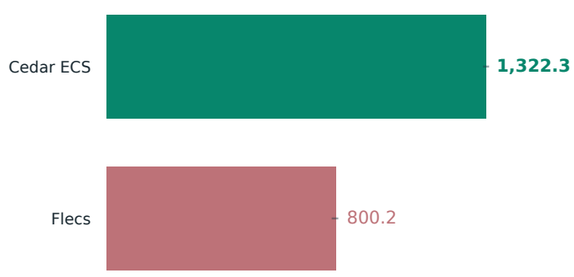
has_mixed
N=1k, T=1, shuffle=1



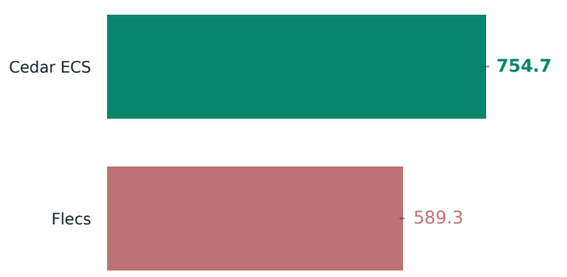
has_mixed
N=100k, T=1



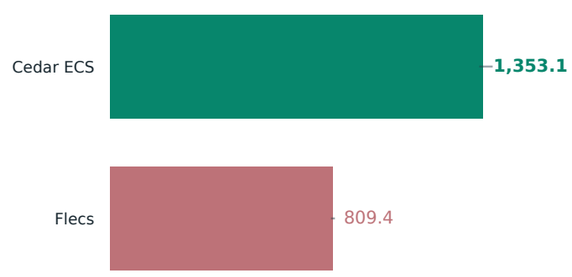
has_mixed
N=100k, T=1, percent=50



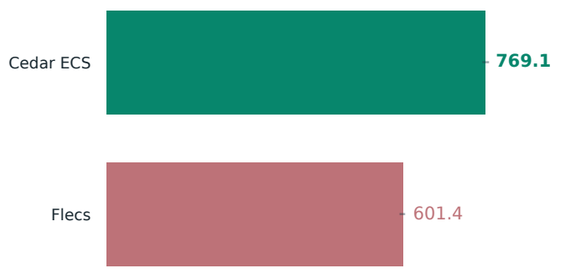
has_mixed
N=100k, T=1, percent=50, shuffle=1



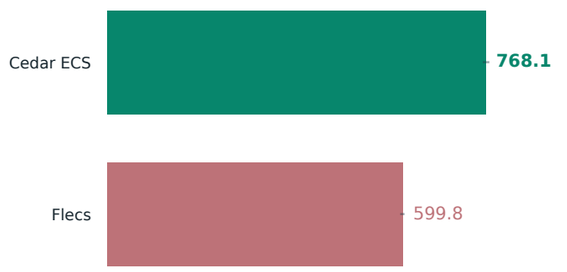
has_mixed
N=100k, T=1, percent=90



has_mixed
N=100k, T=1, percent=90, shuffle=1



has_mixed
N=100k, T=1, shuffle=1



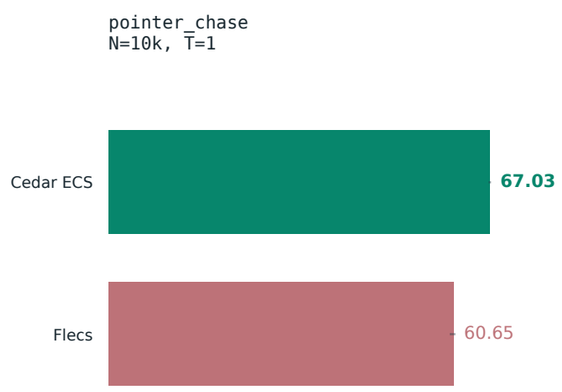
pointer_chase
N=1k, T=1

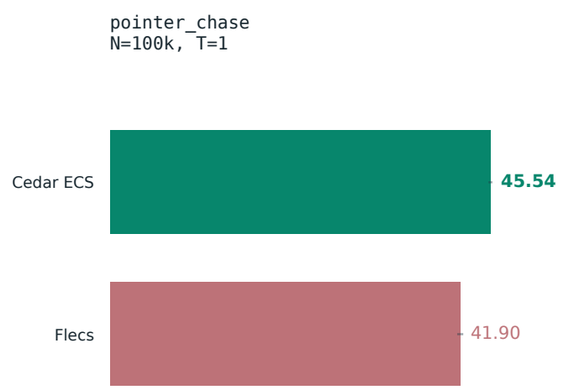


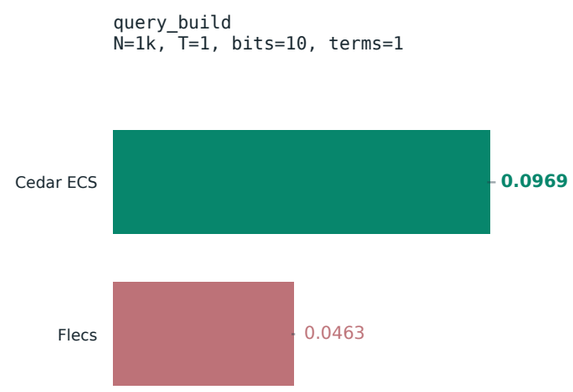
Packed · Cedar vs Flecs

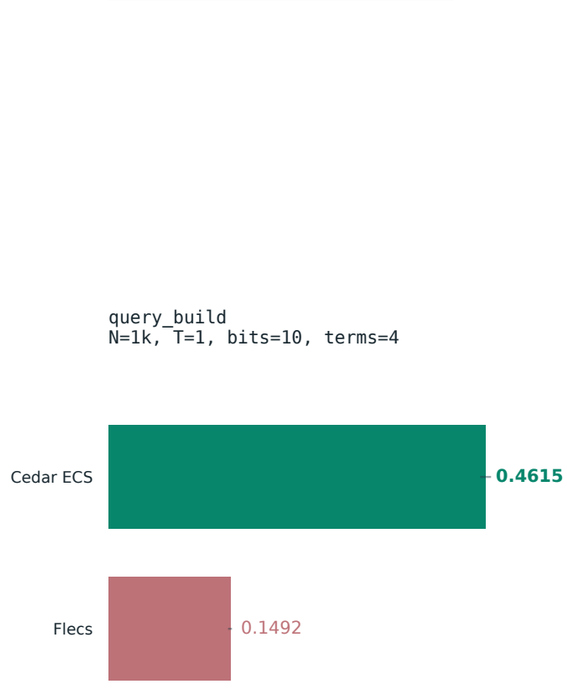
Mop/s ↑

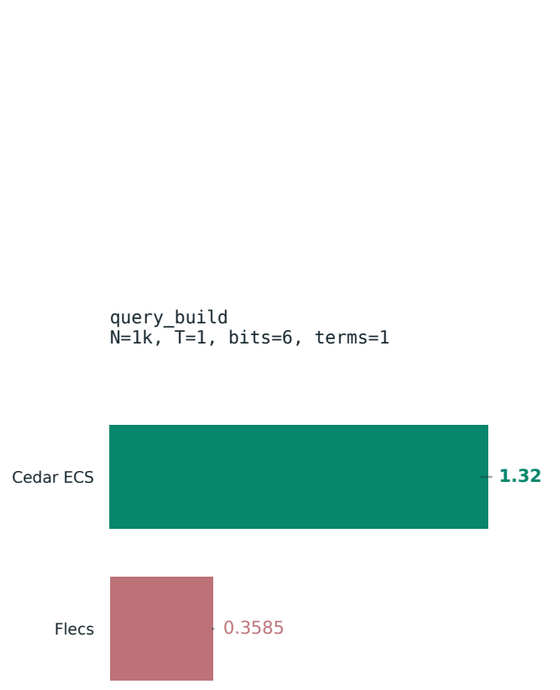
Cedar ECS Flecs

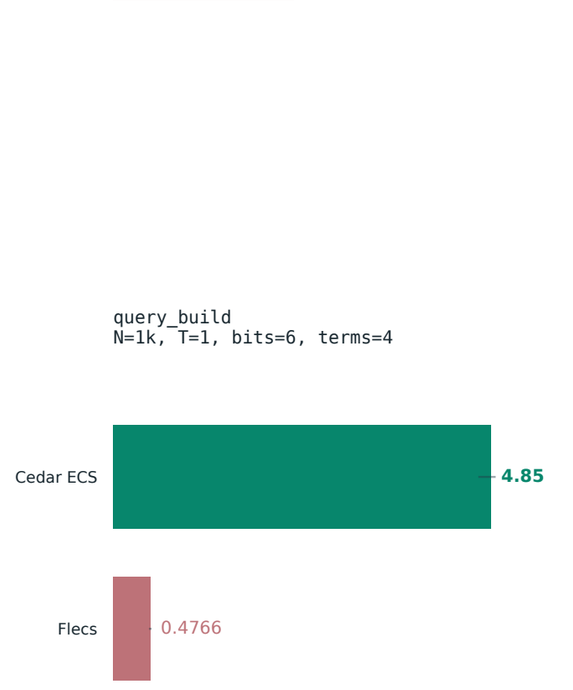


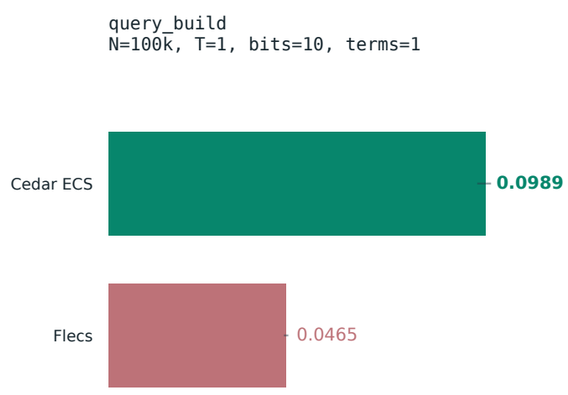


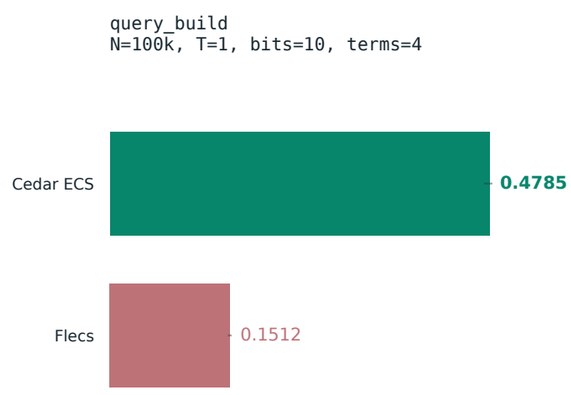


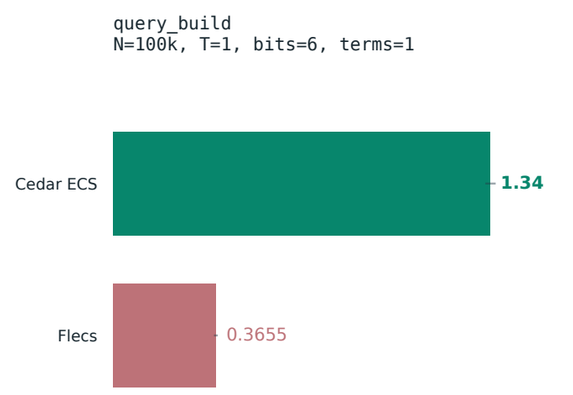


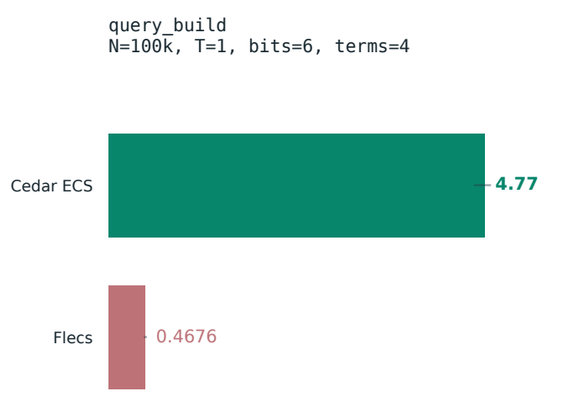


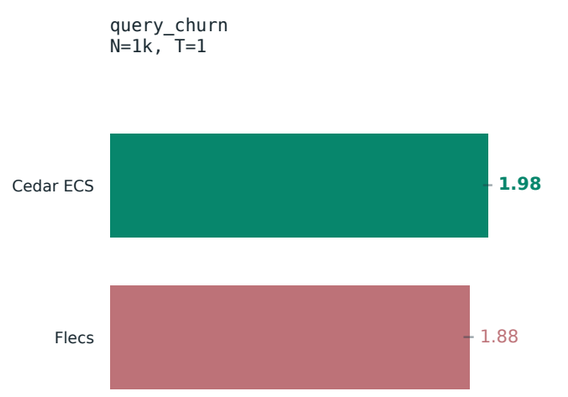


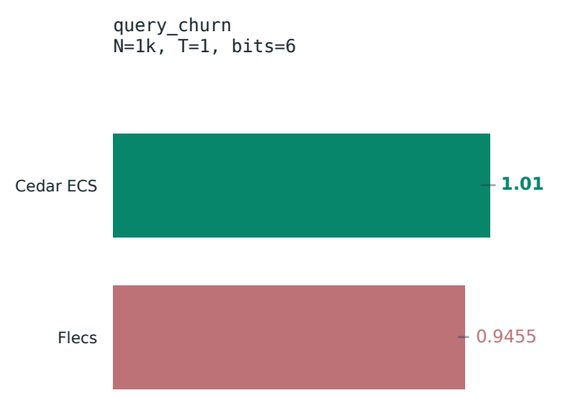












Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

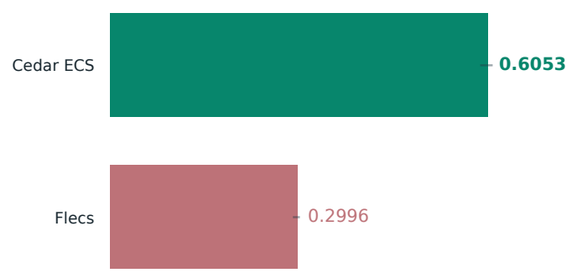
query_churn
N=100k, T=1



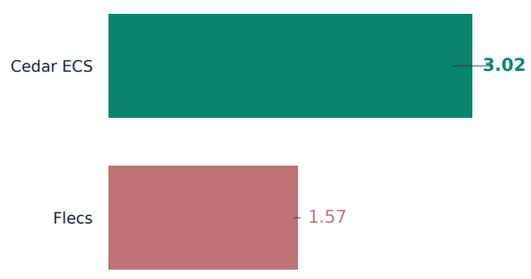
query_churn
N=100k, T=1, bits=6



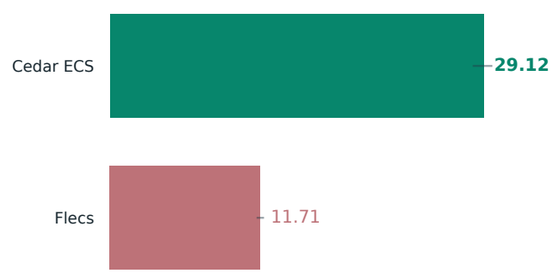
query_data
N=1k, T=1, bits=10, terms=1



query_data
N=1k, T=1, bits=10, terms=4



query_data
N=1k, T=1, bits=10, terms=8



query_data
N=1k, T=1, bits=6, terms=1



query_data
N=1k, T=1, bits=6, terms=4



query_data
N=100k, T=1, bits=10, terms=1



query_data
N=100k, T=1, bits=10, terms=4



query_data
N=100k, T=1, bits=10, terms=8



query_data
N=100k, T=1, bits=6, terms=1



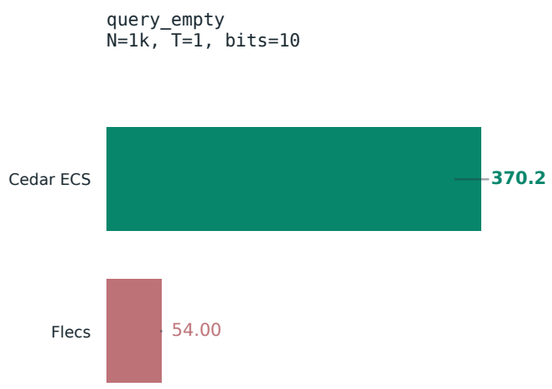
query_data
N=100k, T=1, bits=6, terms=4

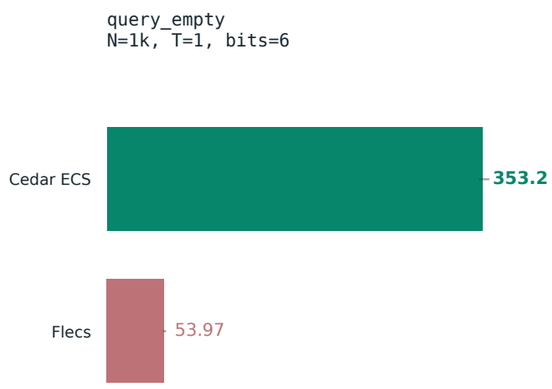


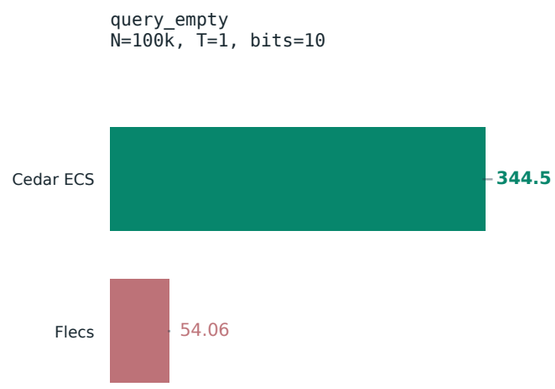
Packed · Cedar vs Flecs

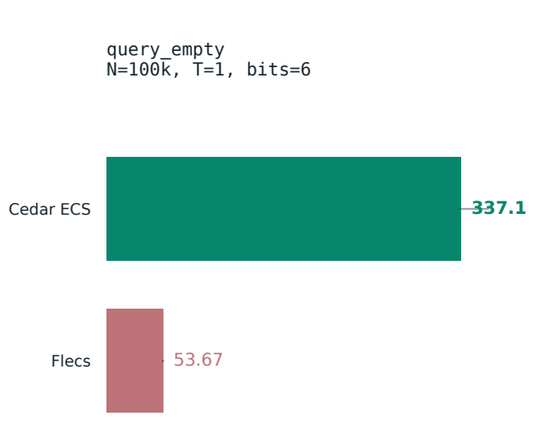
Mop/s ↑

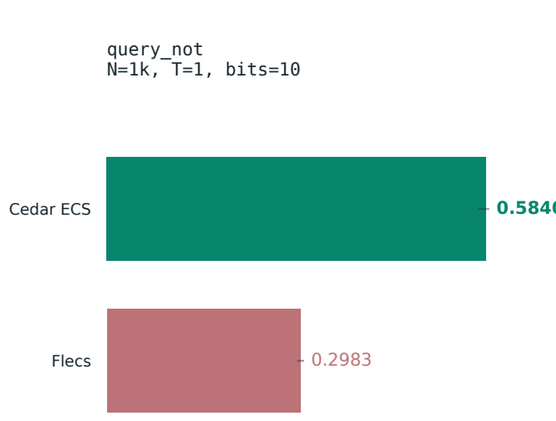
Cedar ECS Flecs

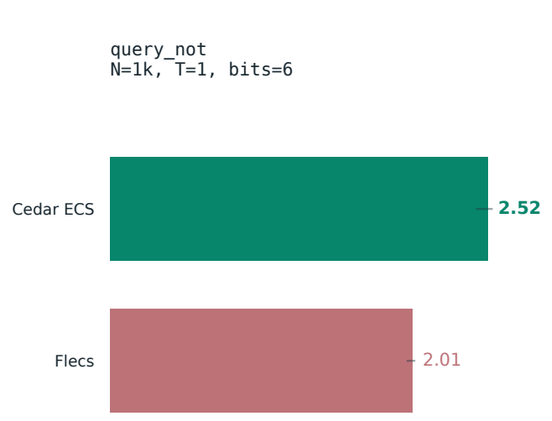


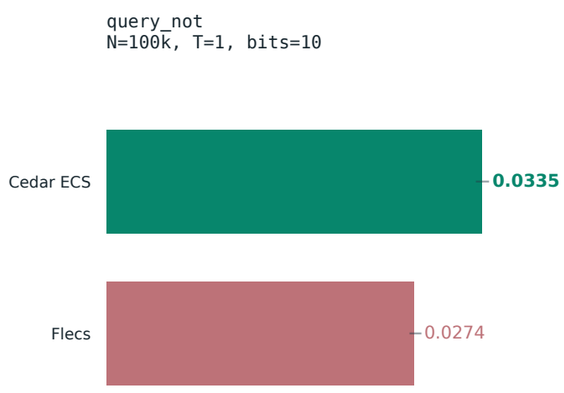


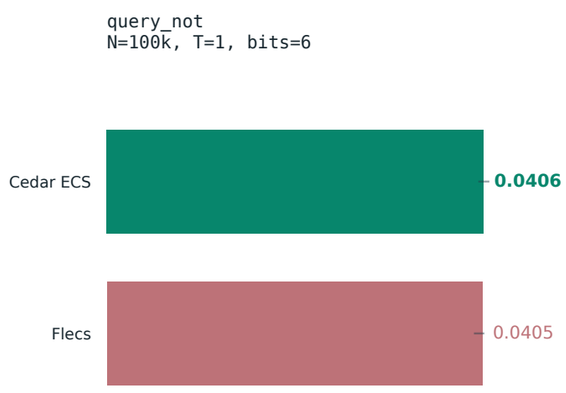


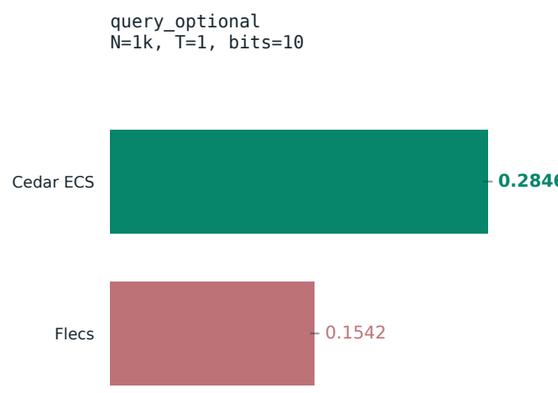


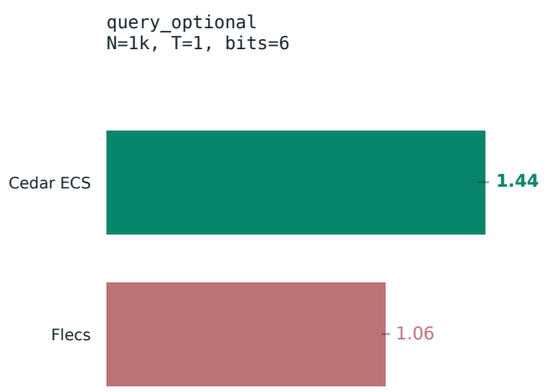


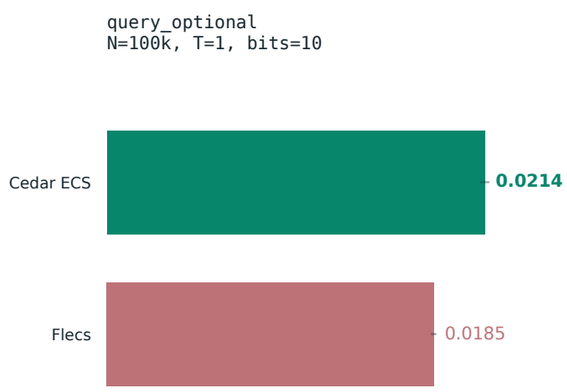


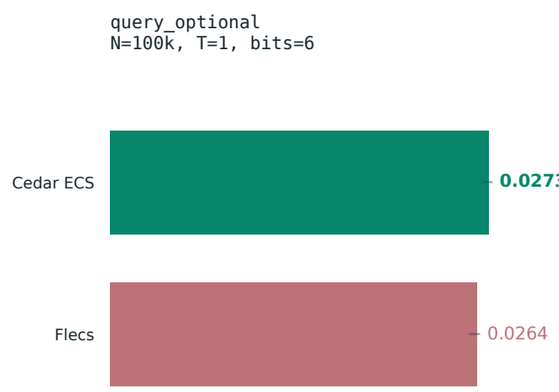












Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

query_plain
N=1k, T=1



query_plain
N=1k, T=1, bits=6



query_plain
N=1k, T=1, width=16



query_plain
N=1k, T=1, width=16, bits=6



query_plain
N=1k, T=1, width=4



query_plain
N=1k, T=1, width=4, bits=6



query_plain
N=1k, T=1, width=8



query_plain
N=1k, T=1, width=8, bits=6



query_plain
N=100k, T=1



query_plain
N=100k, T=1, bits=6



query_plain
N=100k, T=1, width=16



query_plain
N=100k, T=1, width=16, bits=6



Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

query_plain
N=100k, T=1, width=4



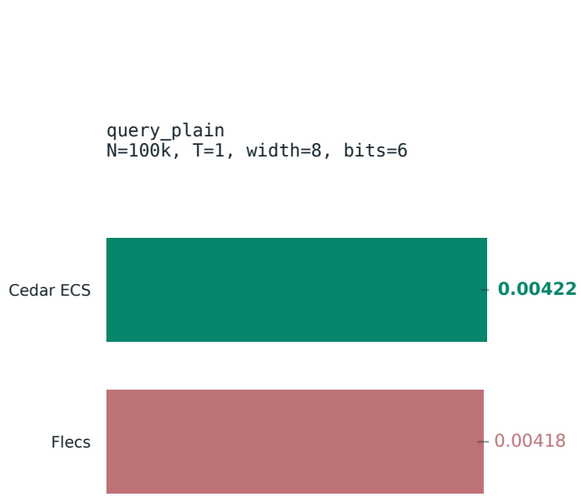
query_plain
N=100k, T=1, width=4, bits=6



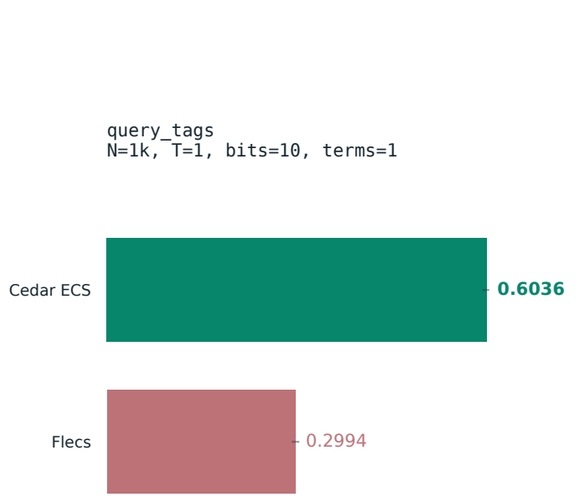
query_plain
N=100k, T=1, width=8



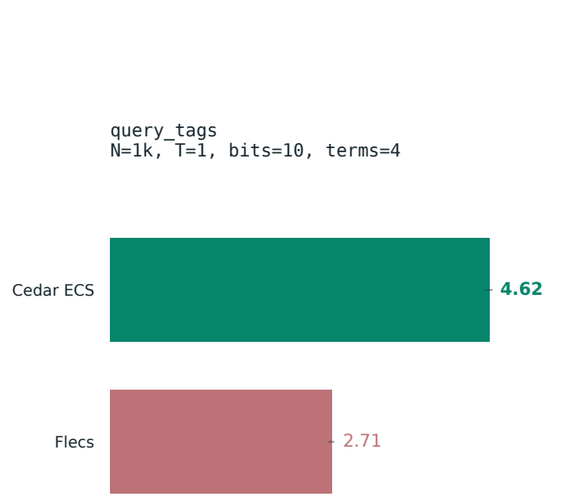
query_plain
N=100k, T=1, width=8, bits=6



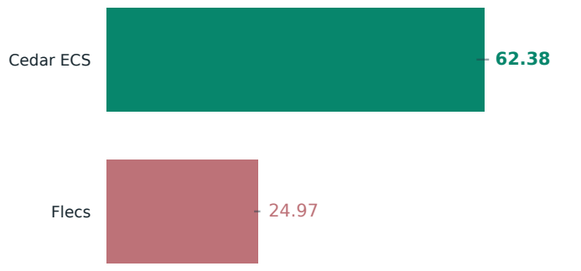
query_tags
N=1k, T=1, bits=10, terms=1



query_tags
N=1k, T=1, bits=10, terms=4



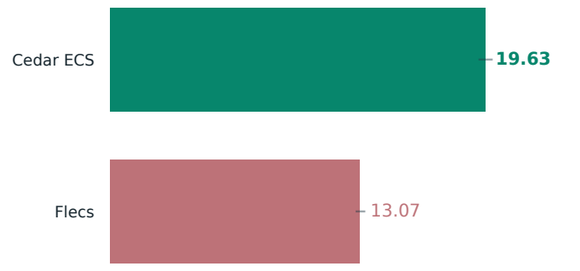
query_tags
N=1k, T=1, bits=10, terms=8



query_tags
N=1k, T=1, bits=6, terms=1



query_tags
N=1k, T=1, bits=6, terms=4



query_tags
N=100k, T=1, bits=10, terms=1



query_tags
N=100k, T=1, bits=10, terms=4



query_tags
N=100k, T=1, bits=10, terms=8



Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

query_tags
N=100k, T=1, bits=6, terms=1



query_tags
N=100k, T=1, bits=6, terms=4



recycle
N=1k, T=1



recycle
N=1k, T=1, bytes=256



recycle
N=1k, T=1, bytes=256, shuffle=1



recycle
N=1k, T=1, shuffle=1



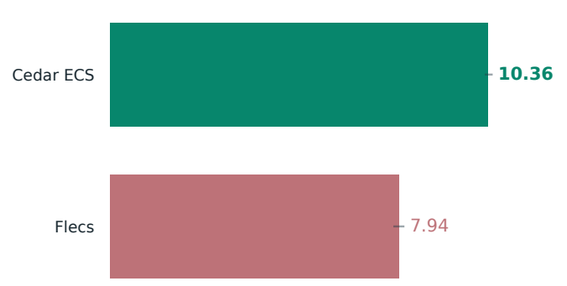
recycle
N=1k, T=1, width=8



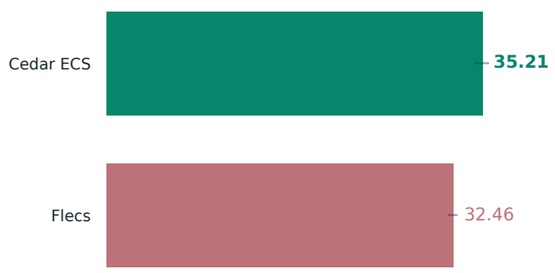
recycle
N=1k, T=1, width=8, bytes=256



recycle
N=1k, T=1, width=8, bytes=256, shuffle=1



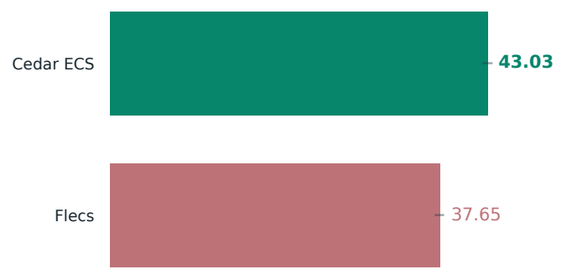
recycle
N=1k, T=1, width=8, shuffle=1



recycle
N=10k, T=1



recycle
N=10k, T=1, bytes=256



Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

recycle
N=10k, T=1, bytes=256, shuffle=1



recycle
N=10k, T=1, shuffle=1



recycle
N=10k, T=1, width=8



recycle
N=10k, T=1, width=8, bytes=256



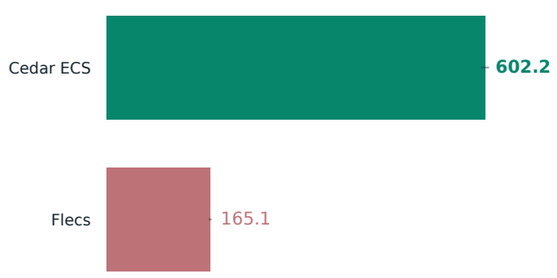
recycle
N=10k, T=1, width=8, bytes=256, shuffle=1



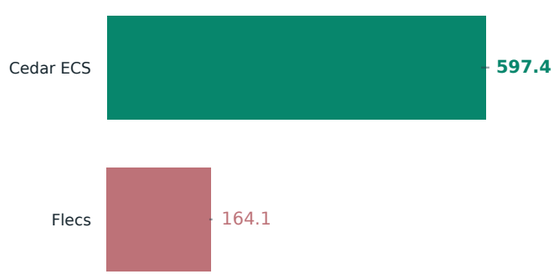
recycle
N=10k, T=1, width=8, shuffle=1



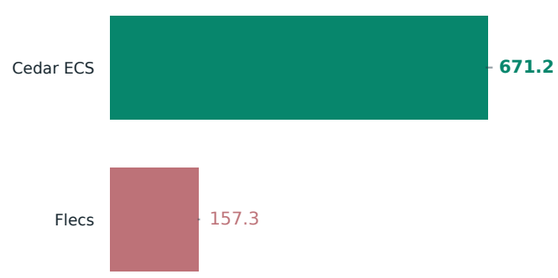
remove_absent
N=1k, T=1



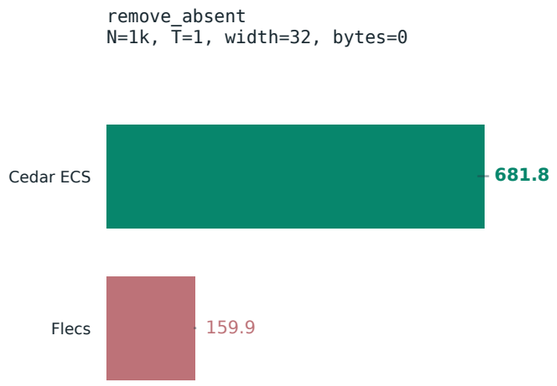
remove_absent
N=1k, T=1, bytes=0



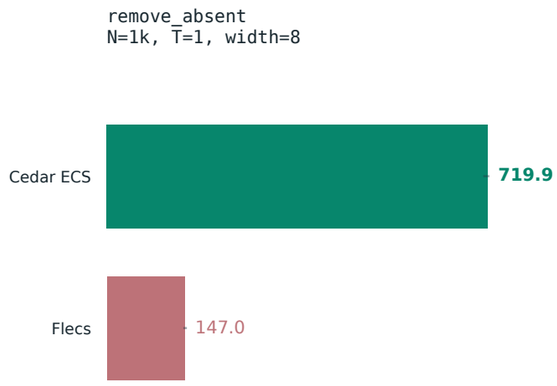
remove_absent
N=1k, T=1, width=32



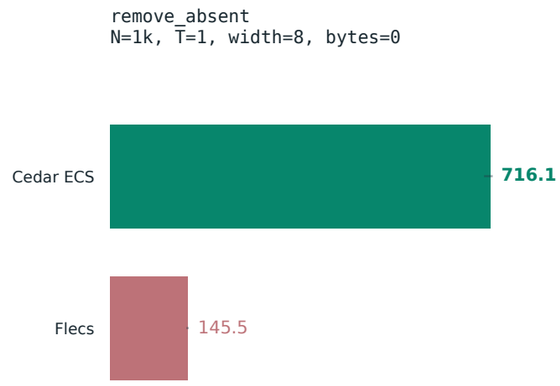
remove_absent
N=1k, T=1, width=32, bytes=0



remove_absent
N=1k, T=1, width=8



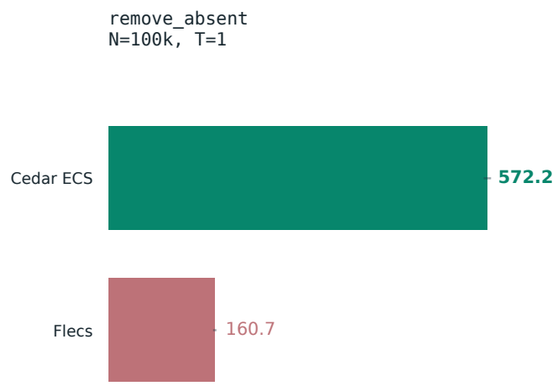
remove_absent
N=1k, T=1, width=8, bytes=0

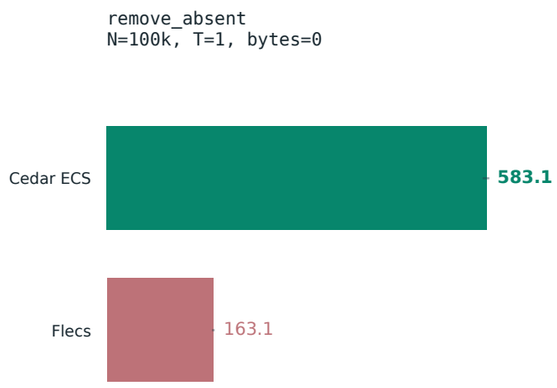


Packed · Cedar vs Flecs

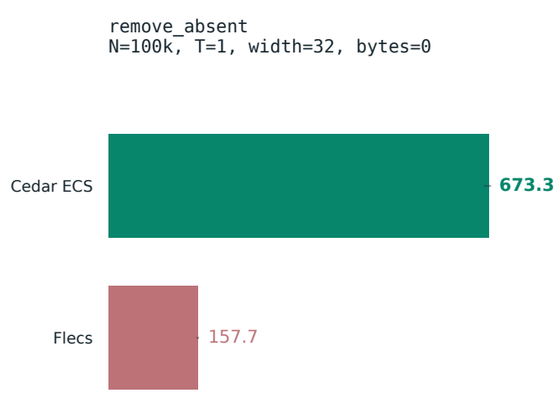
Mop/s ↑

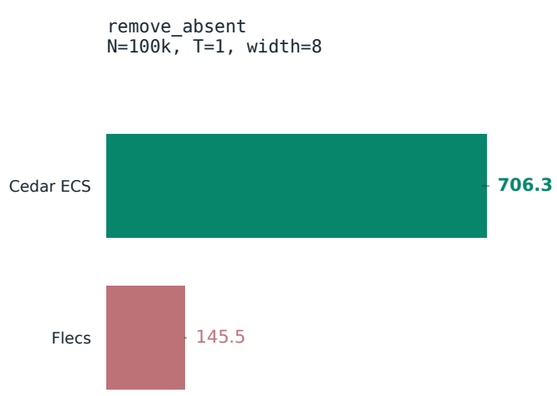
Cedar ECS Flecs

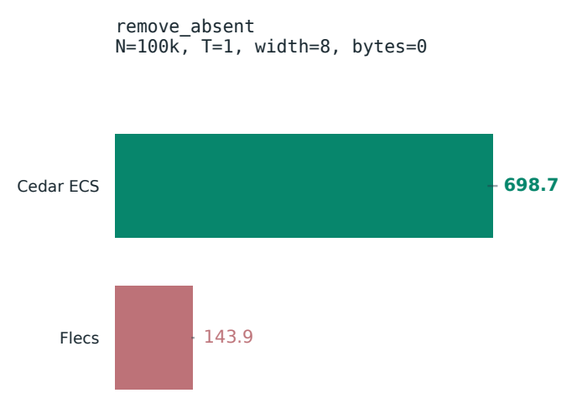


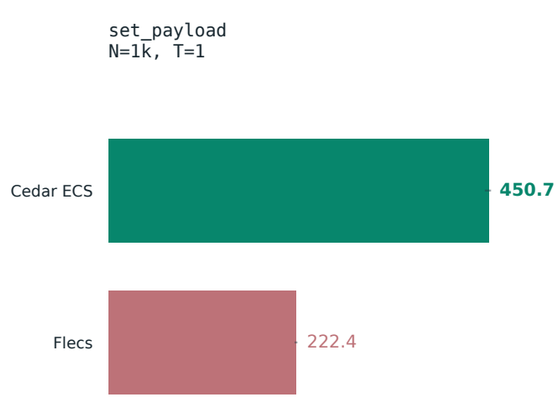


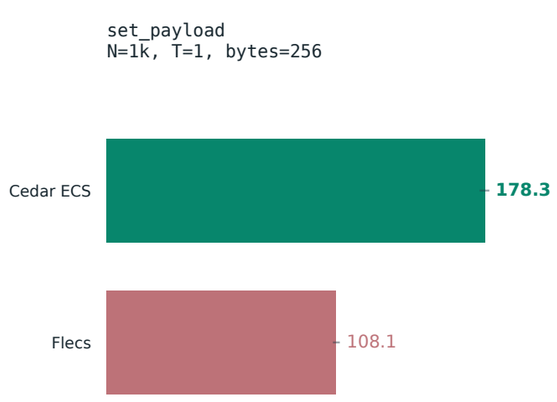


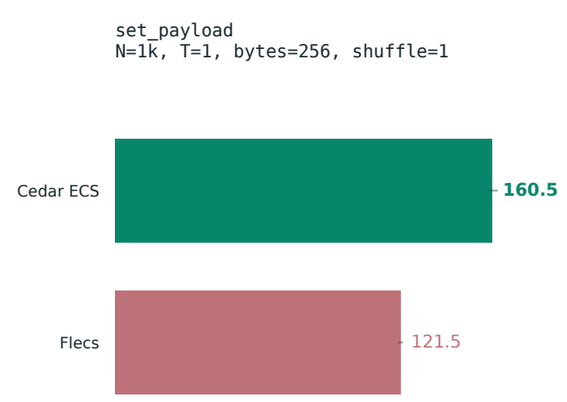


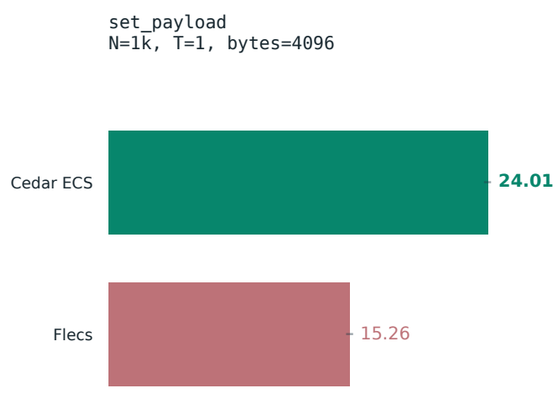


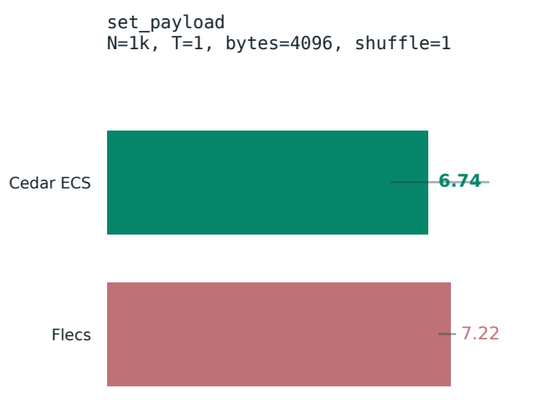


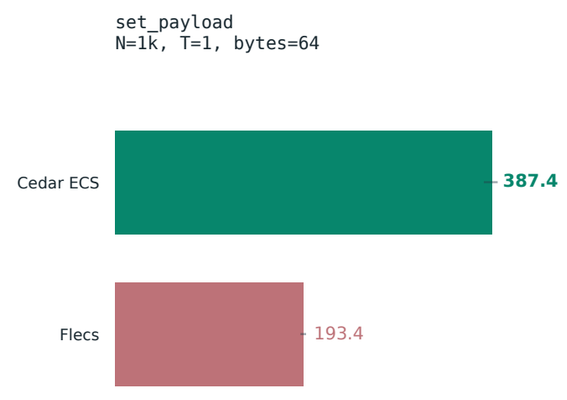








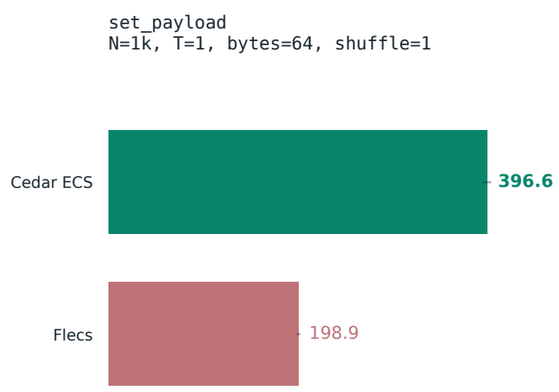


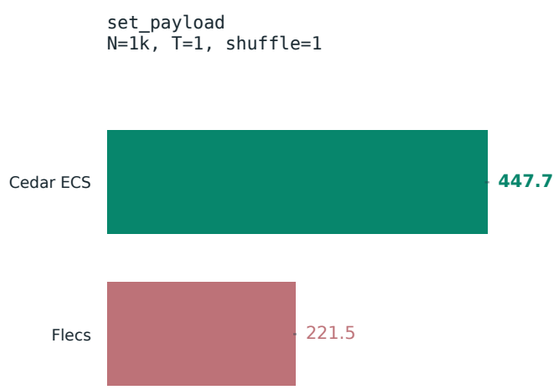


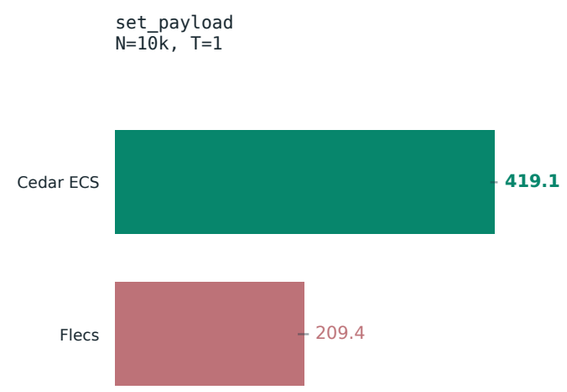
Packed · Cedar vs Flecs

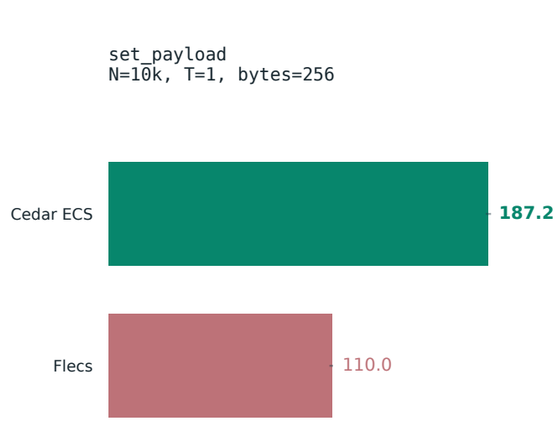
Mop/s ↑

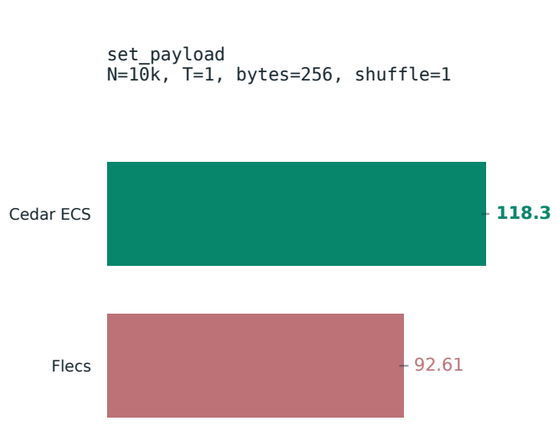
Cedar ECS Flecs

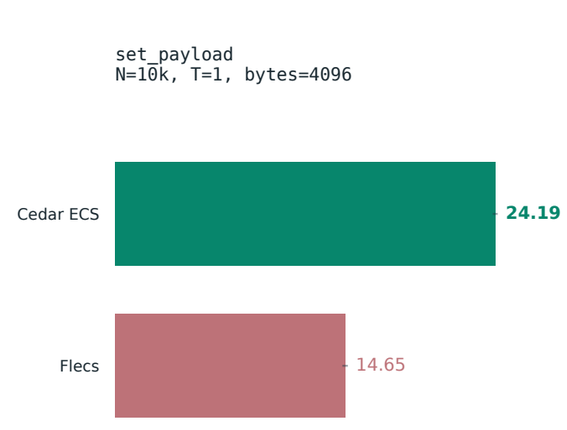


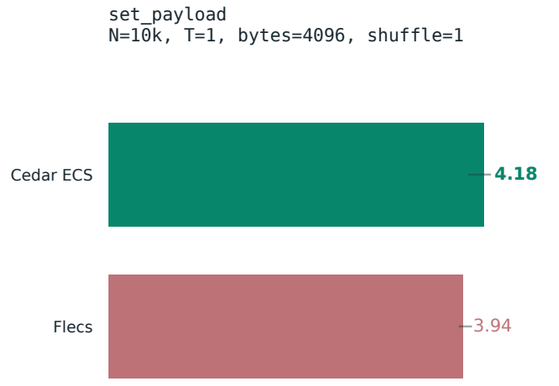


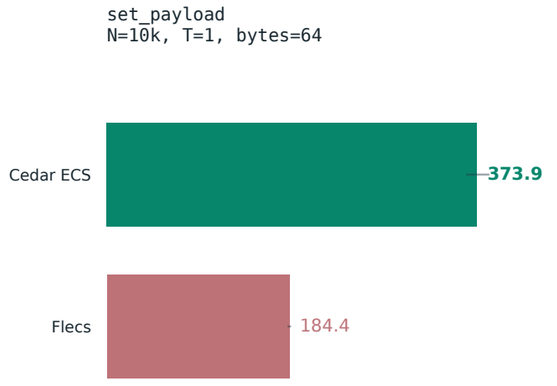


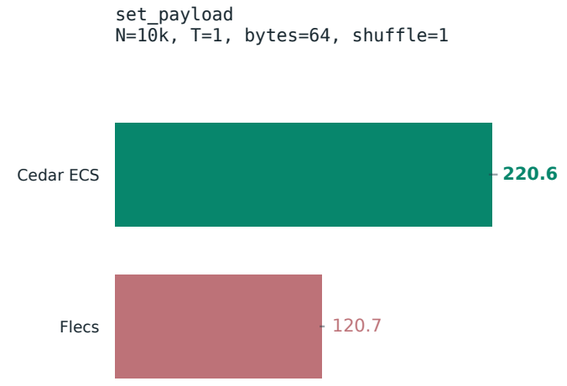


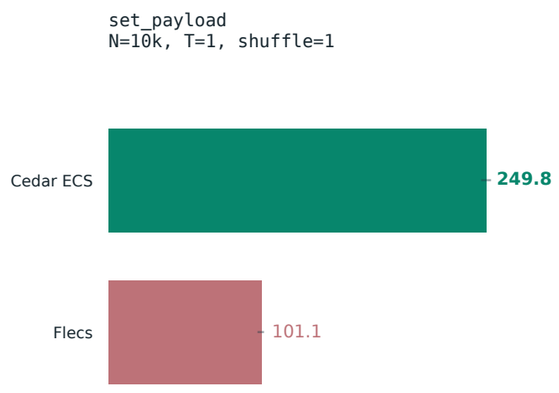


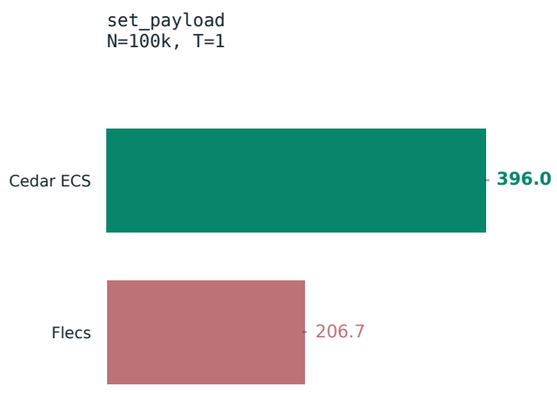


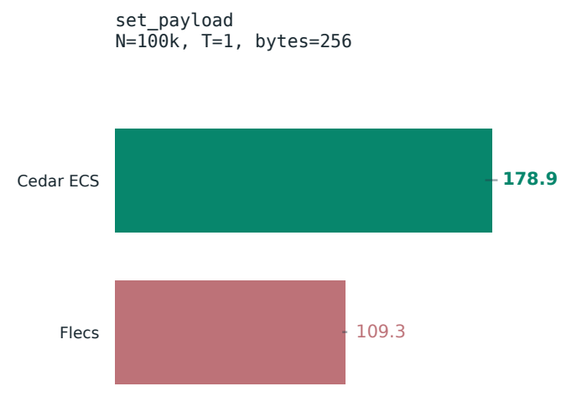










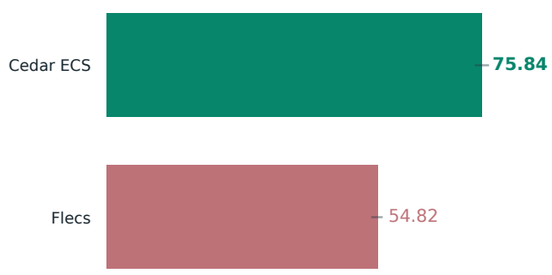


Packed · Cedar vs Flecs

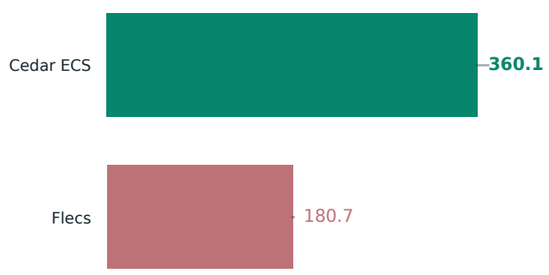
Mop/s ↑

Cedar ECS Flecs

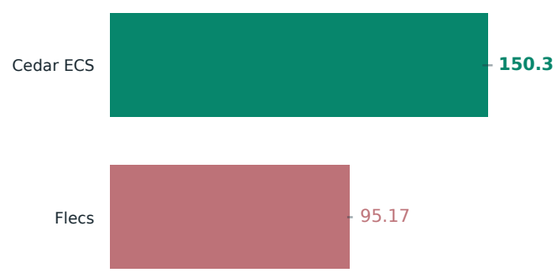
set_payload
N=100k, T=1, bytes=256, shuffle=1



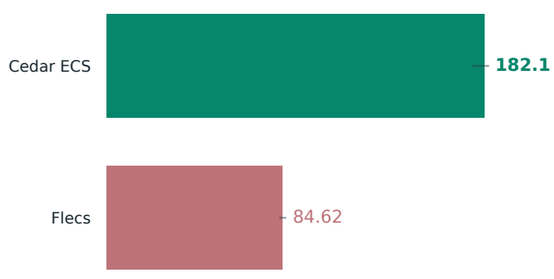
set_payload
N=100k, T=1, bytes=64



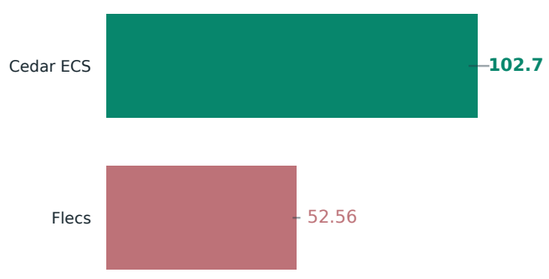
set_payload
N=100k, T=1, bytes=64, shuffle=1



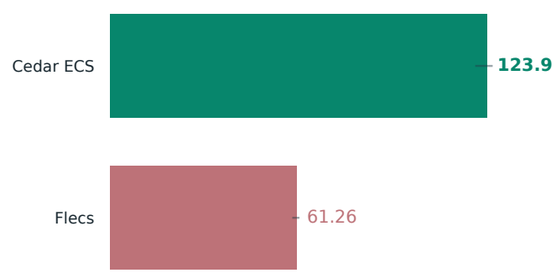
set_payload
N=100k, T=1, shuffle=1



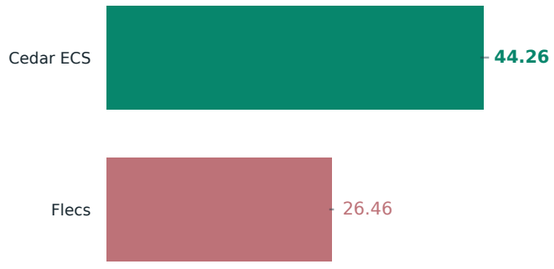
toggle_entity
N=1k, T=1



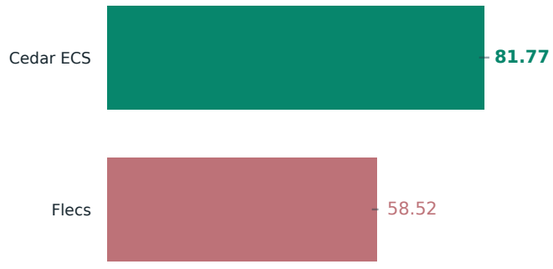
toggle_entity
N=1k, T=1, bytes=0



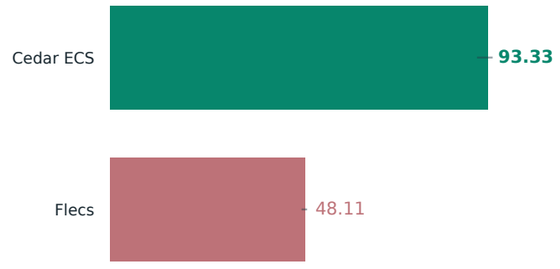
toggle_entity
N=1k, T=1, width=16



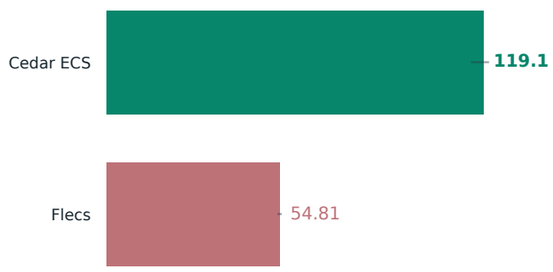
toggle_entity
N=1k, T=1, width=16, bytes=0



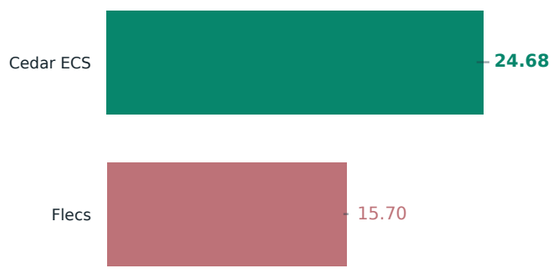
toggle_entity
N=1k, T=1, width=2



toggle_entity
N=1k, T=1, width=2, bytes=0



toggle_entity
N=1k, T=1, width=32



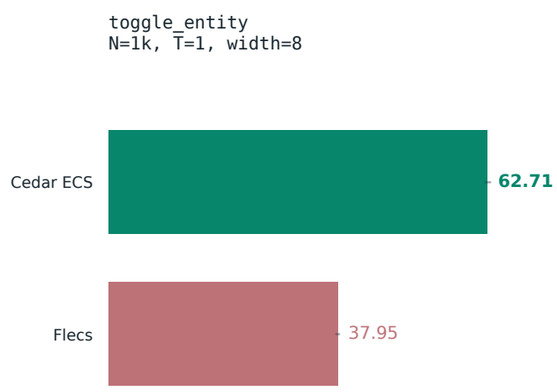
toggle_entity
N=1k, T=1, width=32, bytes=0

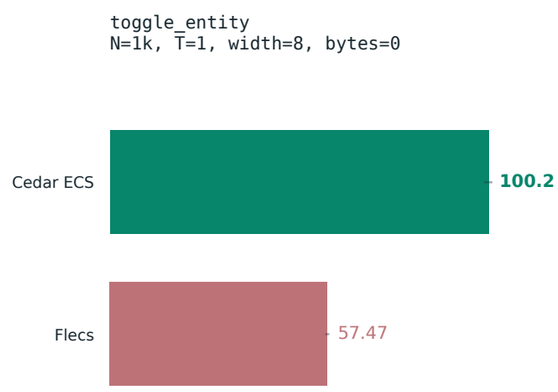


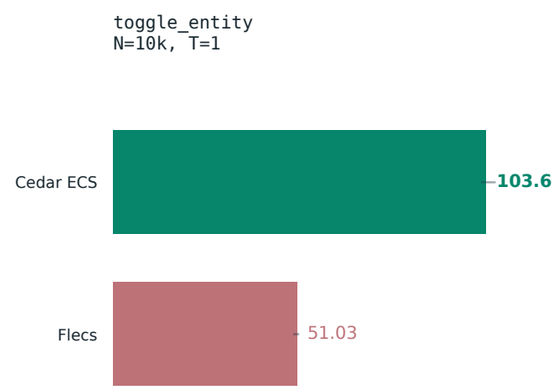
Packed · Cedar vs Flecs

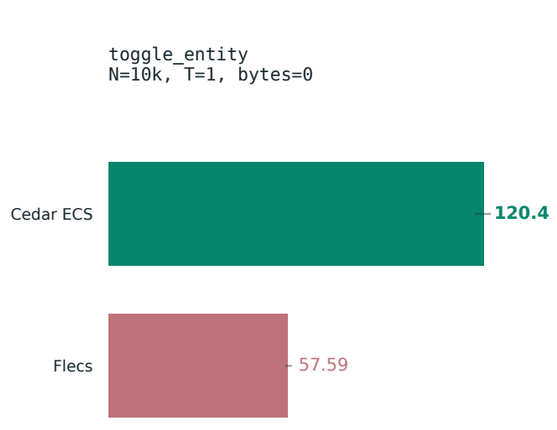
Mop/s ↑

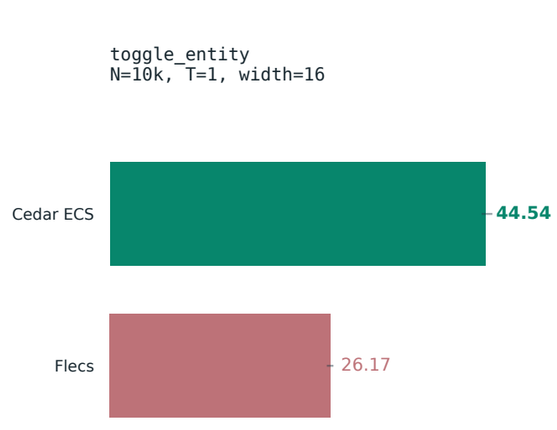
Cedar ECS Flecs

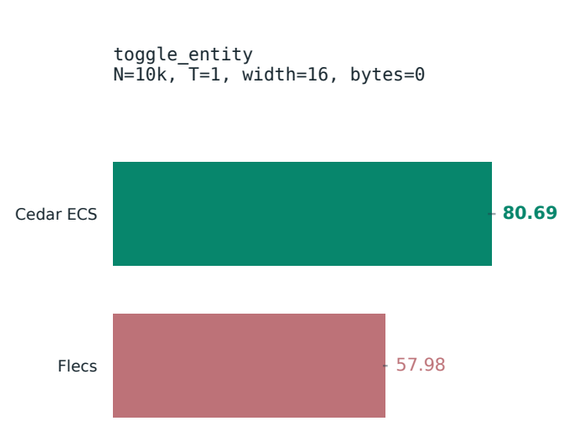


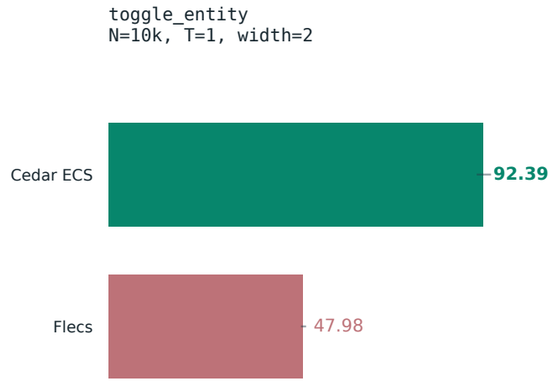


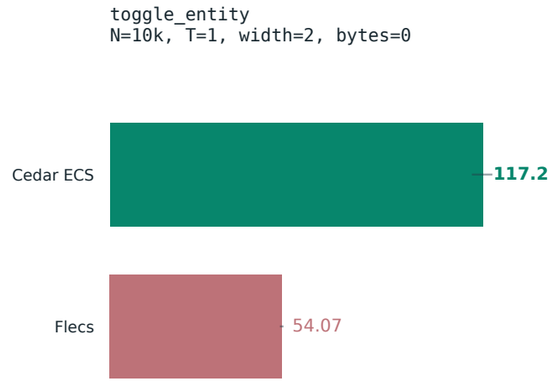


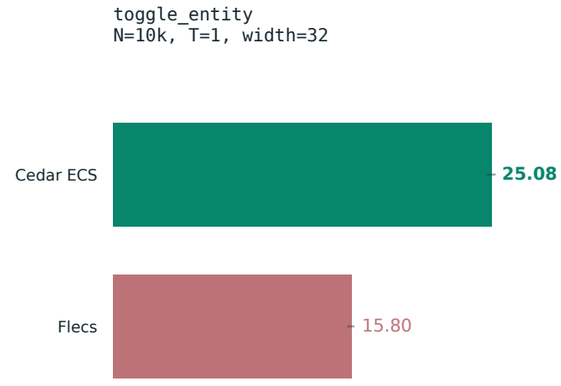


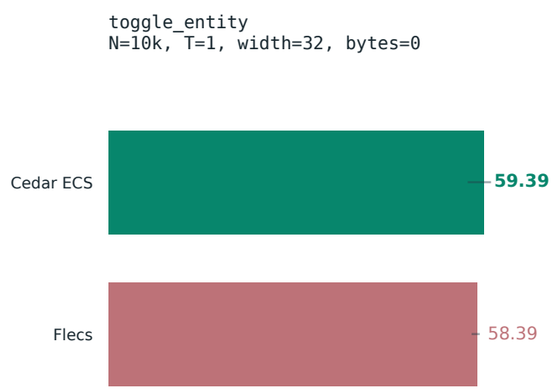


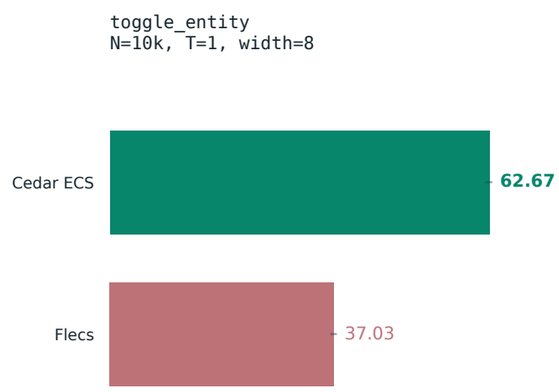


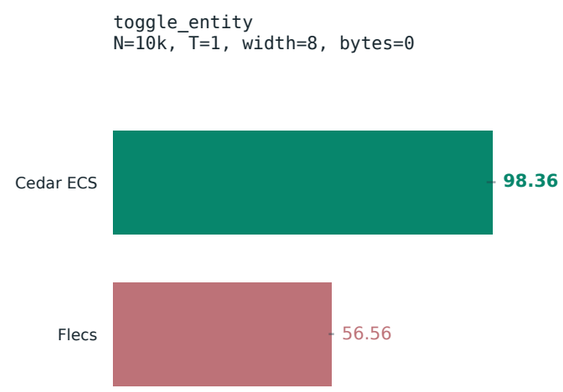








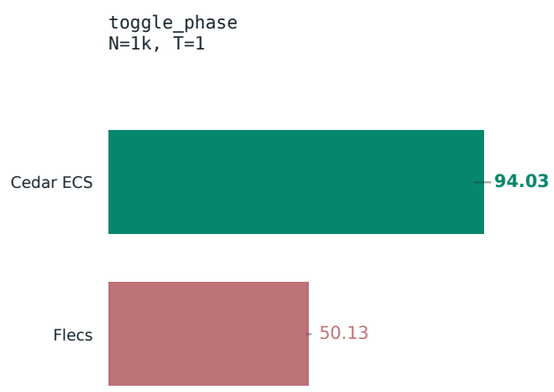


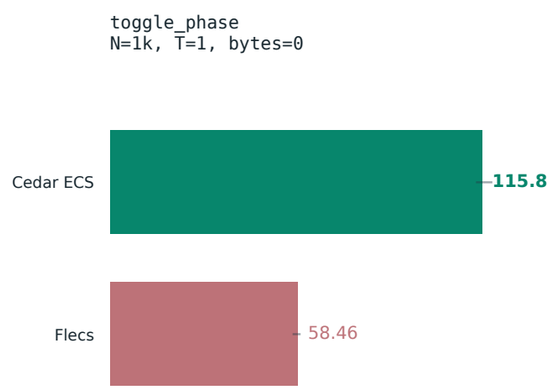


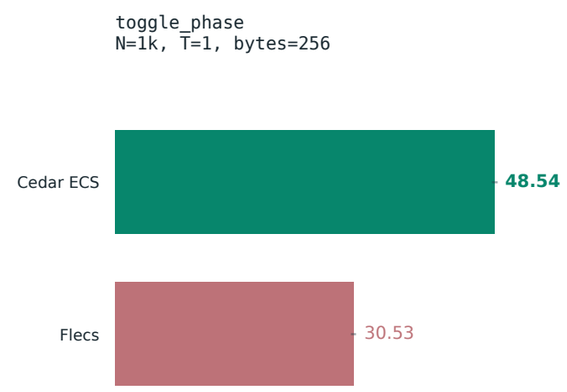
Packed · Cedar vs Flecs

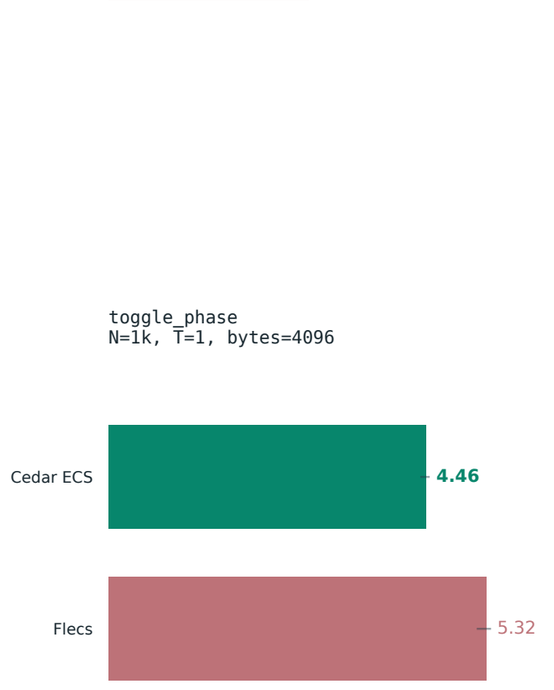
Mop/s ↑

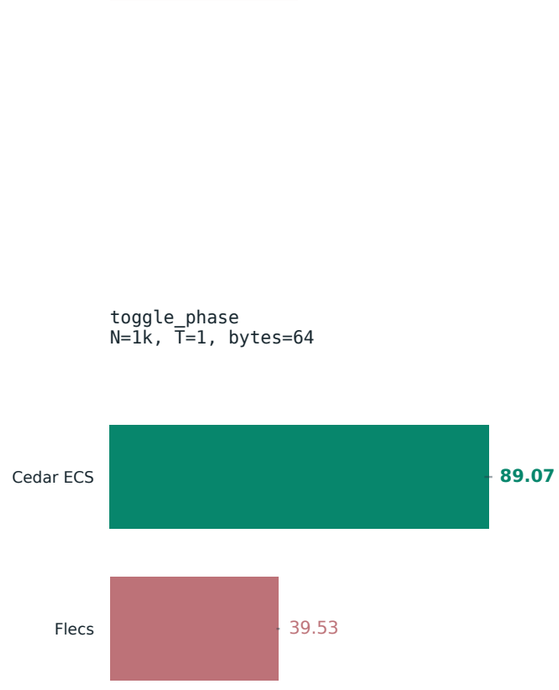
Cedar ECS Flecs

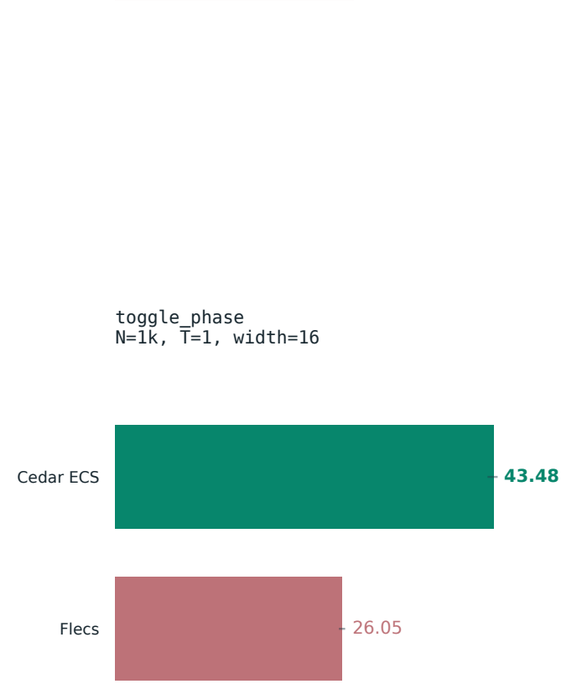


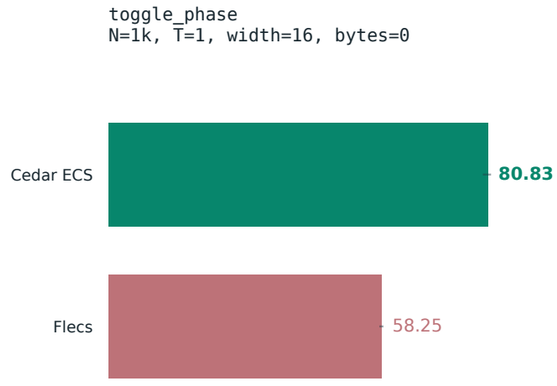


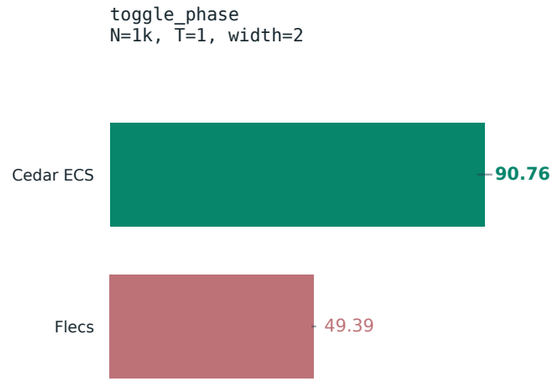


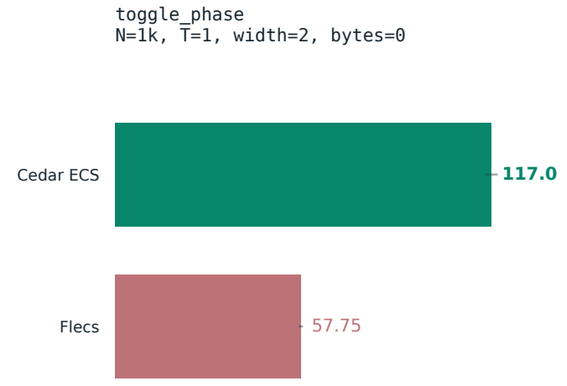


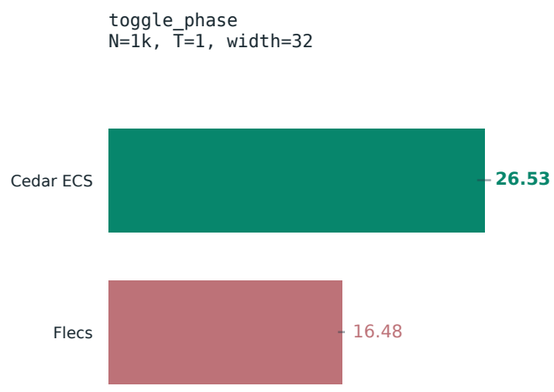




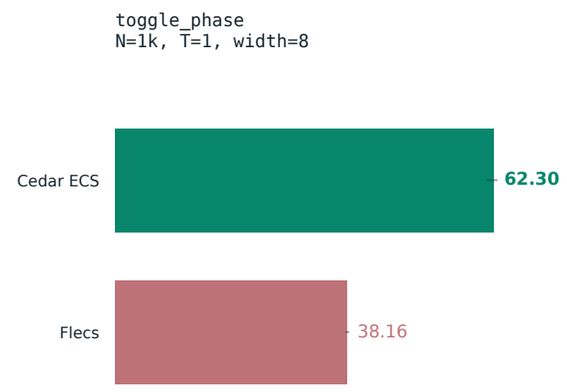








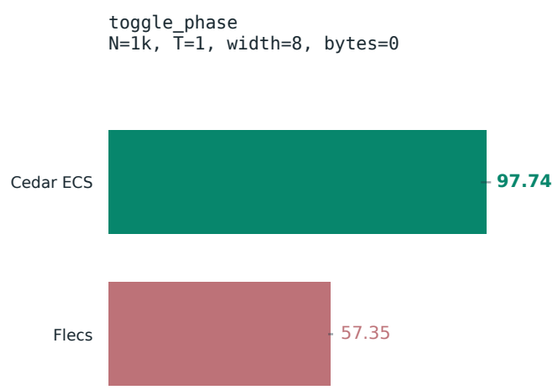


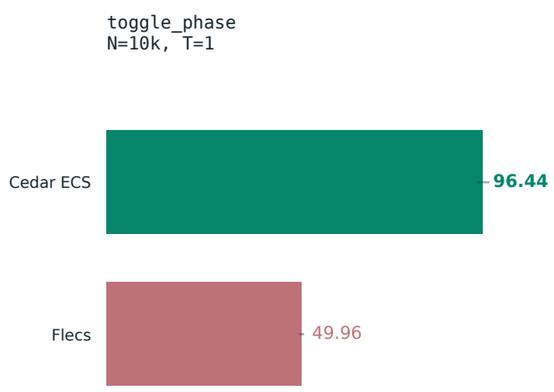


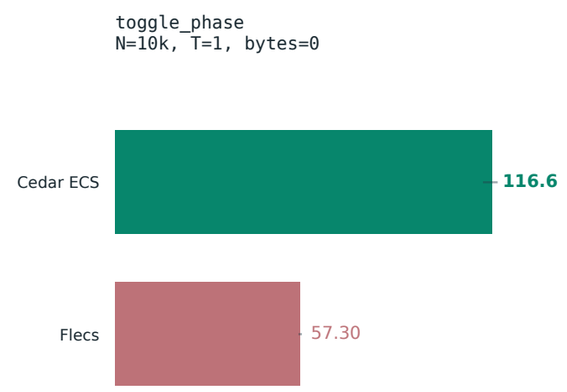
Packed · Cedar vs Flecs

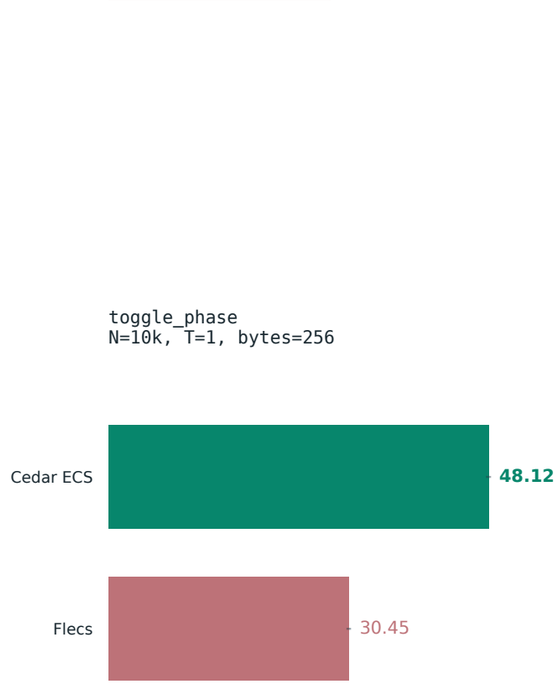
Mop/s ↑

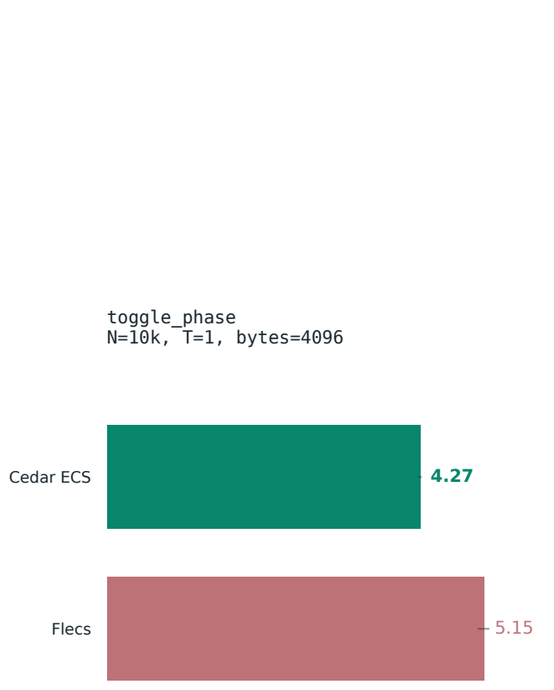
Cedar ECS Flecs

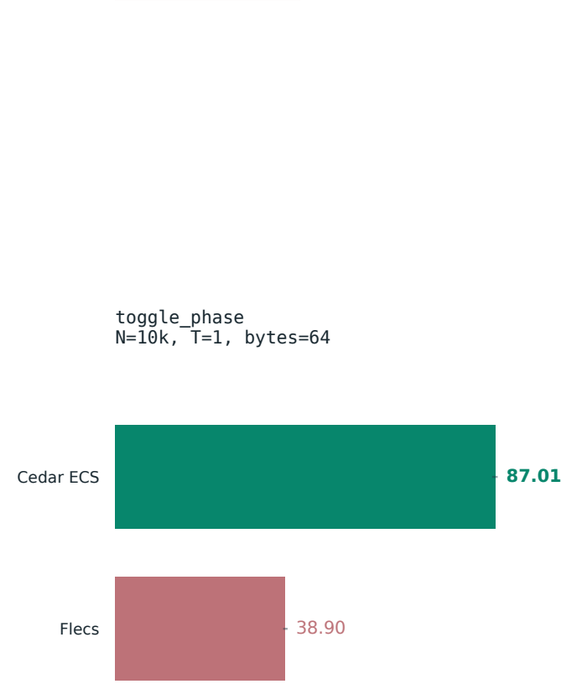


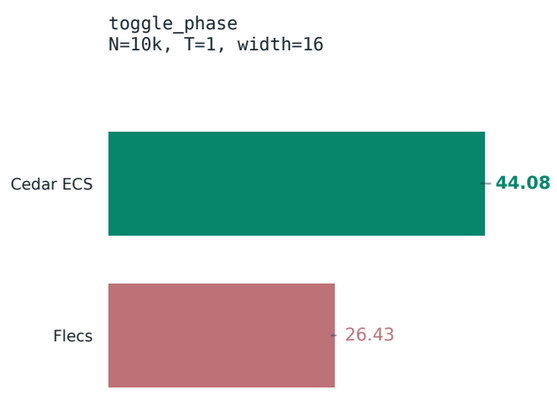


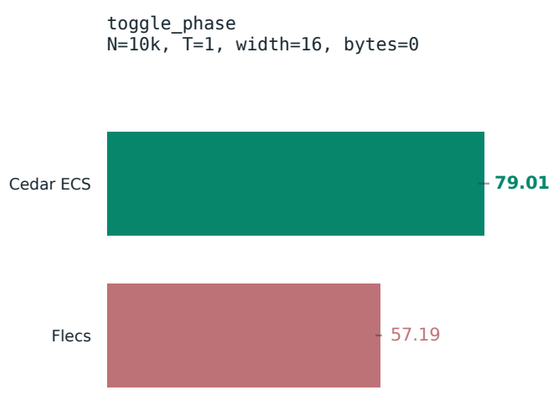


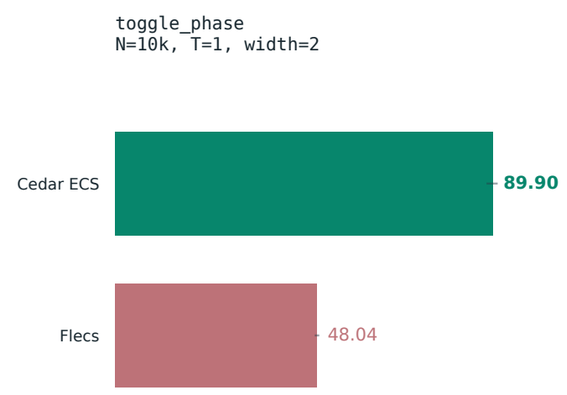


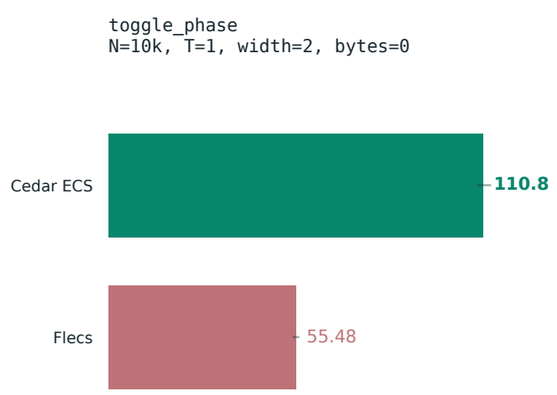


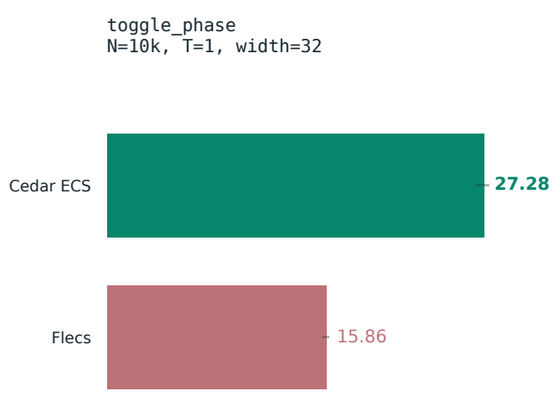


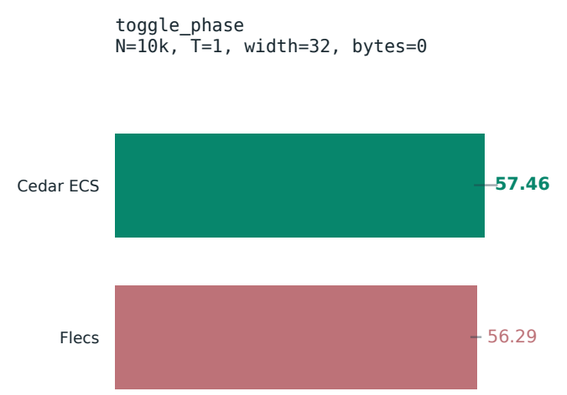








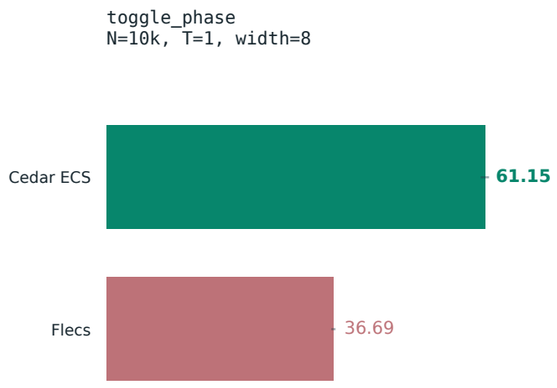


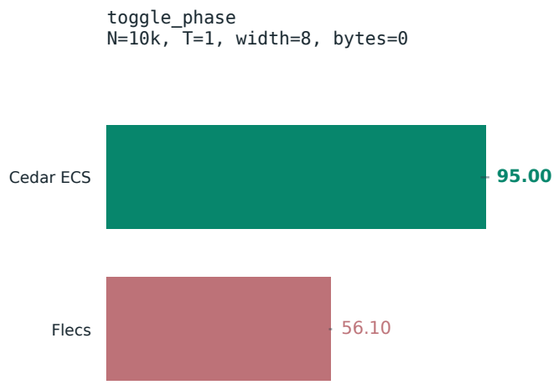


Packed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs



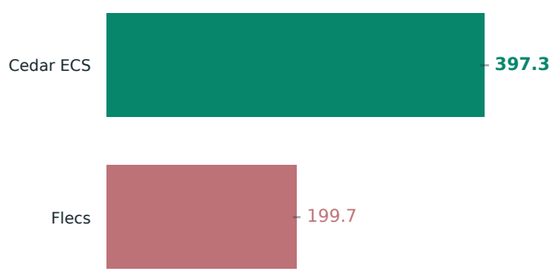


Indexed · Cedar vs Flecs

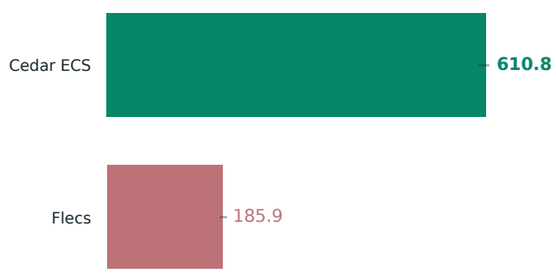
Mop/s ↑

Cedar ECS Flecs

add_existing
N=1k, T=1



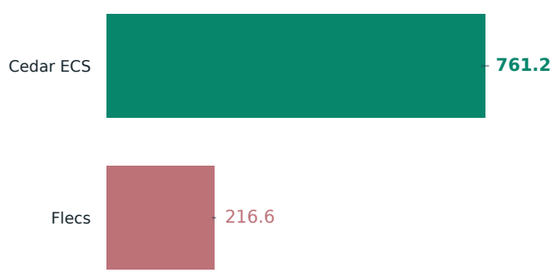
add_existing
N=1k, T=1, bytes=0



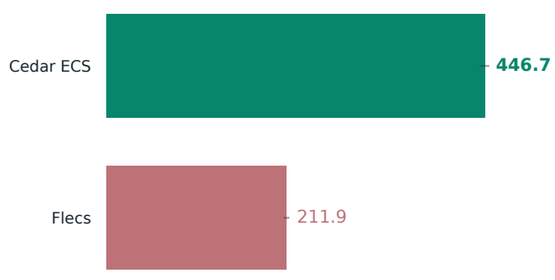
add_existing
N=1k, T=1, width=32



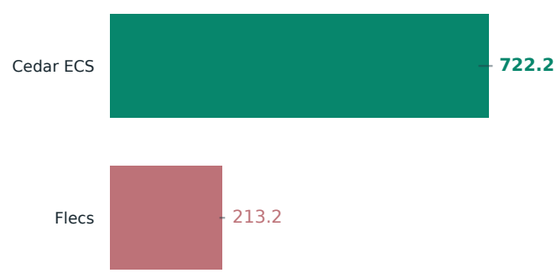
add_existing
N=1k, T=1, width=32, bytes=0



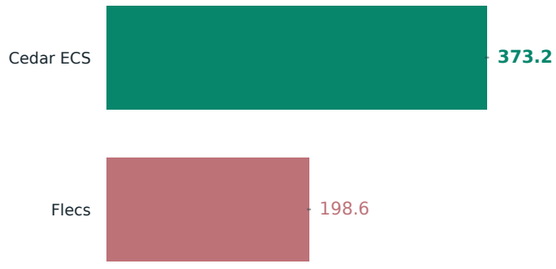
add_existing
N=1k, T=1, width=8



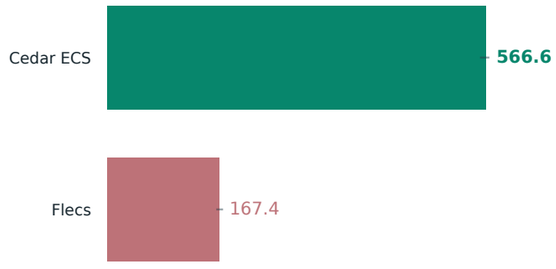
add_existing
N=1k, T=1, width=8, bytes=0



add_existing
N=100k, T=1



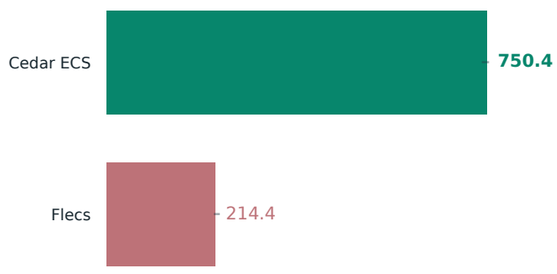
add_existing
N=100k, T=1, bytes=0



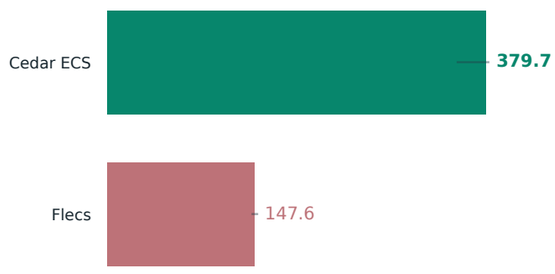
add_existing
N=100k, T=1, width=32



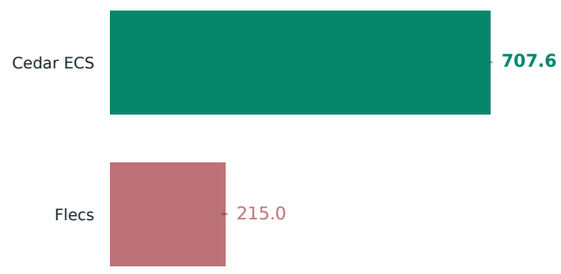
add_existing
N=100k, T=1, width=32, bytes=0



add_existing
N=100k, T=1, width=8



add_existing
N=100k, T=1, width=8, bytes=0



Indexed · Cedar vs Flecs

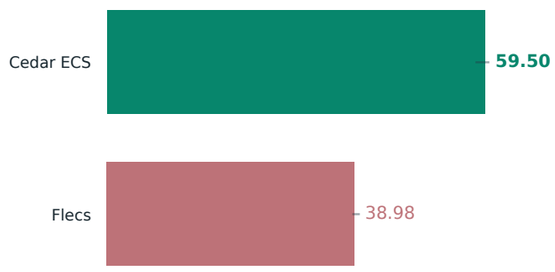
Mop/s ↑

Cedar ECS Flecs

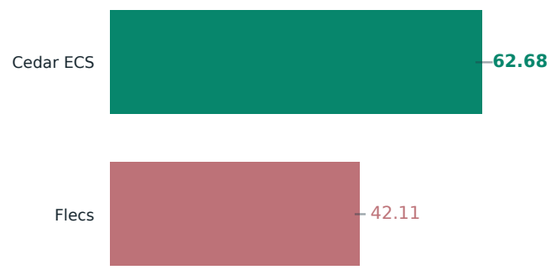
churn
N=10k, T=1, width=2, percent=1, shuffle=1



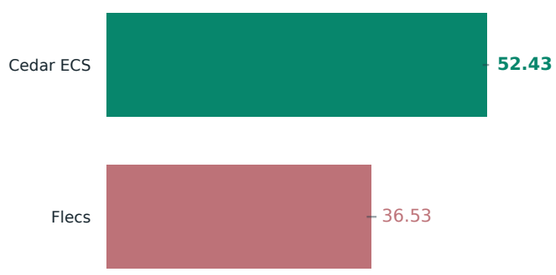
churn
N=10k, T=1, width=2, percent=50, shuffle=1



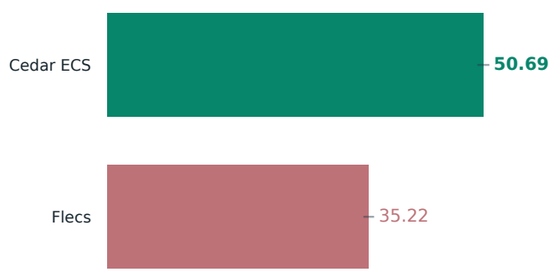
churn
N=10k, T=1, width=2, shuffle=1



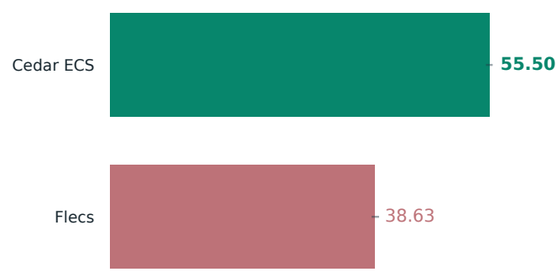
churn
N=100k, T=1, width=2, percent=1, shuffle=1



churn
N=100k, T=1, width=2, percent=50, shuffle=1



churn
N=100k, T=1, width=2, shuffle=1



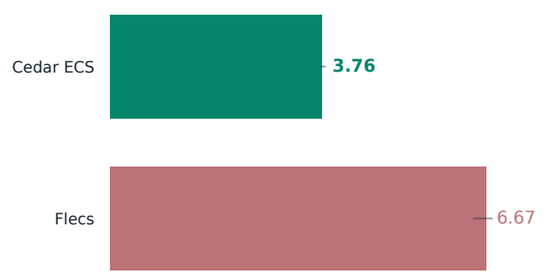
create
N=1k, T=1



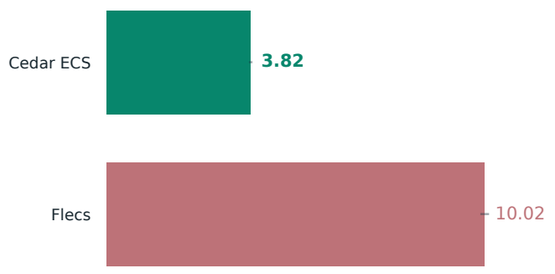
create
N=1k, T=1, reserve=1



create
N=1k, T=1, width=32



create
N=1k, T=1, width=32, reserve=1



create
N=1k, T=1, width=8



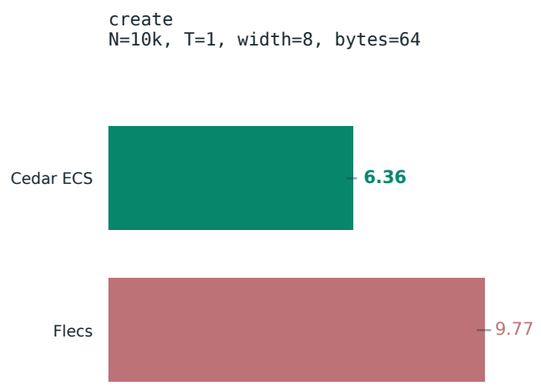
create
N=1k, T=1, width=8, reserve=1



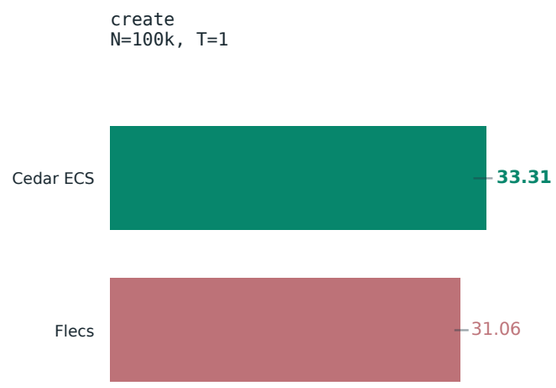
Indexed · Cedar vs Flecs

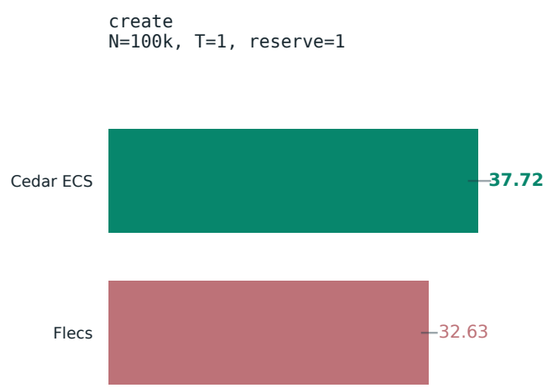
Mop/s ↑

Cedar ECS Flecs



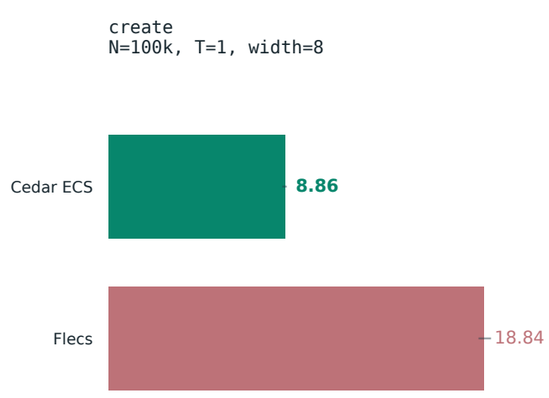


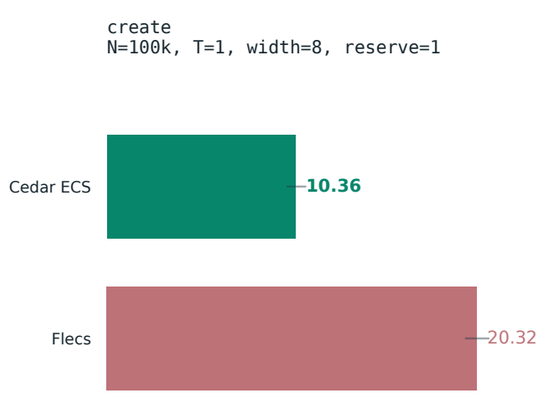


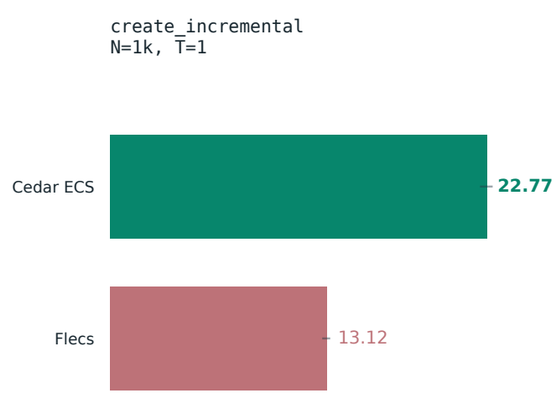


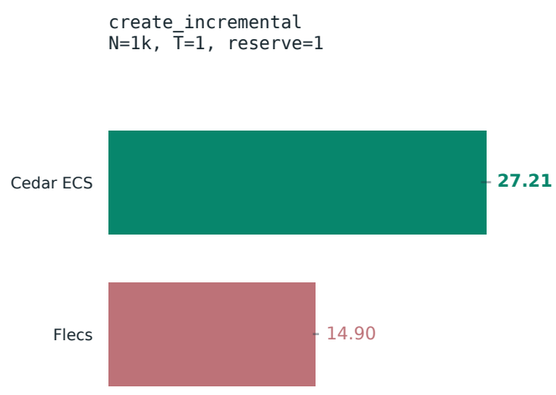


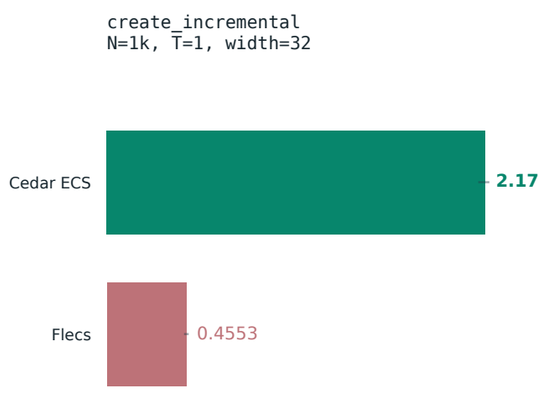


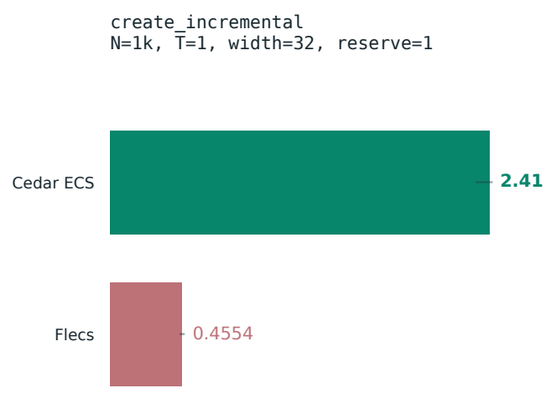










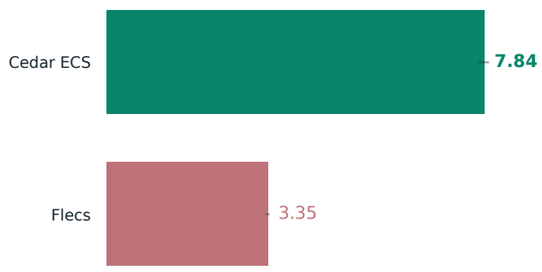


Indexed · Cedar vs Flecs

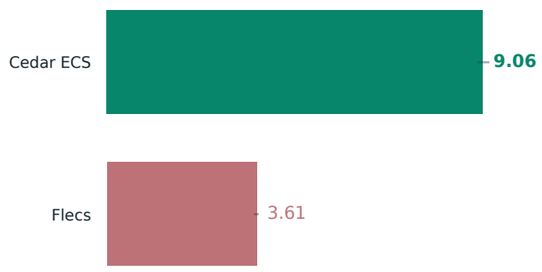
Mop/s ↑

Cedar ECS Flecs

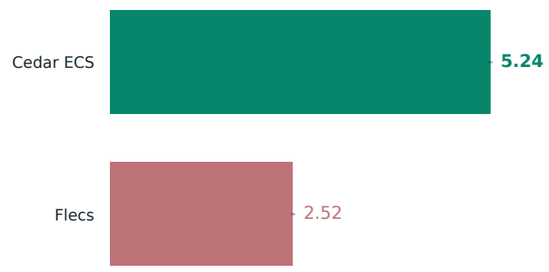
create_incremental
N=1k, T=1, width=8



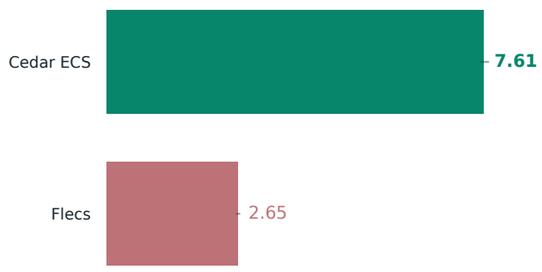
create_incremental
N=1k, T=1, width=8, reserve=1



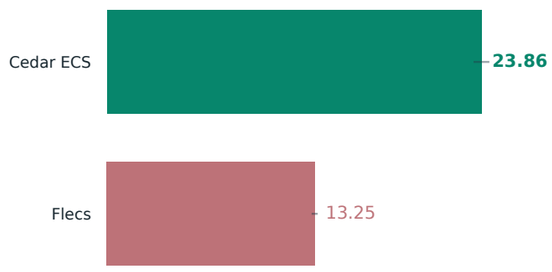
create_incremental
N=10k, T=1, width=8, bytes=64



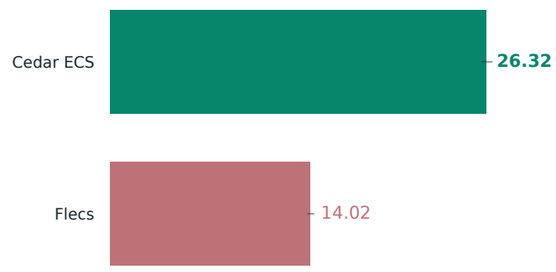
create_incremental
N=10k, T=1, width=8, bytes=64, reserve=1



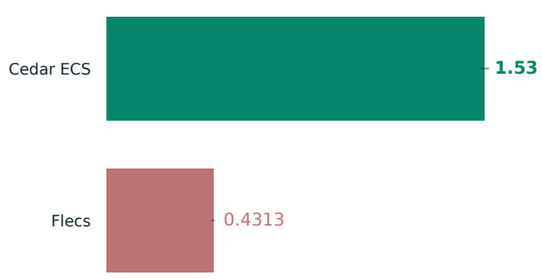
create_incremental
N=100k, T=1



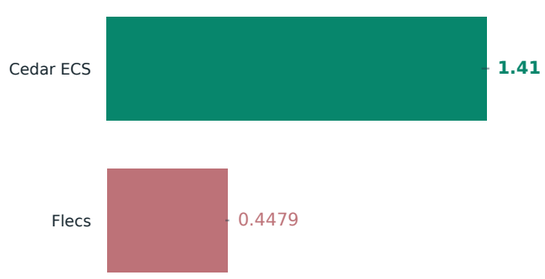
create_incremental
N=100k, T=1, reserve=1



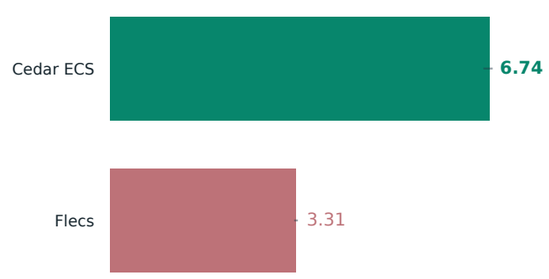
create_incremental
N=100k, T=1, width=32



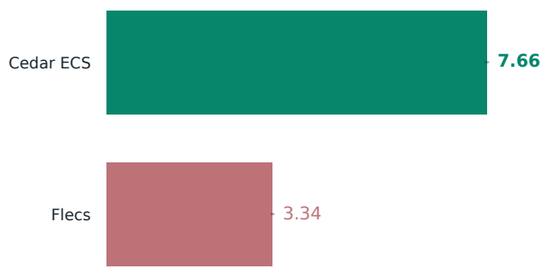
create_incremental
N=100k, T=1, width=32, reserve=1



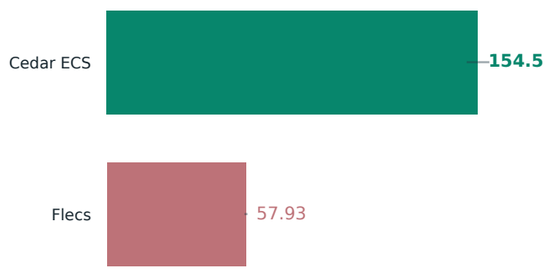
create_incremental
N=100k, T=1, width=8



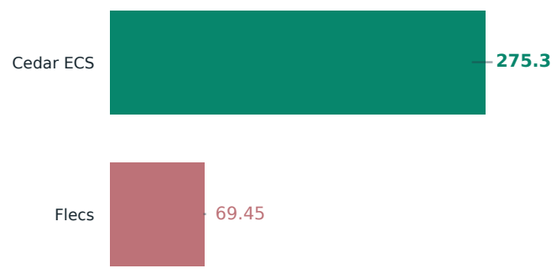
create_incremental
N=100k, T=1, width=8, reserve=1



defer_cancel
N=1k, T=1



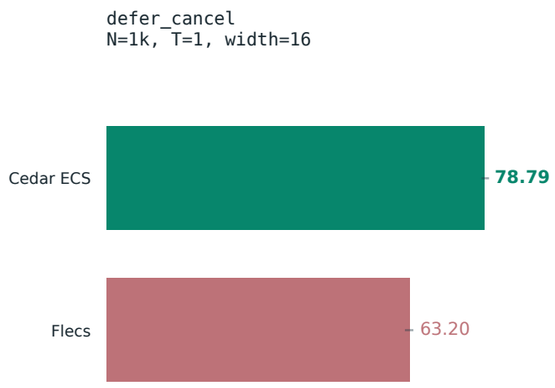
defer_cancel
N=1k, T=1, bytes=0

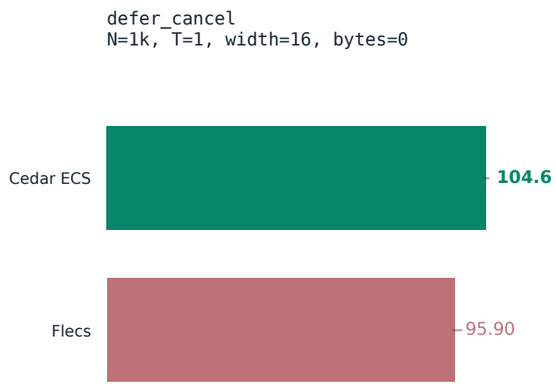


Indexed · Cedar vs Flecs

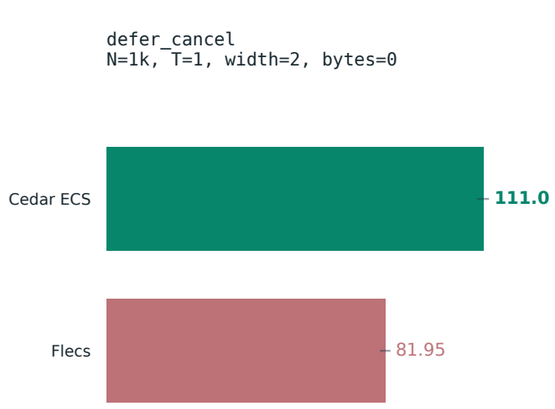
Mop/s ↑

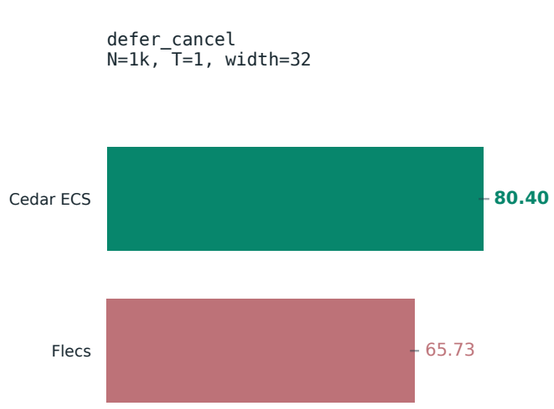
Cedar ECS Flecs

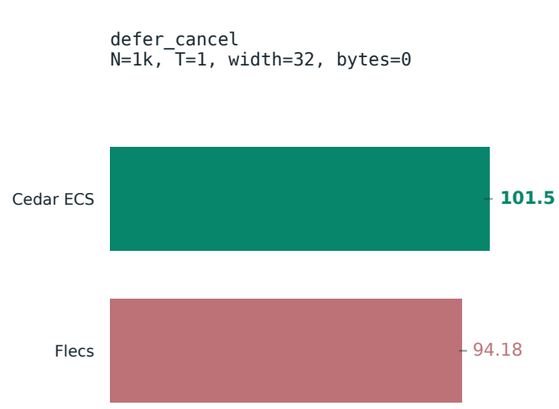


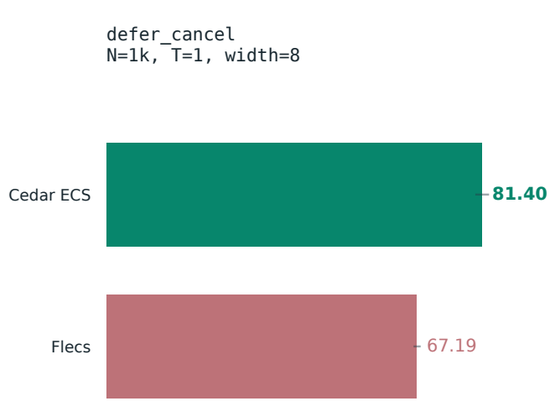


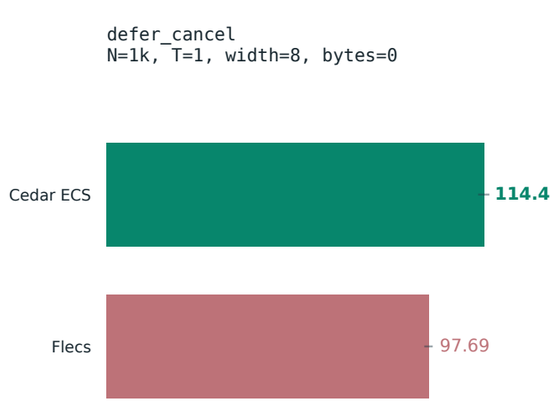


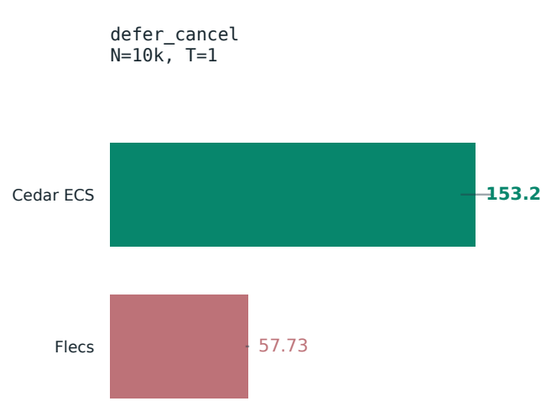


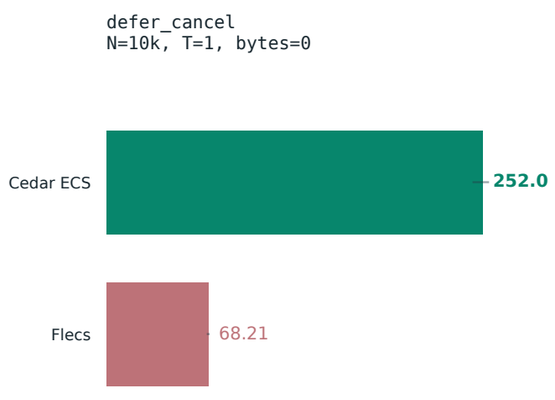


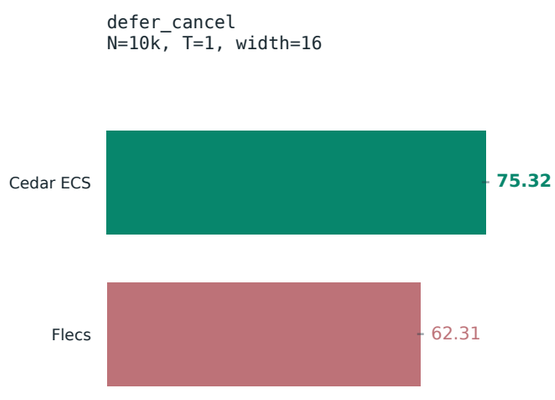


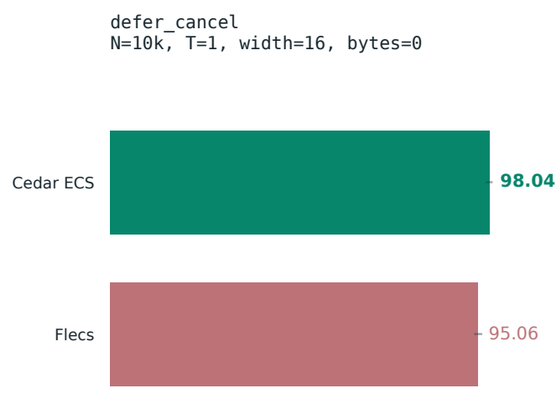








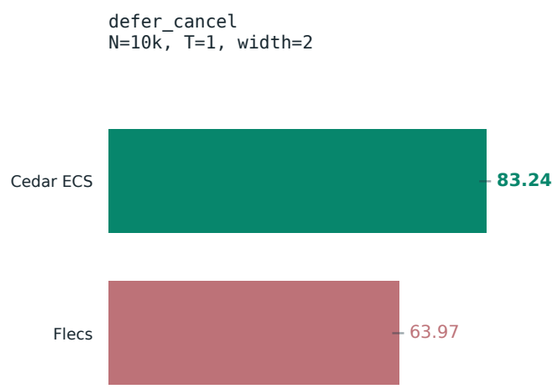


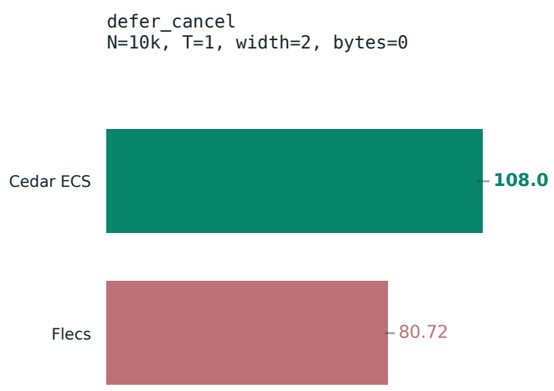


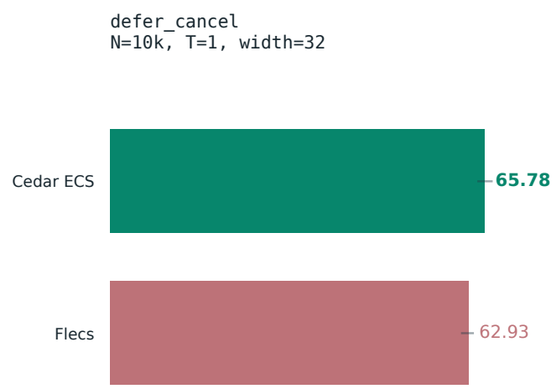
Indexed · Cedar vs Flecs

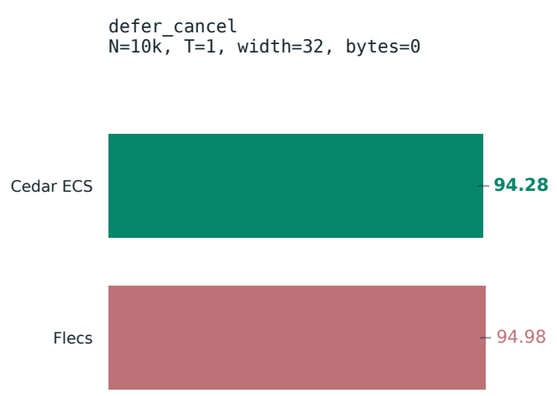
Mop/s ↑

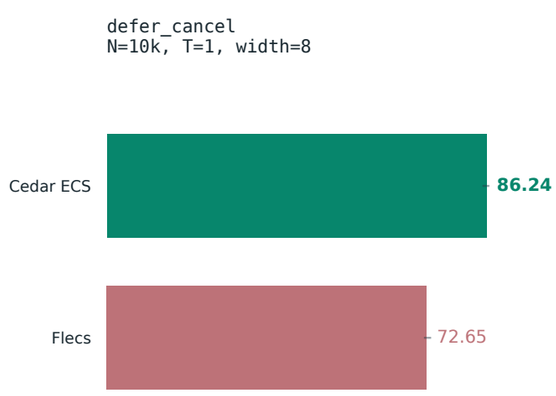
Cedar ECS Flecs

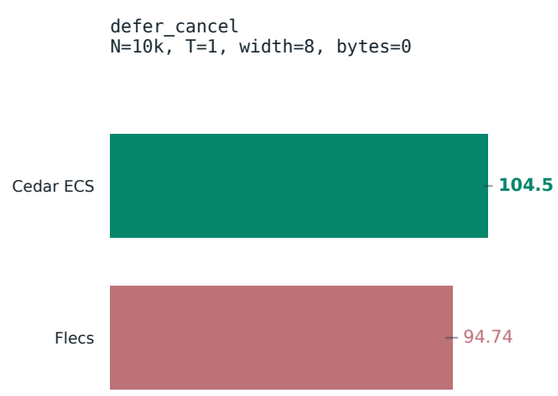


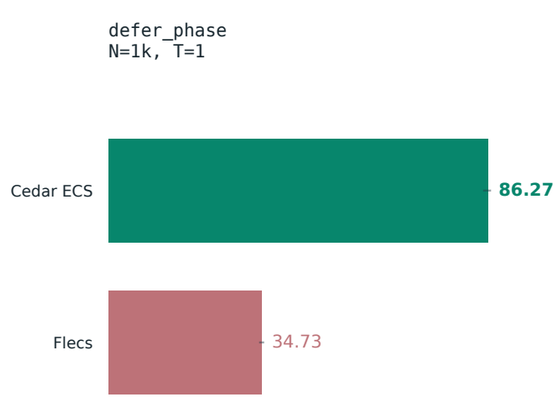


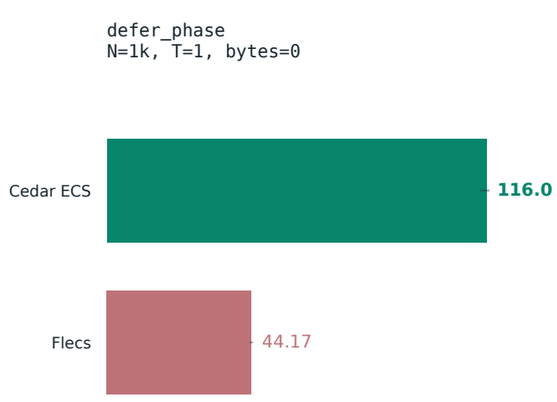


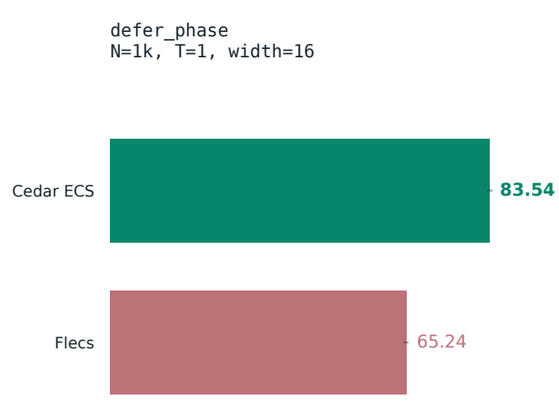


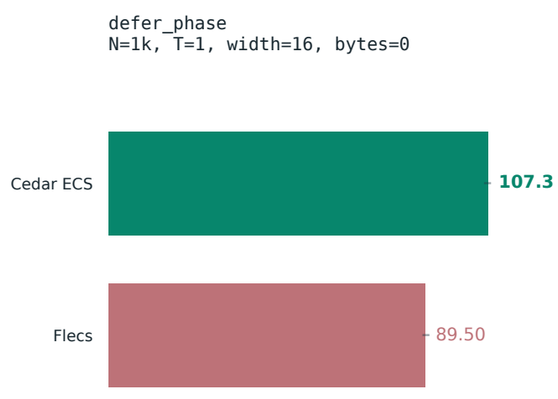


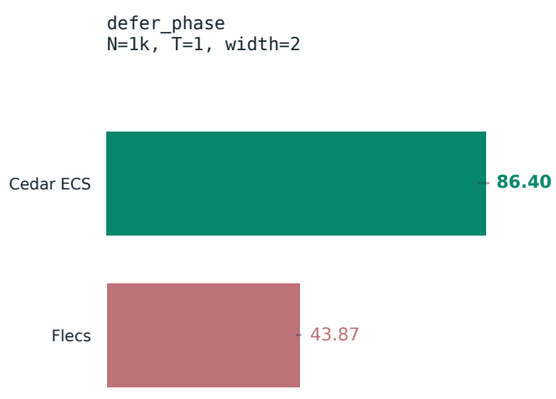


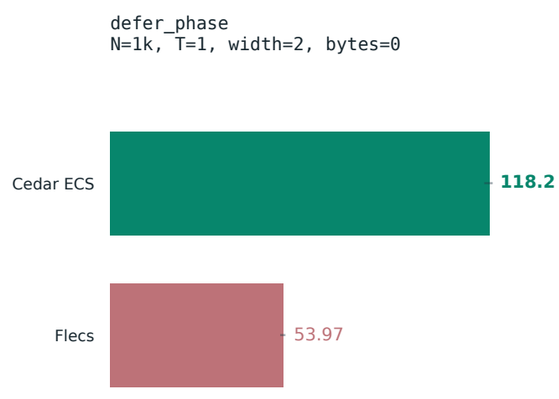








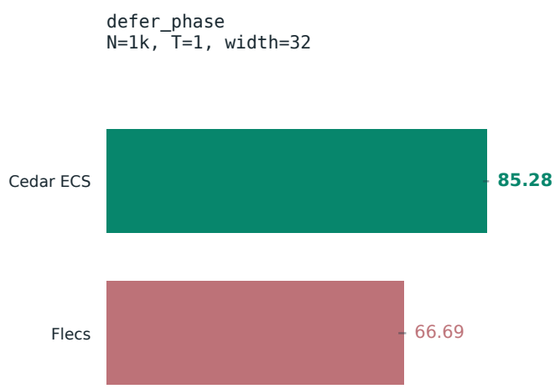


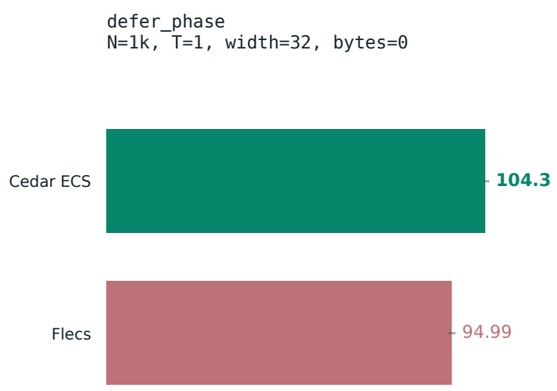


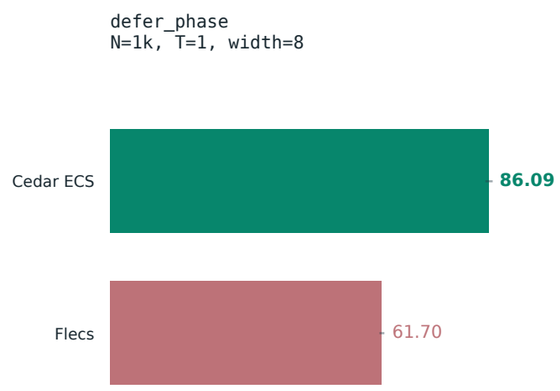
Indexed · Cedar vs Flecs

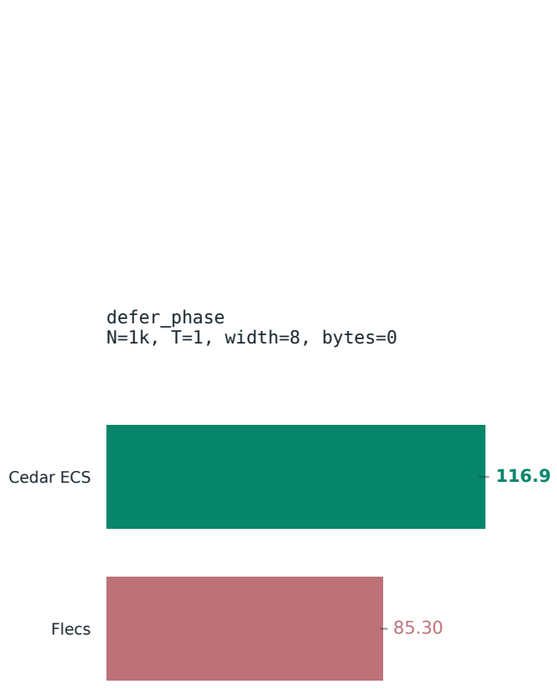
Mop/s ↑

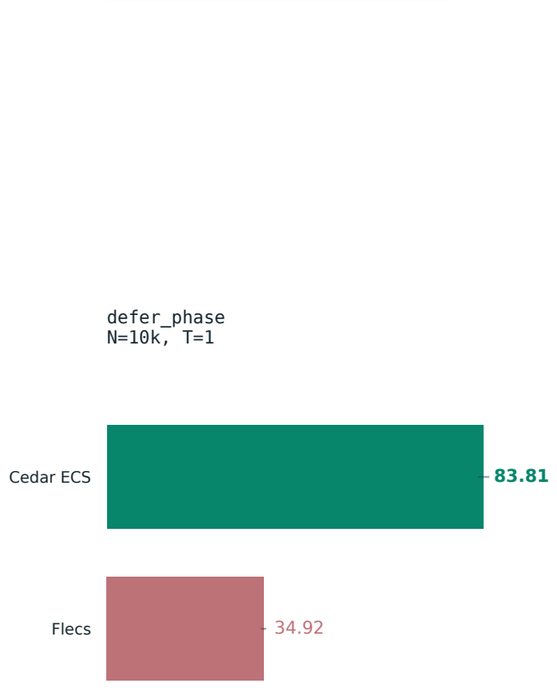
Cedar ECS Flecs

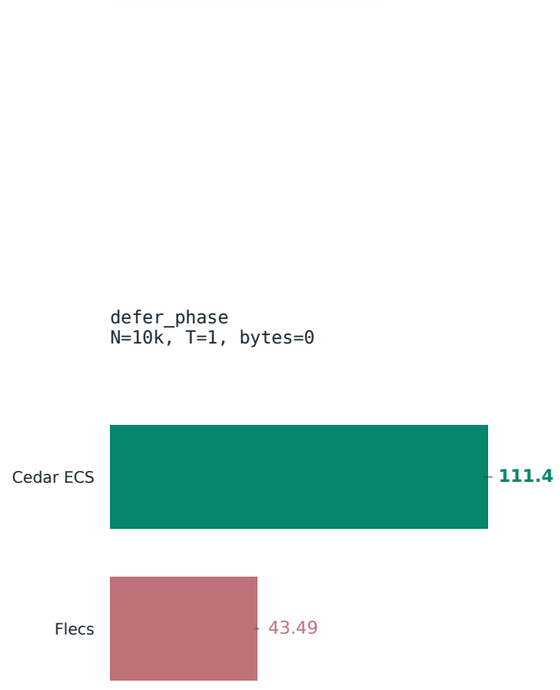


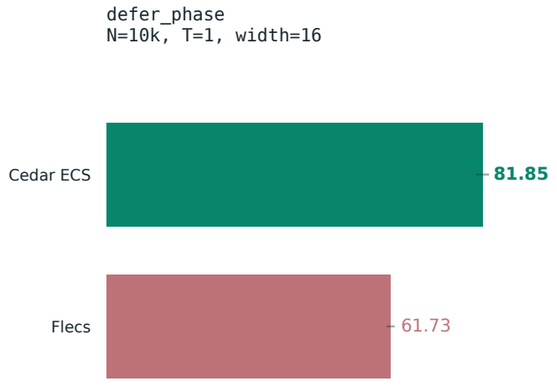


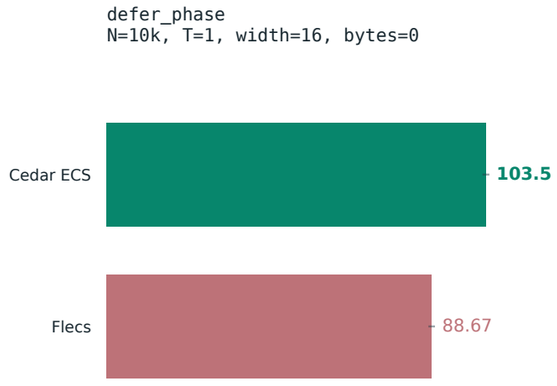


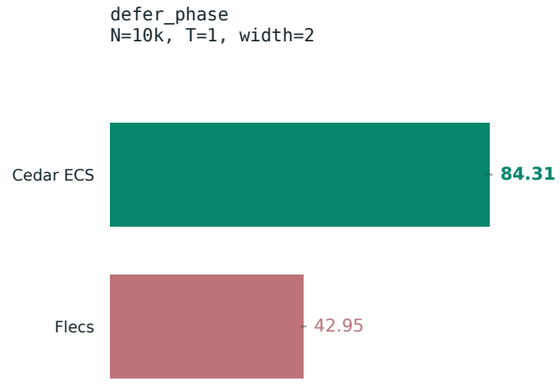


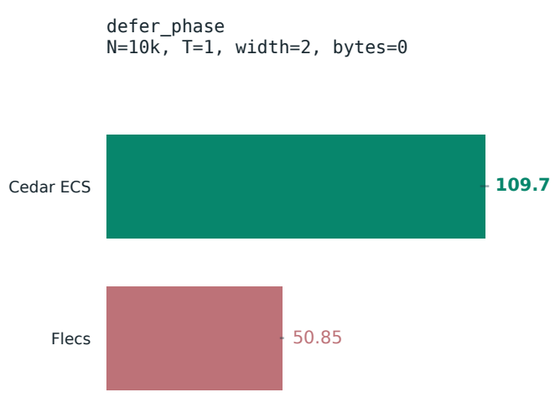


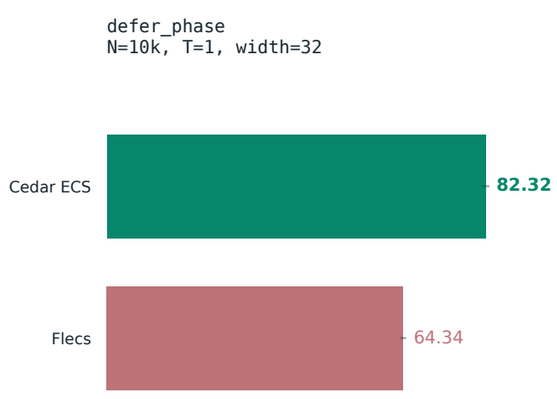


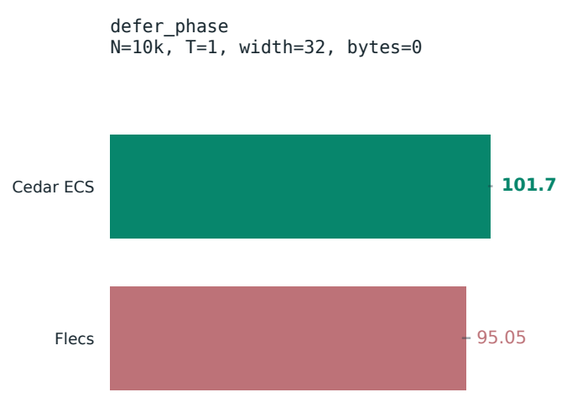












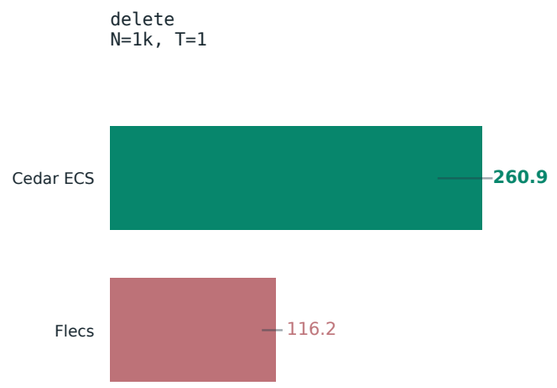
Indexed · Cedar vs Flecs

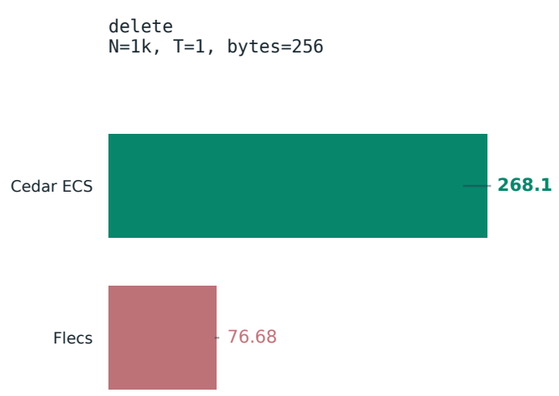
Mop/s ↑

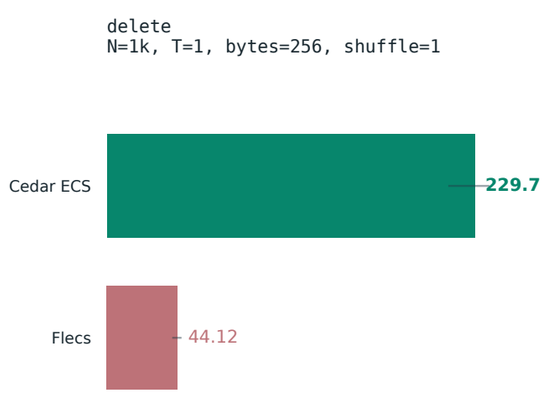
Cedar ECS Flecs

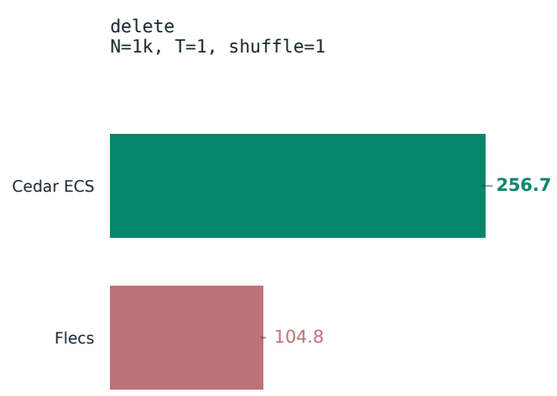


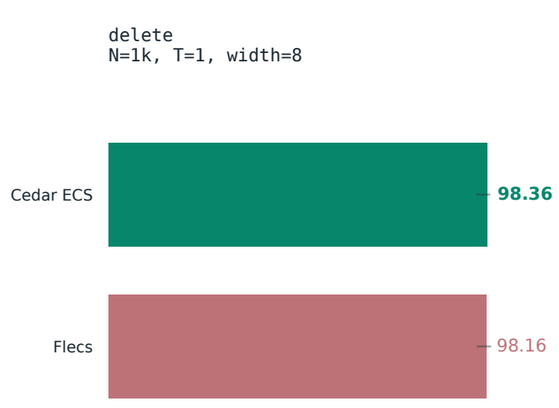


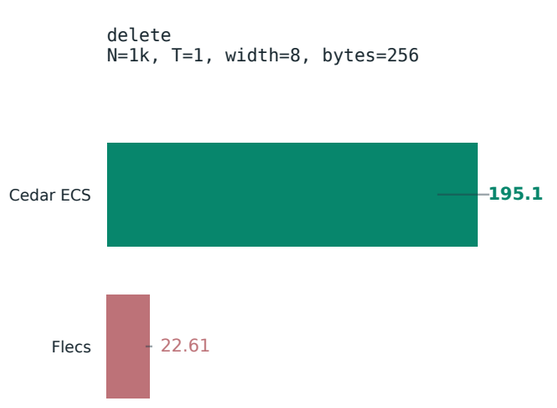


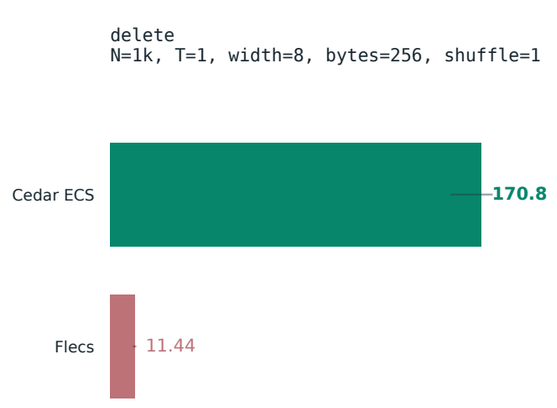


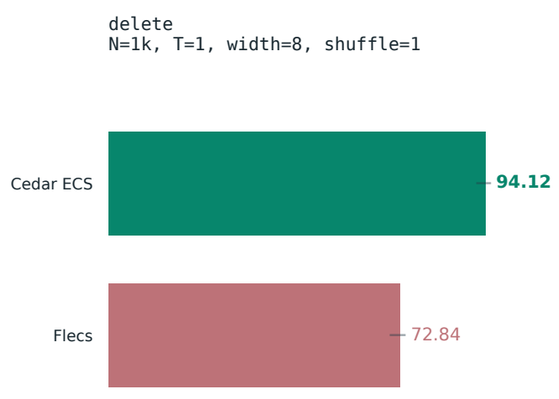


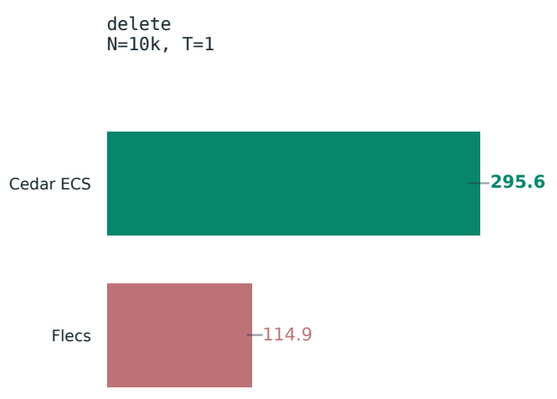


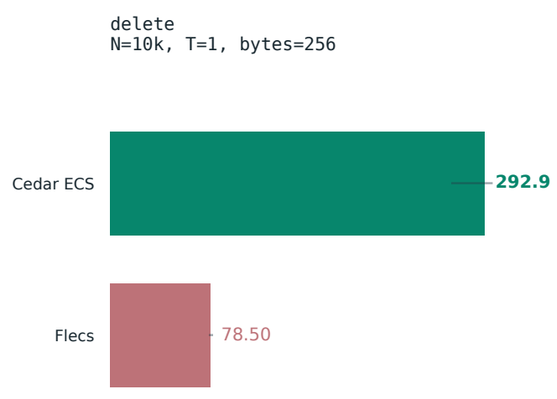










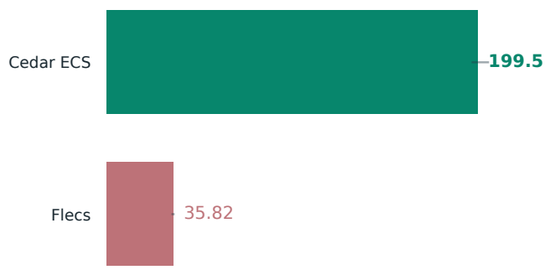


Indexed · Cedar vs Flecs

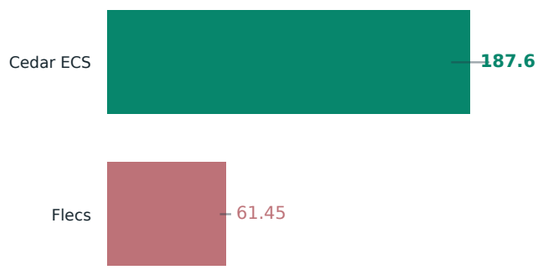
Mop/s ↑

Cedar ECS Flecs

delete
N=10k, T=1, bytes=256, shuffle=1



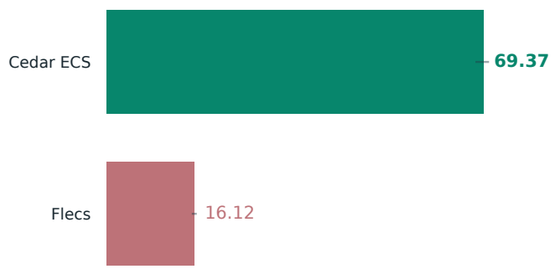
delete
N=10k, T=1, shuffle=1



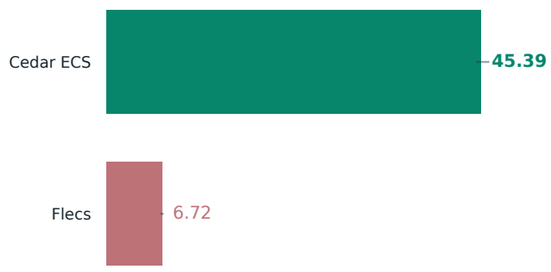
delete
N=10k, T=1, width=8



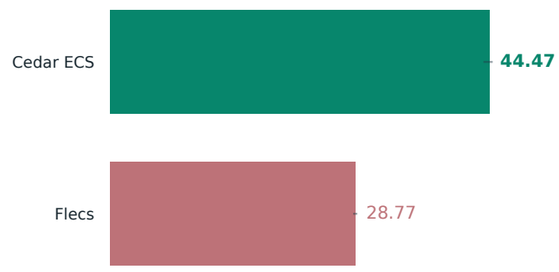
delete
N=10k, T=1, width=8, bytes=256



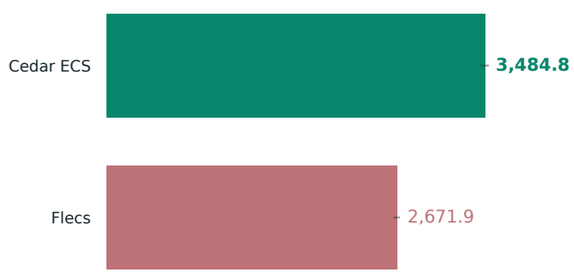
delete
N=10k, T=1, width=8, bytes=256, shuffle=1



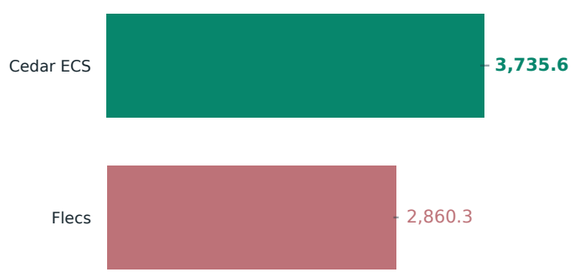
delete
N=10k, T=1, width=8, shuffle=1



frame
N=1k, T=1



frame
N=1k, T=1, grain=16384



frame
N=1k, T=1, grain=64



frame
N=1k, T=1, width=4



frame
N=1k, T=1, width=4, sys=32



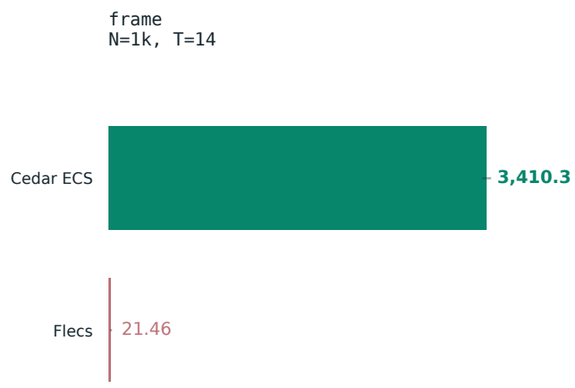
frame
N=1k, T=1, width=4, sys=8

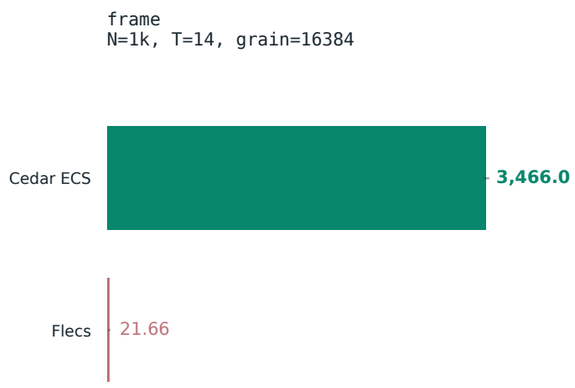


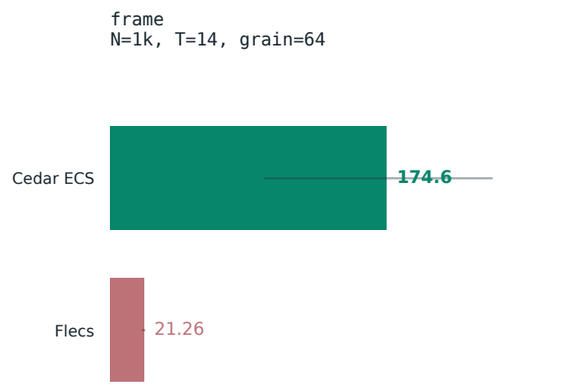
Indexed · Cedar vs Flecs

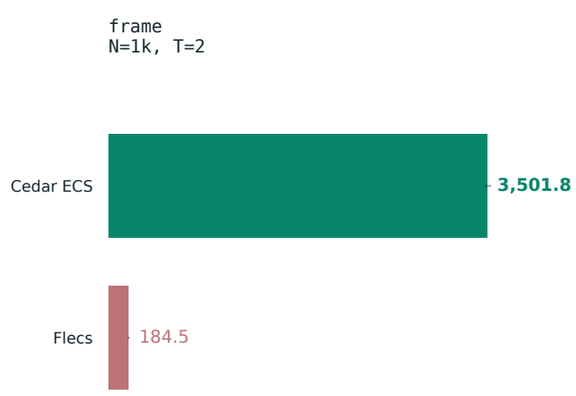
Mop/s ↑

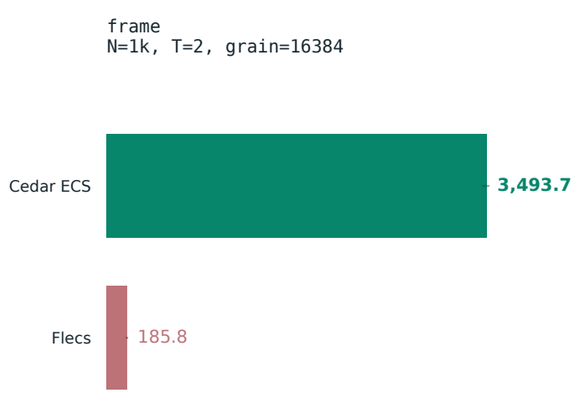
Cedar ECS Flecs

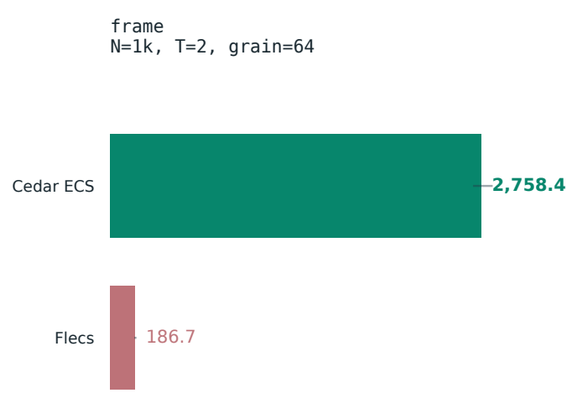


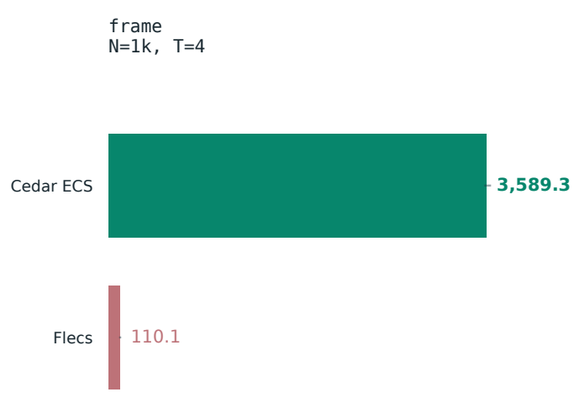


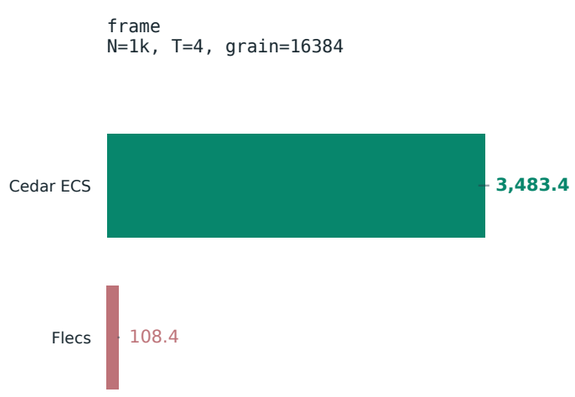


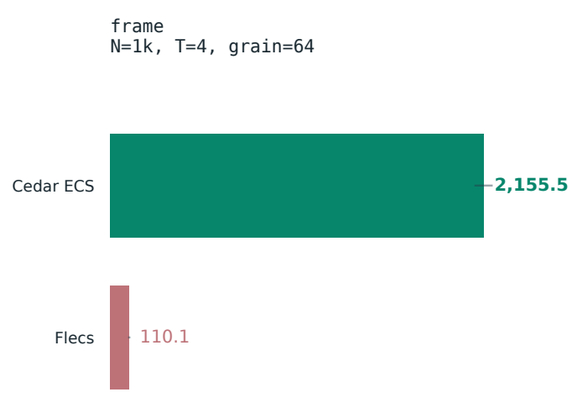


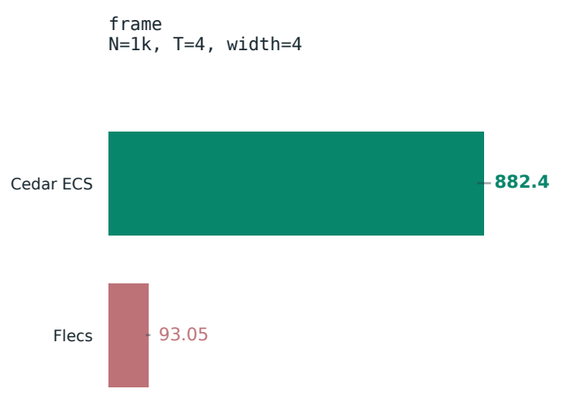


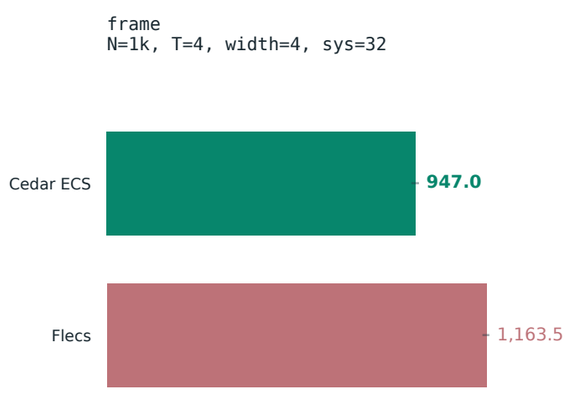


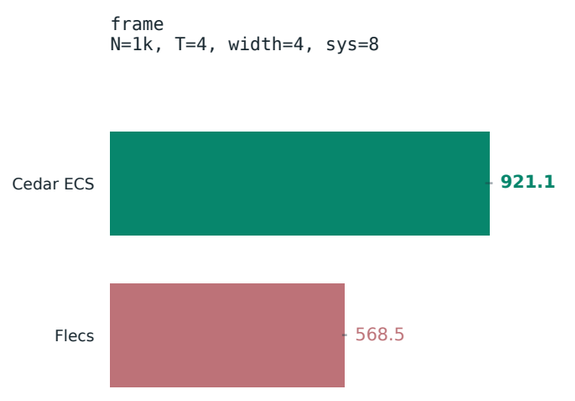








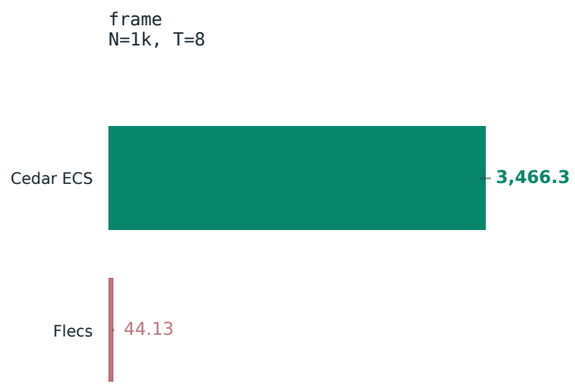


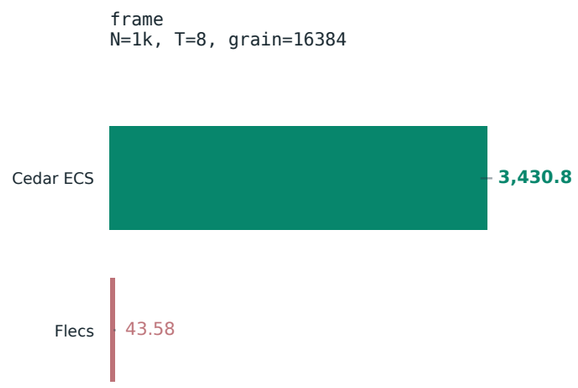


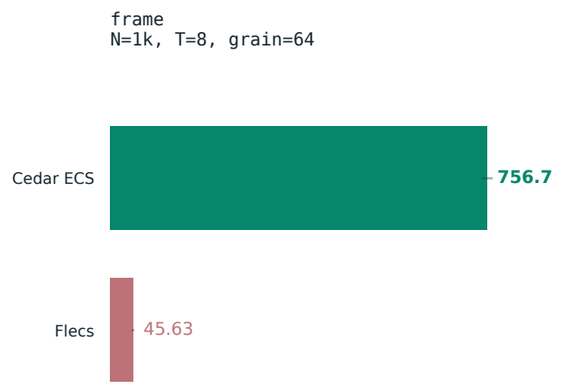
Indexed · Cedar vs Flecs

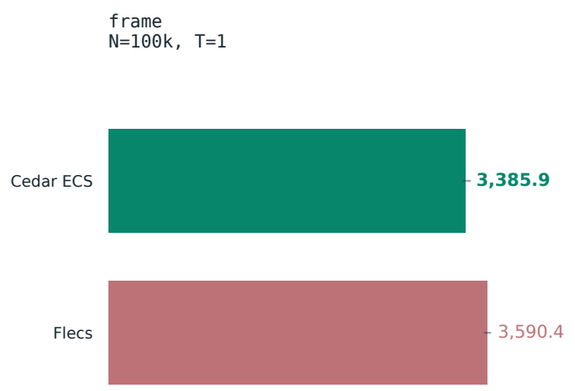
Mop/s ↑

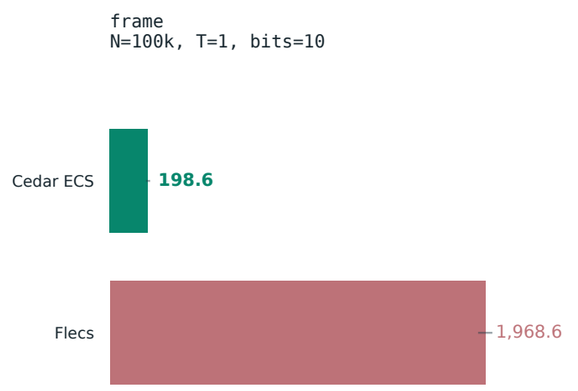
Cedar ECS Flecs

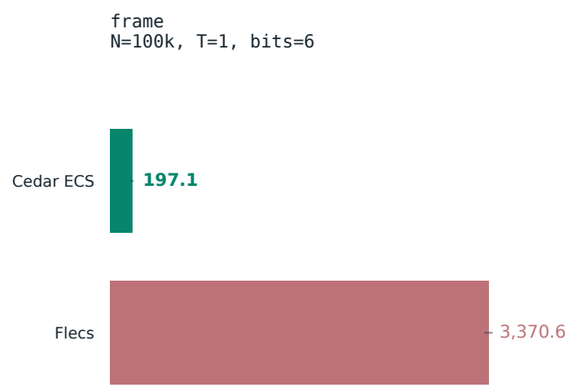


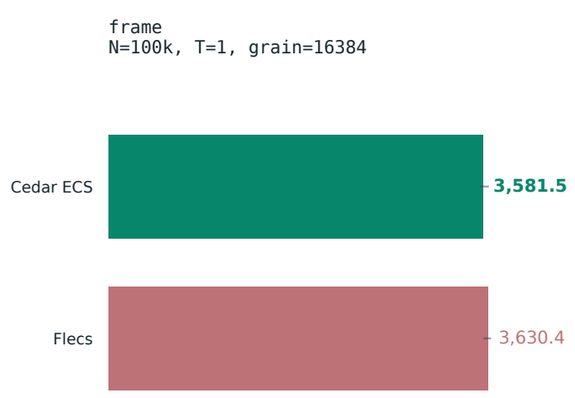


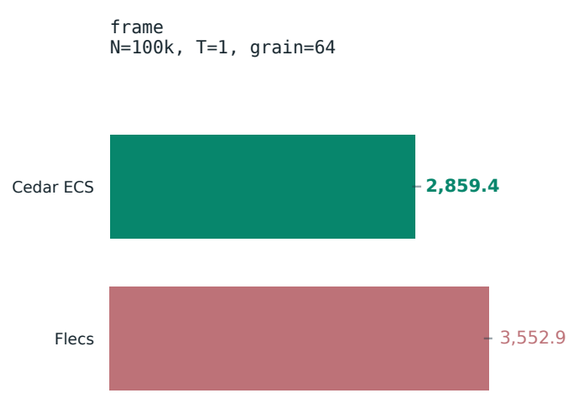


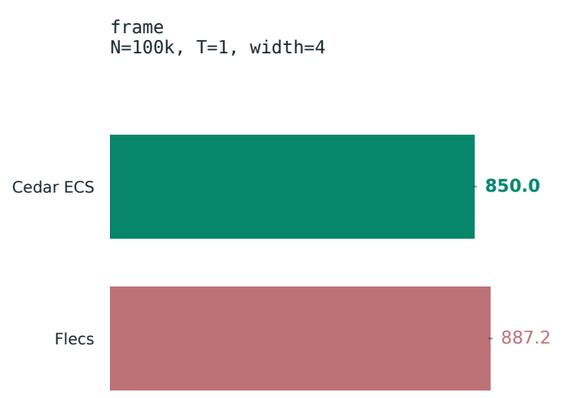


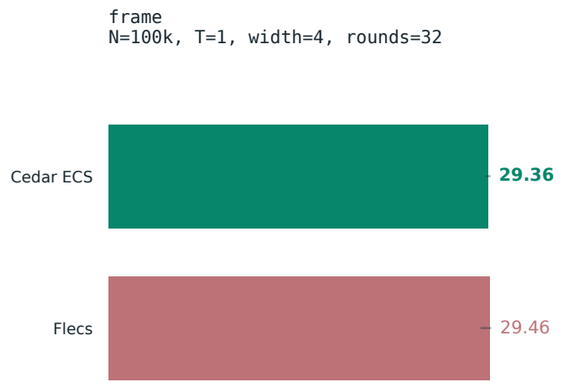


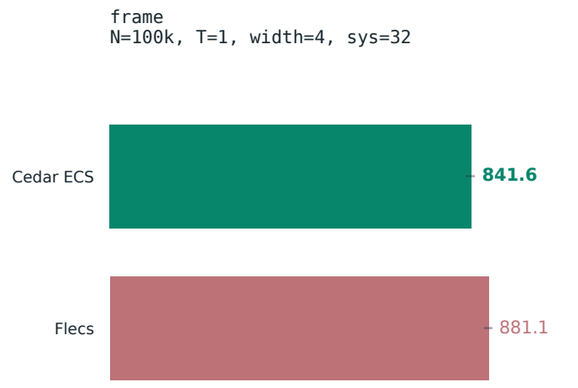


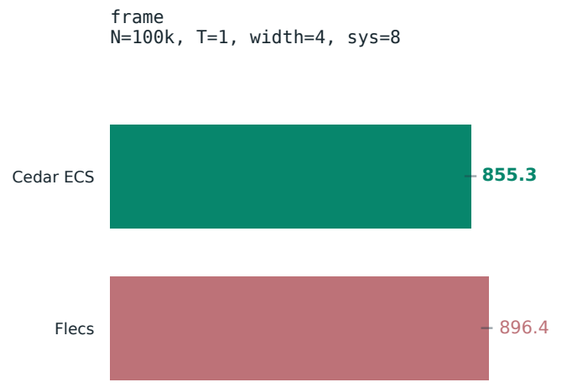












Indexed · Cedar vs Flecs

Mop/s ↑

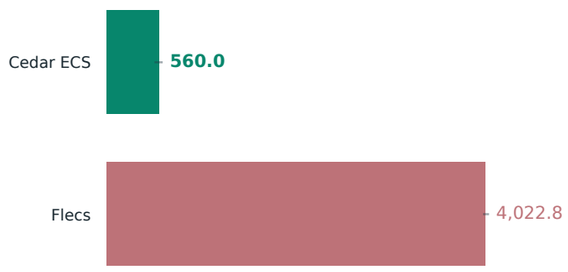
Cedar ECS Flecs

Indexed · Cedar vs Flecs

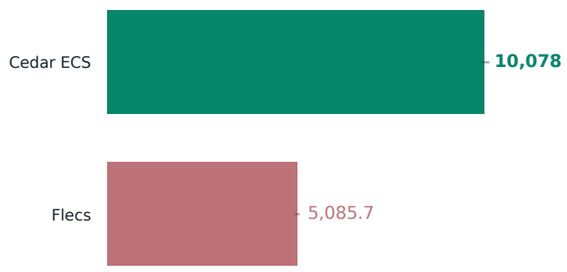
Mop/s ↑

Cedar ECS Flecs

frame
N=100k, T=4, bits=6



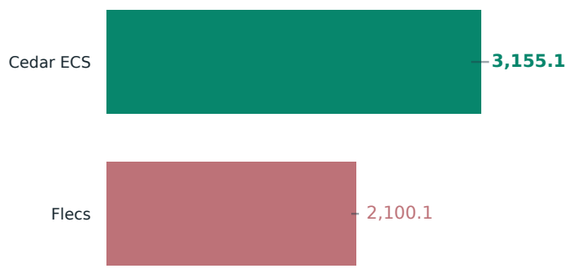
frame
N=100k, T=4, grain=16384



frame
N=100k, T=4, grain=64



frame
N=100k, T=4, width=4



frame
N=100k, T=4, width=4, rounds=32



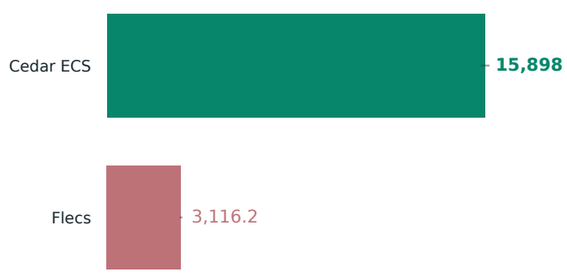
frame
N=100k, T=4, width=4, sys=32



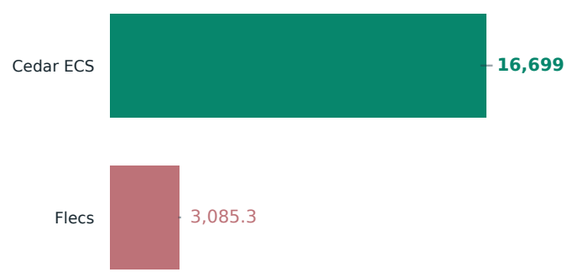
frame
N=100k, T=4, width=4, sys=8



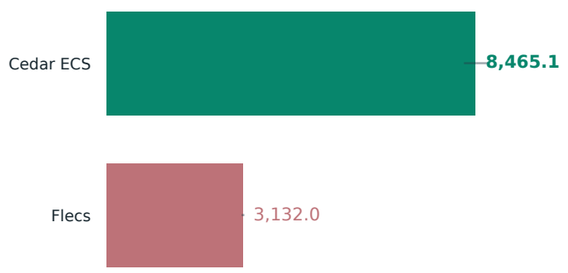
frame
N=100k, T=8



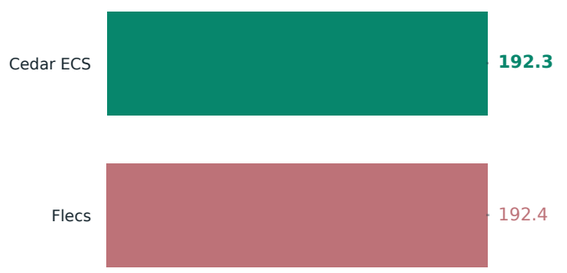
frame
N=100k, T=8, grain=16384



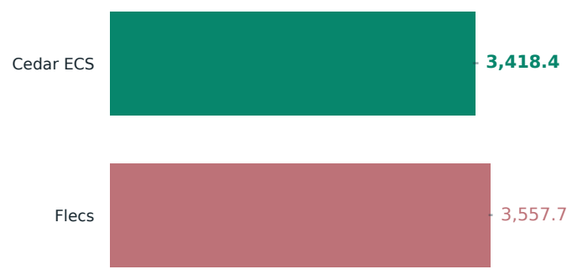
frame
N=100k, T=8, grain=64



frame
N=100k, T=8, width=4, rounds=32



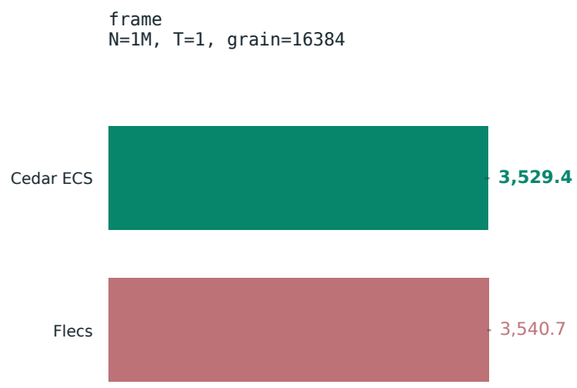
frame
N=1M, T=1

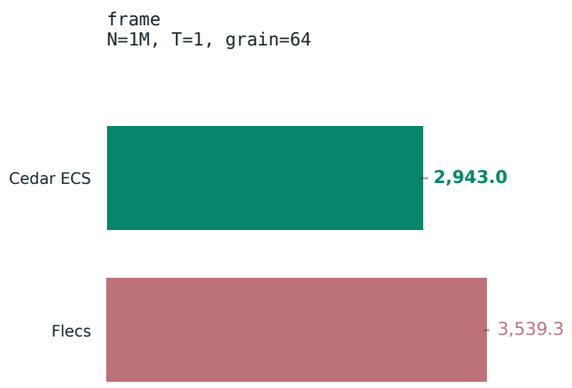


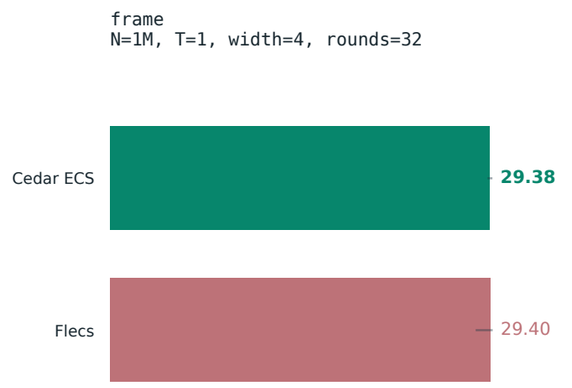
Indexed · Cedar vs Flecs

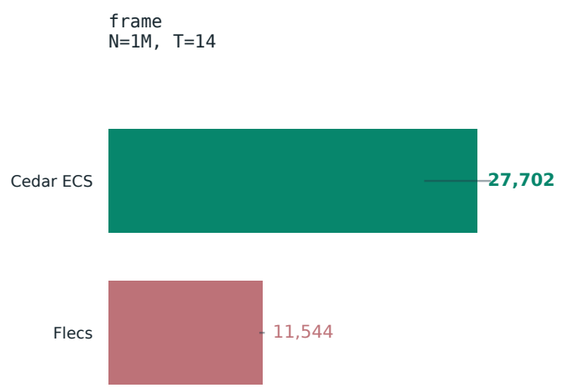
Mop/s ↑

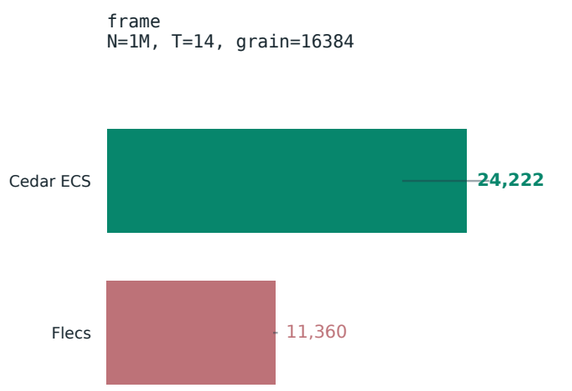
Cedar ECS Flecs

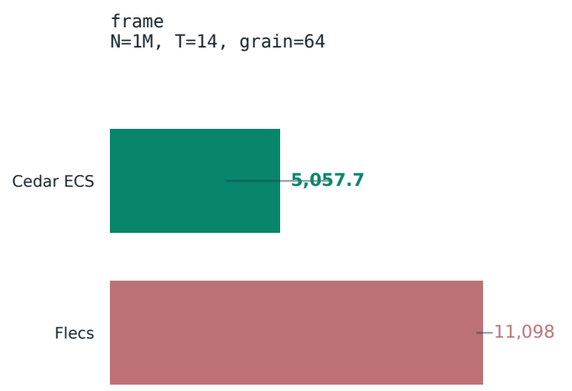


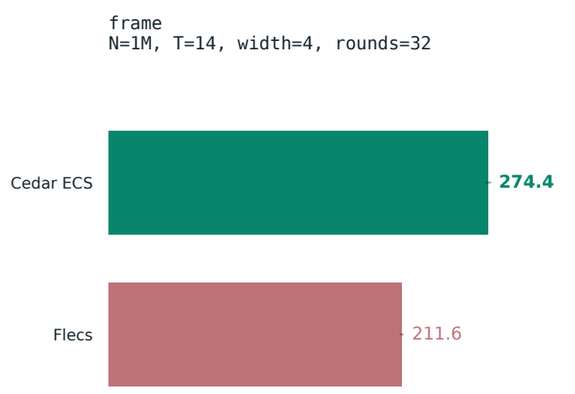


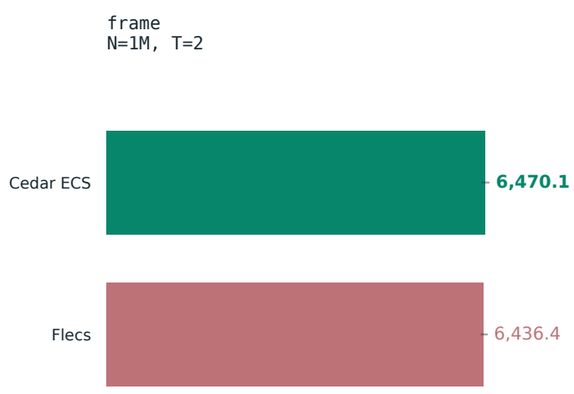


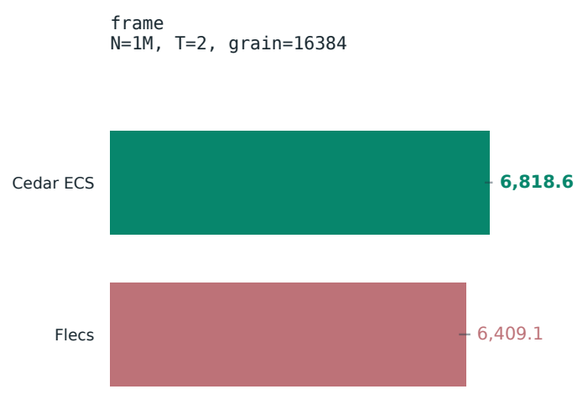




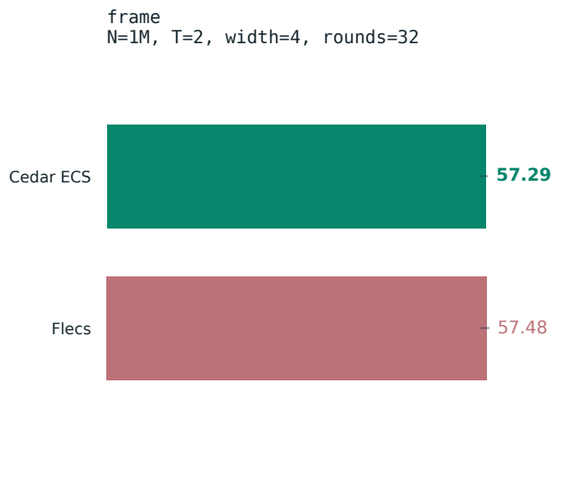


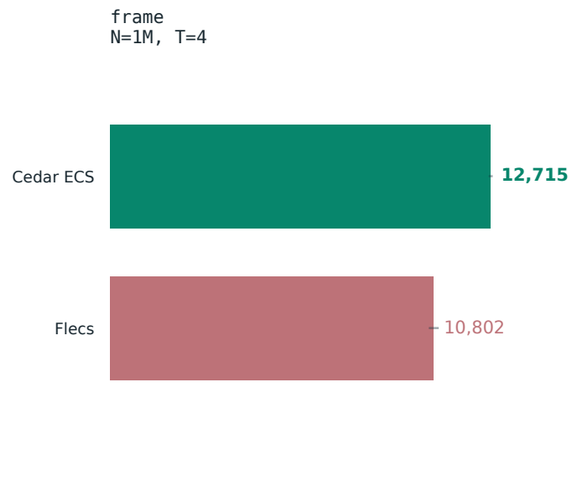








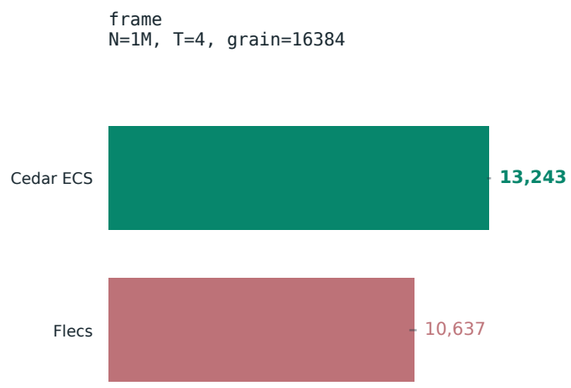


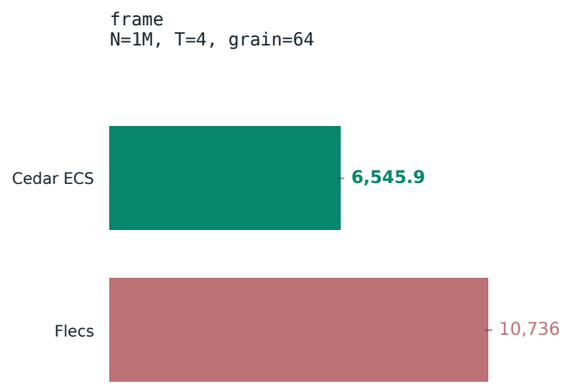


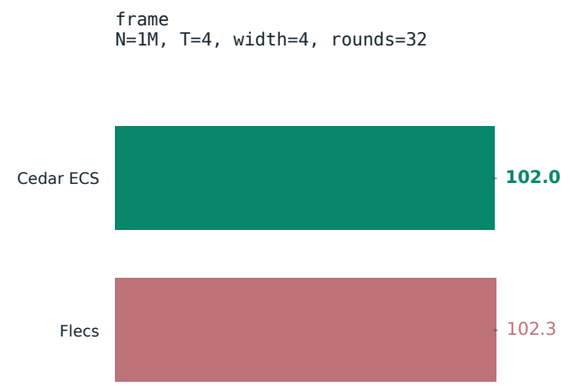
Indexed · Cedar vs Flecs

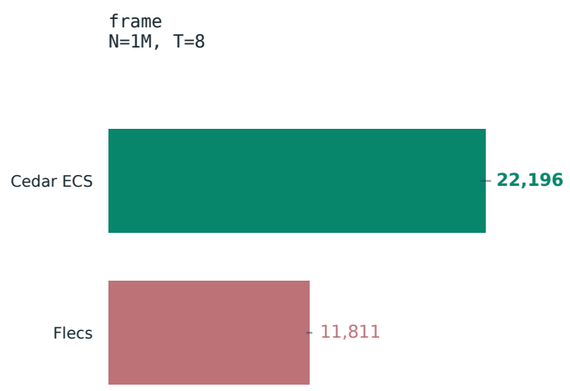
Mop/s ↑

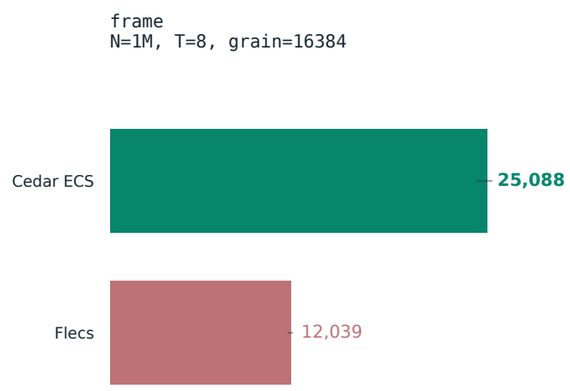
Cedar ECS Flecs

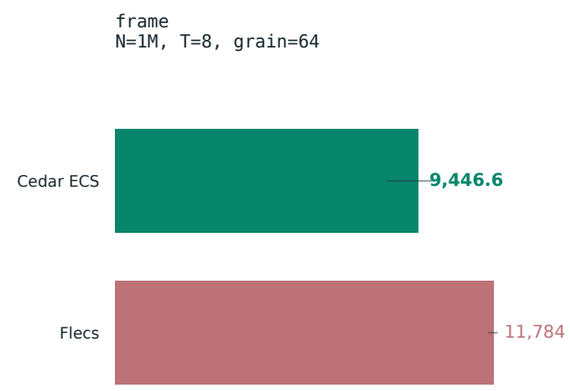


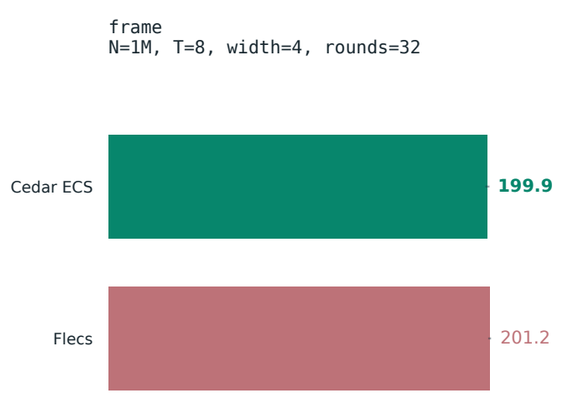


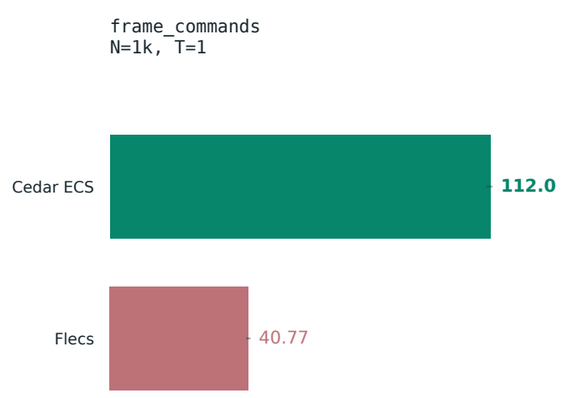


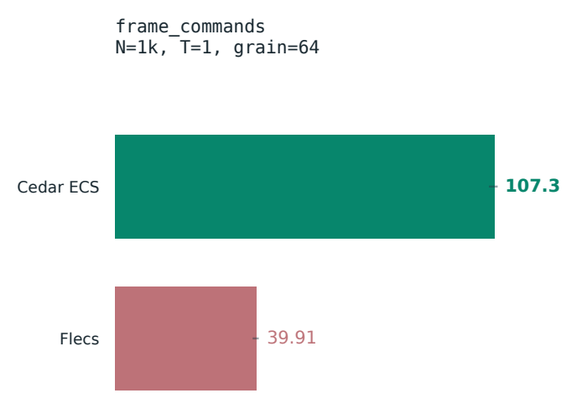


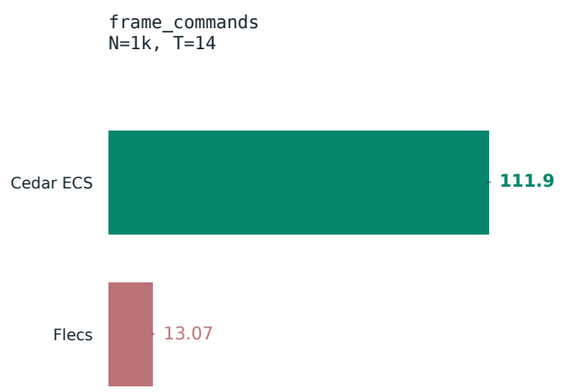


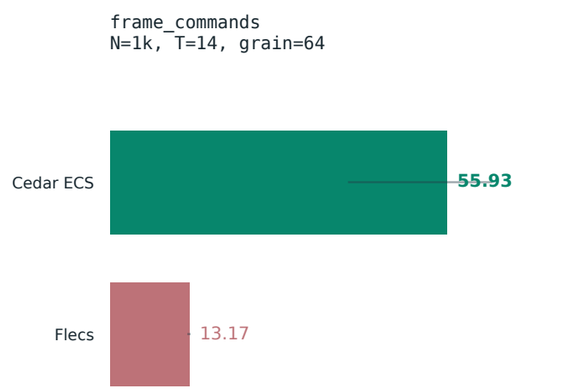


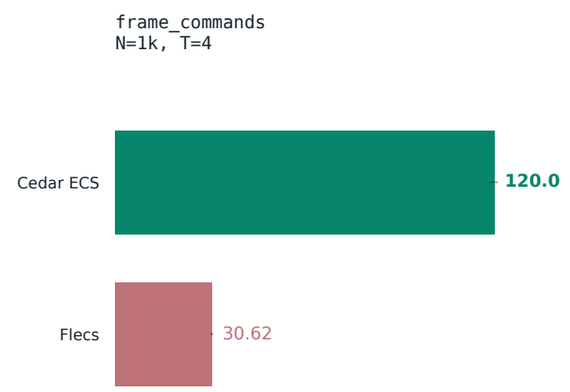








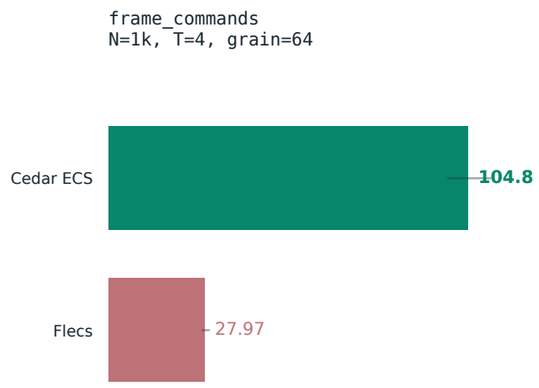


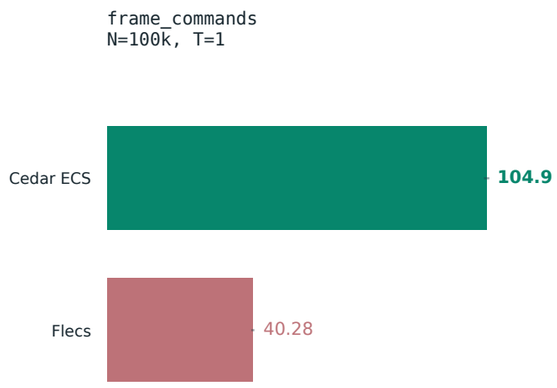


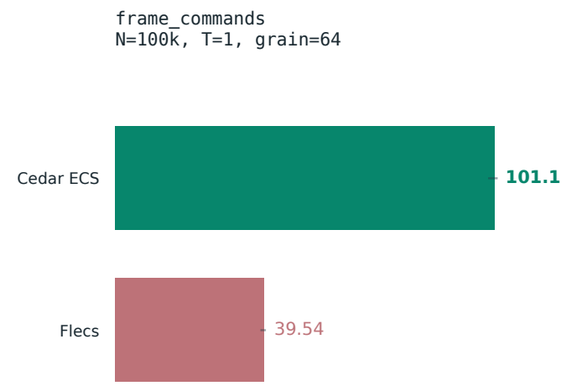
Indexed · Cedar vs Flecs

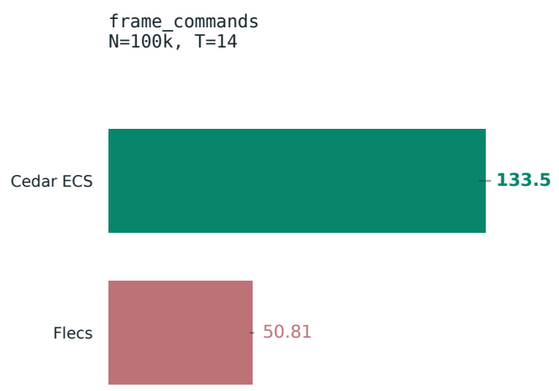
Mop/s ↑

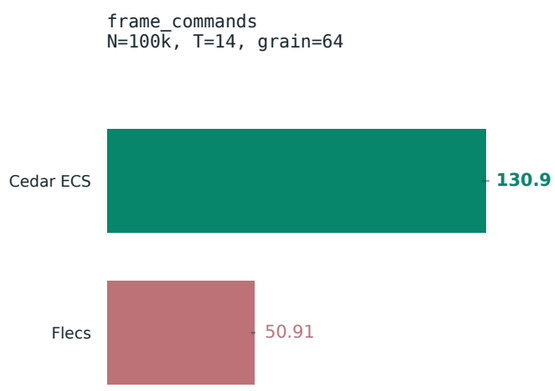
Cedar ECS Flecs

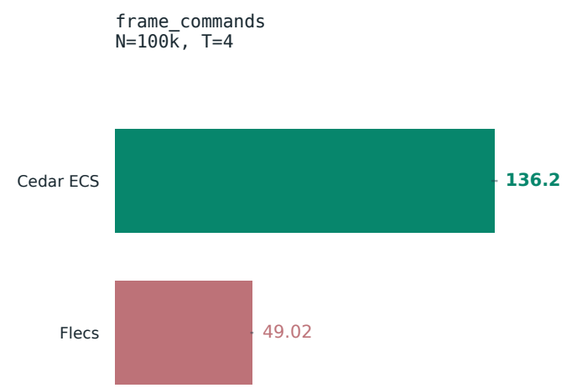


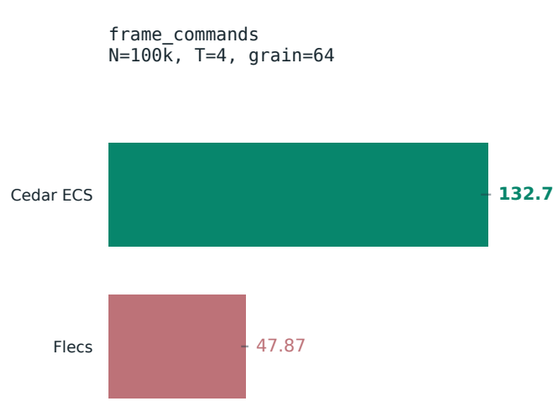


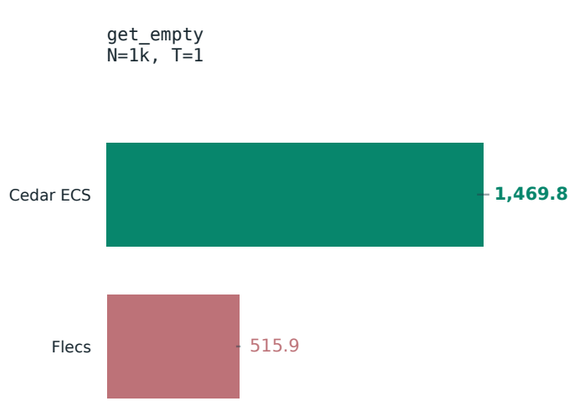




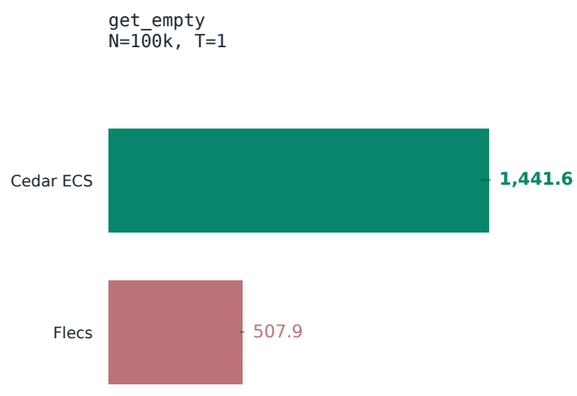


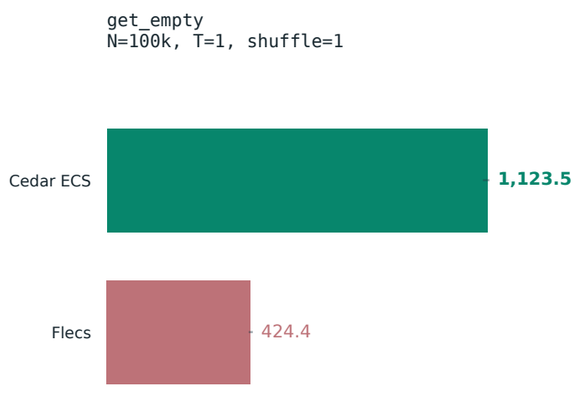


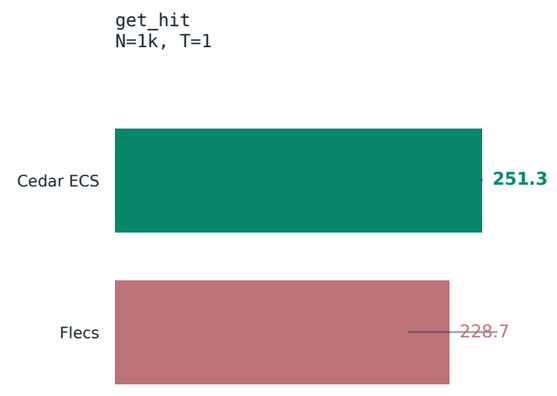








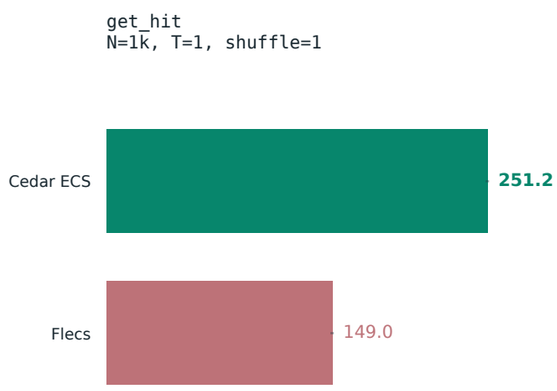


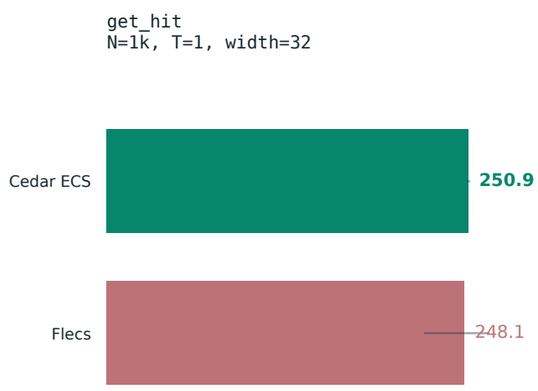


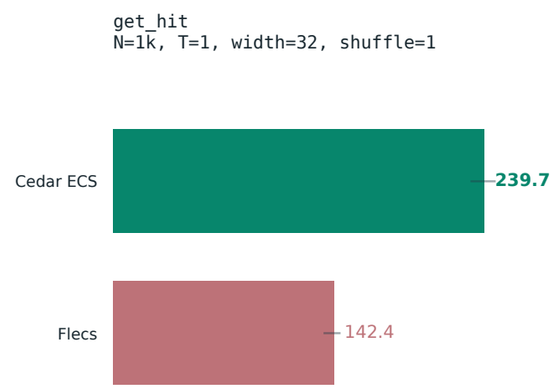
Indexed · Cedar vs Flecs

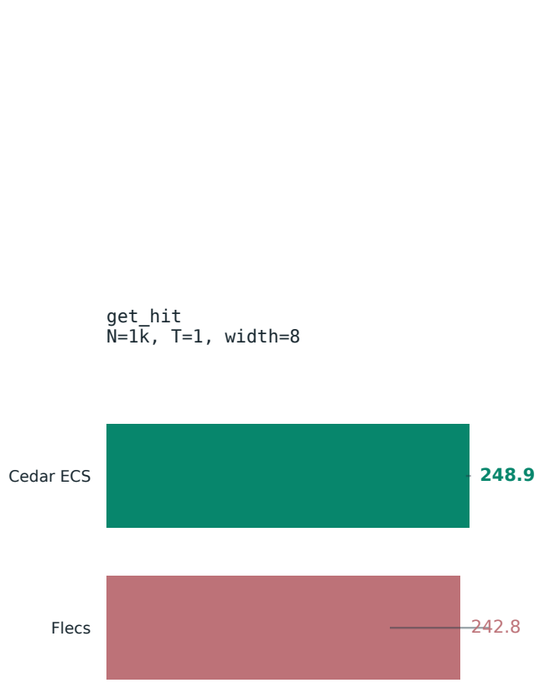
Mop/s ↑

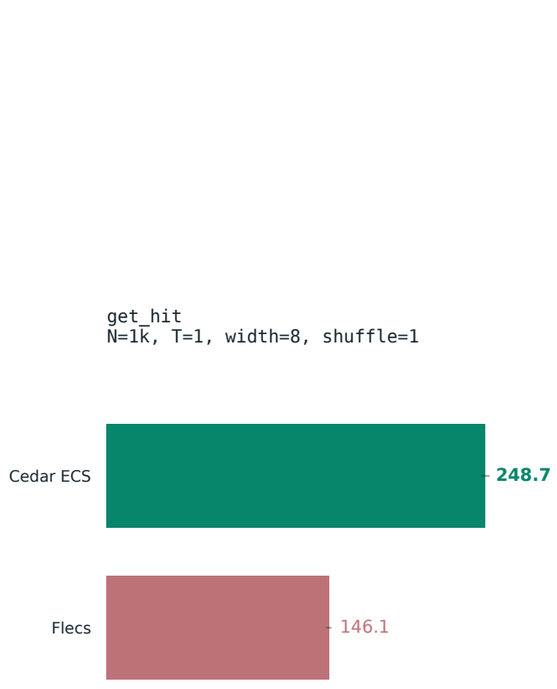
Cedar ECS Flecs

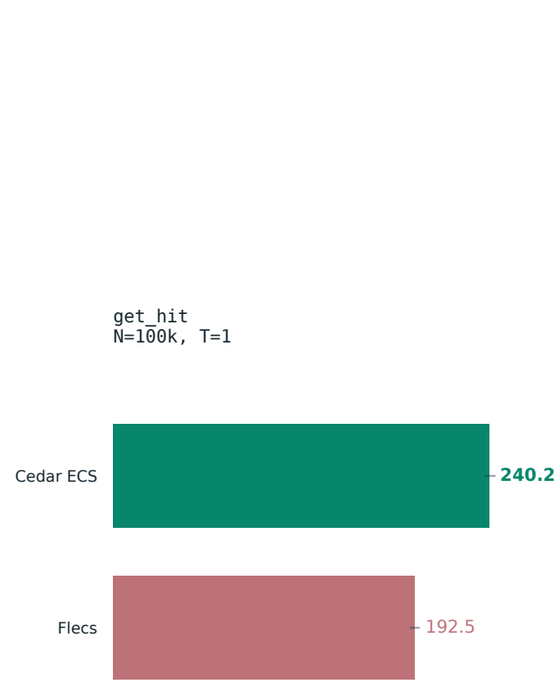


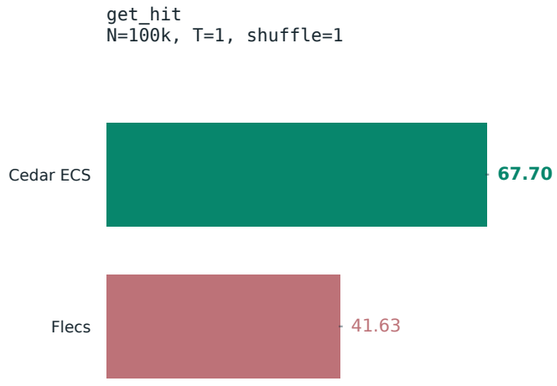


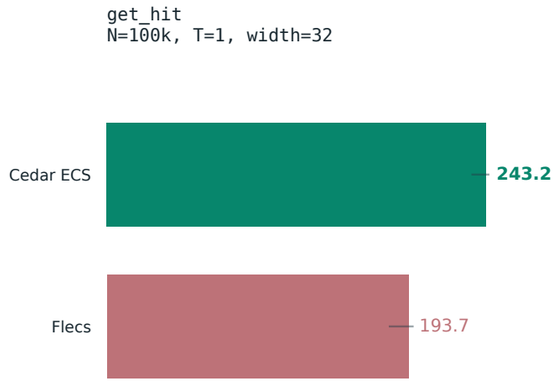








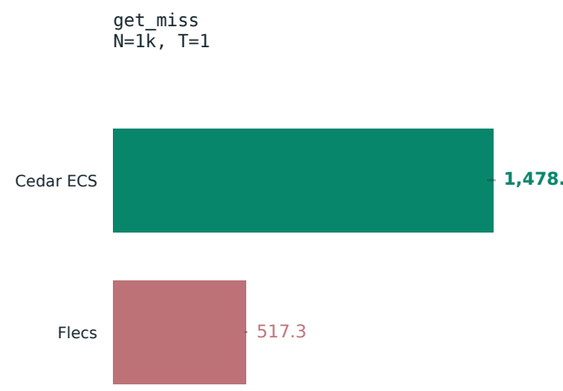










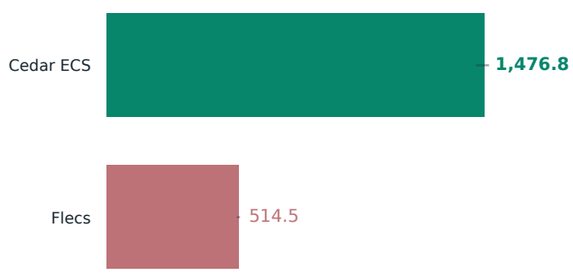


Indexed · Cedar vs Flecs

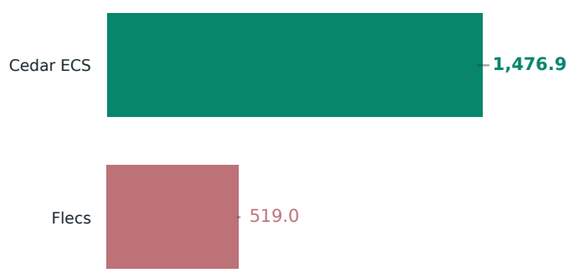
Mop/s ↑

Cedar ECS Flecs

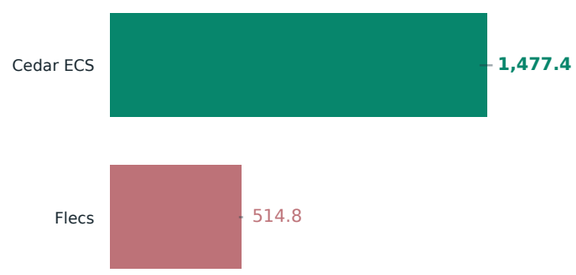
get_miss
N=1k, T=1, shuffle=1



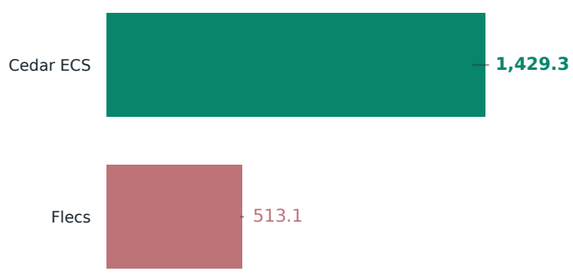
get_miss
N=1k, T=1, width=32



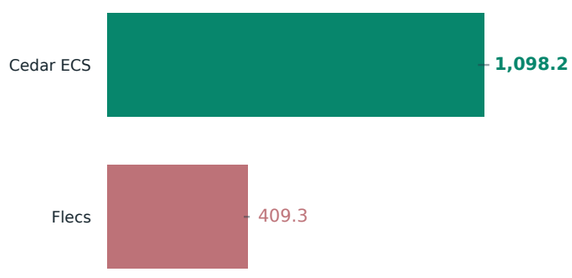
get_miss
N=1k, T=1, width=32, shuffle=1



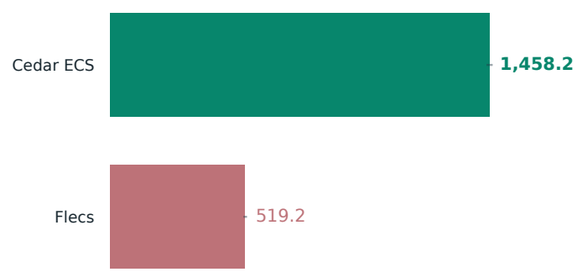
get_miss
N=100k, T=1



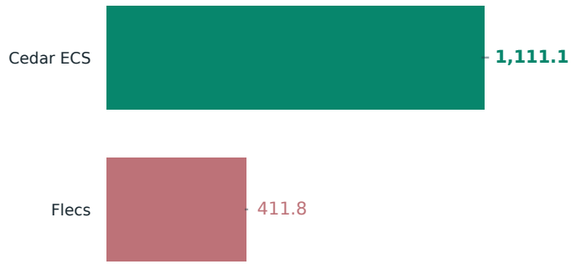
get_miss
N=100k, T=1, shuffle=1



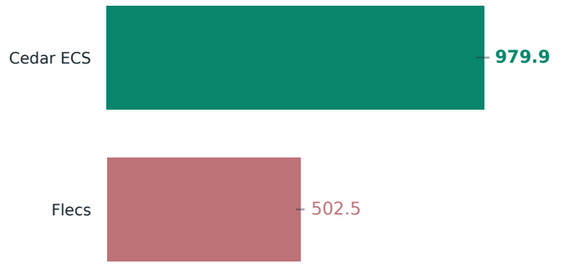
get_miss
N=100k, T=1, width=32



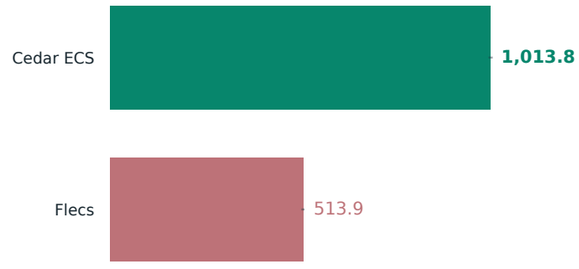
get_miss
N=100k, T=1, width=32, shuffle=1



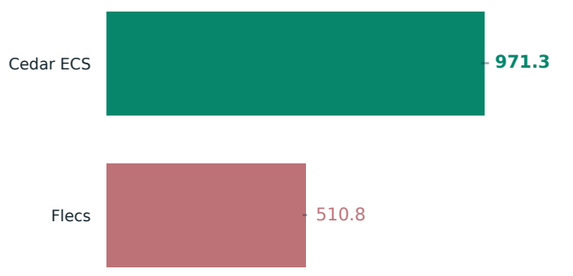
get_stale
N=1k, T=1



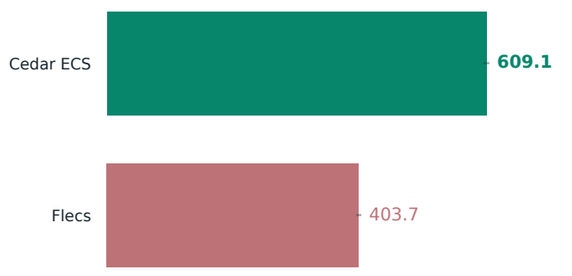
get_stale
N=1k, T=1, shuffle=1



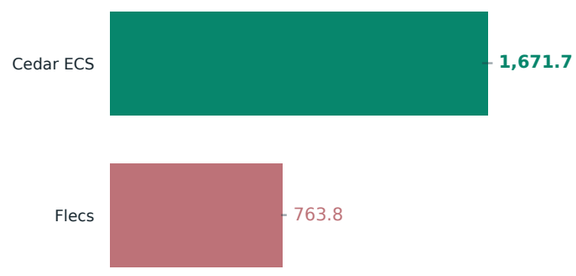
get_stale
N=100k, T=1



get_stale
N=100k, T=1, shuffle=1



has_mixed
N=1k, T=1

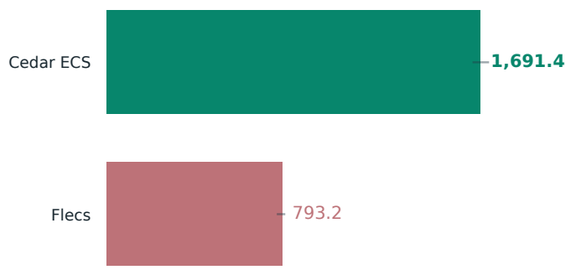


Indexed · Cedar vs Flecs

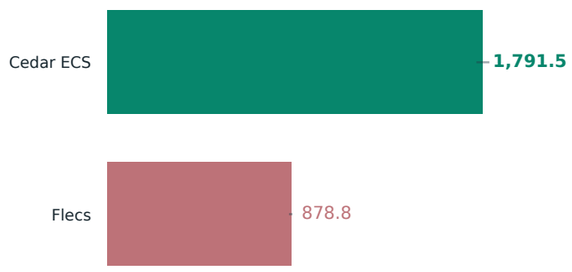
Mop/s ↑

Cedar ECS Flecs

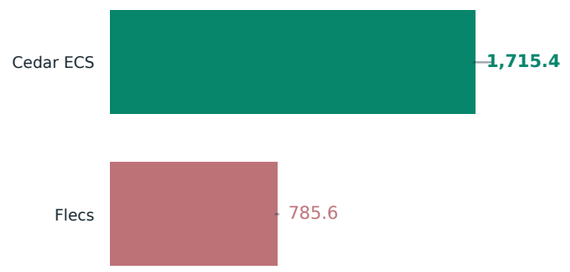
has_mixed
N=1k, T=1, percent=50



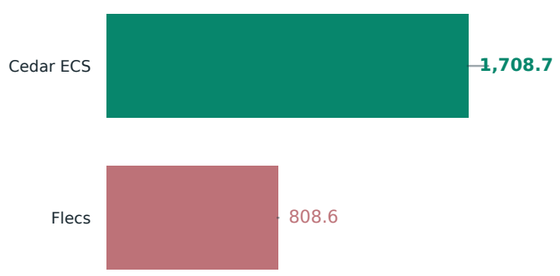
has_mixed
N=1k, T=1, percent=50, shuffle=1



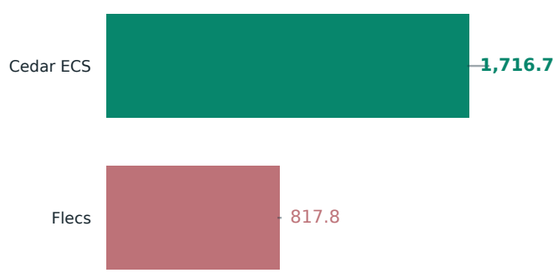
has_mixed
N=1k, T=1, percent=90



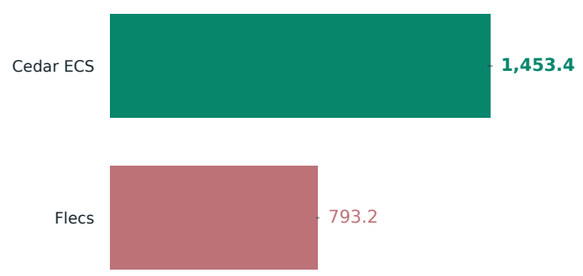
has_mixed
N=1k, T=1, percent=90, shuffle=1



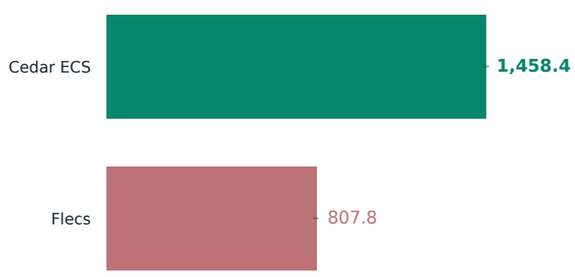
has_mixed
N=1k, T=1, shuffle=1



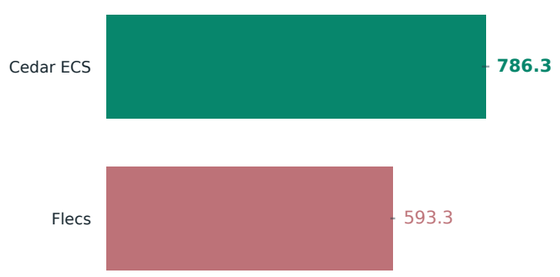
has_mixed
N=100k, T=1



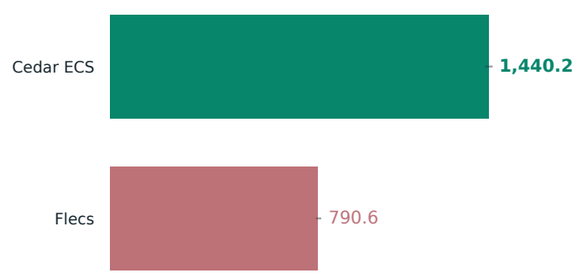
has_mixed
N=100k, T=1, percent=50



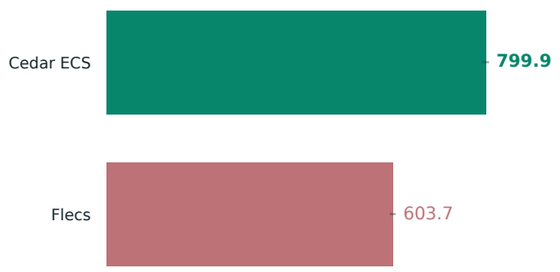
has_mixed
N=100k, T=1, percent=50, shuffle=1



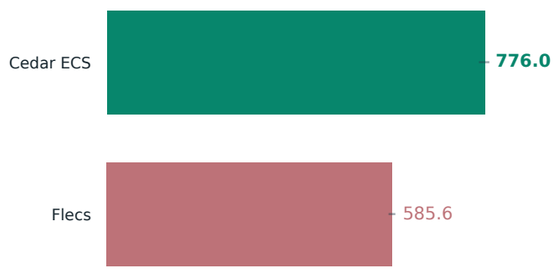
has_mixed
N=100k, T=1, percent=90



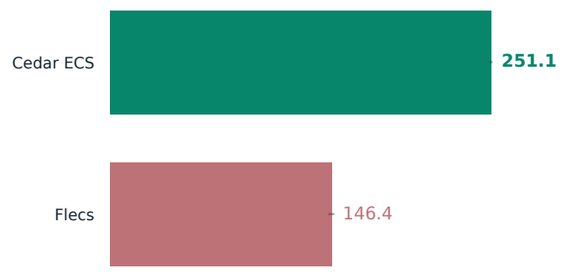
has_mixed
N=100k, T=1, percent=90, shuffle=1



has_mixed
N=100k, T=1, shuffle=1



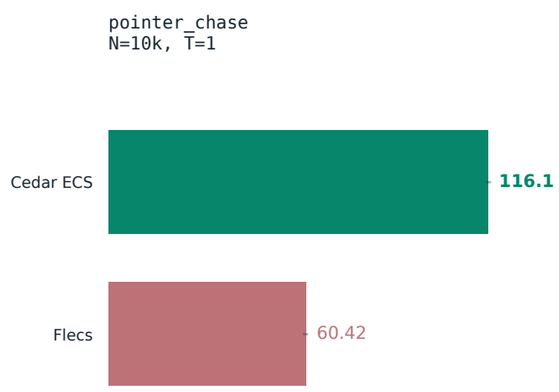
pointer_chase
N=1k, T=1

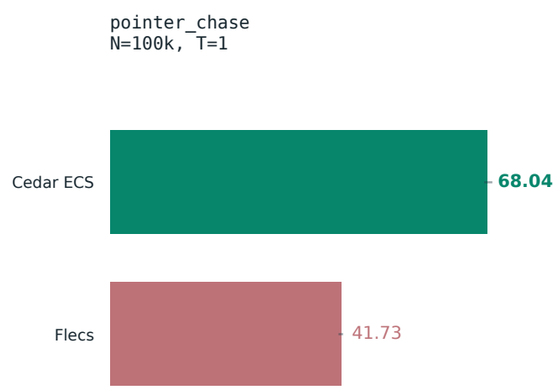


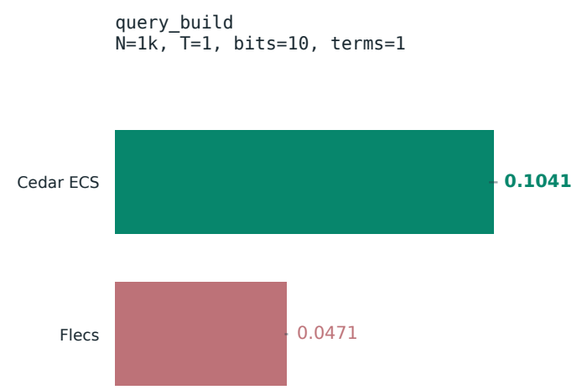
Indexed · Cedar vs Flecs

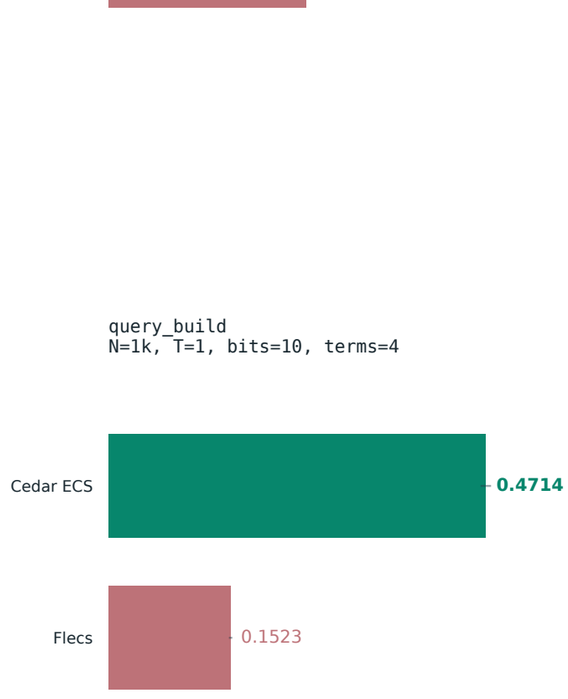
Mop/s ↑

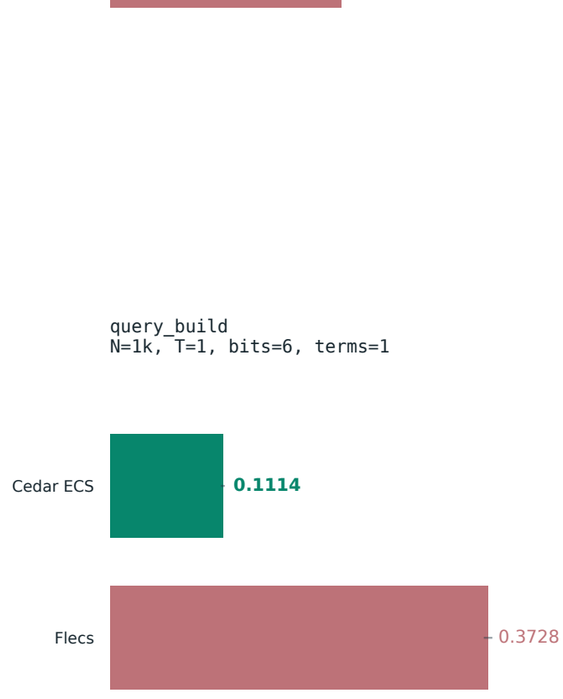
Cedar ECS Flecs

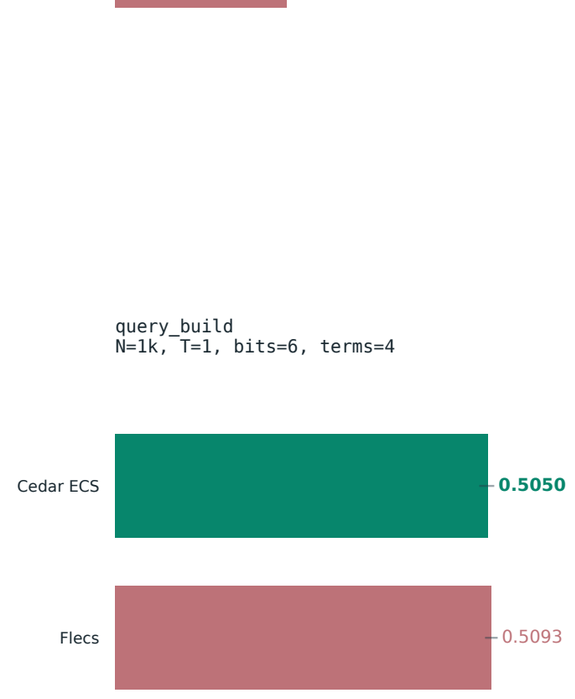


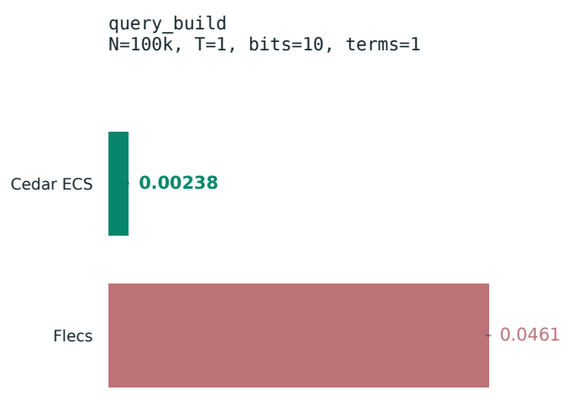


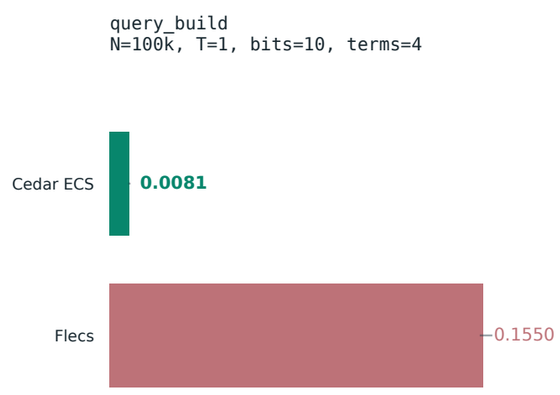




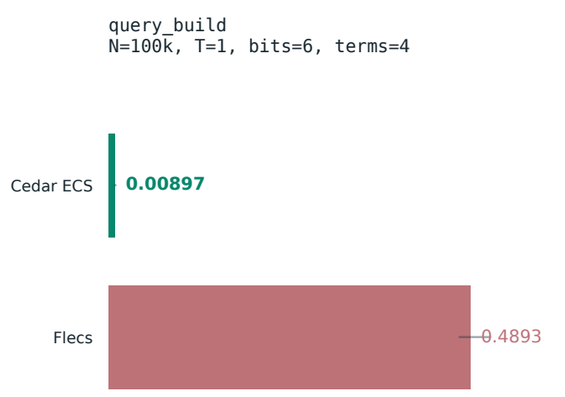


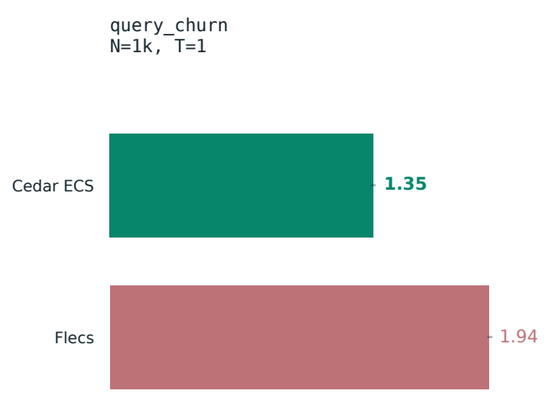


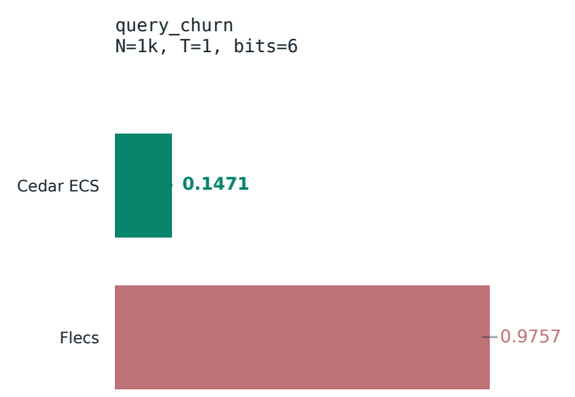








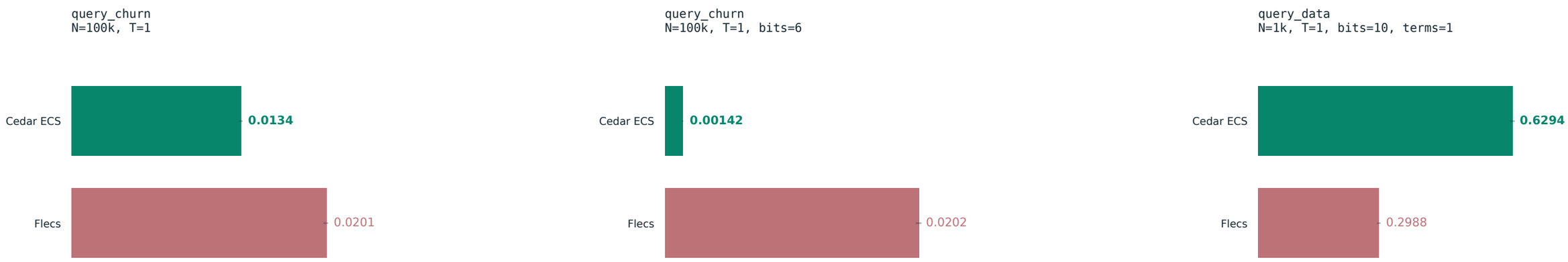




Indexed · Cedar vs Flecs

Mop/s ↑

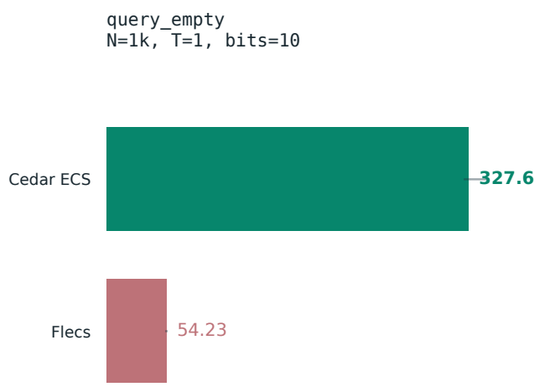
Cedar ECS Flecs

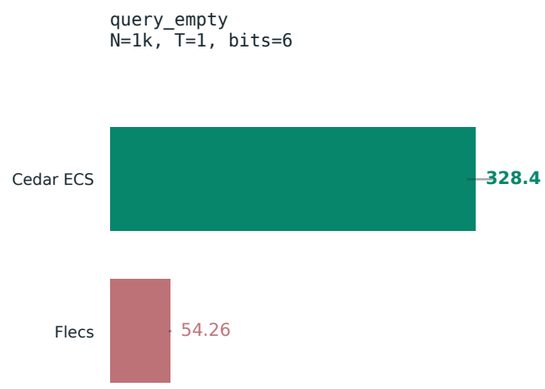


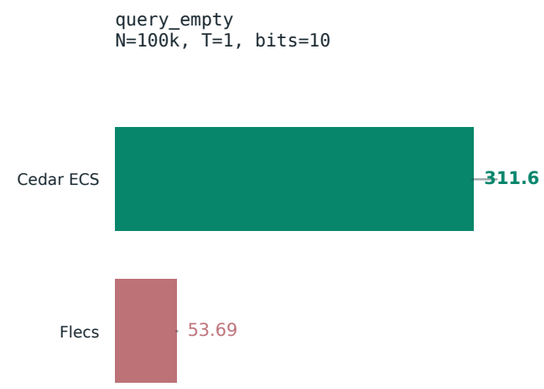
Indexed · Cedar vs Flecs

Mop/s ↑

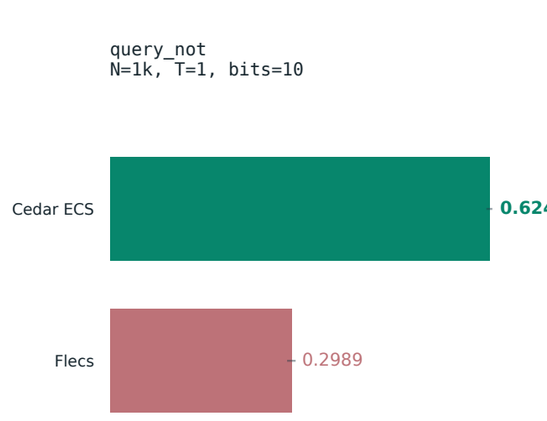
Cedar ECS Flecs

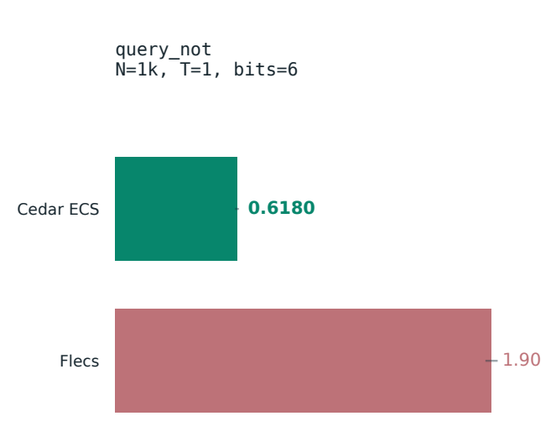


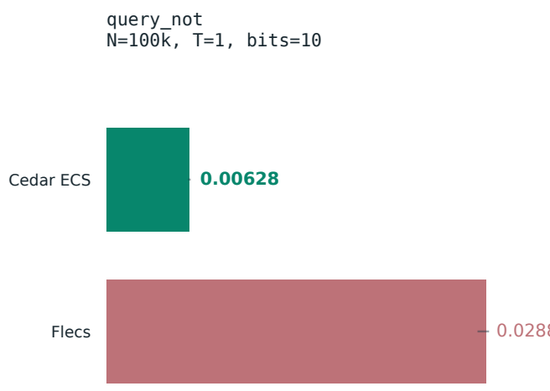


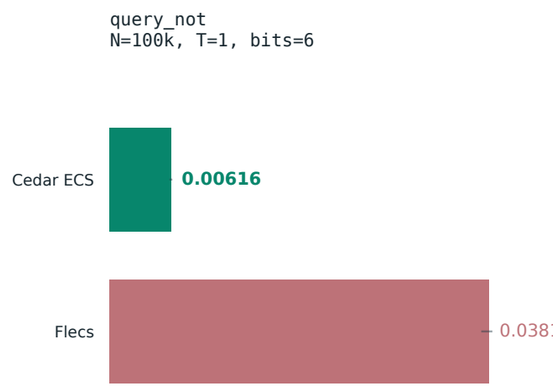


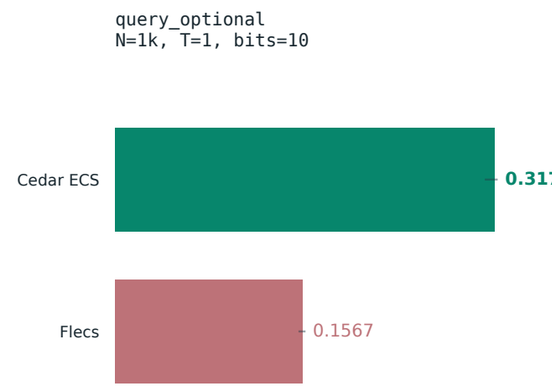


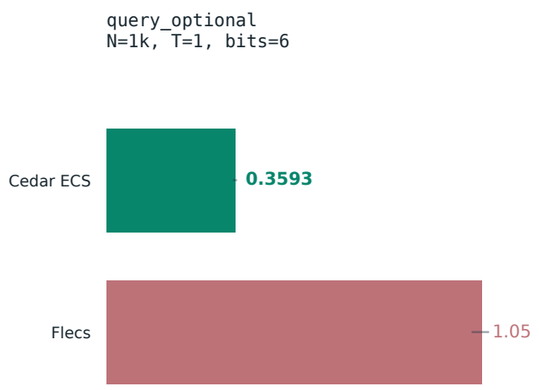


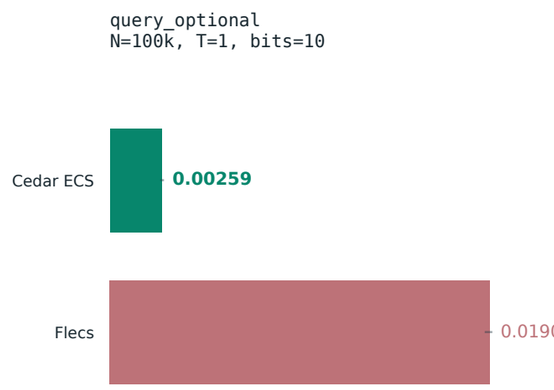


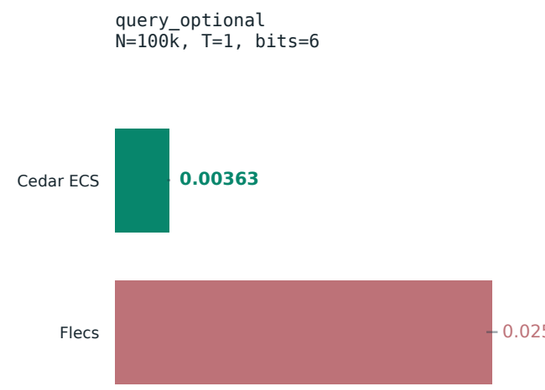












Indexed · Cedar vs Flecs

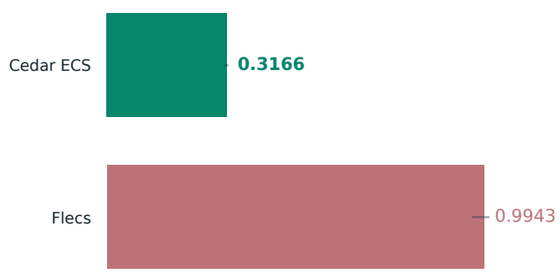
Mop/s ↑

Cedar ECS Flecs

query_plain
N=1k, T=1



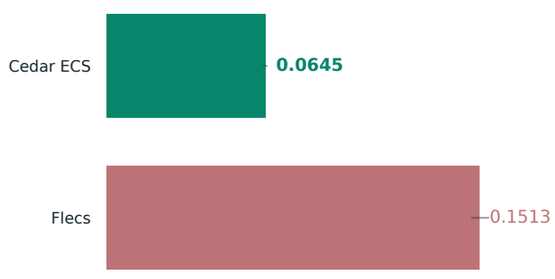
query_plain
N=1k, T=1, bits=6



query_plain
N=1k, T=1, width=16



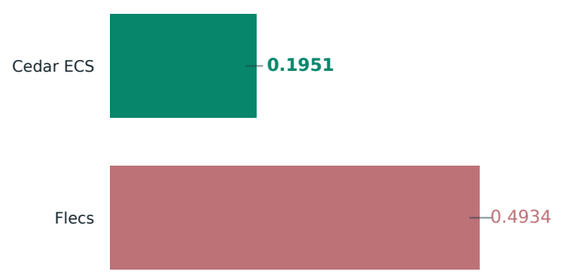
query_plain
N=1k, T=1, width=16, bits=6



query_plain
N=1k, T=1, width=4



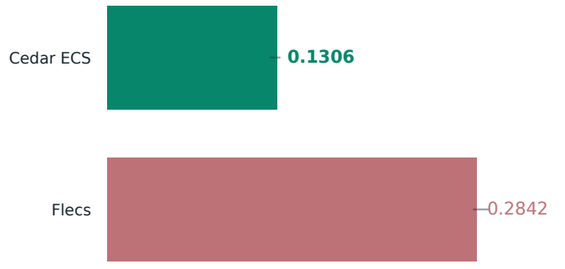
query_plain
N=1k, T=1, width=4, bits=6



query_plain
N=1k, T=1, width=8



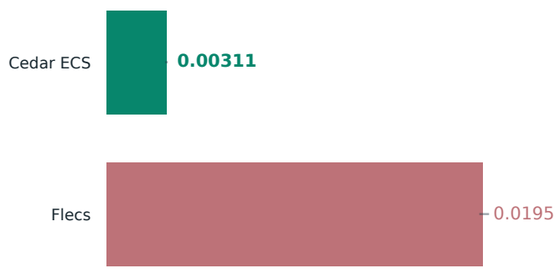
query_plain
N=1k, T=1, width=8, bits=6



query_plain
N=100k, T=1



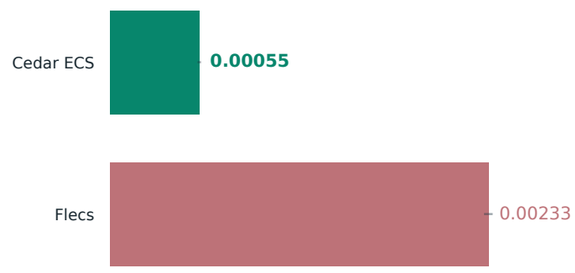
query_plain
N=100k, T=1, bits=6



query_plain
N=100k, T=1, width=16



query_plain
N=100k, T=1, width=16, bits=6



Indexed · Cedar vs Flecs

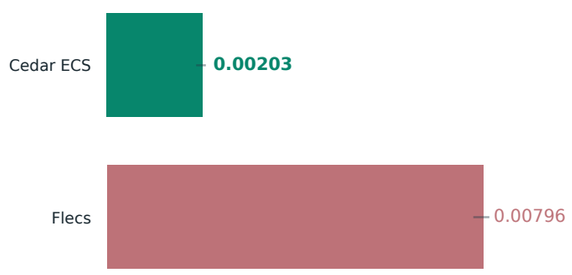
Mop/s ↑

Cedar ECS Flecs

query_plain
N=100k, T=1, width=4



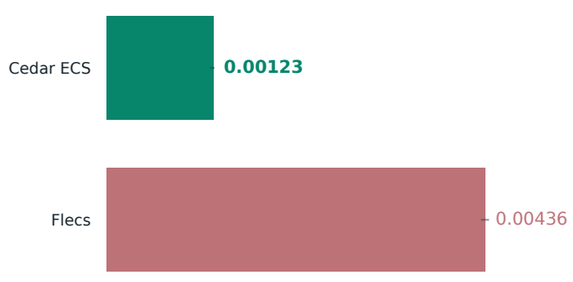
query_plain
N=100k, T=1, width=4, bits=6



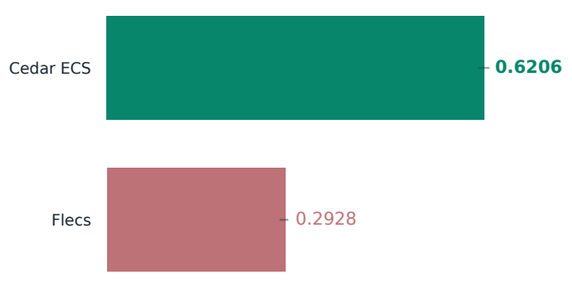
query_plain
N=100k, T=1, width=8



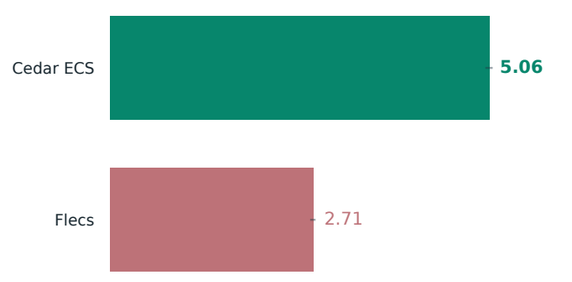
query_plain
N=100k, T=1, width=8, bits=6



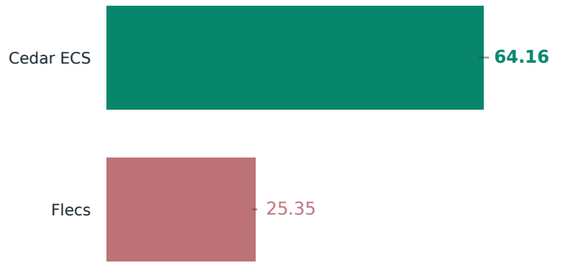
query_tags
N=1k, T=1, bits=10, terms=1



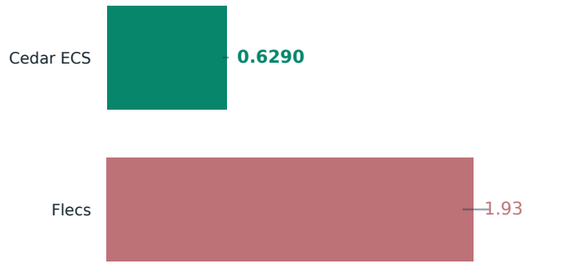
query_tags
N=1k, T=1, bits=10, terms=4



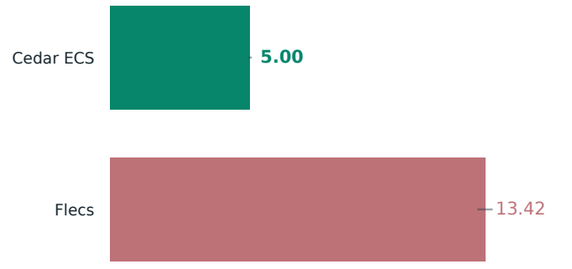
query_tags
N=1k, T=1, bits=10, terms=8



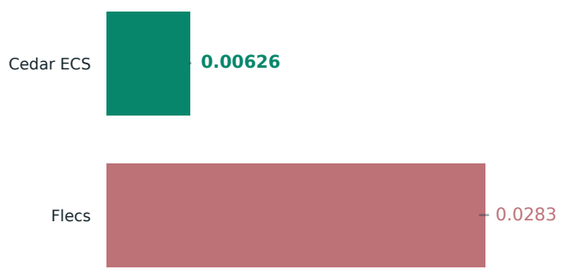
query_tags
N=1k, T=1, bits=6, terms=1



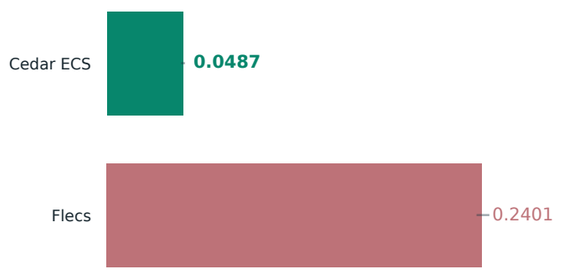
query_tags
N=1k, T=1, bits=6, terms=4



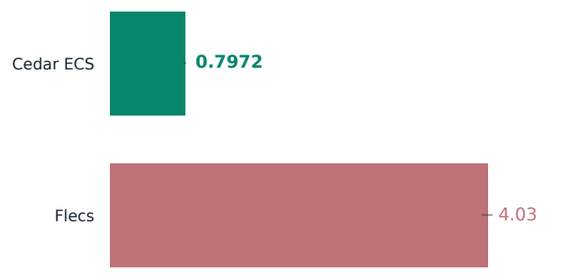
query_tags
N=100k, T=1, bits=10, terms=1



query_tags
N=100k, T=1, bits=10, terms=4



query_tags
N=100k, T=1, bits=10, terms=8

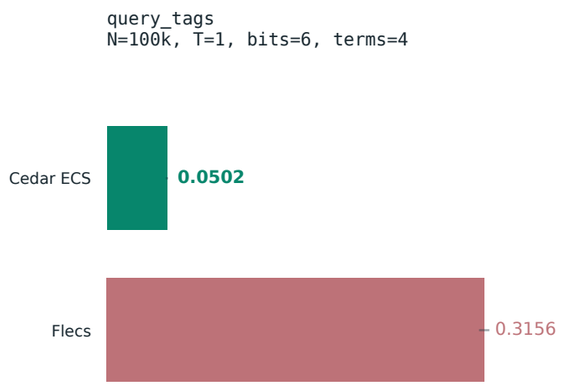


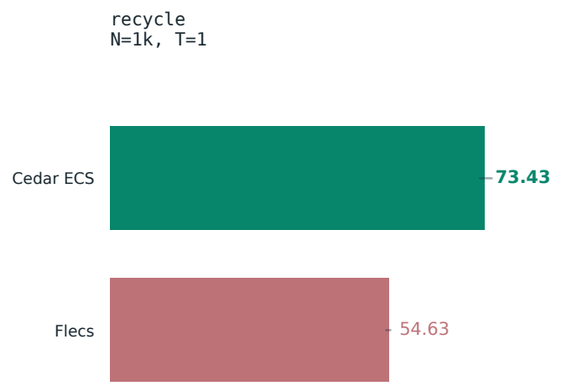
Indexed · Cedar vs Flecs

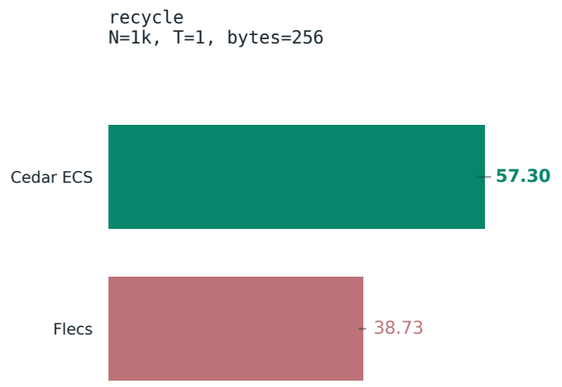
Mop/s ↑

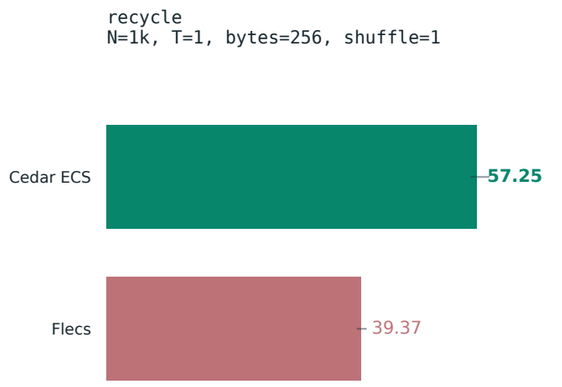
Cedar ECS Flecs

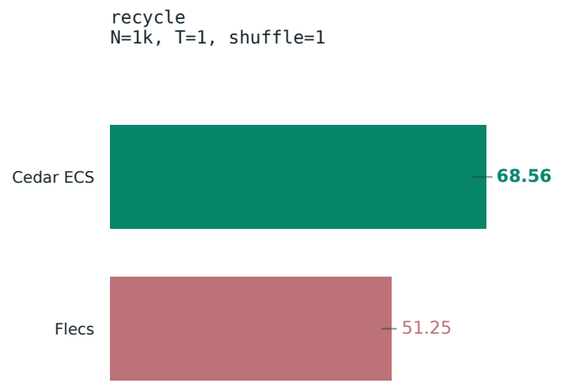


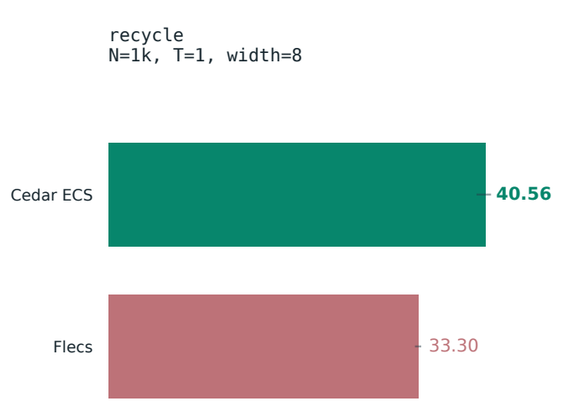


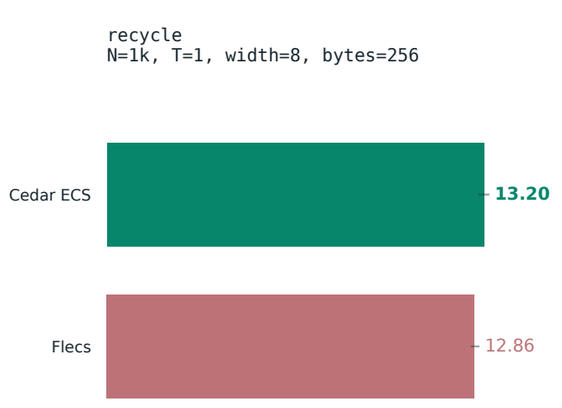


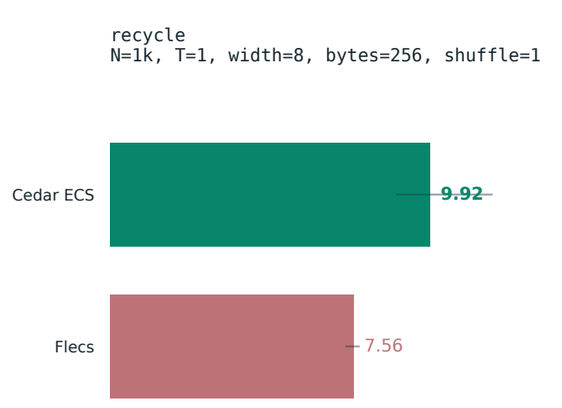


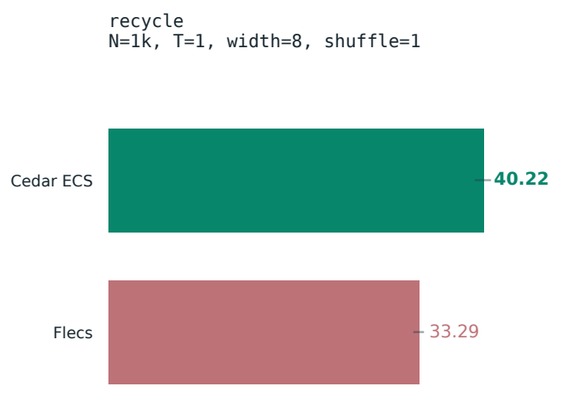


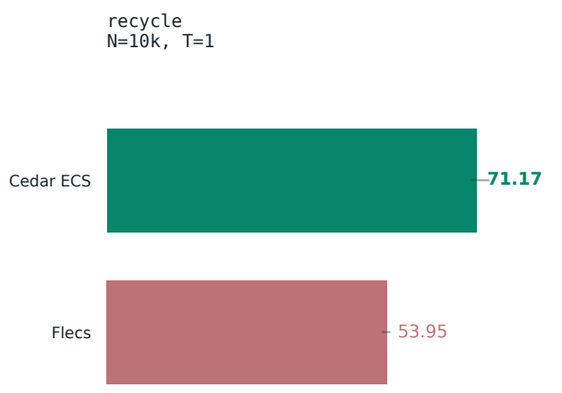


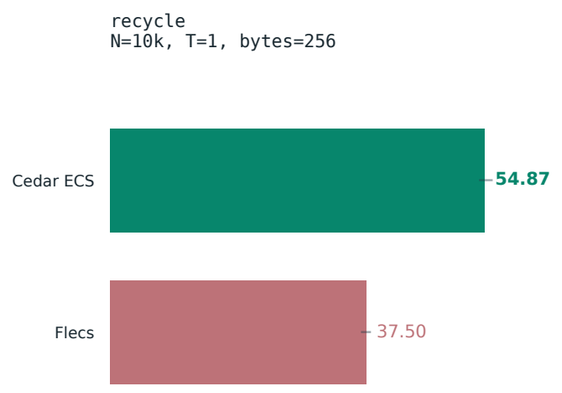










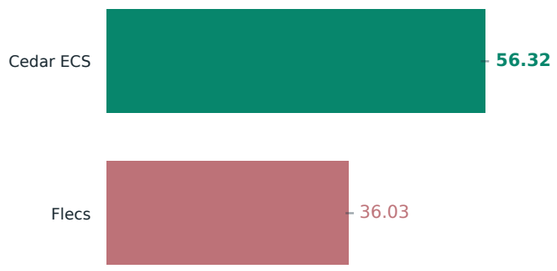


Indexed · Cedar vs Flecs

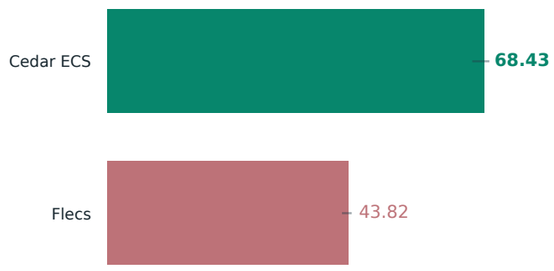
Mop/s ↑

Cedar ECS Flecs

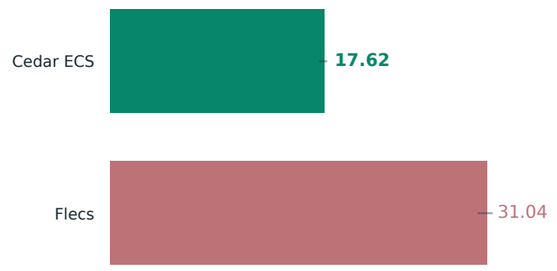
recycle
N=10k, T=1, bytes=256, shuffle=1



recycle
N=10k, T=1, shuffle=1



recycle
N=10k, T=1, width=8



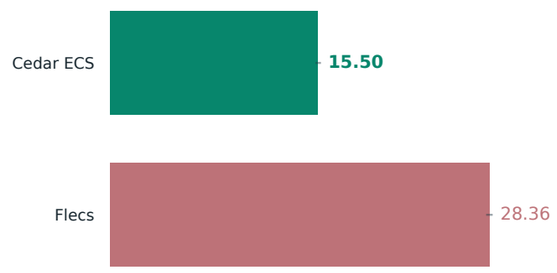
recycle
N=10k, T=1, width=8, bytes=256



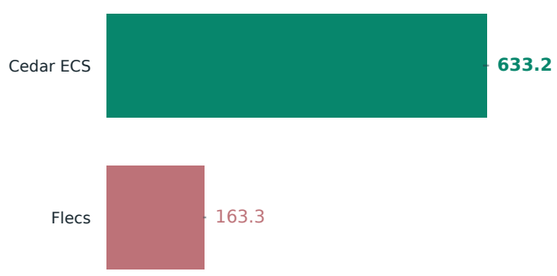
recycle
N=10k, T=1, width=8, bytes=256, shuffle=1



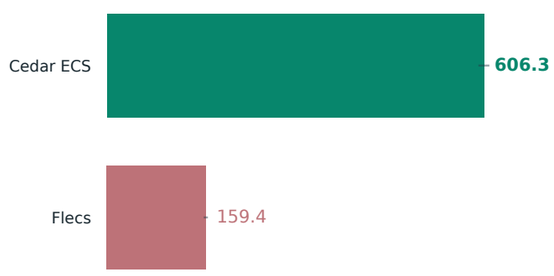
recycle
N=10k, T=1, width=8, shuffle=1



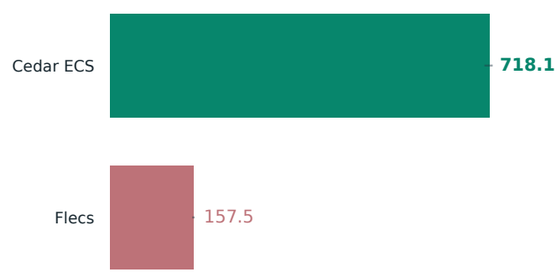
remove_absent
N=1k, T=1



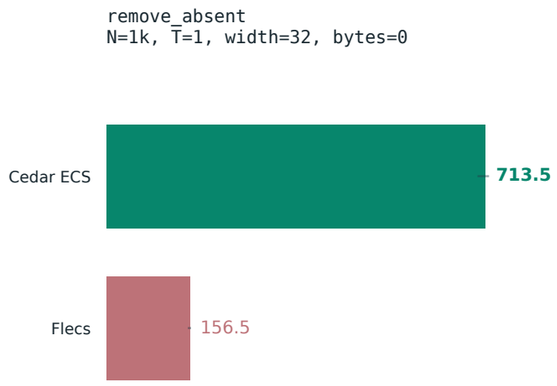
remove_absent
N=1k, T=1, bytes=0



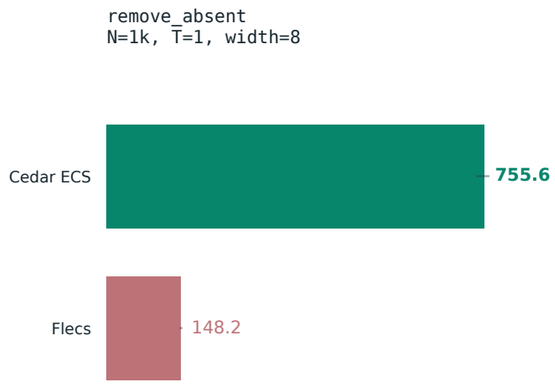
remove_absent
N=1k, T=1, width=32



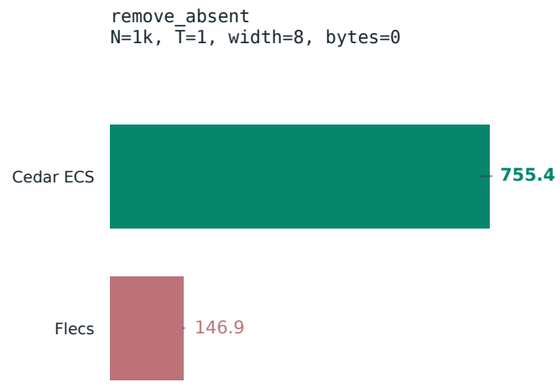
remove_absent
N=1k, T=1, width=32, bytes=0



remove_absent
N=1k, T=1, width=8



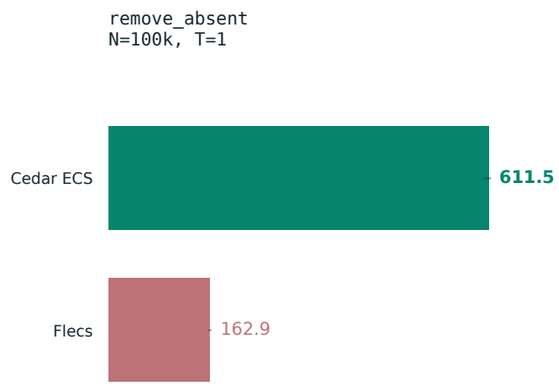
remove_absent
N=1k, T=1, width=8, bytes=0

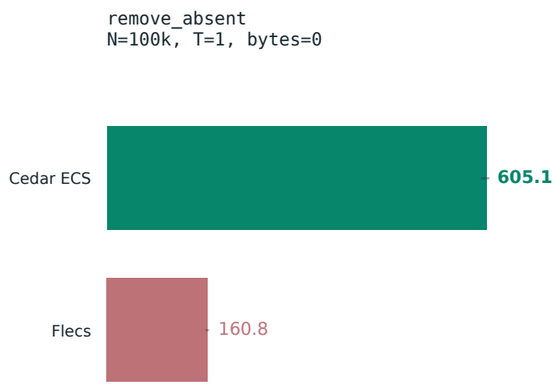


Indexed · Cedar vs Flecs

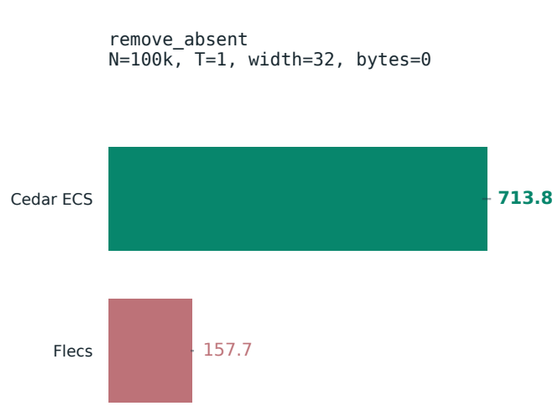
Mop/s ↑

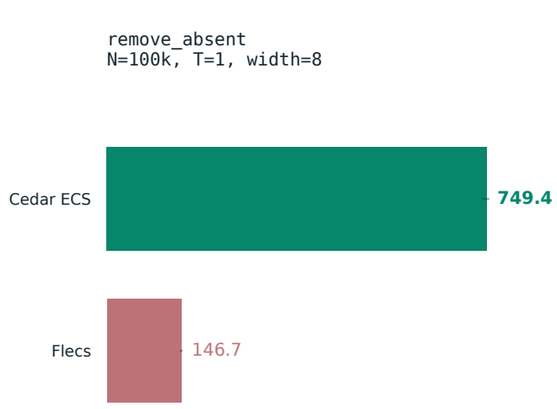
Cedar ECS Flecs

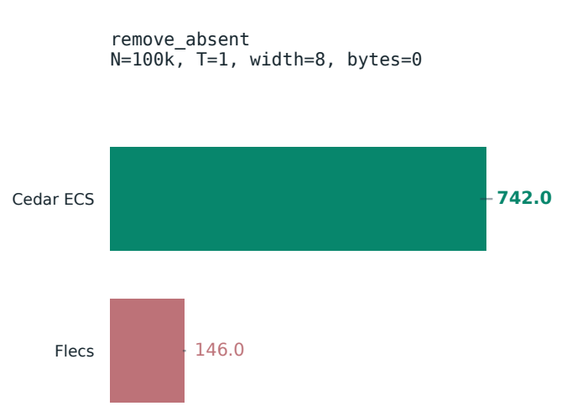


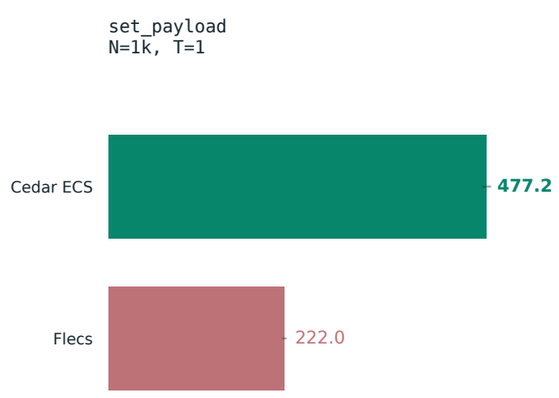


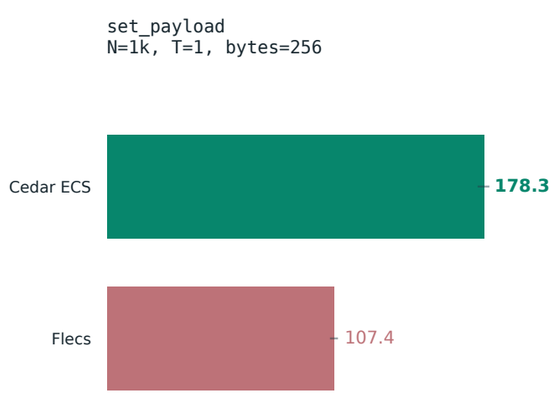


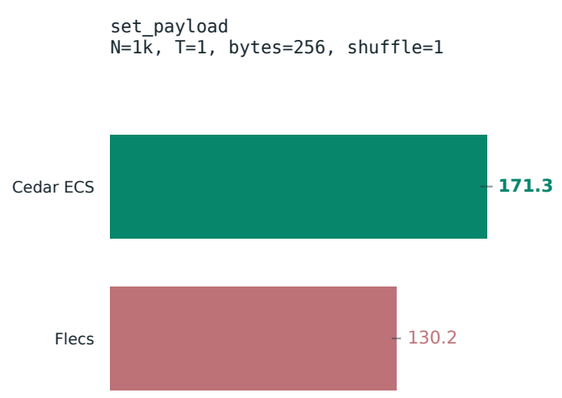


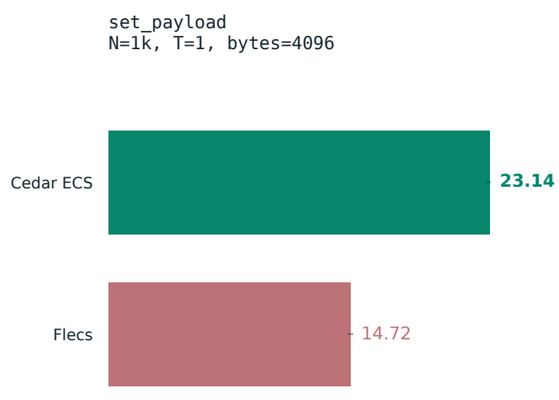


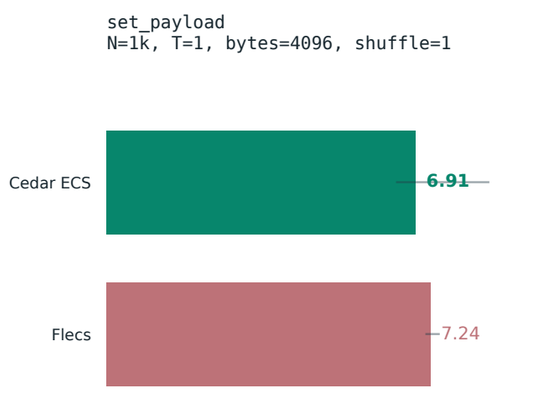


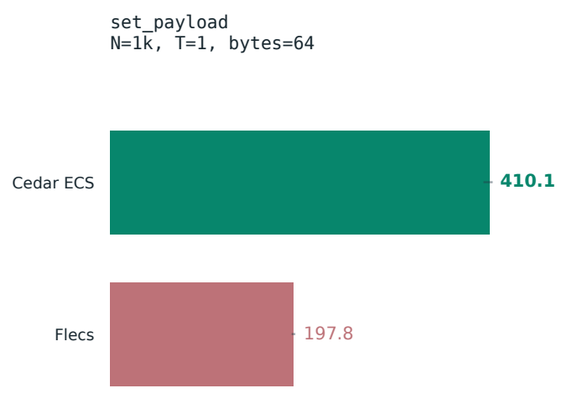








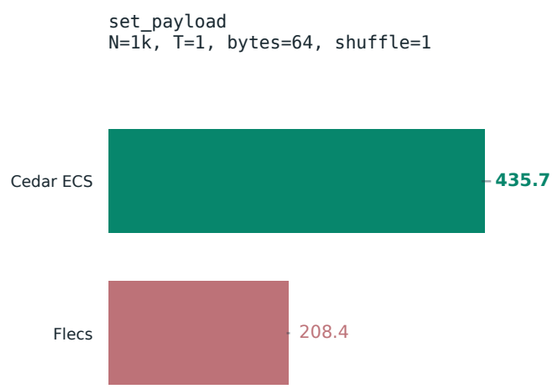


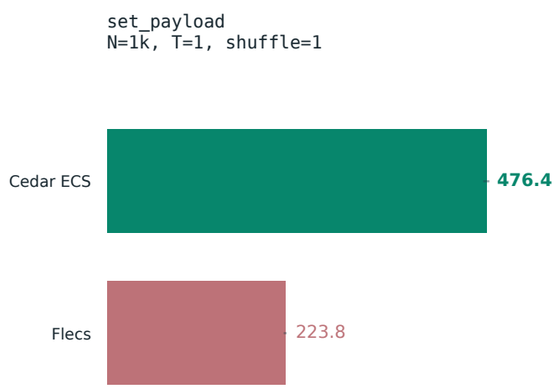


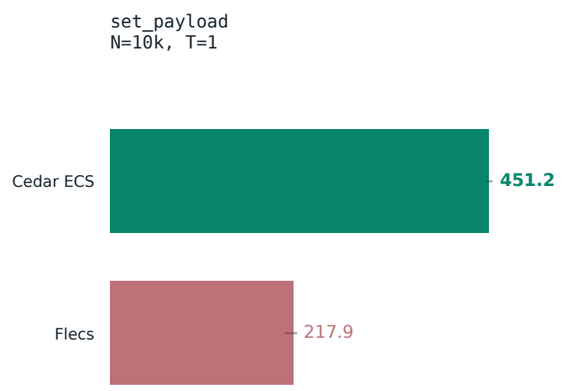
Indexed · Cedar vs Flecs

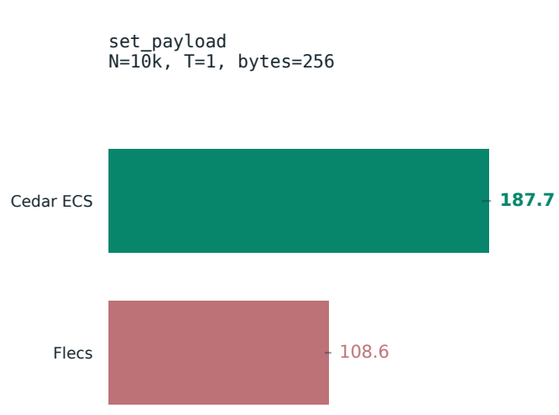
Mop/s ↑

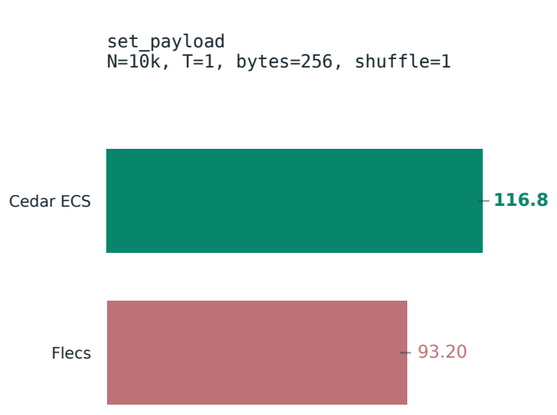
Cedar ECS Flecs

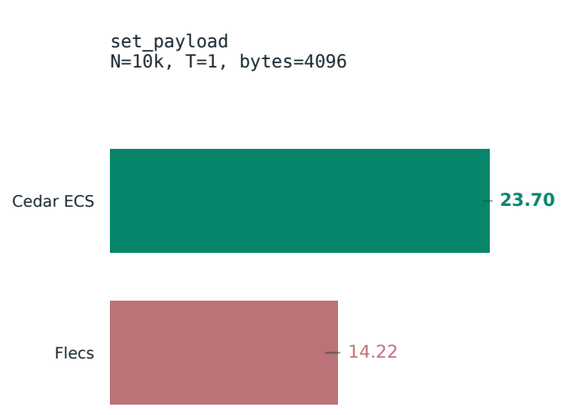


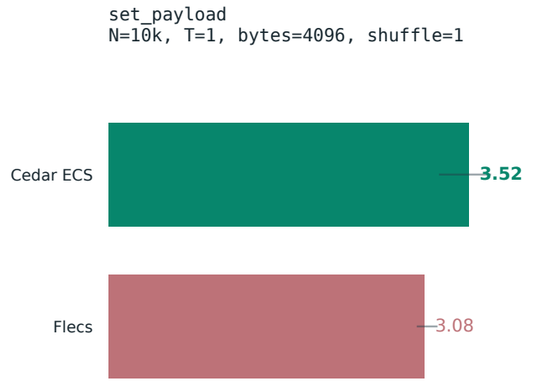


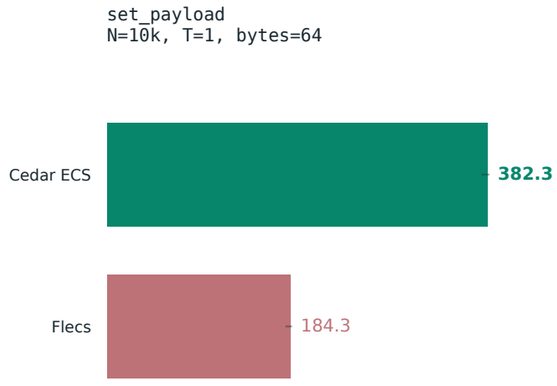


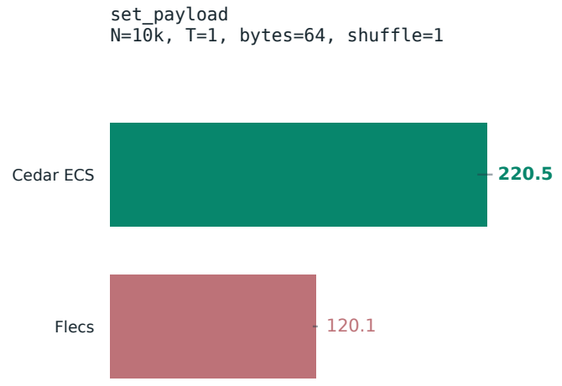


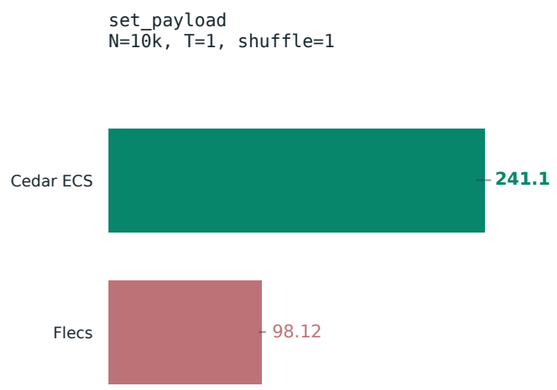


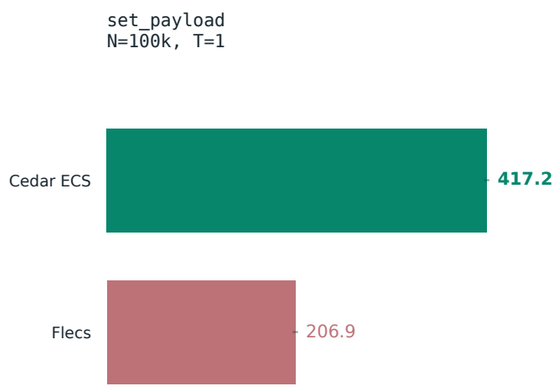


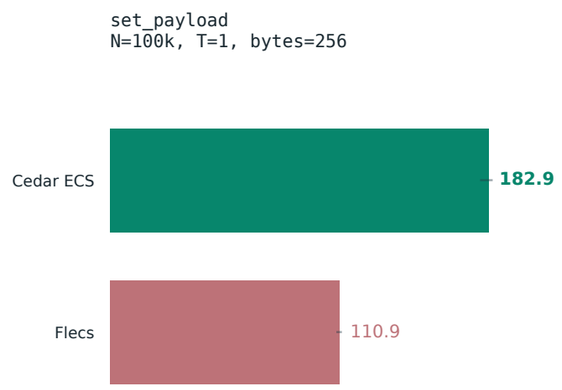








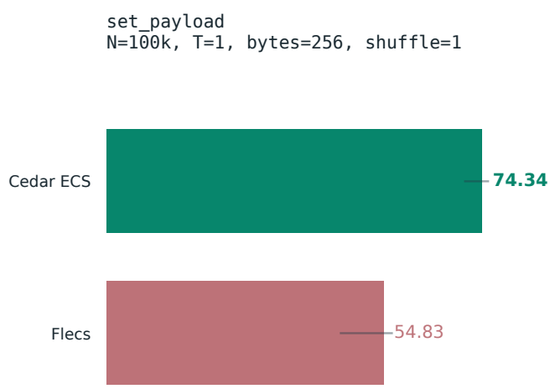


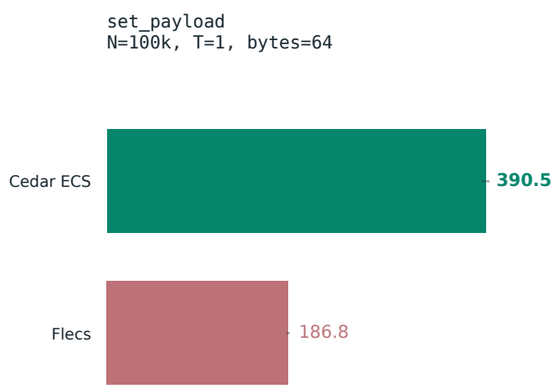


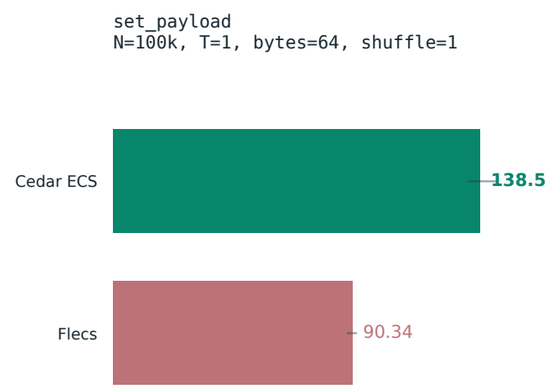
Indexed · Cedar vs Flecs

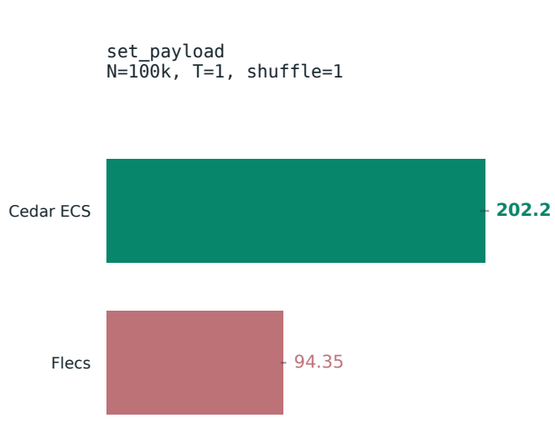
Mop/s ↑

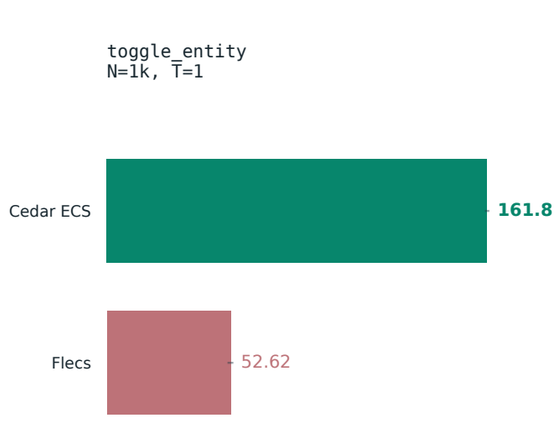
Cedar ECS Flecs

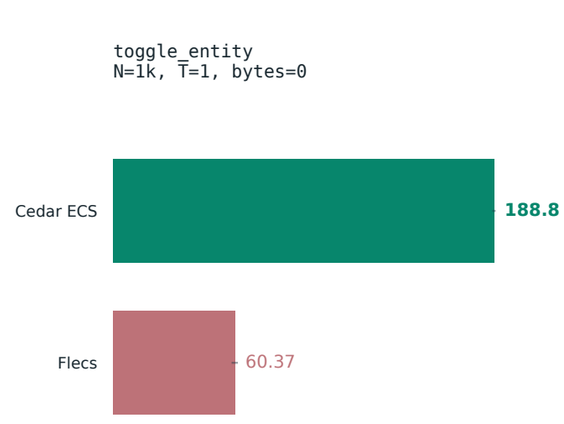


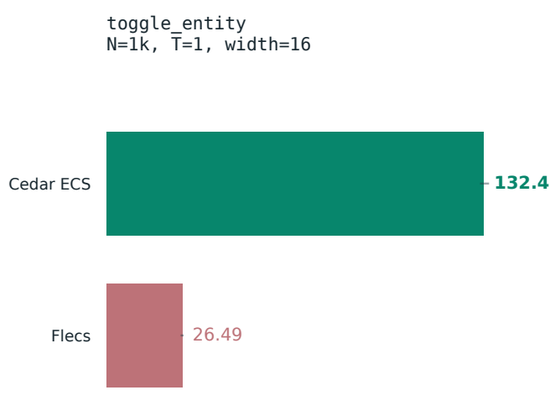


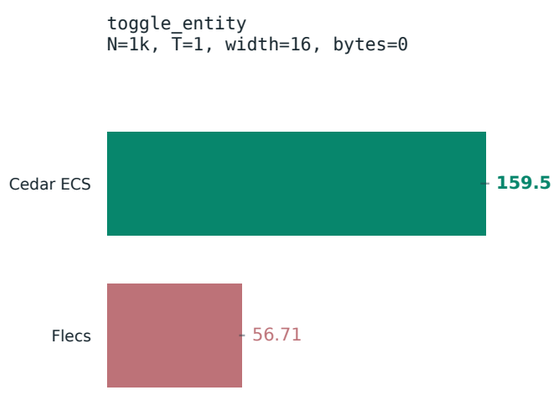


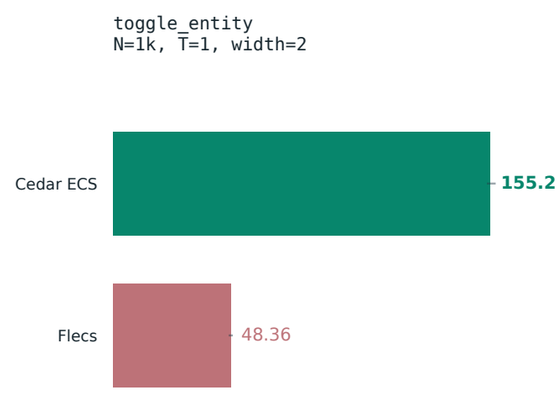


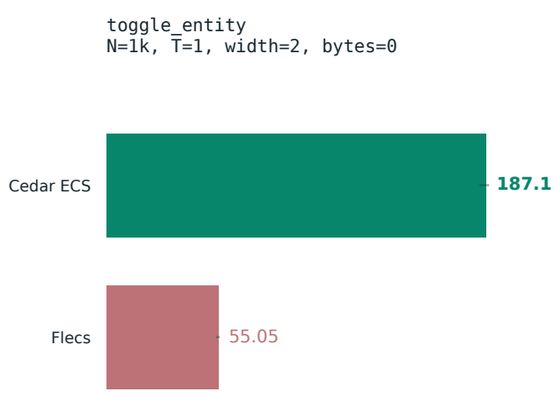


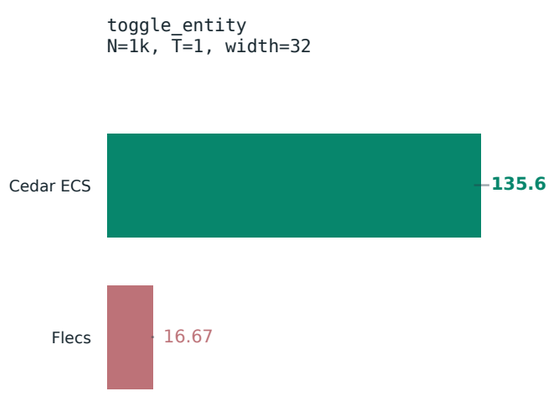


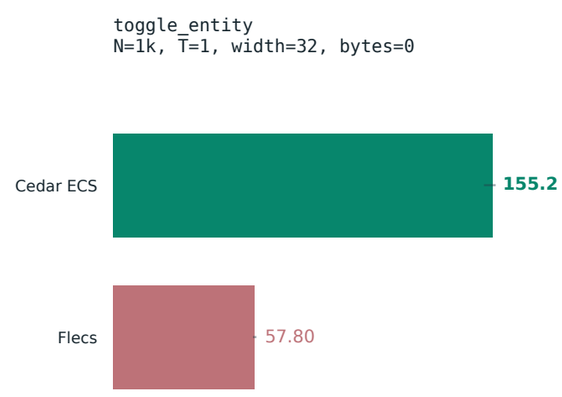










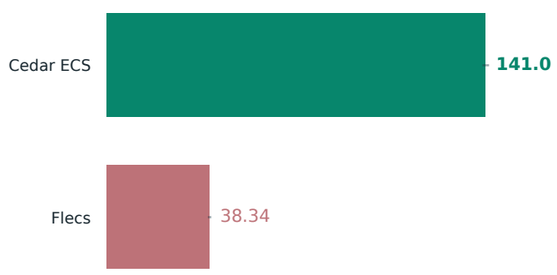


Indexed · Cedar vs Flecs

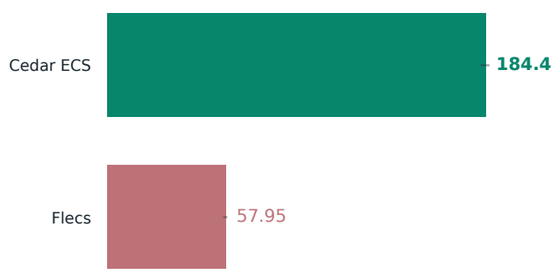
Mop/s ↑

Cedar ECS Flecs

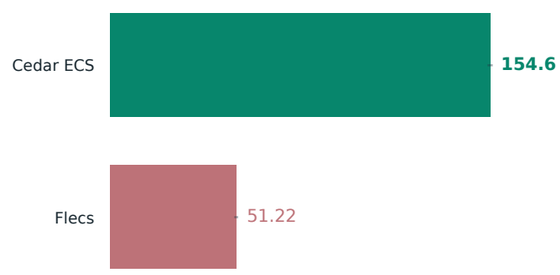
toggle_entity
N=1k, T=1, width=8



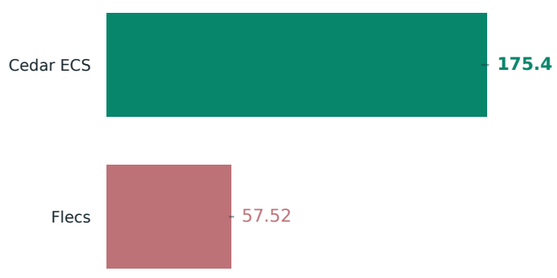
toggle_entity
N=1k, T=1, width=8, bytes=0



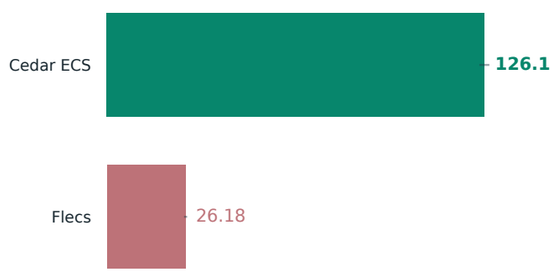
toggle_entity
N=10k, T=1



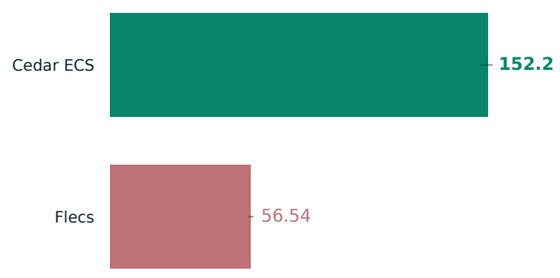
toggle_entity
N=10k, T=1, bytes=0



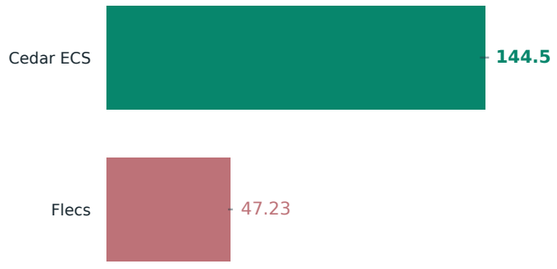
toggle_entity
N=10k, T=1, width=16



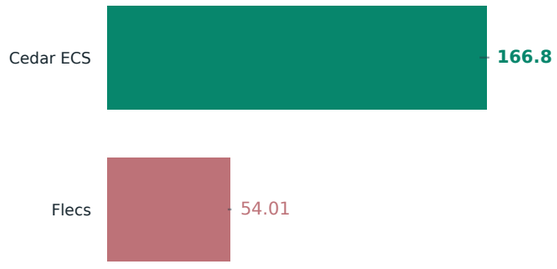
toggle_entity
N=10k, T=1, width=16, bytes=0



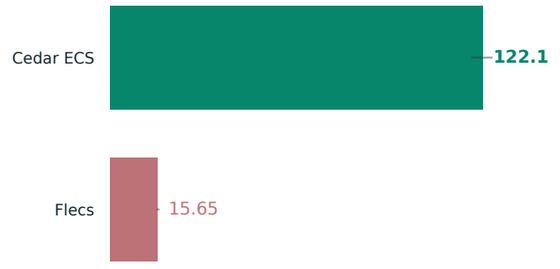
toggle_entity
N=10k, T=1, width=2



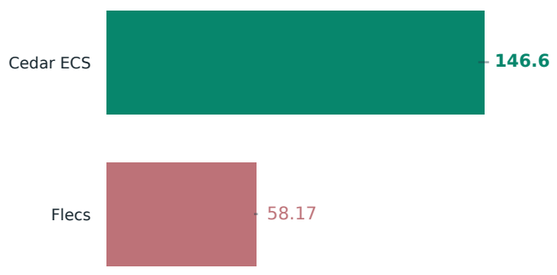
toggle_entity
N=10k, T=1, width=2, bytes=0



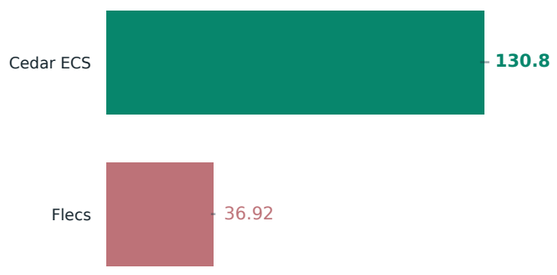
toggle_entity
N=10k, T=1, width=32



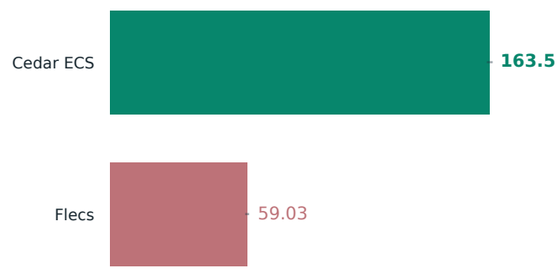
toggle_entity
N=10k, T=1, width=32, bytes=0



toggle_entity
N=10k, T=1, width=8



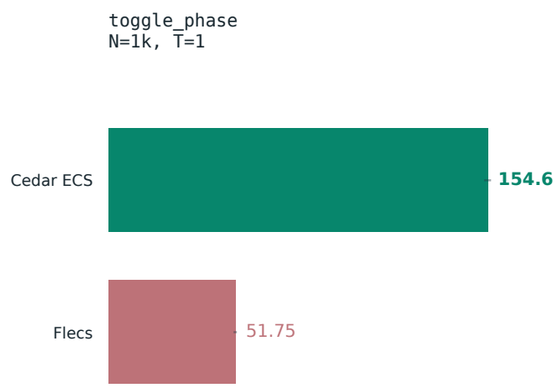
toggle_entity
N=10k, T=1, width=8, bytes=0

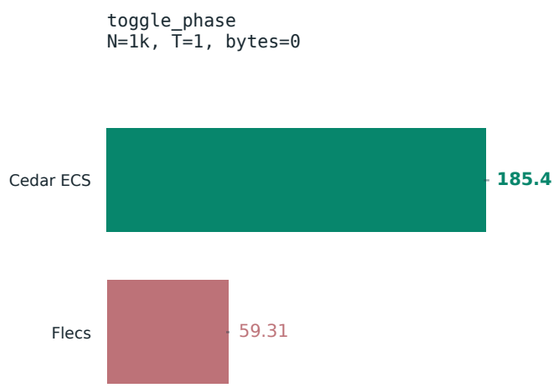


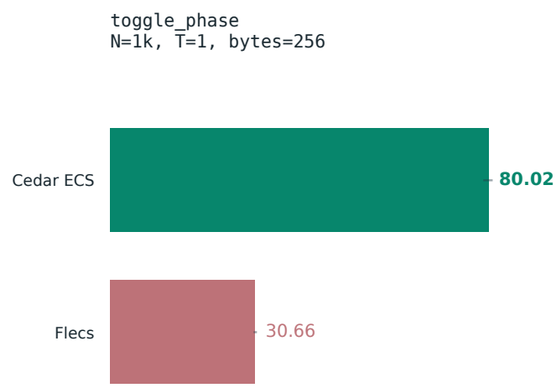
Indexed · Cedar vs Flecs

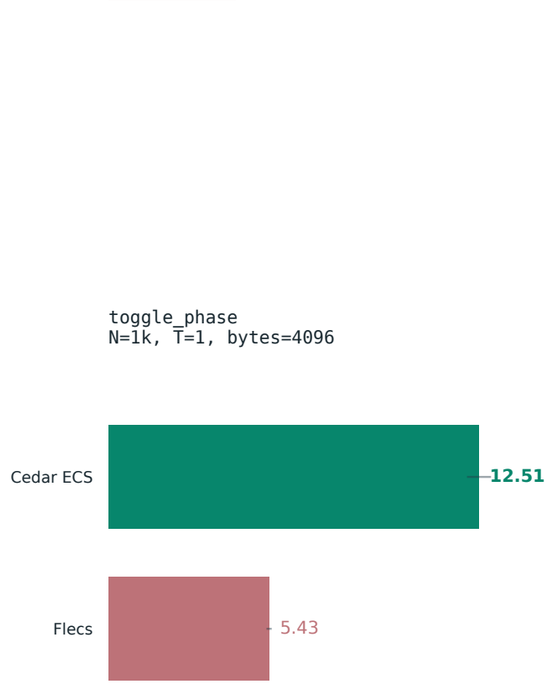
Mop/s ↑

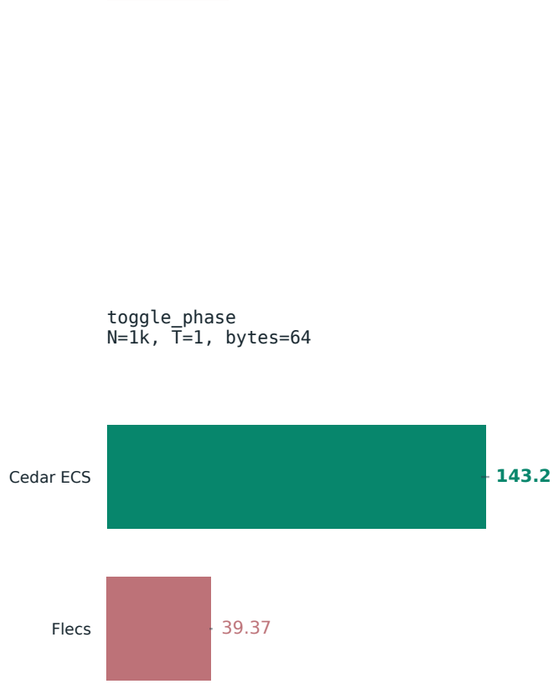
Cedar ECS Flecs

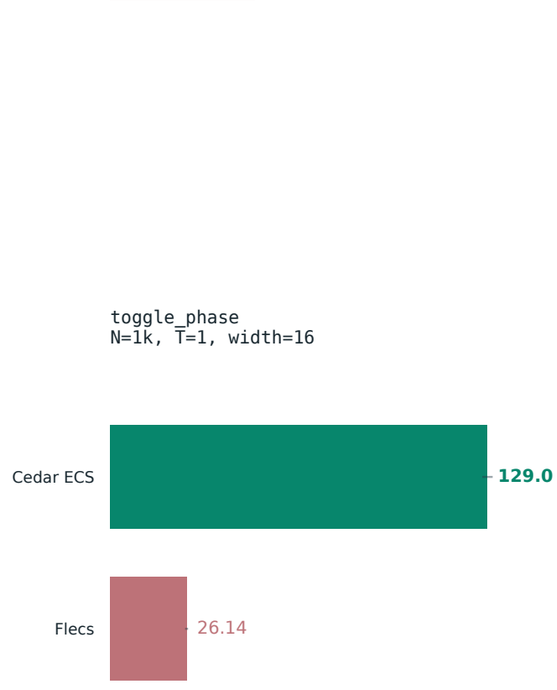


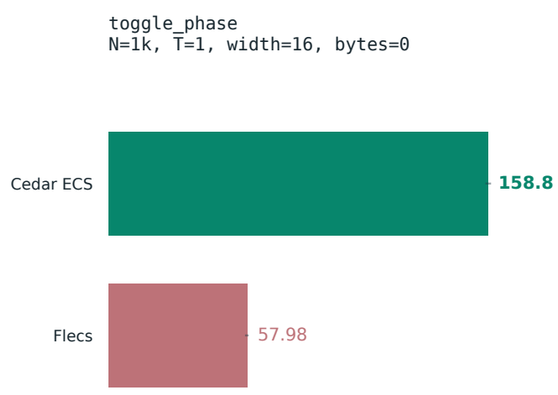


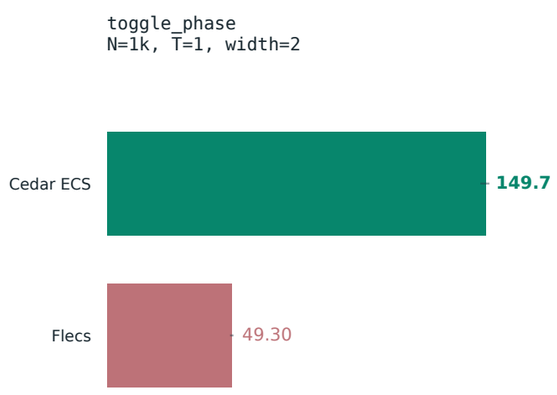


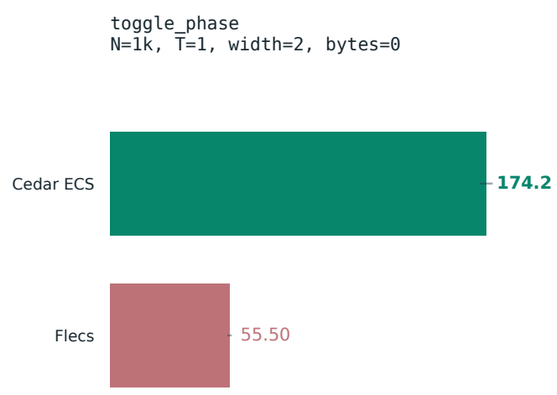


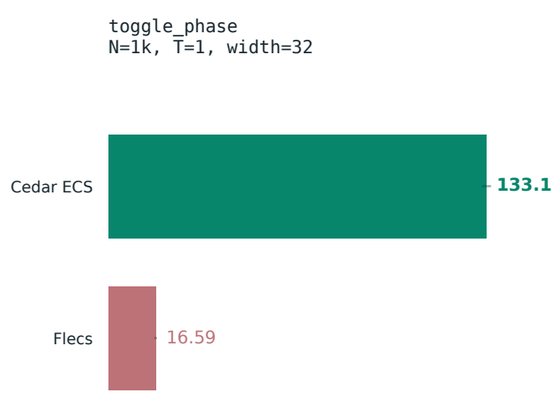


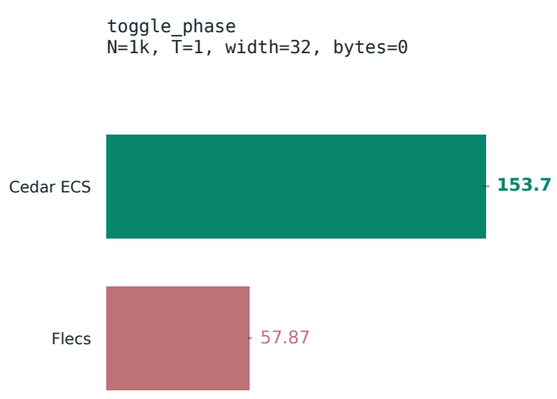


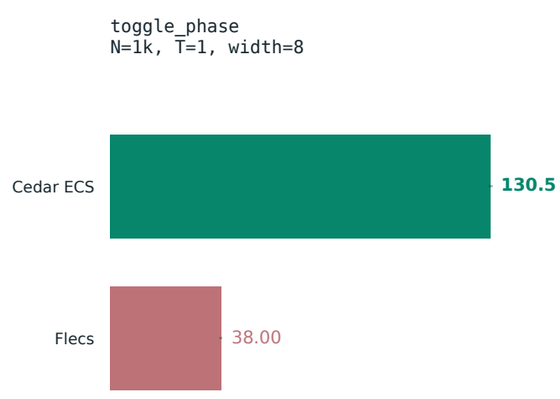








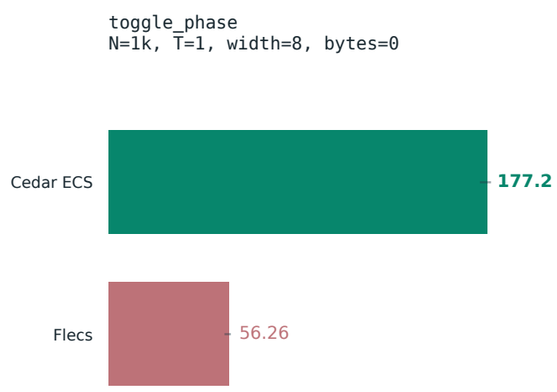


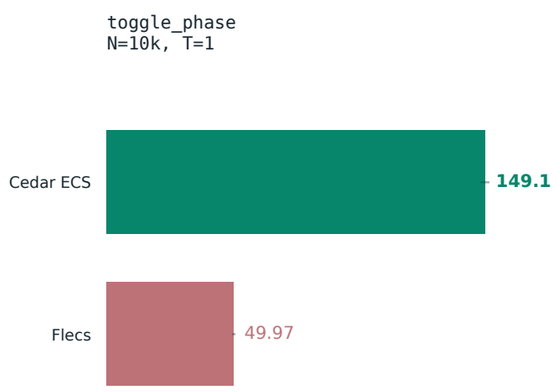


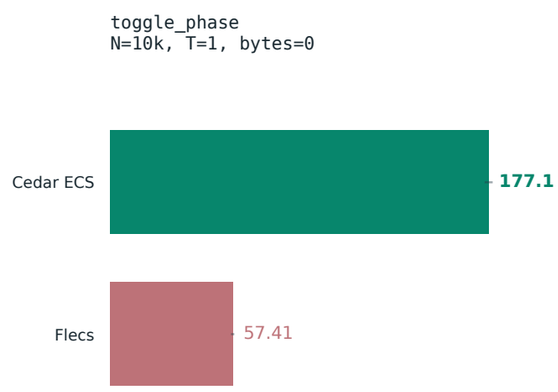
Indexed · Cedar vs Flecs

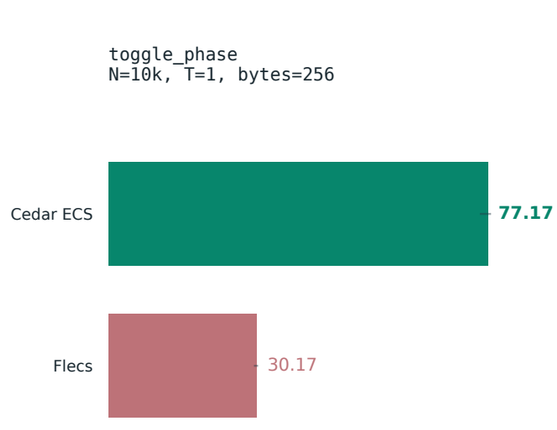
Mop/s ↑

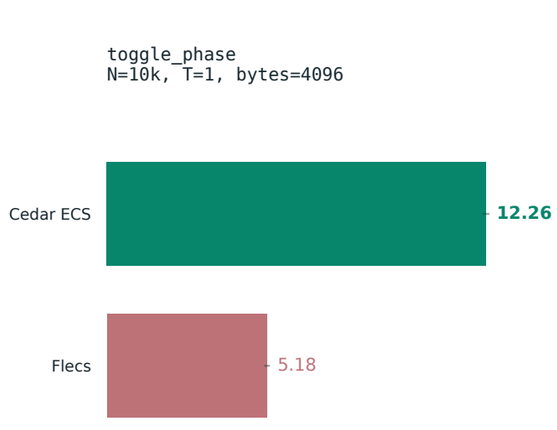
Cedar ECS Flecs

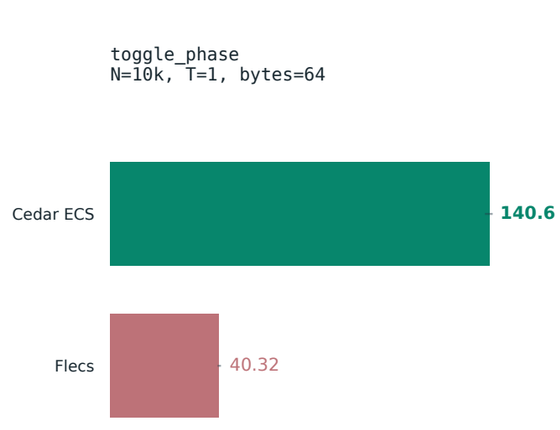


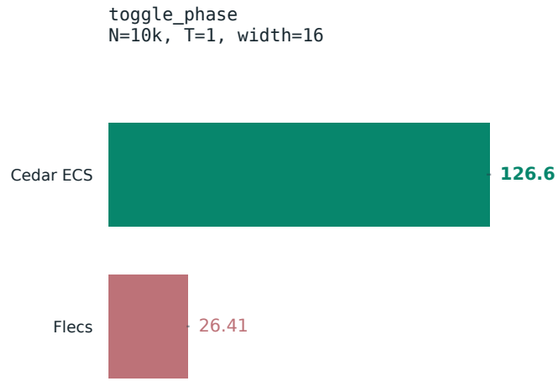


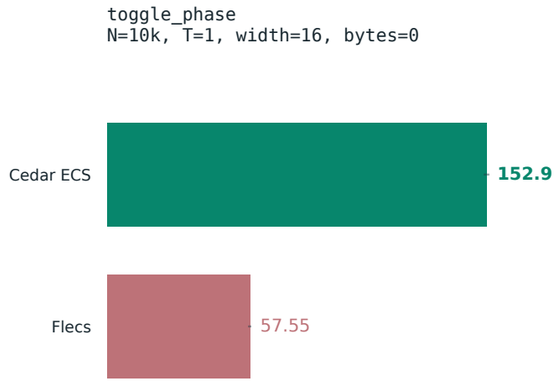


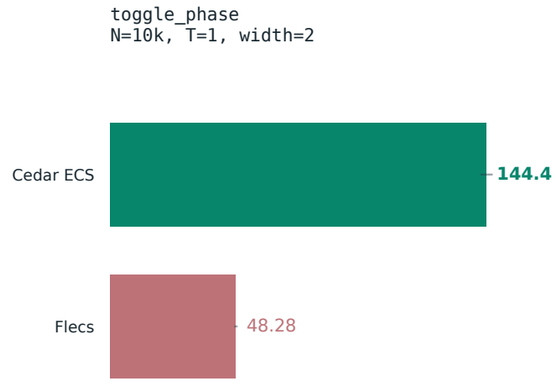


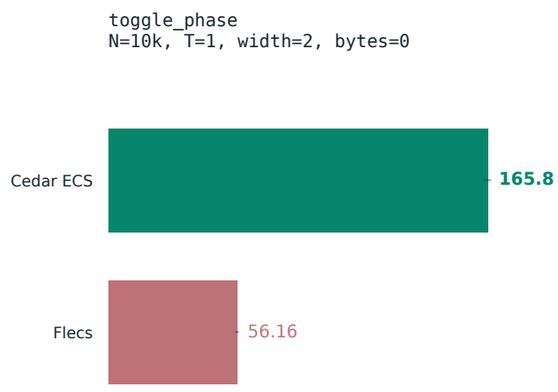


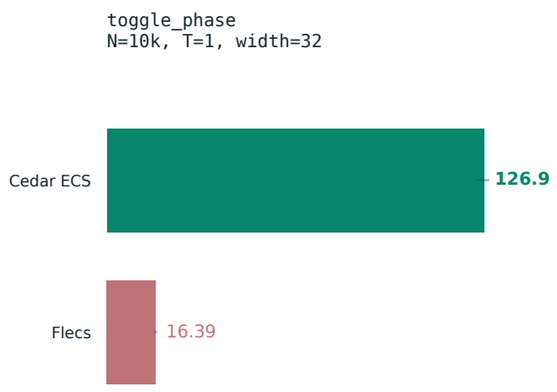


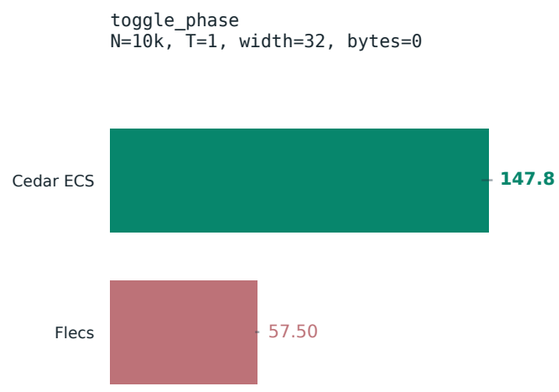








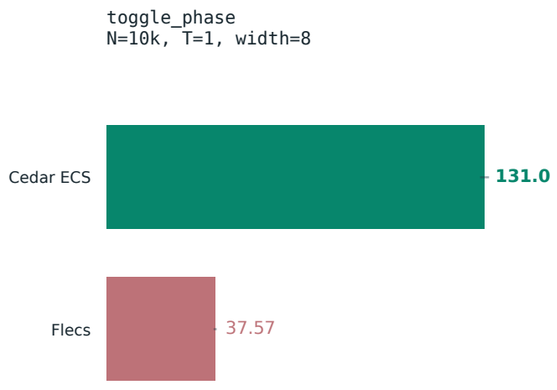


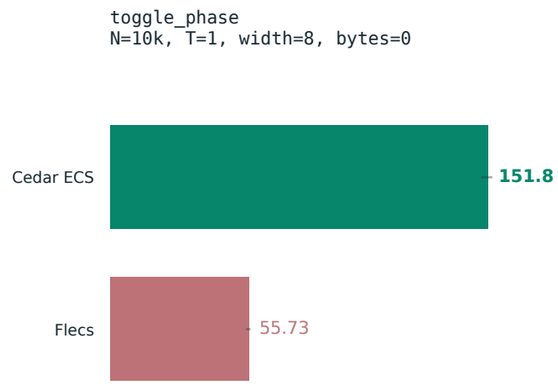


Indexed · Cedar vs Flecs

Mop/s ↑

Cedar ECS Flecs

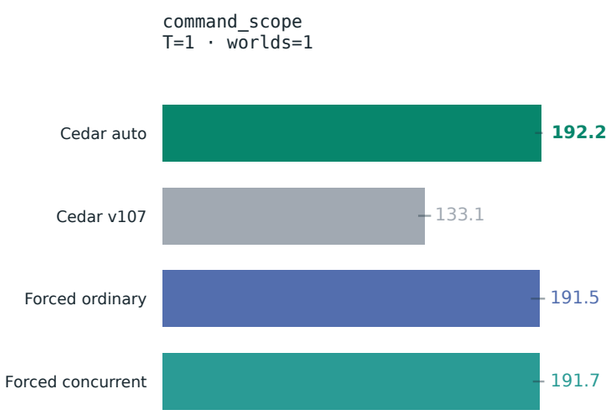


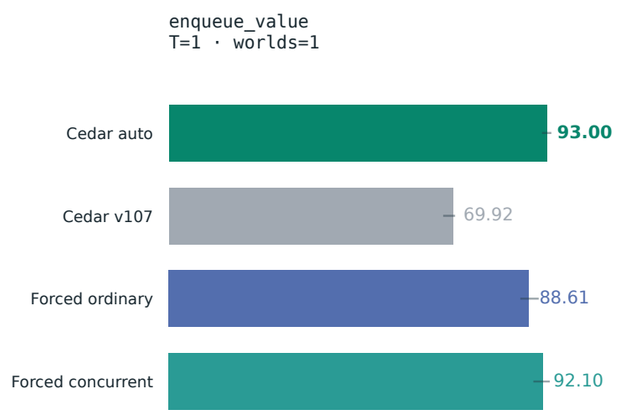


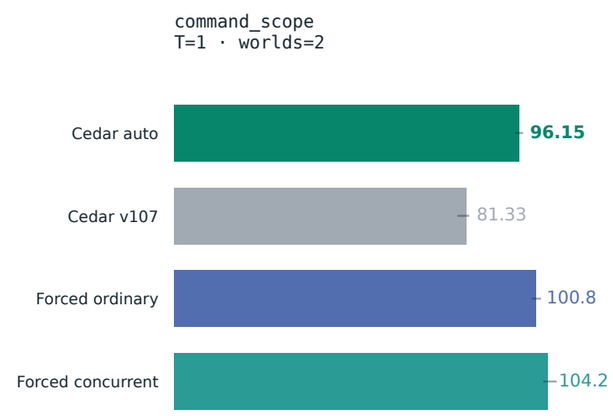
Concurrent registry

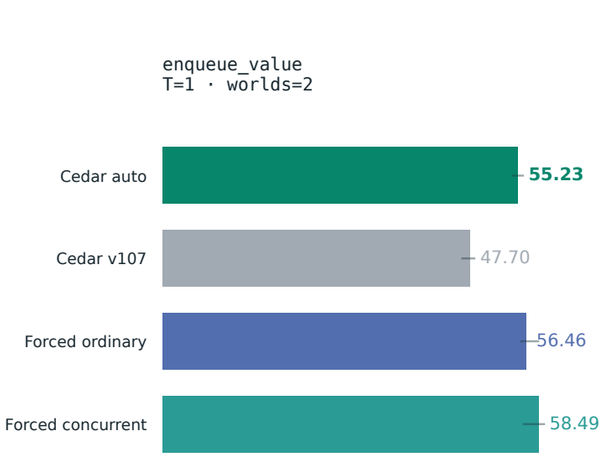
Mop/s ↑

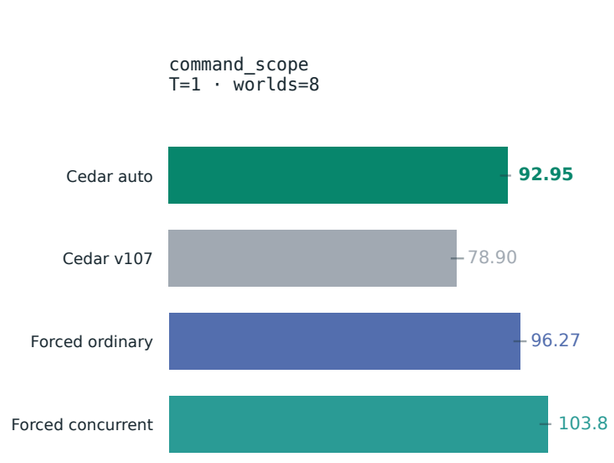
Cedar auto · Cedar v107 · Forced ordinary · Forced concurrent

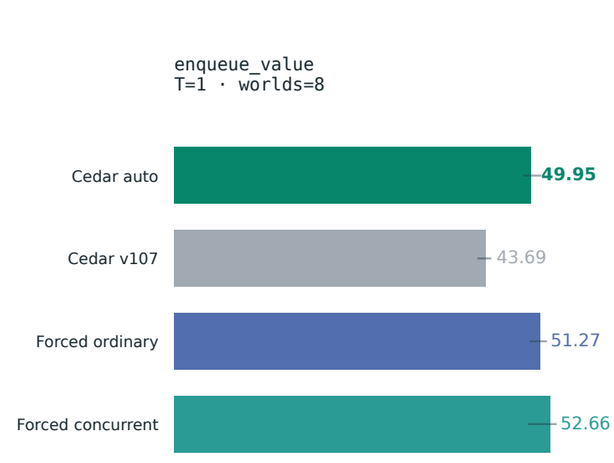


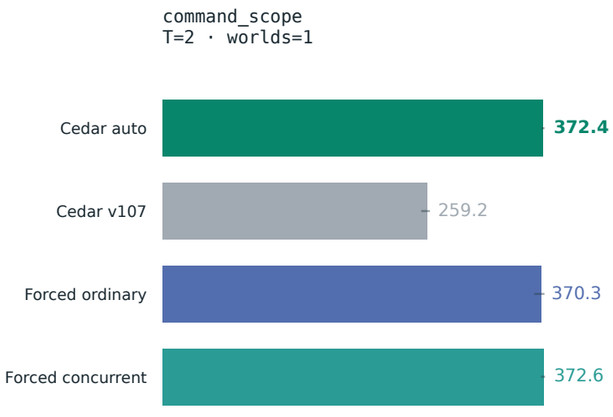


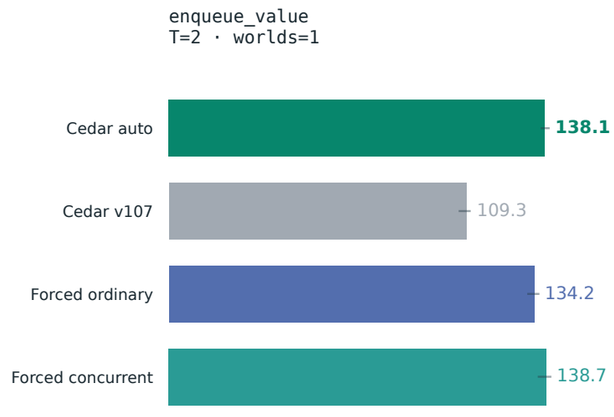


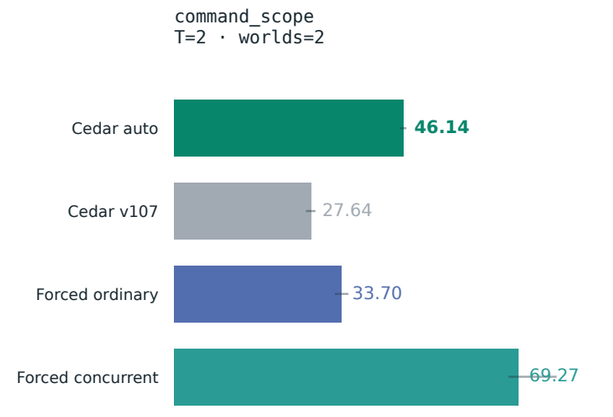


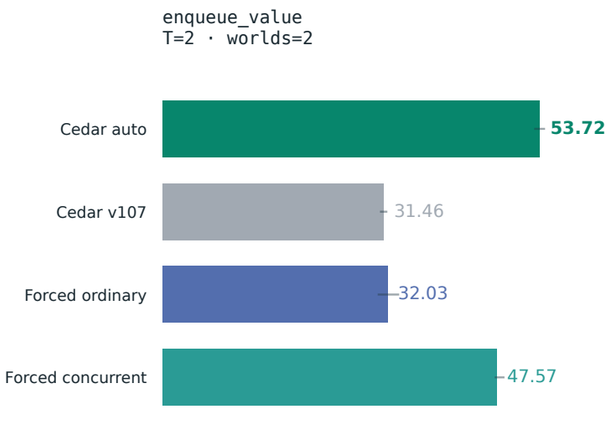


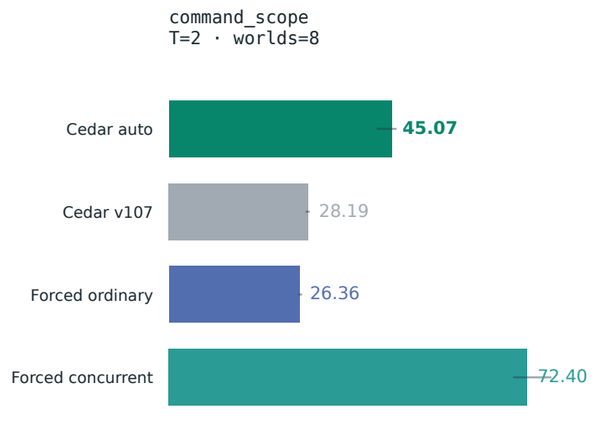


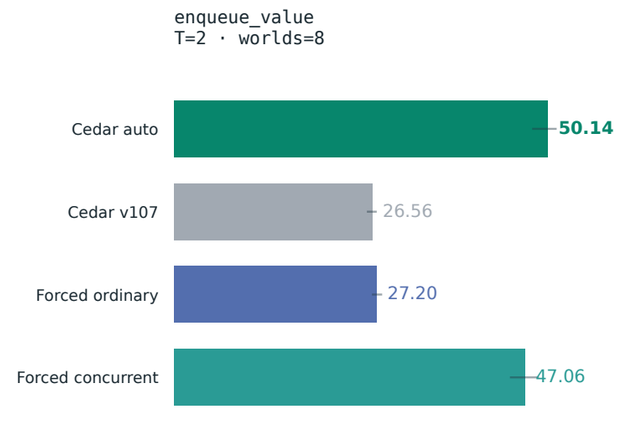








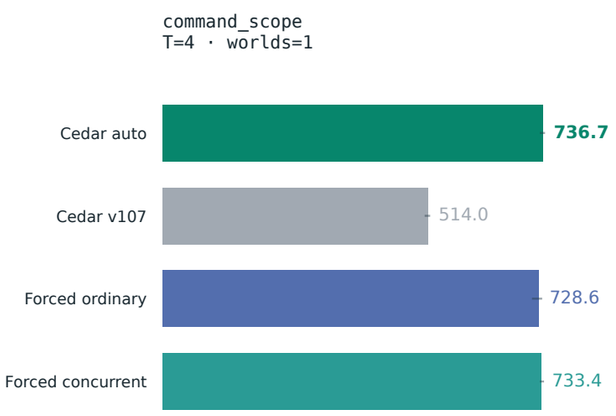


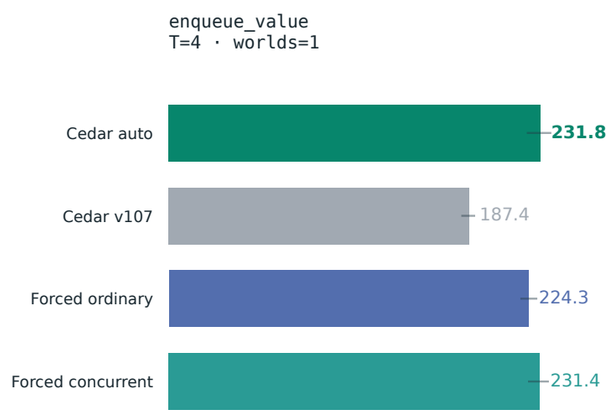


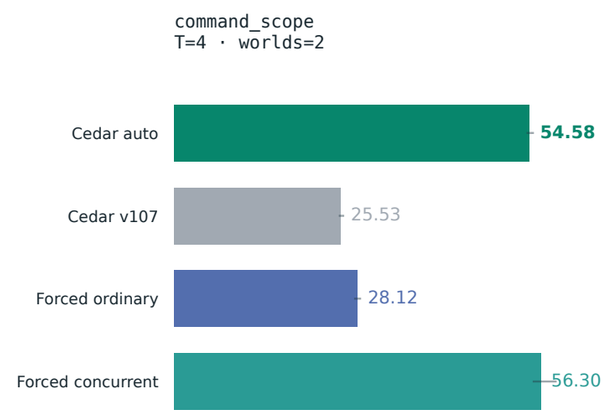
Concurrent registry

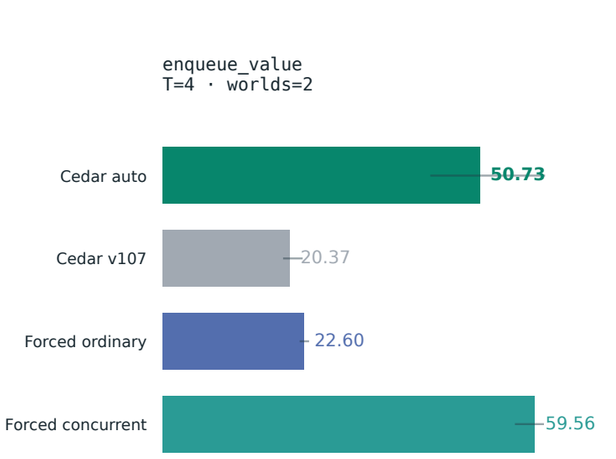
Mop/s ↑

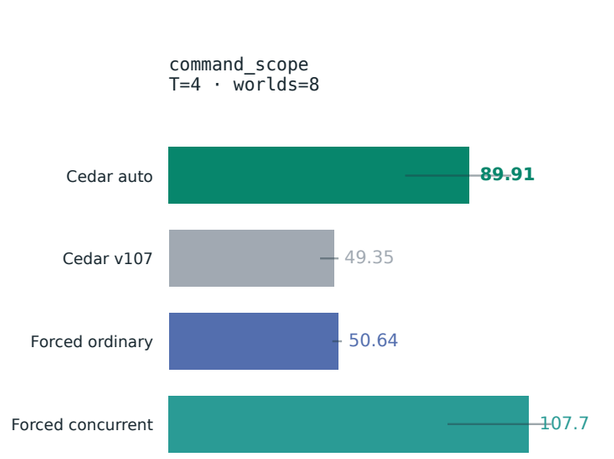
Cedar auto Cedar v107 Forced ordinary Forced concurrent

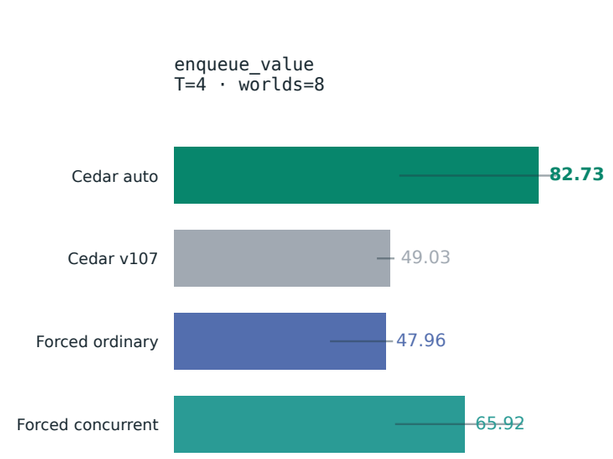


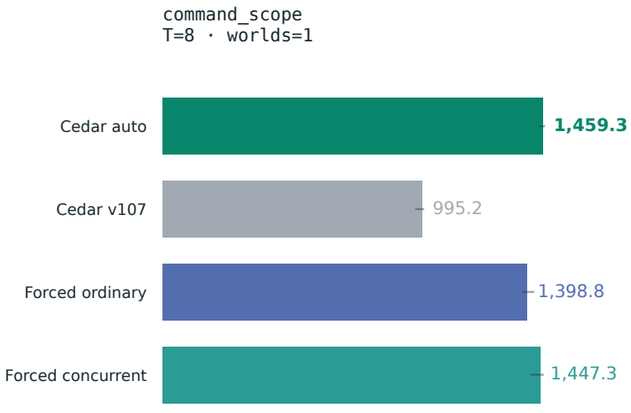


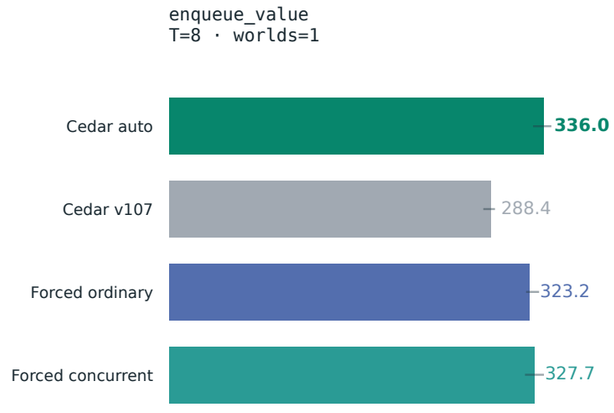


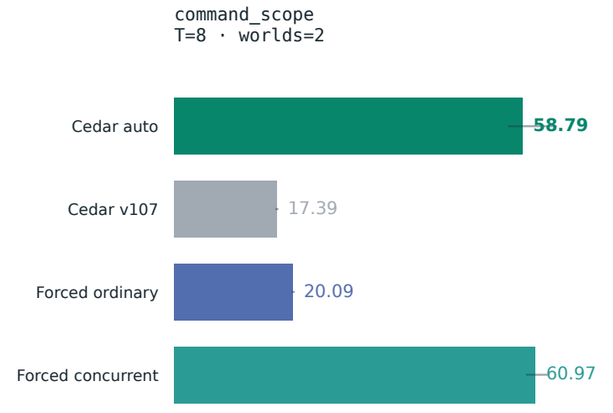


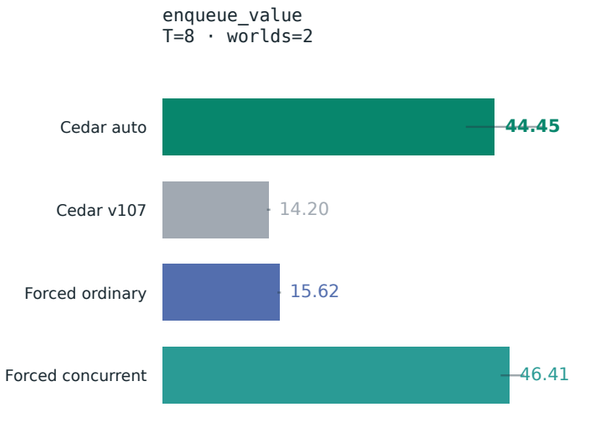


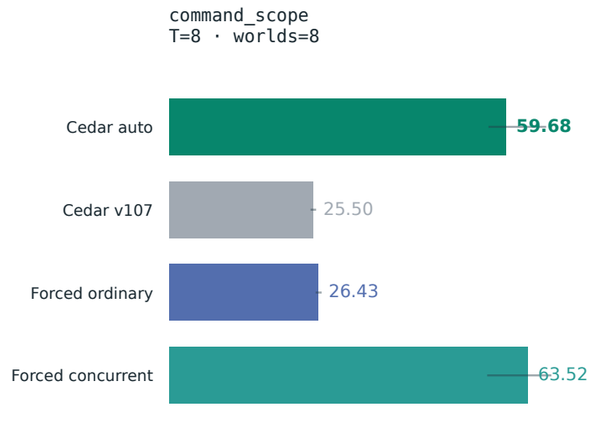


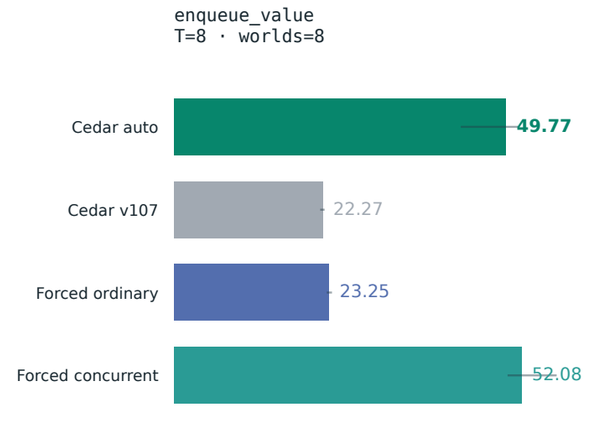












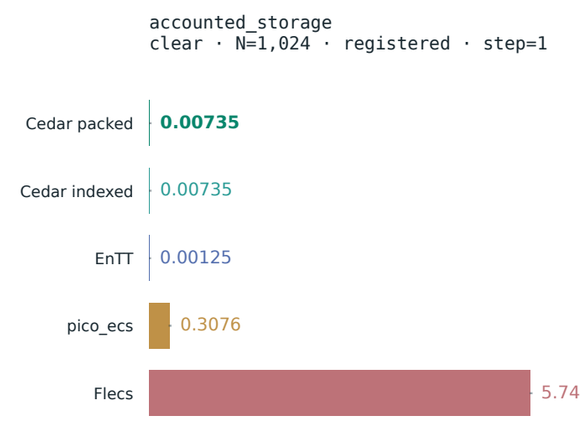
Memory comparison

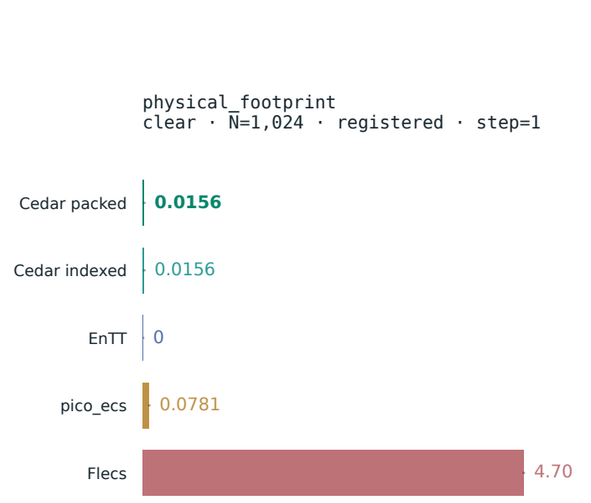
MiB ↓

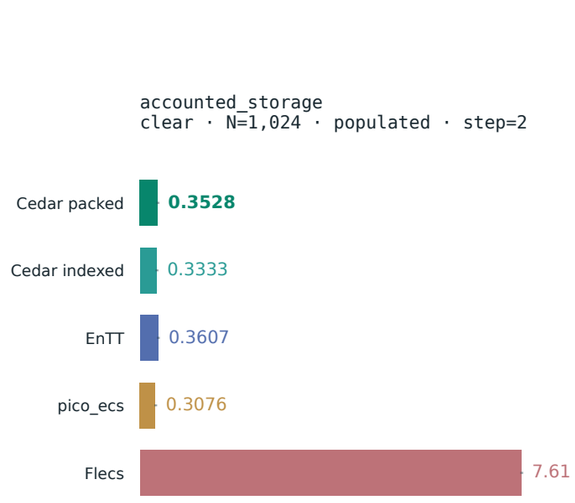
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

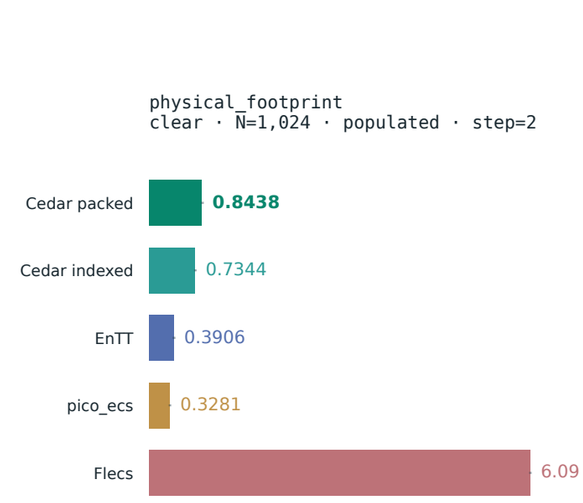


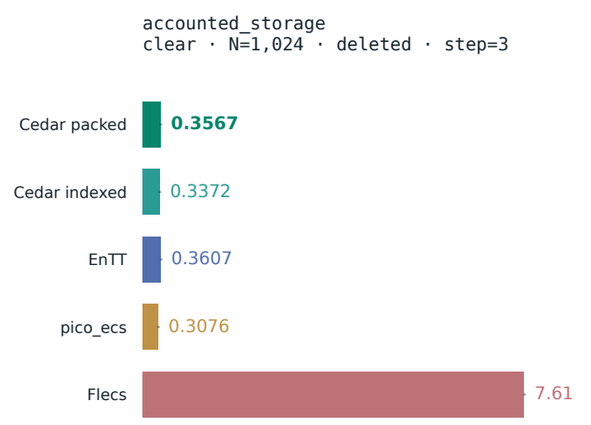


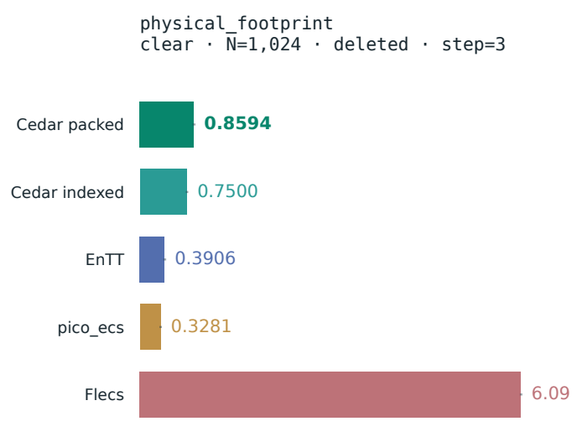




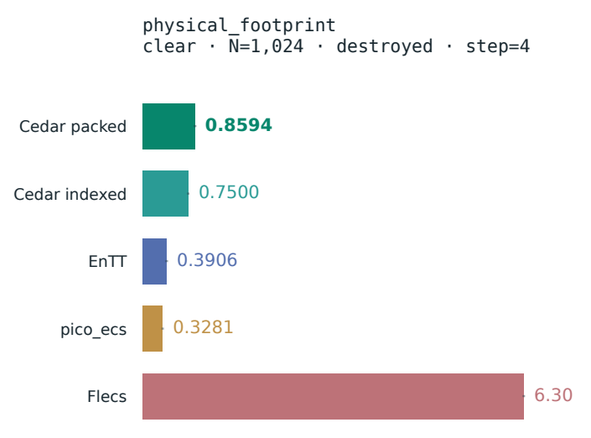












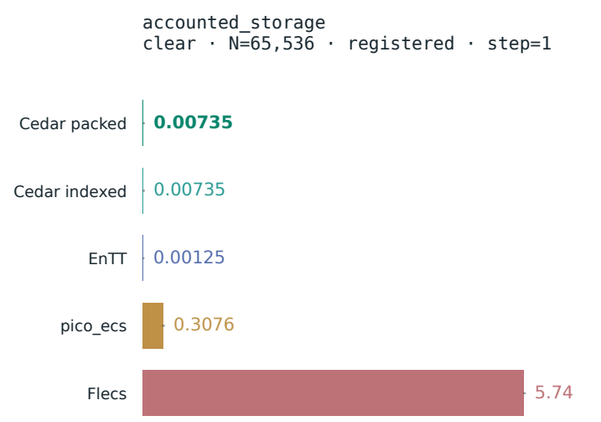


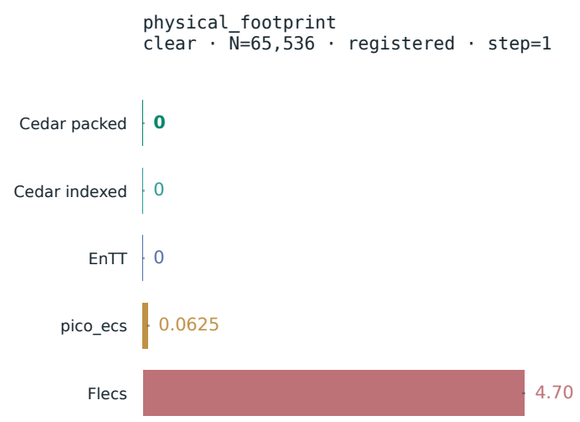


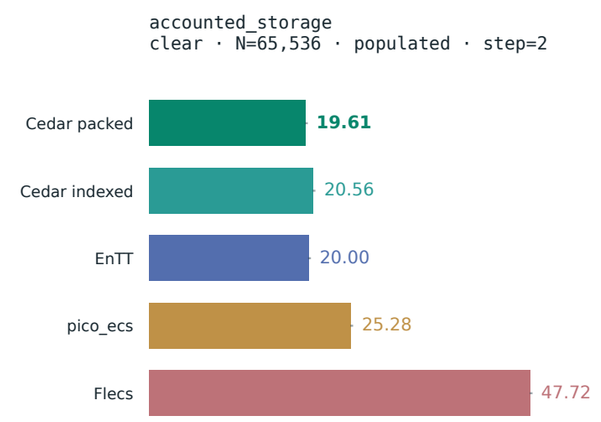
Memory comparison

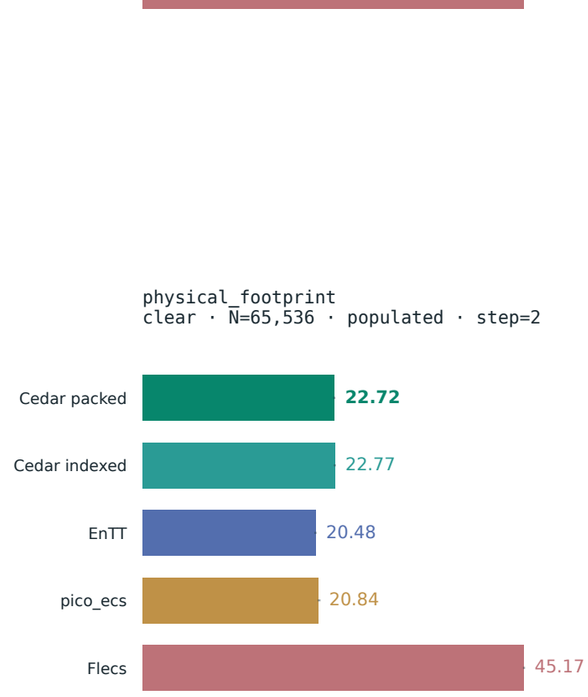
MiB ↓

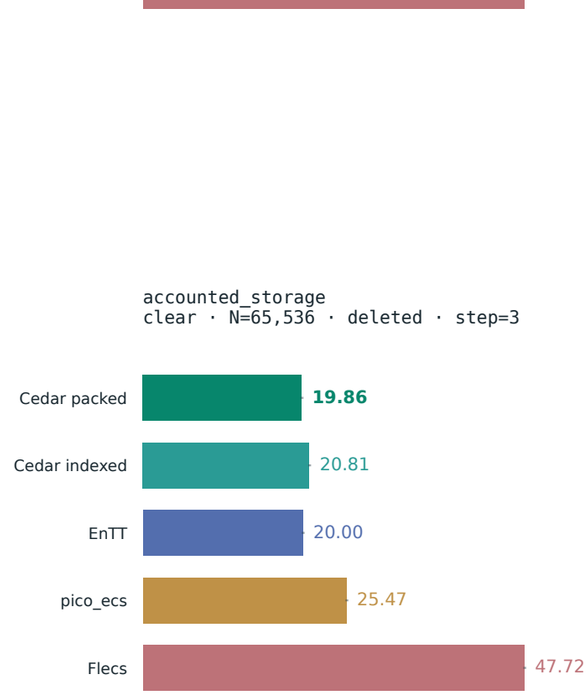
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

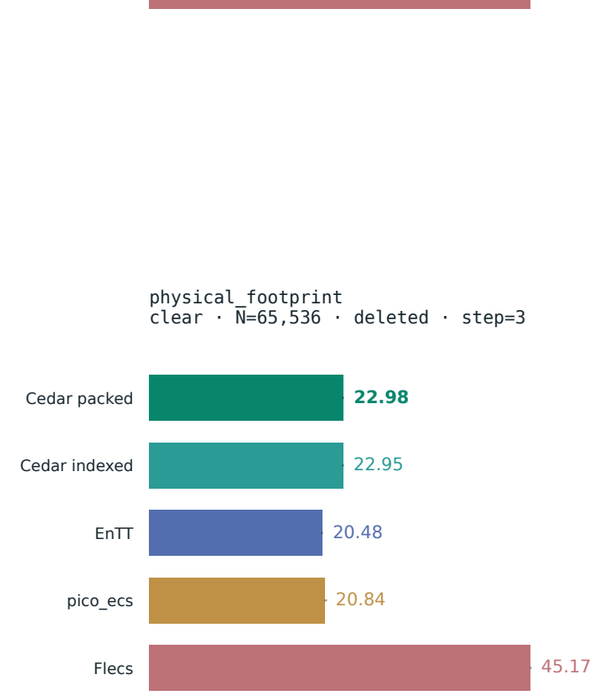


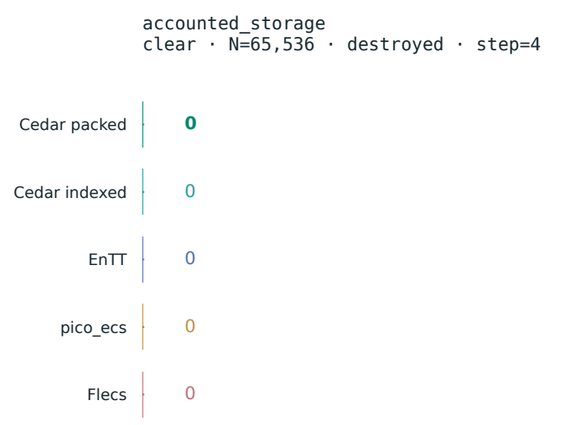


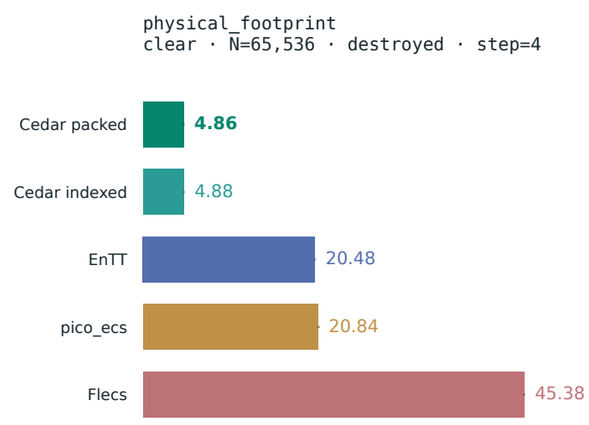






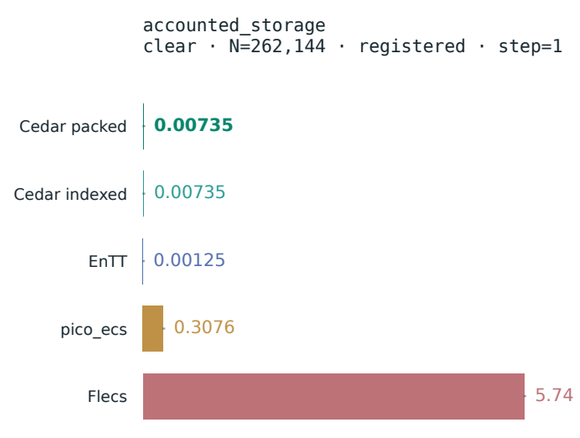


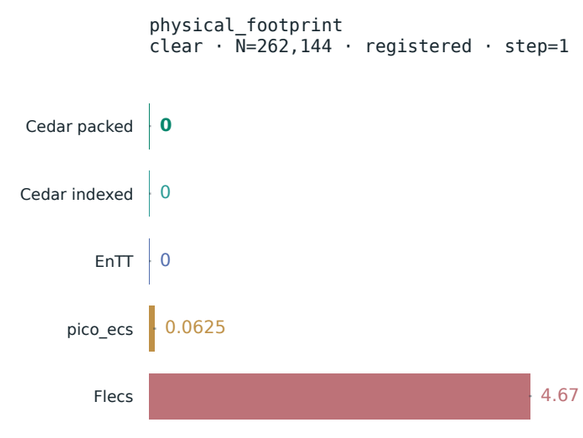








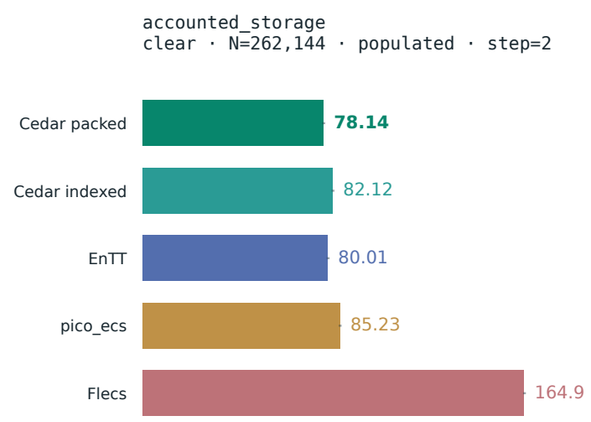


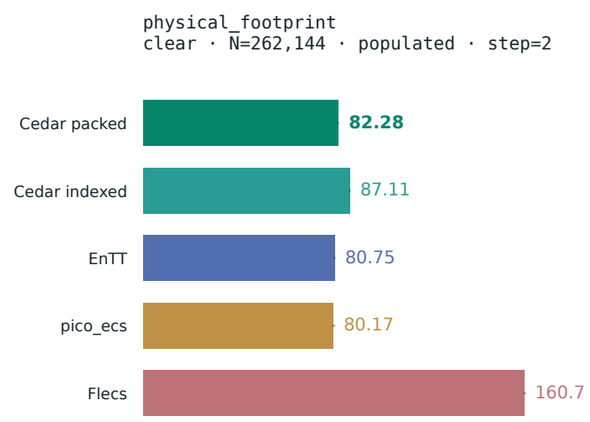


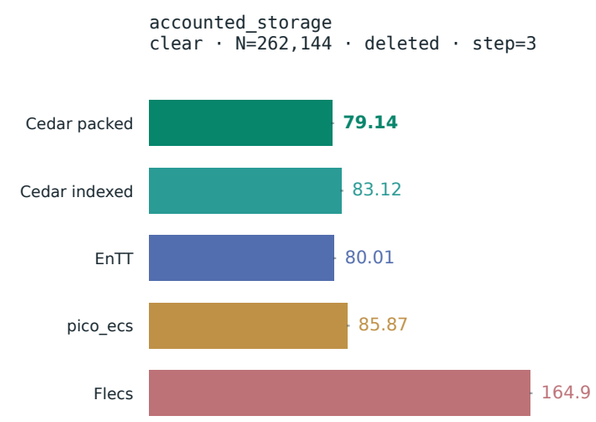
Memory comparison

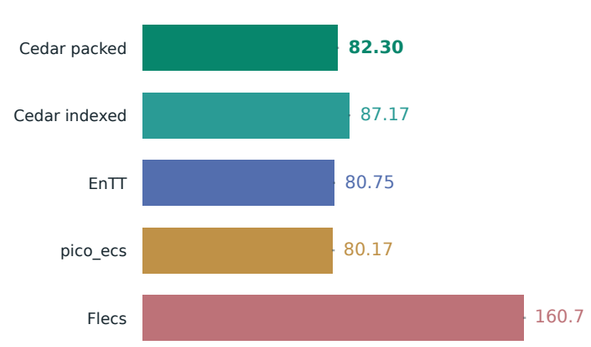
MiB ↓

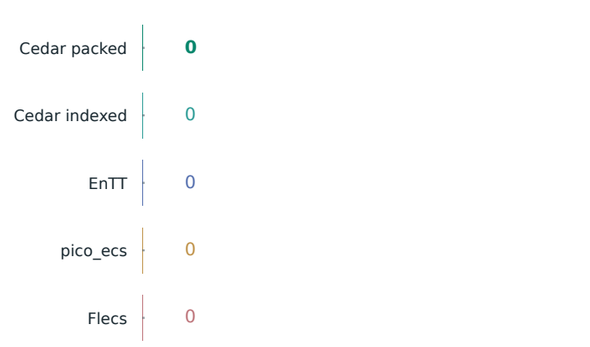
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

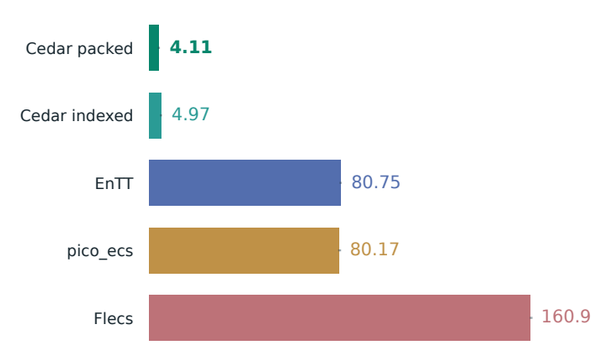


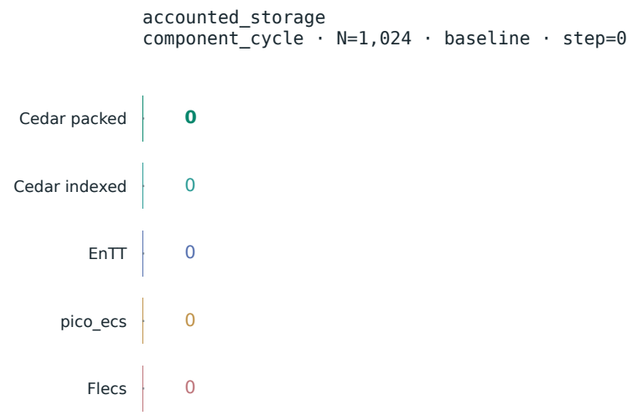


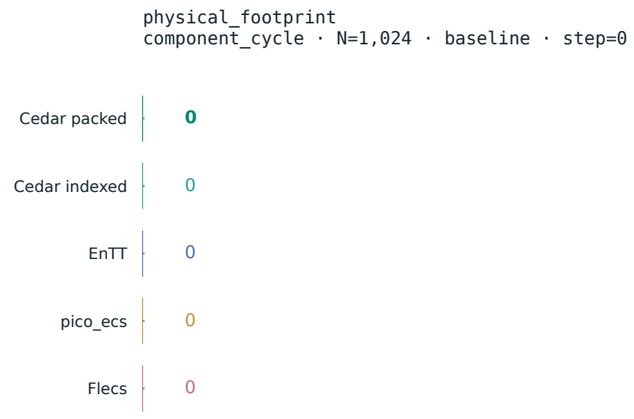


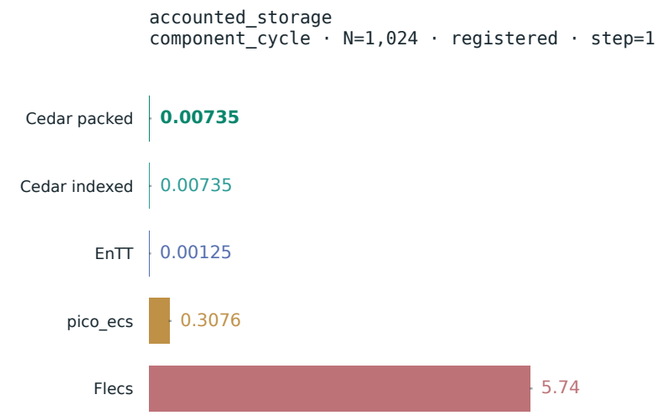


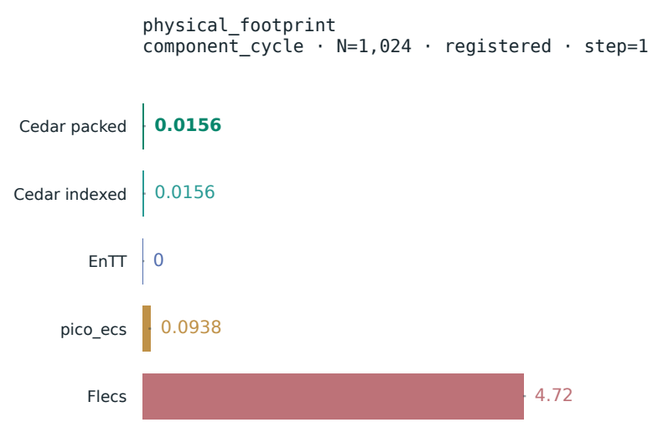


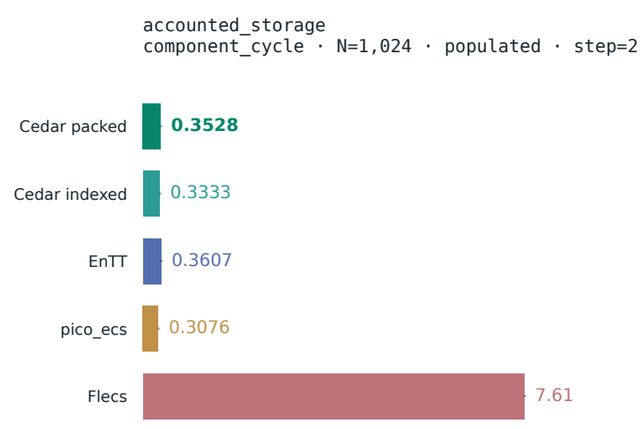


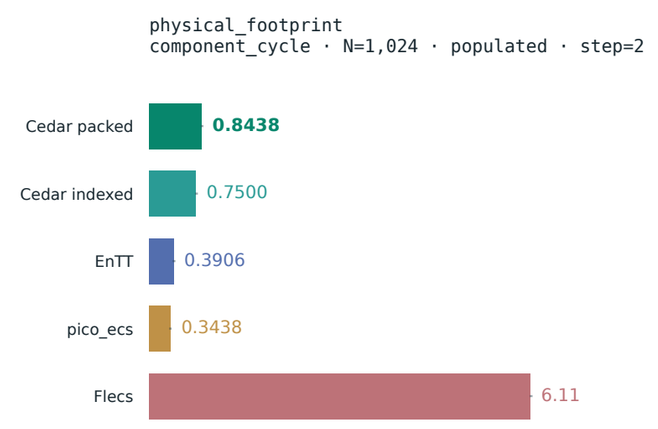








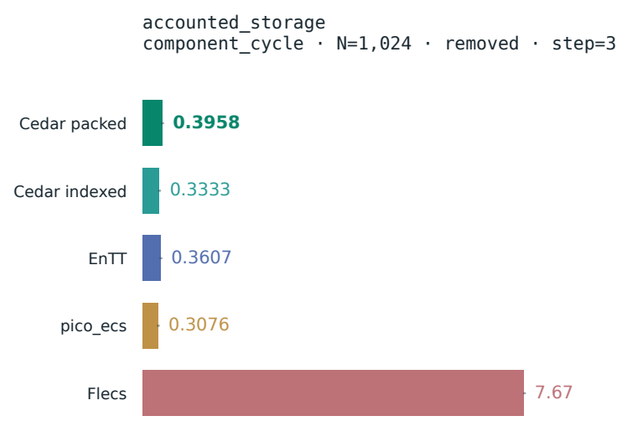


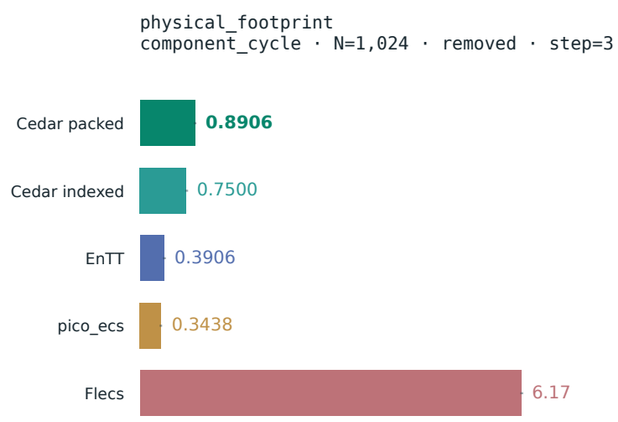


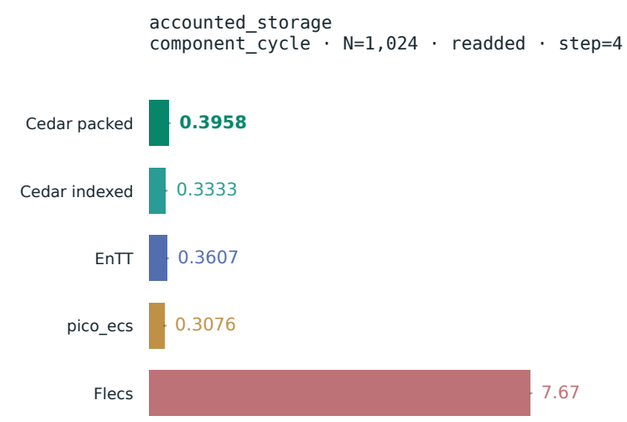
Memory comparison

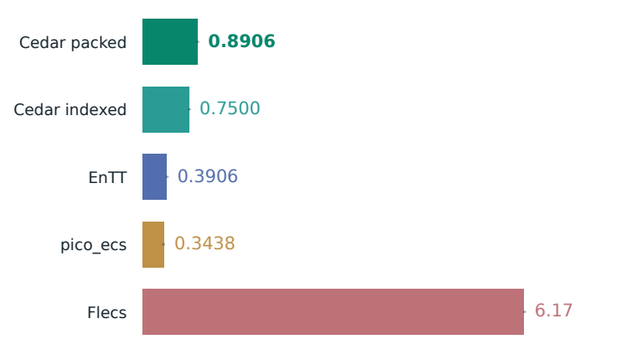
MiB ↓

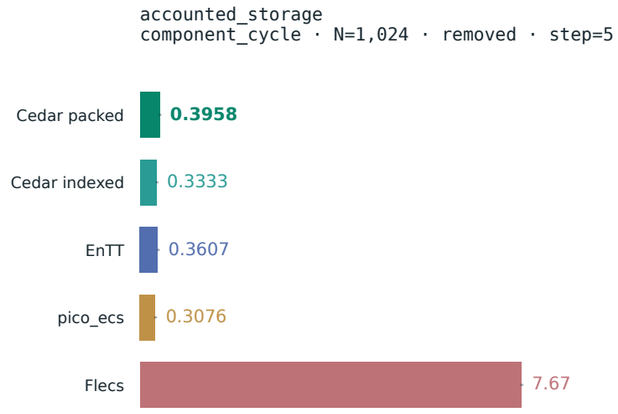
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

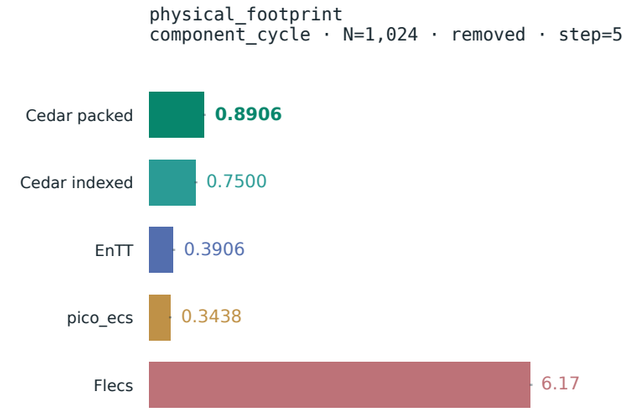


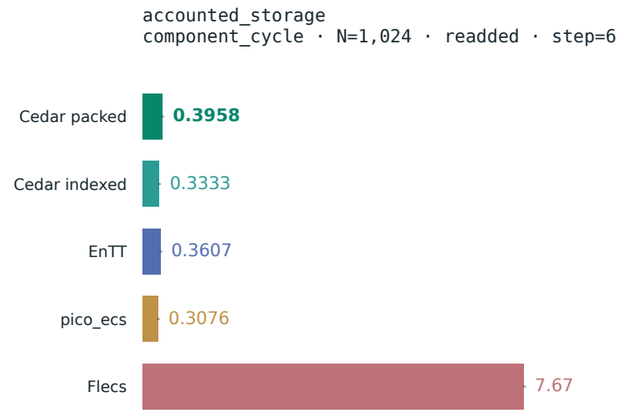


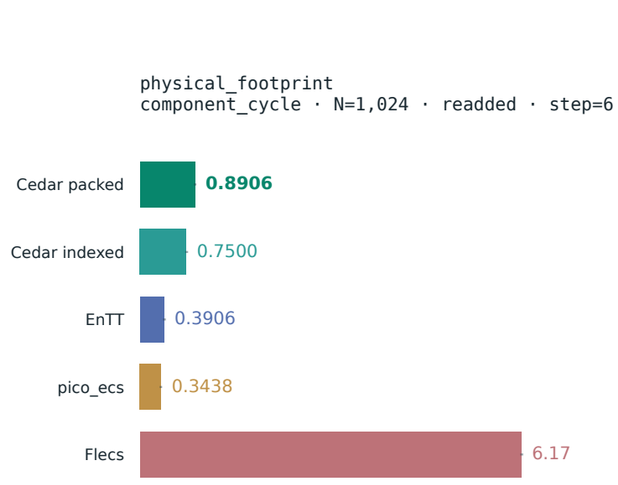


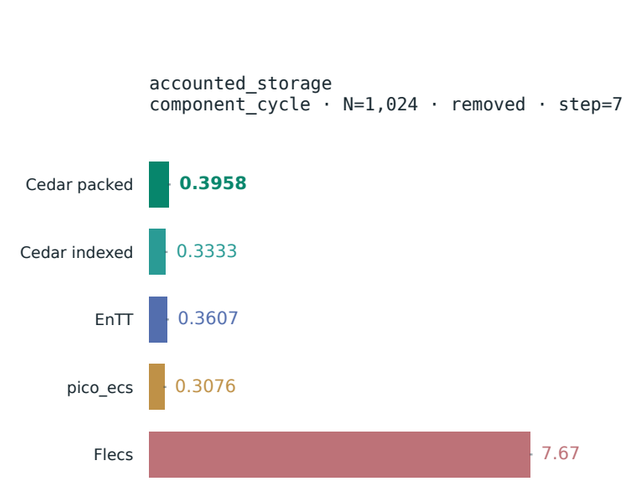


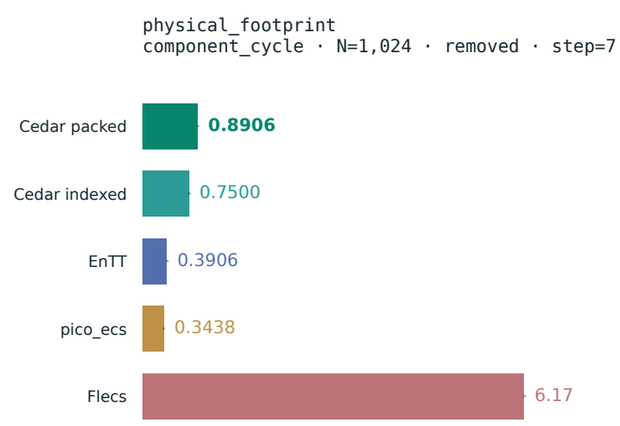


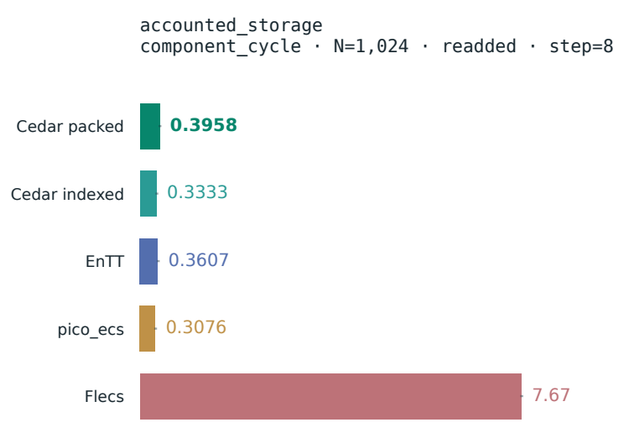


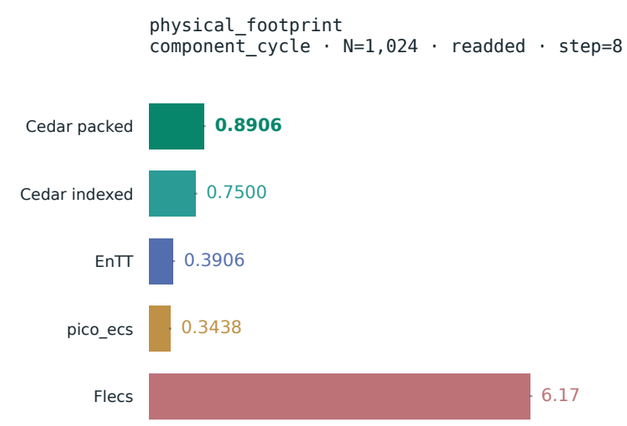








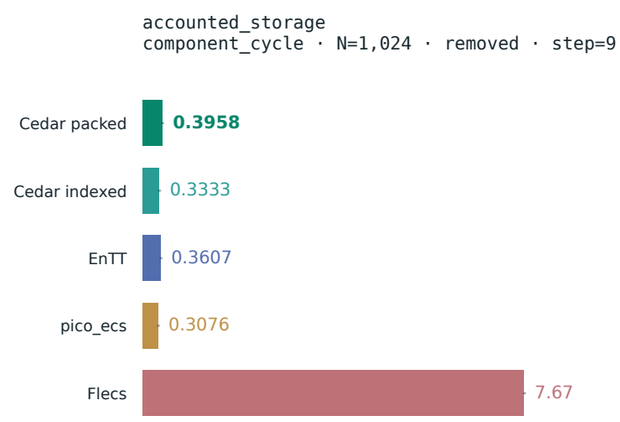


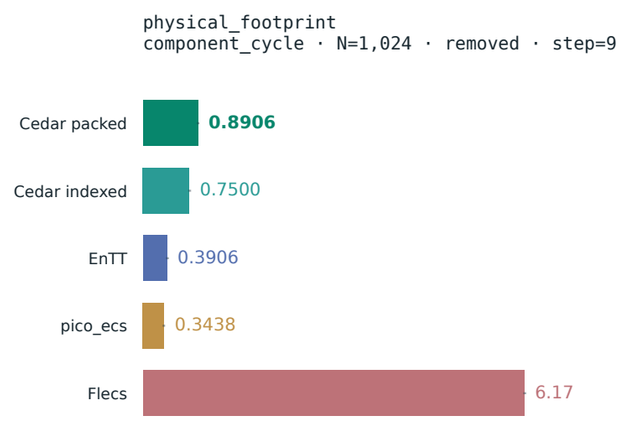


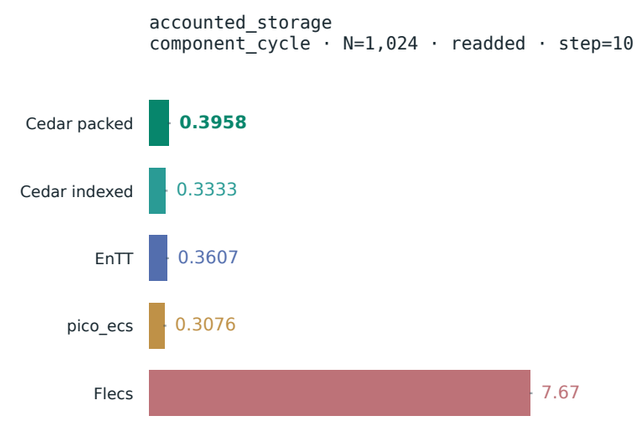
Memory comparison

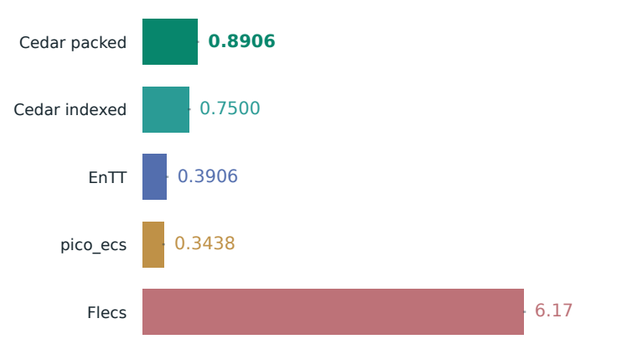
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

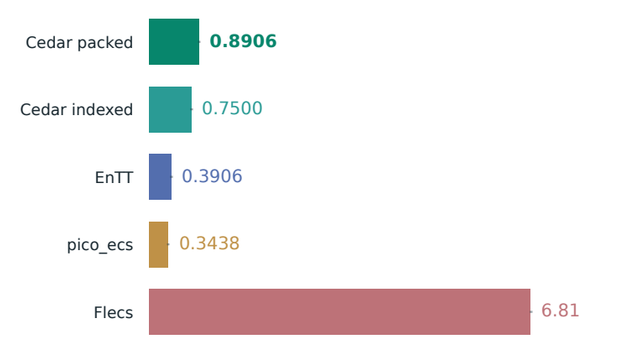


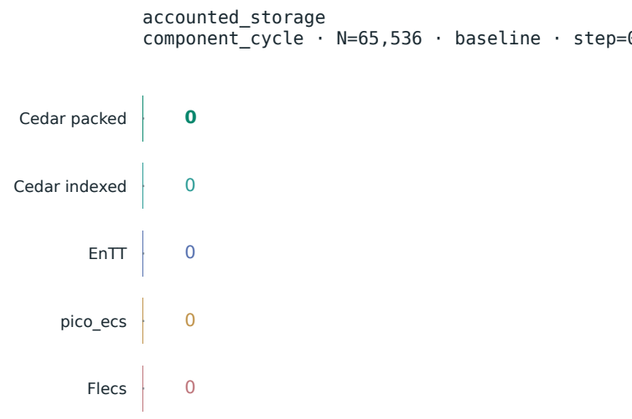


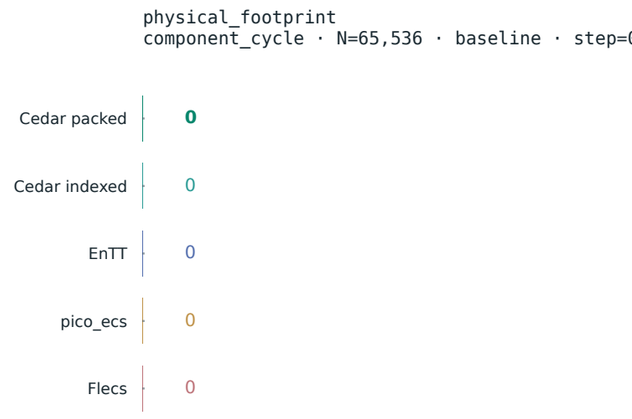


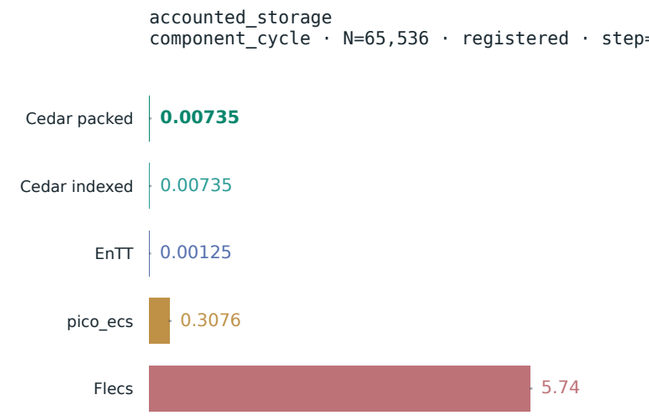


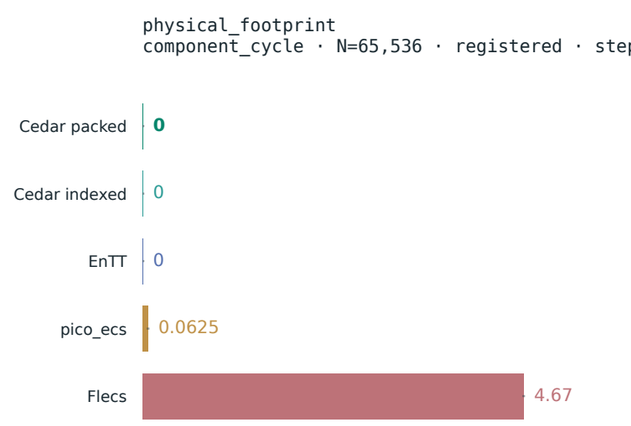


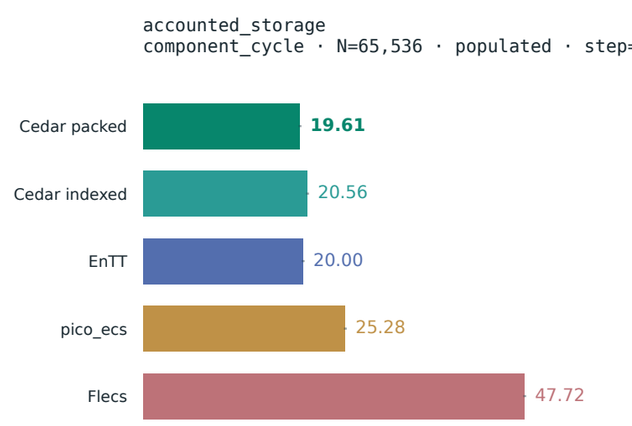


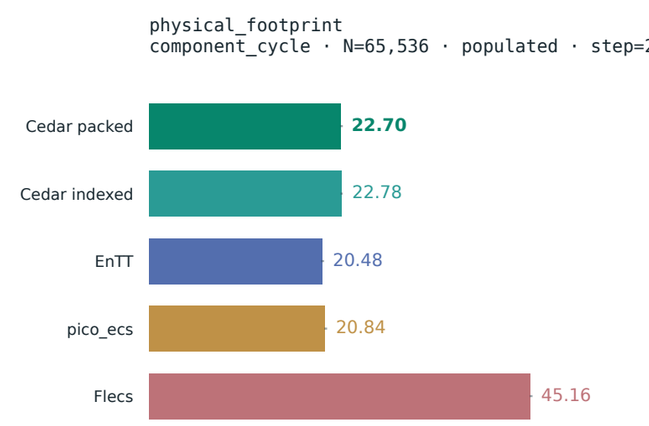








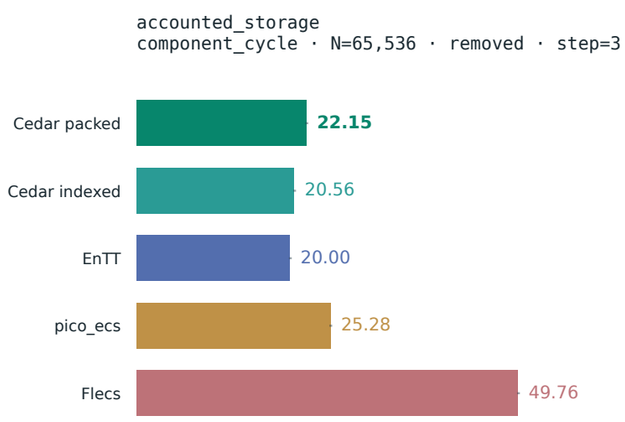


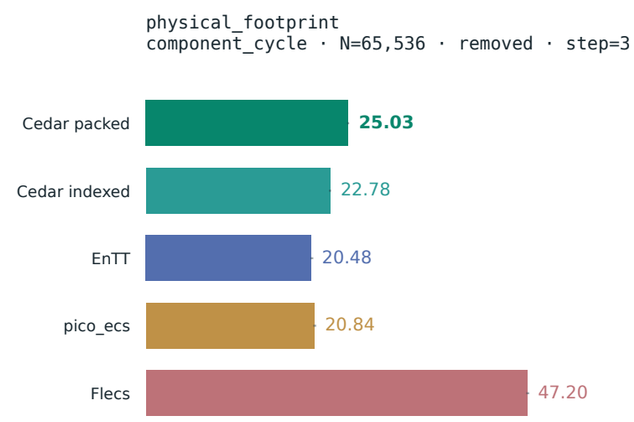


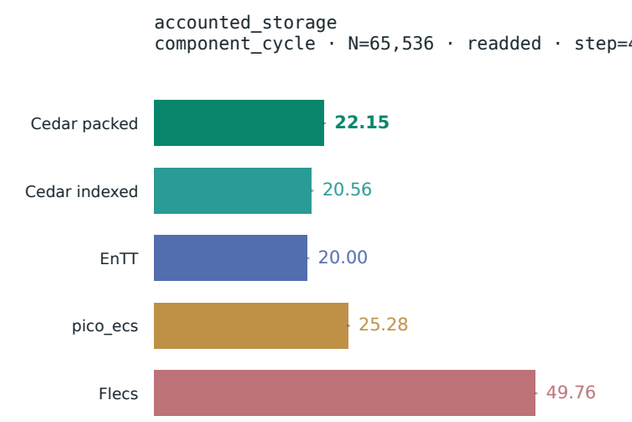
Memory comparison

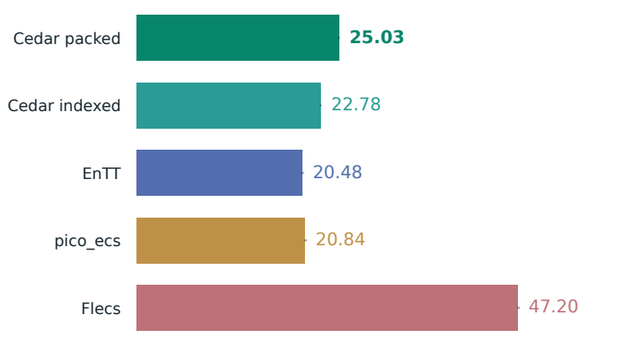
MiB ↓

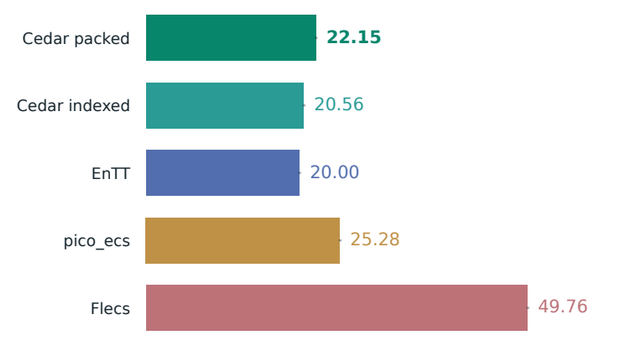
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

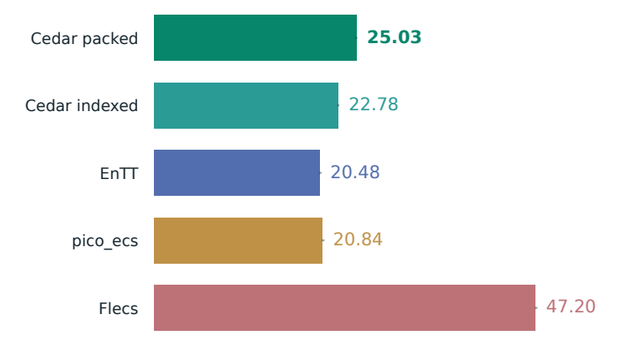


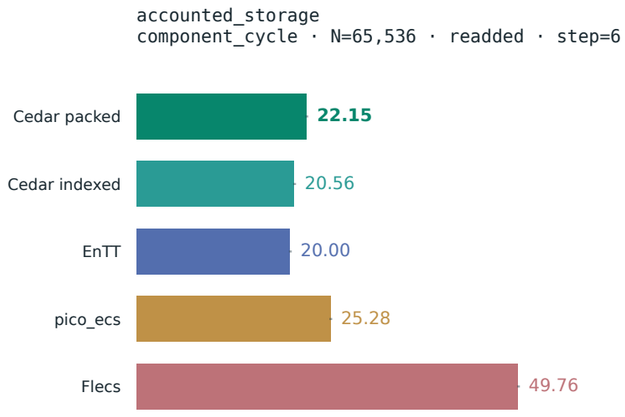


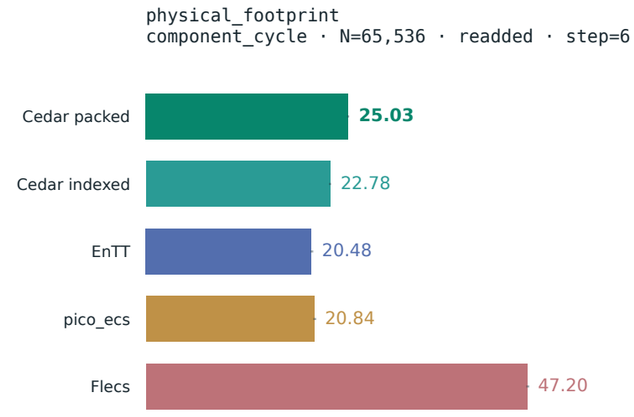


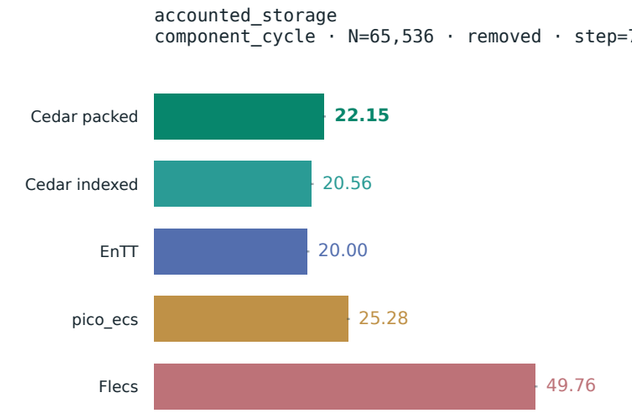


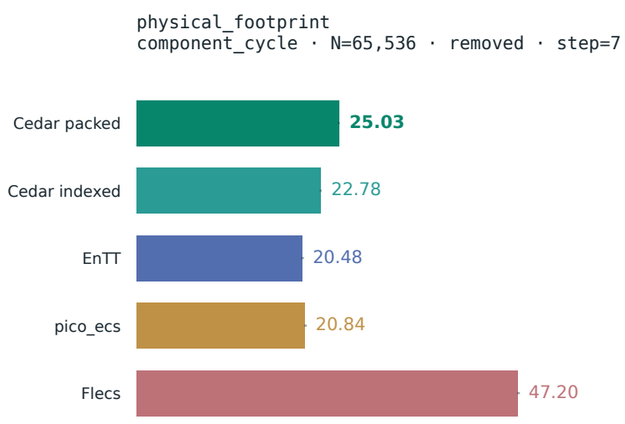


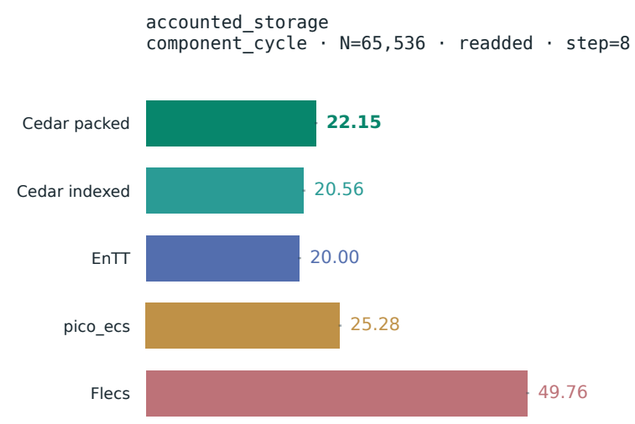


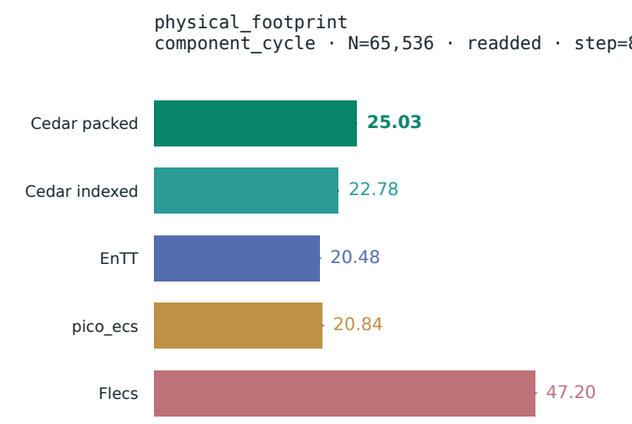








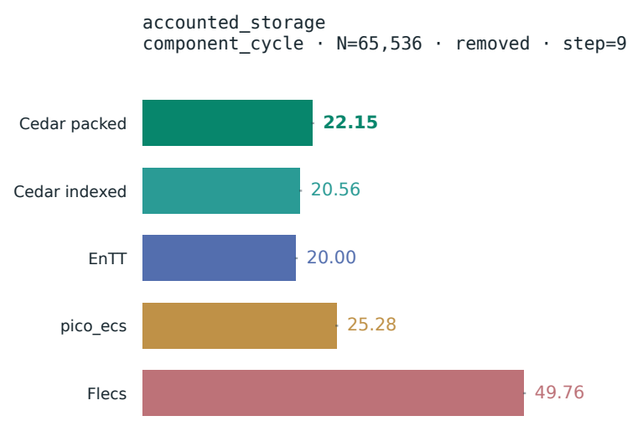


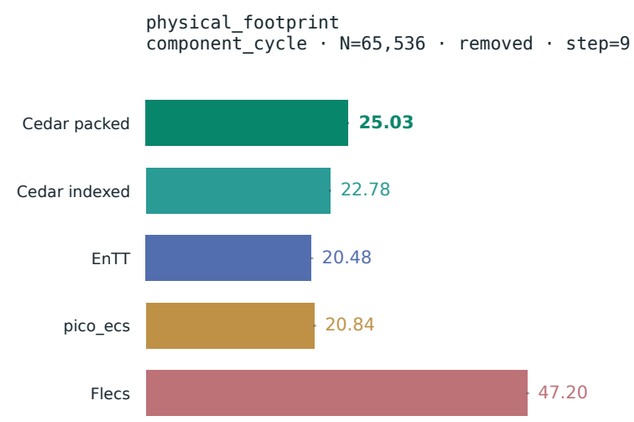


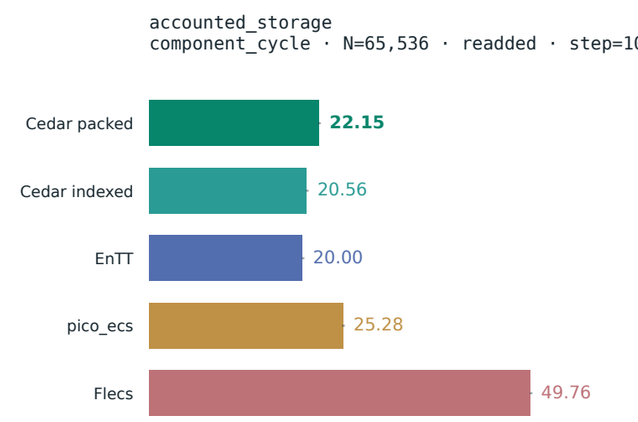
Memory comparison

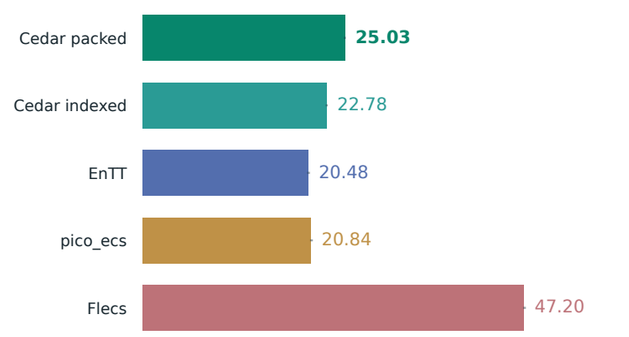
MiB ↓

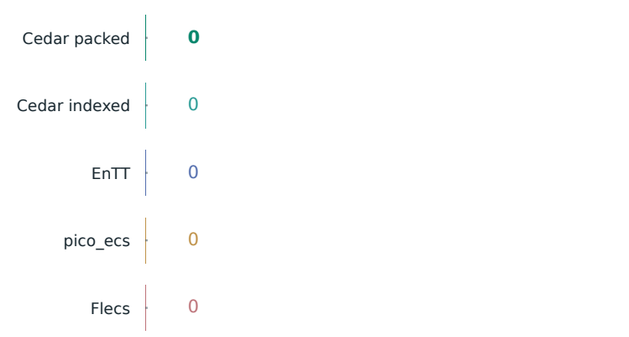
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

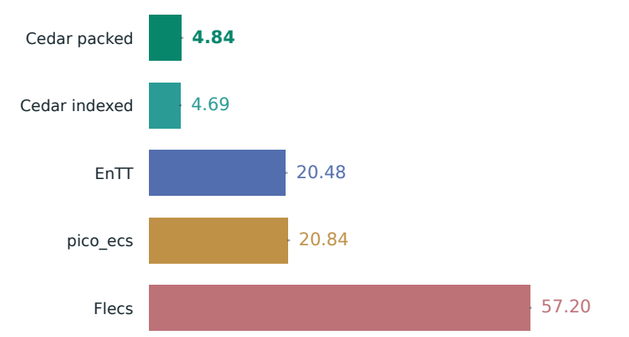


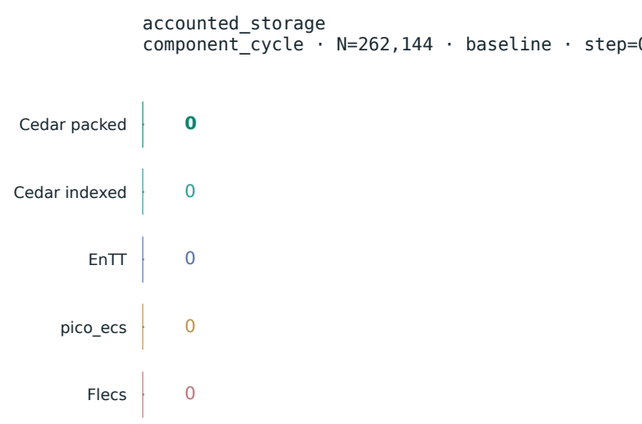


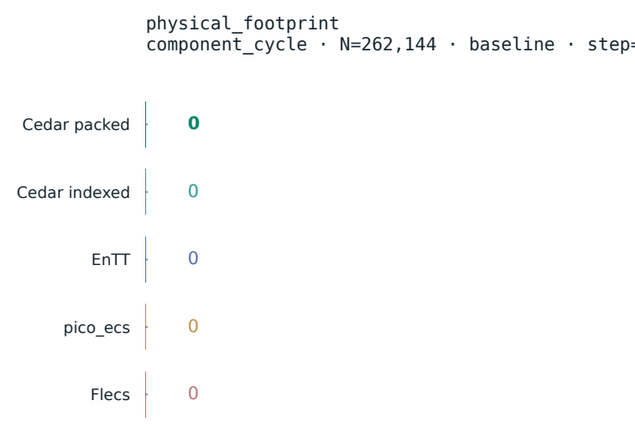


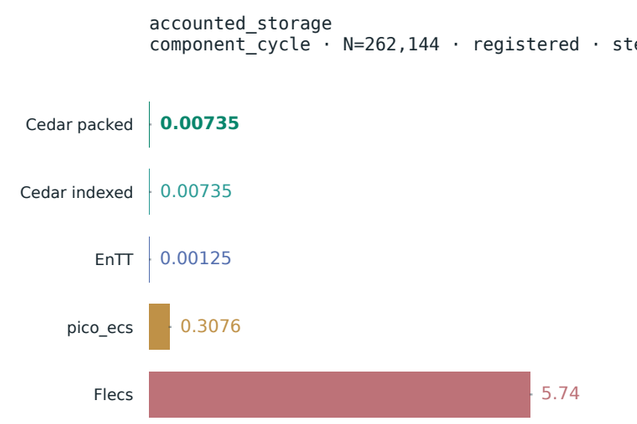


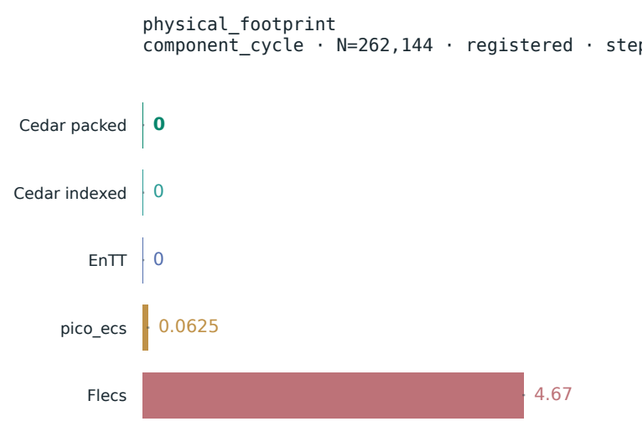


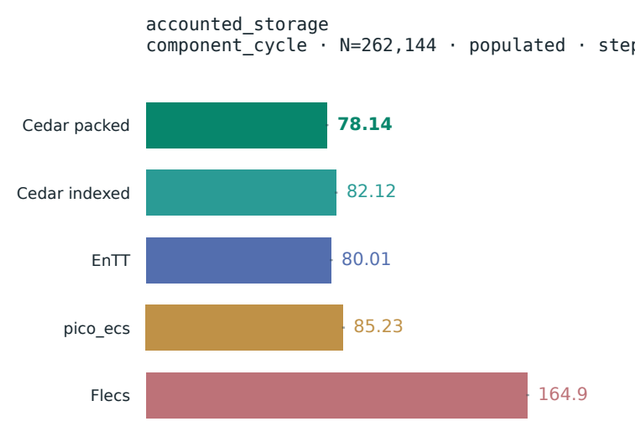


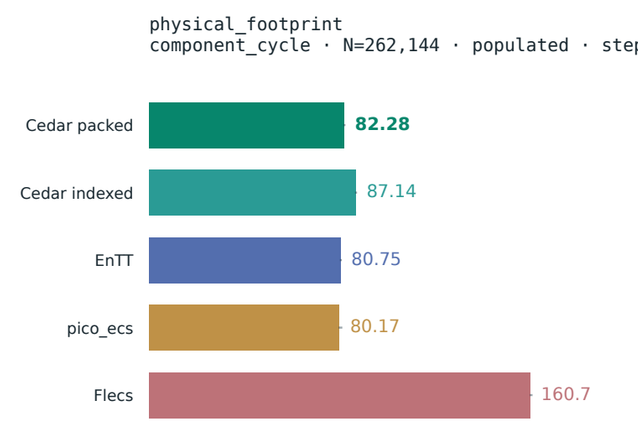








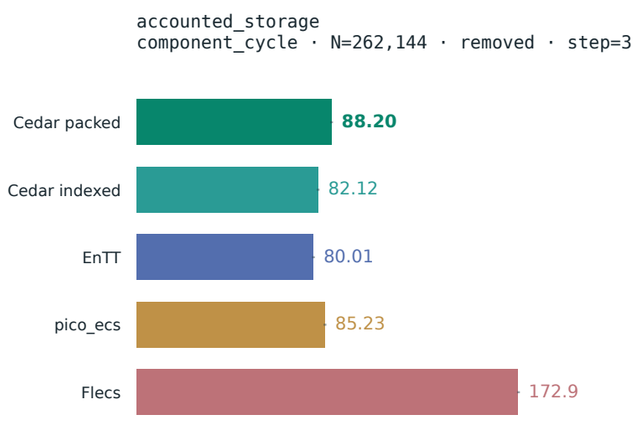


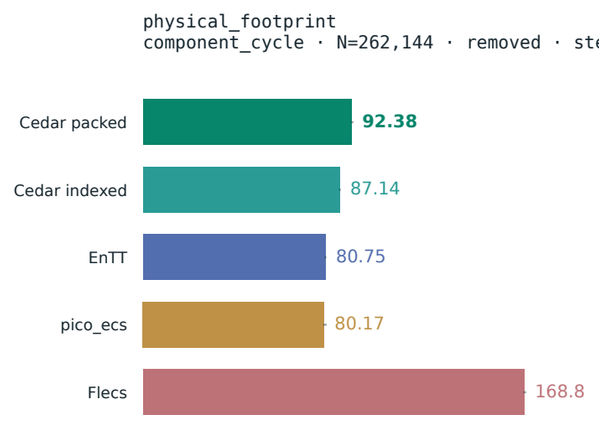


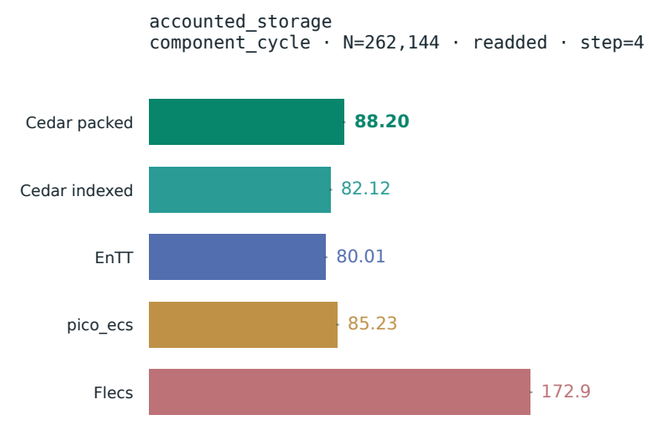
Memory comparison

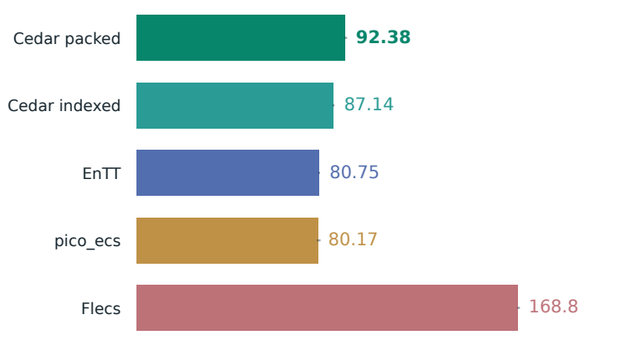
MiB ↓

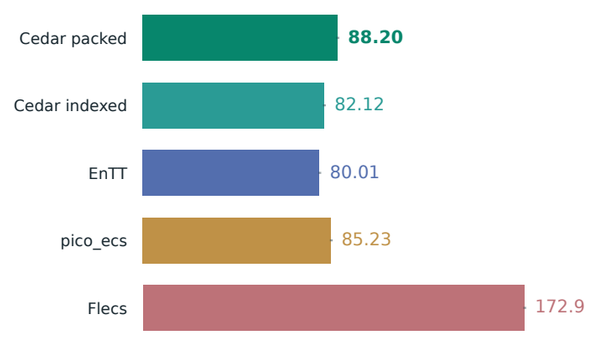
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

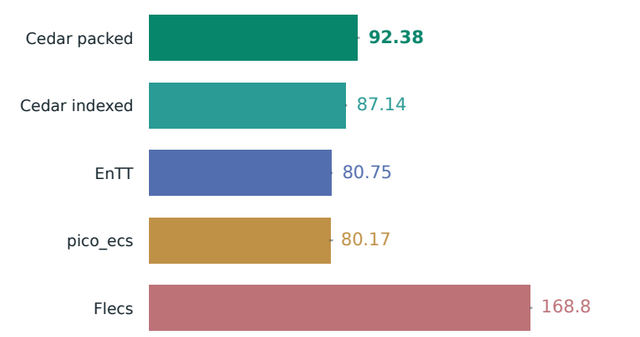


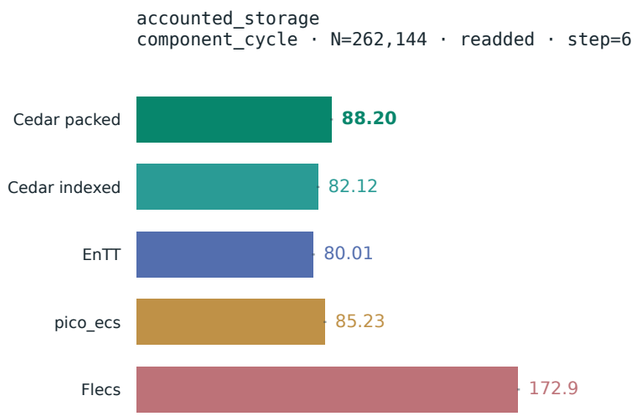


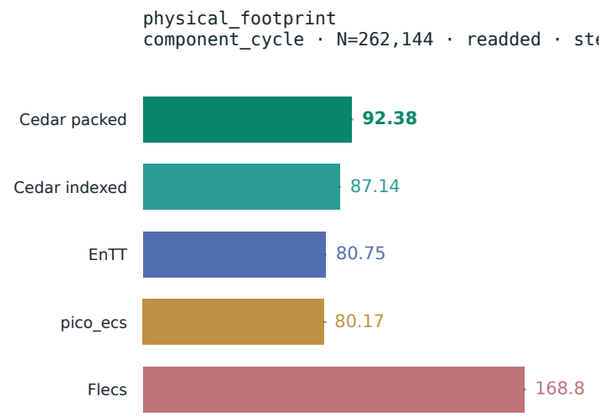


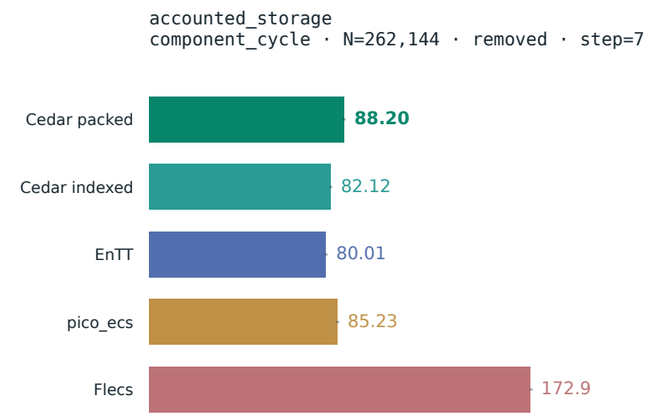


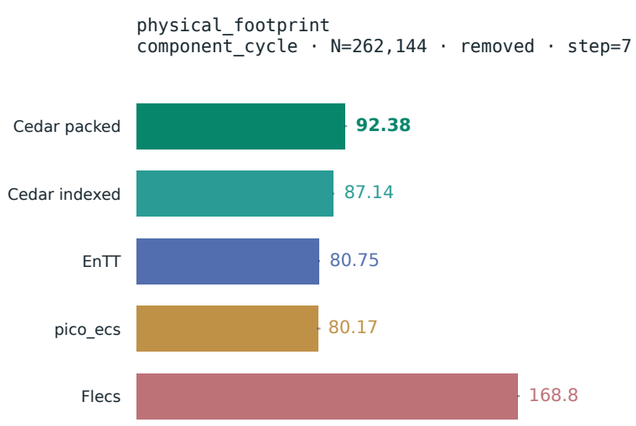


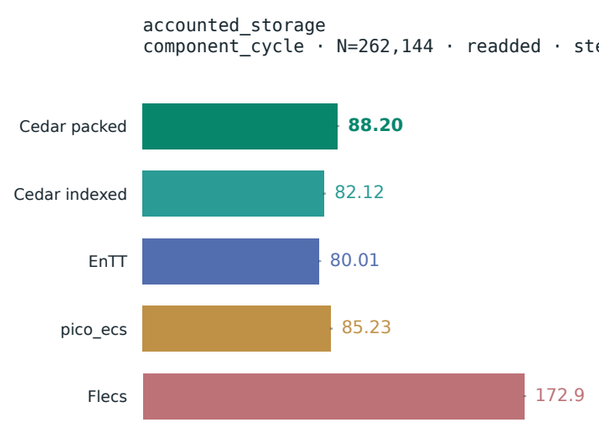


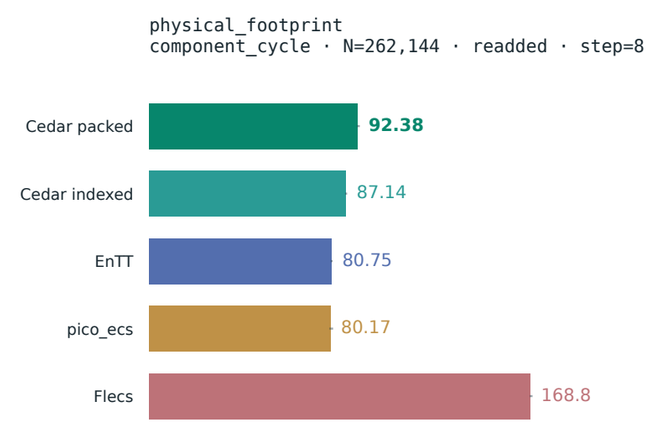










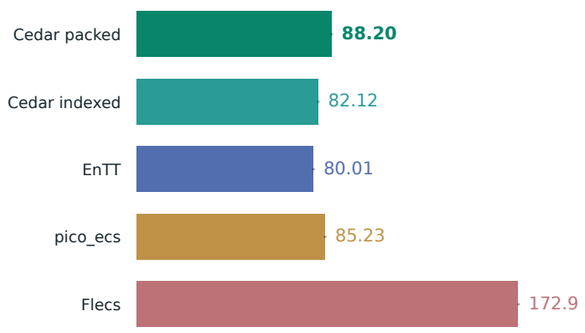


Memory comparison

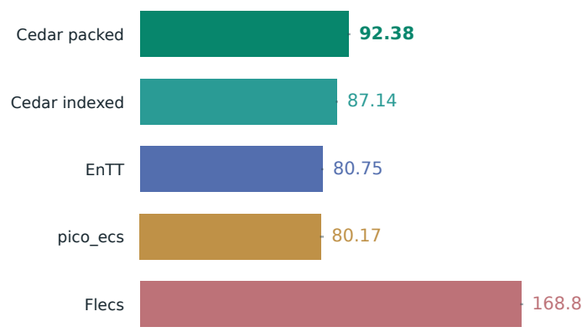
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

accounted_storage
component_cycle · N=262,144 · removed · step=9

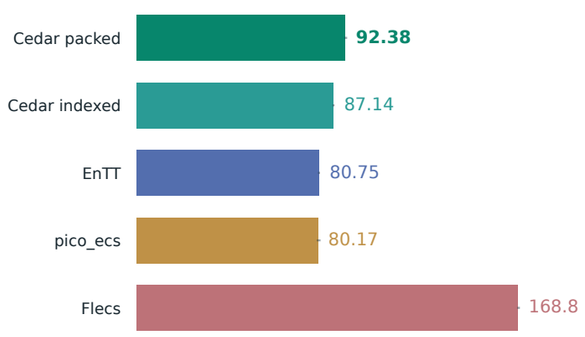


physical_footprint
component_cycle · N=262,144 · removed · step=9

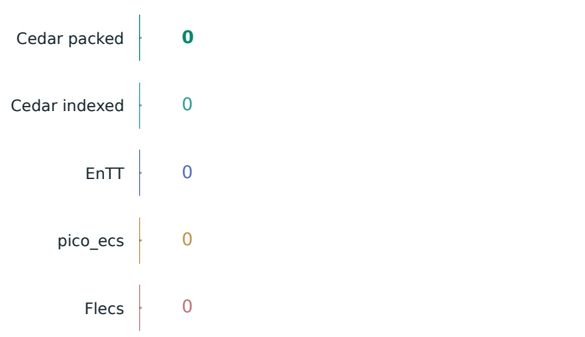


accounted_storage
component_cycle · N=262,144 · readded · step=10

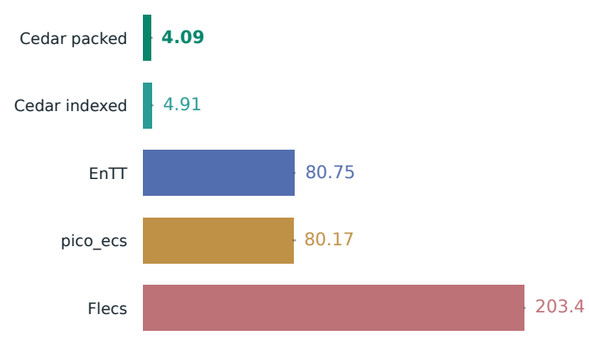
physical_footprint
component_cycle · N=262,144 · readded · step=10



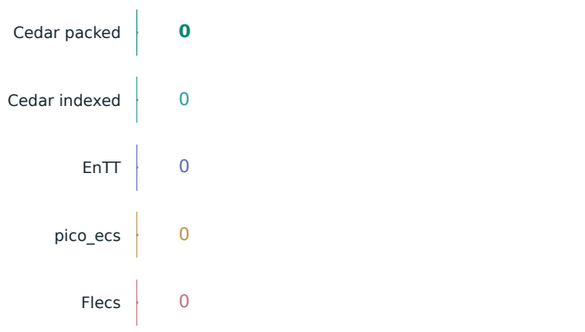
accounted_storage
component_cycle · N=262,144 · destroyed · step=11



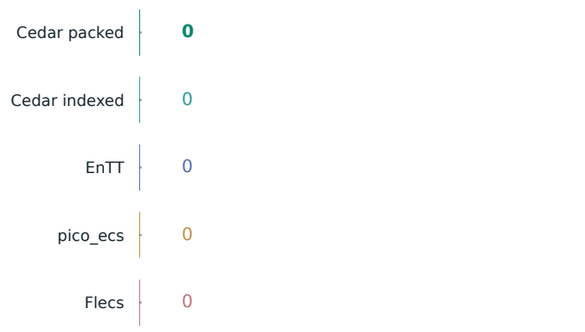
physical_footprint
component_cycle · N=262,144 · destroyed · step=11



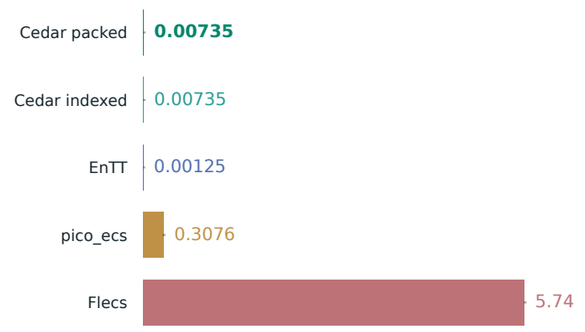
accounted_storage
delete_95 · N=1,024 · baseline · step=0



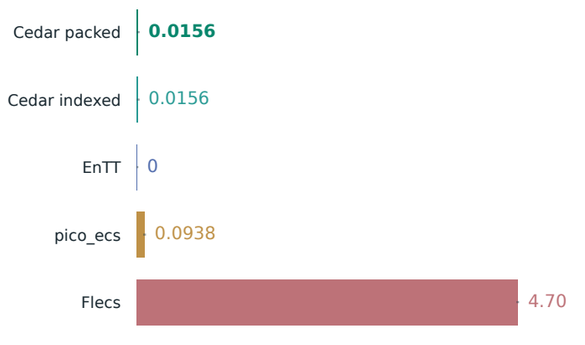
physical_footprint
delete_95 · N=1,024 · baseline · step=0



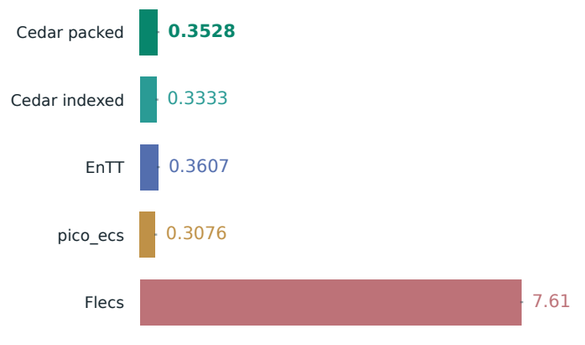
accounted_storage
delete_95 · N=1,024 · registered · step=1



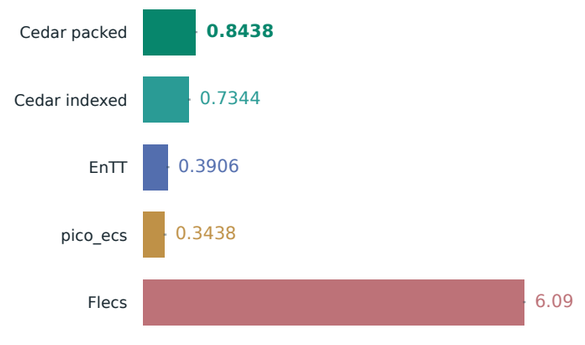
physical_footprint
delete_95 · N=1,024 · registered · step=1



accounted_storage
delete_95 · N=1,024 · populated · step=2



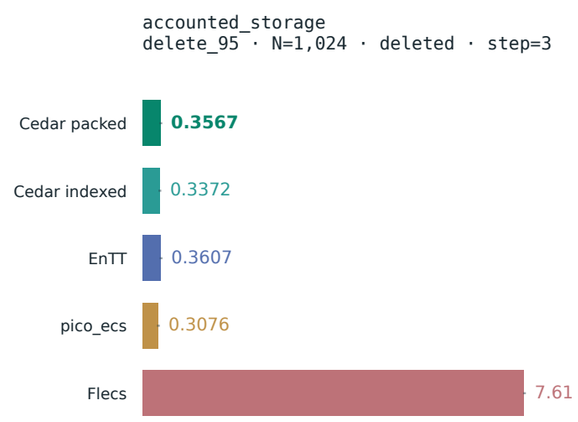
physical_footprint
delete_95 · N=1,024 · populated · step=2

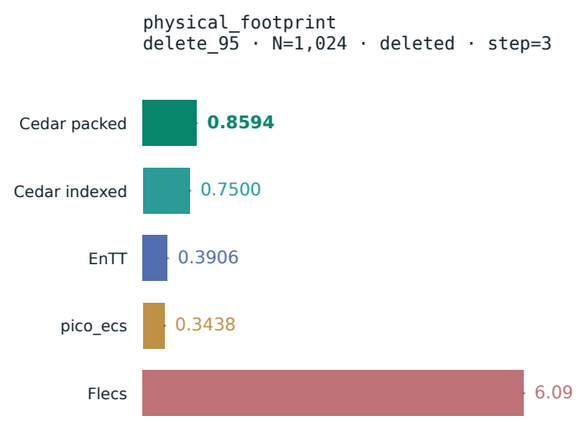


Memory comparison

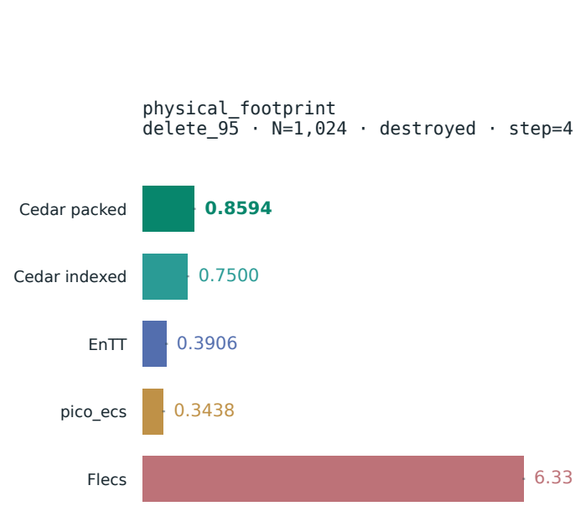
MiB ↓

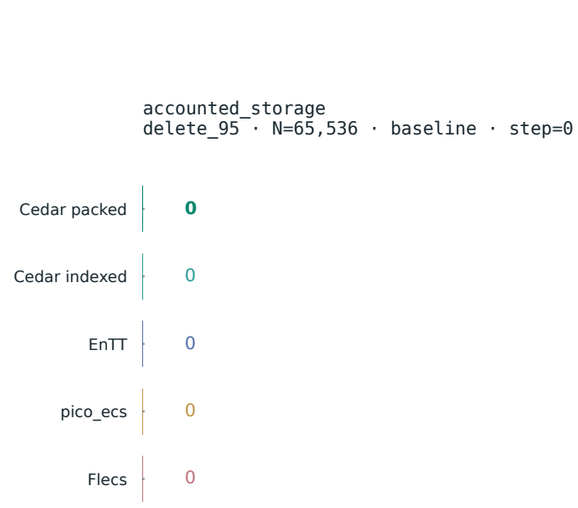
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs



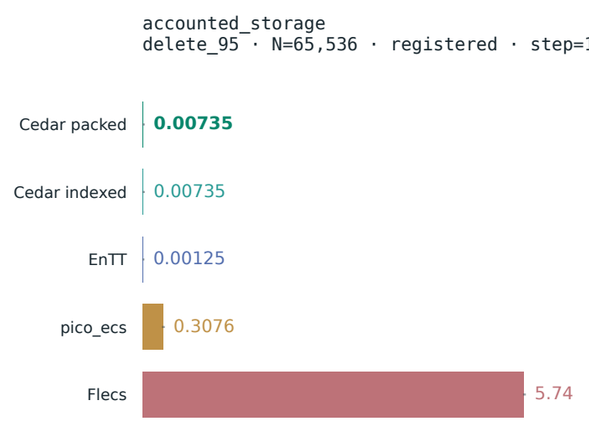


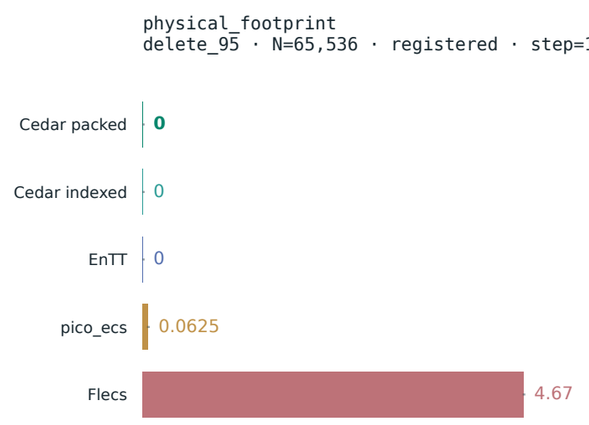


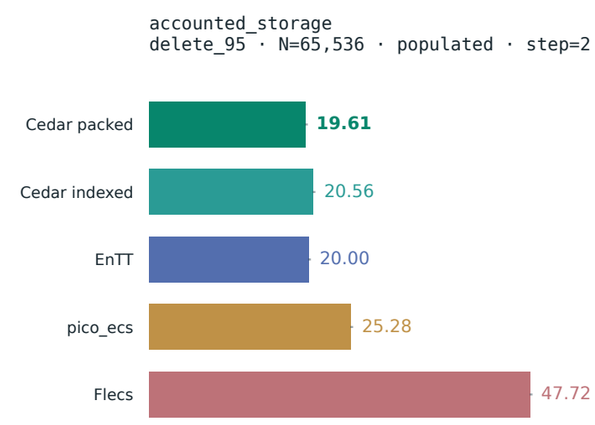


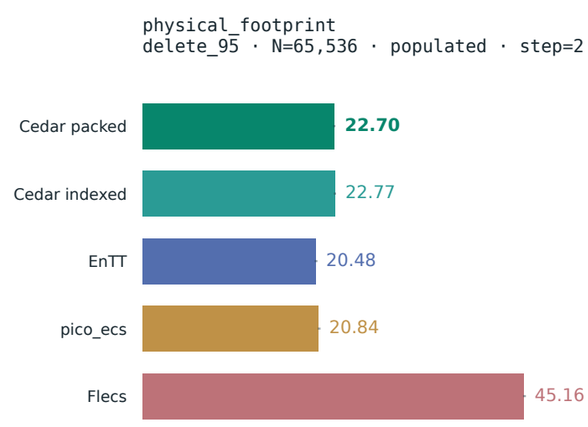


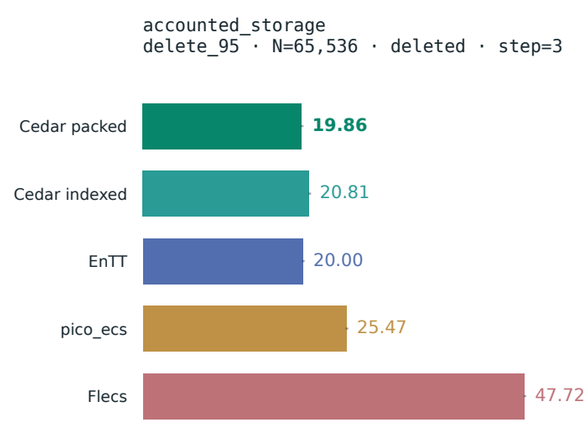


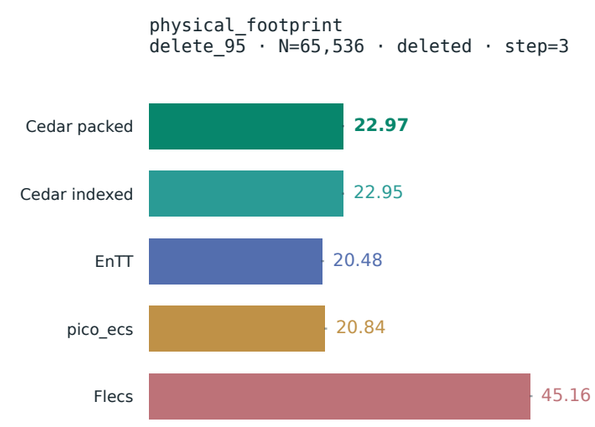










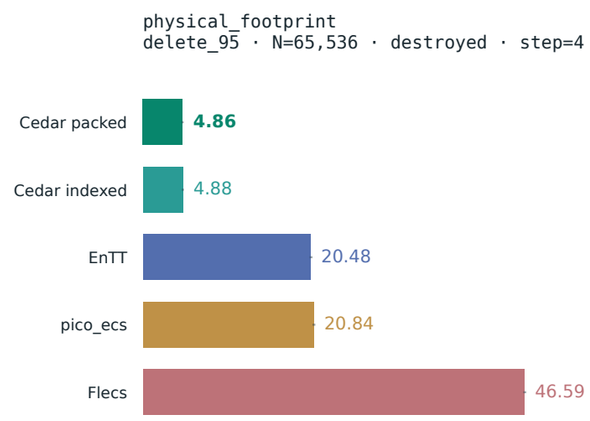


Memory comparison

MiB ↓

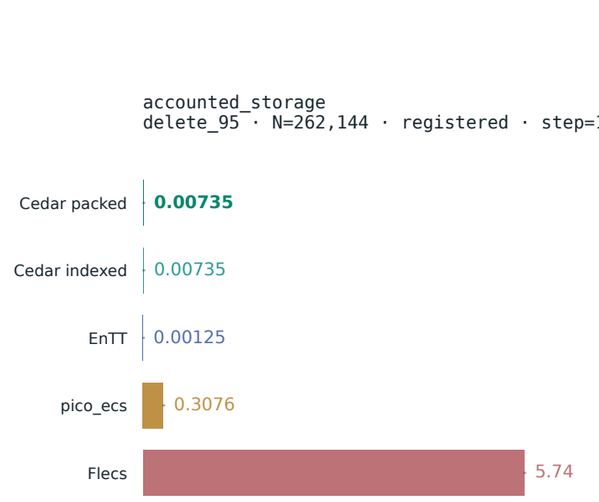
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

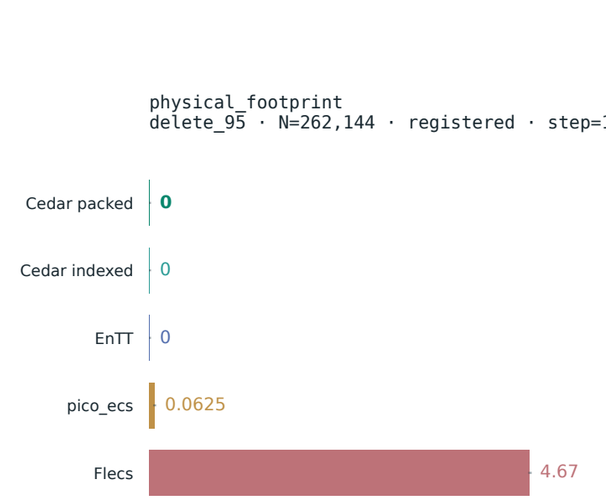


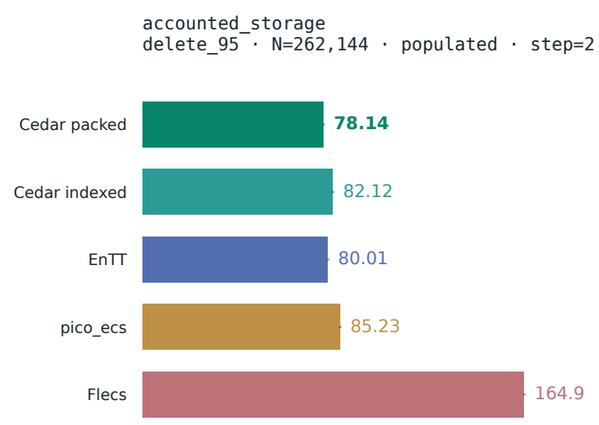


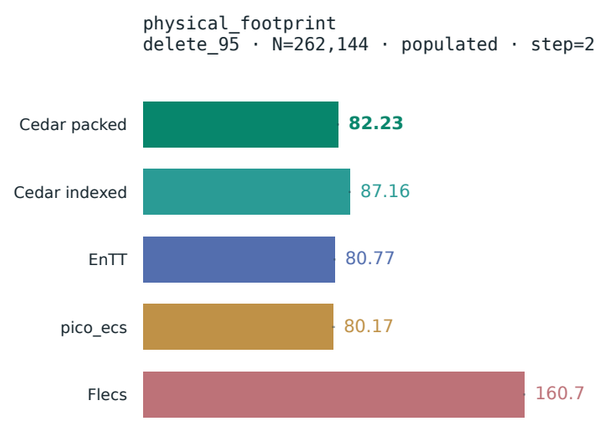


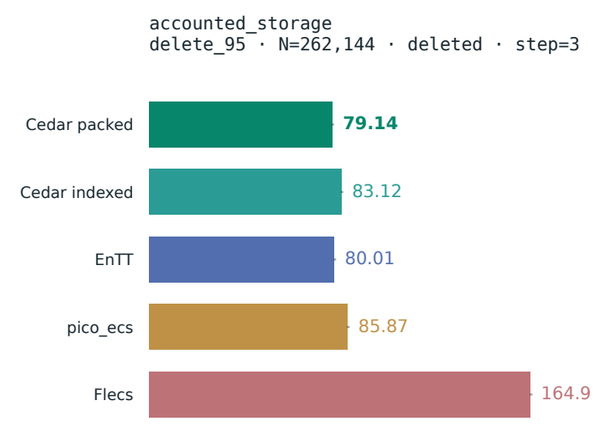


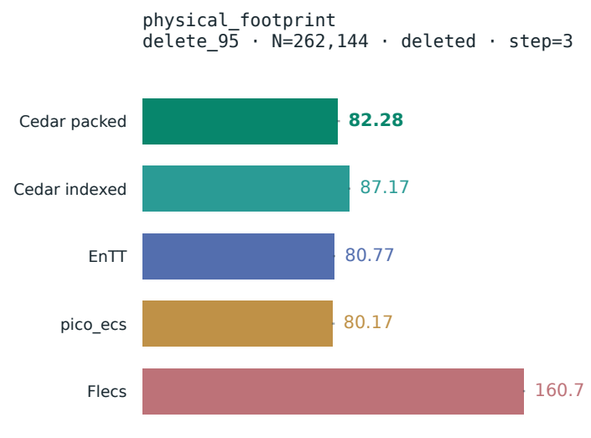




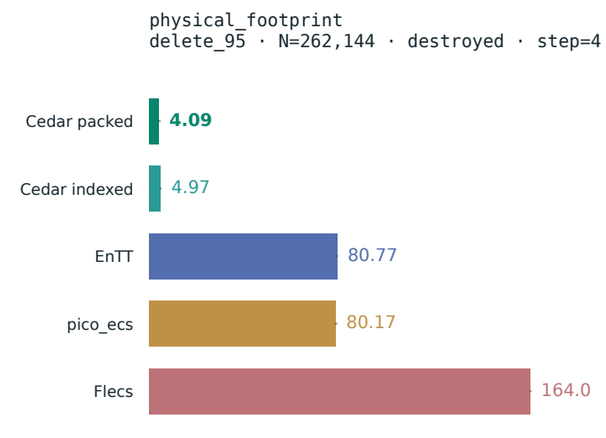












Memory comparison

MiB ↓

Cedar packed Cedar indexed EnTT pico_ecs Flecs

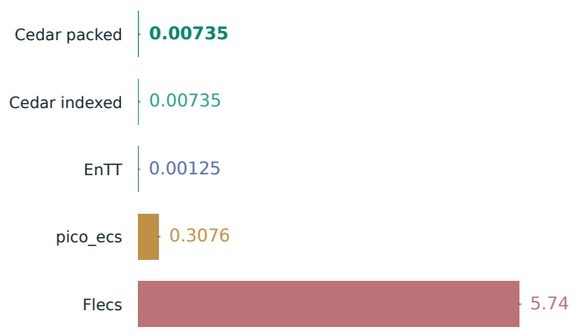
accounted_storage
delete_99 · N=1,024 · baseline · step=0



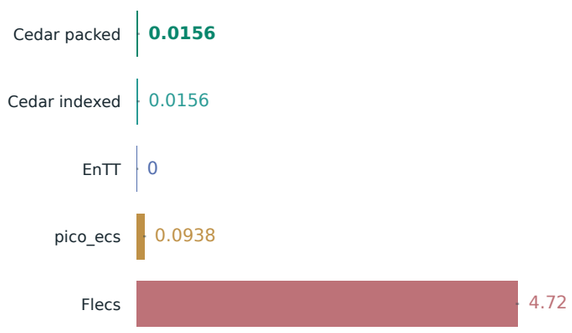
physical_footprint
delete_99 · N=1,024 · baseline · step=0



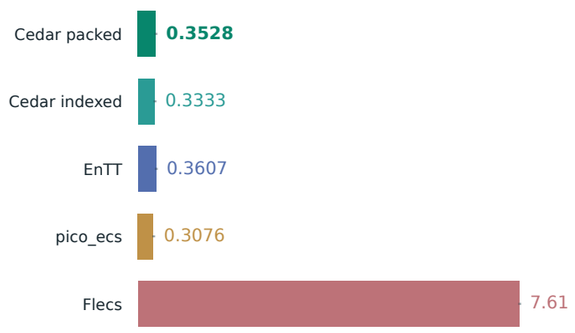
accounted_storage
delete_99 · N=1,024 · registered · step=1



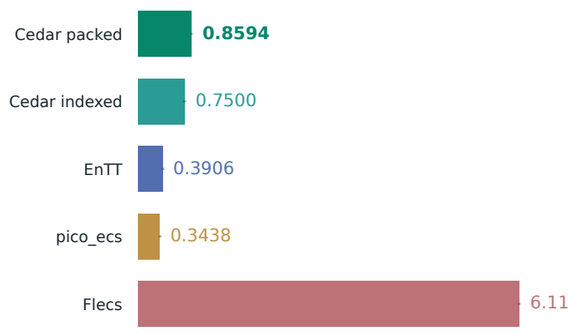
physical_footprint
delete_99 · N=1,024 · registered · step=1



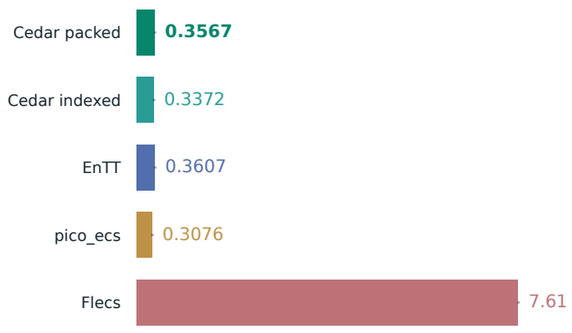
accounted_storage
delete_99 · N=1,024 · populated · step=2



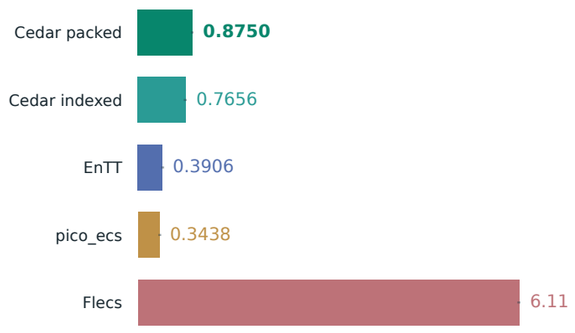
physical_footprint
delete_99 · N=1,024 · populated · step=2



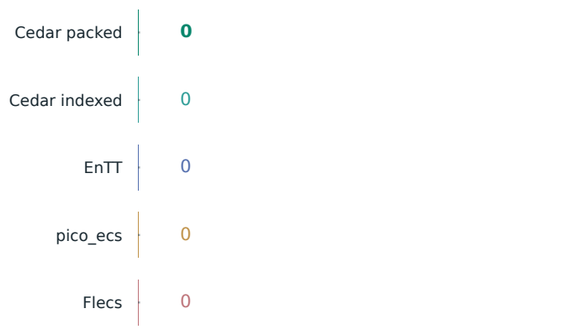
accounted_storage
delete_99 · N=1,024 · deleted · step=3



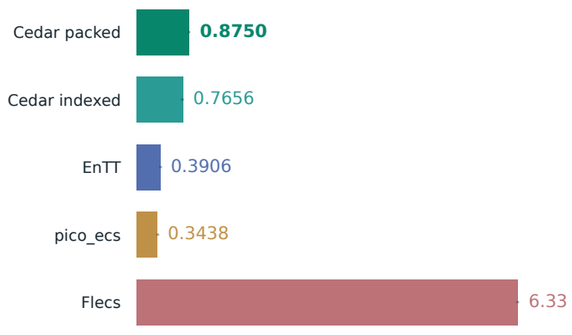
physical_footprint
delete_99 · N=1,024 · deleted · step=3



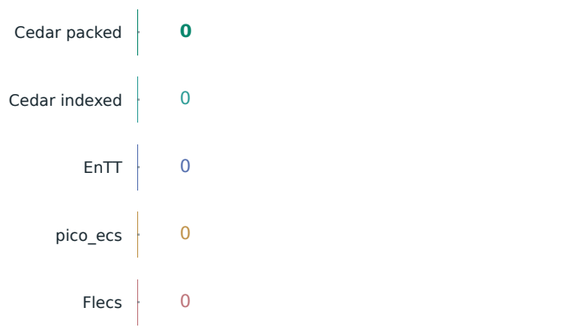
accounted_storage
delete_99 · N=1,024 · destroyed · step=4



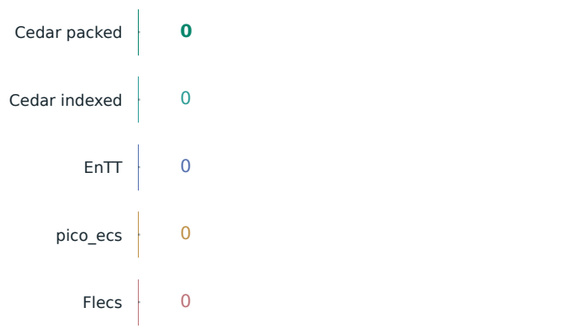
physical_footprint
delete_99 · N=1,024 · destroyed · step=4



accounted_storage
delete_99 · N=65,536 · baseline · step=0



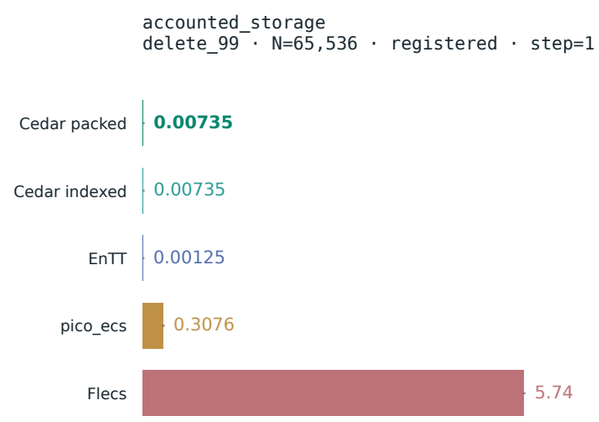
physical_footprint
delete_99 · N=65,536 · baseline · step=0

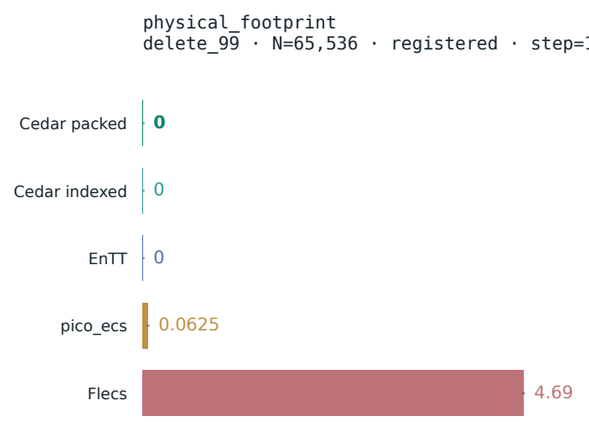


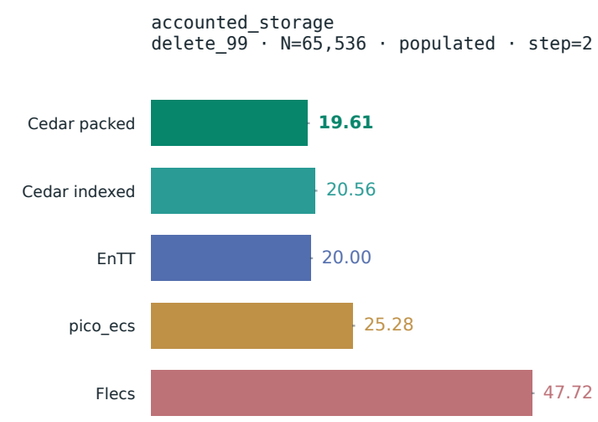
Memory comparison

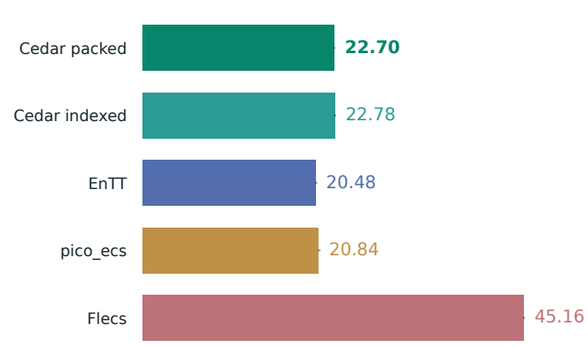
MiB ↓

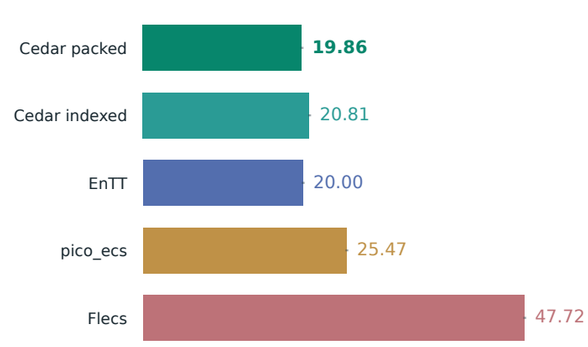
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

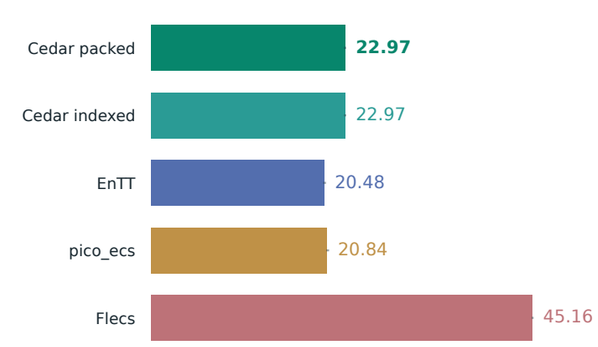


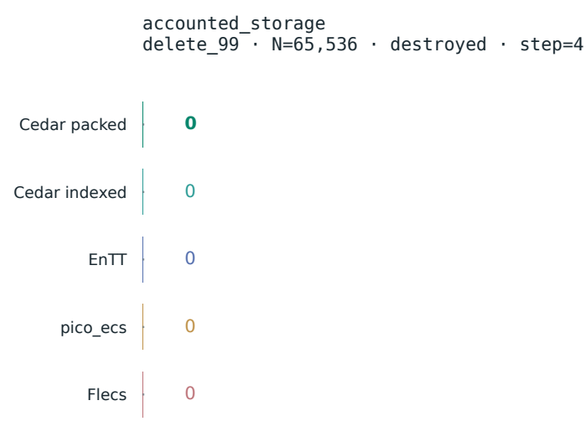


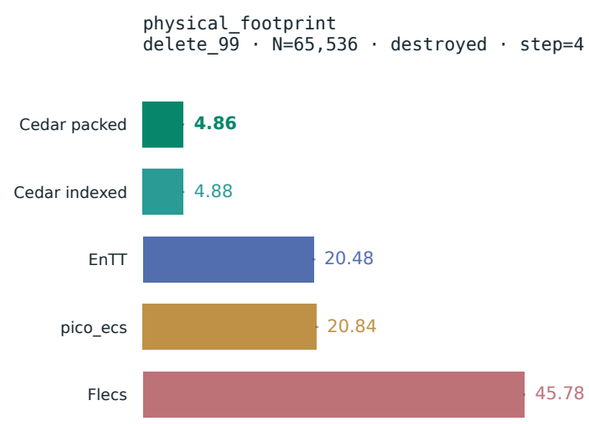






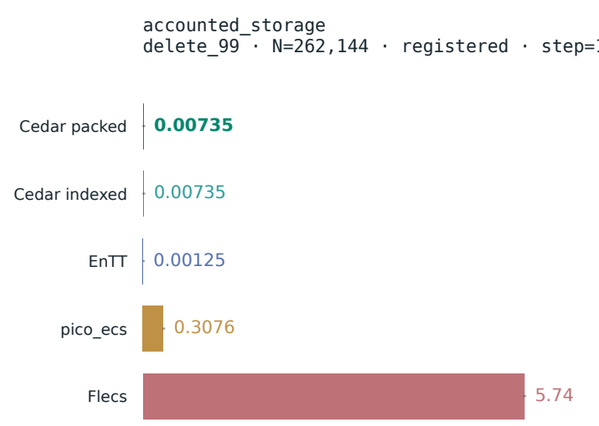










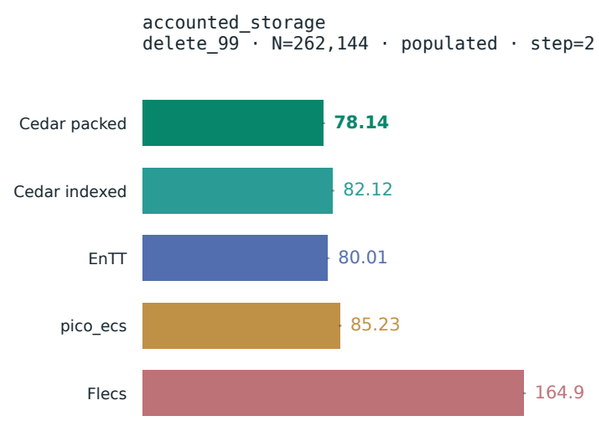


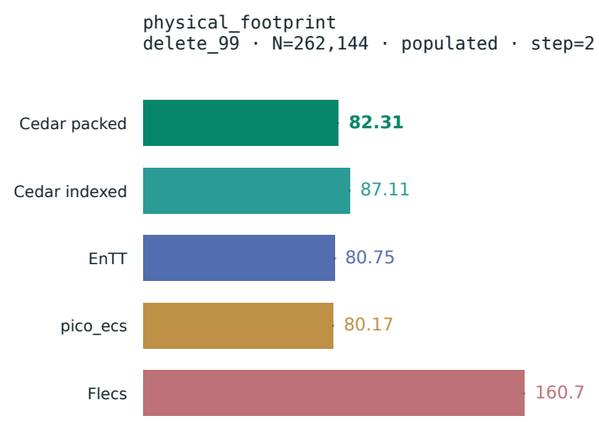


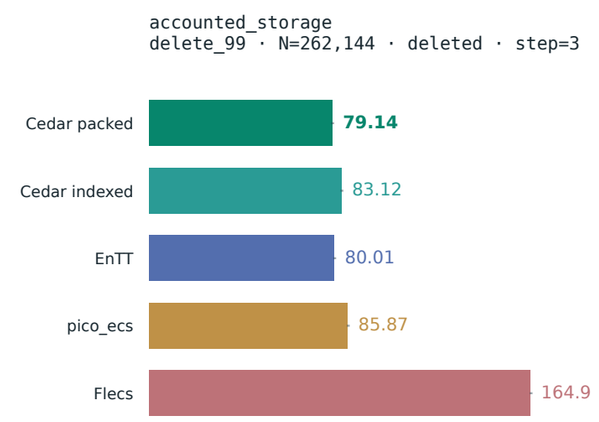
Memory comparison

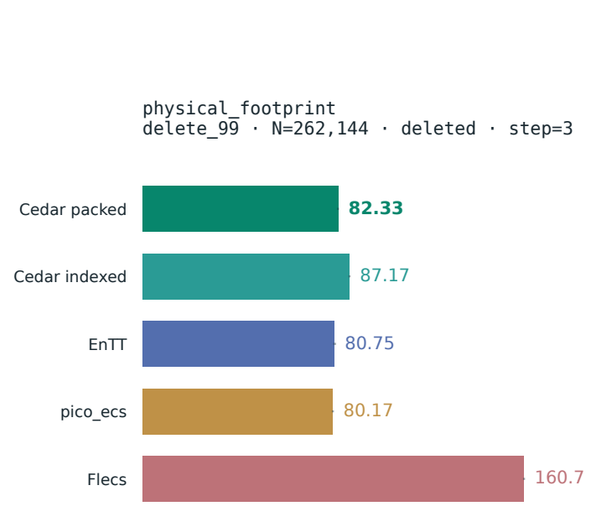
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

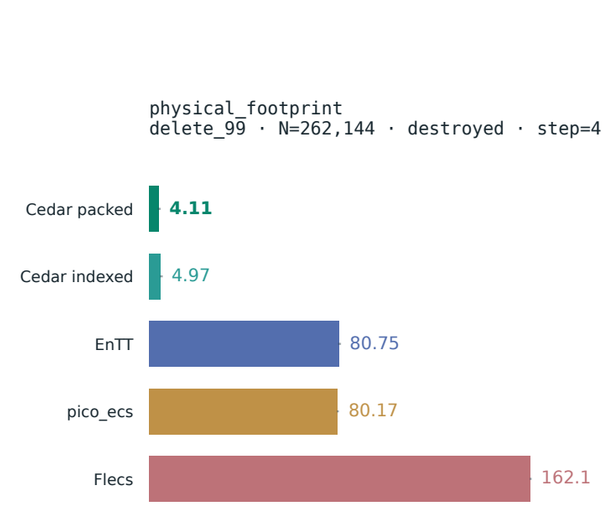






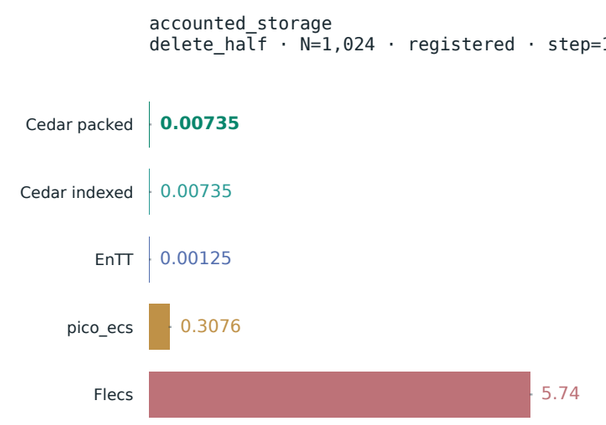


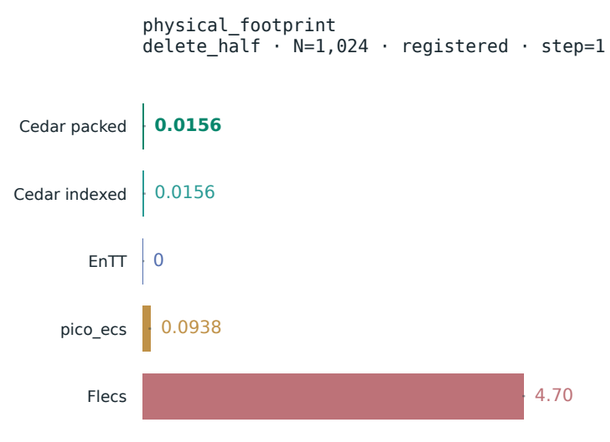


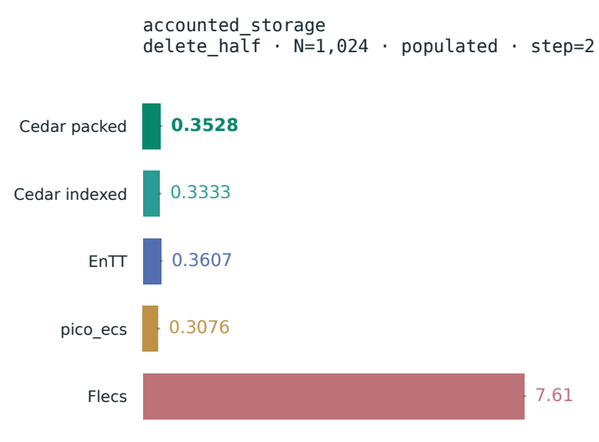


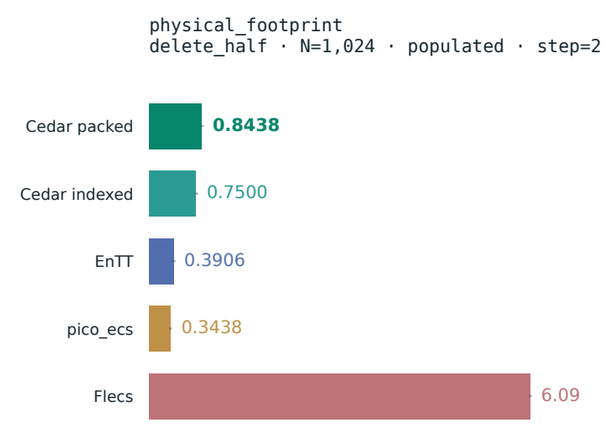








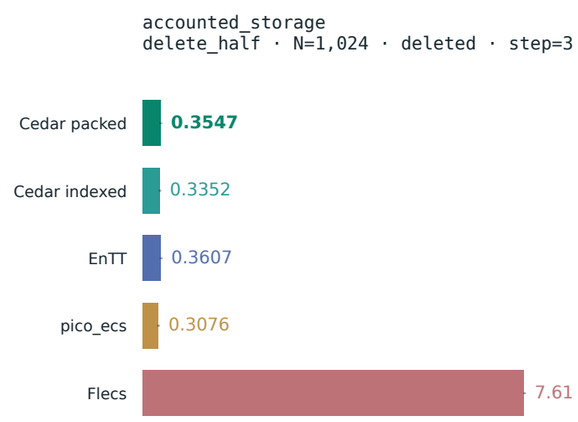


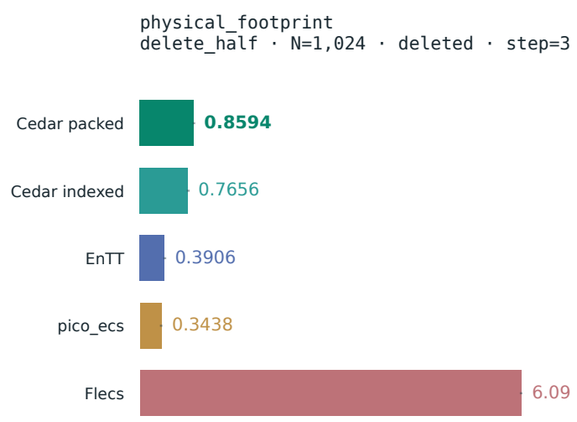


Memory comparison

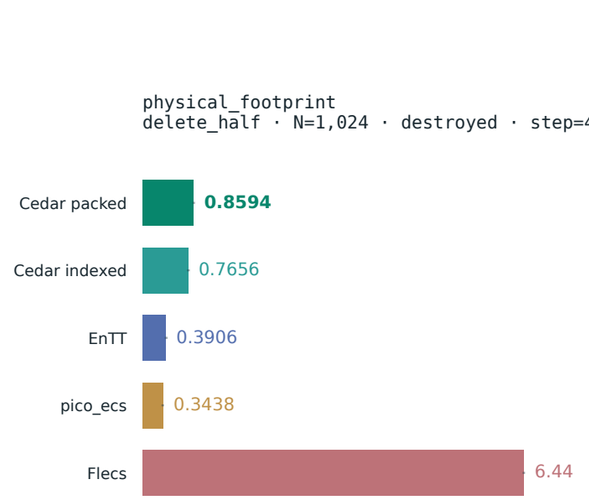
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

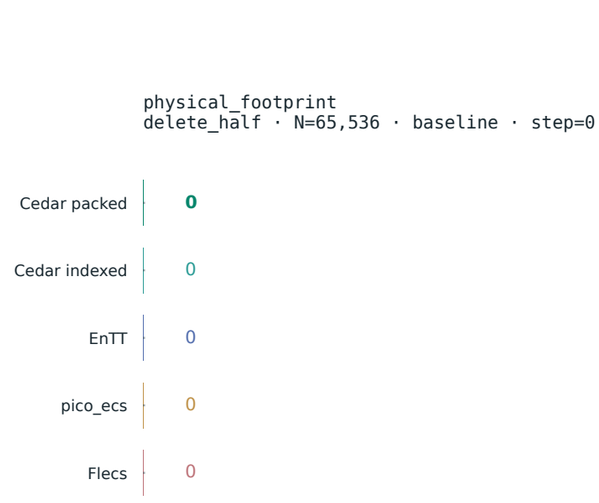


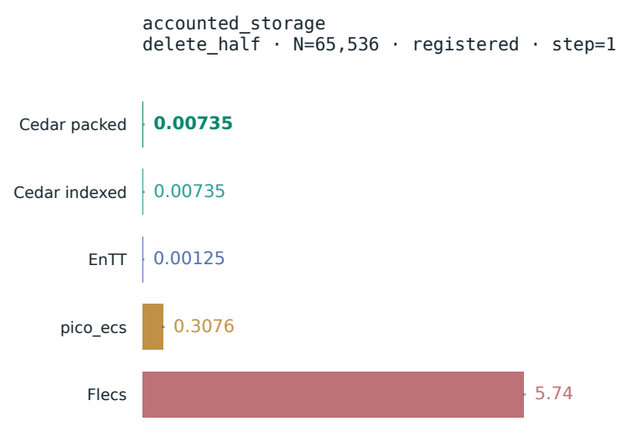


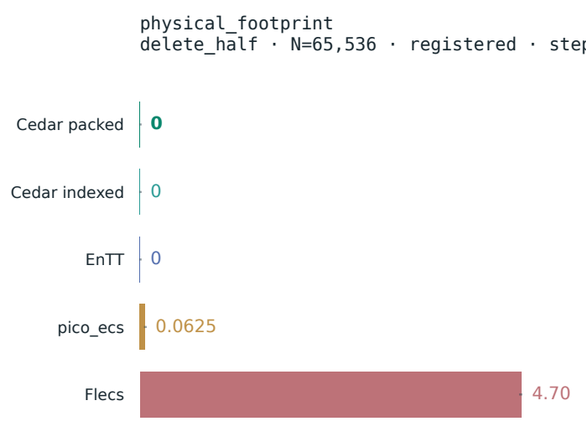


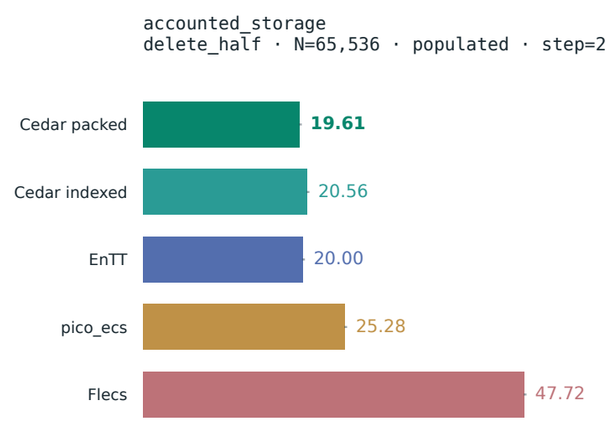


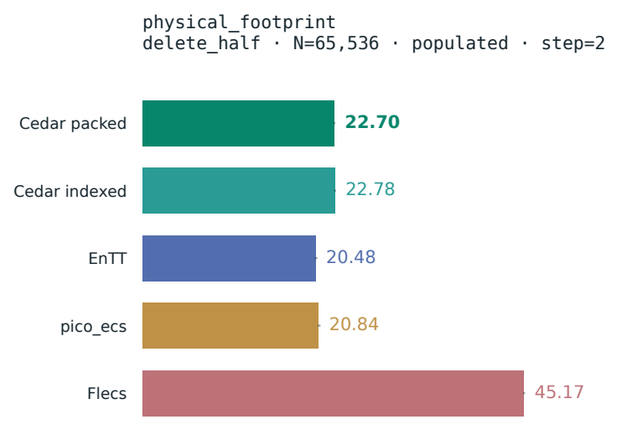


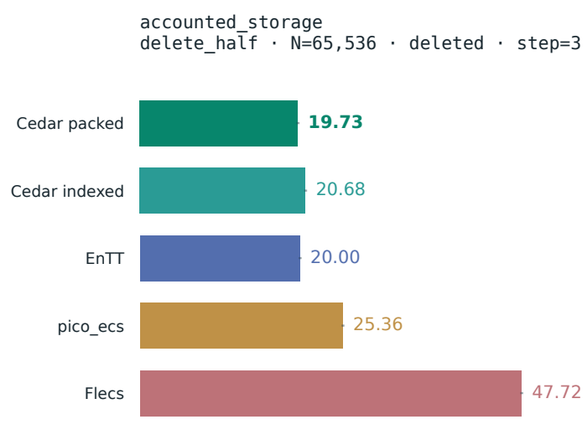


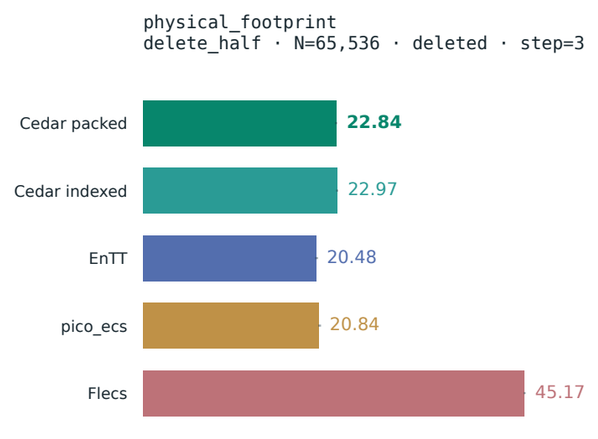








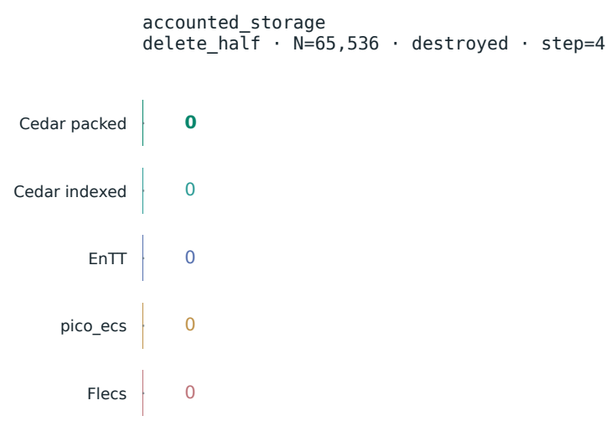


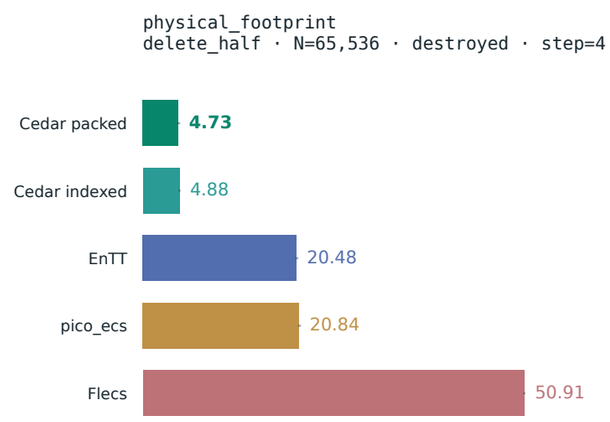


Memory comparison

MiB ↓

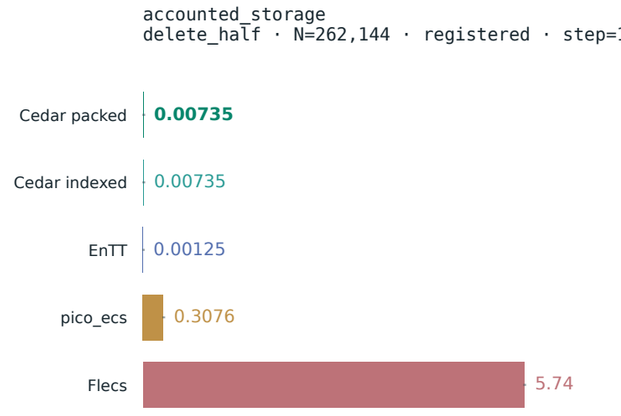
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

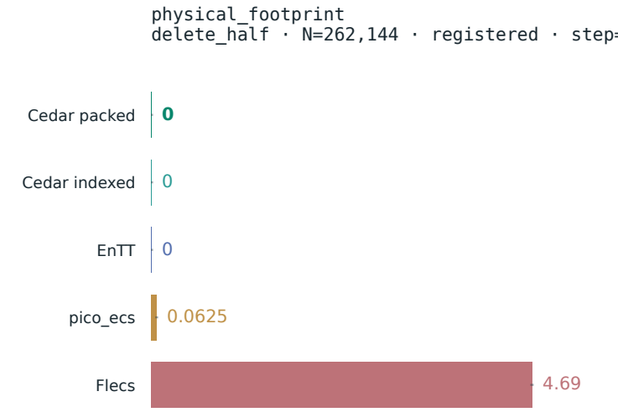


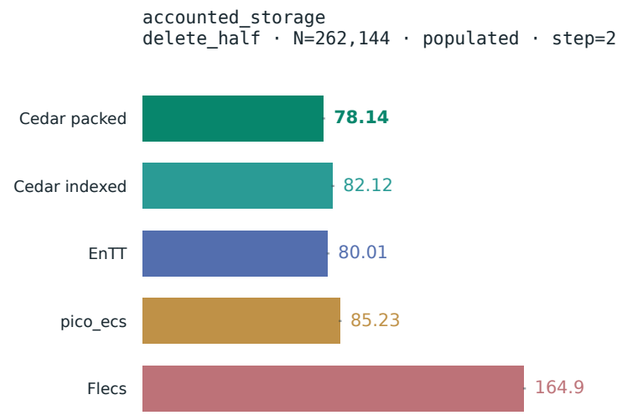


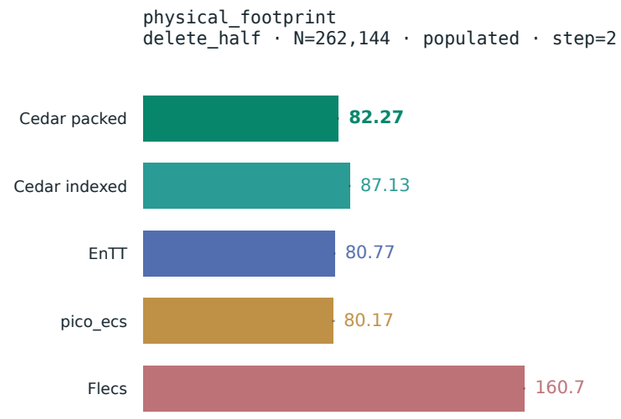


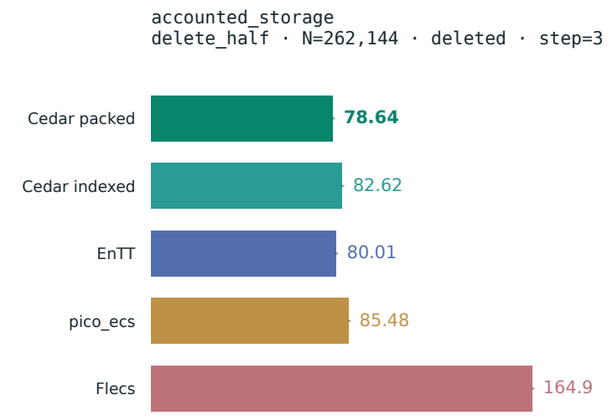


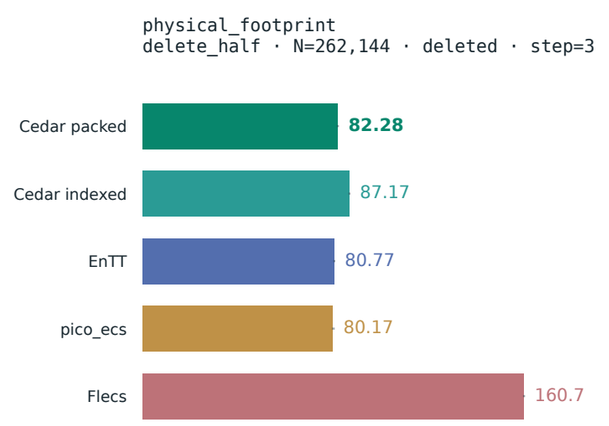




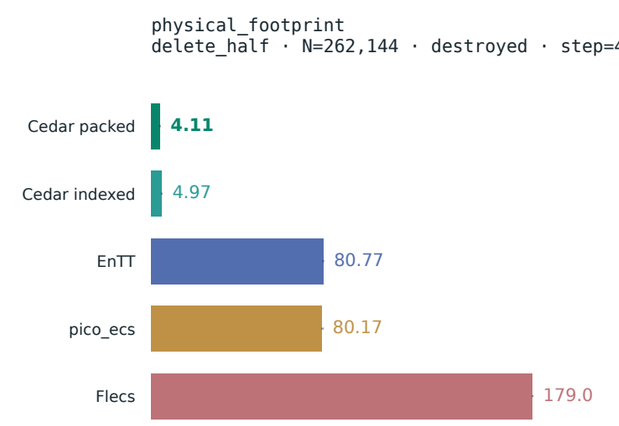










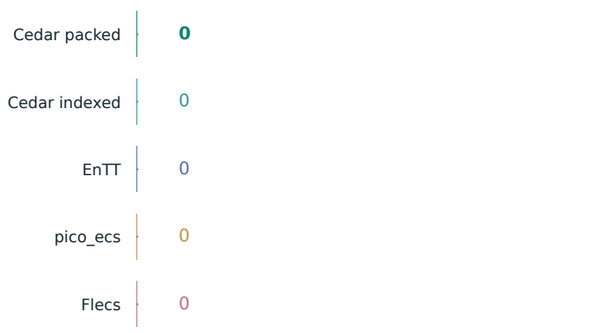


Memory comparison

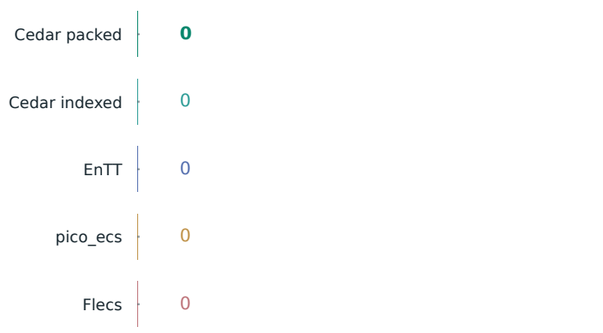
MiB ↓

Cedar packed Cedar indexed EnTT pico_ecs Flecs

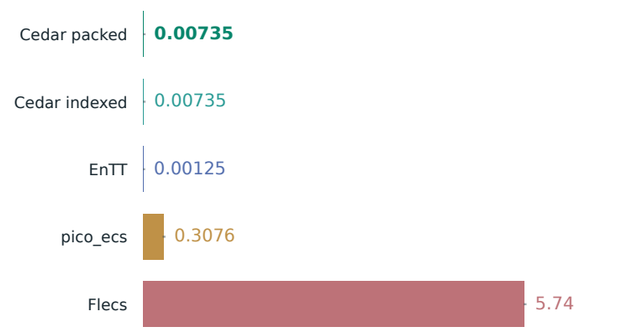
accounted_storage
dense_payload · N=1,024 · baseline · step=0



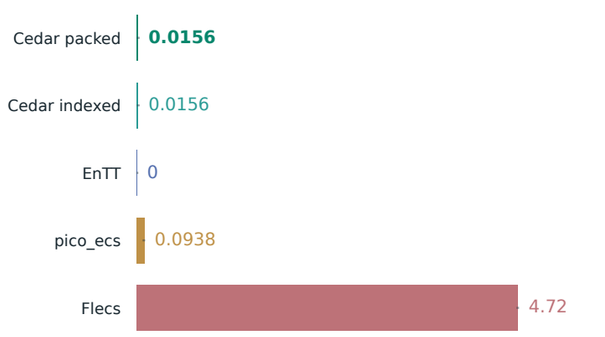
physical_footprint
dense_payload · N=1,024 · baseline · step=0



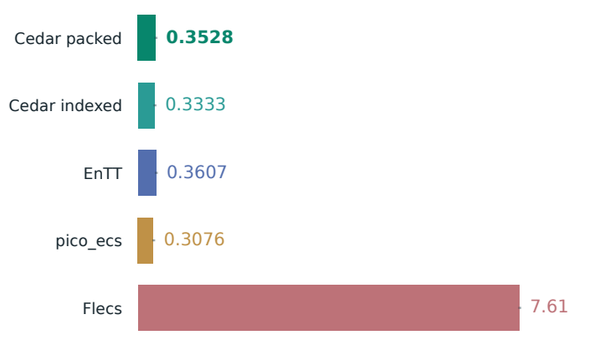
accounted_storage
dense_payload · N=1,024 · registered · step=1



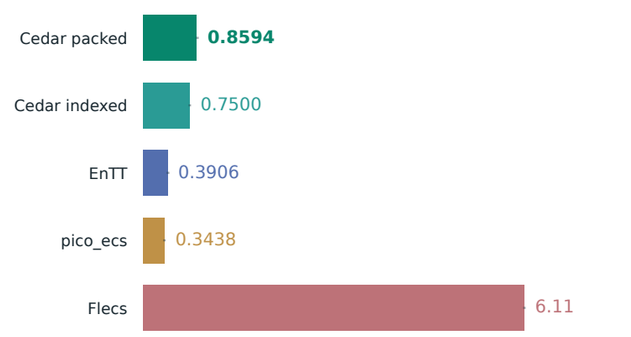
physical_footprint
dense_payload · N=1,024 · registered · step=1



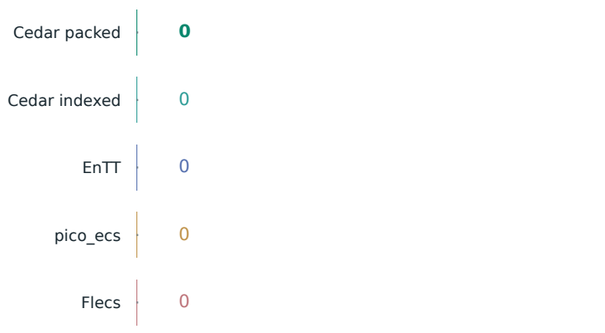
accounted_storage
dense_payload · N=1,024 · populated · step=2



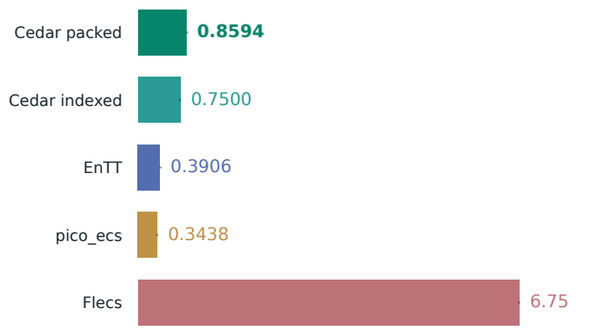
physical_footprint
dense_payload · N=1,024 · populated · step=2



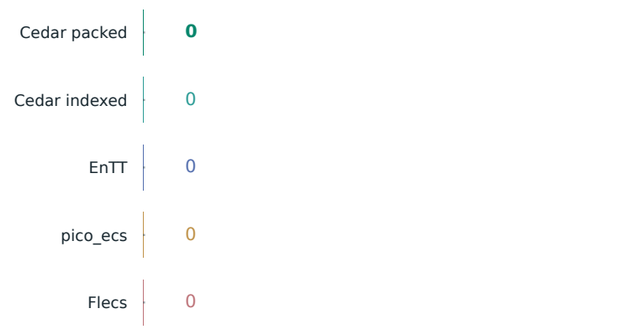
accounted_storage
dense_payload · N=1,024 · destroyed · step=3



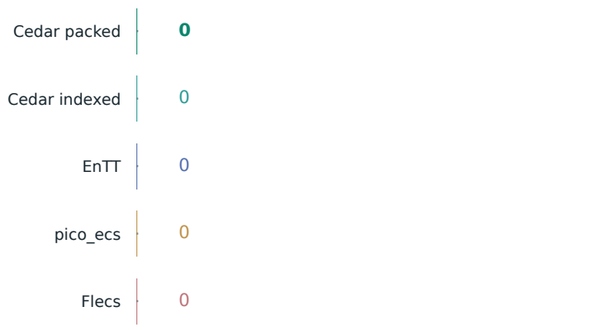
physical_footprint
dense_payload · N=1,024 · destroyed · step=3



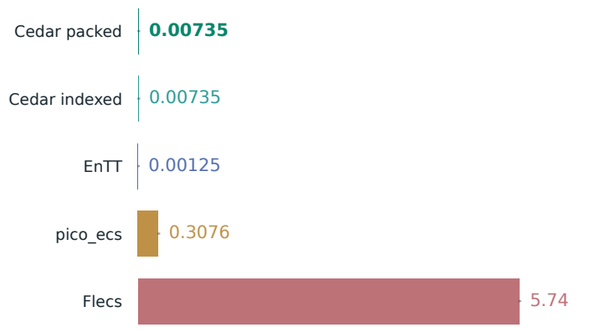
accounted_storage
dense_payload · N=65,536 · baseline · step=0



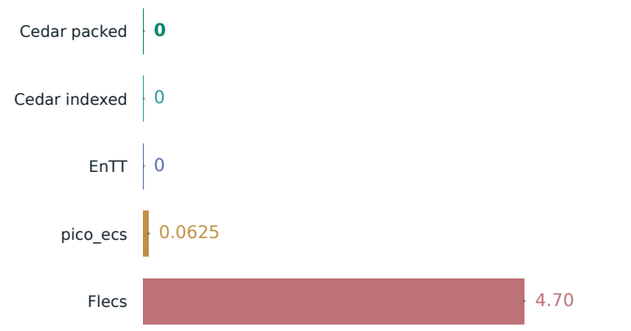
physical_footprint
dense_payload · N=65,536 · baseline · step=0



accounted_storage
dense_payload · N=65,536 · registered · step=1



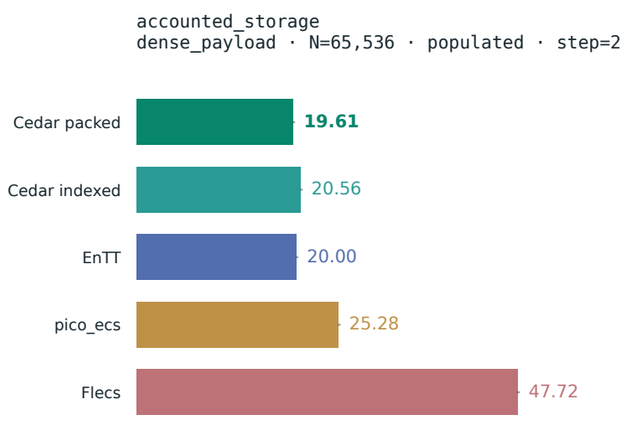
physical_footprint
dense_payload · N=65,536 · registered · step=1

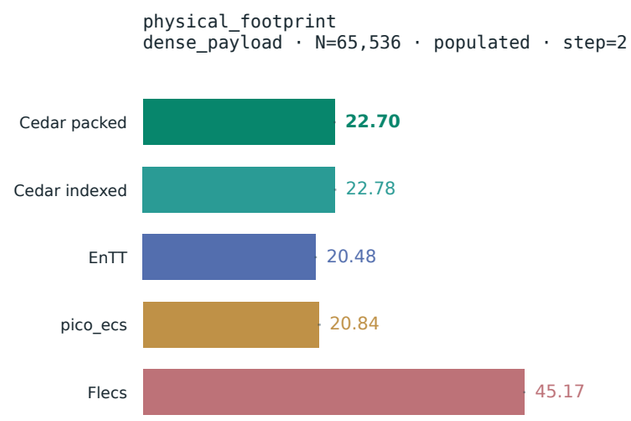


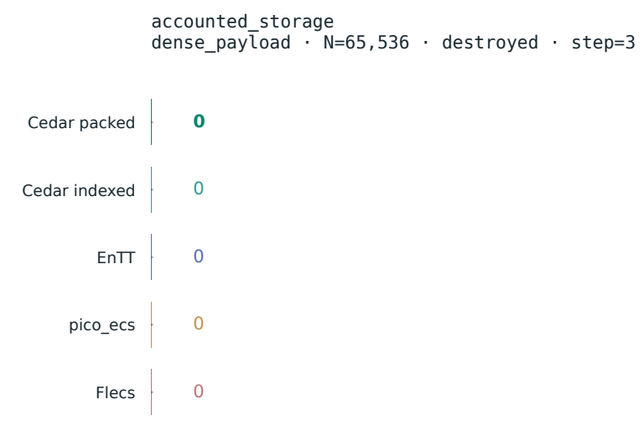
Memory comparison

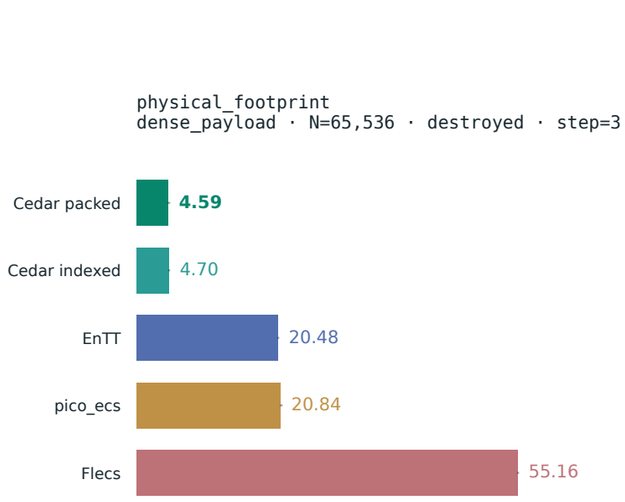
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs



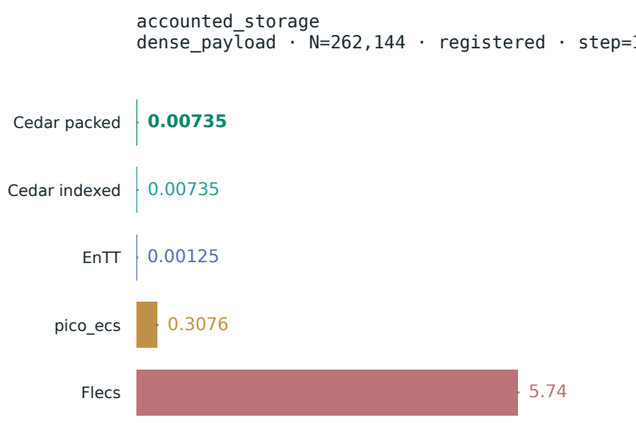


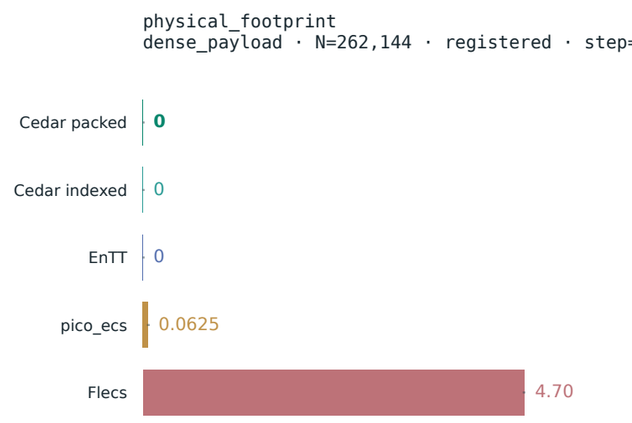


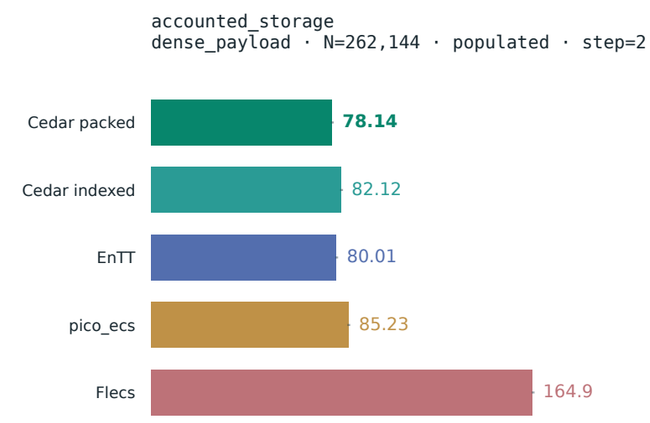


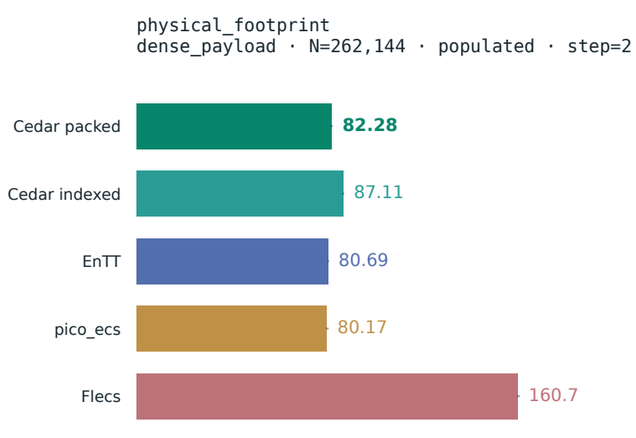




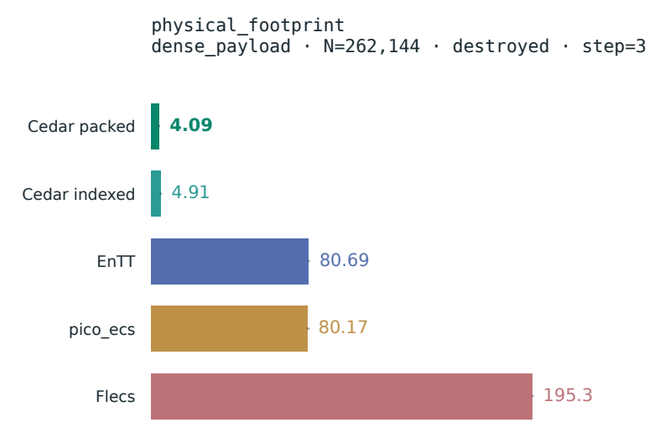












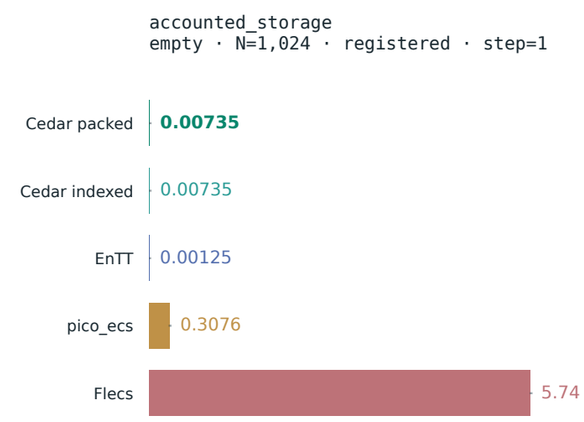
Memory comparison

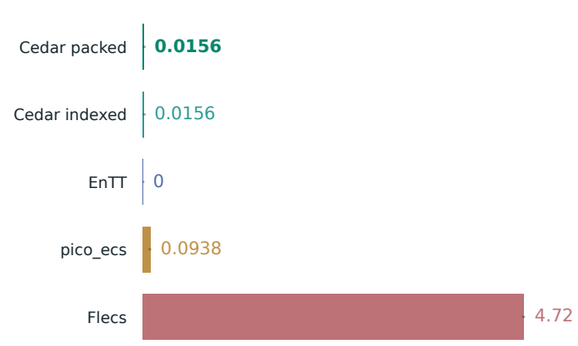
MiB ↓

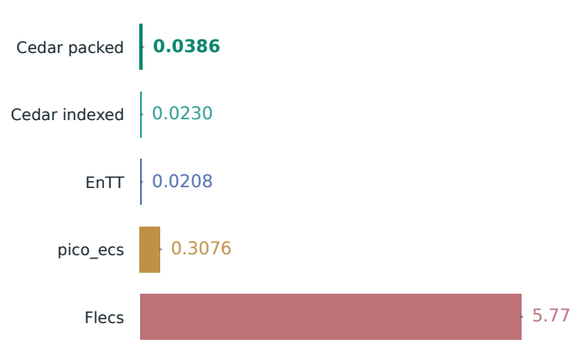
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

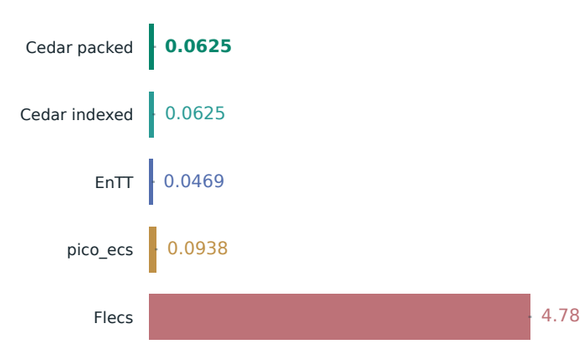










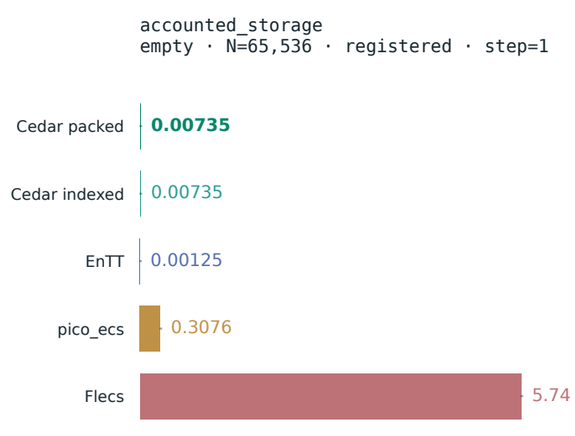


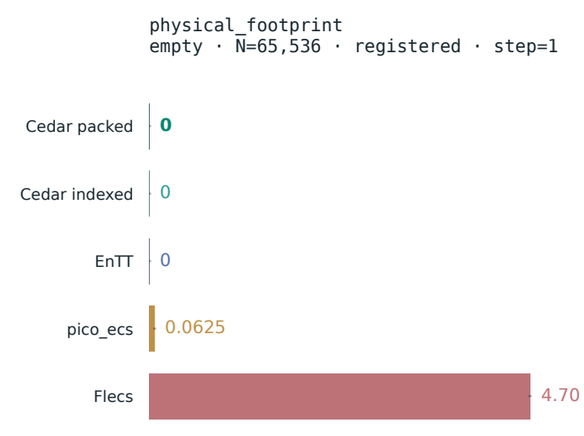








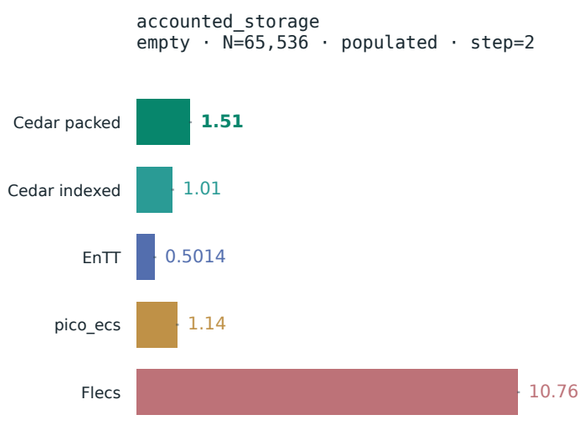


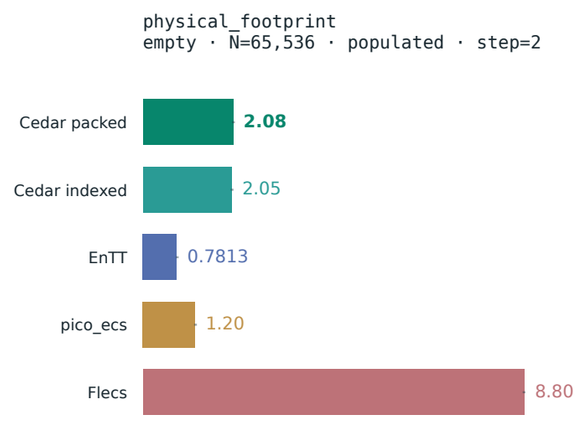


Memory comparison

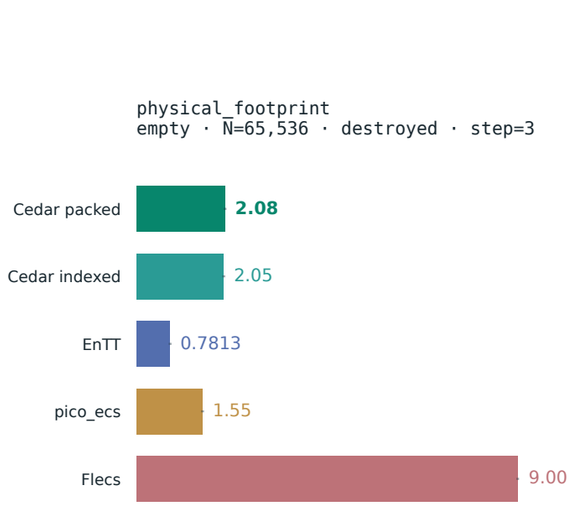
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs



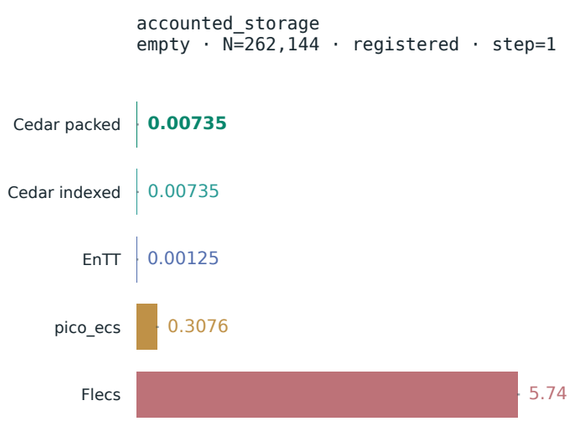


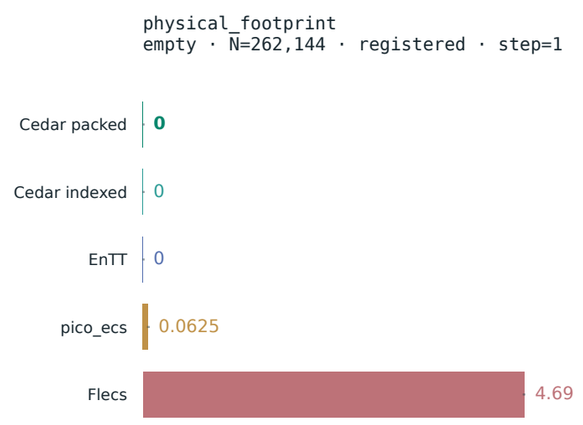


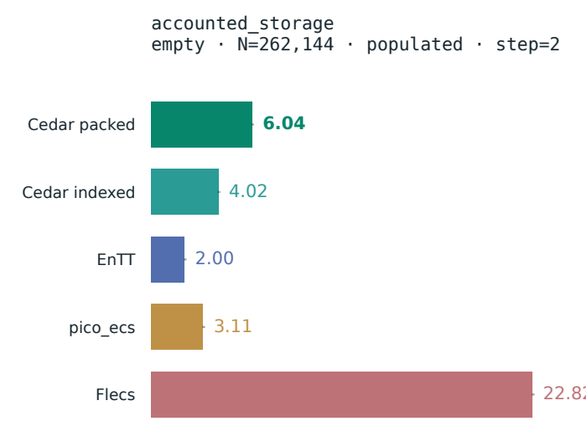


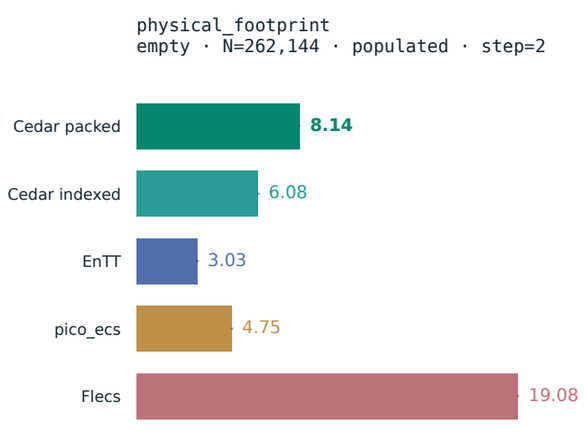




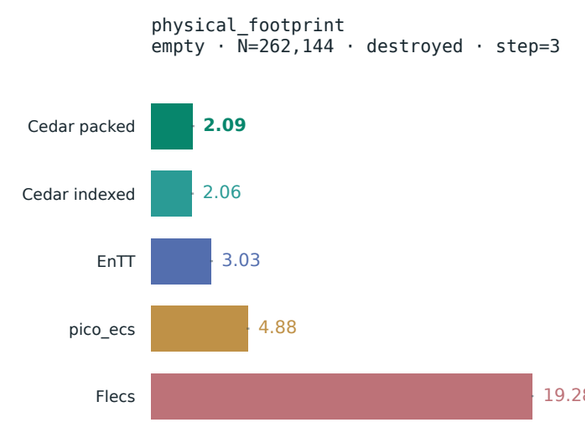












Memory comparison

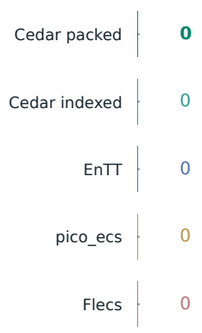
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

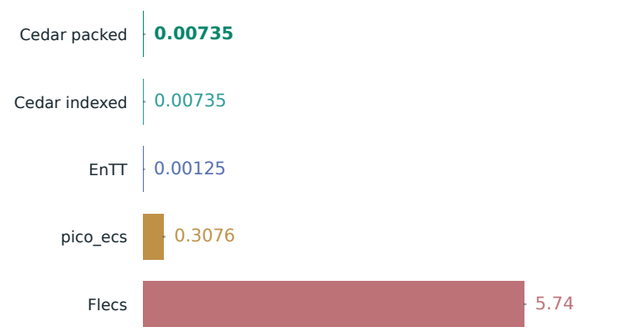
accounted_storage
empty_refill · N=1,024 · baseline · step=0



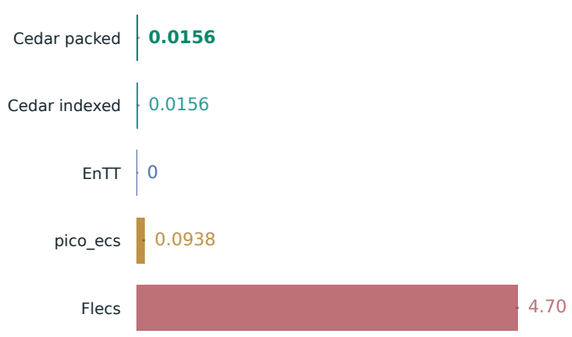
physical_footprint
empty_refill · N=1,024 · baseline · step=0



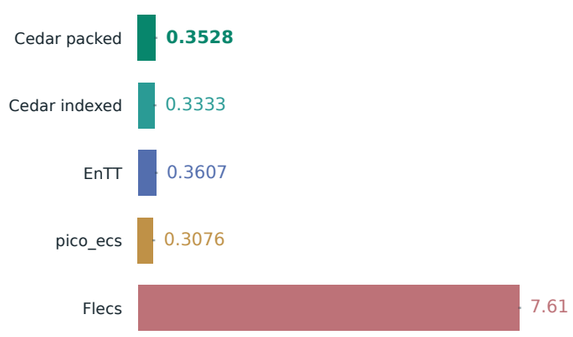
accounted_storage
empty_refill · N=1,024 · registered · step=0



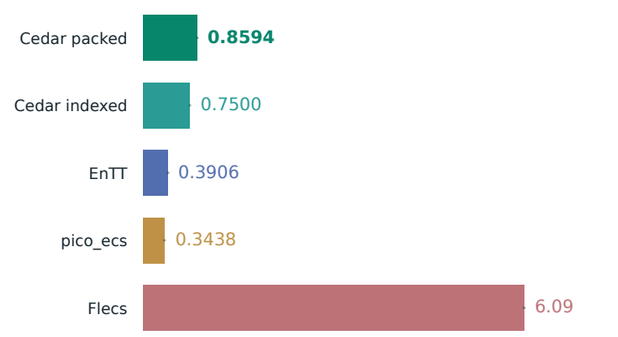
physical_footprint
empty_refill · N=1,024 · registered · step=1



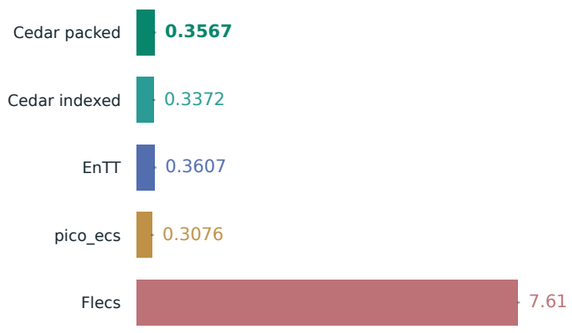
accounted_storage
empty_refill · N=1,024 · populated · step=2



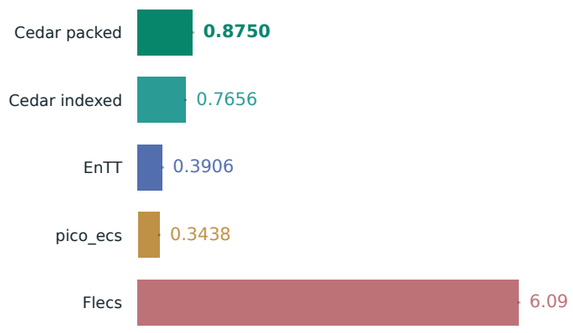
physical_footprint
empty_refill · N=1,024 · populated · step=2



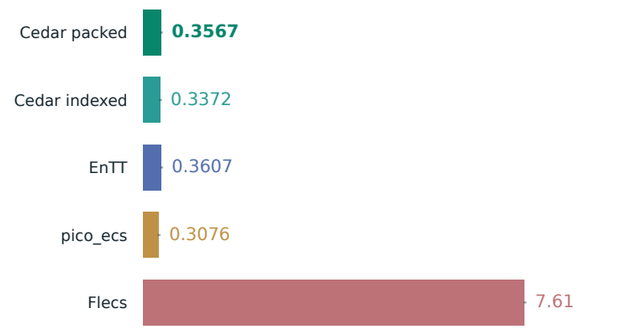
accounted_storage
empty_refill · N=1,024 · deleted · step=3



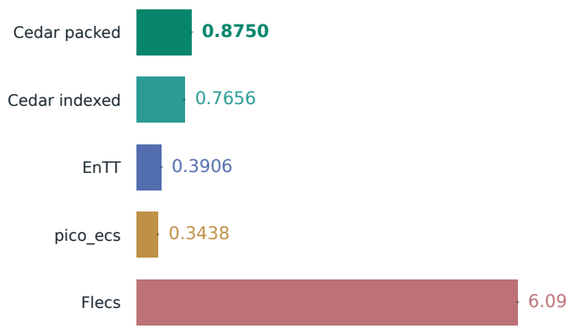
physical_footprint
empty_refill · N=1,024 · deleted · step=3



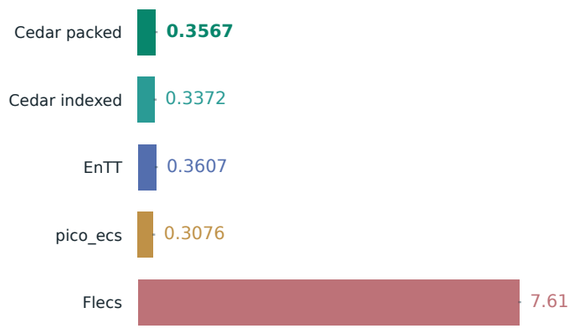
accounted_storage
empty_refill · N=1,024 · regrown · step=4



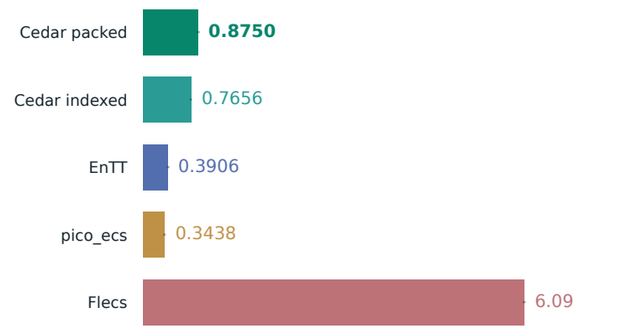
physical_footprint
empty_refill · N=1,024 · regrown · step=4



accounted_storage
empty_refill · N=1,024 · deleted · step=5



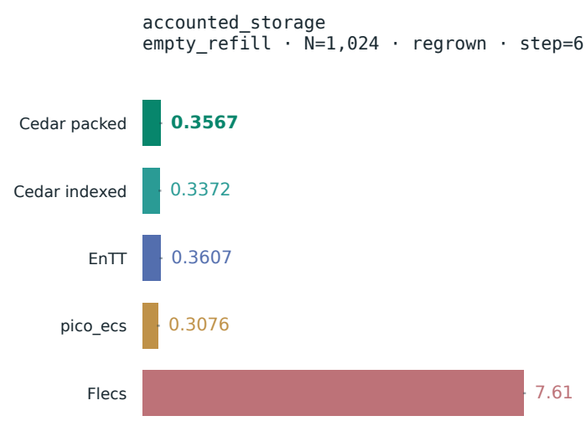
physical_footprint
empty_refill · N=1,024 · deleted · step=5

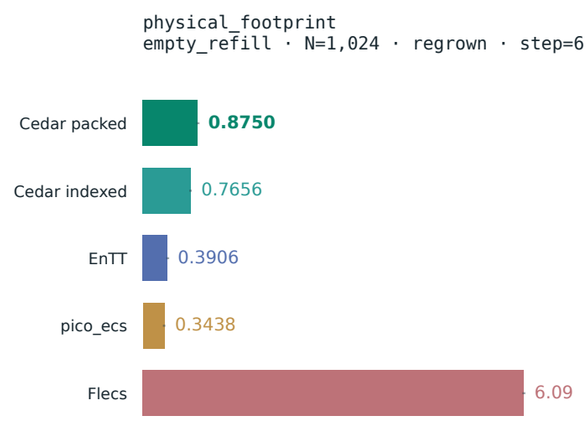


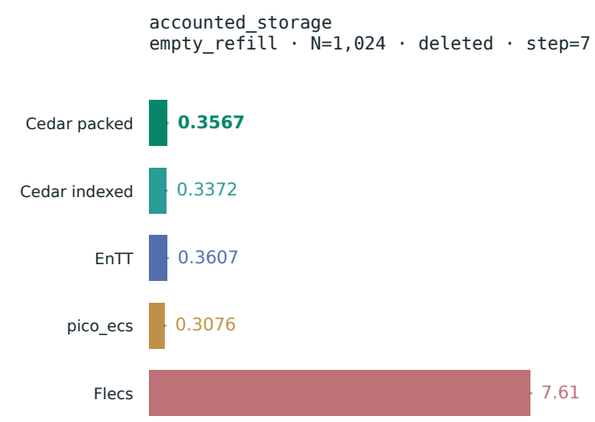
Memory comparison

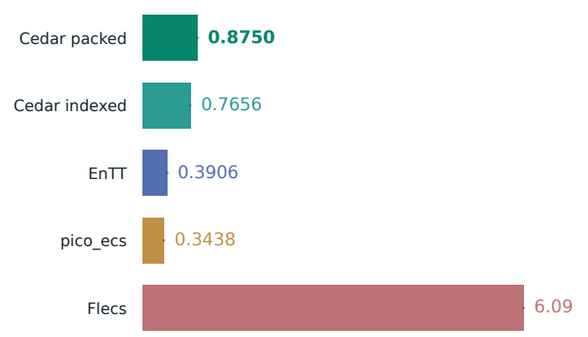
MiB ↓

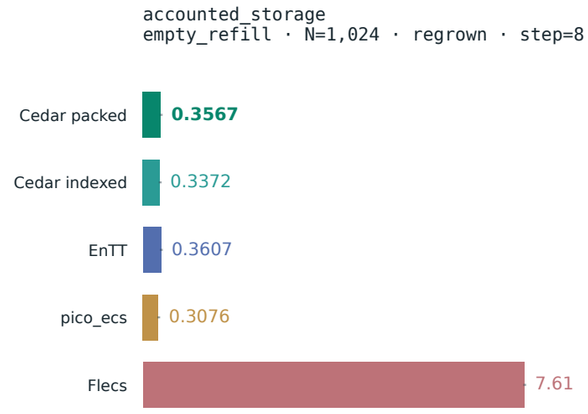
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

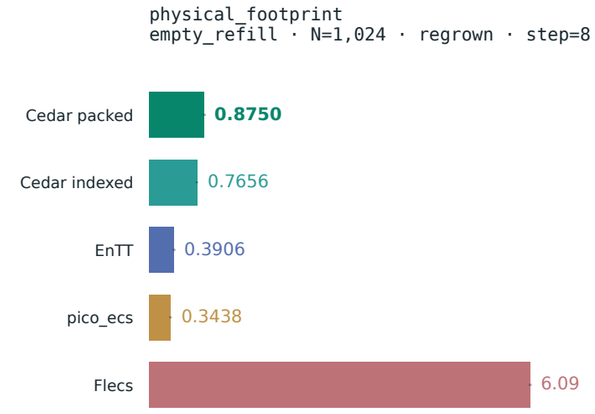


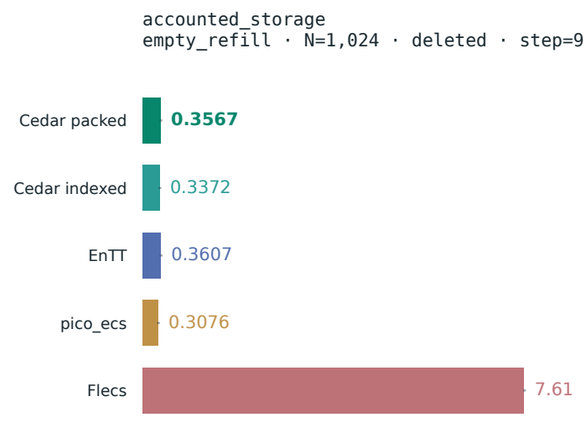


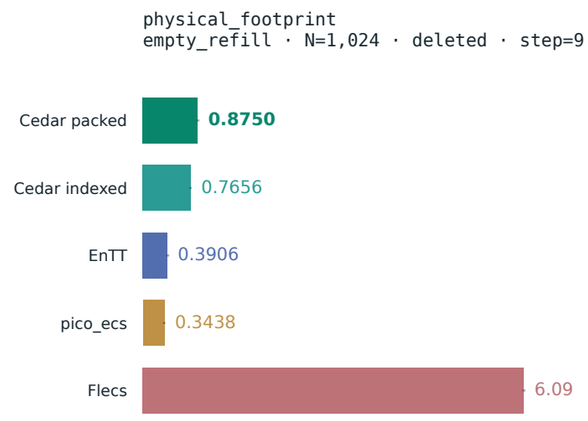


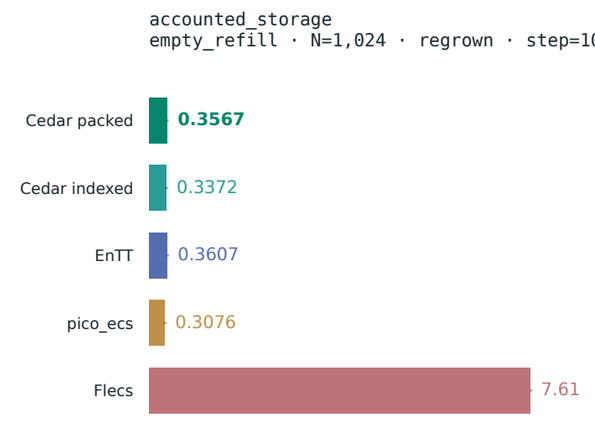


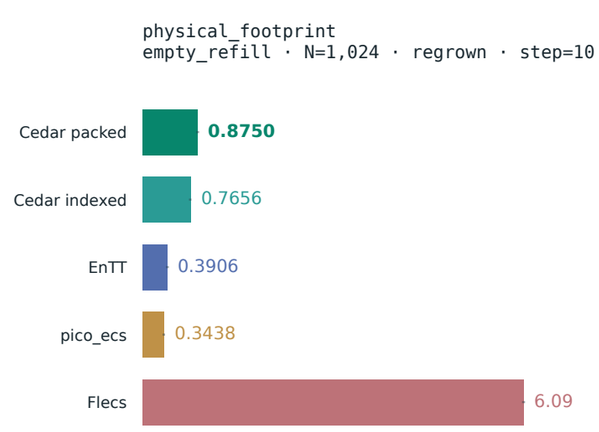




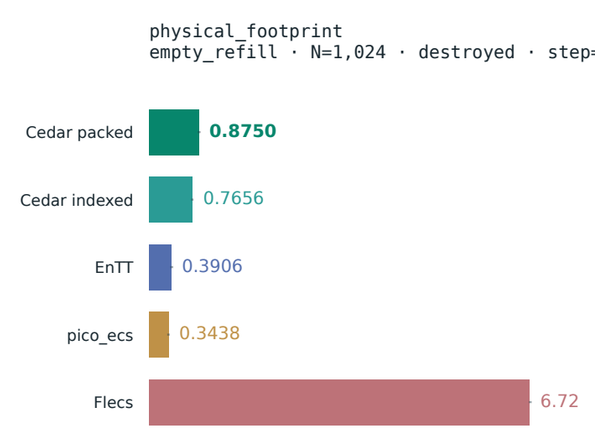








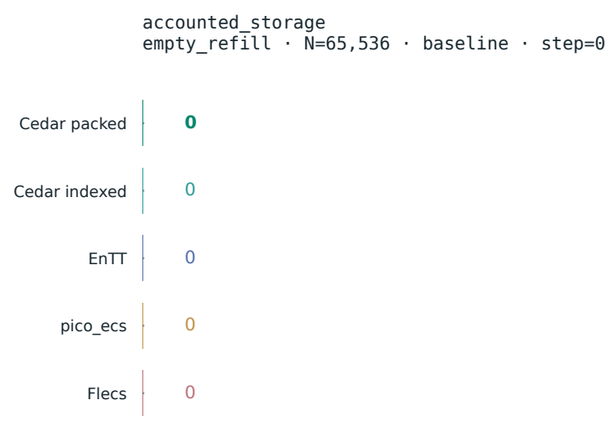


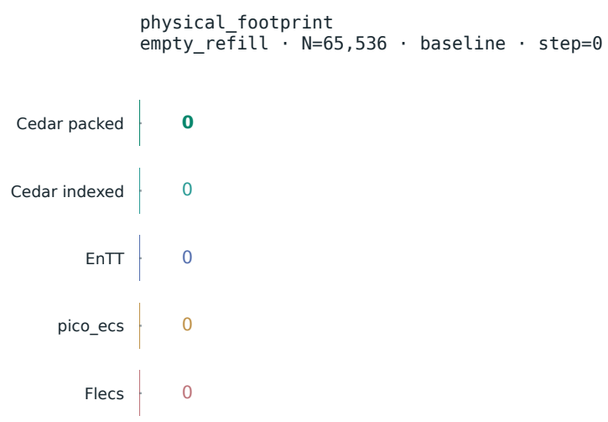


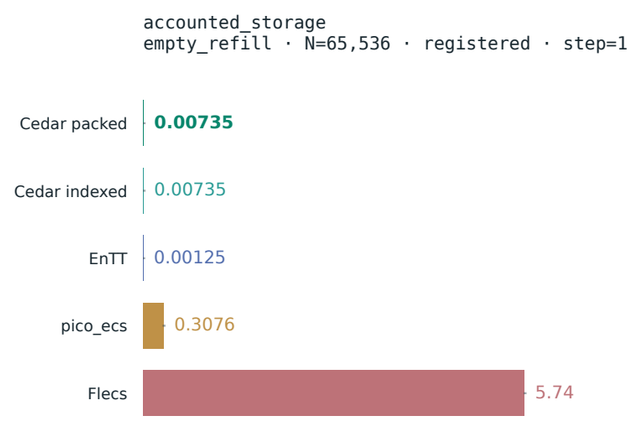
Memory comparison

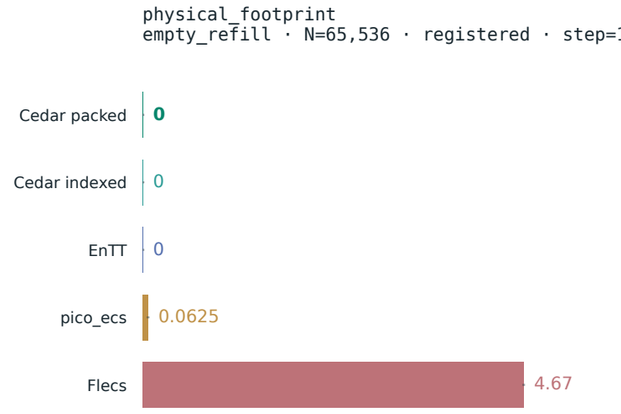
MiB ↓

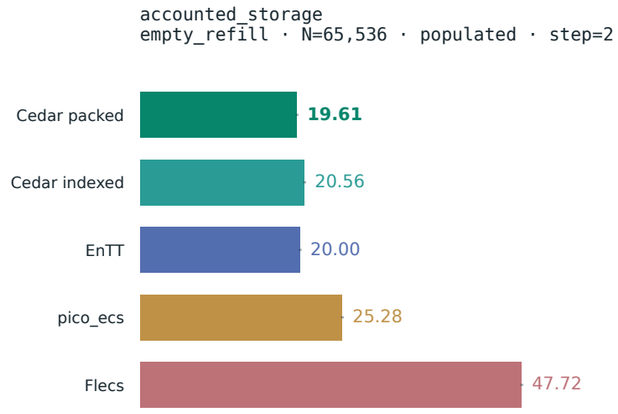
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

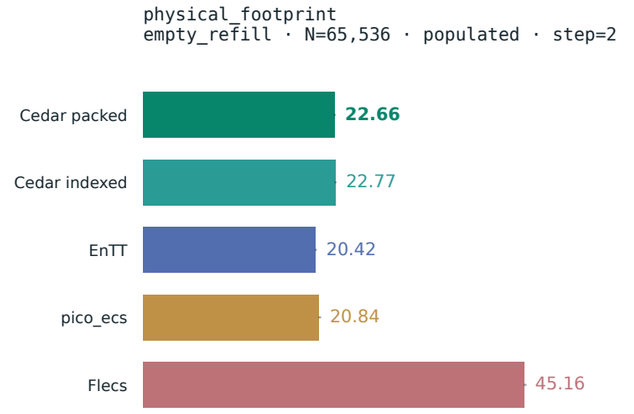


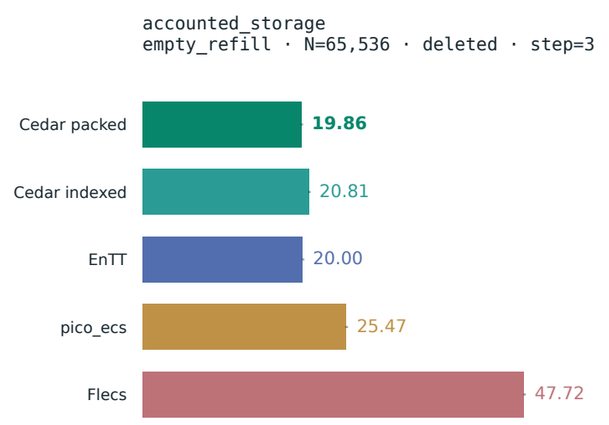


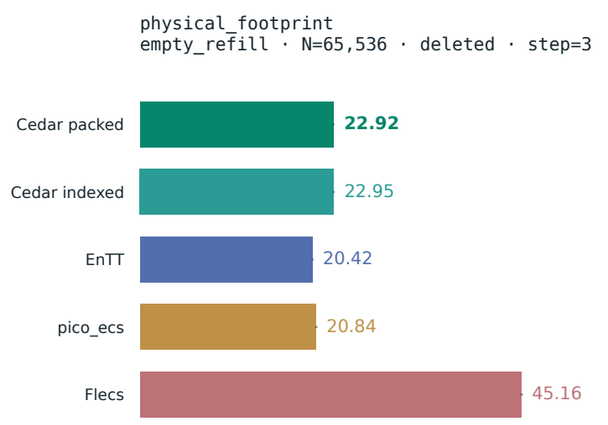


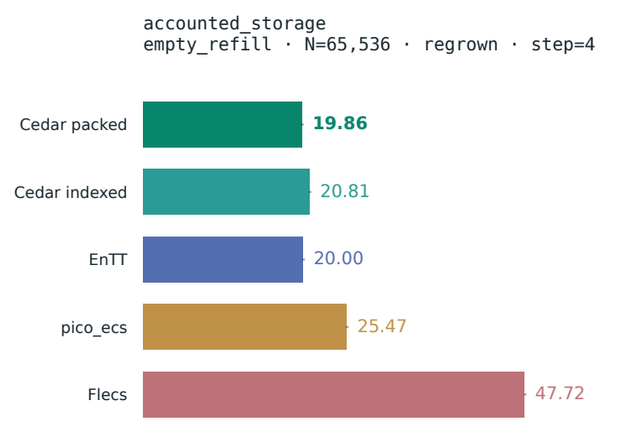


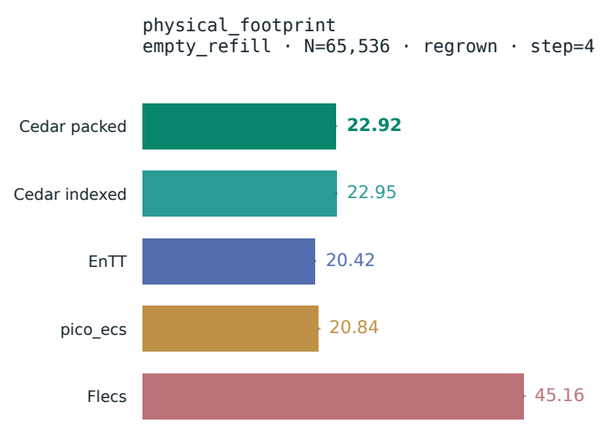


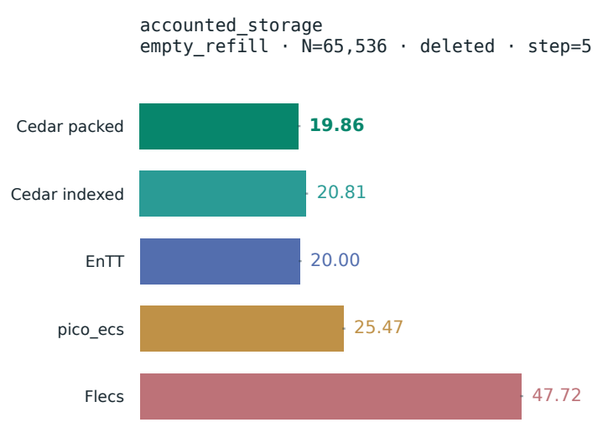


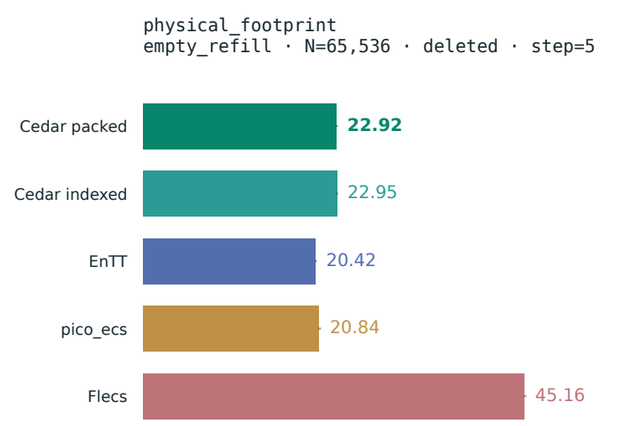










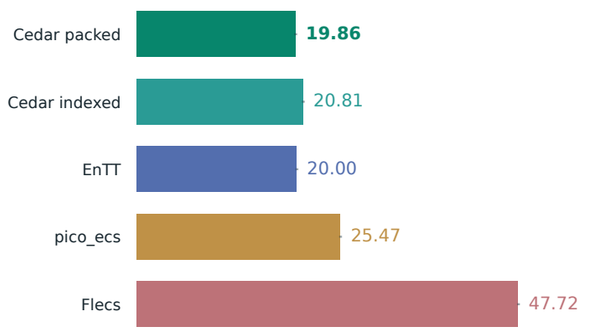


Memory comparison

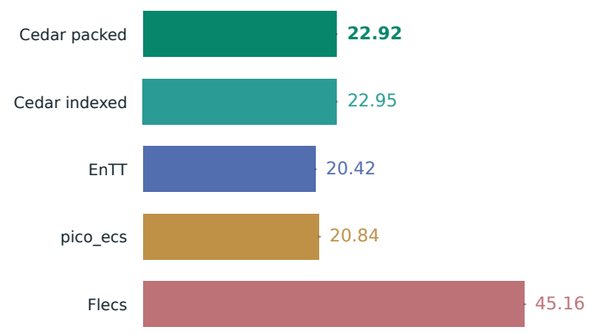
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

accounted_storage
empty_refill · N=65,536 · regrown · step=6

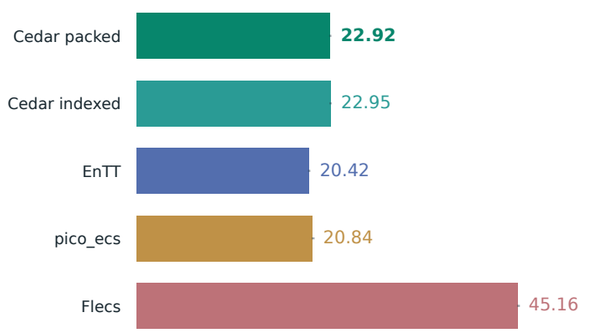


physical_footprint
empty_refill · N=65,536 · regrown · step=6

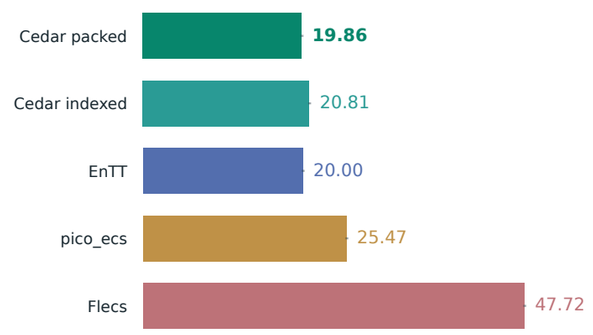


accounted_storage
empty_refill · N=65,536 · deleted · step=7

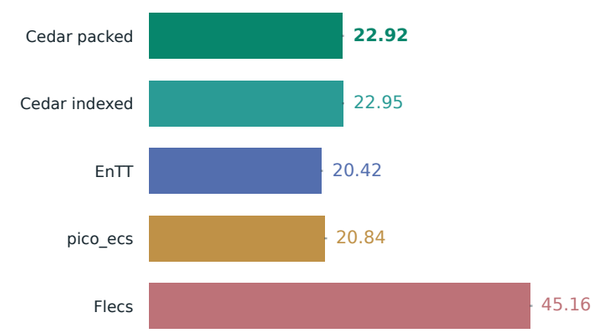
physical_footprint
empty_refill · N=65,536 · deleted · step=7



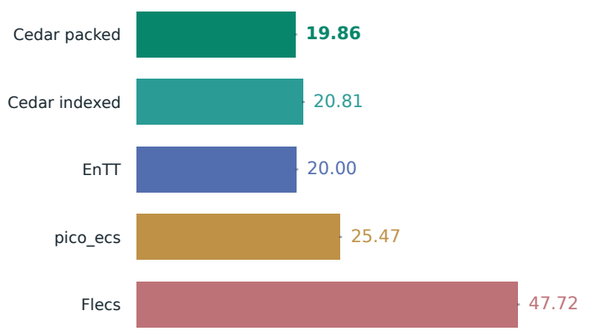
accounted_storage
empty_refill · N=65,536 · regrown · step=8



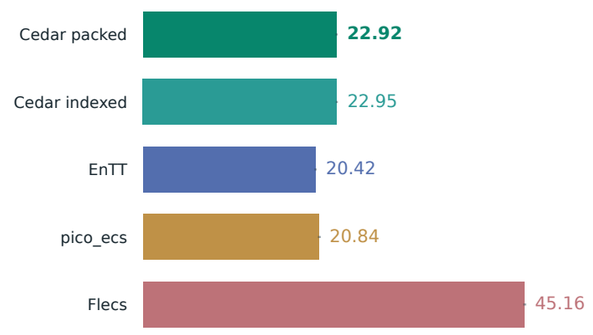
physical_footprint
empty_refill · N=65,536 · regrown · step=8



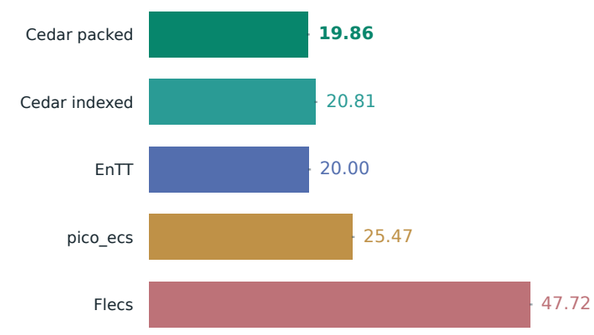
accounted_storage
empty_refill · N=65,536 · deleted · step=9



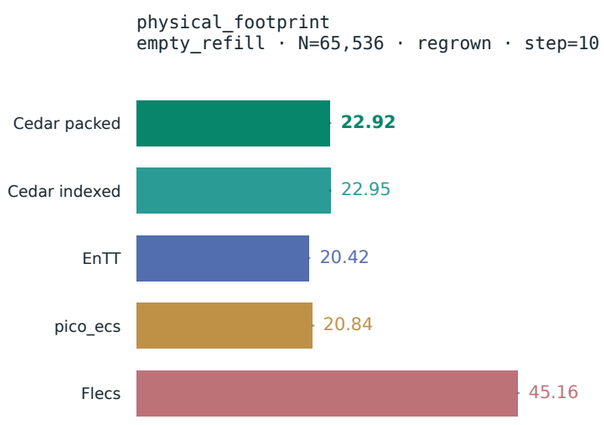
physical_footprint
empty_refill · N=65,536 · deleted · step=9



accounted_storage
empty_refill · N=65,536 · regrown · step=10



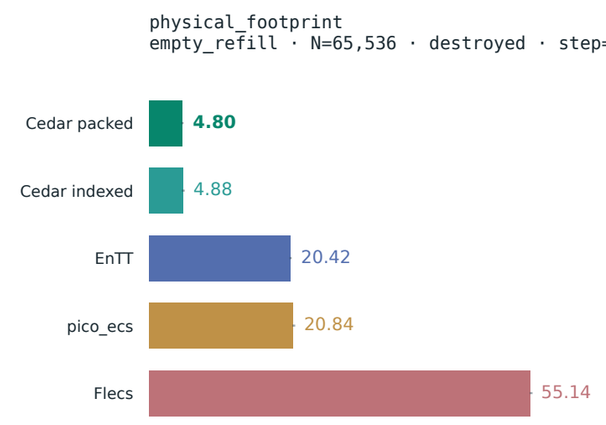
physical_footprint
empty_refill · N=65,536 · regrown · step=10



accounted_storage
empty_refill · N=65,536 · destroyed · step=11



physical_footprint
empty_refill · N=65,536 · destroyed · step=11



Memory comparison

MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

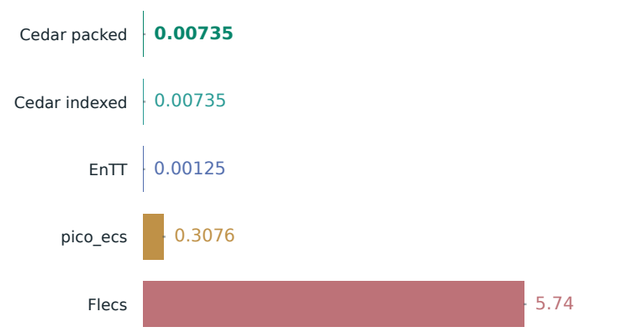
accounted_storage
empty_refill · N=262,144 · baseline · step=0



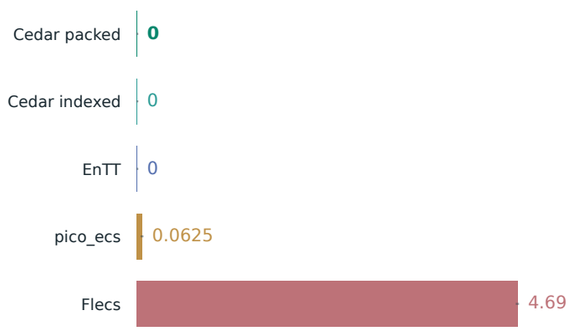
physical_footprint
empty_refill · N=262,144 · baseline · step=0



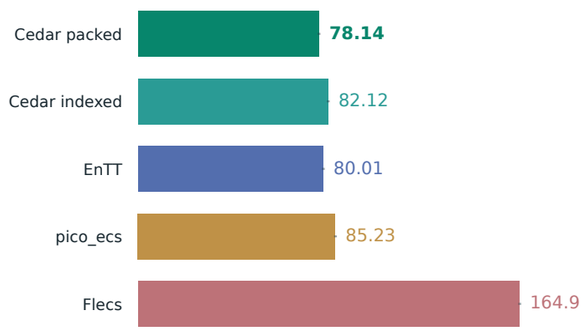
accounted_storage
empty_refill · N=262,144 · registered · step=1



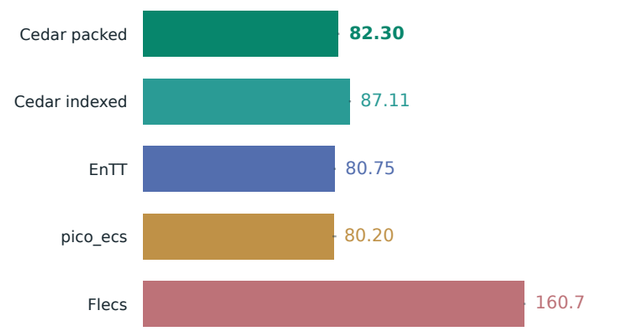
physical_footprint
empty_refill · N=262,144 · registered · step=1



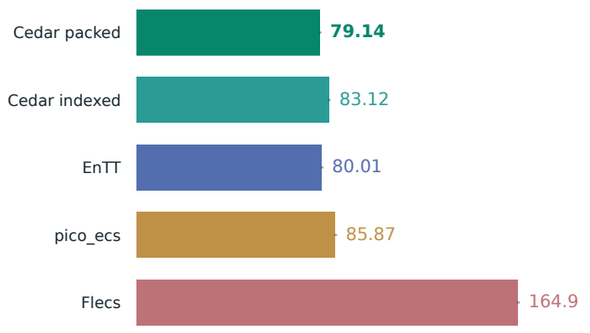
accounted_storage
empty_refill · N=262,144 · populated · step=2



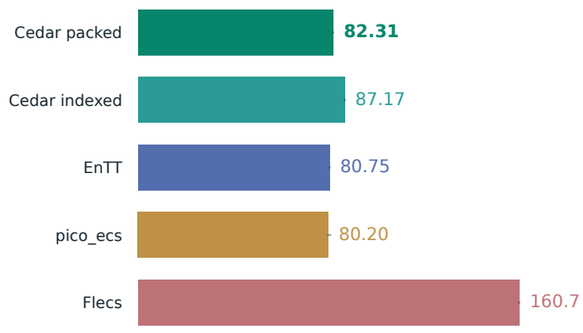
physical_footprint
empty_refill · N=262,144 · populated · step=2



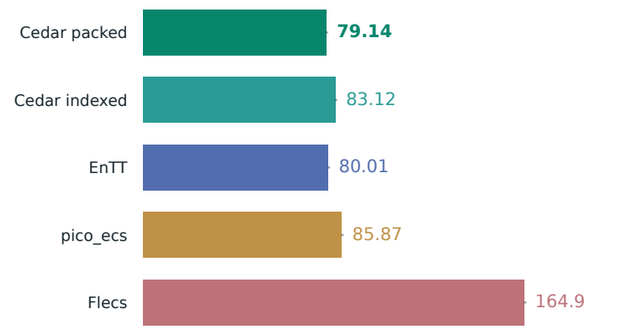
accounted_storage
empty_refill · N=262,144 · deleted · step=3



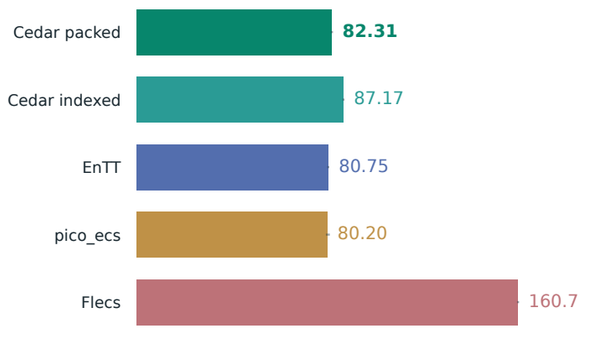
physical_footprint
empty_refill · N=262,144 · deleted · step=3



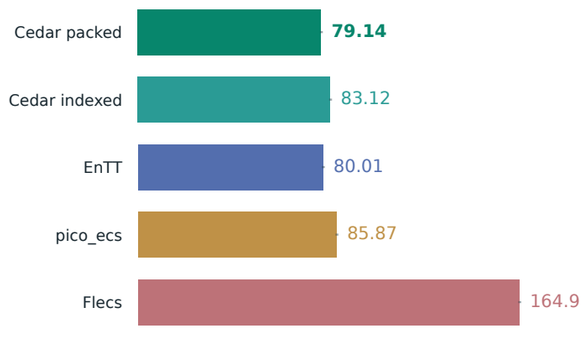
accounted_storage
empty_refill · N=262,144 · regrown · step=4



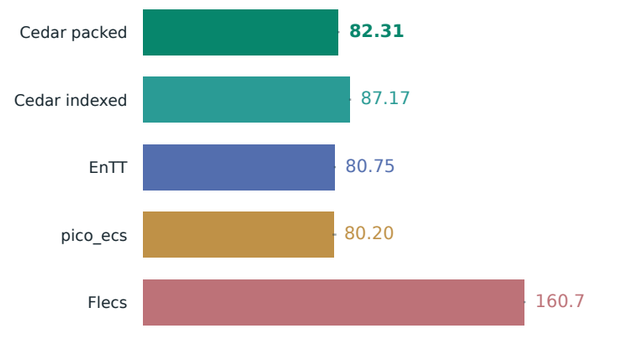
physical_footprint
empty_refill · N=262,144 · regrown · step=4



accounted_storage
empty_refill · N=262,144 · deleted · step=5



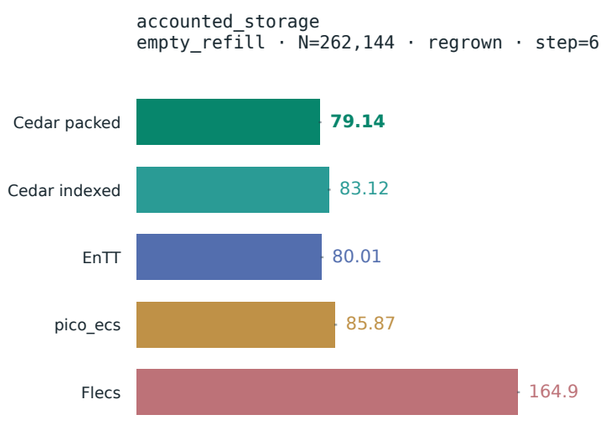
physical_footprint
empty_refill · N=262,144 · deleted · step=5

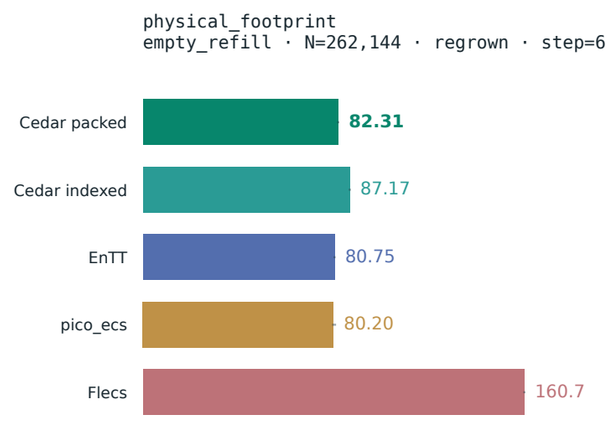


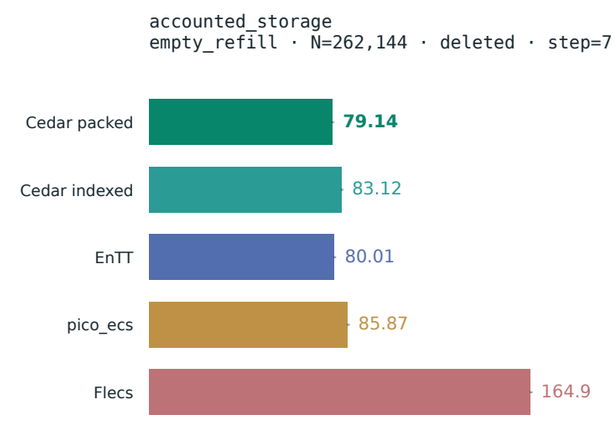
Memory comparison

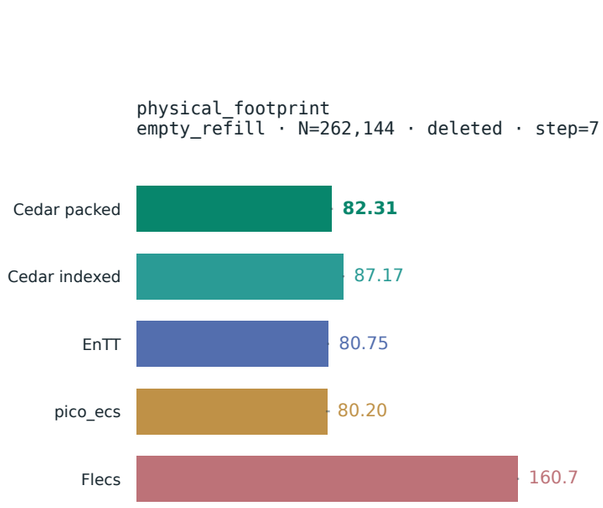
MiB ↓

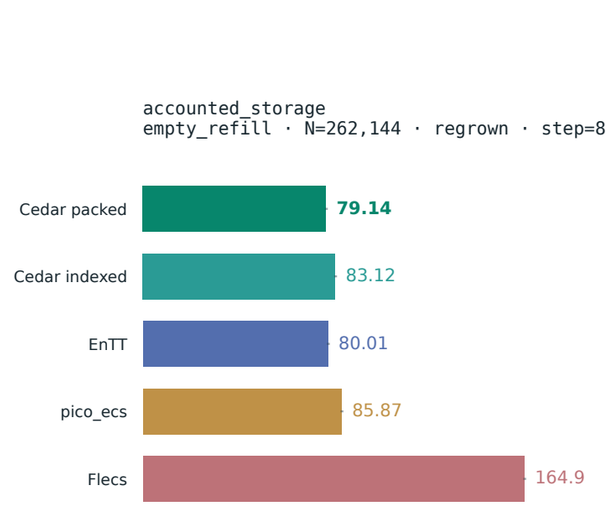
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

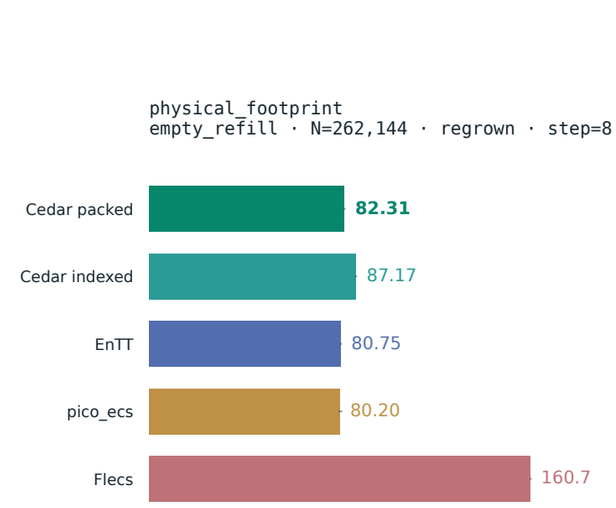


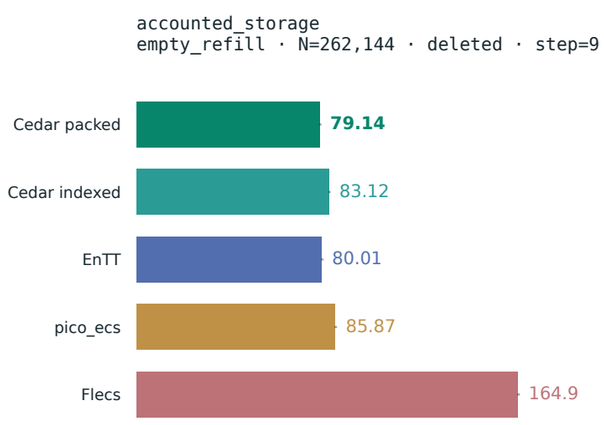


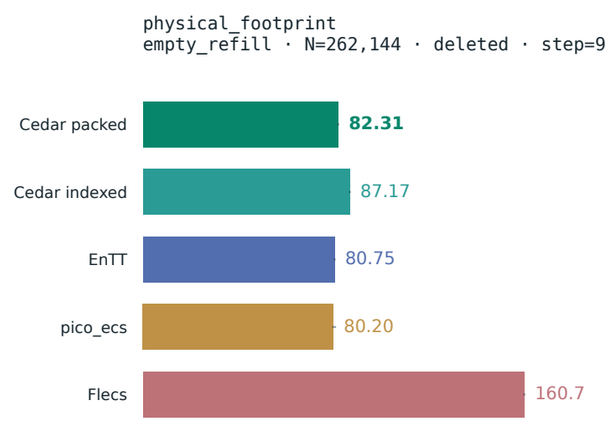


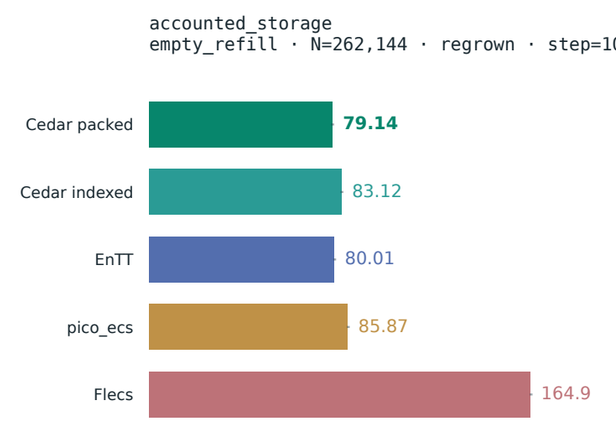


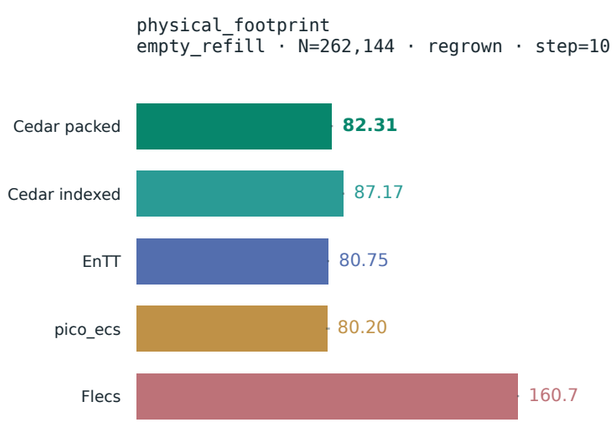




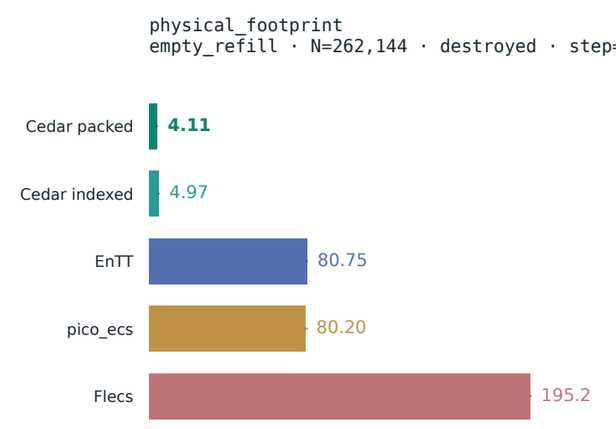








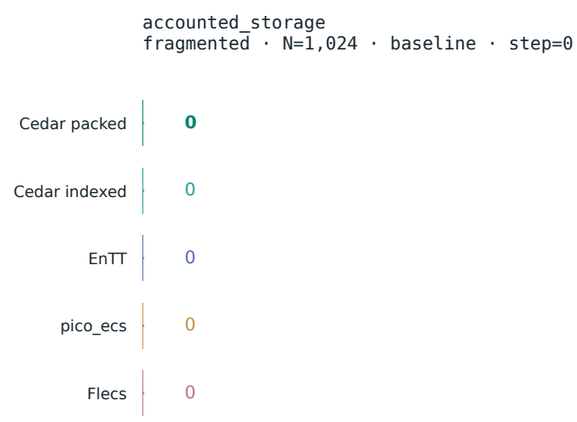




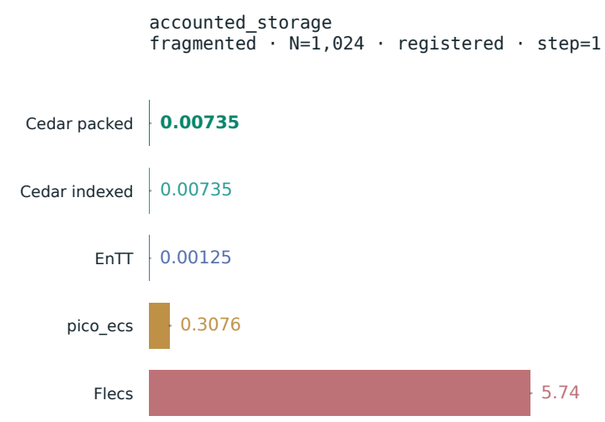
Memory comparison

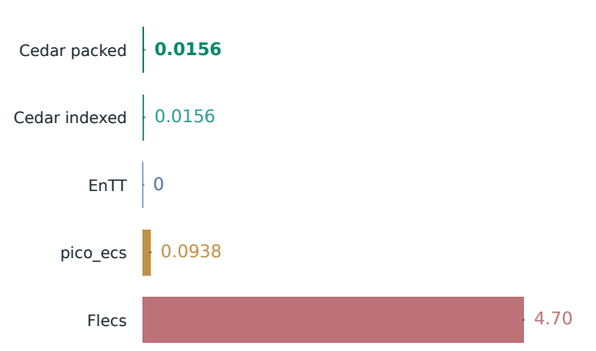
MiB ↓

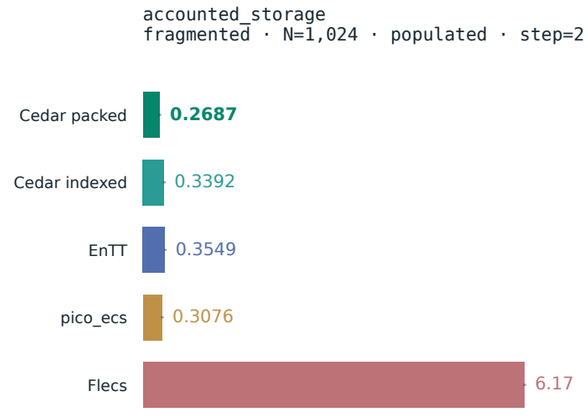
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

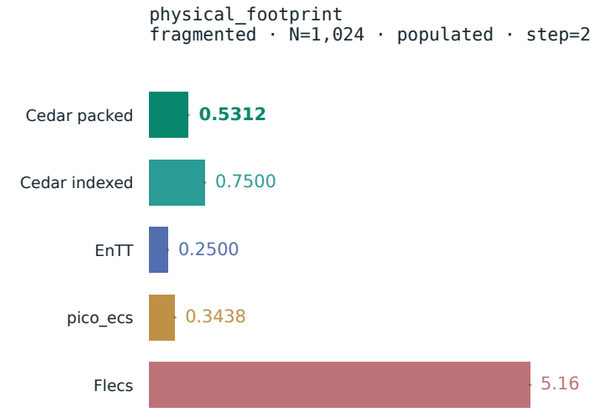


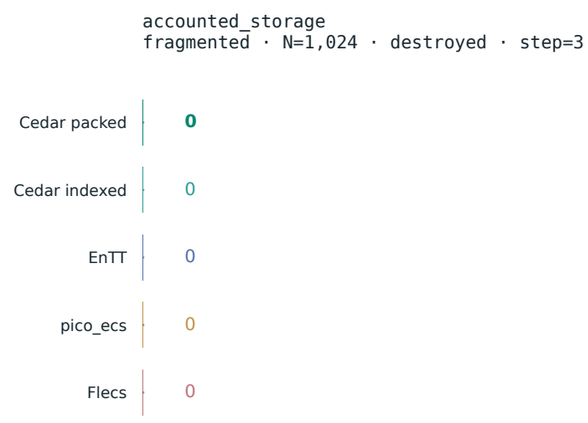


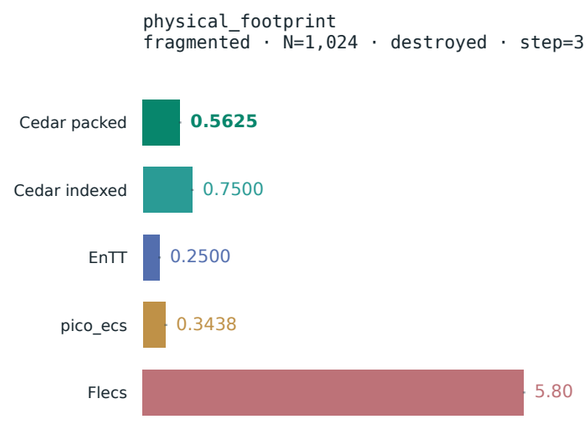


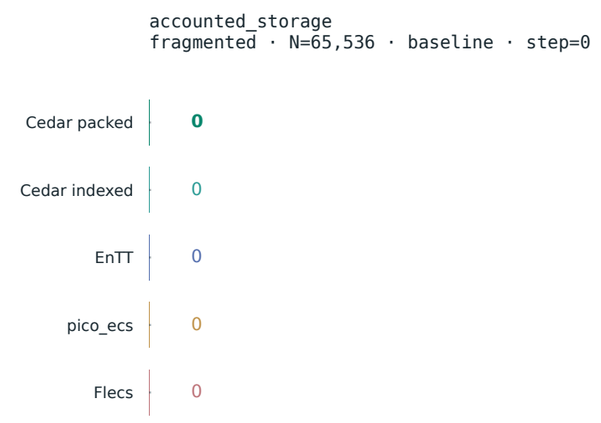




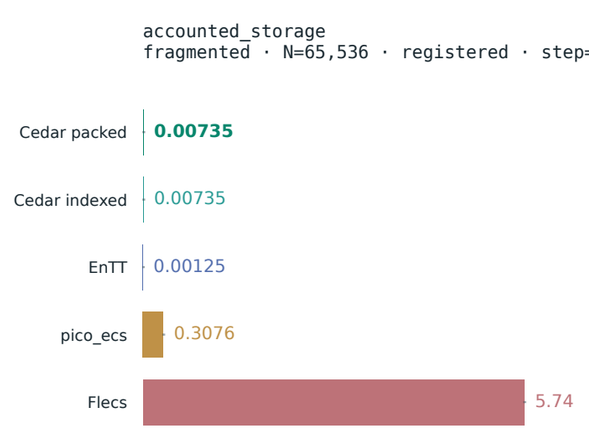


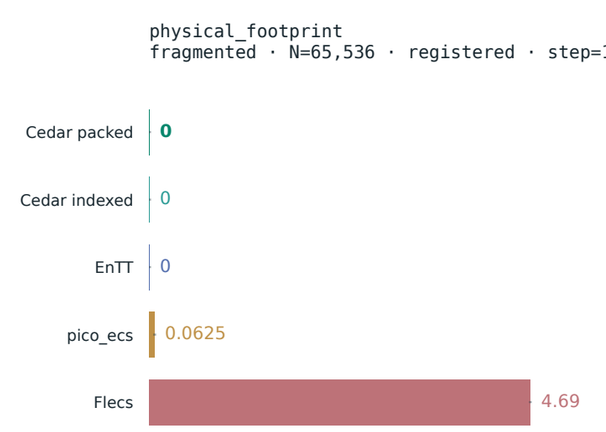








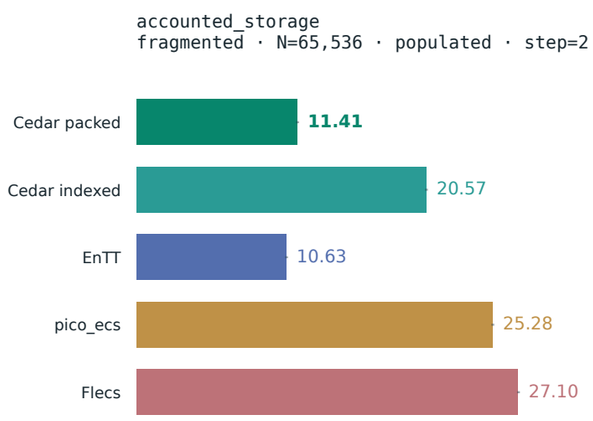


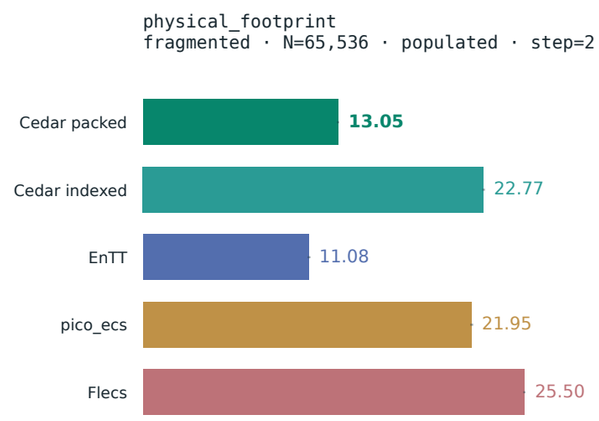


Memory comparison

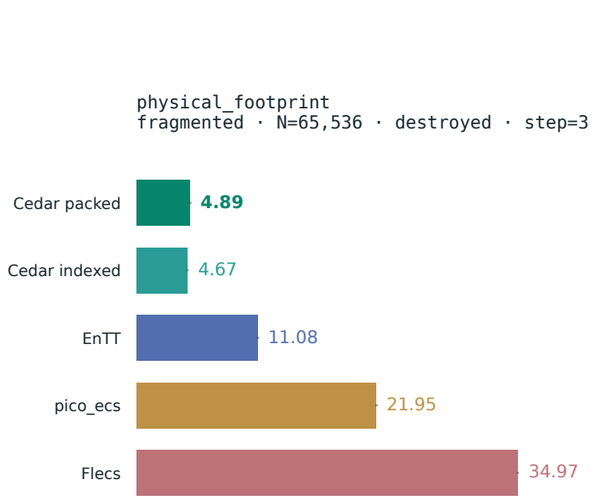
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs



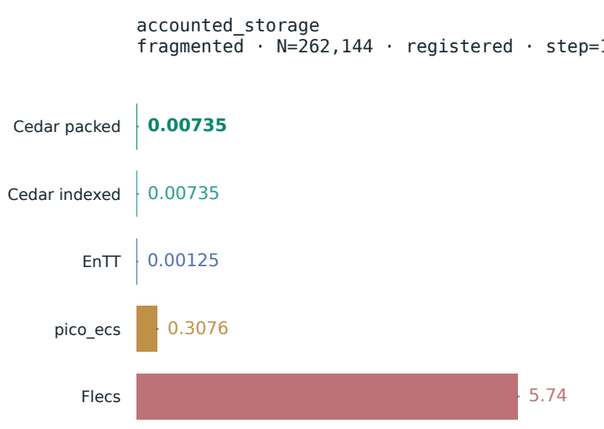


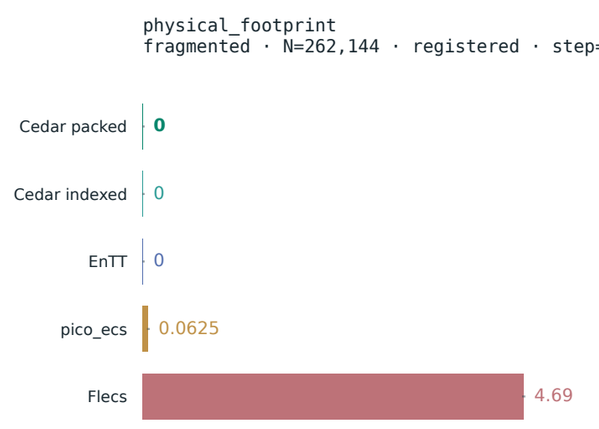


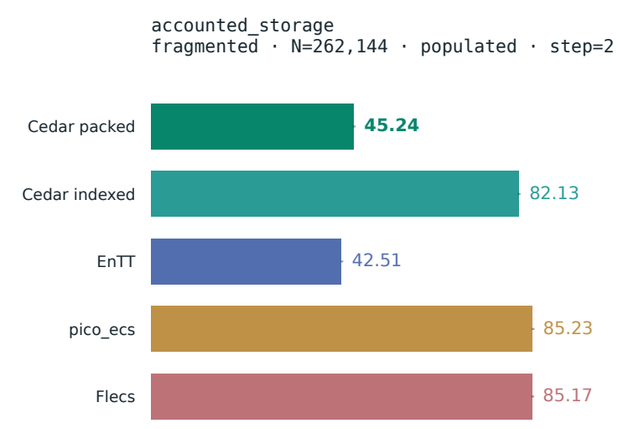


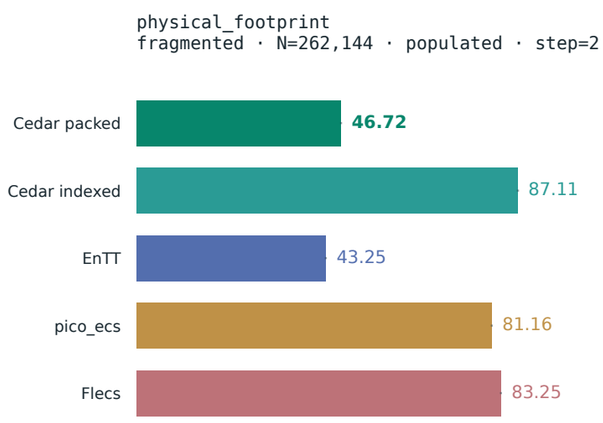




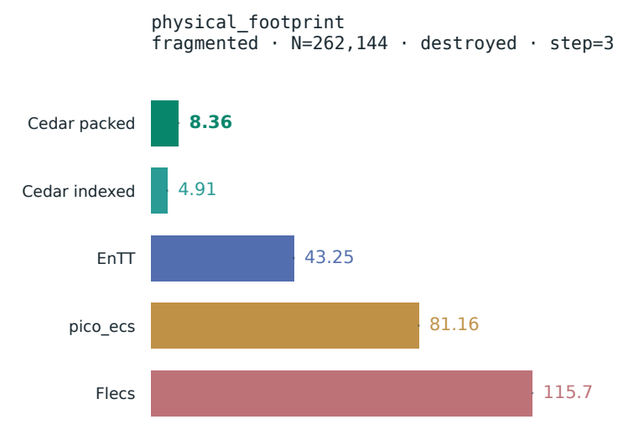










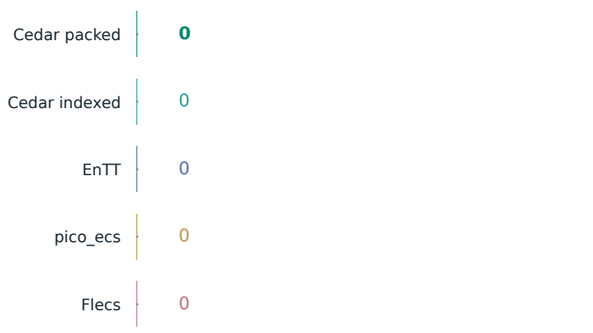


Memory comparison

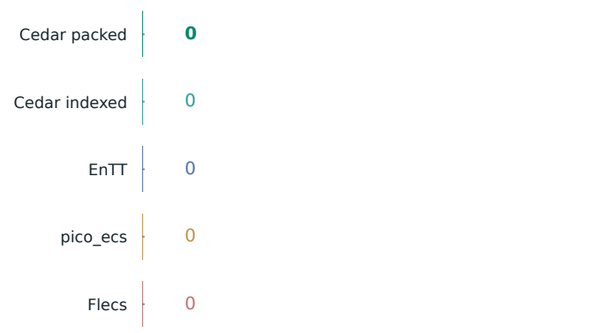
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

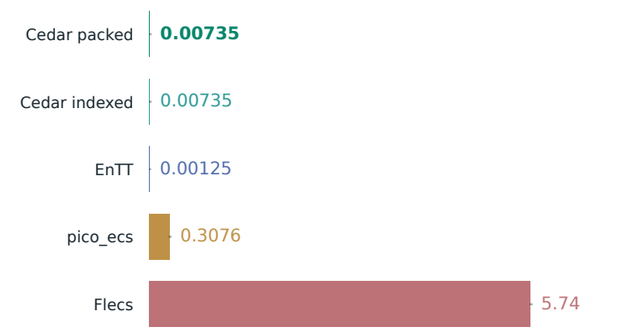
accounted_storage
growth_steps · N=1,024 · baseline · step=0



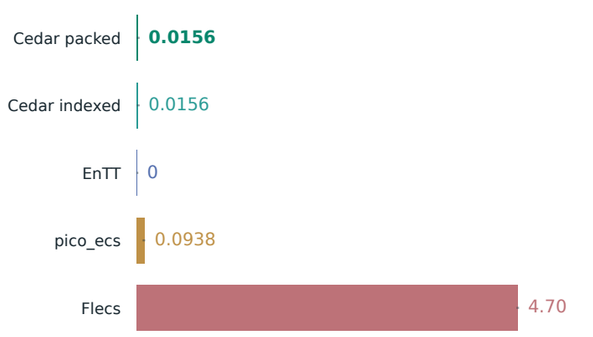
physical_footprint
growth_steps · N=1,024 · baseline · step=0



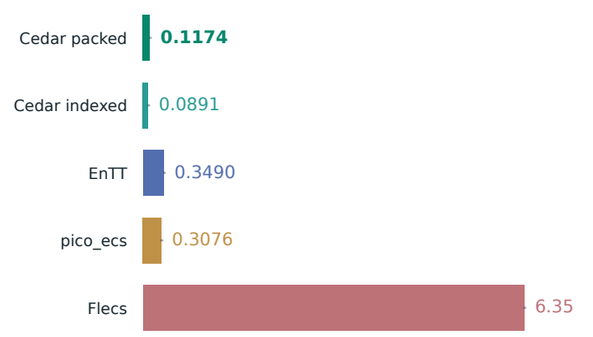
accounted_storage
growth_steps · N=1,024 · registered · step=1



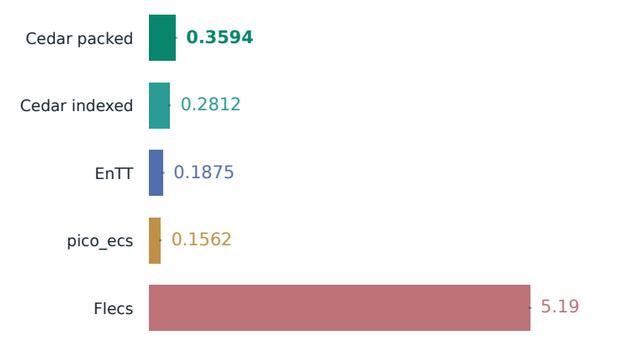
physical_footprint
growth_steps · N=1,024 · registered · step=1



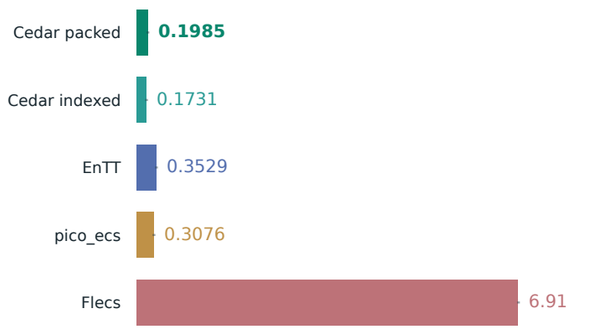
accounted_storage
growth_steps · N=1,024 · grown_25 · step=2



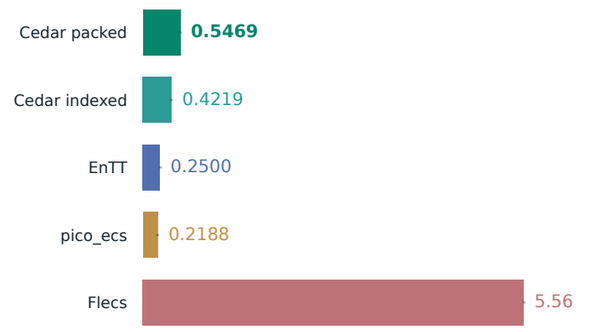
physical_footprint
growth_steps · N=1,024 · grown_25 · step=2



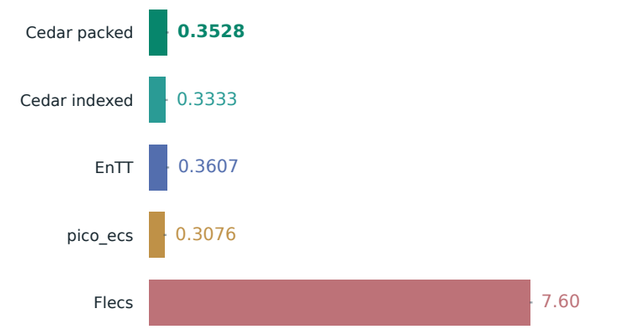
accounted_storage
growth_steps · N=1,024 · grown_50 · step=3



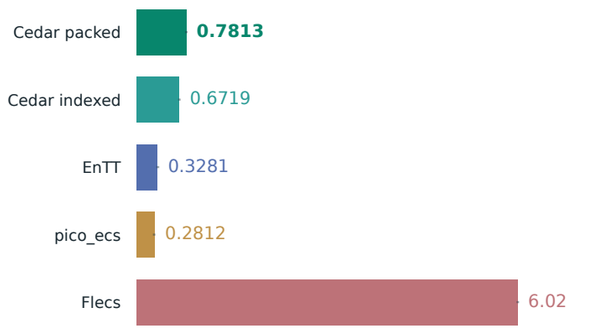
physical_footprint
growth_steps · N=1,024 · grown_50 · step=3



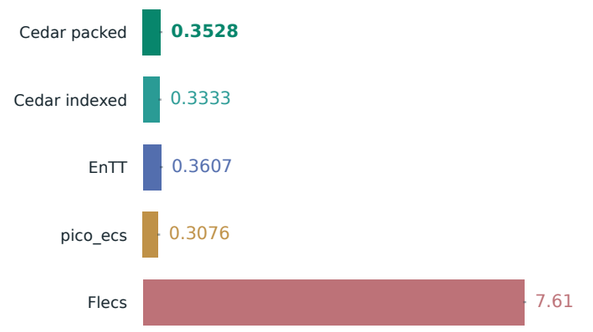
accounted_storage
growth_steps · N=1,024 · grown_75 · step=4



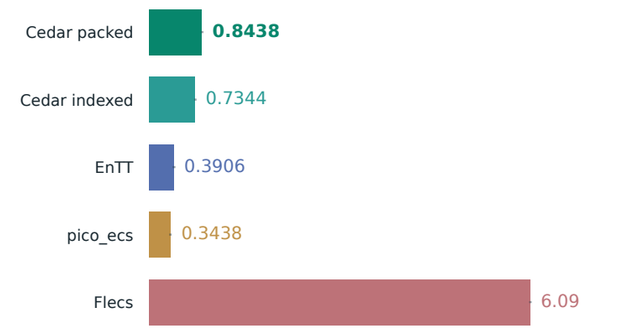
physical_footprint
growth_steps · N=1,024 · grown_75 · step=4



accounted_storage
growth_steps · N=1,024 · populated · step=5



physical_footprint
growth_steps · N=1,024 · populated · step=5

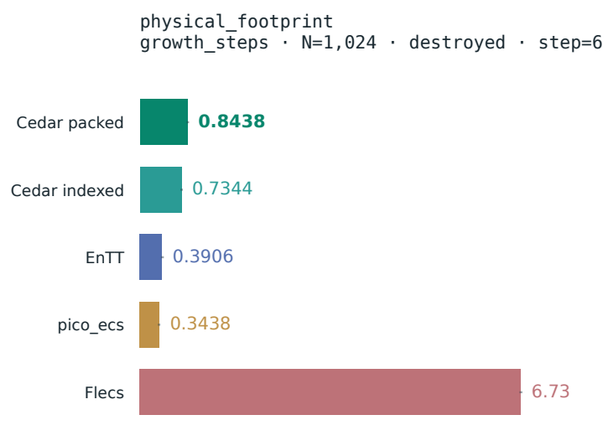


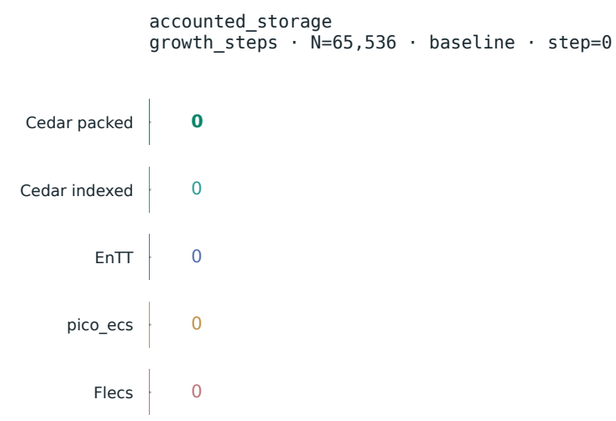
Memory comparison

MiB ↓

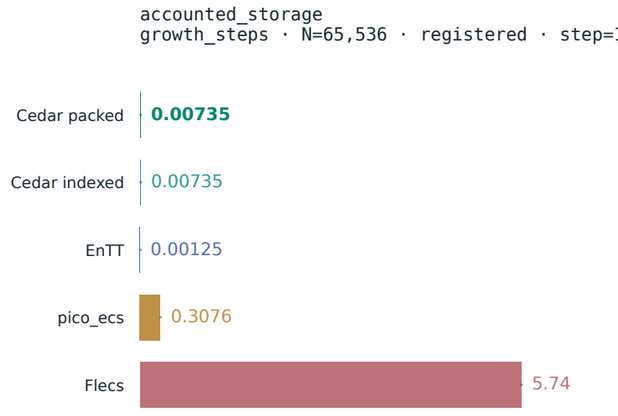
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

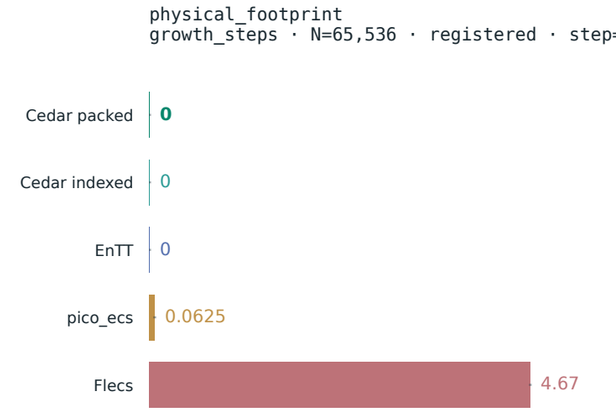


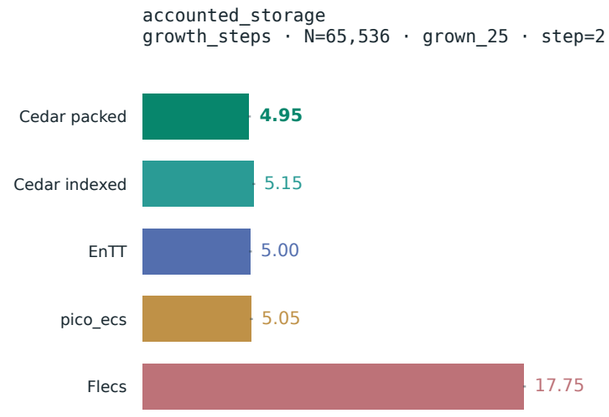


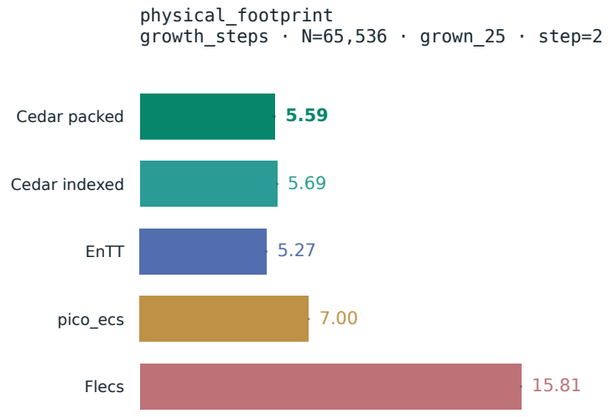


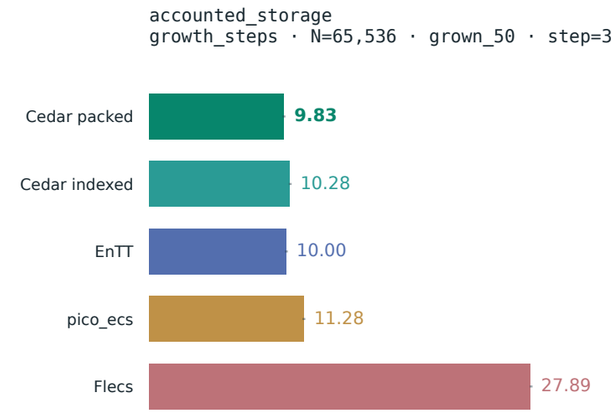


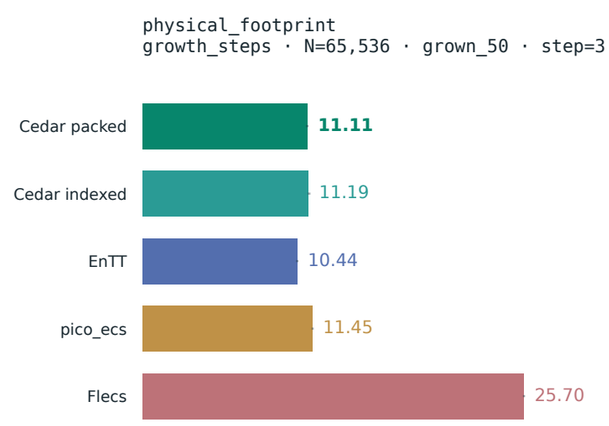


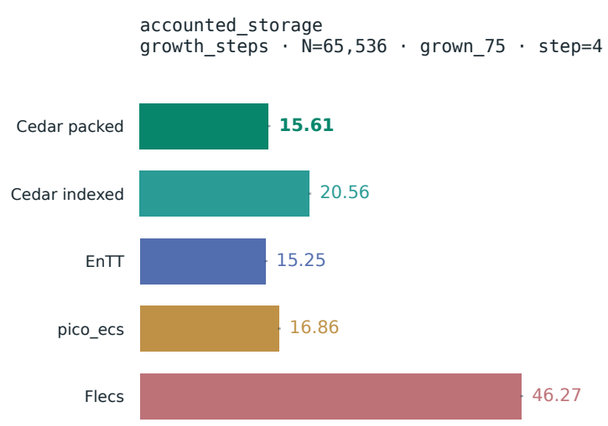


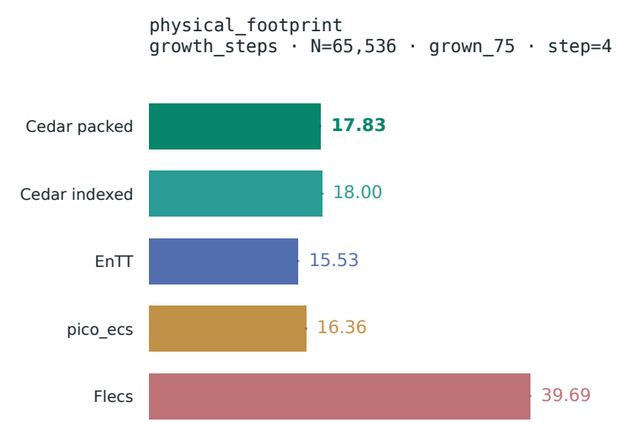








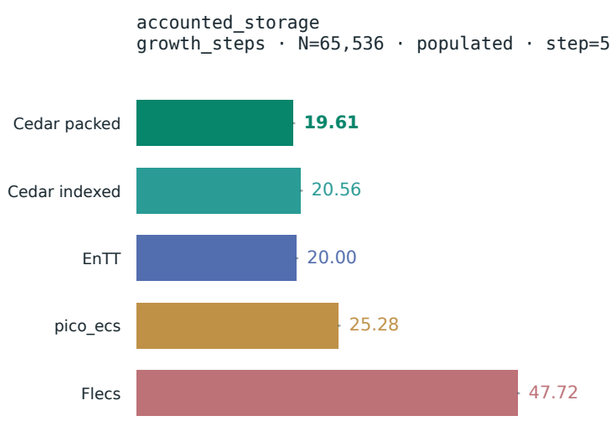


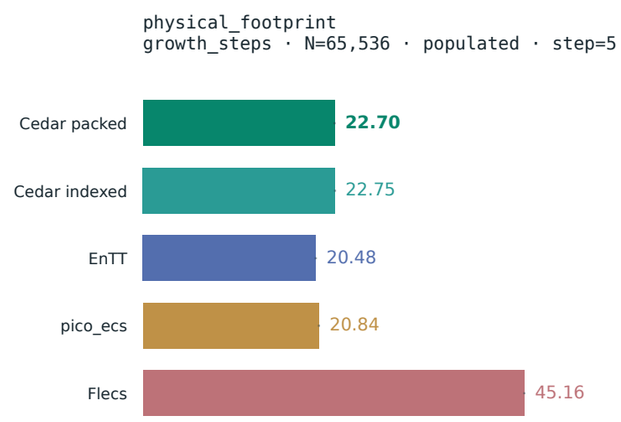


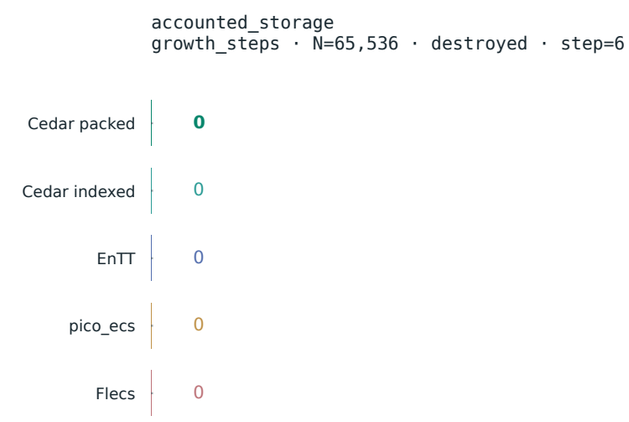
Memory comparison

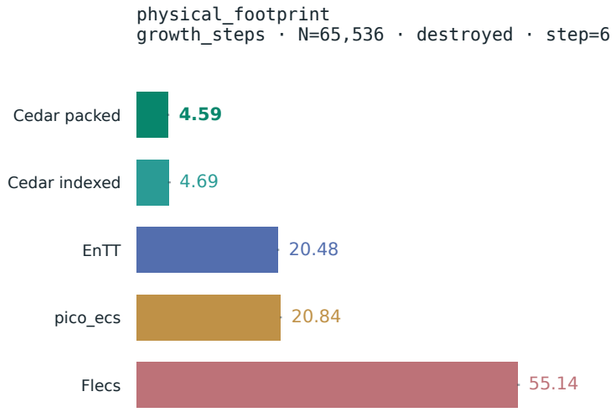
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

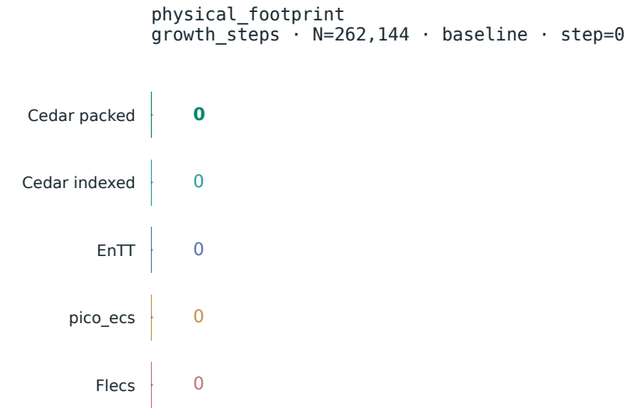


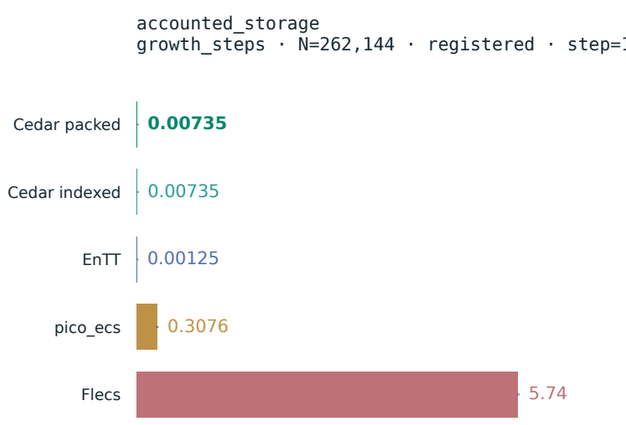


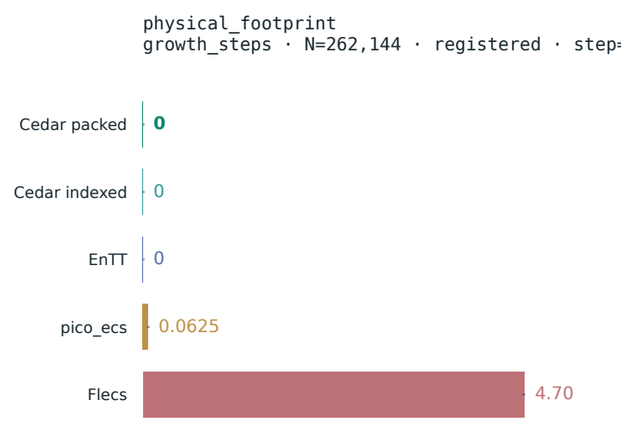


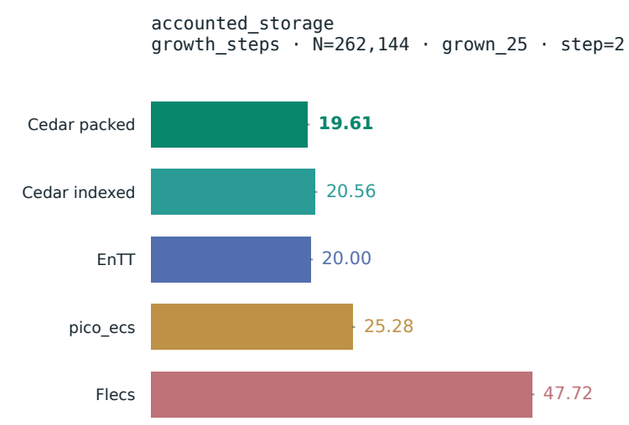


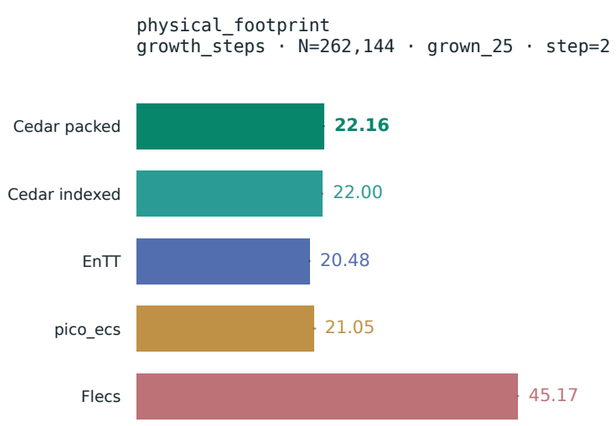


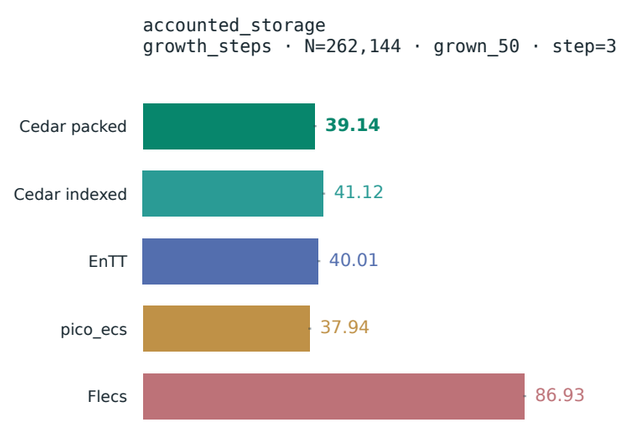


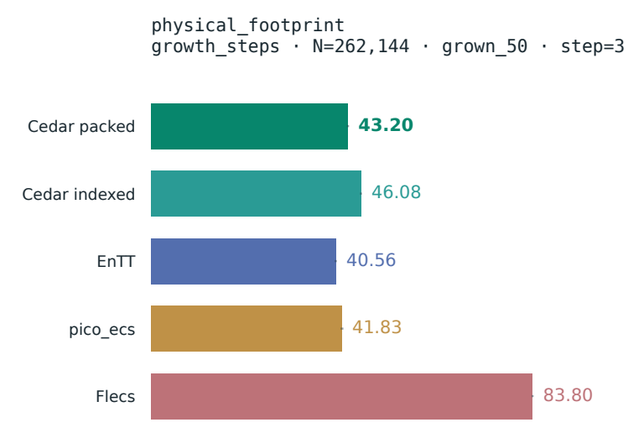








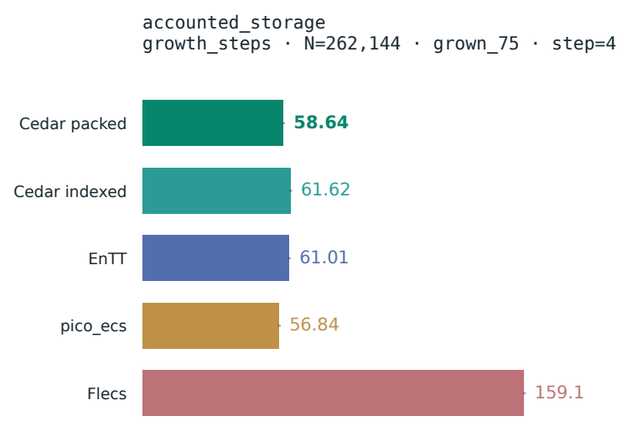


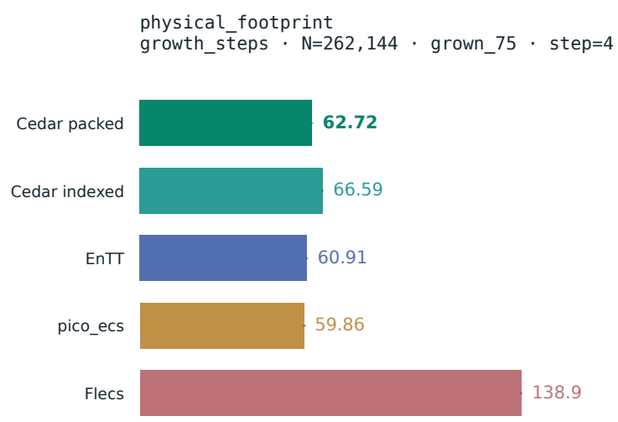


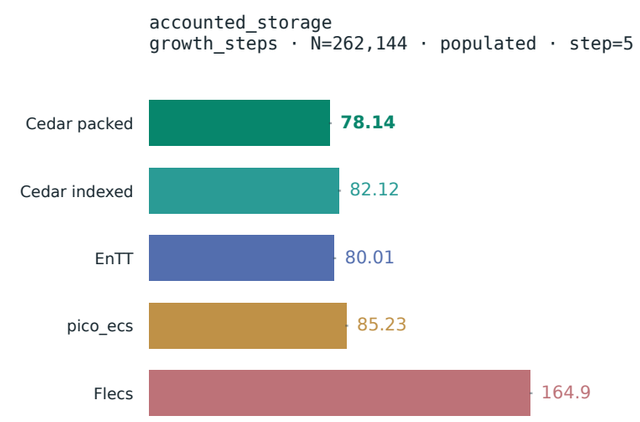
Memory comparison

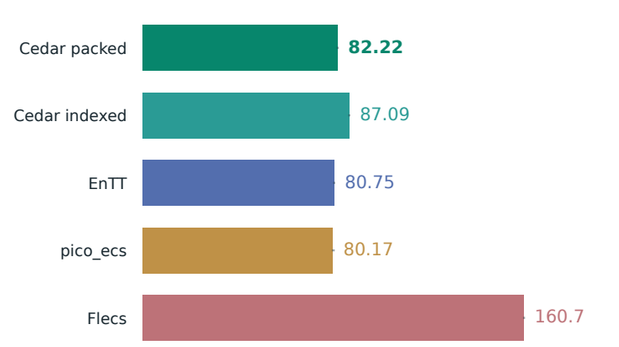
MiB ↓

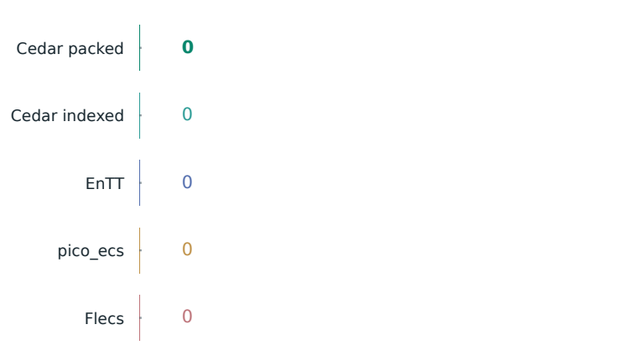
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

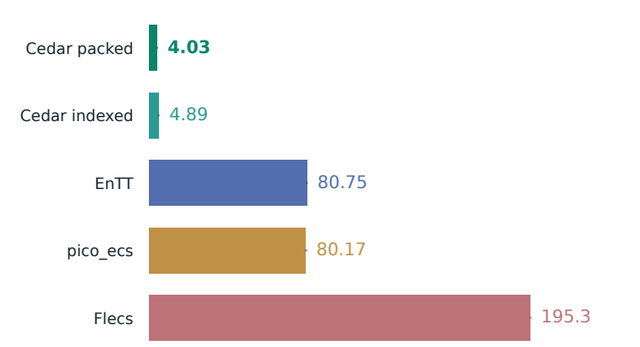


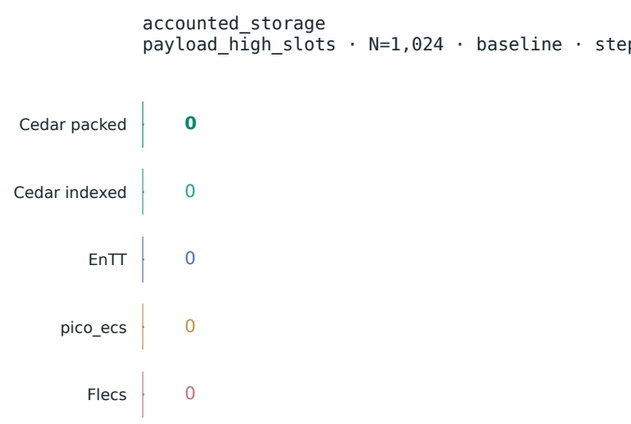




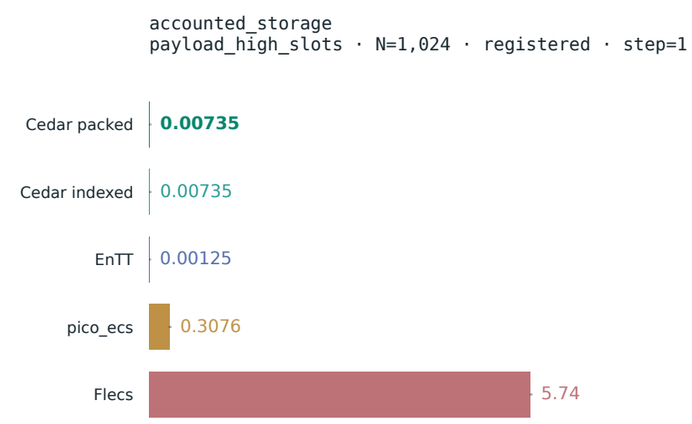


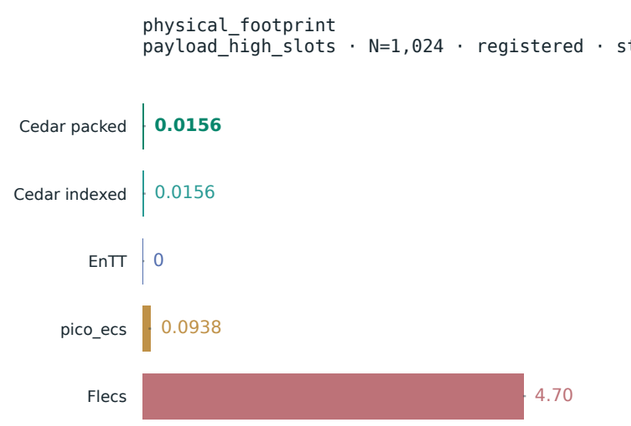


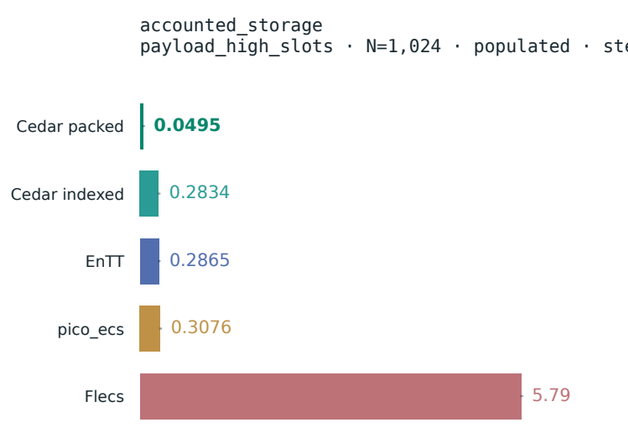


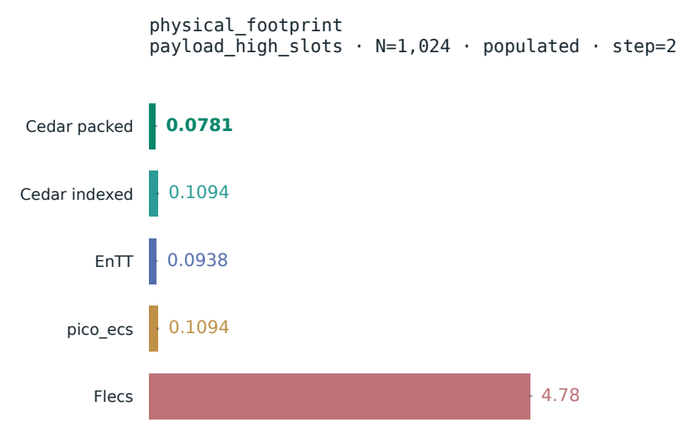










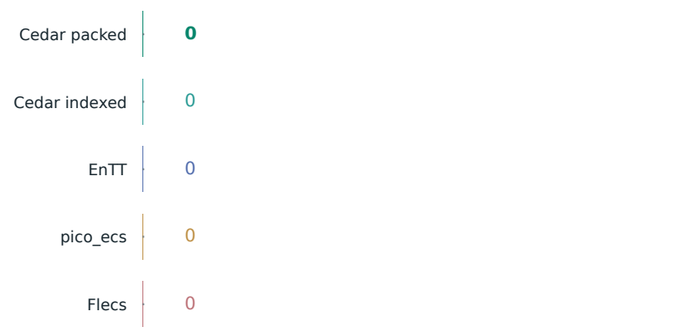


Memory comparison

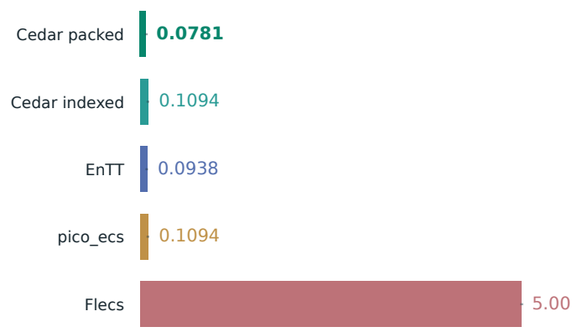
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

accounted_storage
payload_high_slots · N=1,024 · destroyed · step=3



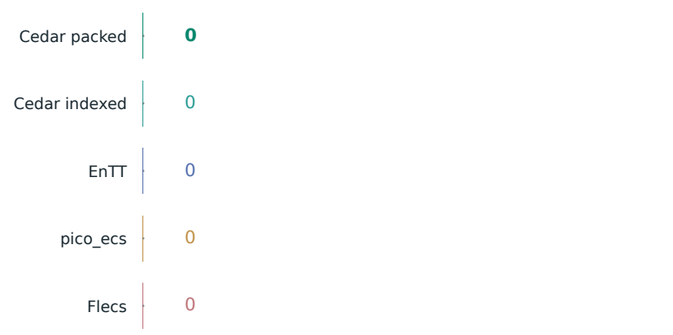
physical_footprint
payload_high_slots · N=1,024 · destroyed · step=3



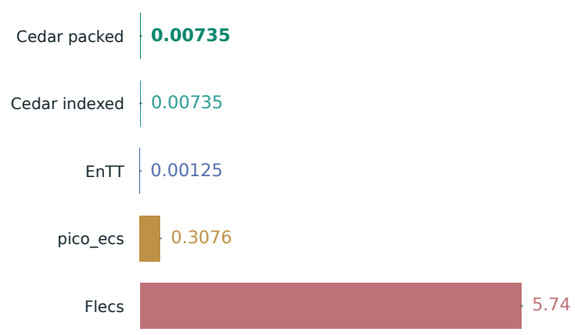
accounted_storage
payload_high_slots · N=65,536 · baseline · step=0



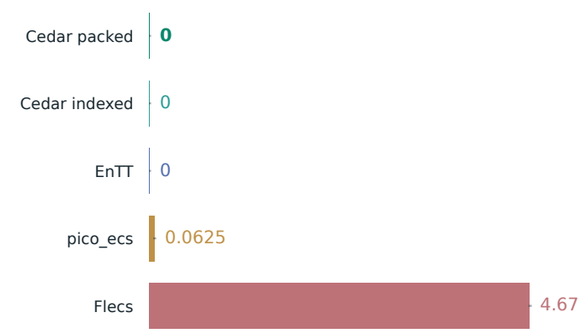
physical_footprint
payload_high_slots · N=65,536 · baseline · step=0



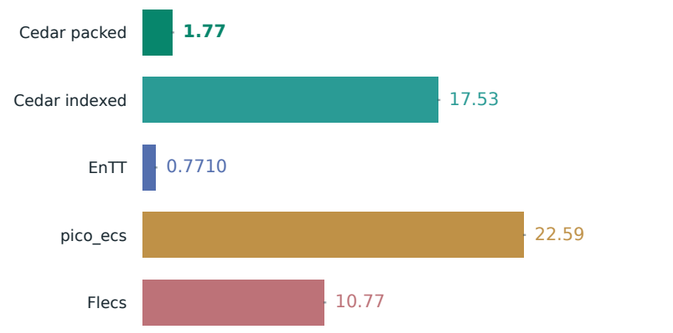
accounted_storage
payload_high_slots · N=65,536 · registered · step=1



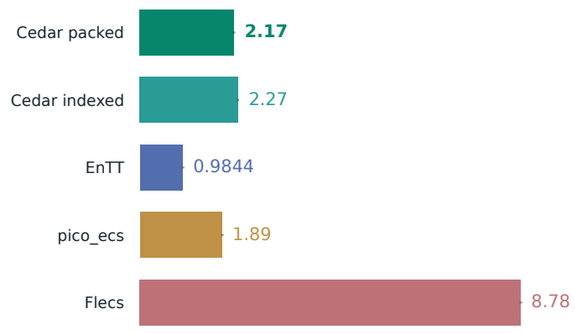
physical_footprint
payload_high_slots · N=65,536 · registered · step=1



accounted_storage
payload_high_slots · N=65,536 · populated · step=2



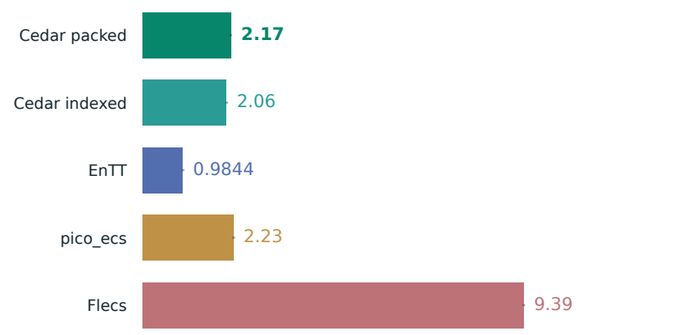
physical_footprint
payload_high_slots · N=65,536 · populated · step=2



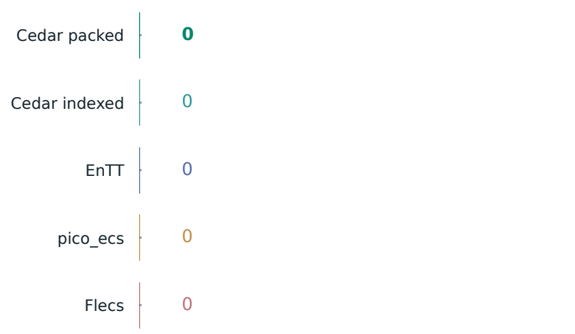
accounted_storage
payload_high_slots · N=65,536 · destroyed · step=3



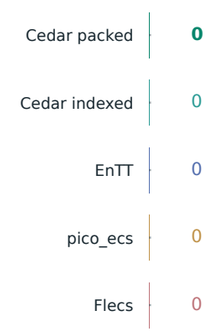
physical_footprint
payload_high_slots · N=65,536 · destroyed · step=3



accounted_storage
payload_high_slots · N=262,144 · baseline · step=0



physical_footprint
payload_high_slots · N=262,144 · baseline · step=0

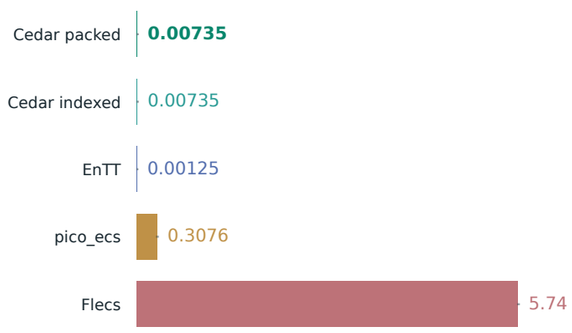


Memory comparison

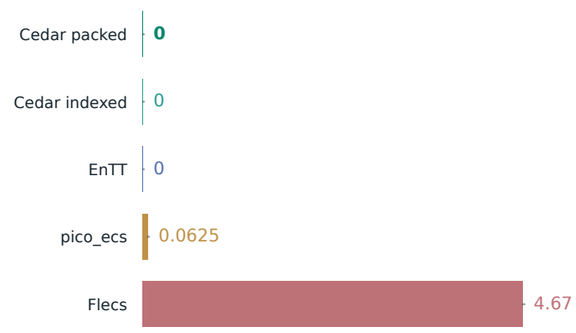
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

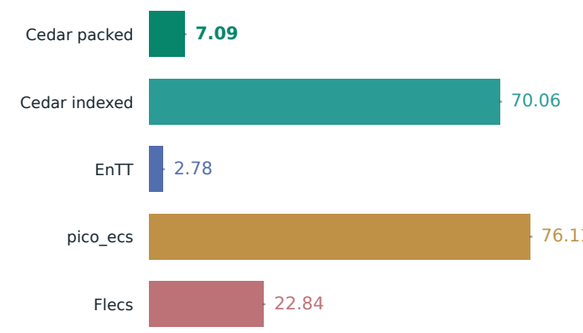
accounted_storage
payload_high_slots · N=262,144 · registered · step=1



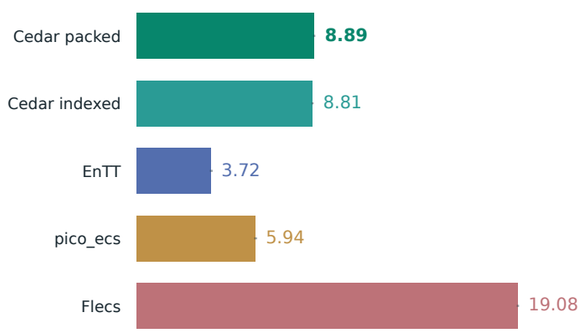
physical_footprint
payload_high_slots · N=262,144 · registered · step=1



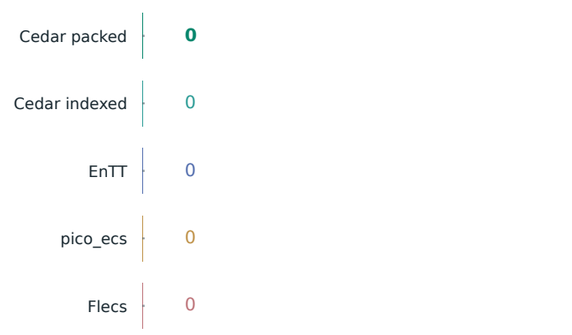
accounted_storage
payload_high_slots · N=262,144 · populated · step=2



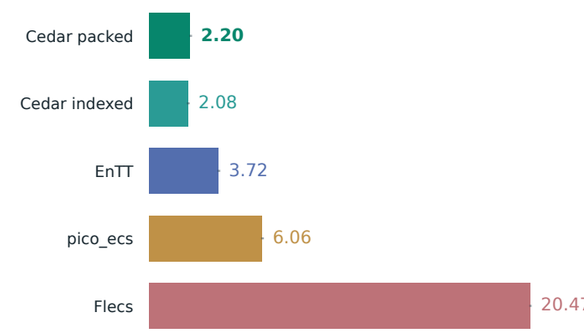
physical_footprint
payload_high_slots · N=262,144 · populated · step=2



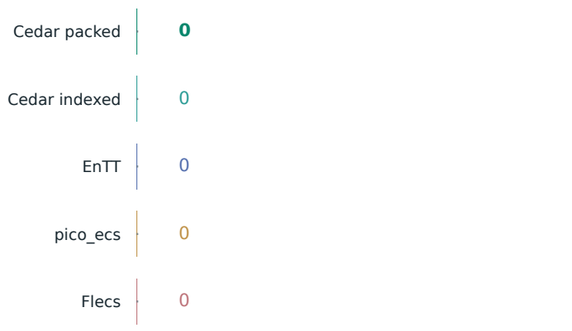
accounted_storage
payload_high_slots · N=262,144 · destroyed · step=3



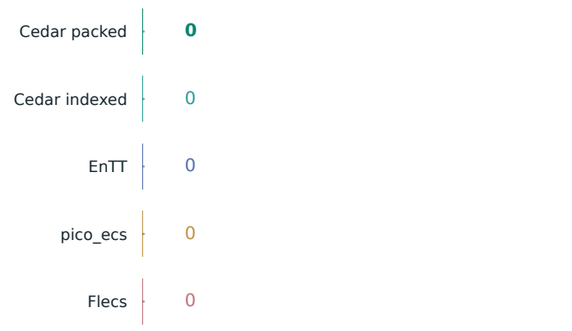
physical_footprint
payload_high_slots · N=262,144 · destroyed · step=3



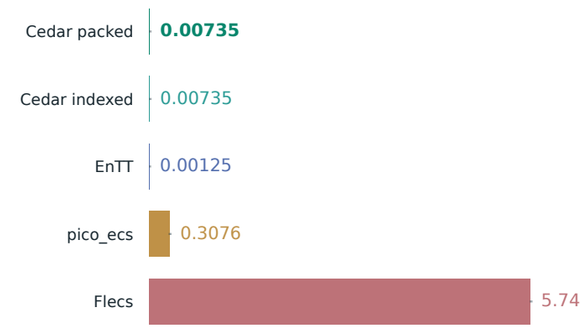
accounted_storage
payload_low_slots · N=1,024 · baseline · step=0



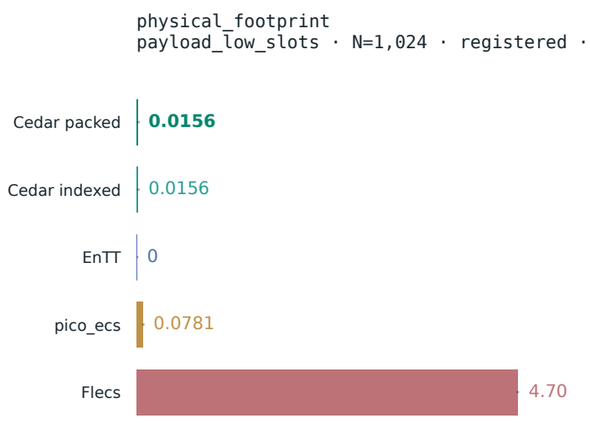
physical_footprint
payload_low_slots · N=1,024 · baseline · step=0



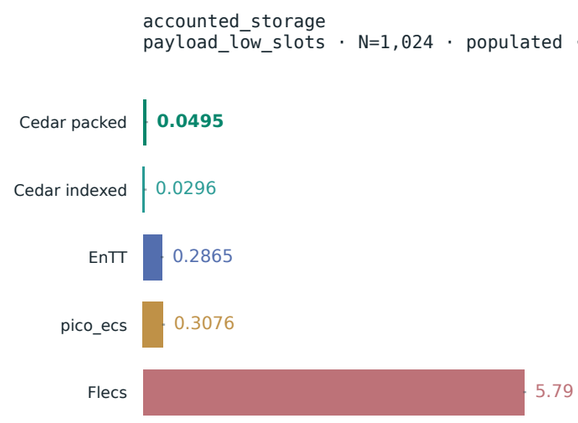
accounted_storage
payload_low_slots · N=1,024 · registered · step=1



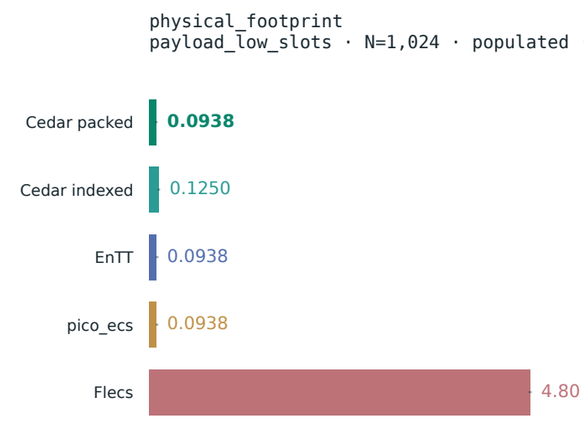
physical_footprint
payload_low_slots · N=1,024 · registered · step=1



accounted_storage
payload_low_slots · N=1,024 · populated · step=2



physical_footprint
payload_low_slots · N=1,024 · populated · step=2

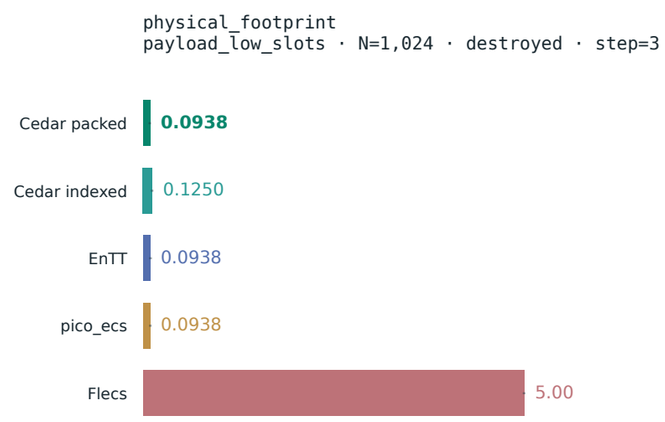


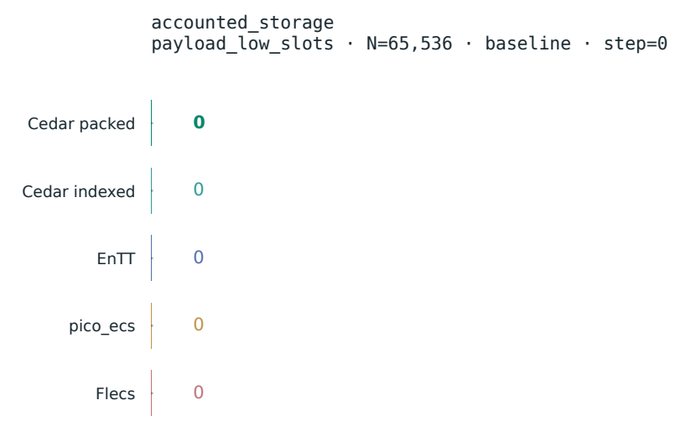
Memory comparison

MiB ↓

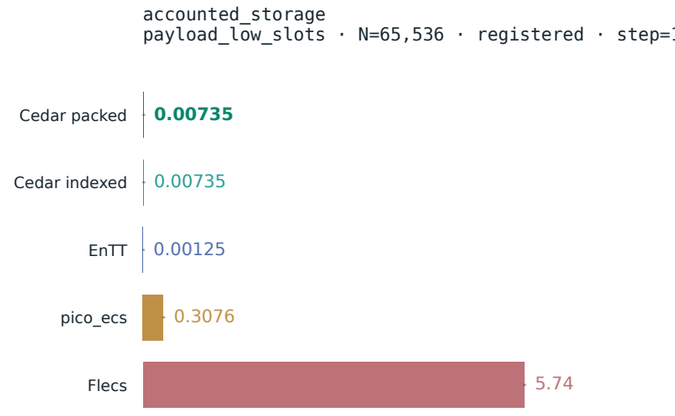
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

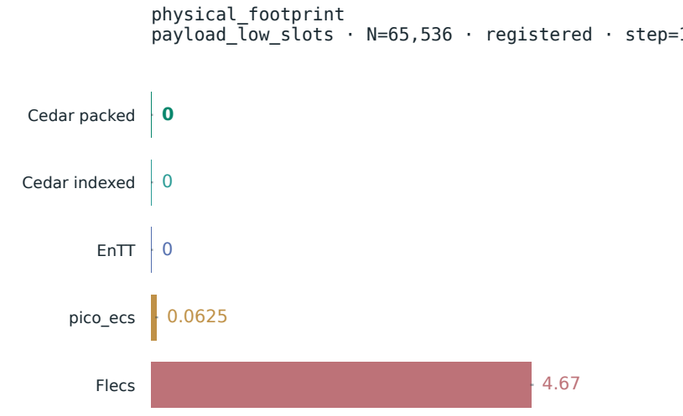


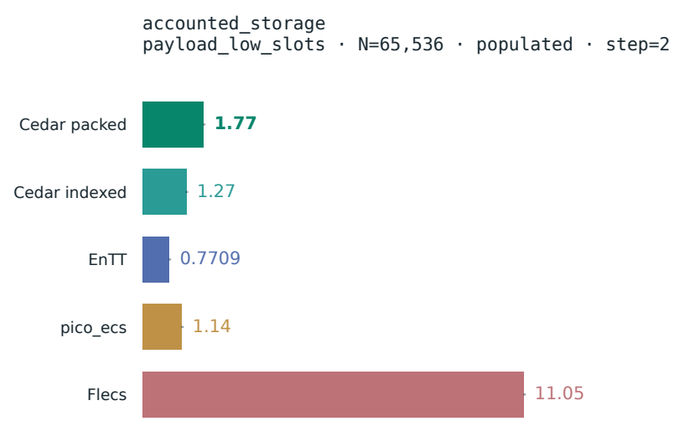


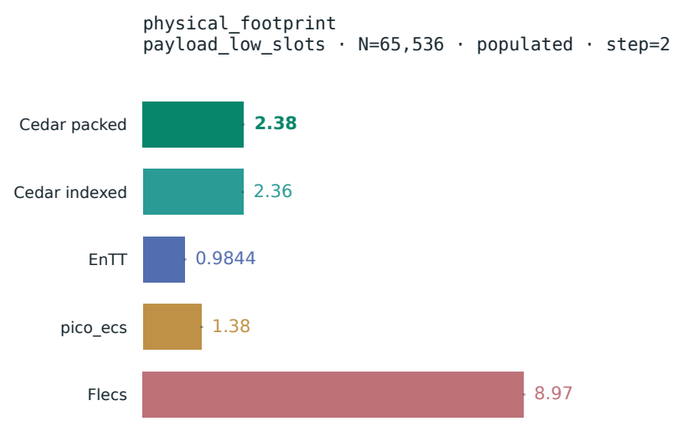




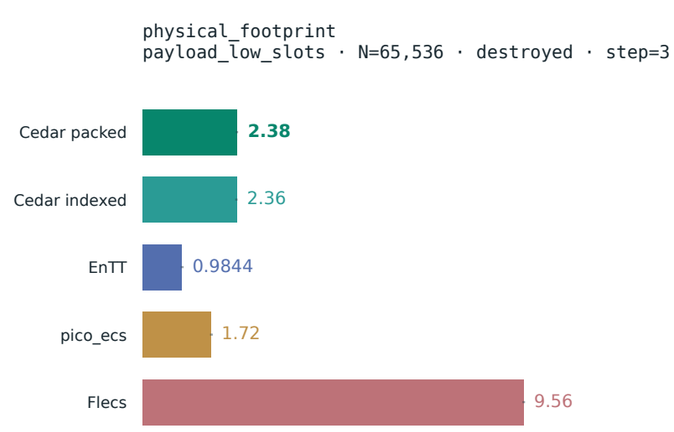




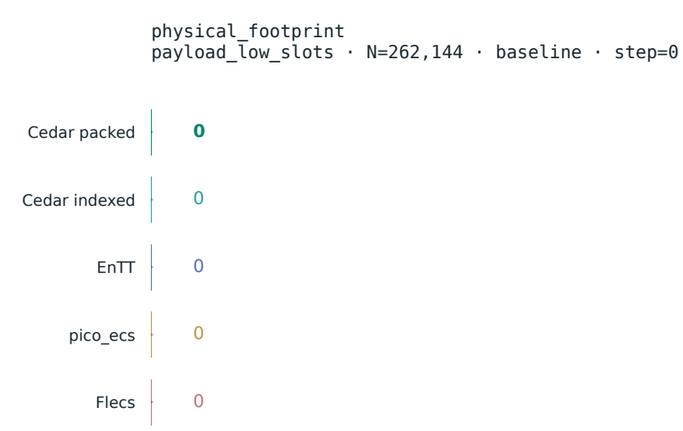








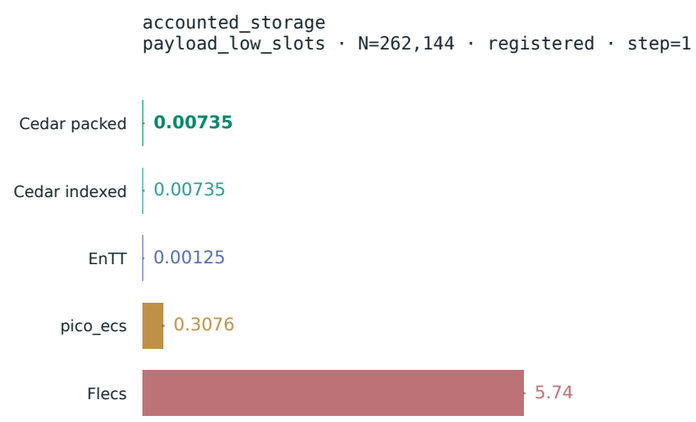


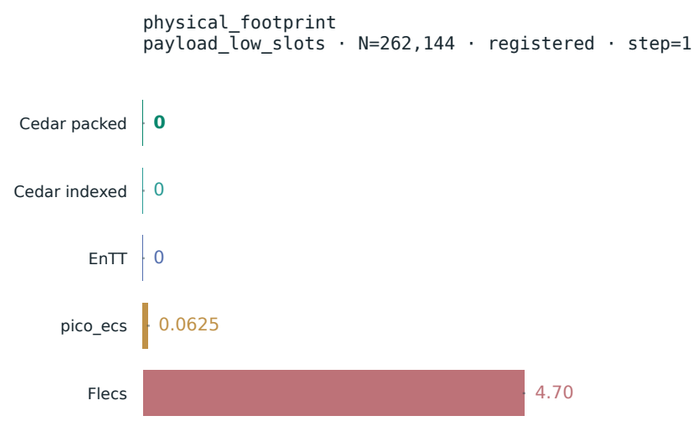


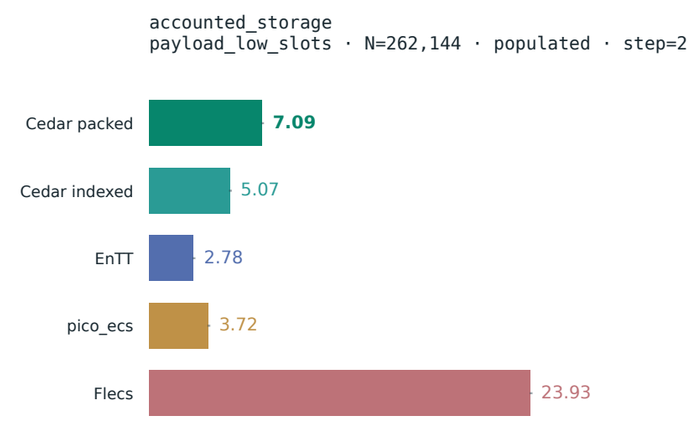
Memory comparison

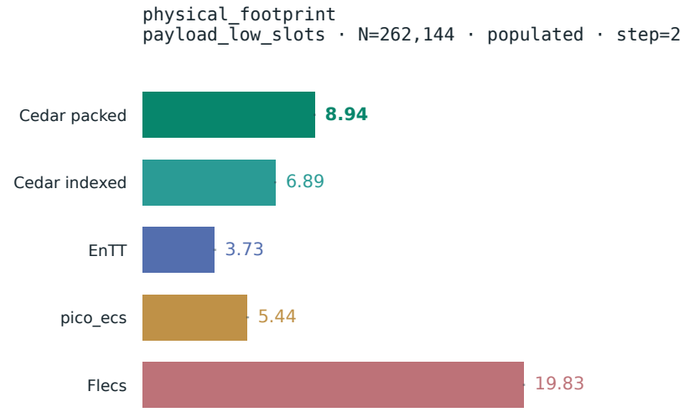
MiB ↓

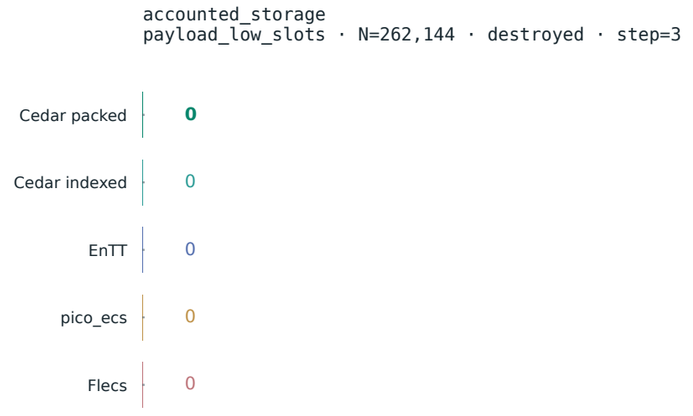
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

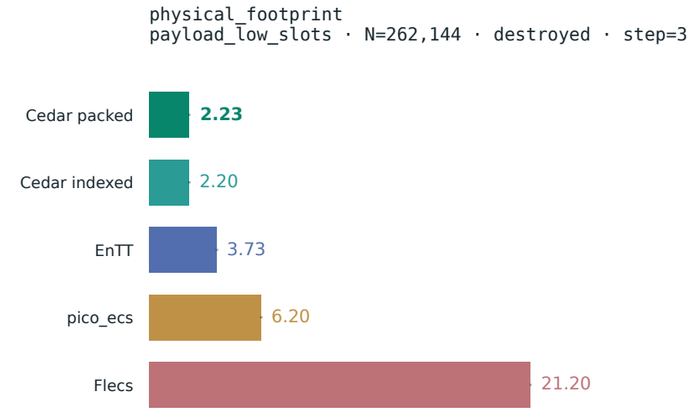




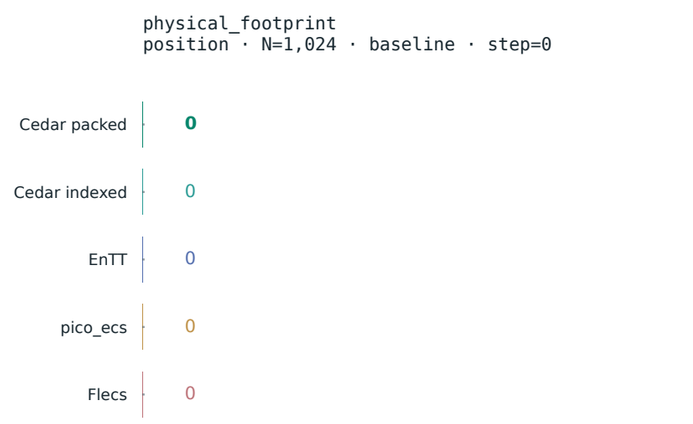


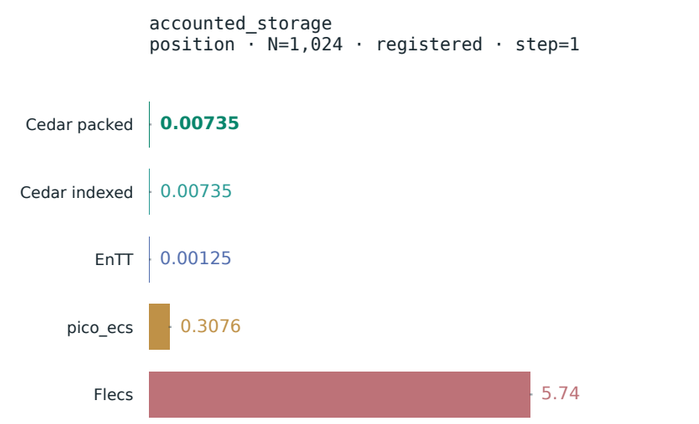


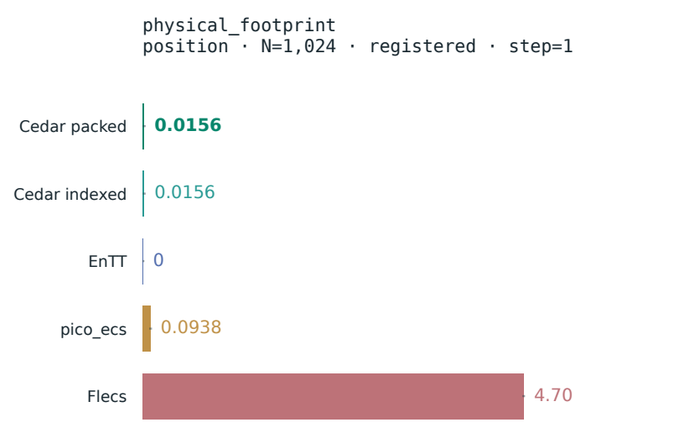


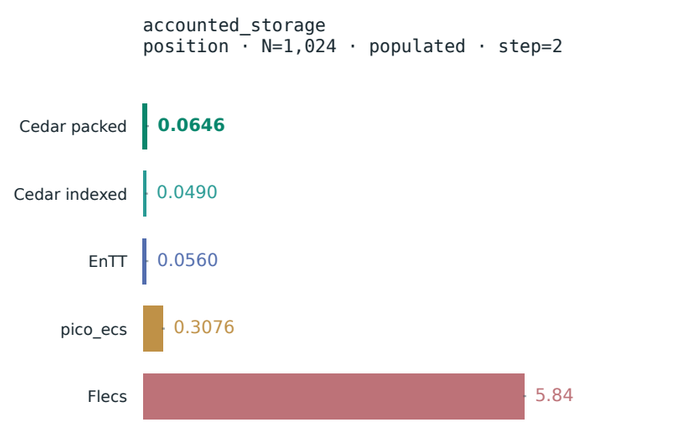


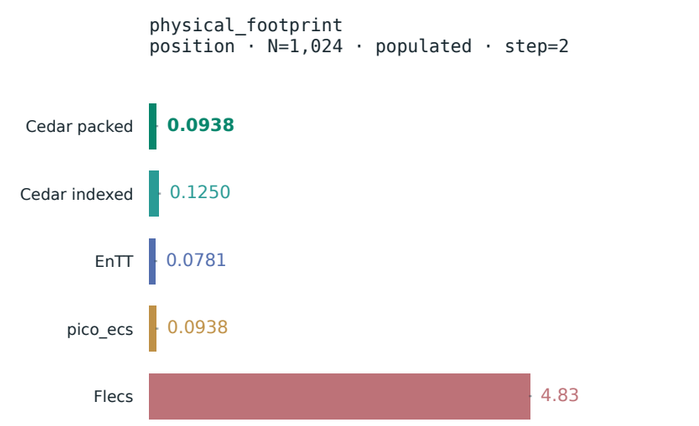








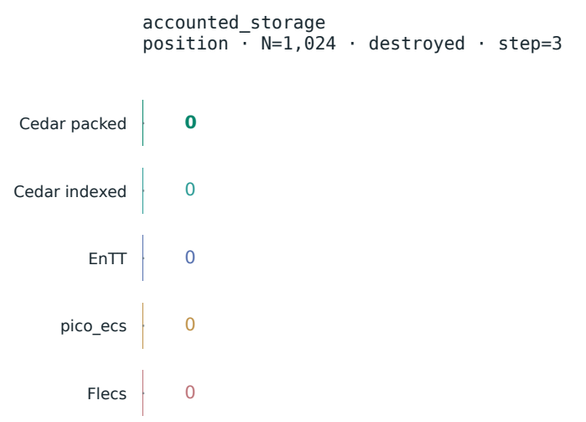


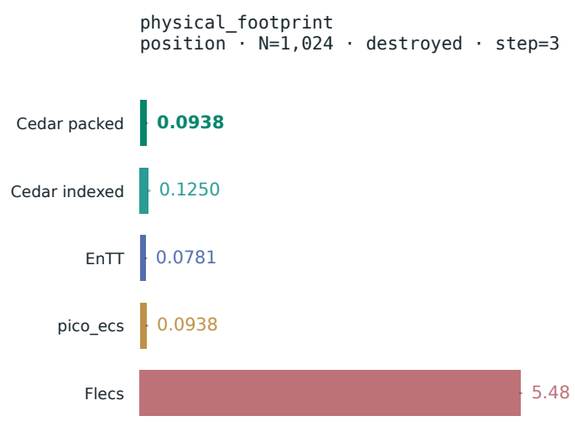


Memory comparison

MiB ↓

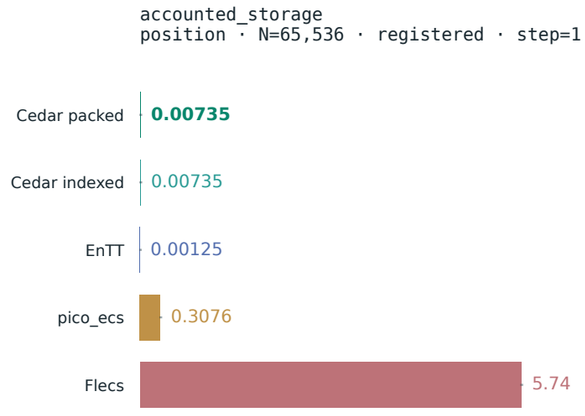
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

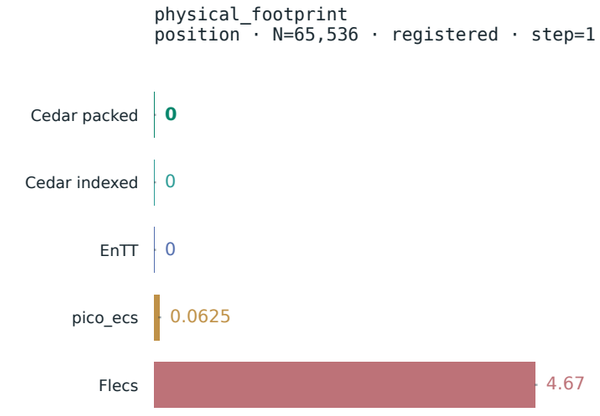


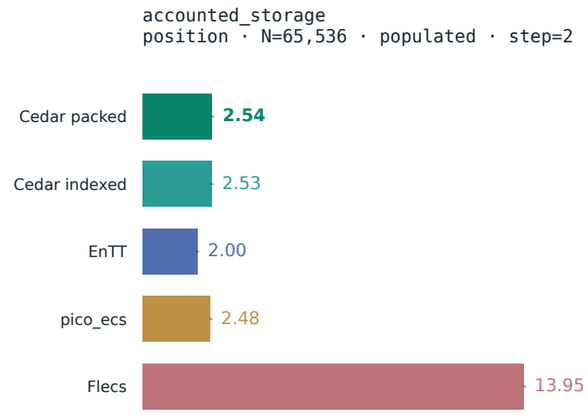


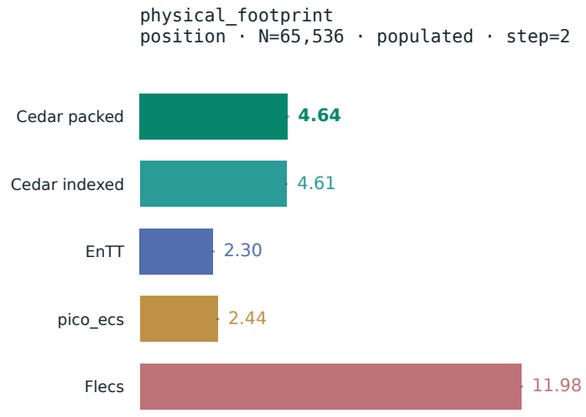




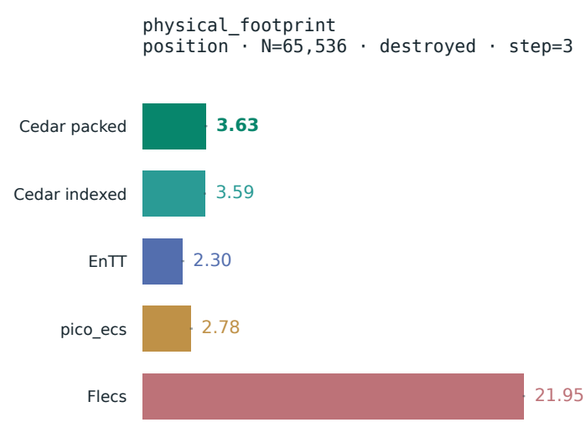












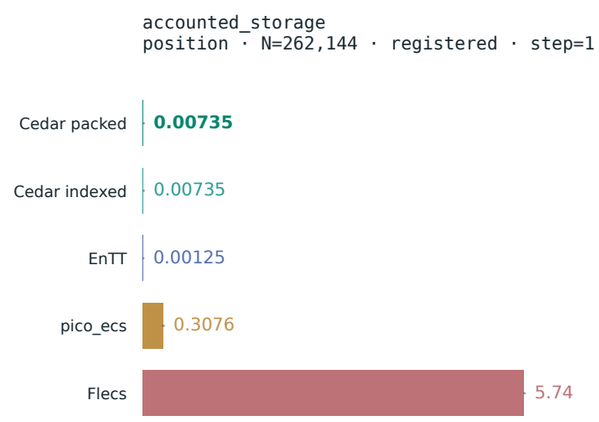


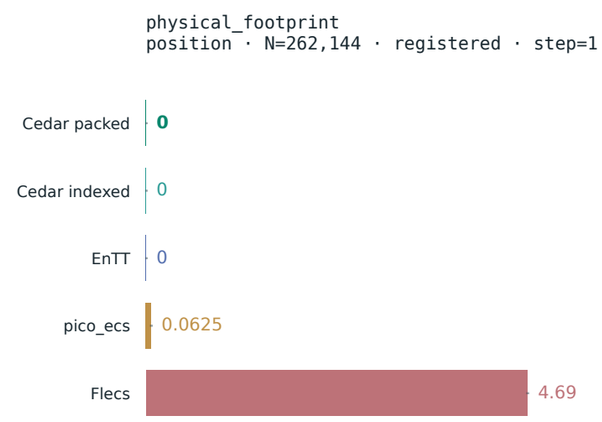


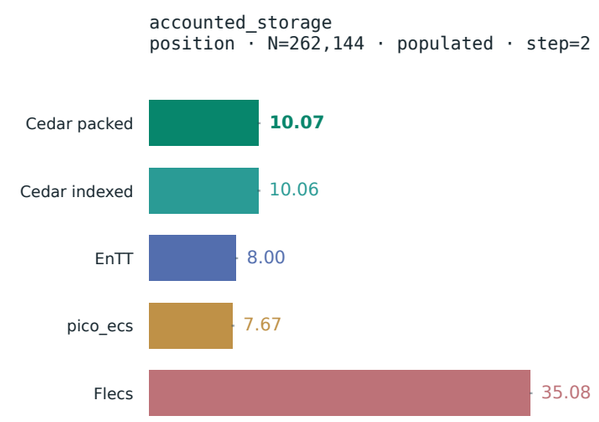
Memory comparison

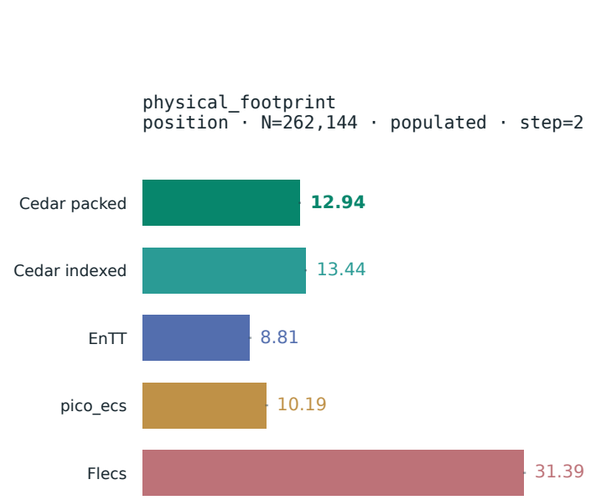
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

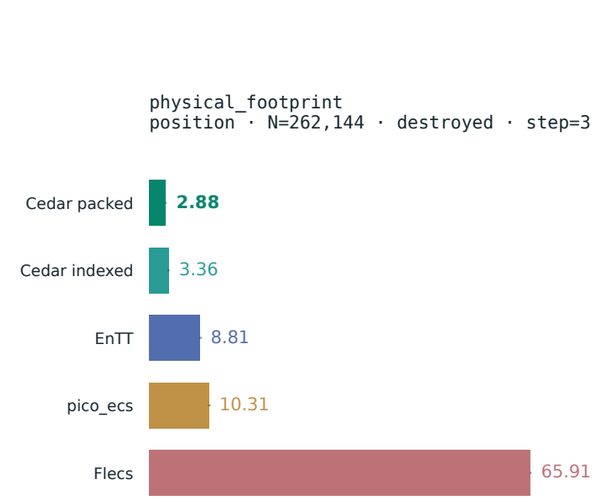






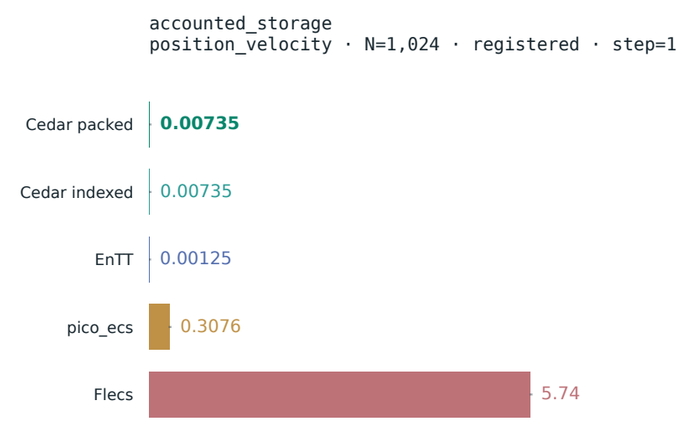


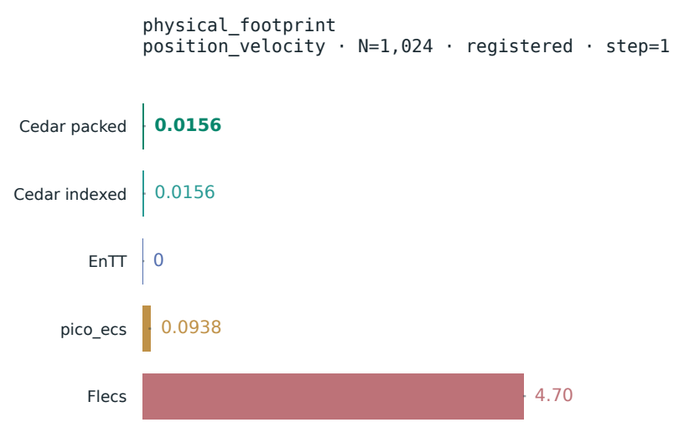


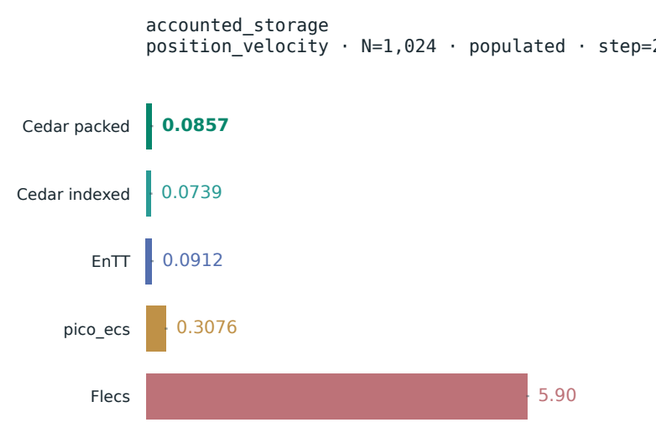


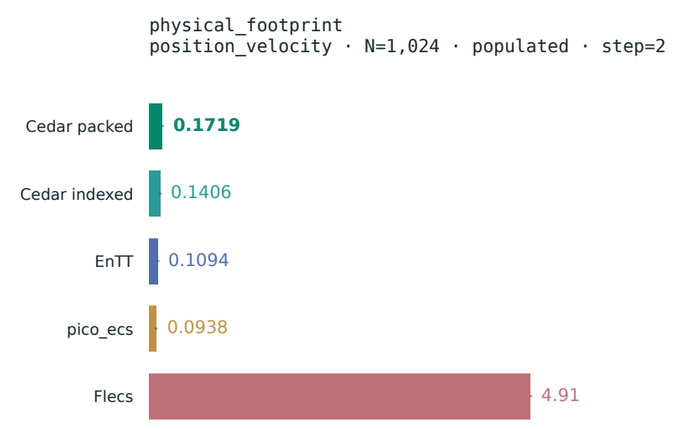










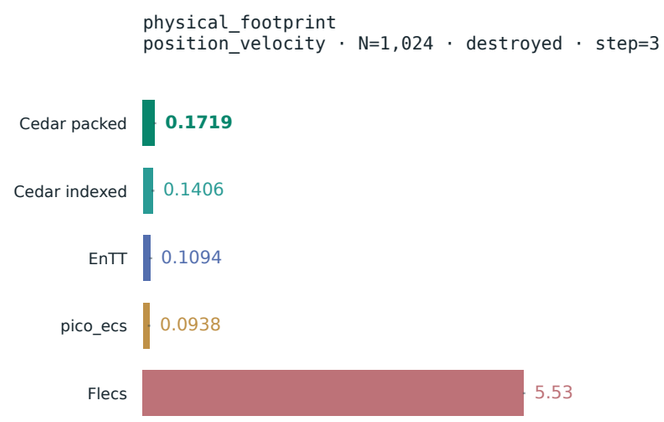


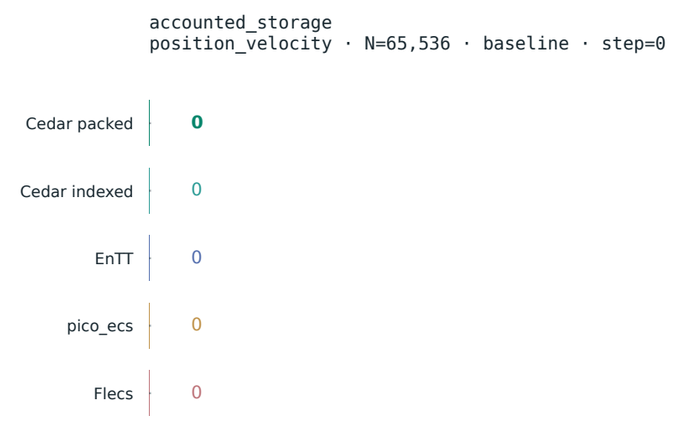
Memory comparison

MiB ↓

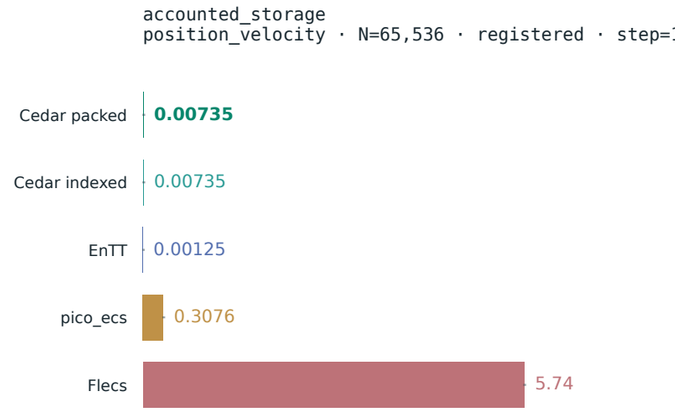
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

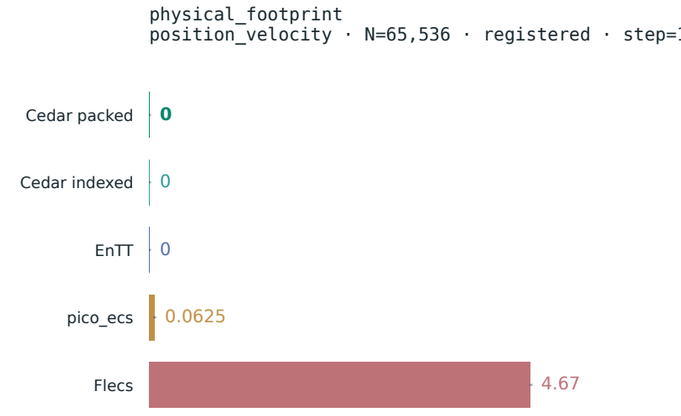


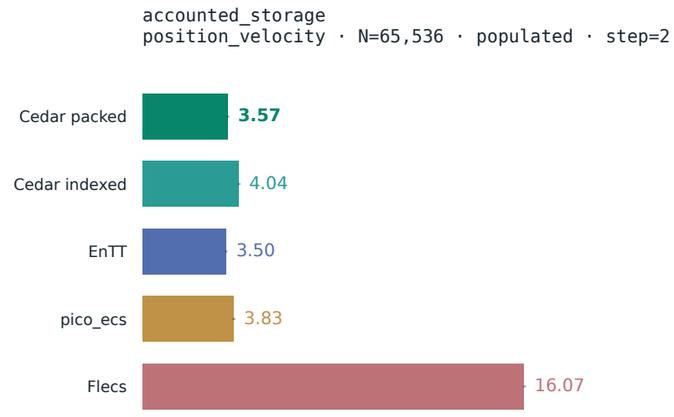


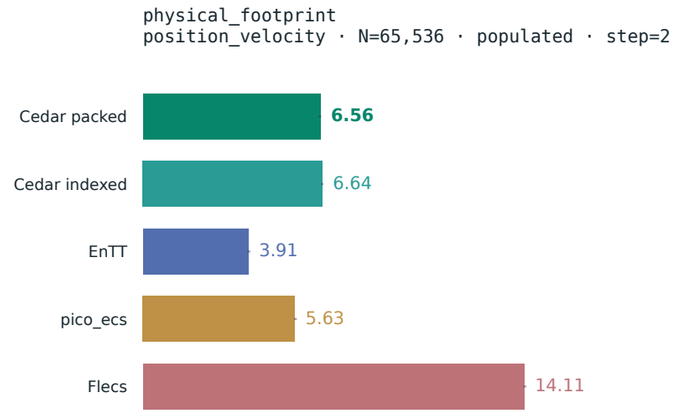




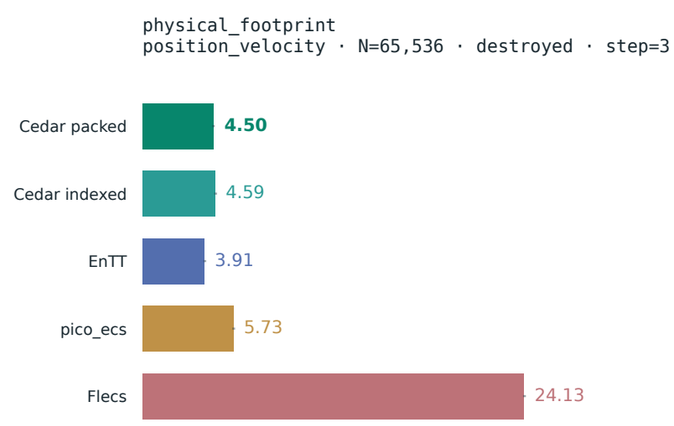




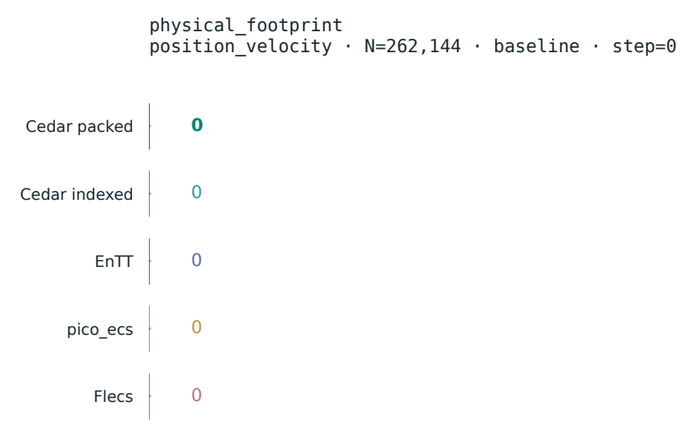








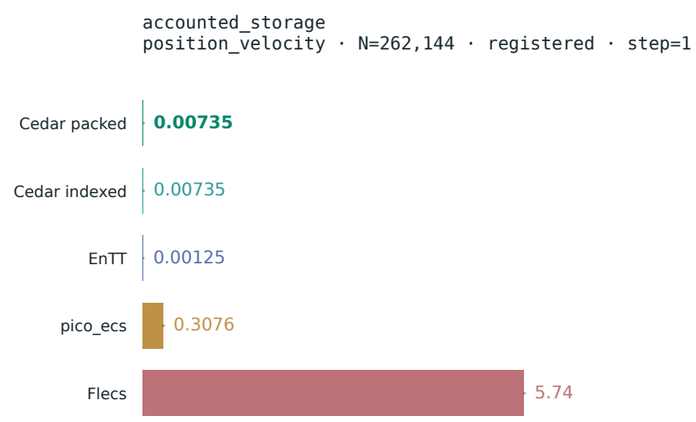


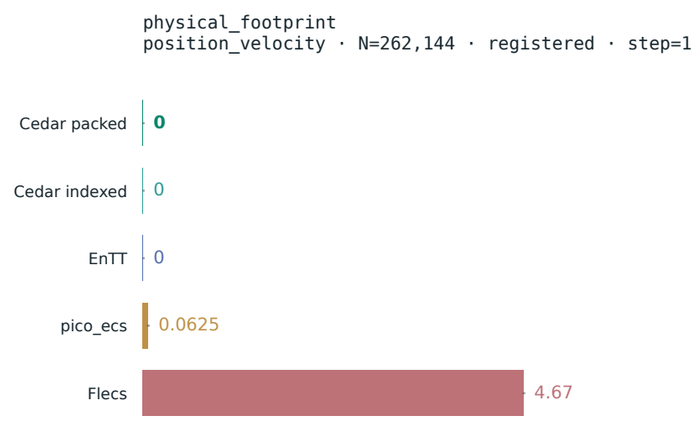


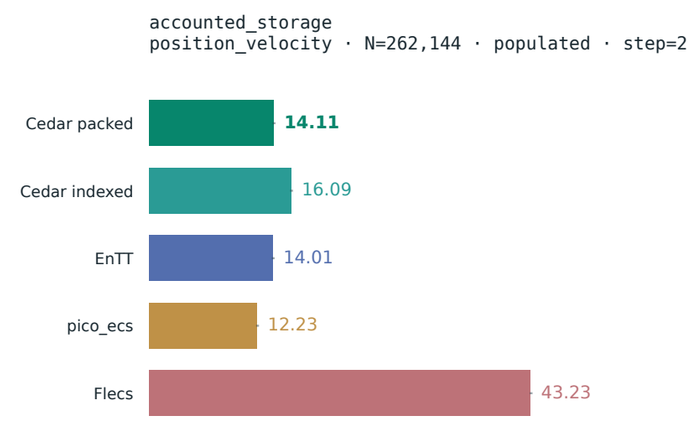
Memory comparison

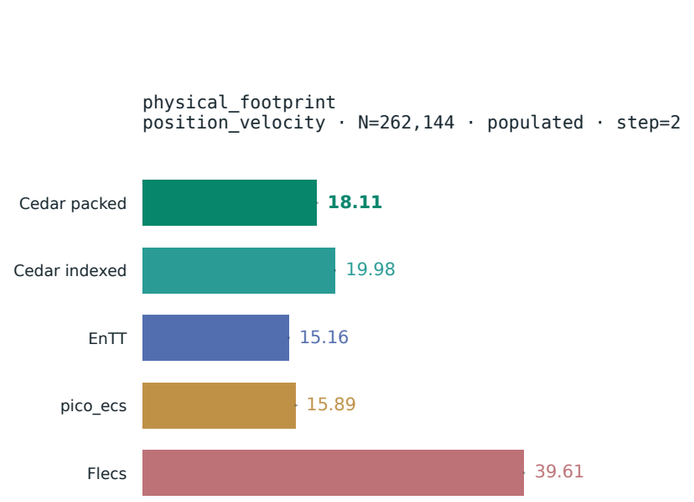
MiB ↓

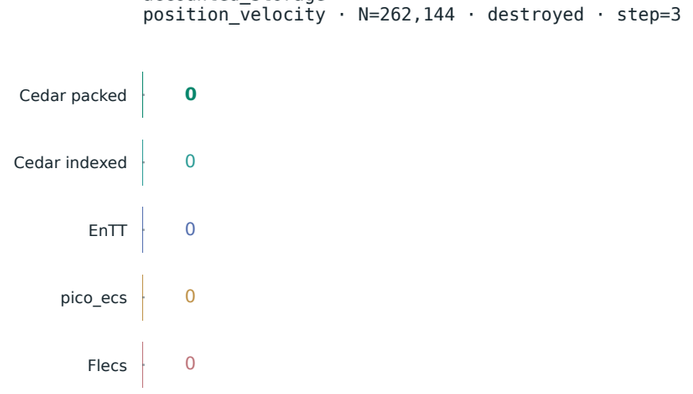
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

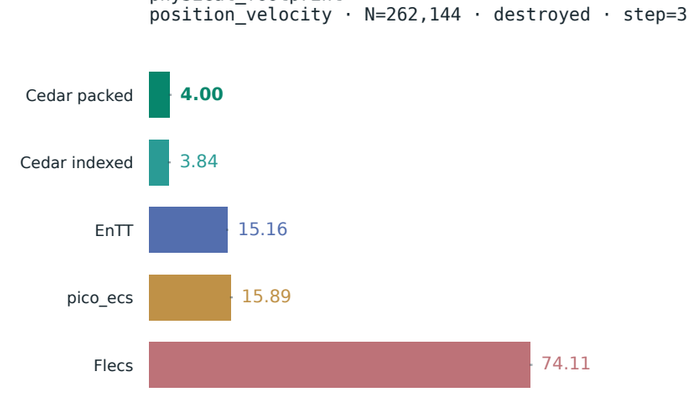




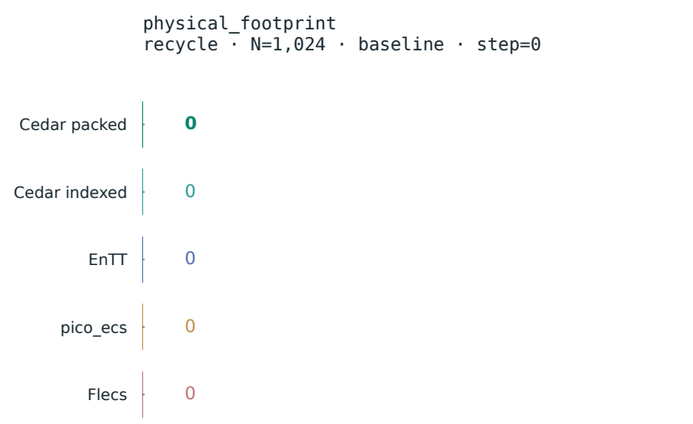


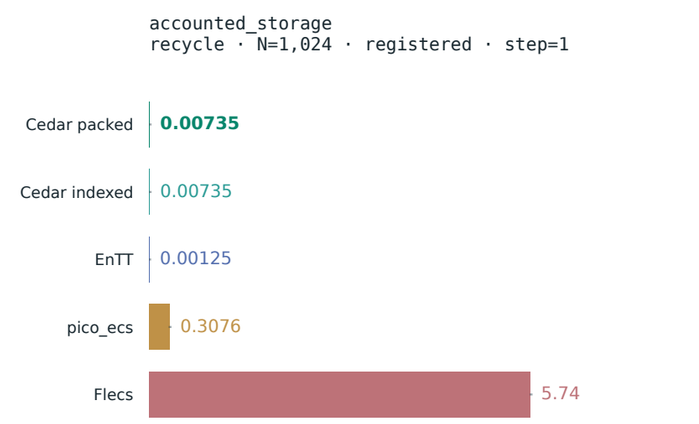


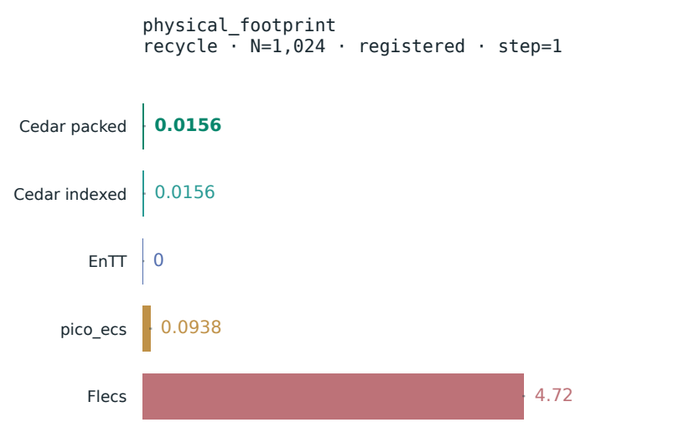


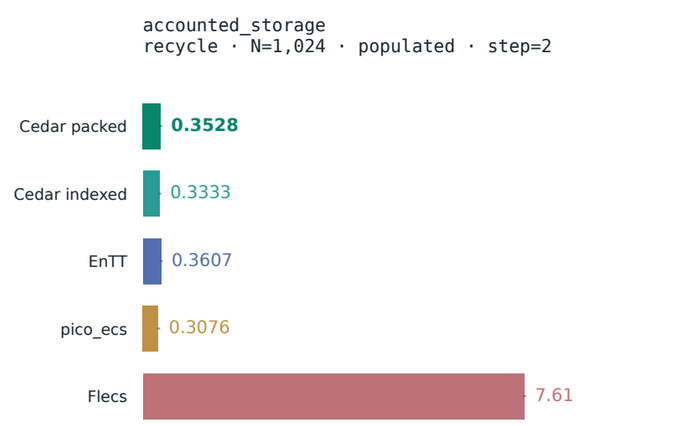


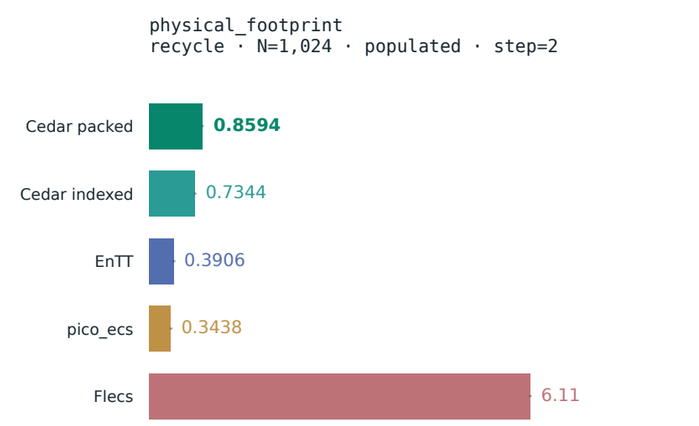








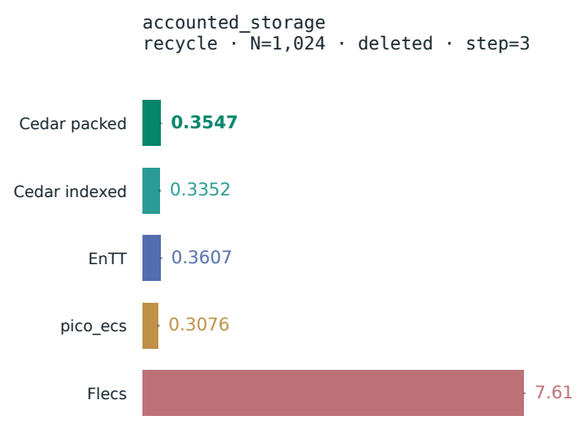


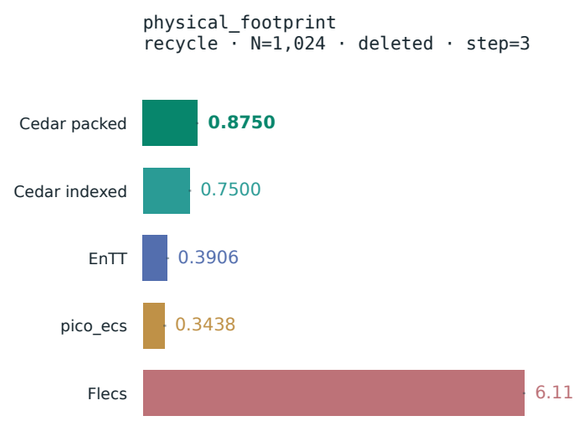


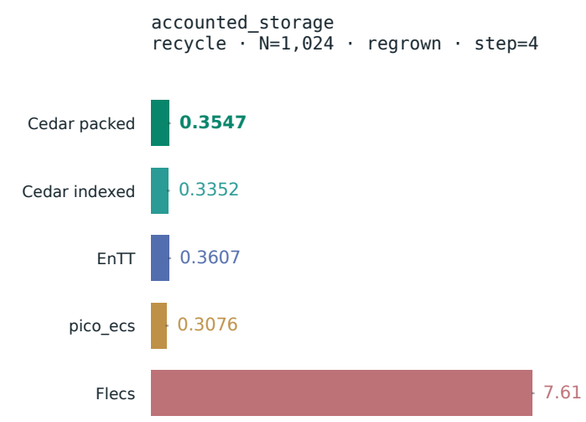
Memory comparison

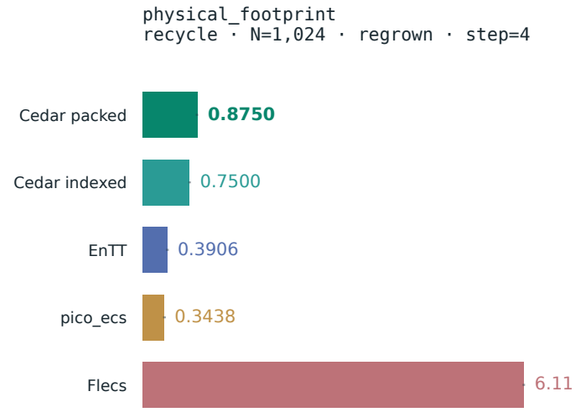
MiB ↓

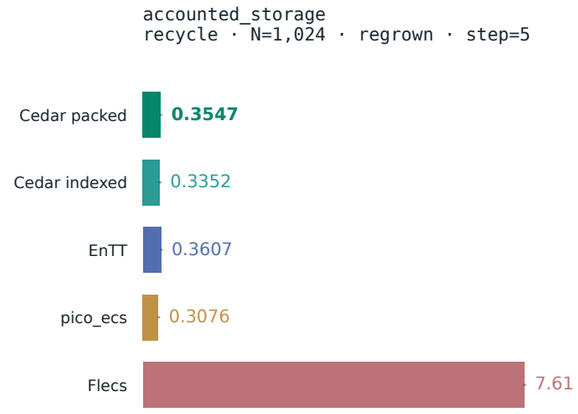
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

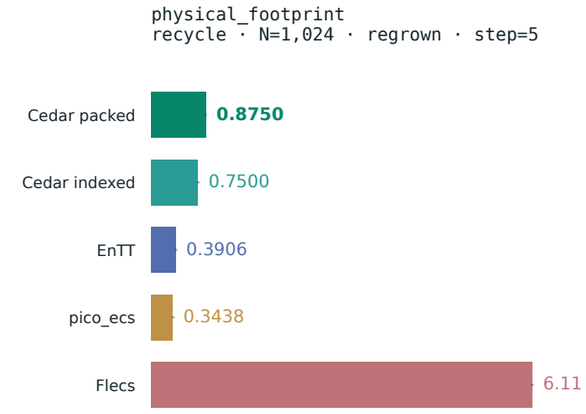


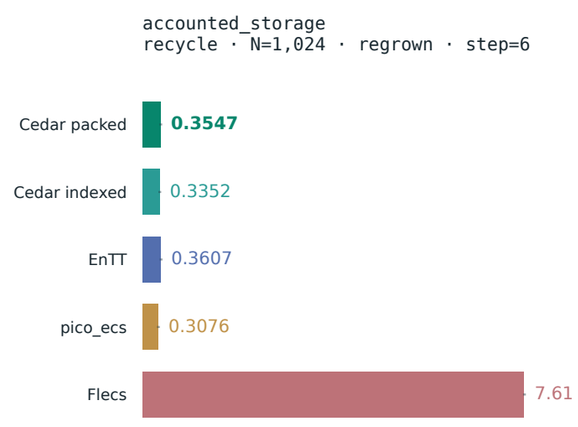


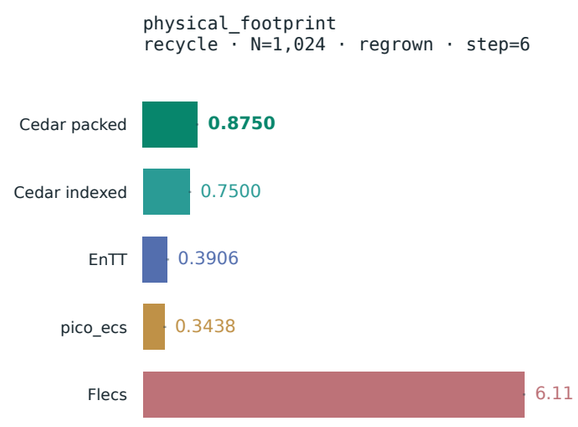


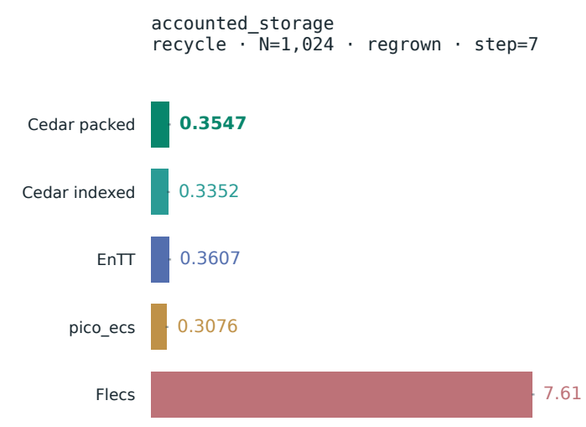


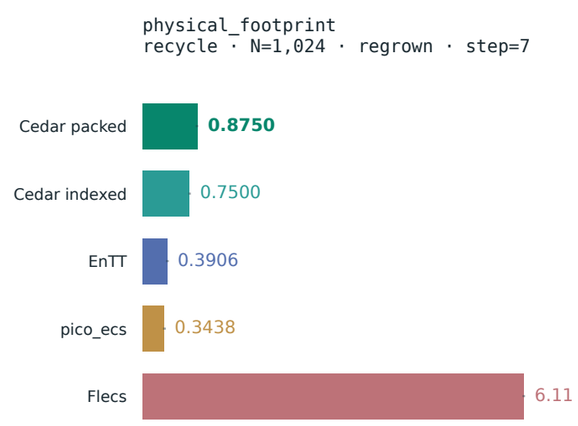


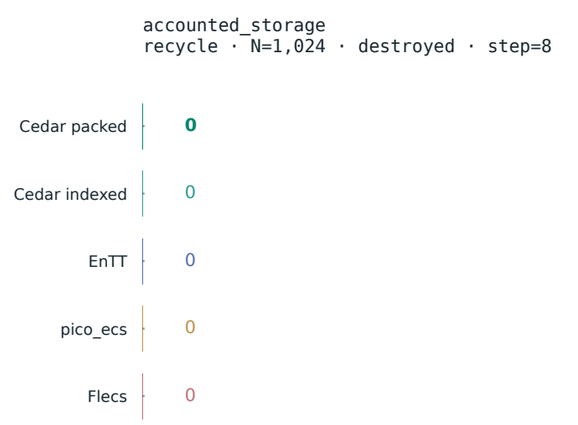


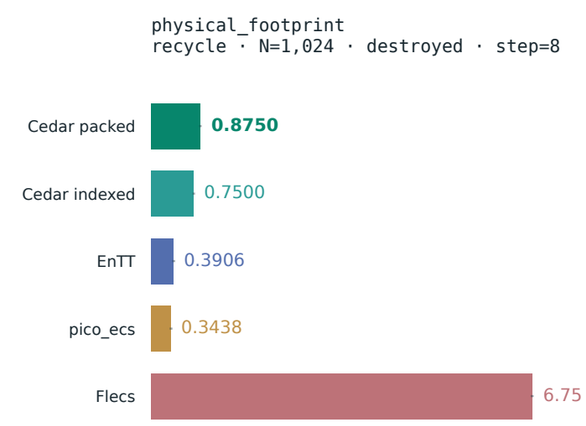








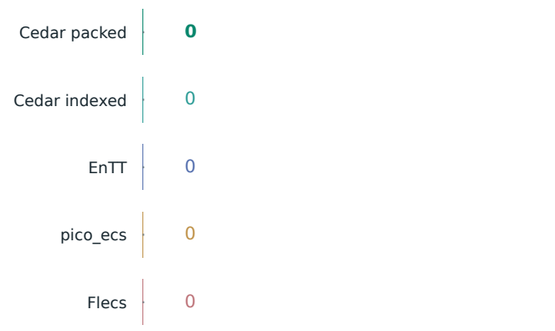


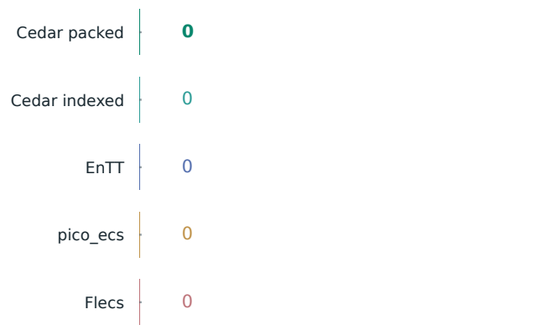


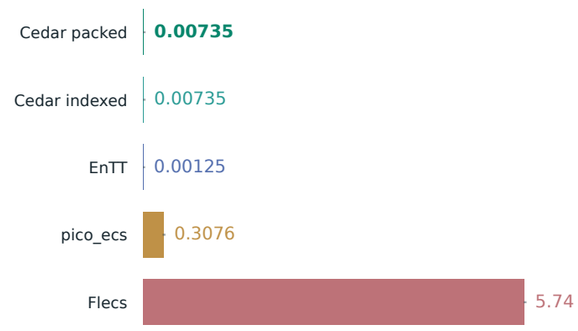
Memory comparison

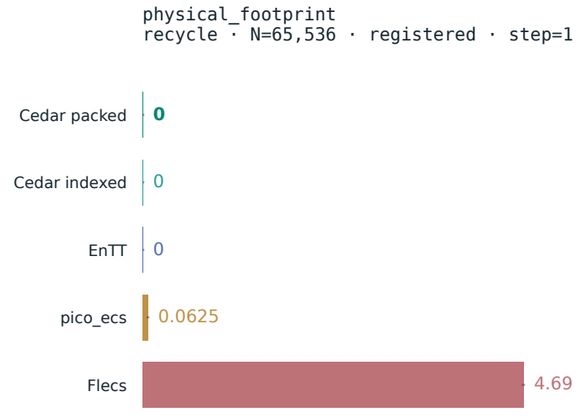
MiB ↓

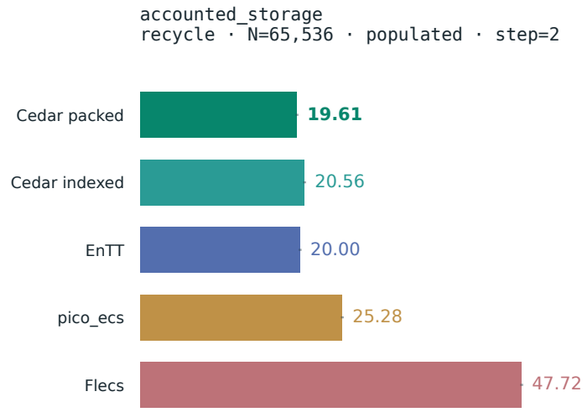
Cedar packed Cedar indexed EnTT pico_ecs Flecs

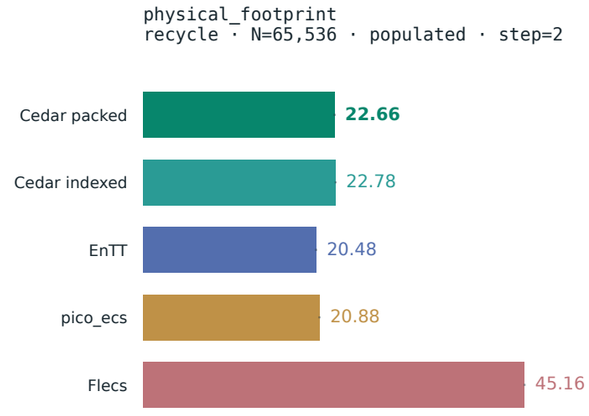


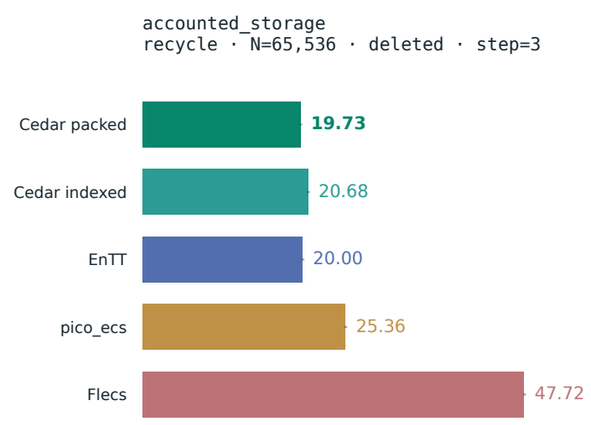


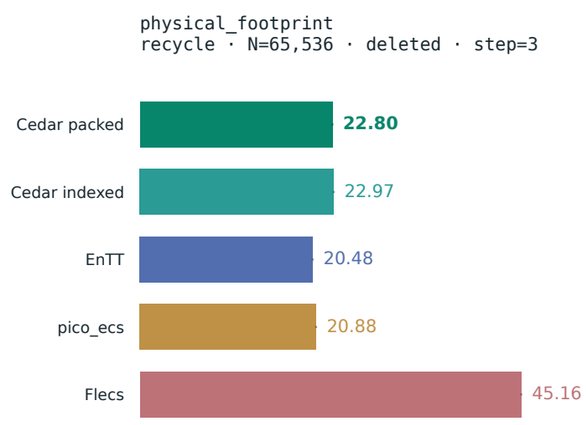


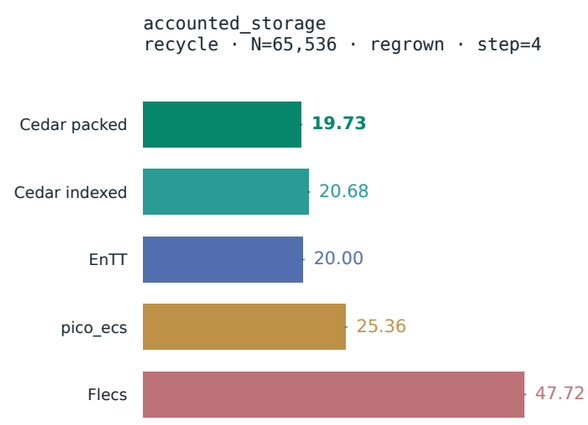


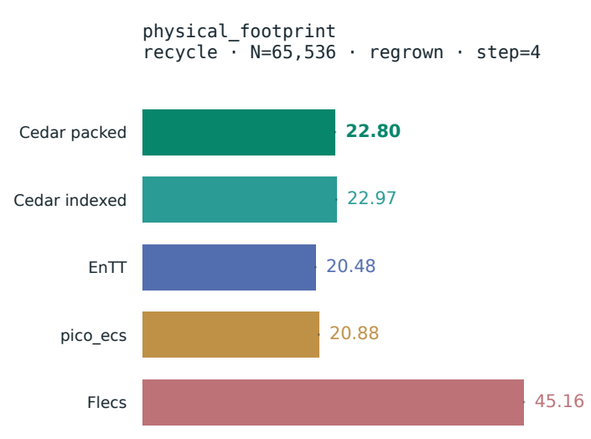


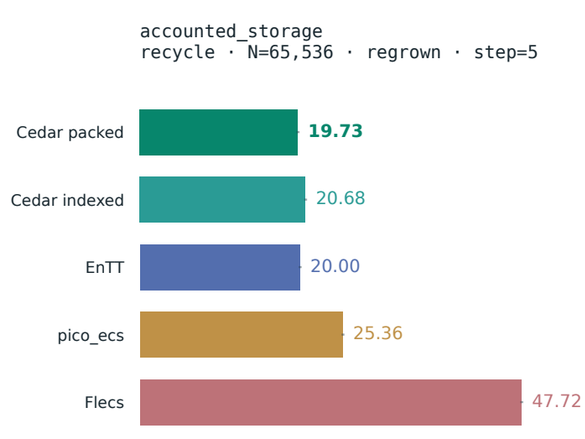


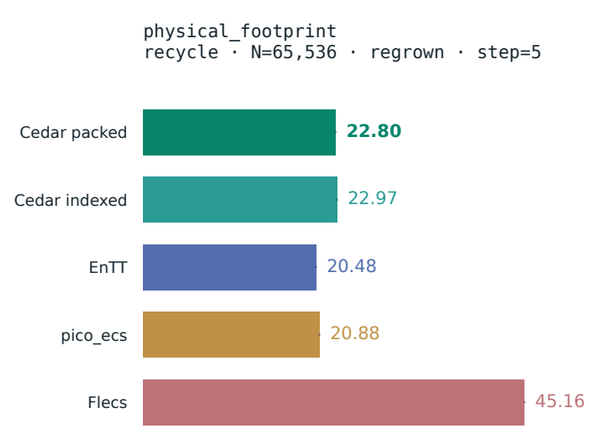








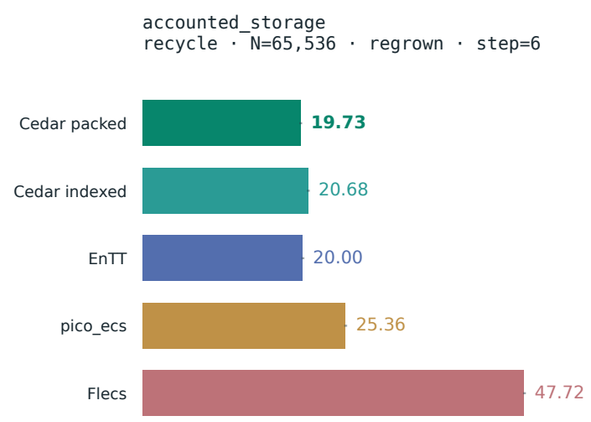


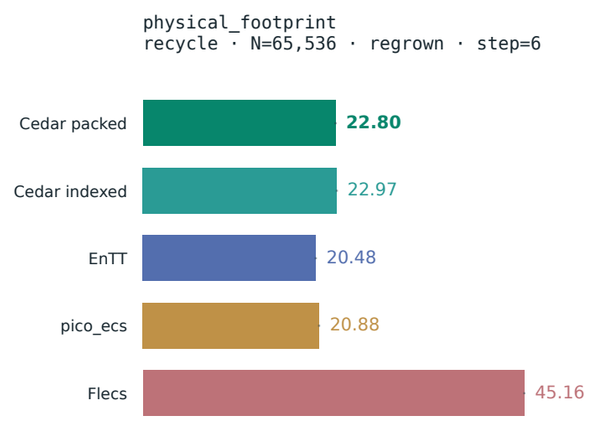


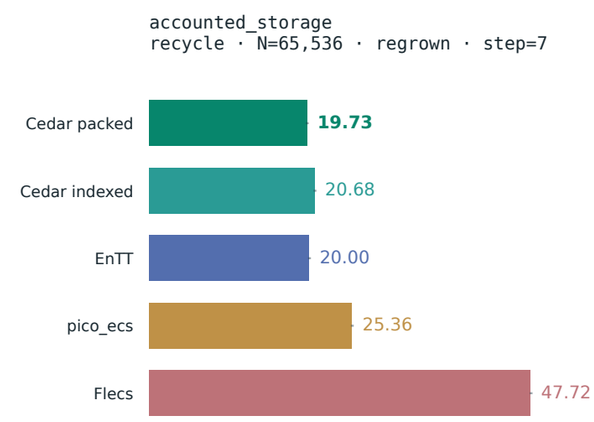
Memory comparison

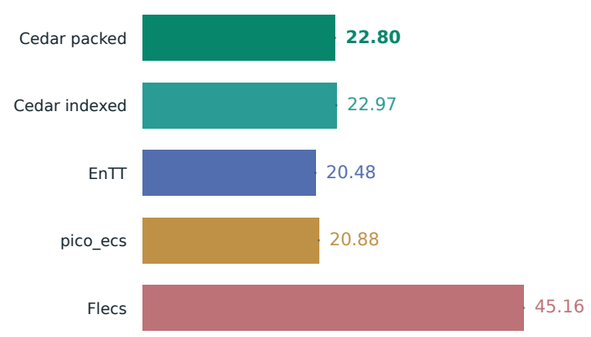
MiB ↓

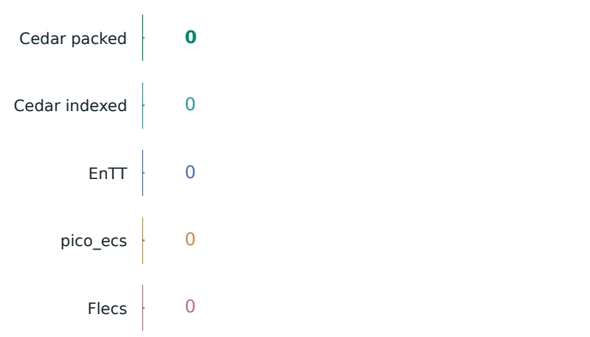
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

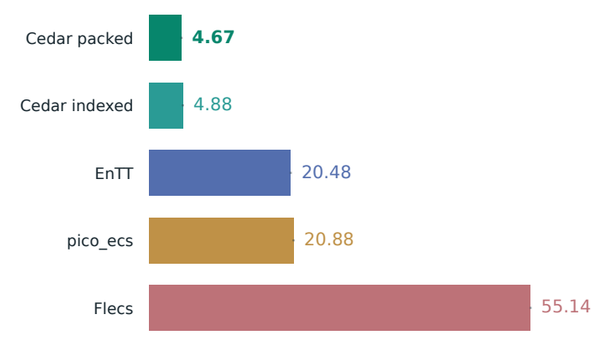






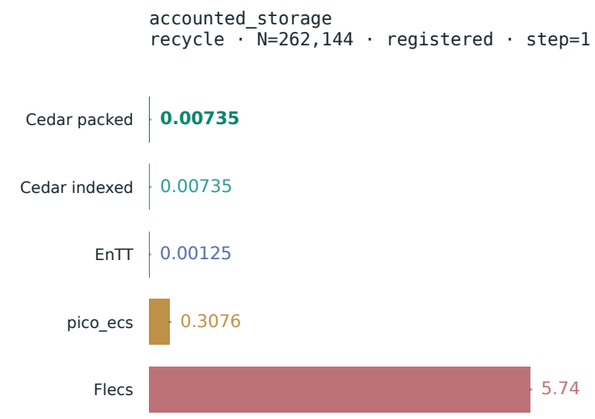


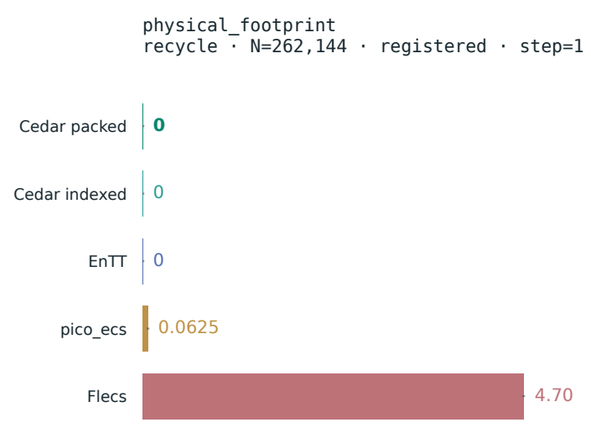


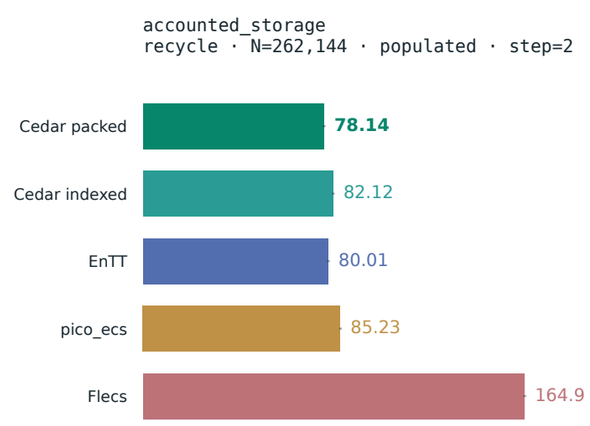


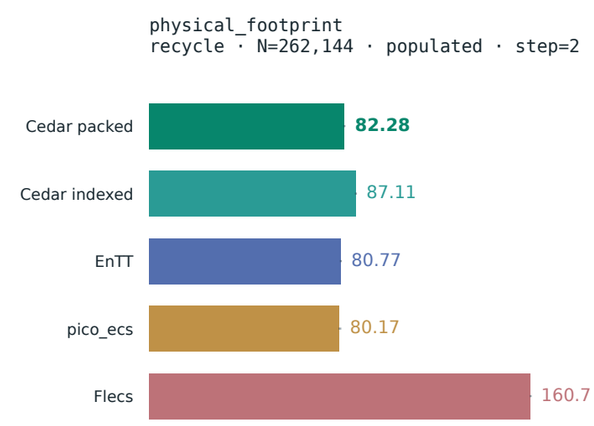








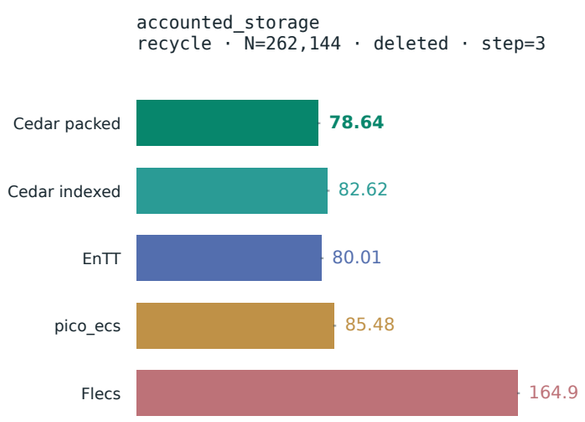


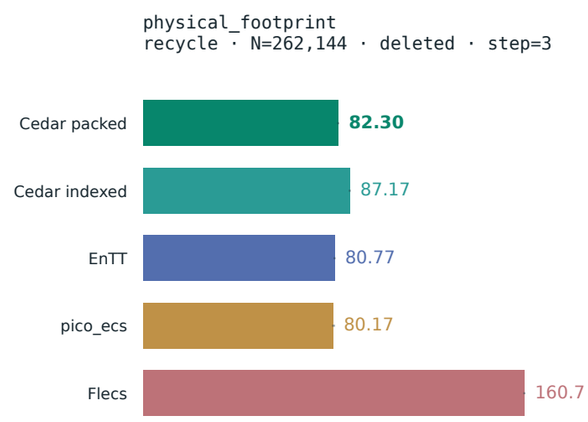


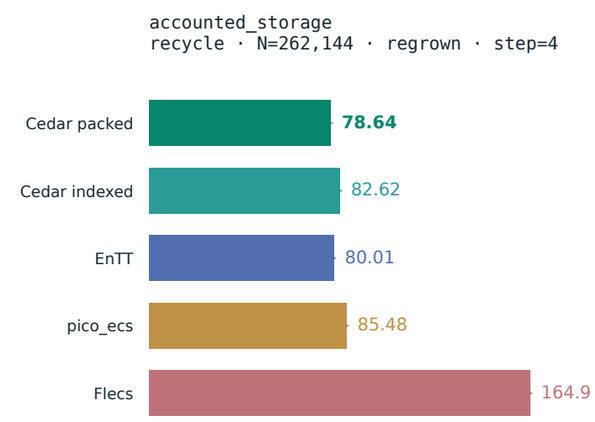
Memory comparison

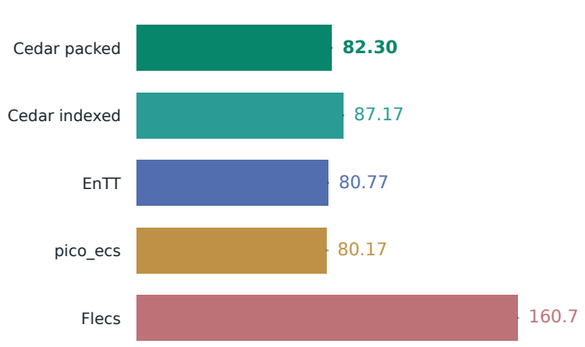
MiB ↓

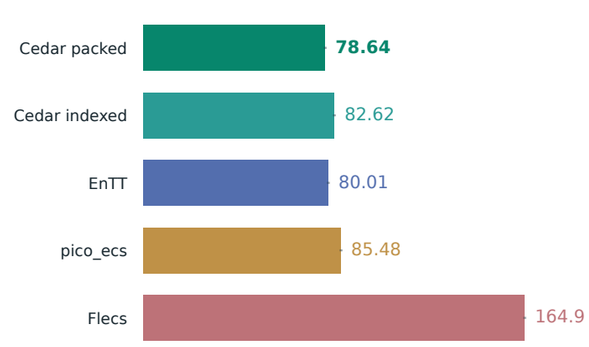
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

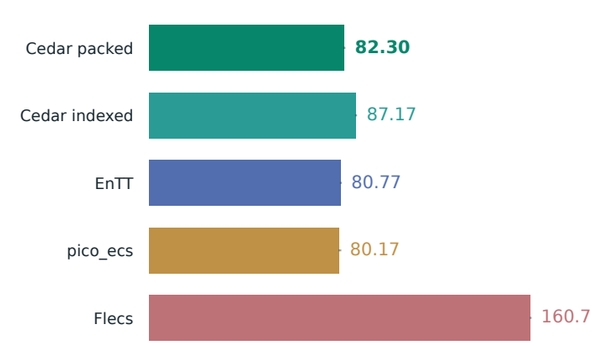


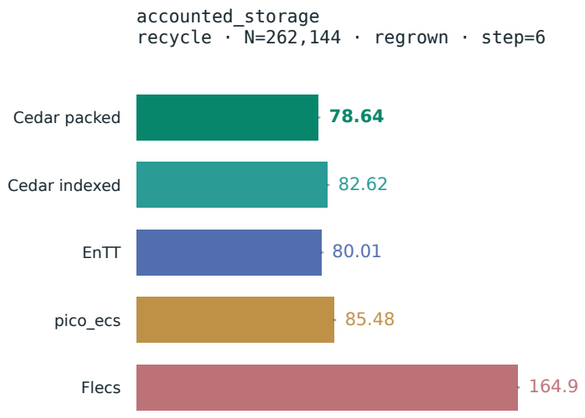


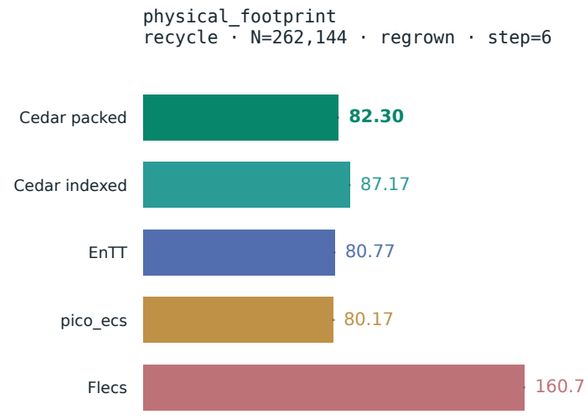


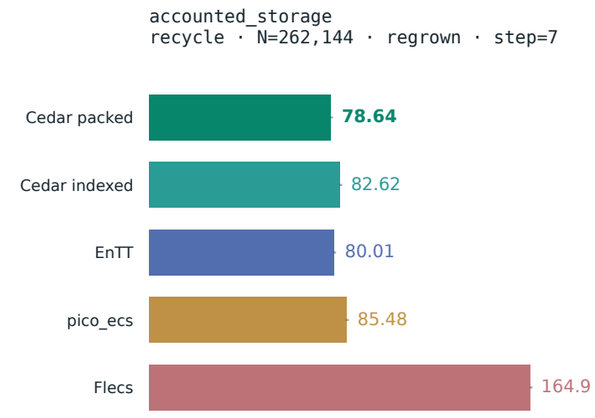


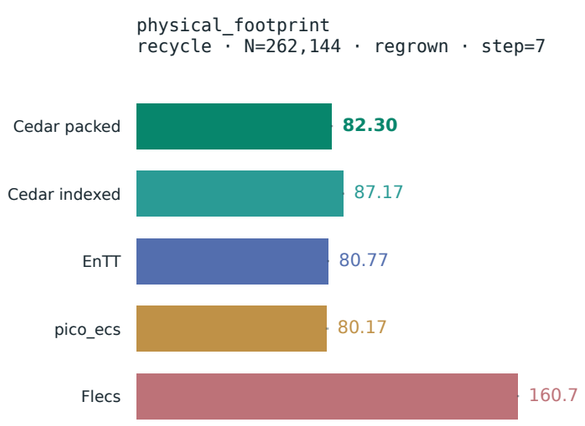


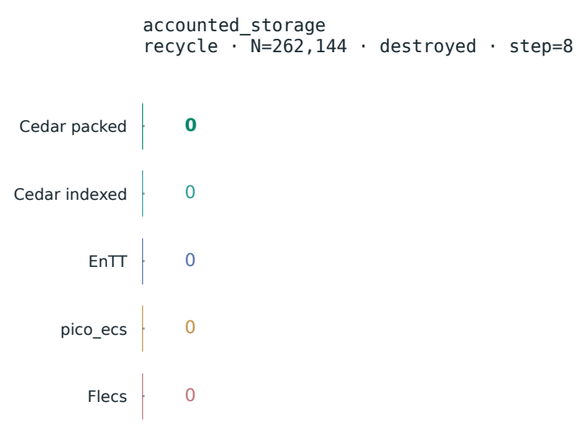


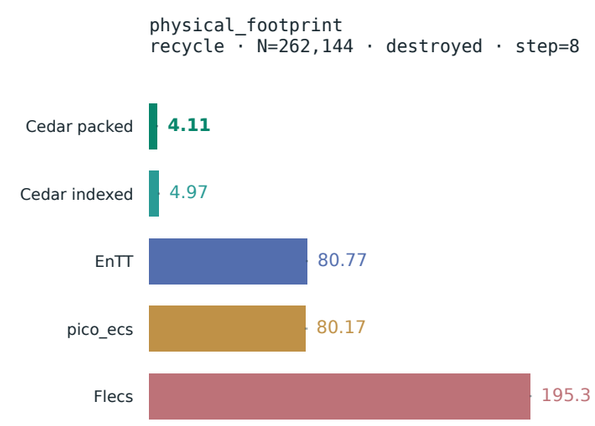










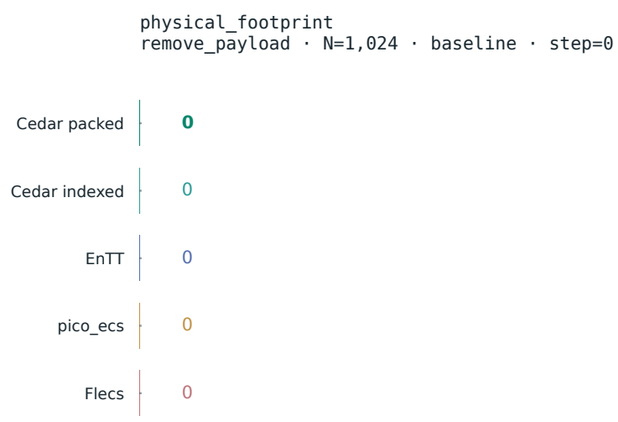


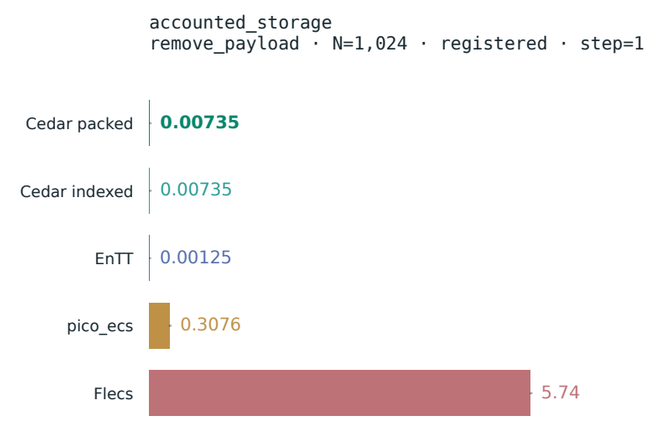
Memory comparison

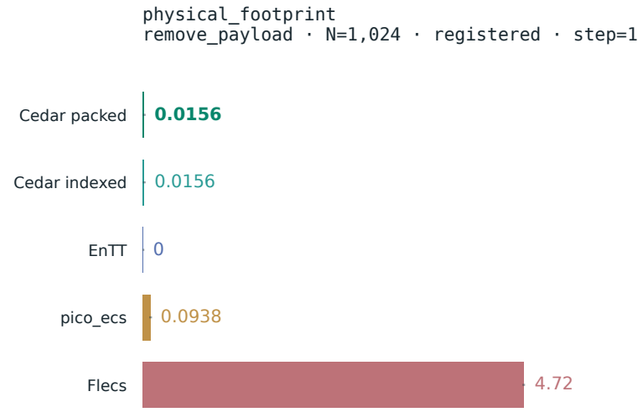
MiB ↓

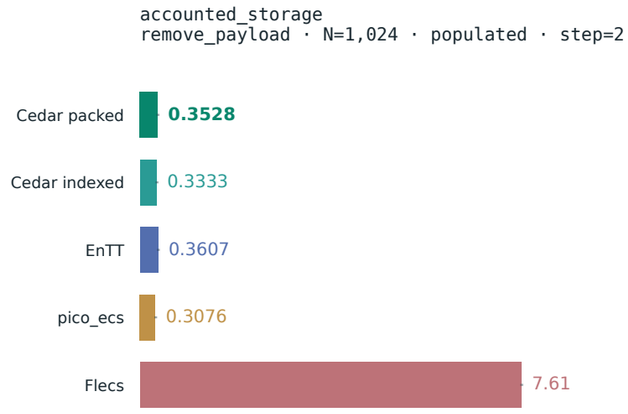
Cedar packed Cedar indexed EnTT pico_ecs Flecs

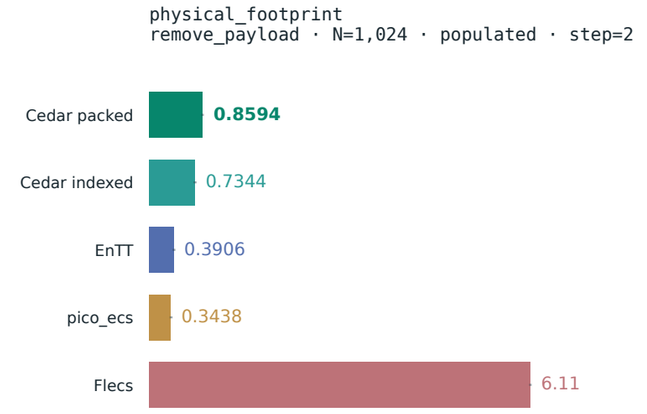


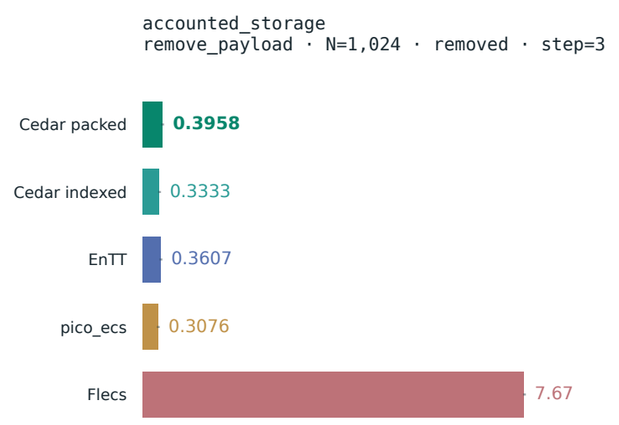


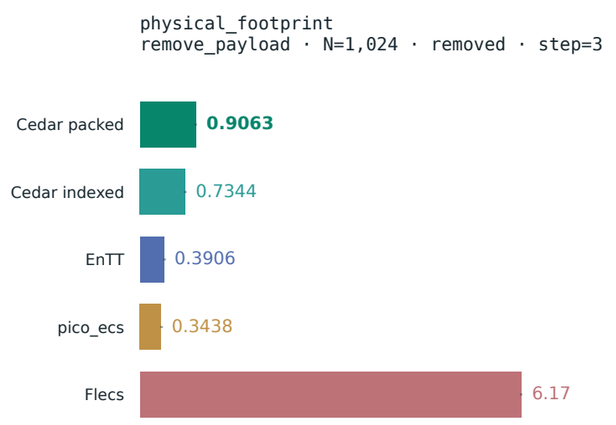




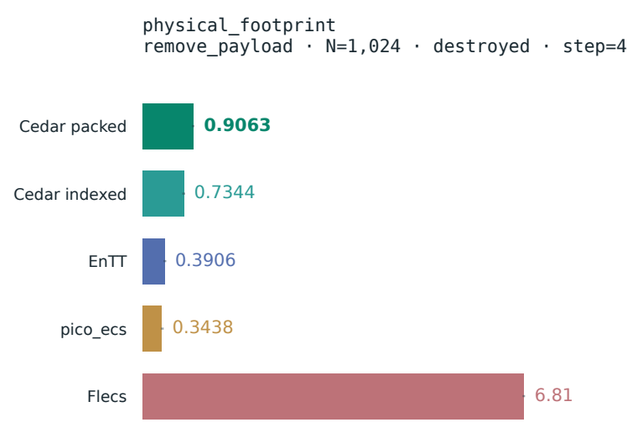


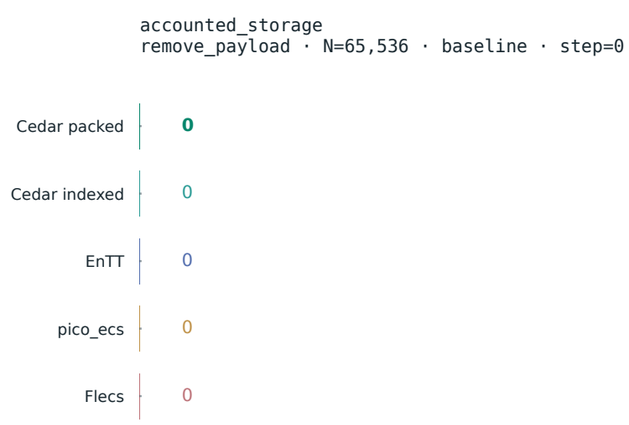


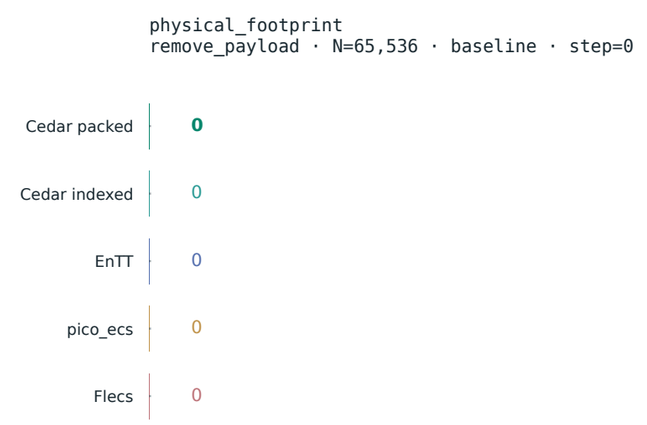








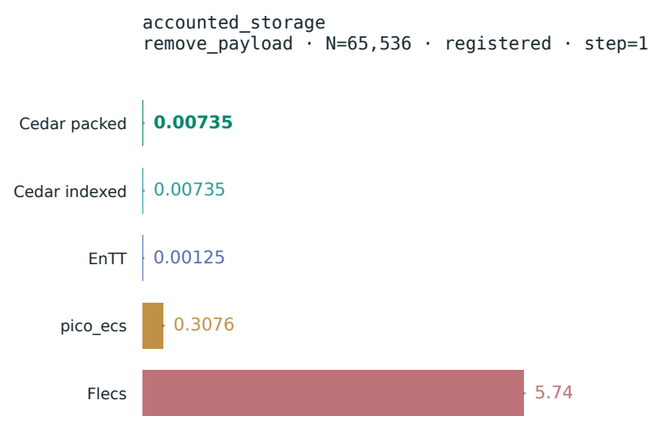


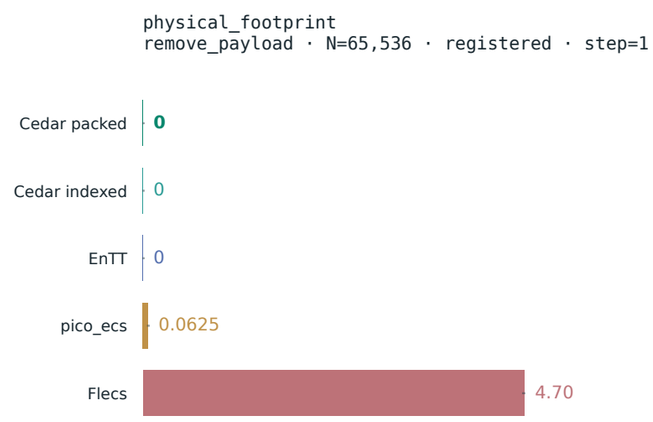


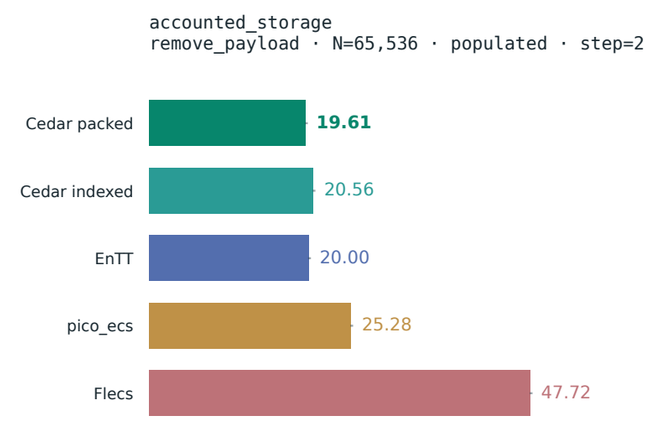
Memory comparison

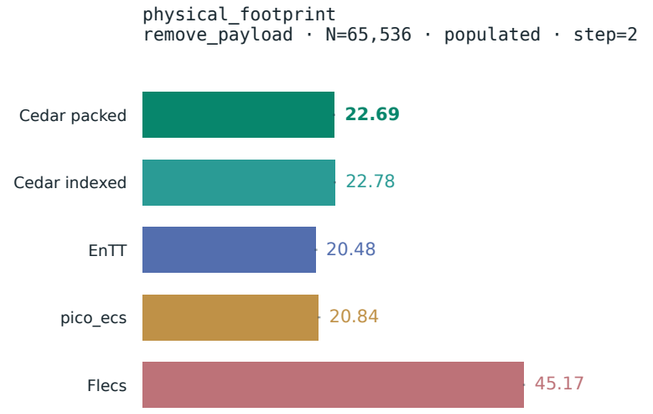
MiB ↓

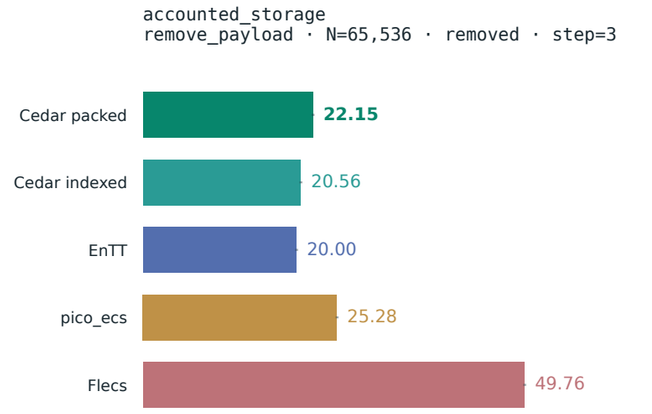
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

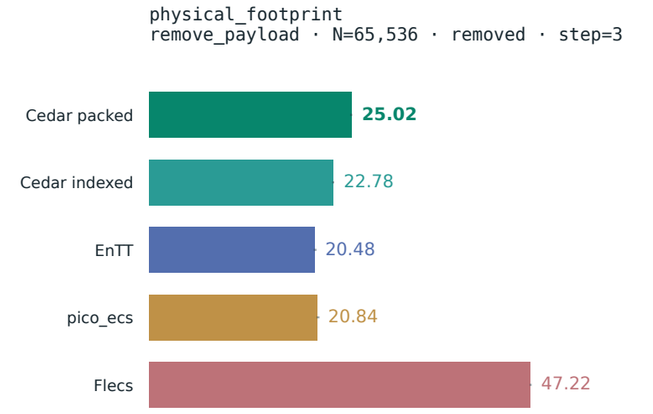


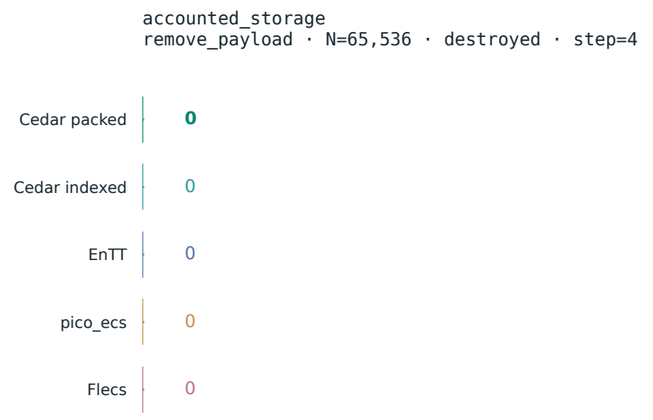


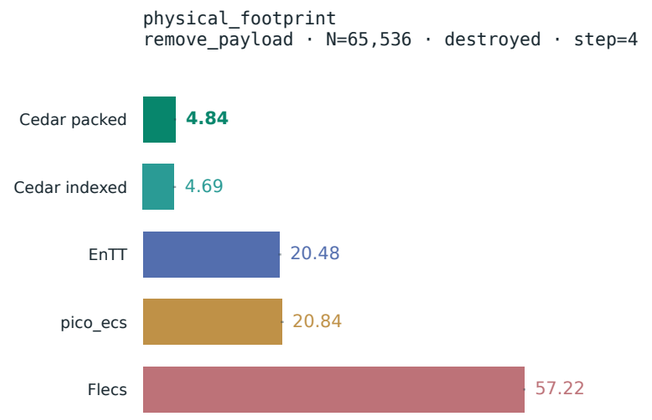


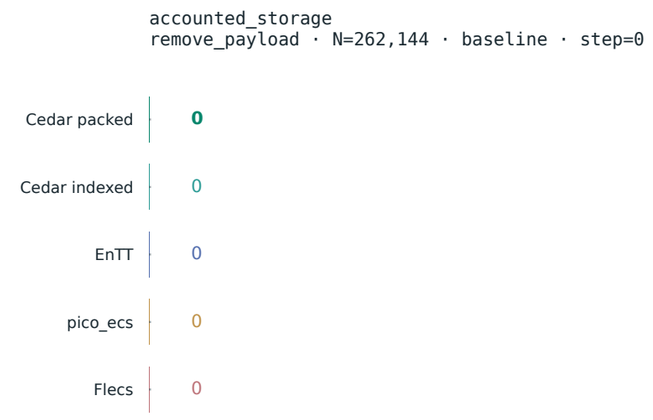




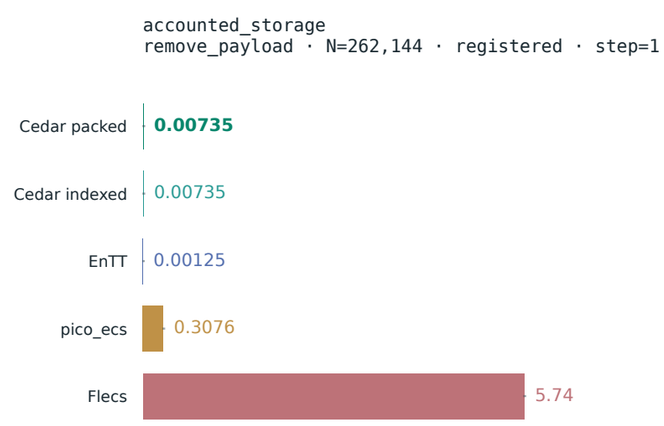


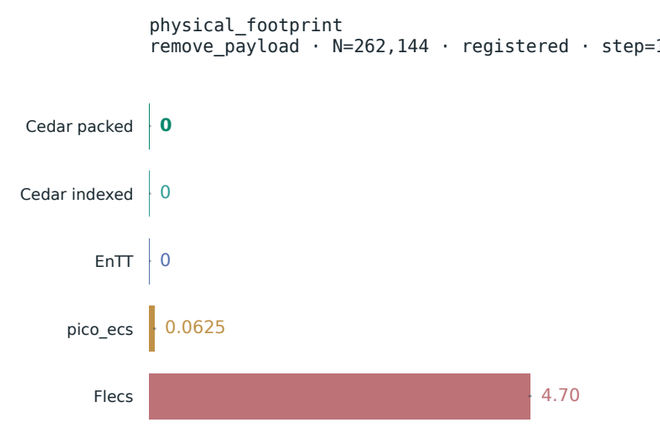








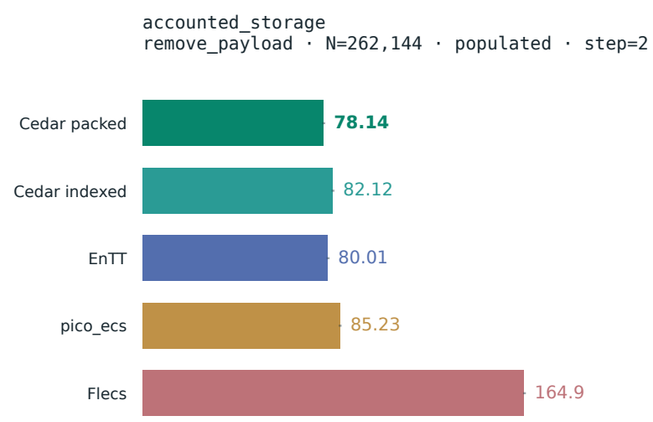


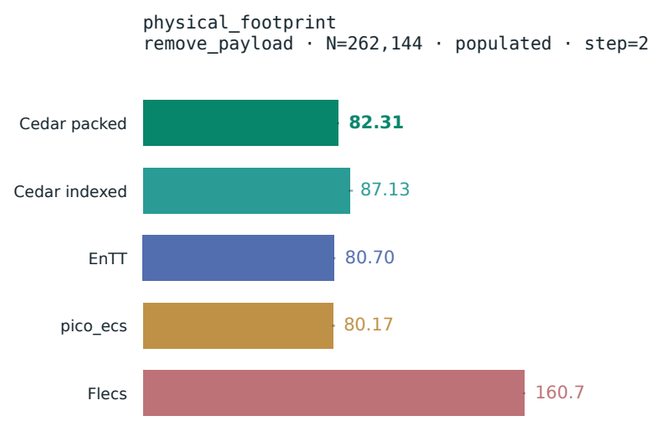


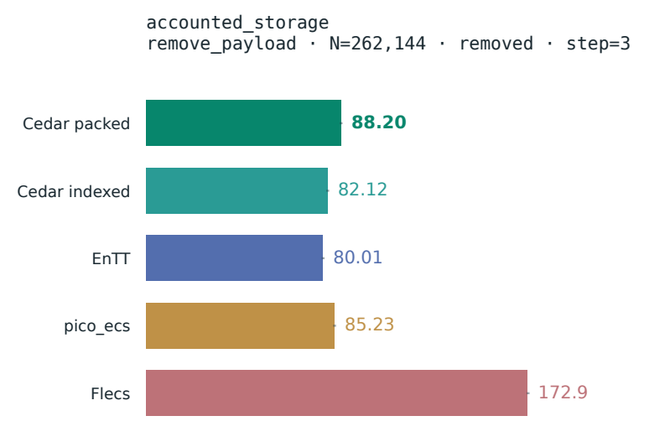
Memory comparison

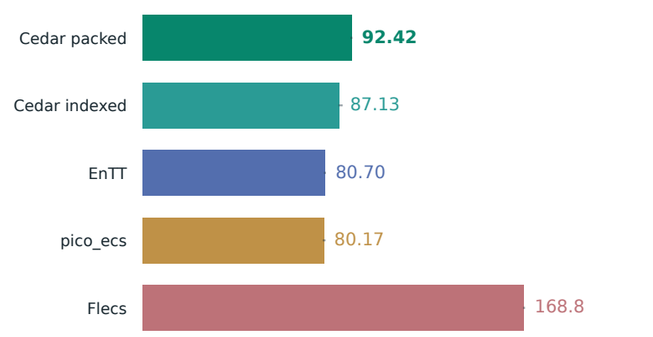
MiB ↓

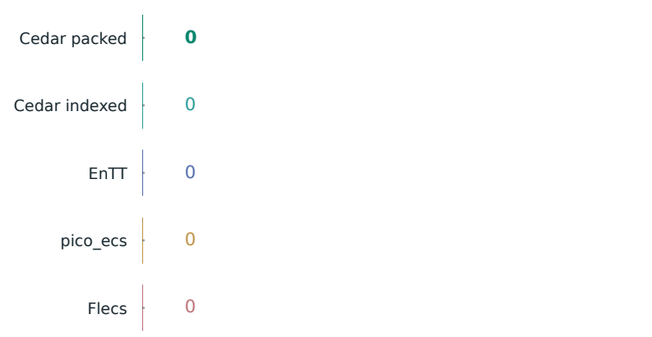
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

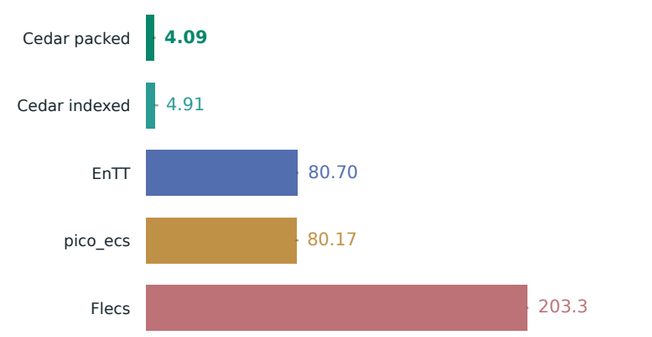






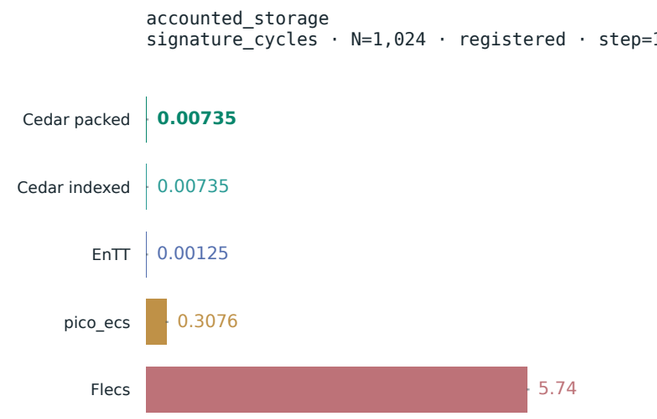


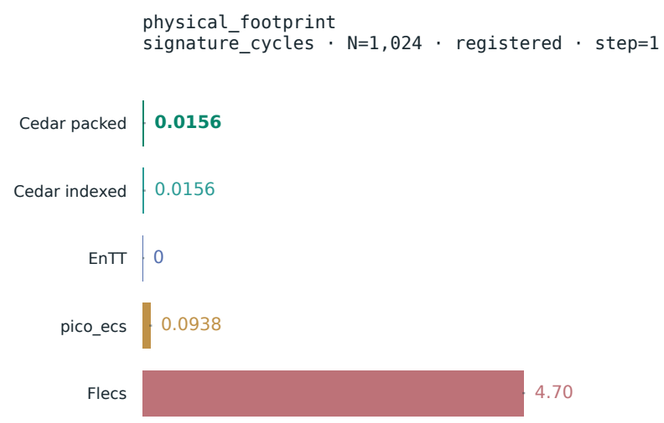


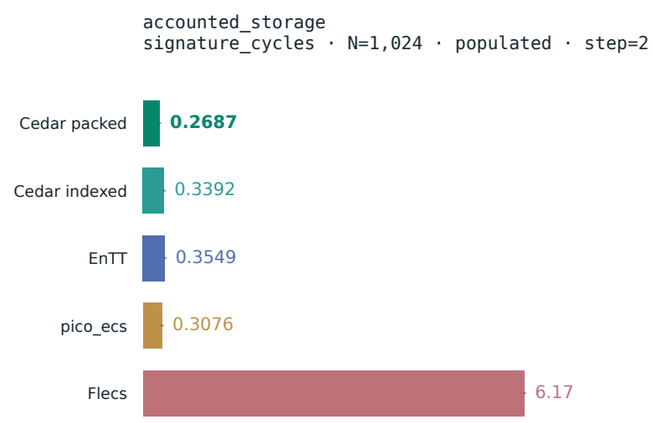


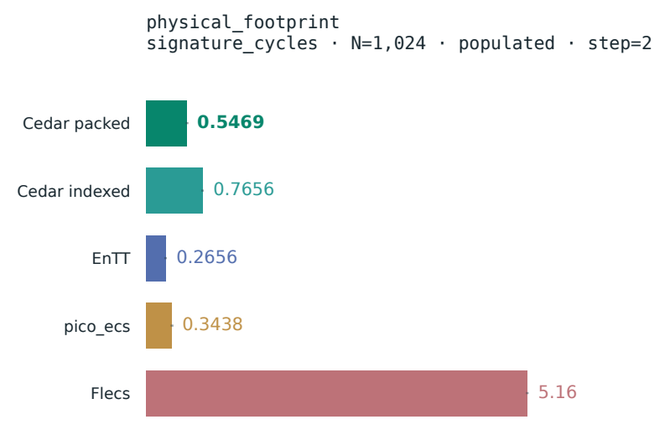








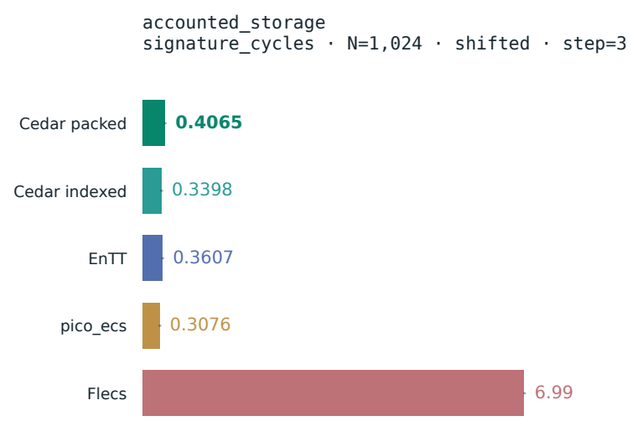


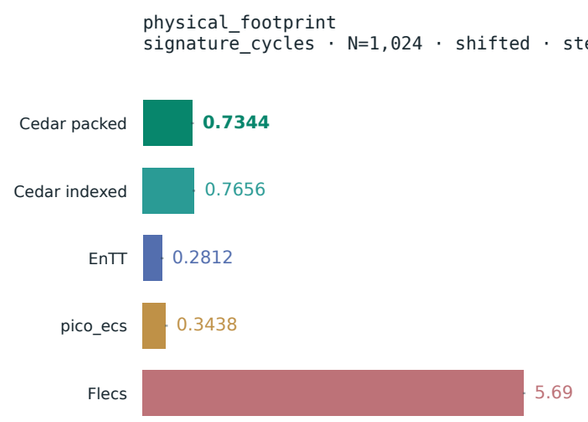


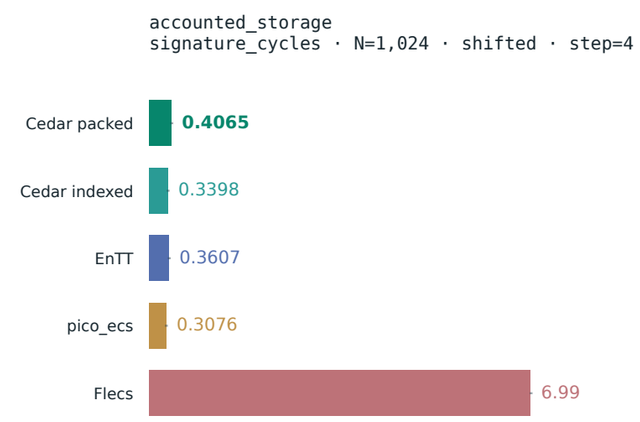
Memory comparison

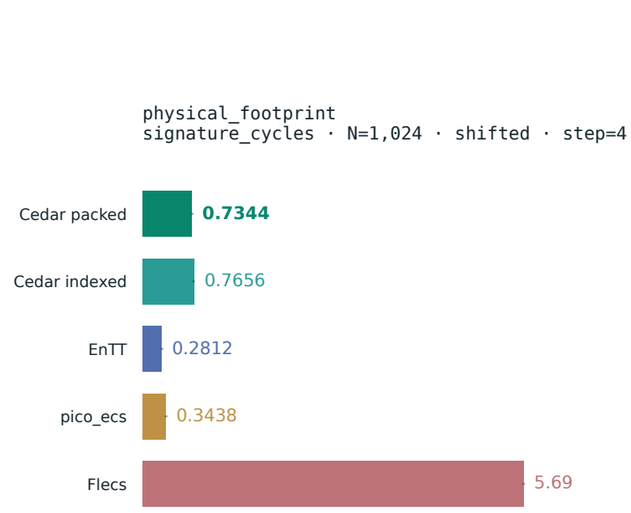
MiB ↓

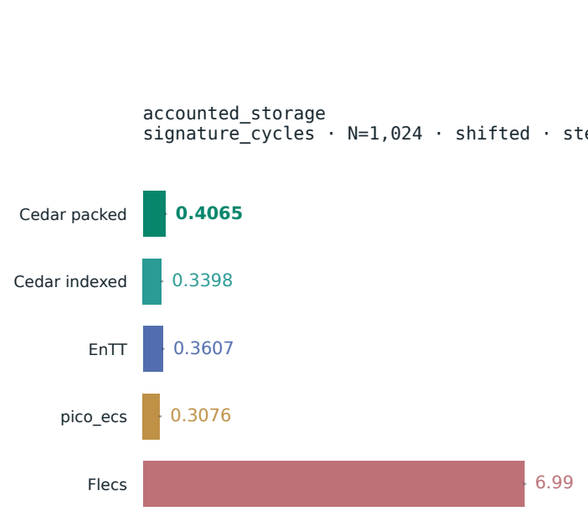
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

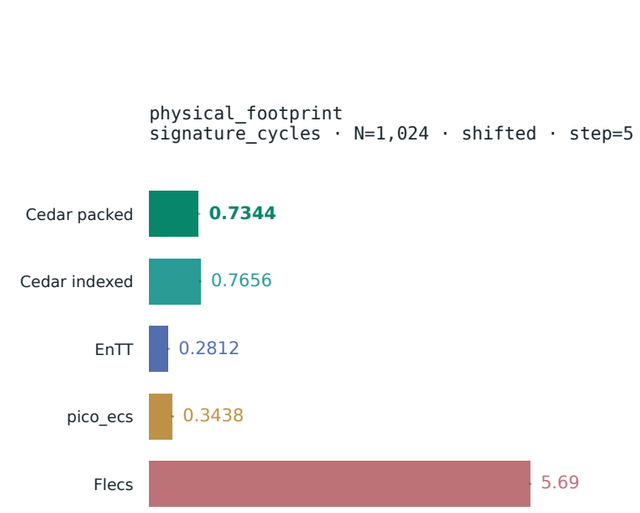


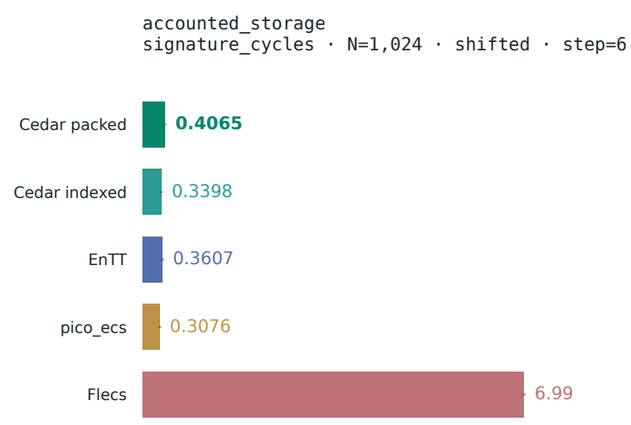


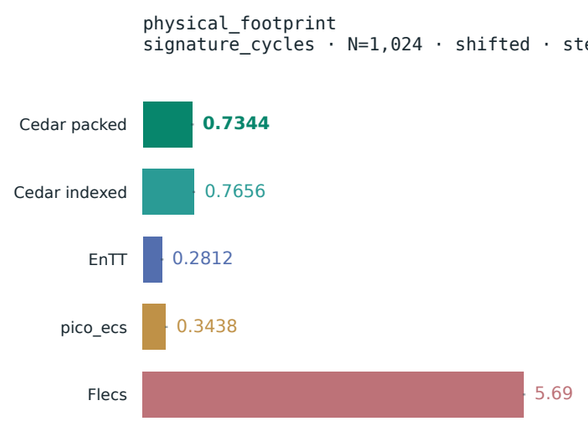




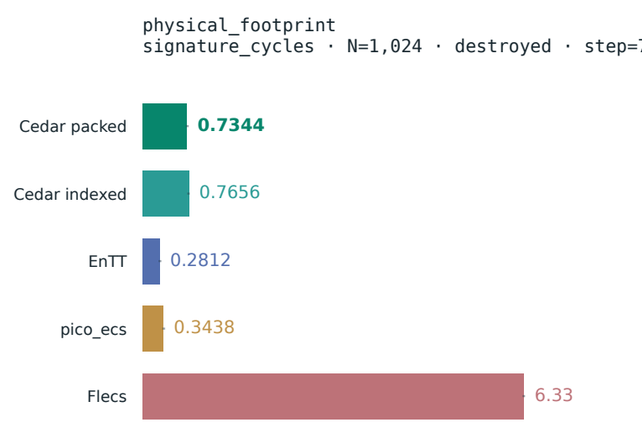


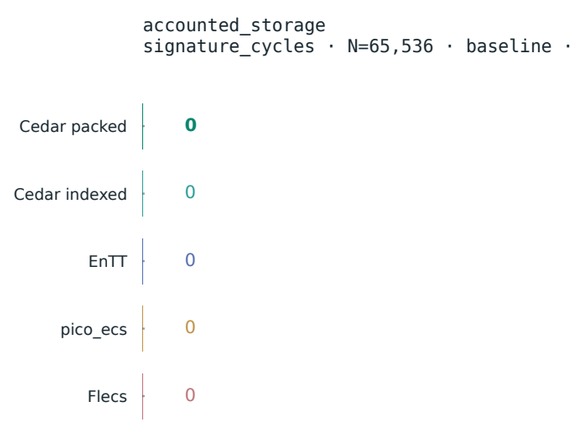


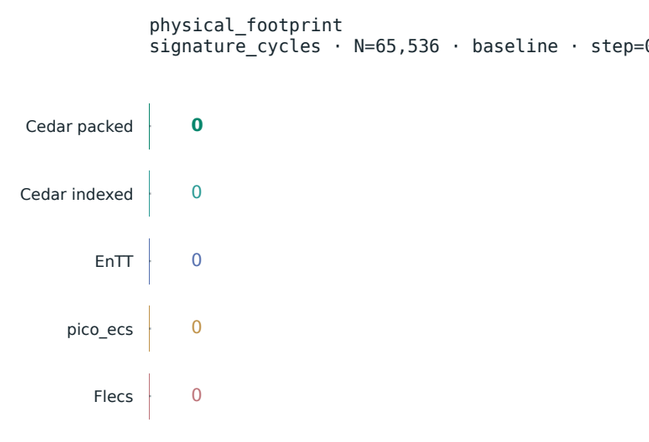








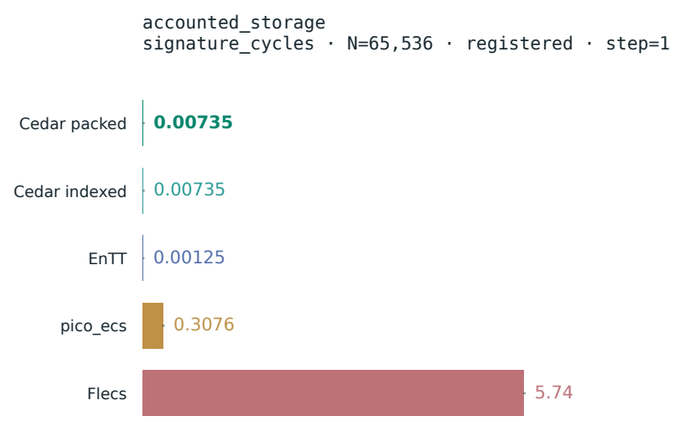


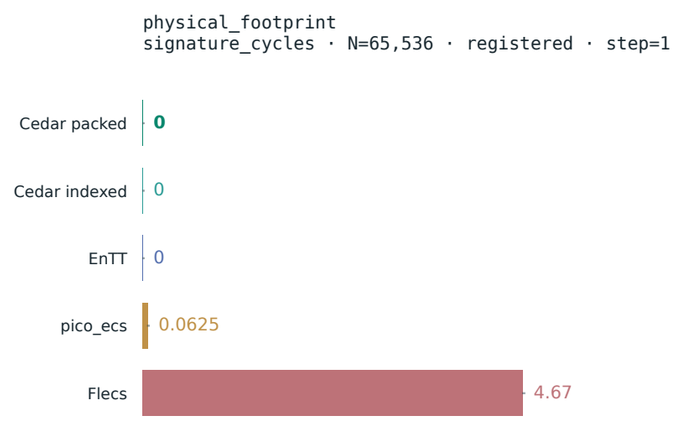


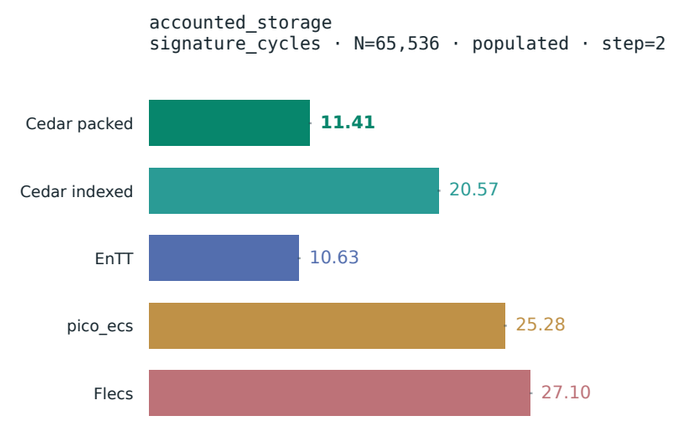
Memory comparison

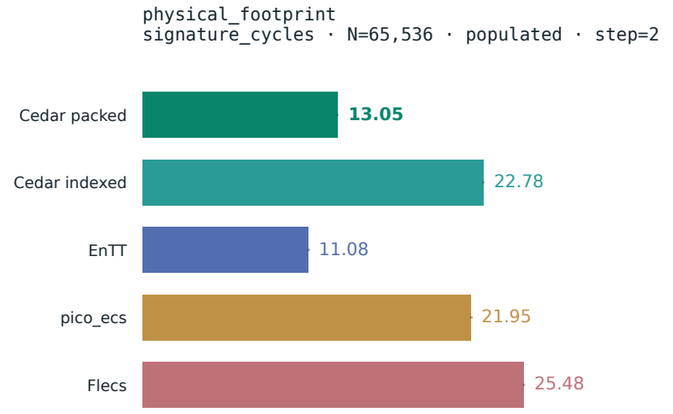
MiB ↓

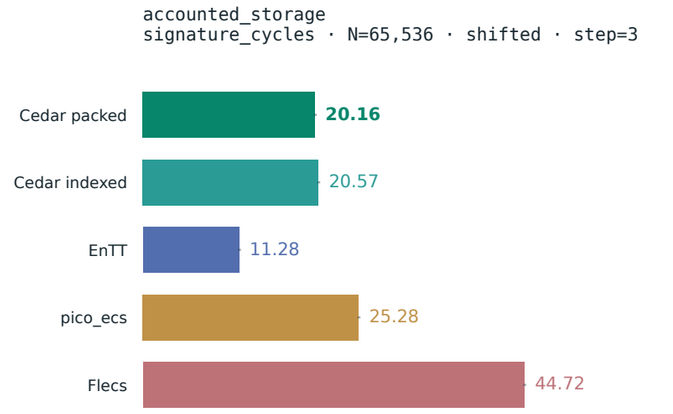
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

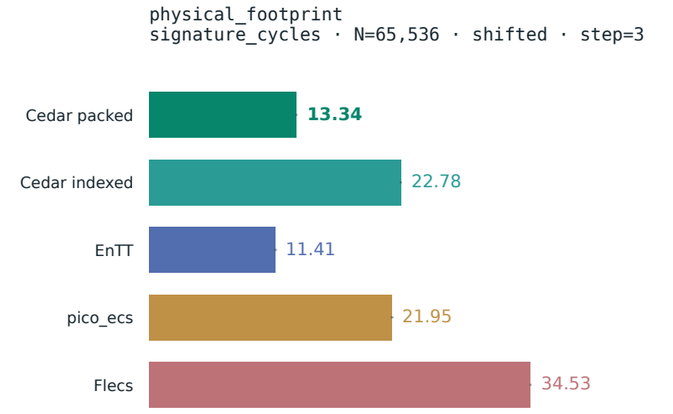


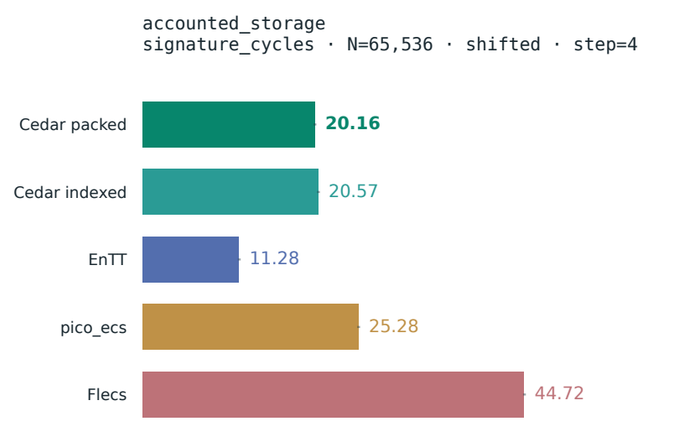


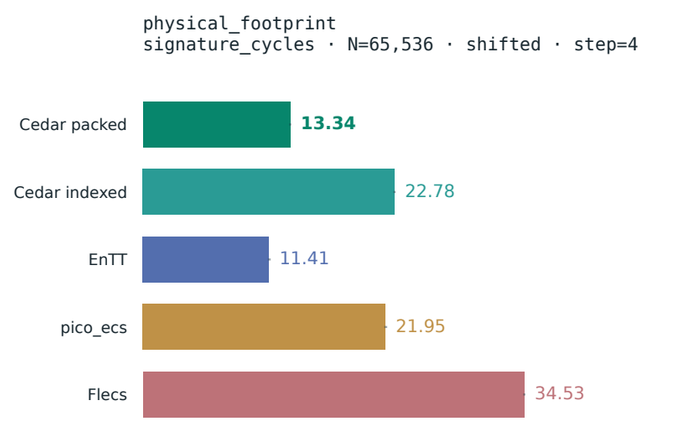


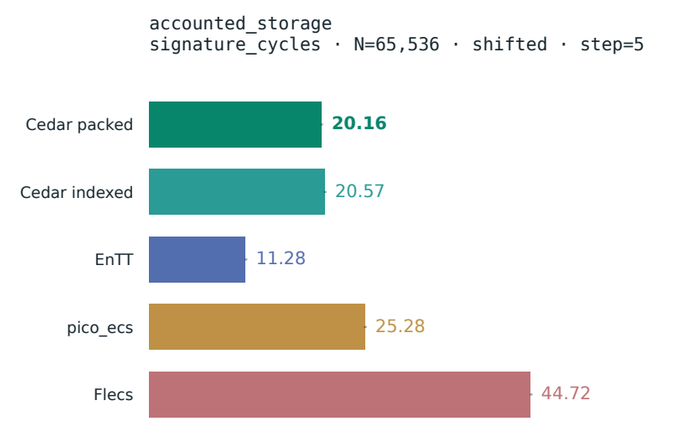


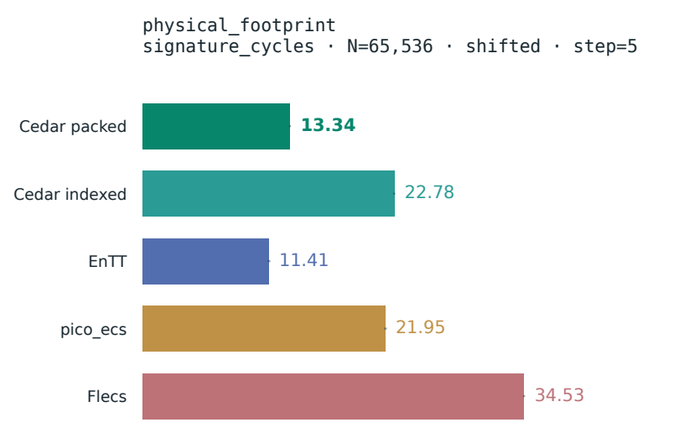


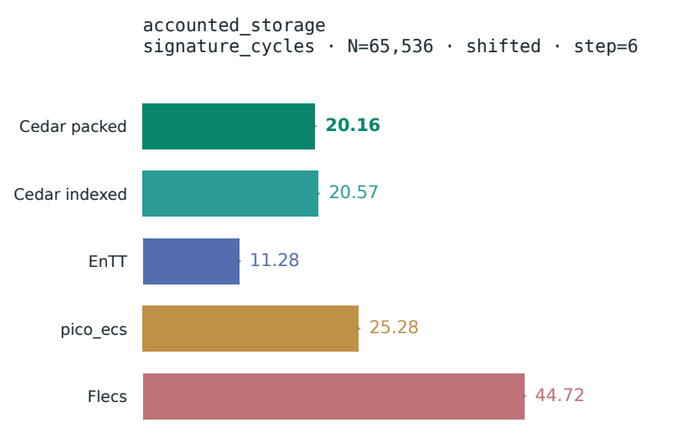


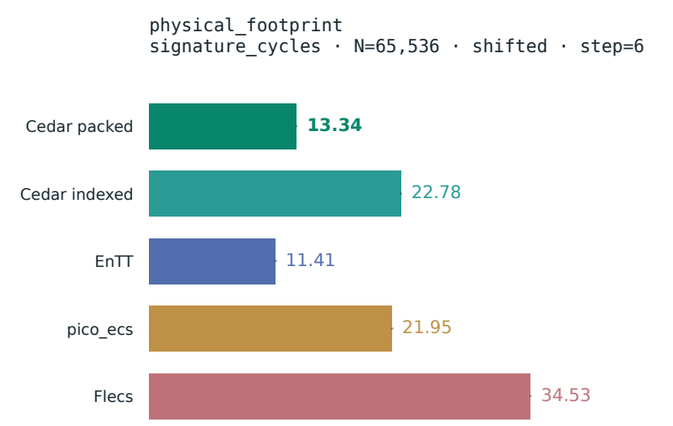








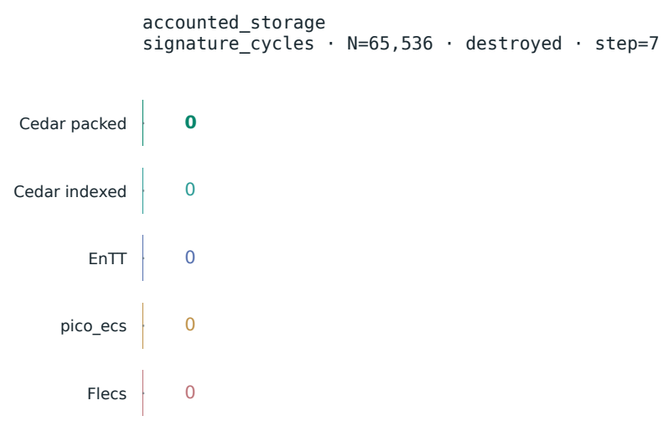


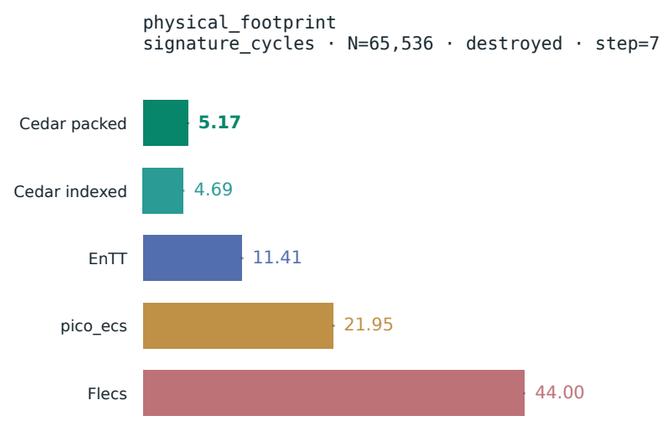


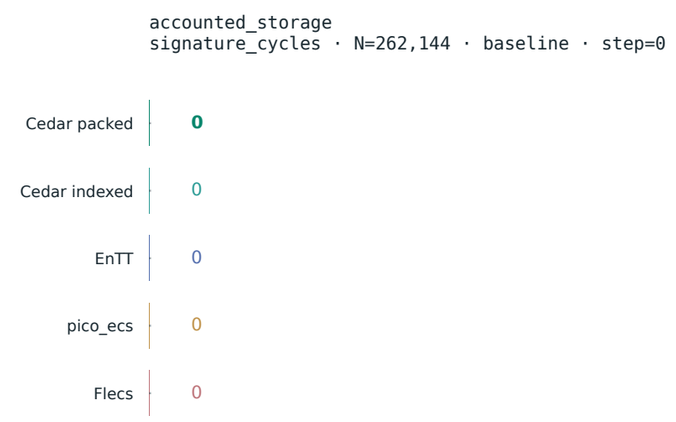
Memory comparison

MiB ↓

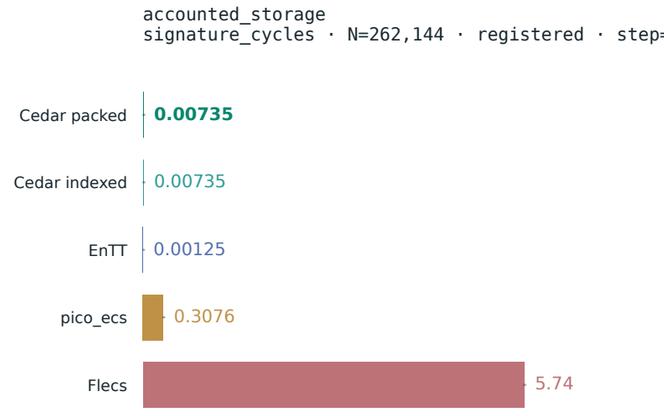
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

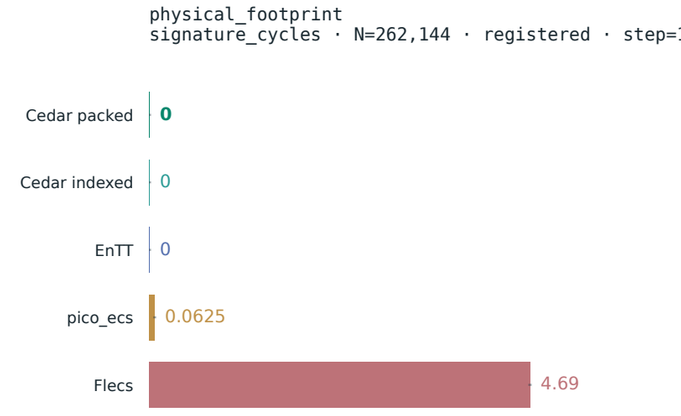


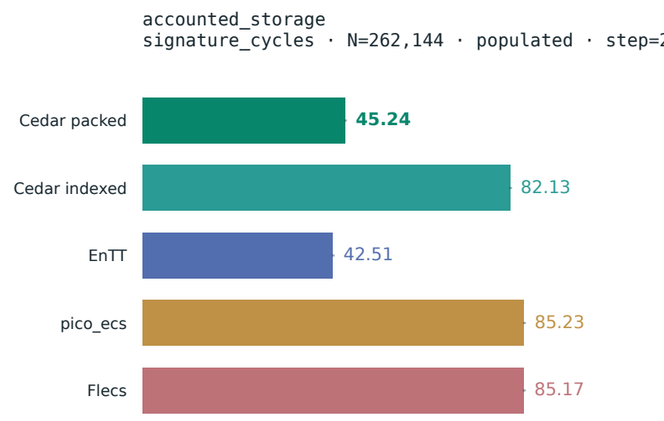


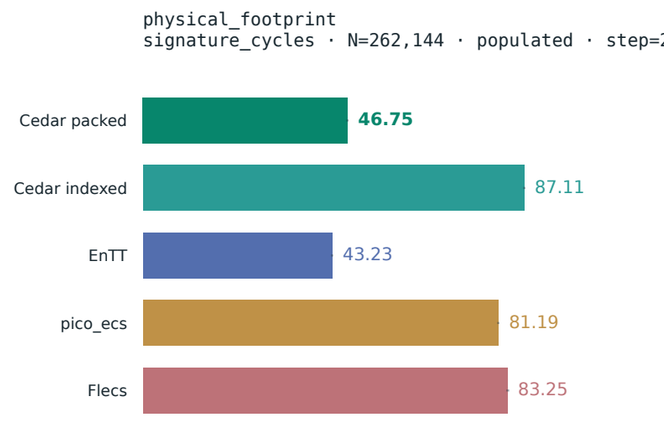


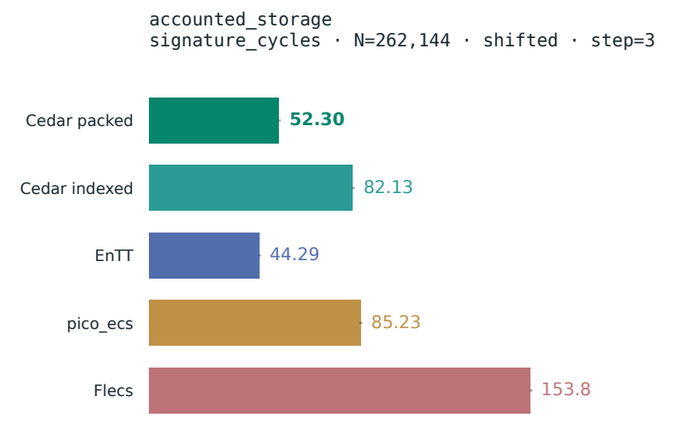


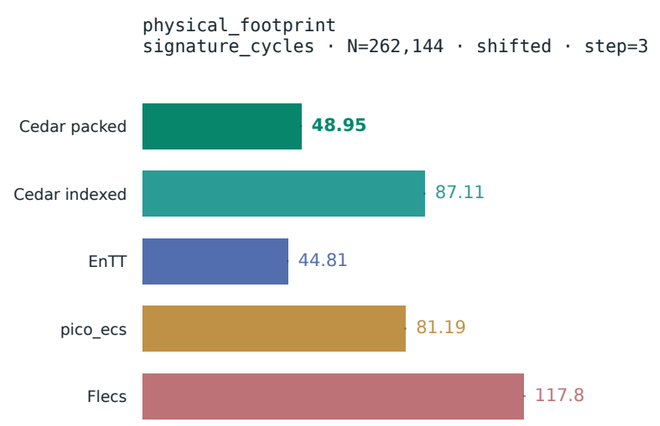


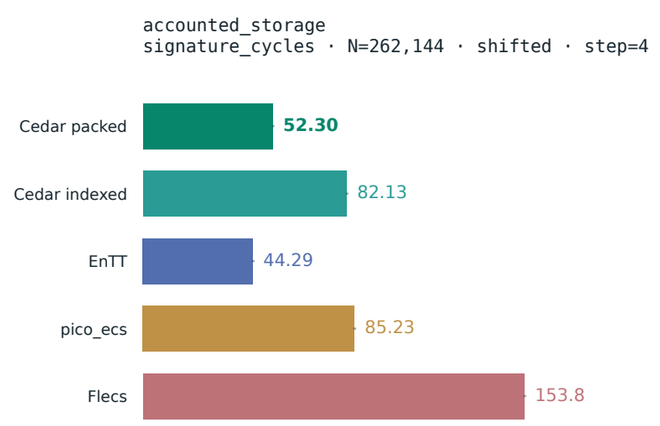


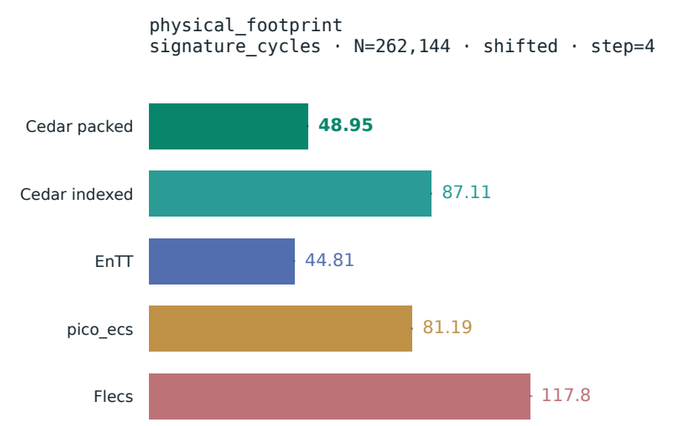








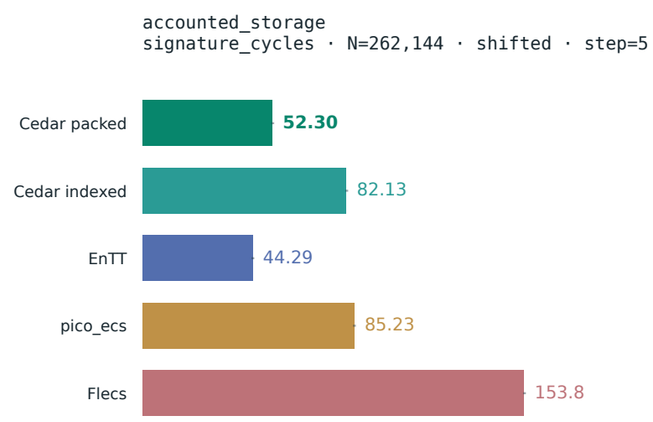


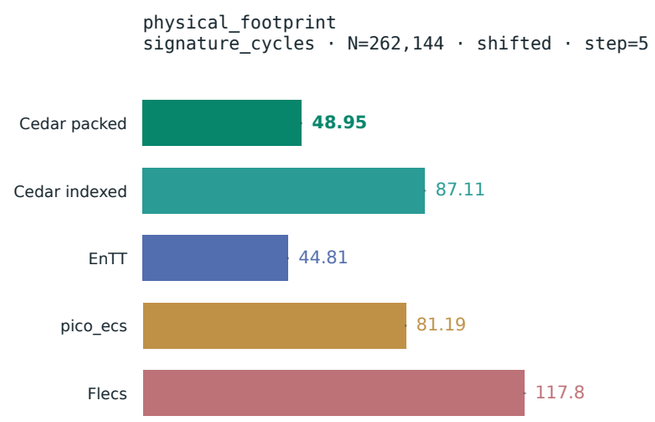


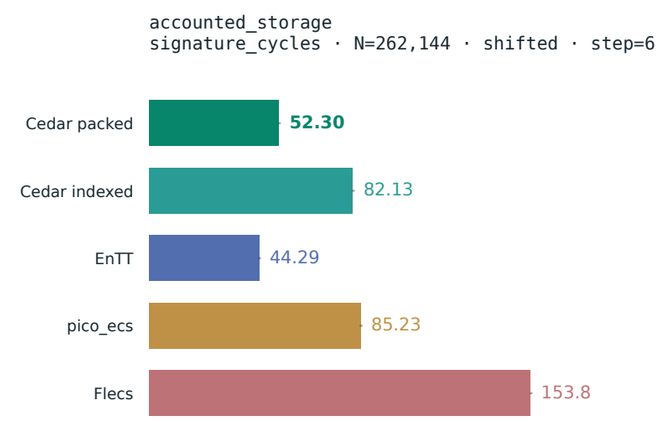
Memory comparison

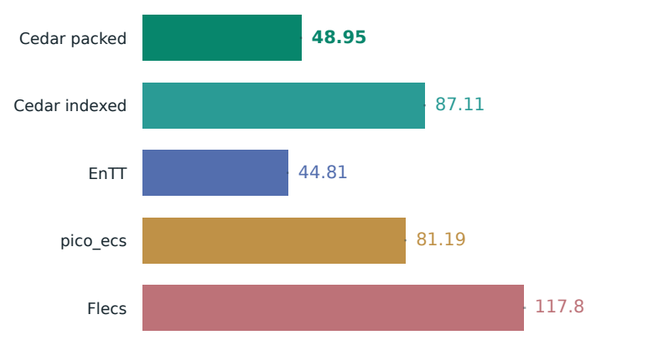
MiB ↓

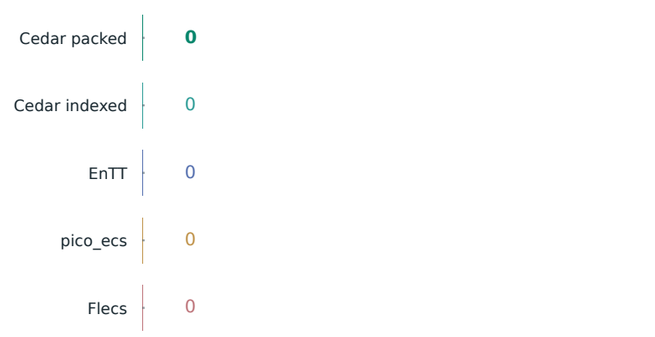
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

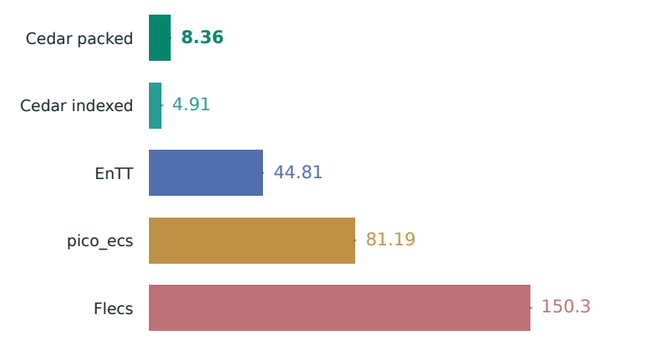






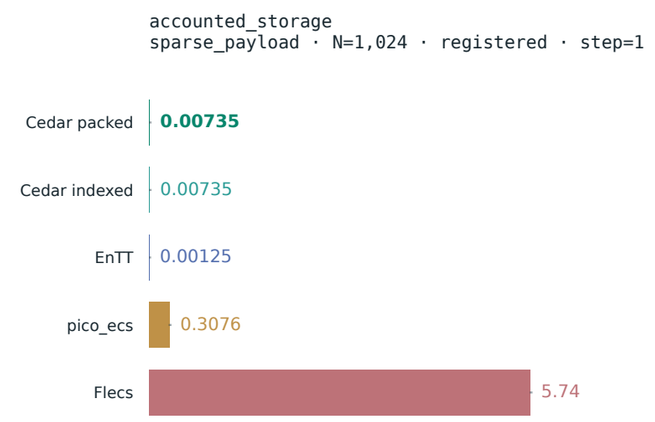


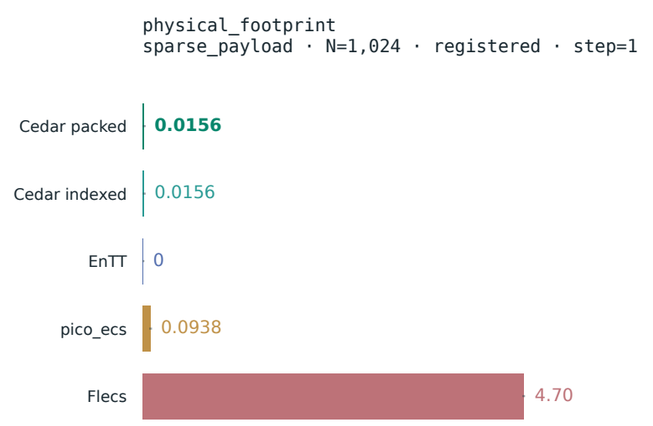


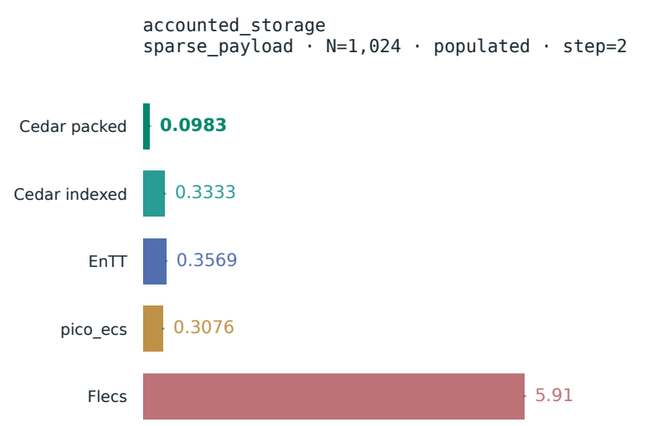


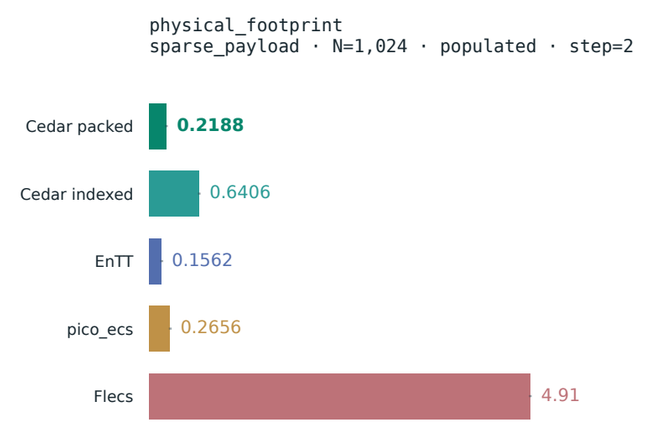










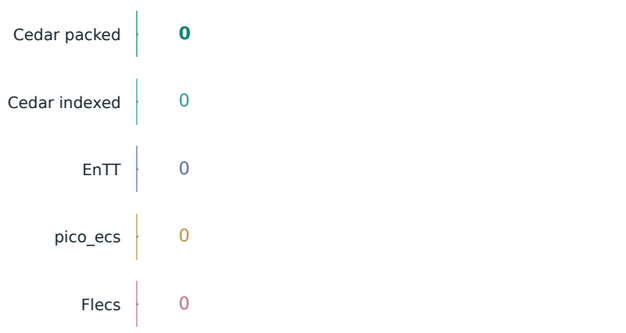


Memory comparison

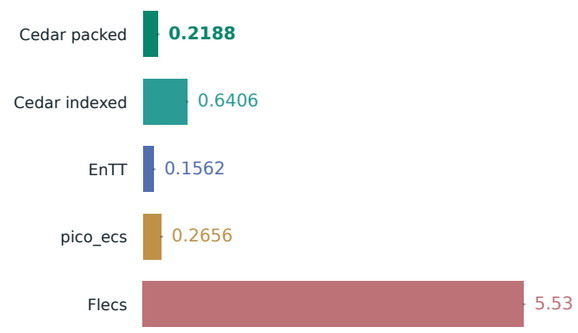
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

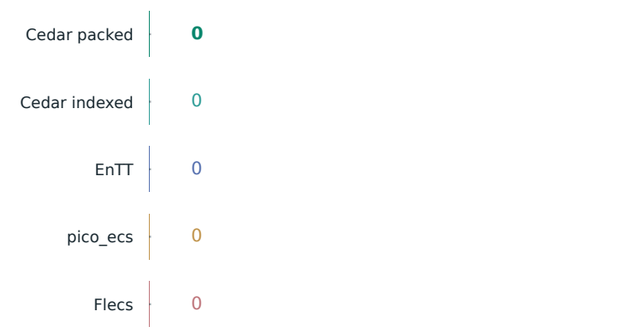
accounted_storage
sparse_payload · N=1,024 · destroyed · step=3



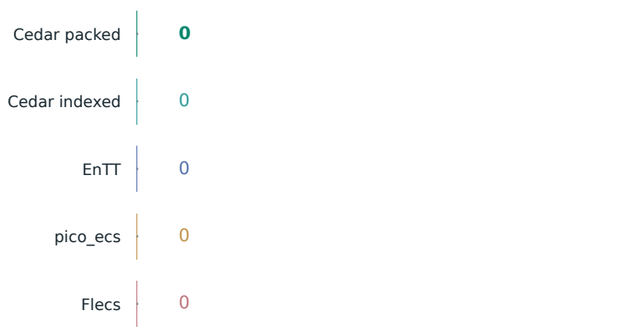
physical_footprint
sparse_payload · N=1,024 · destroyed · step=3



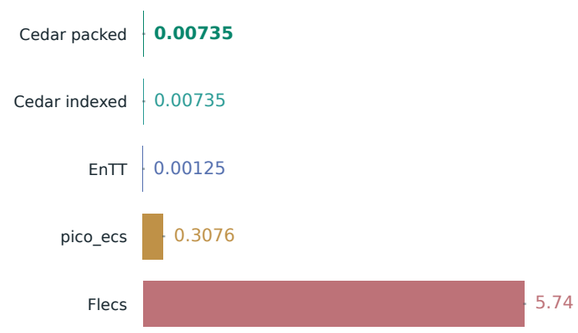
accounted_storage
sparse_payload · N=65,536 · destroyed · step=3



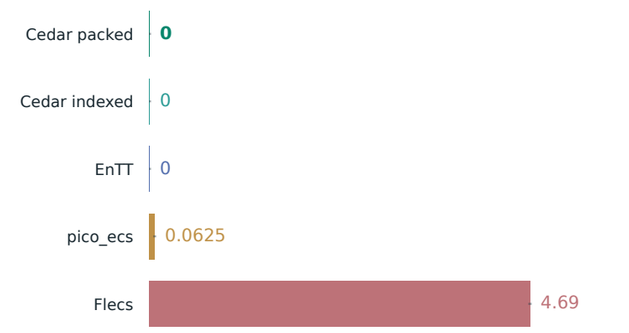
physical_footprint
sparse_payload · N=65,536 · baseline · step=0



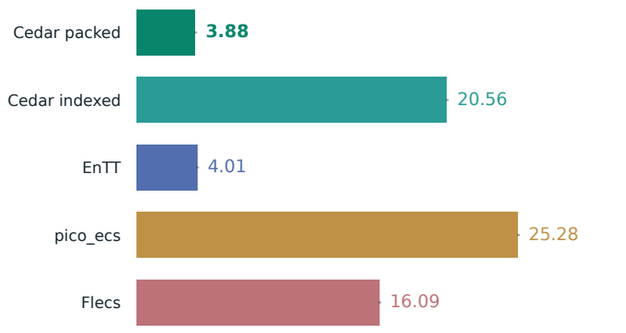
accounted_storage
sparse_payload · N=65,536 · registered · step=1



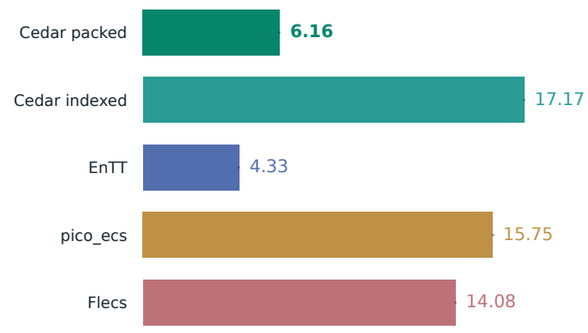
physical_footprint
sparse_payload · N=65,536 · registered · step=1



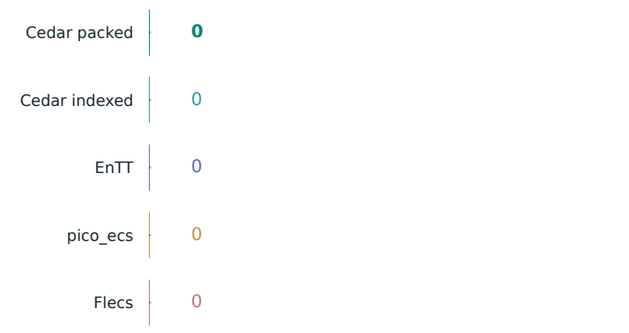
accounted_storage
sparse_payload · N=65,536 · populated · step=2



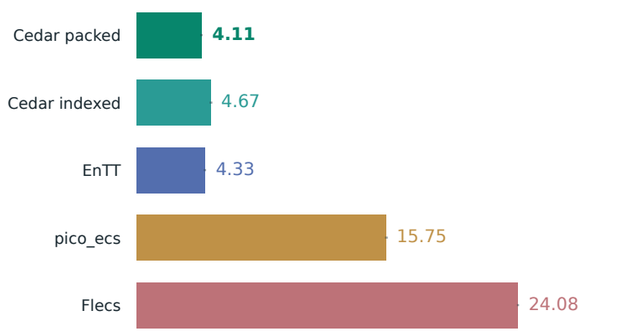
physical_footprint
sparse_payload · N=65,536 · populated · step=2



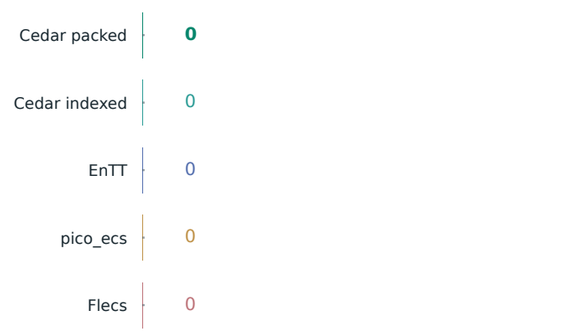
accounted_storage
sparse_payload · N=65,536 · destroyed · step=3



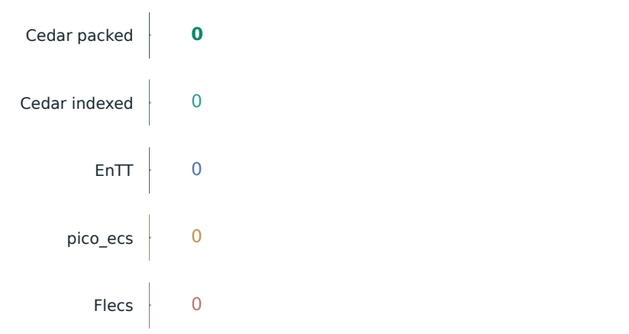
physical_footprint
sparse_payload · N=65,536 · destroyed · step=3



accounted_storage
sparse_payload · N=262,144 · baseline · step=0



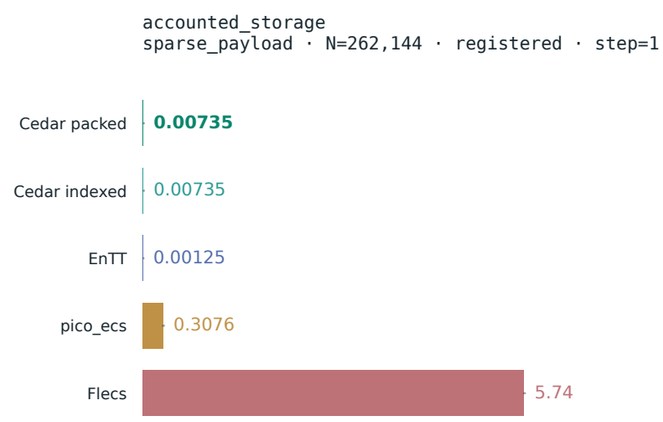
physical_footprint
sparse_payload · N=262,144 · baseline · step=0

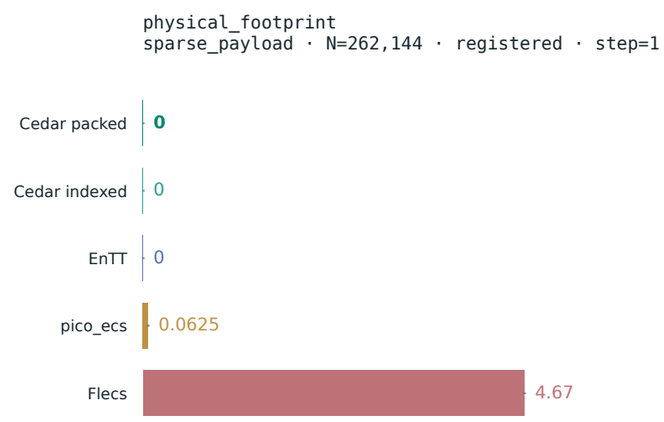


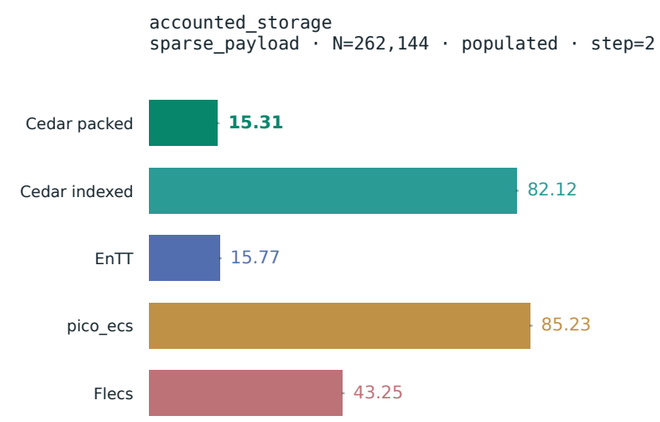
Memory comparison

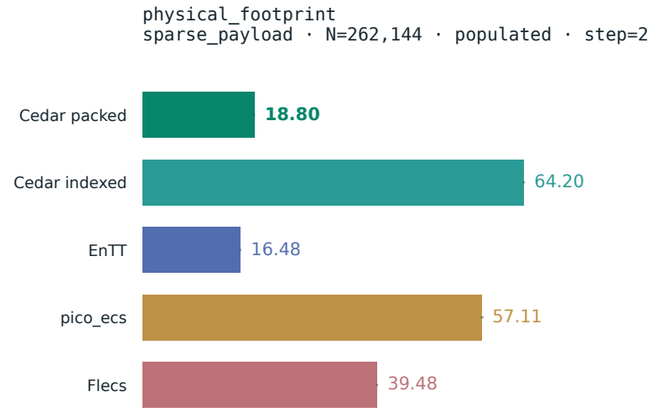
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

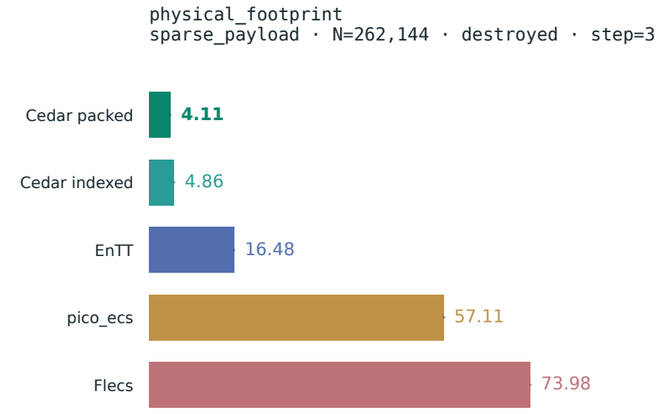






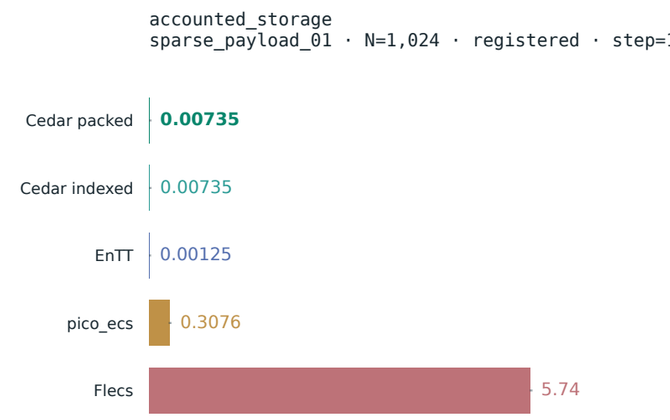


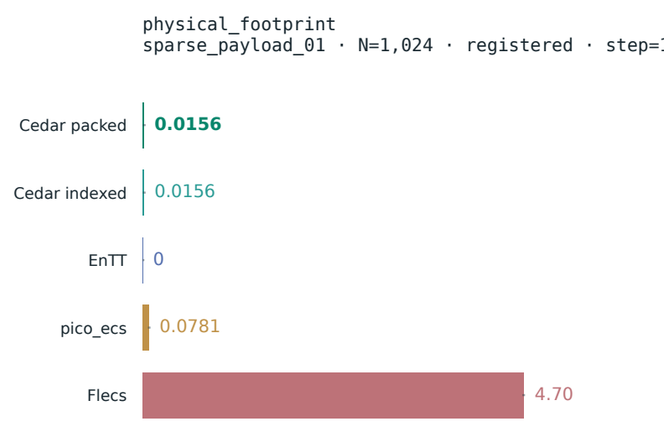


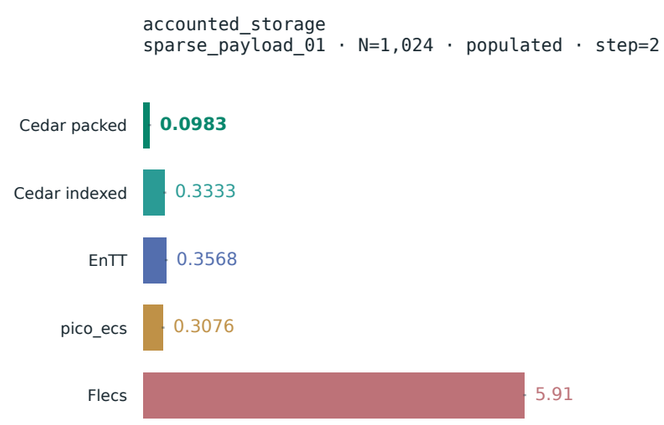


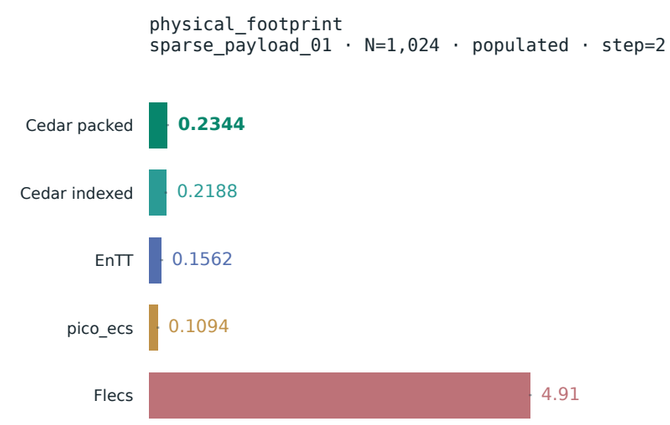










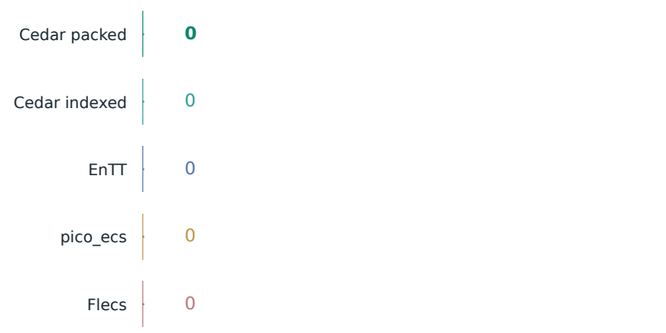


Memory comparison

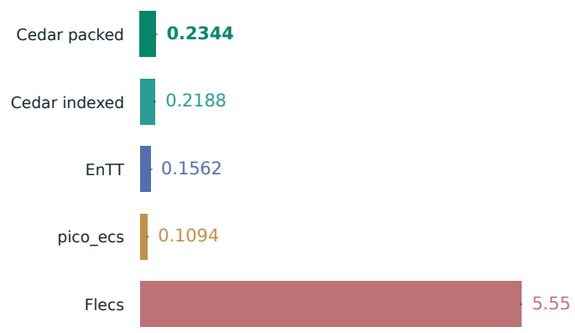
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

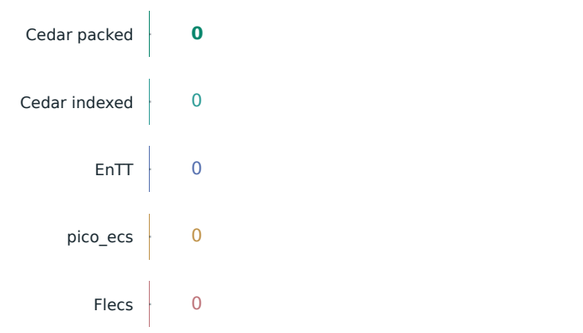
accounted_storage
sparse_payload_01 · N=1,024 · destroyed · step=3



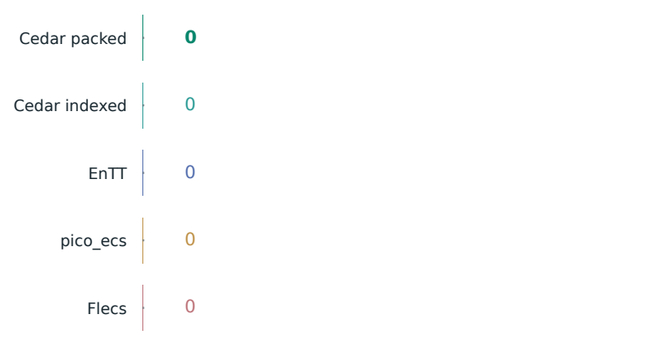
physical_footprint
sparse_payload_01 · N=1,024 · destroyed · step=3



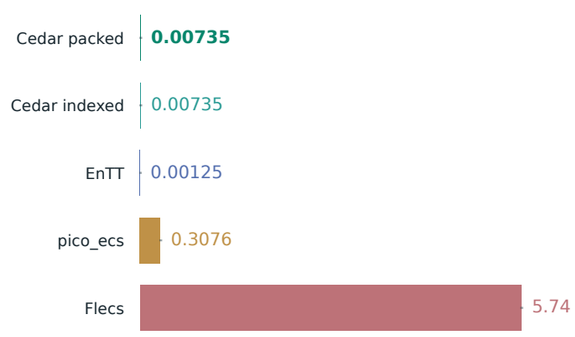
accounted_storage
sparse_payload_01 · N=65,536 · baseline · step=0



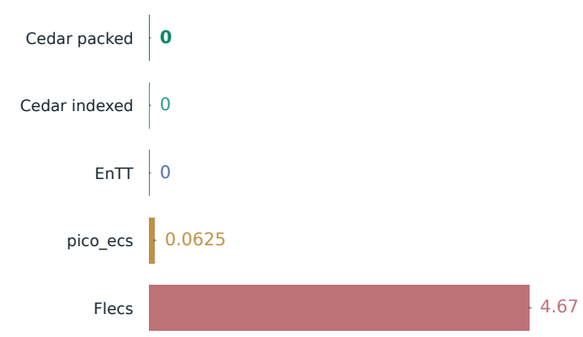
physical_footprint
sparse_payload_01 · N=65,536 · baseline · step=0



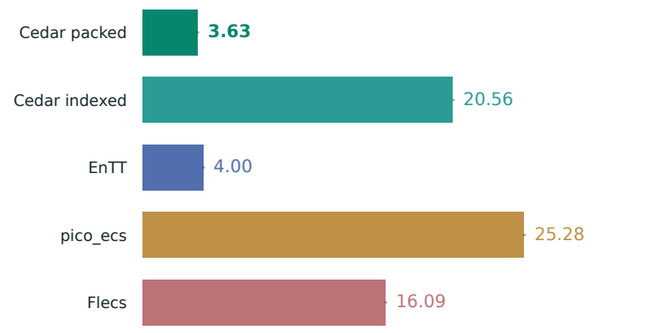
accounted_storage
sparse_payload_01 · N=65,536 · registered · step=1



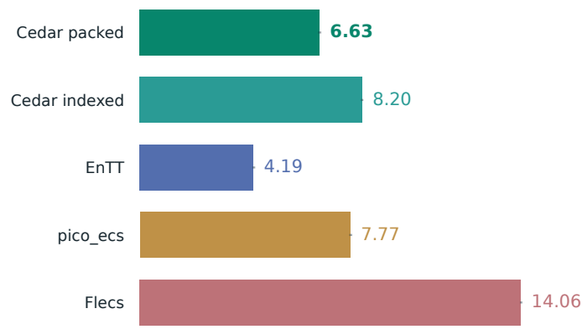
physical_footprint
sparse_payload_01 · N=65,536 · registered · step=1



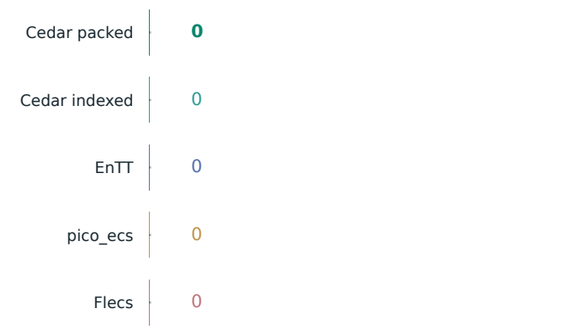
accounted_storage
sparse_payload_01 · N=65,536 · populated · step=2



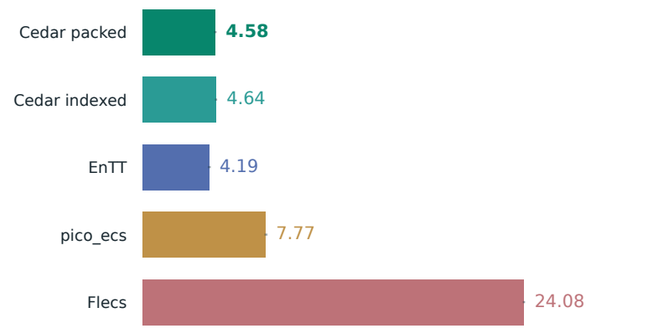
physical_footprint
sparse_payload_01 · N=65,536 · populated · step=2



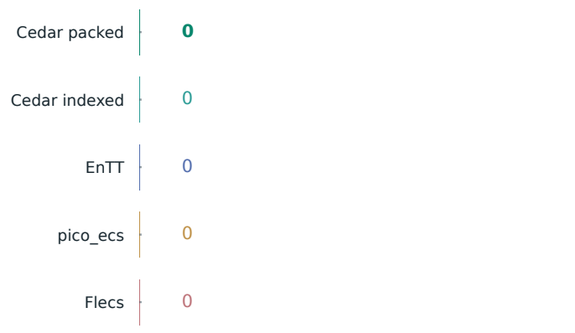
accounted_storage
sparse_payload_01 · N=65,536 · destroyed · step=3



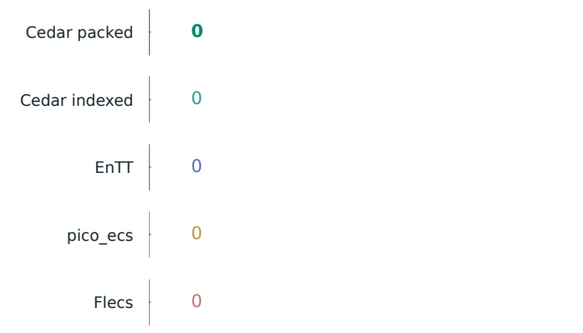
physical_footprint
sparse_payload_01 · N=65,536 · destroyed · step=3



accounted_storage
sparse_payload_01 · N=262,144 · baseline · step=0



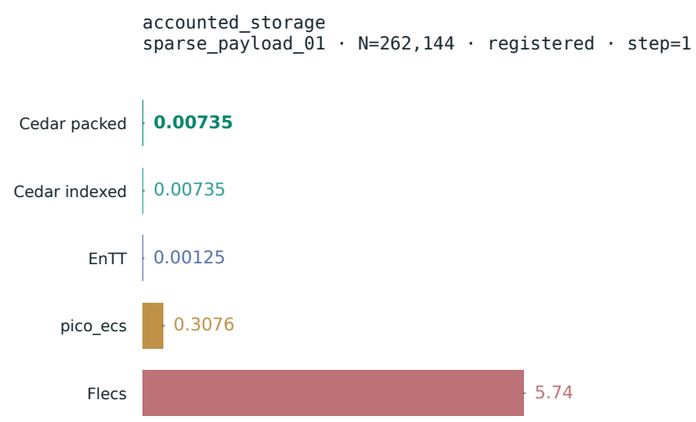
physical_footprint
sparse_payload_01 · N=262,144 · baseline · step=0

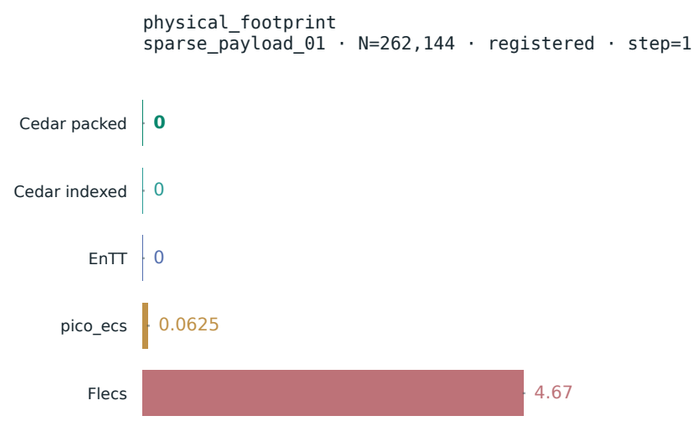


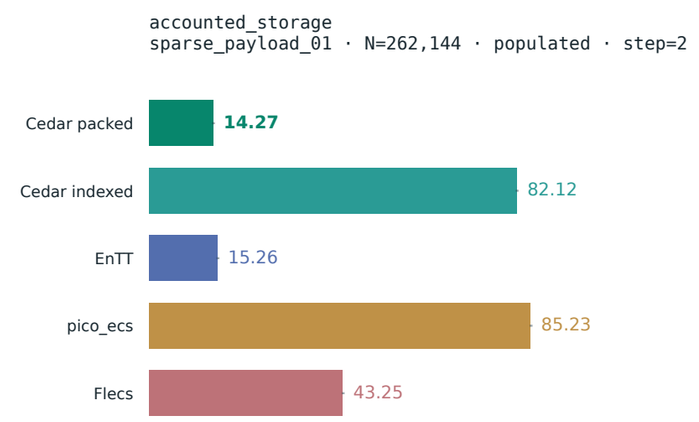
Memory comparison

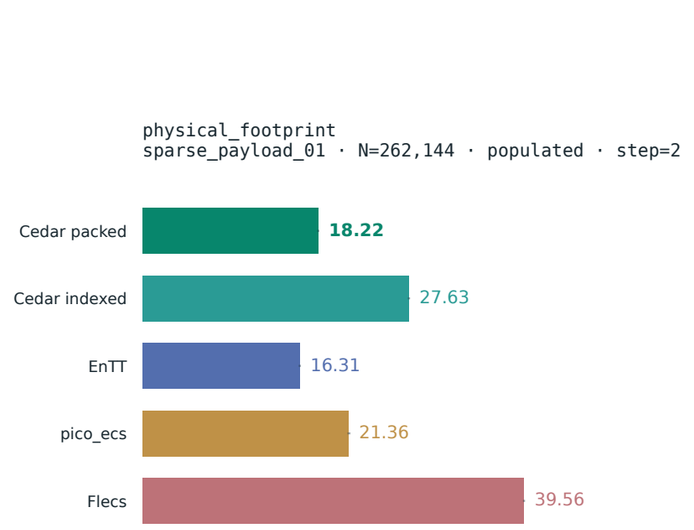
MiB ↓

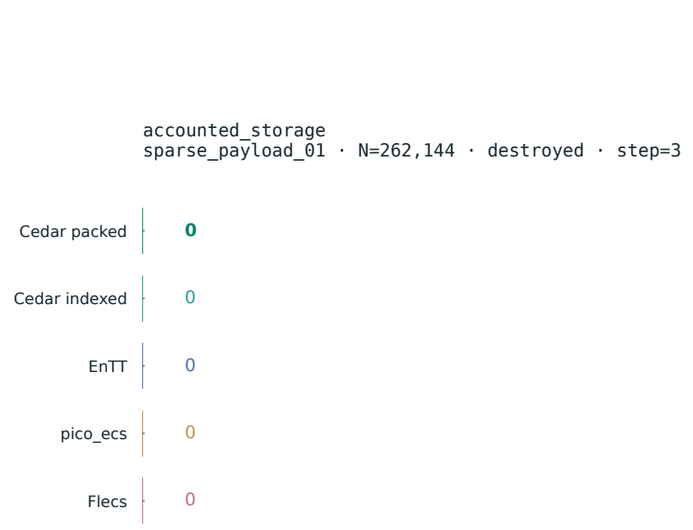
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

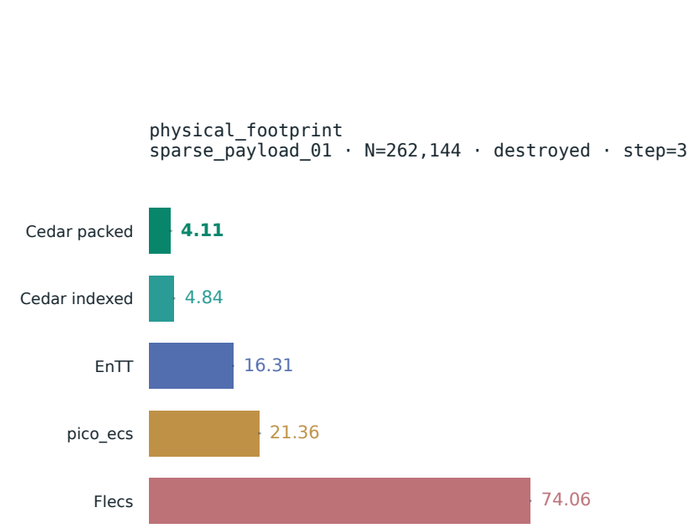


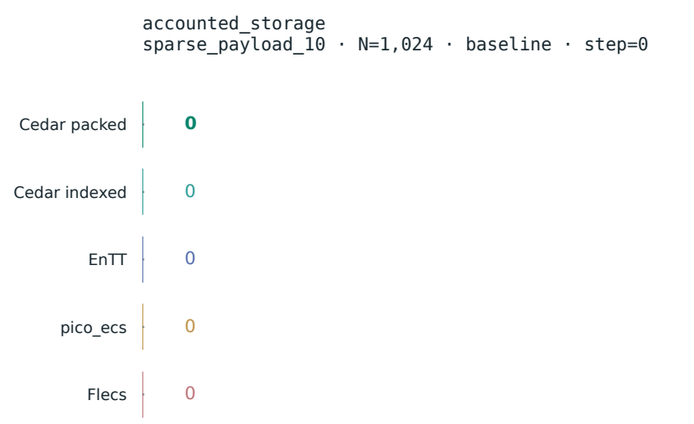


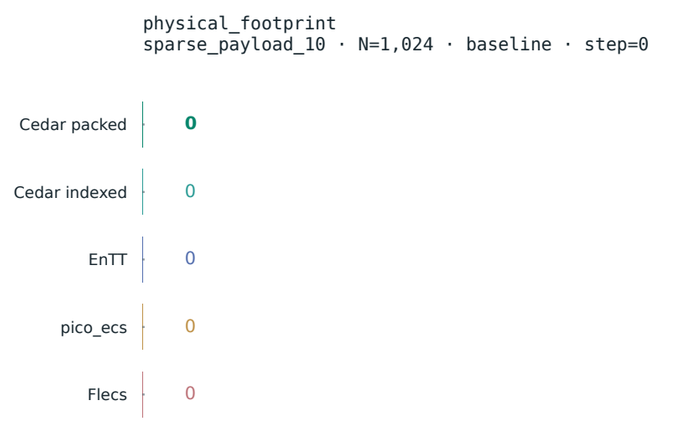


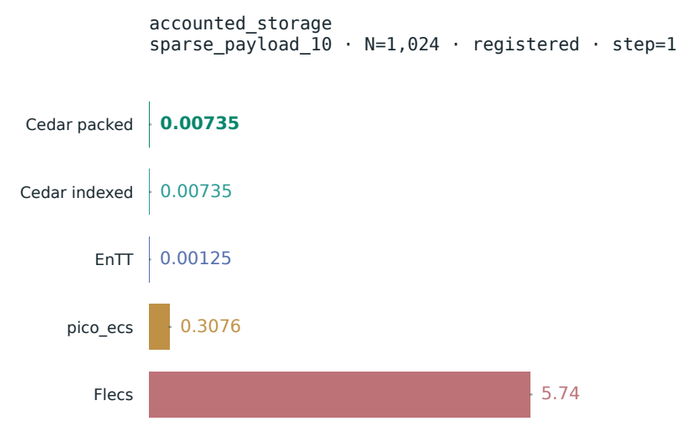


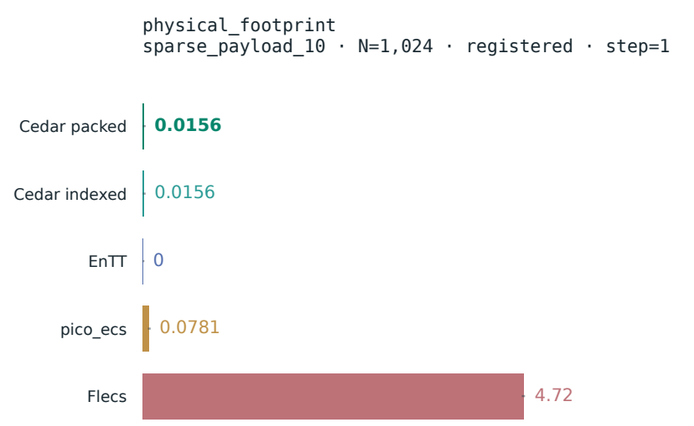


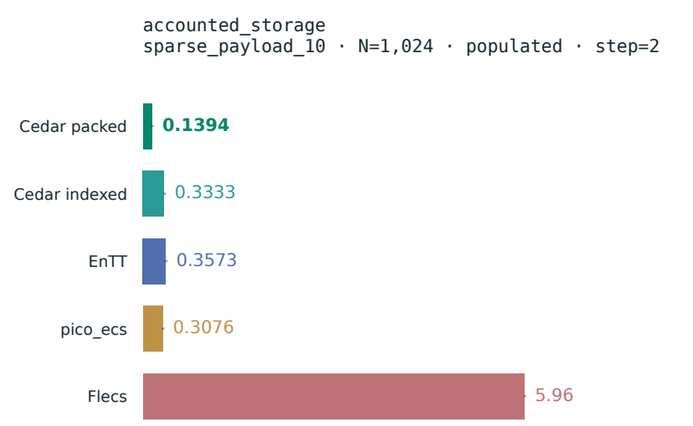


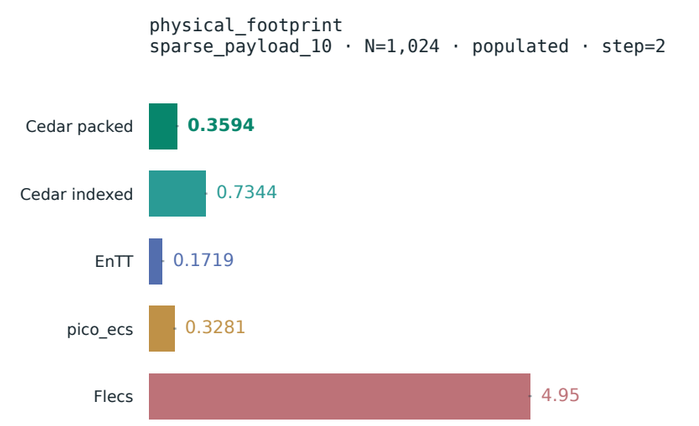










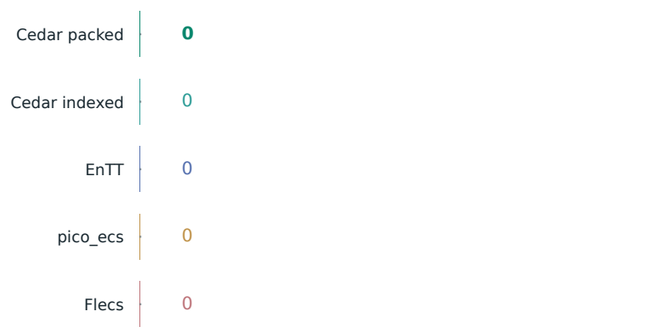


Memory comparison

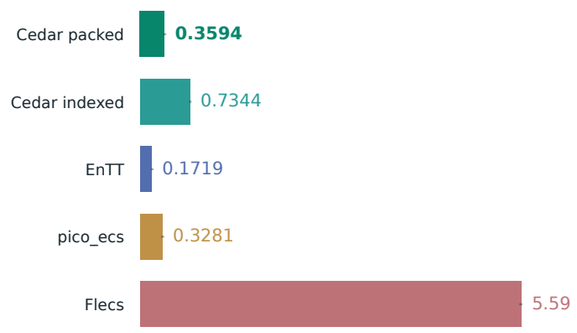
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

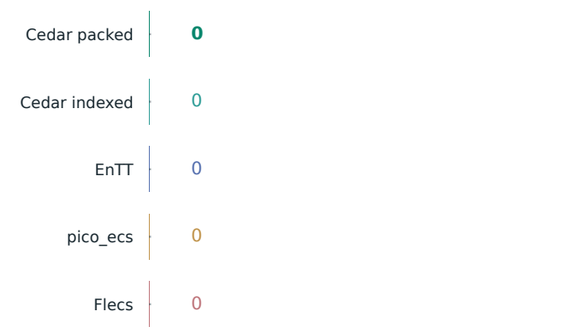
accounted_storage
sparse_payload_10 · N=1,024 · destroyed · step=3



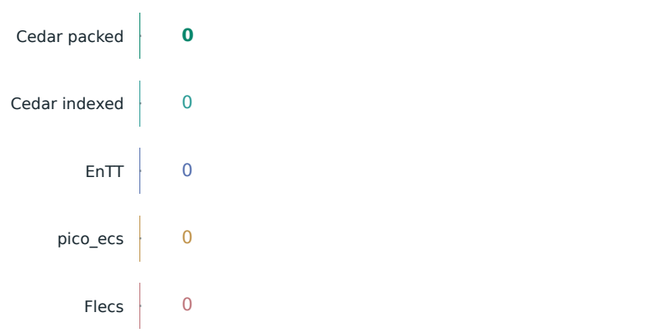
physical_footprint
sparse_payload_10 · N=1,024 · destroyed · step=3



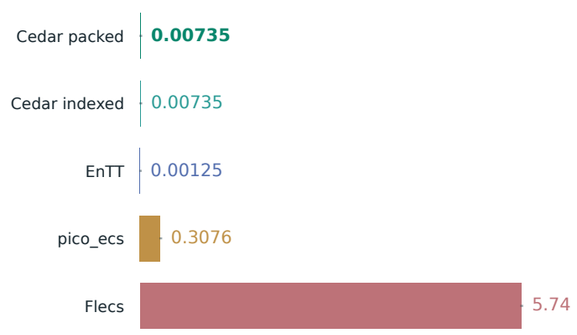
accounted_storage
sparse_payload_10 · N=65,536 · destroyed · step=3



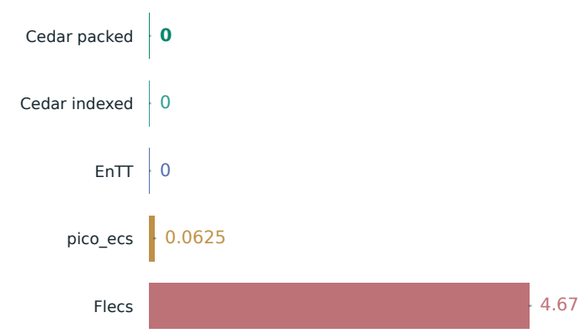
physical_footprint
sparse_payload_10 · N=65,536 · baseline · step=0



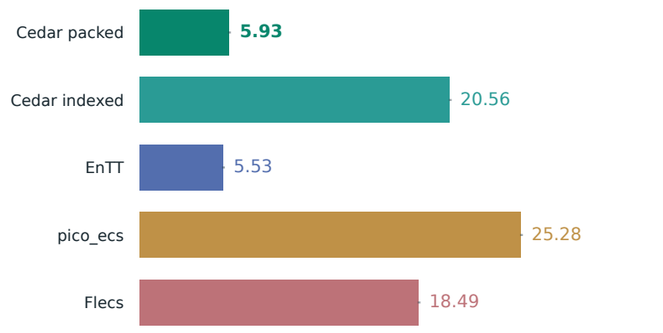
accounted_storage
sparse_payload_10 · N=65,536 · registered · step=1



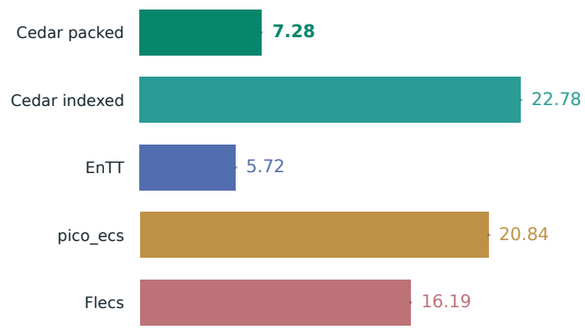
physical_footprint
sparse_payload_10 · N=65,536 · registered · step=1



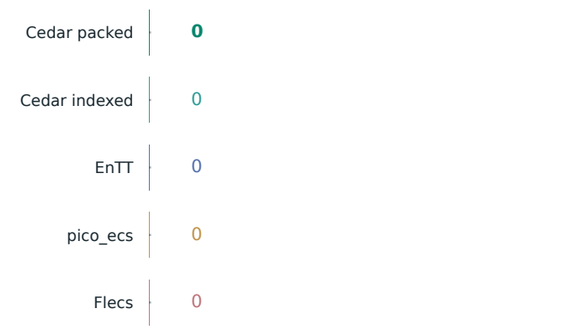
accounted_storage
sparse_payload_10 · N=65,536 · populated · step=2



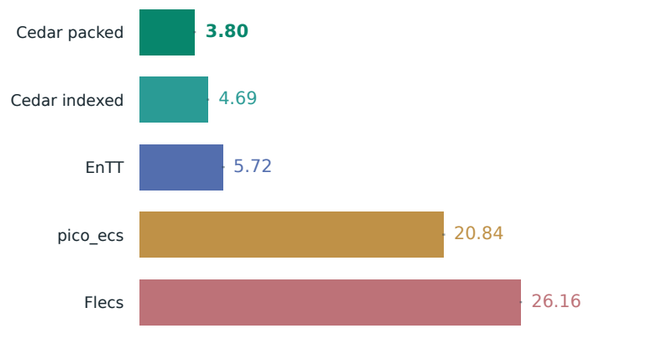
physical_footprint
sparse_payload_10 · N=65,536 · populated · step=2



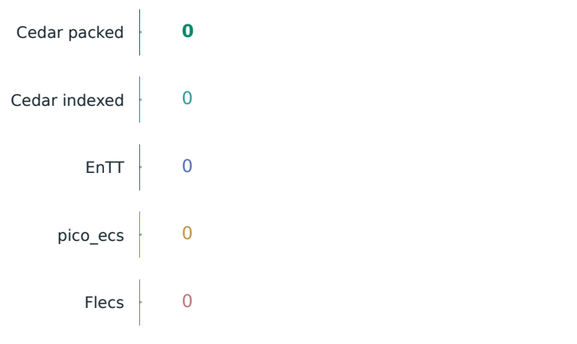
accounted_storage
sparse_payload_10 · N=65,536 · destroyed · step=3



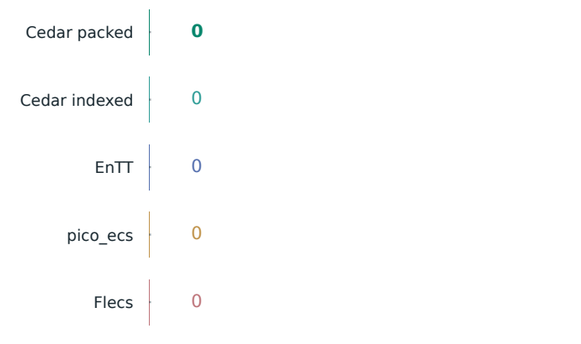
physical_footprint
sparse_payload_10 · N=65,536 · destroyed · step=3



accounted_storage
sparse_payload_10 · N=262,144 · baseline · step=0



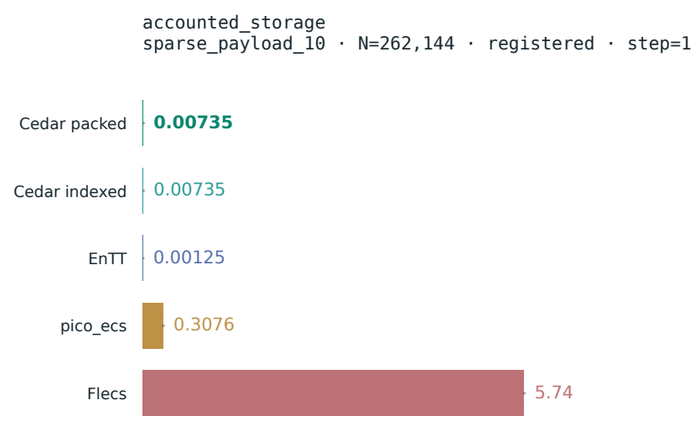
physical_footprint
sparse_payload_10 · N=262,144 · baseline · step=0

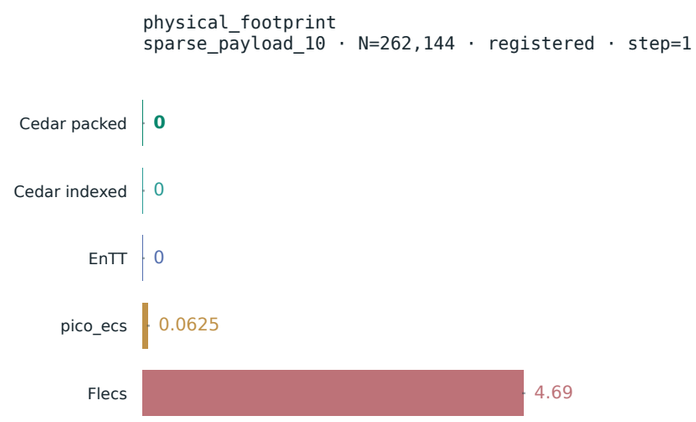


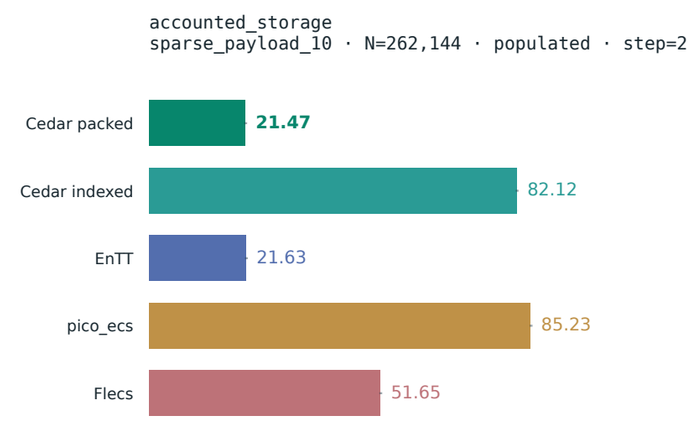
Memory comparison

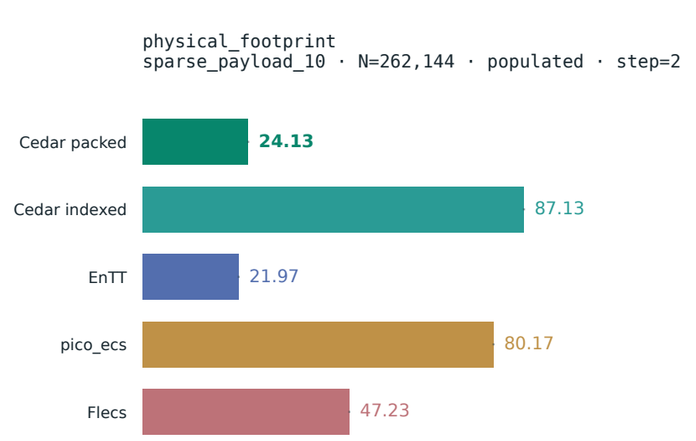
MiB ↓

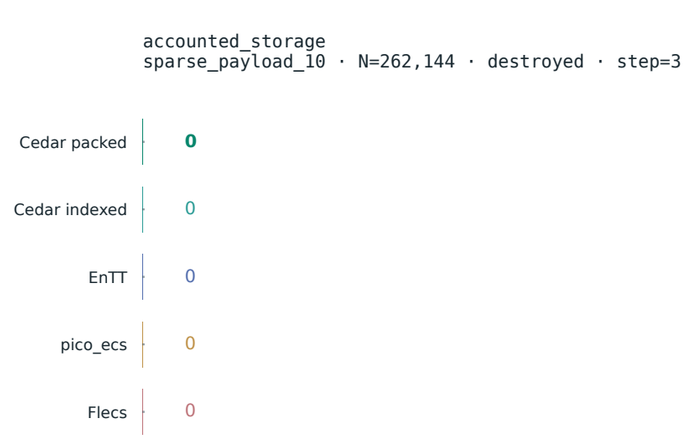
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

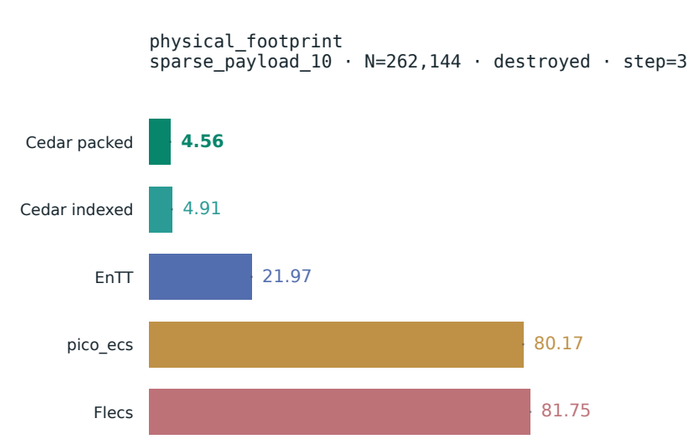


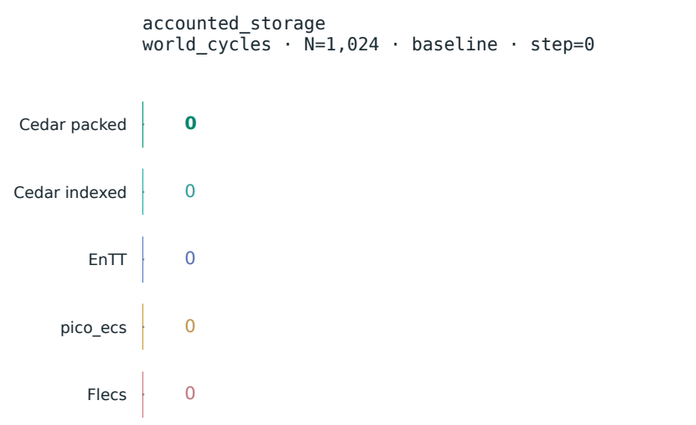


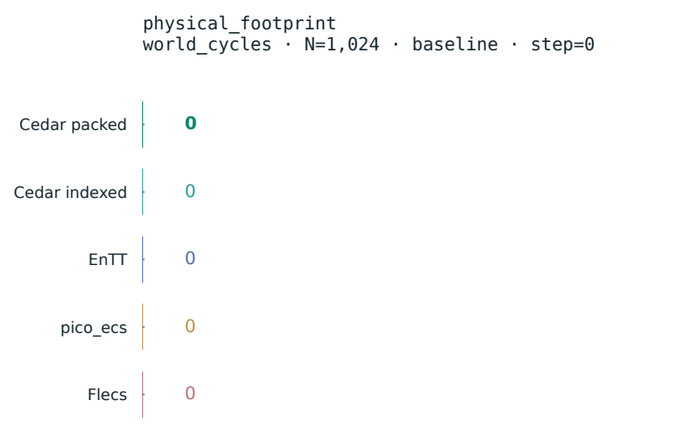


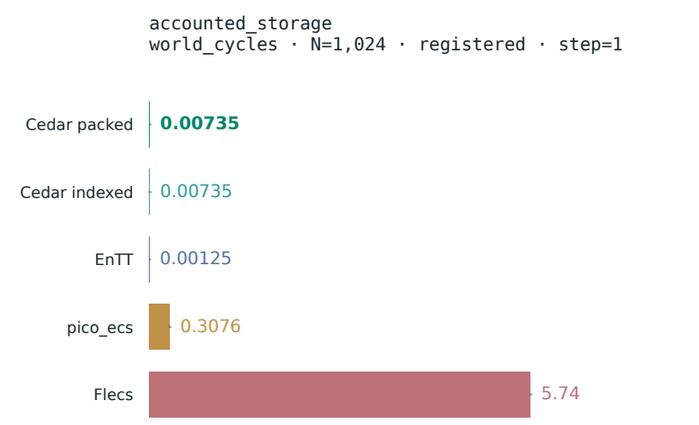


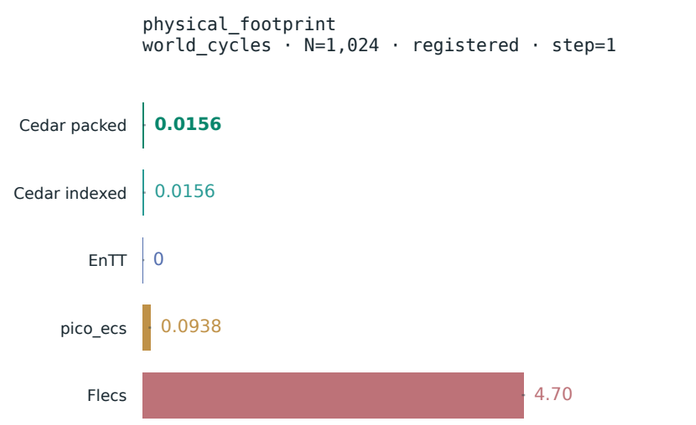


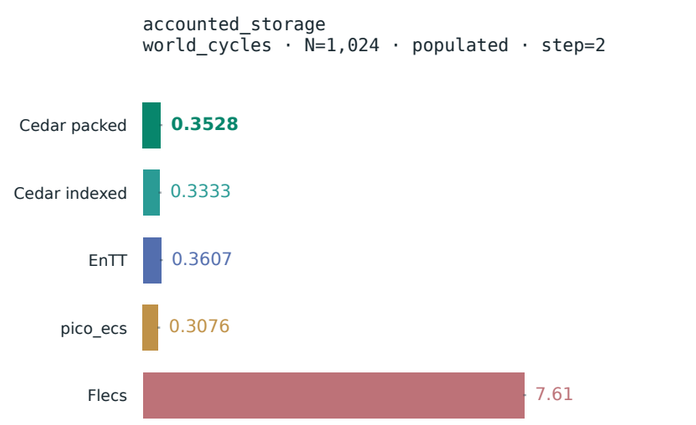


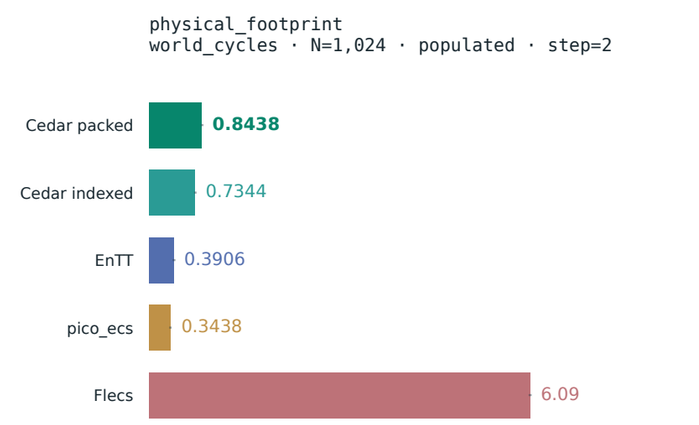








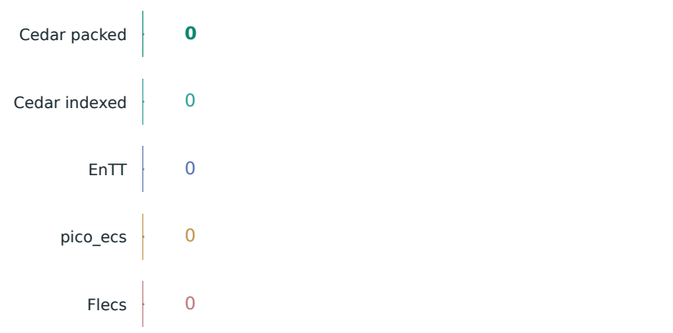


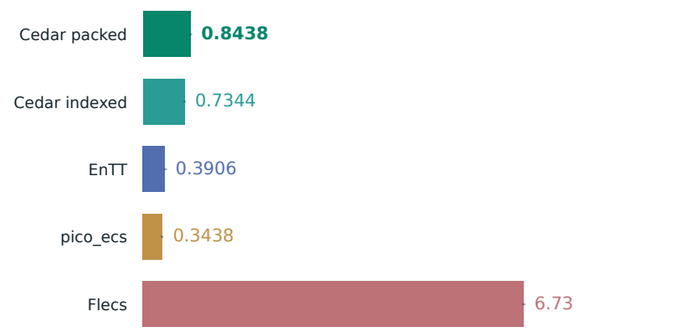


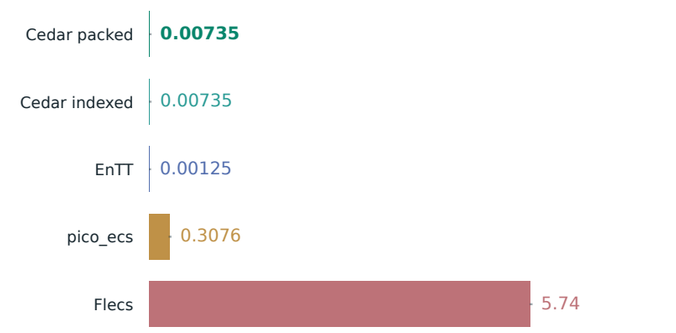
Memory comparison

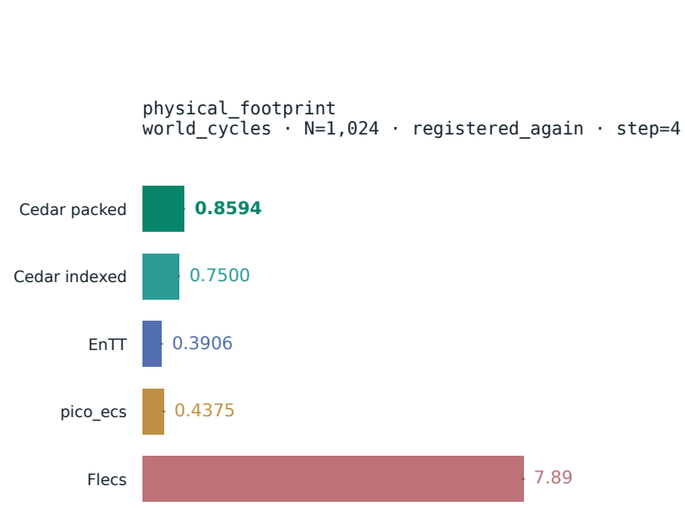
MiB ↓

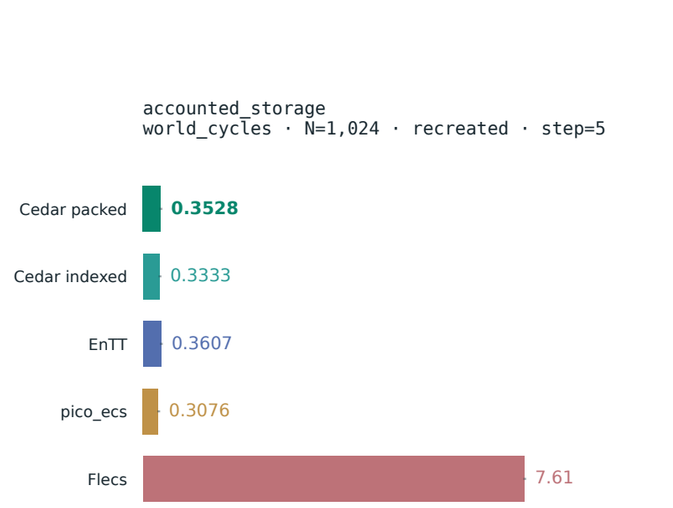
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

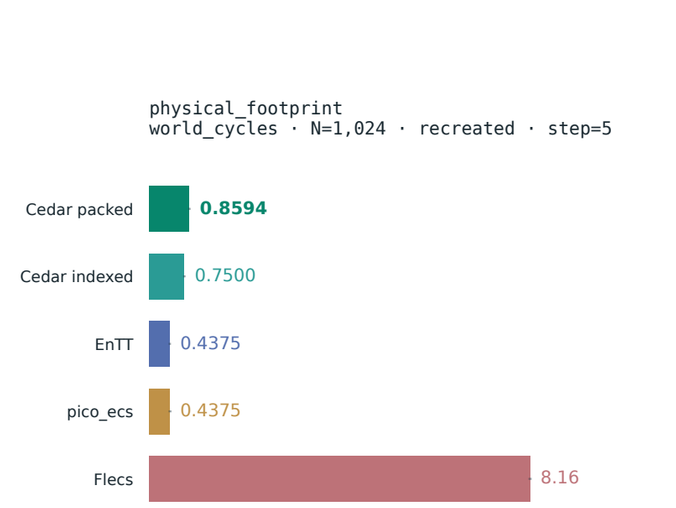




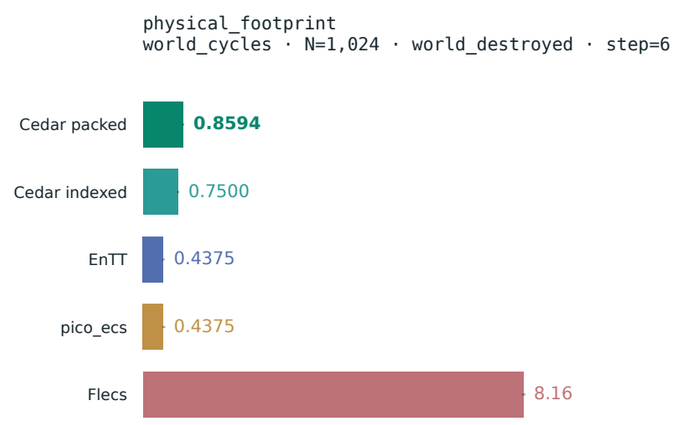


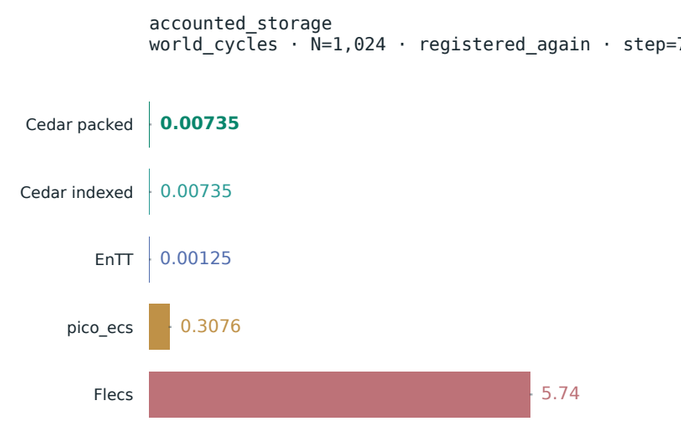


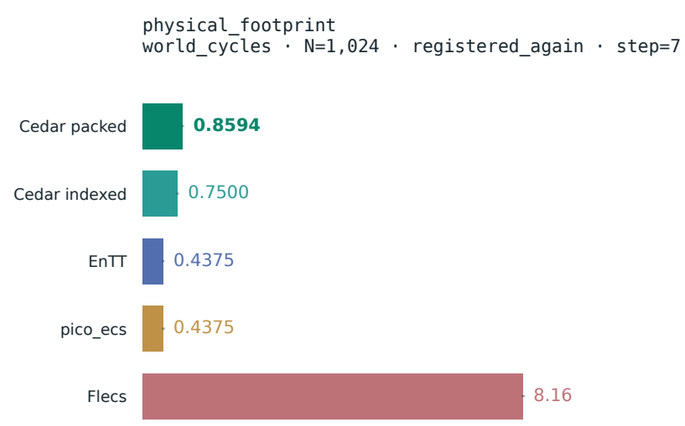


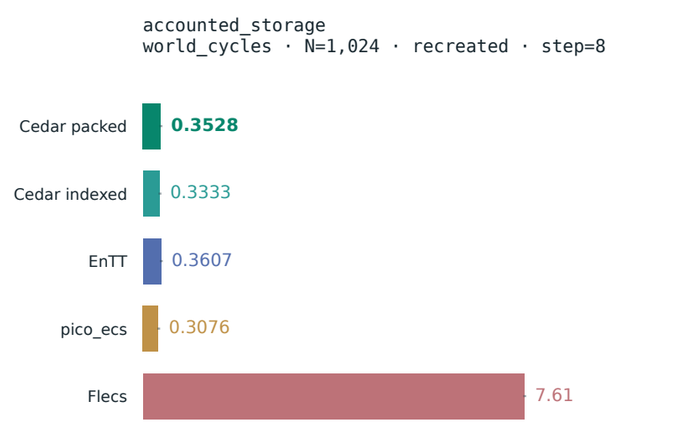


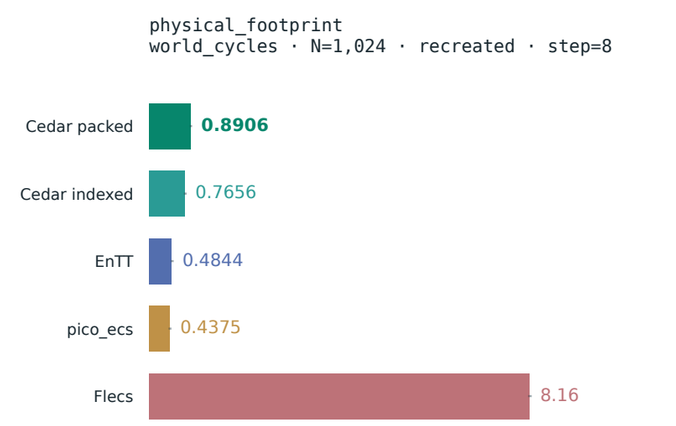










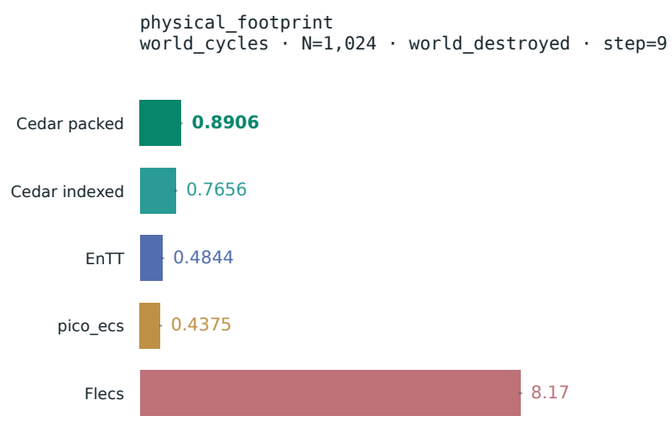


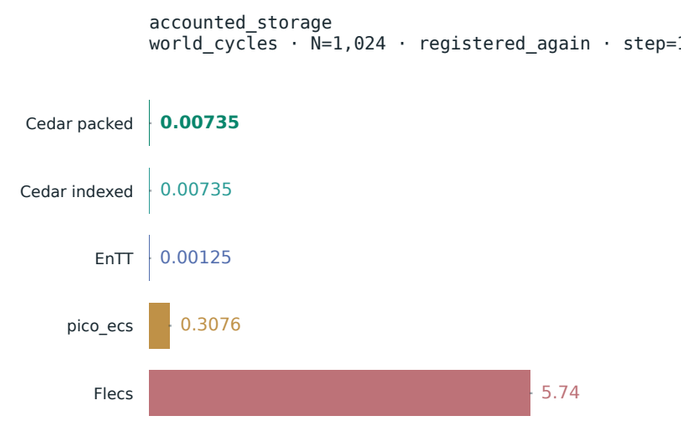
Memory comparison

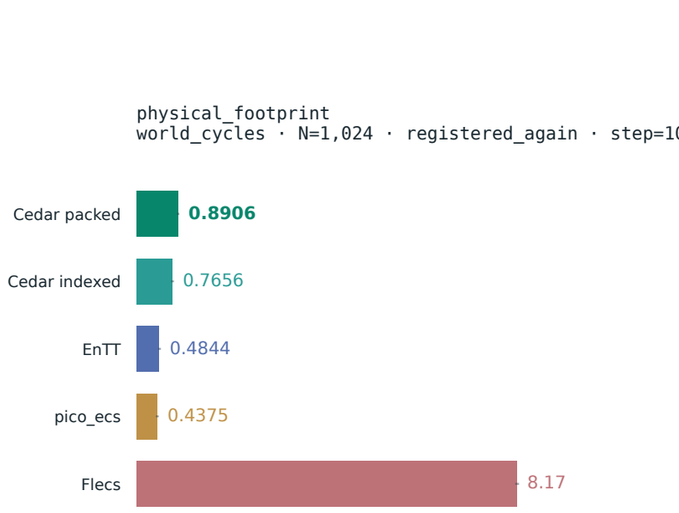
MiB ↓

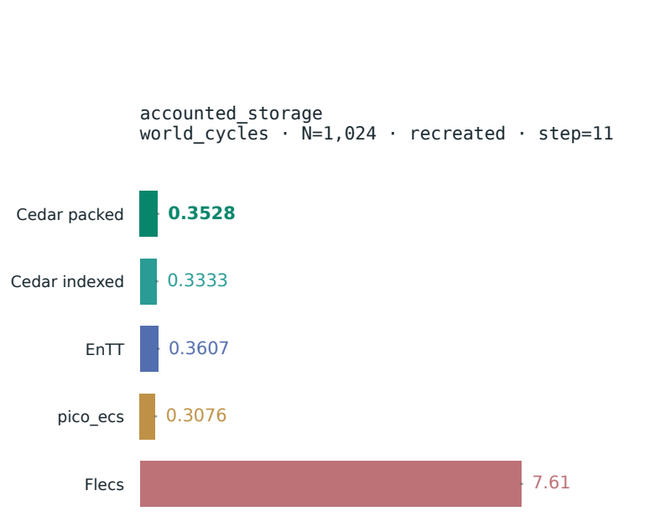
Cedar packed Cedar indexed EnTT pico_ecs Flecs

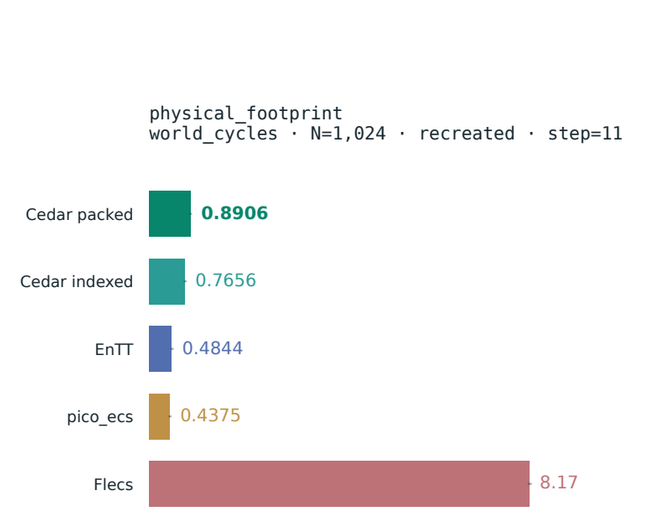




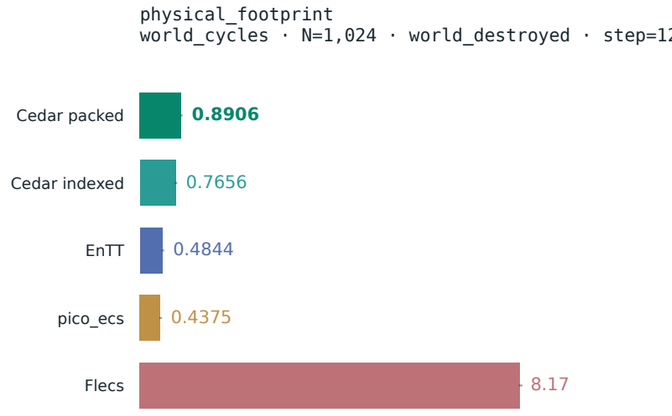


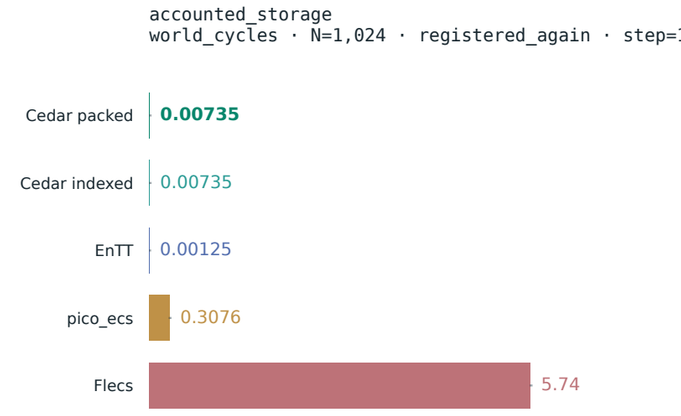


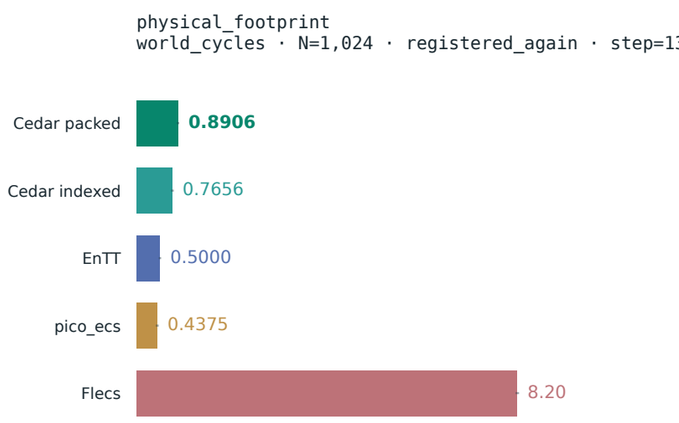


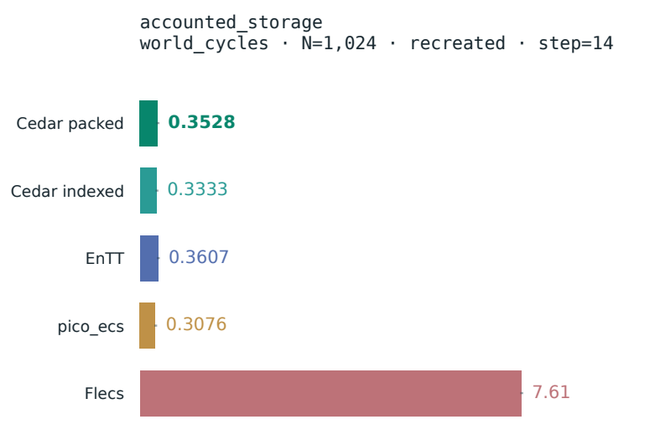


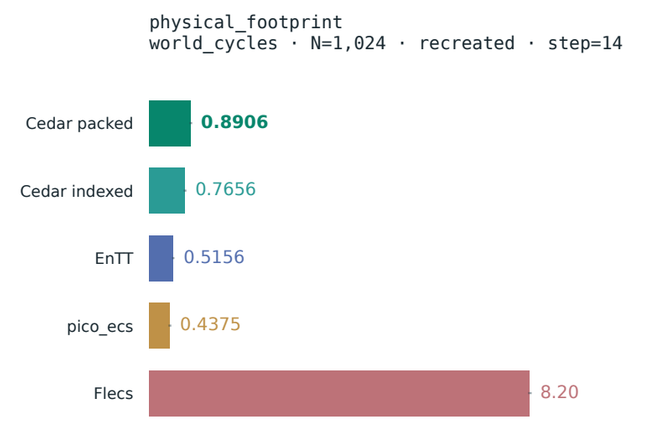








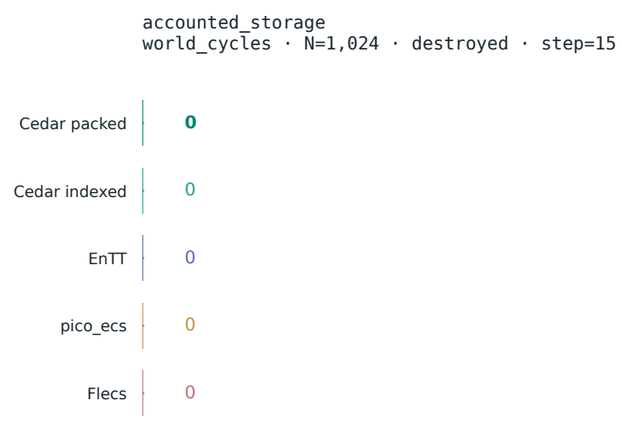


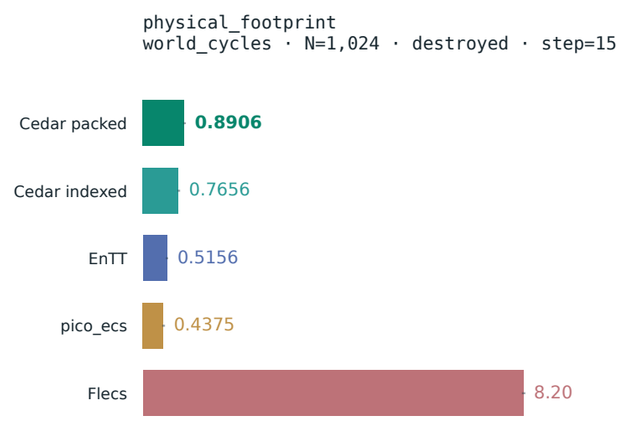


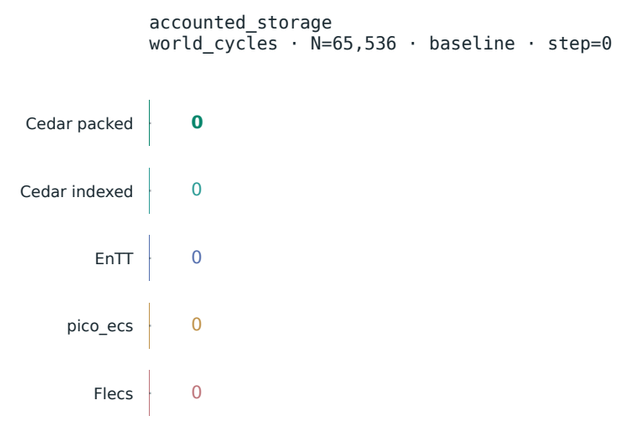
Memory comparison

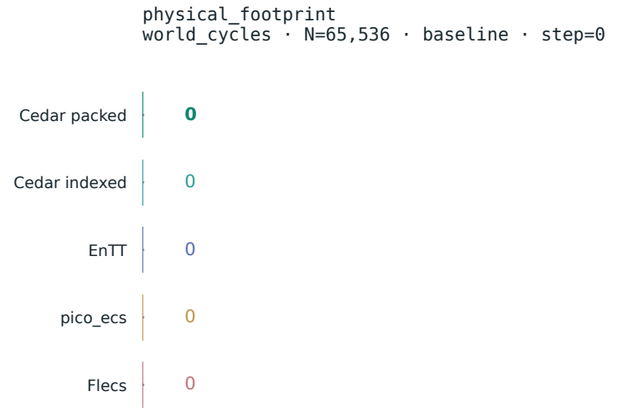
MiB ↓

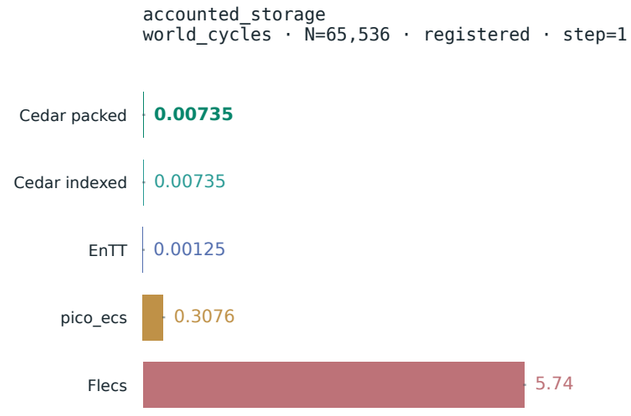
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

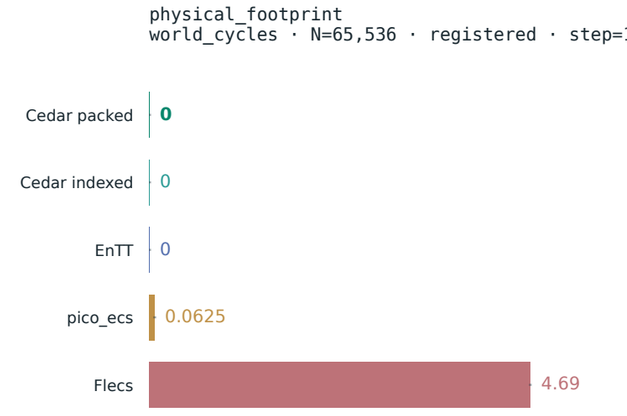


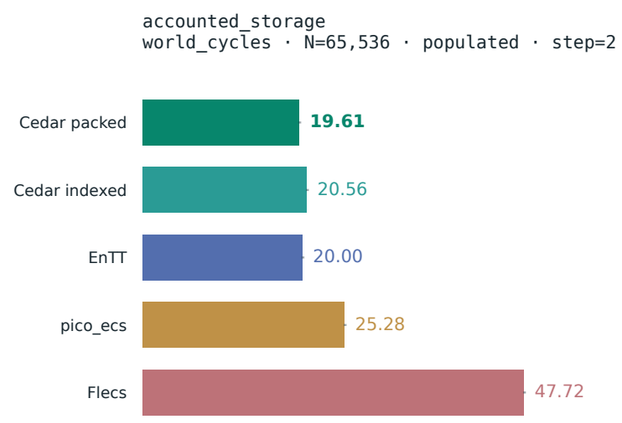


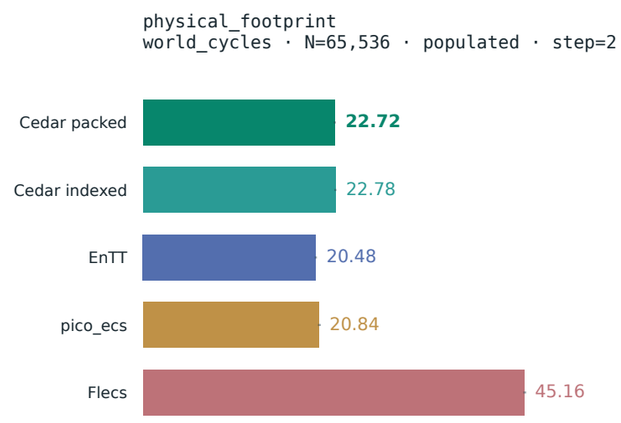




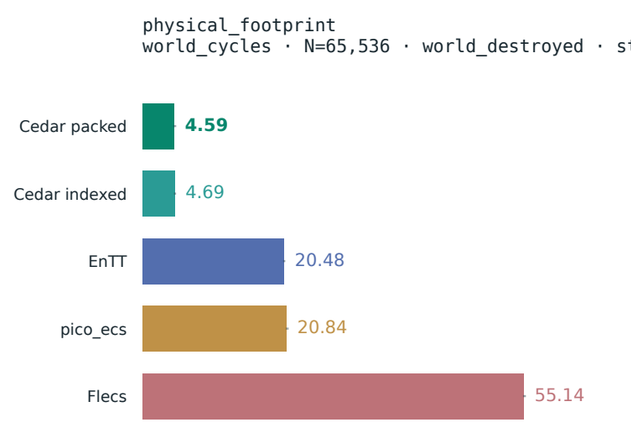


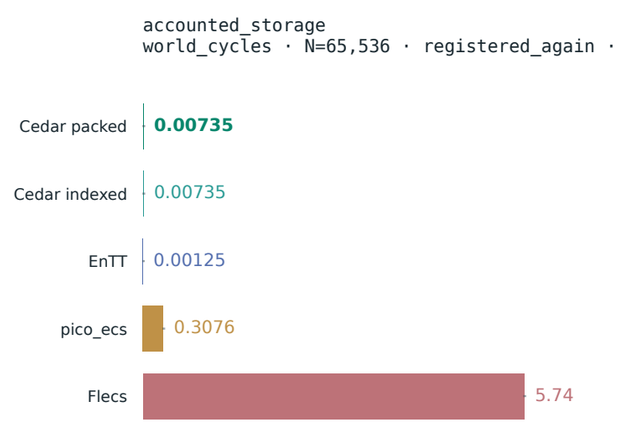


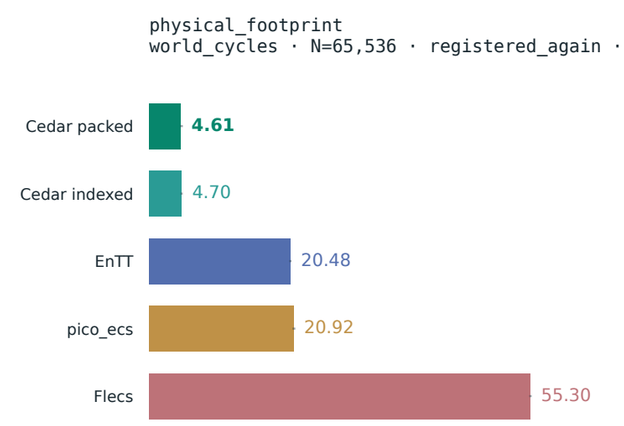










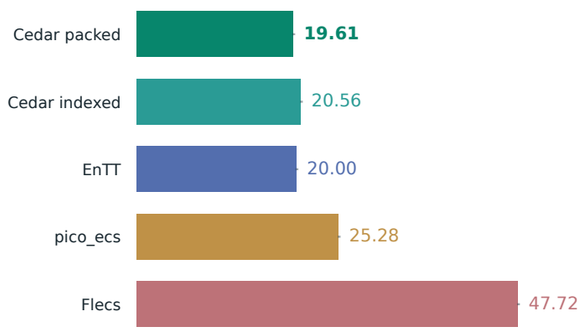


Memory comparison

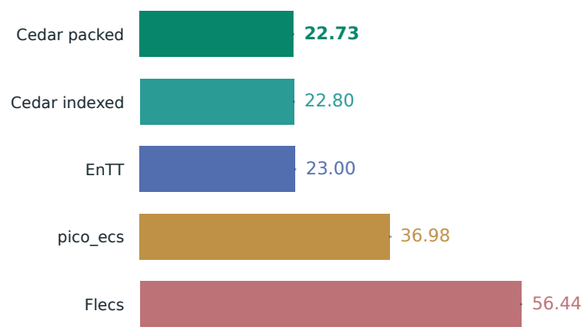
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

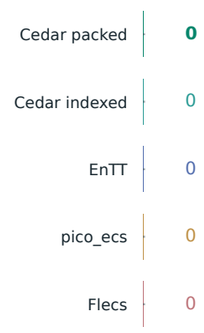
accounted_storage
world_cycles · N=65,536 · recreated · step=5



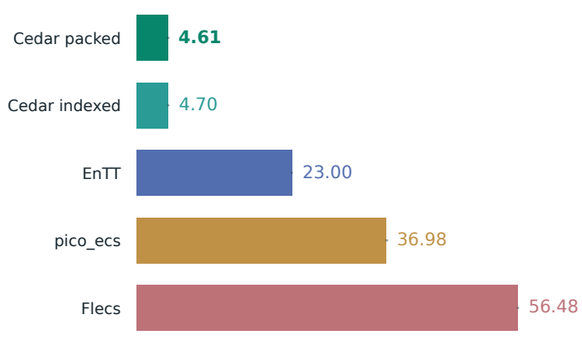
physical_footprint
world_cycles · N=65,536 · recreated · step=5



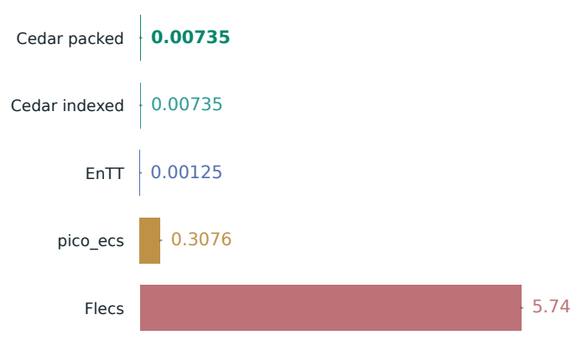
accounted_storage
world_cycles · N=65,536 · world_destroyed · step=6



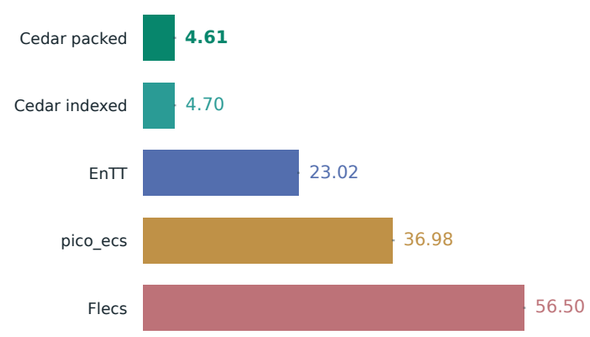
physical_footprint
world_cycles · N=65,536 · world_destroyed · step=6



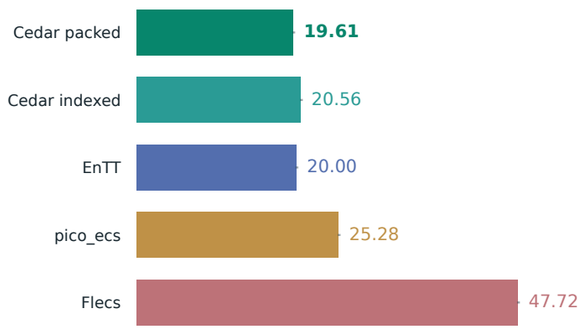
accounted_storage
world_cycles · N=65,536 · registered_again · step=7



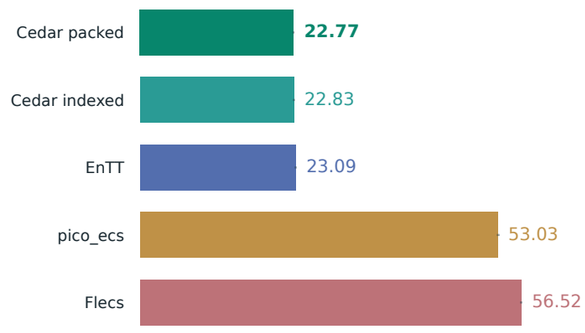
physical_footprint
world_cycles · N=65,536 · registered_again · step=7



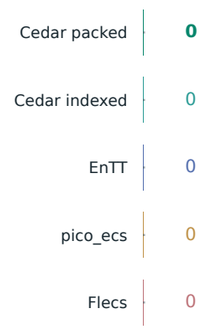
accounted_storage
world_cycles · N=65,536 · recreated · step=8



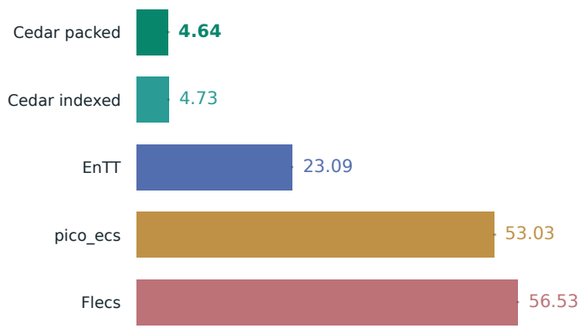
physical_footprint
world_cycles · N=65,536 · recreated · step=8



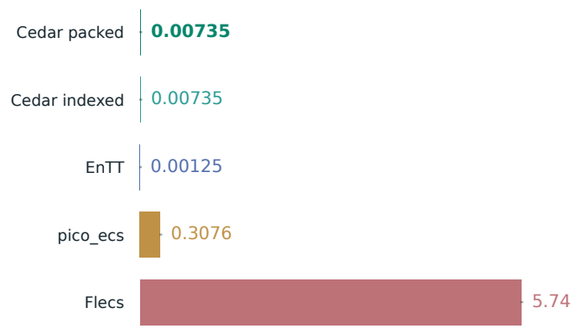
accounted_storage
world_cycles · N=65,536 · world_destroyed · step=9



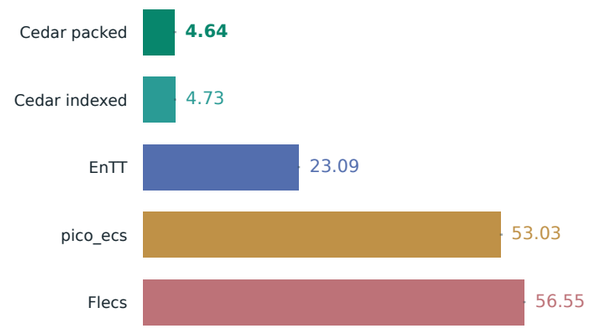
physical_footprint
world_cycles · N=65,536 · world_destroyed · step=9



accounted_storage
world_cycles · N=65,536 · registered_again · step=10



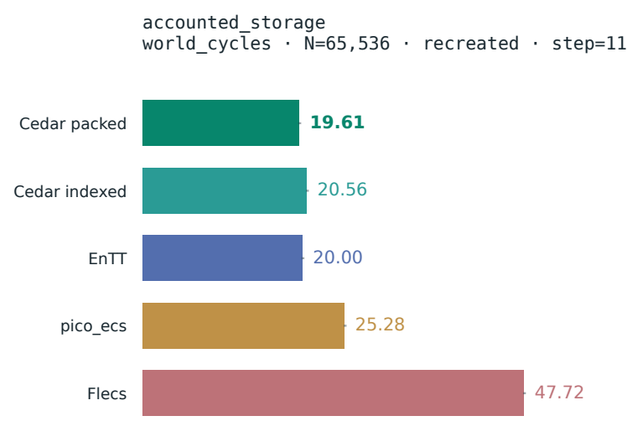
physical_footprint
world_cycles · N=65,536 · registered_again · step=1

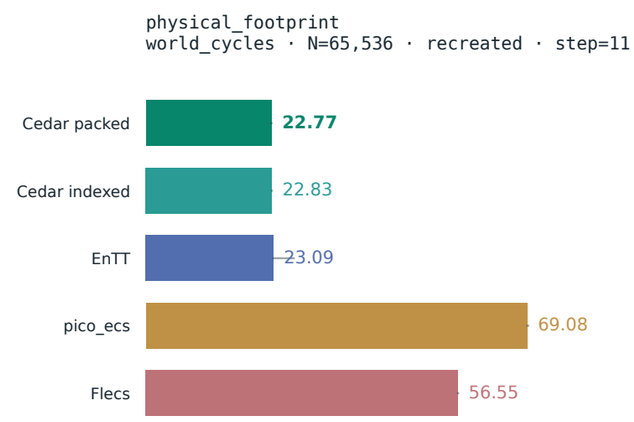


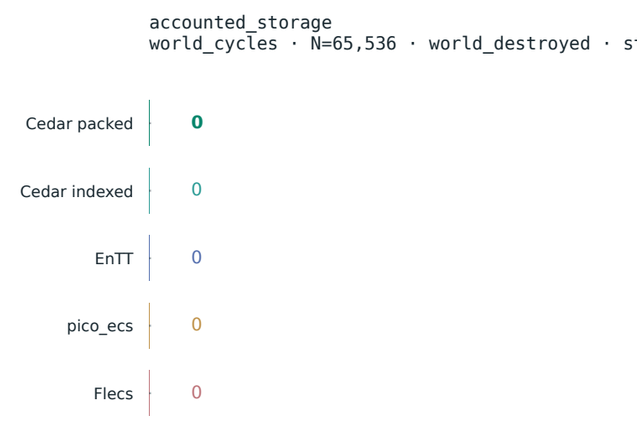
Memory comparison

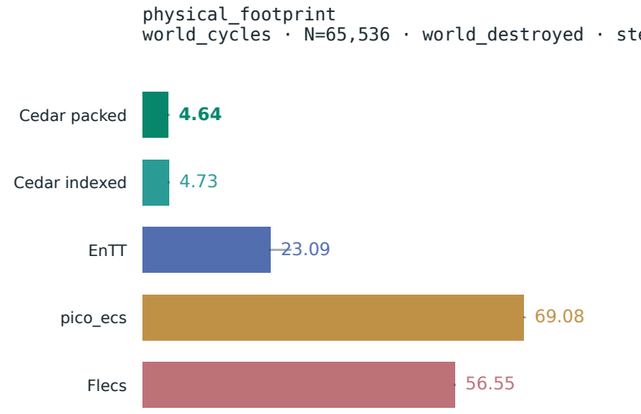
MiB ↓

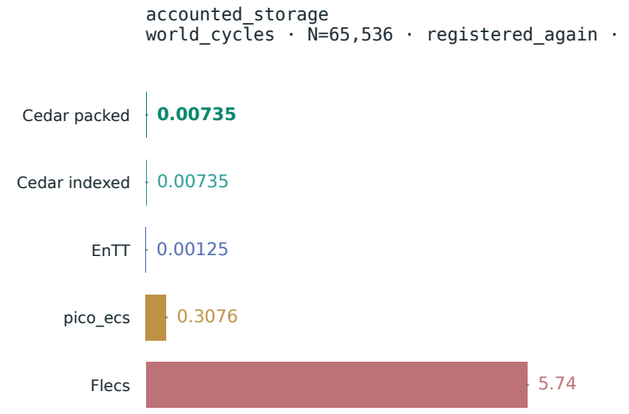
Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

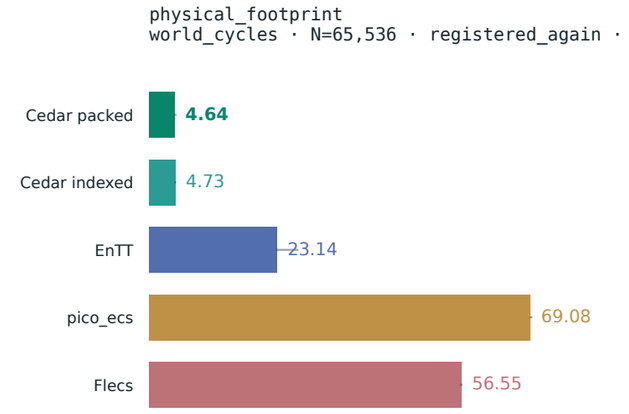


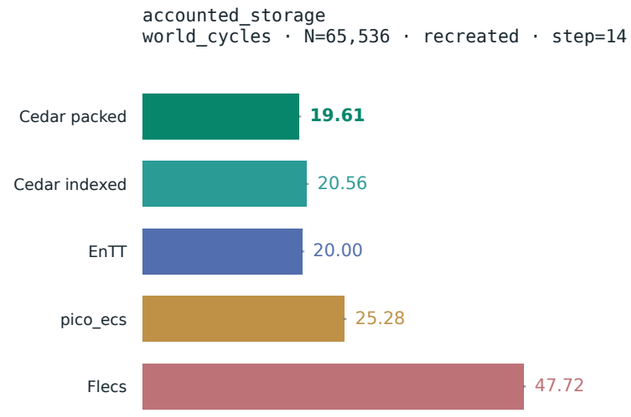


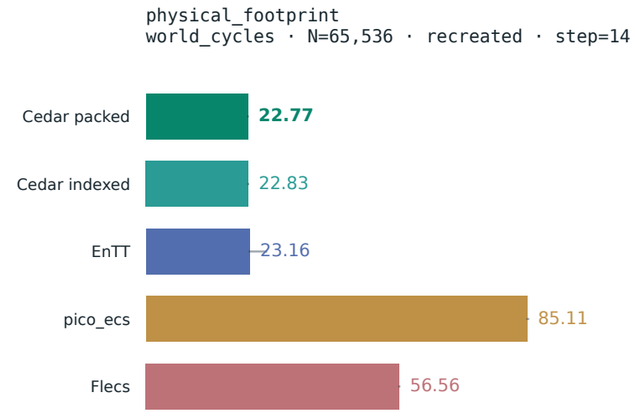


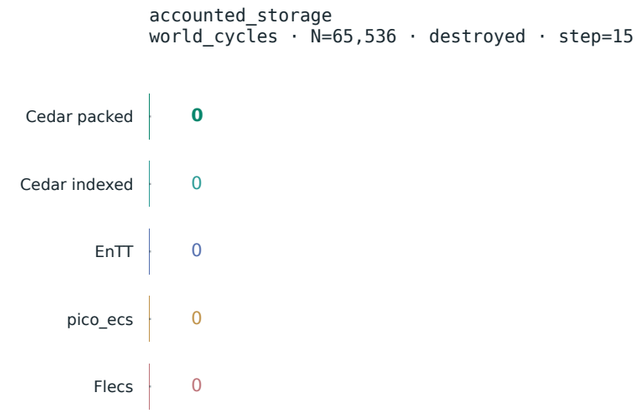


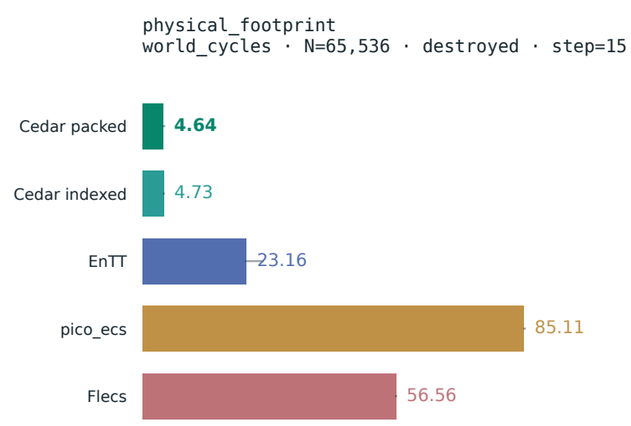




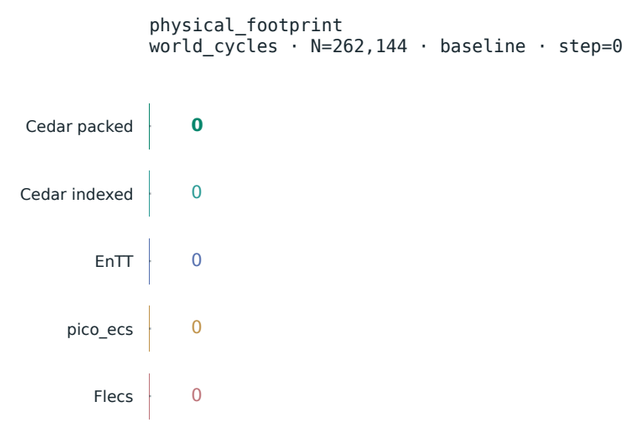










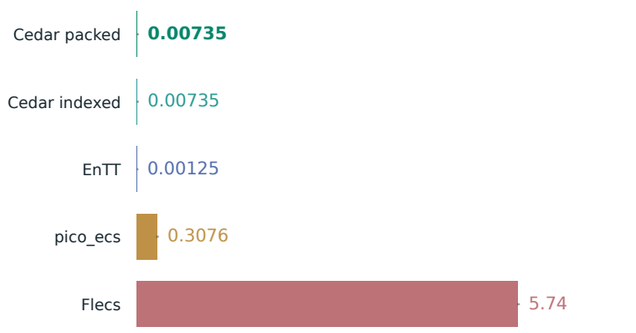


Memory comparison

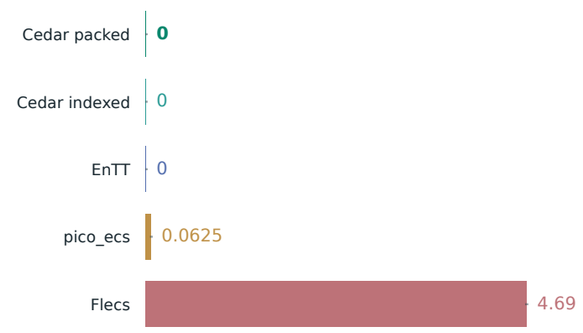
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

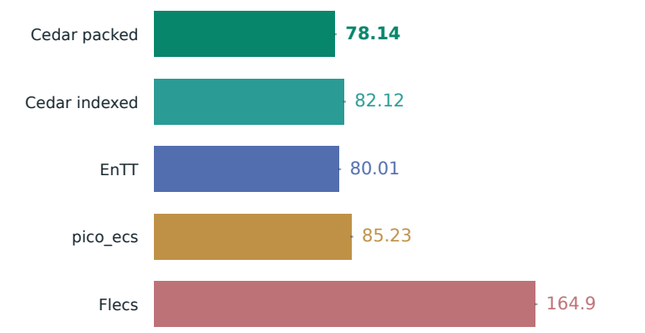
accounted_storage
world_cycles · N=262,144 · registered · step=1



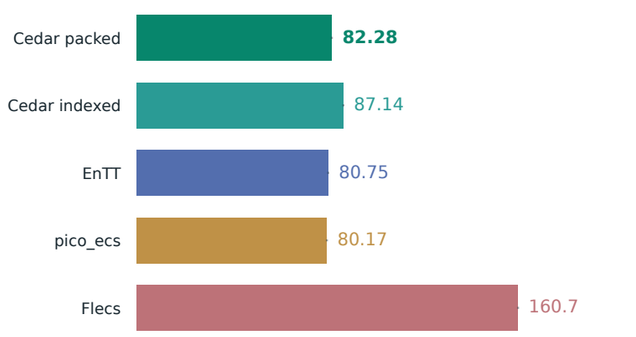
physical_footprint
world_cycles · N=262,144 · registered · step=1



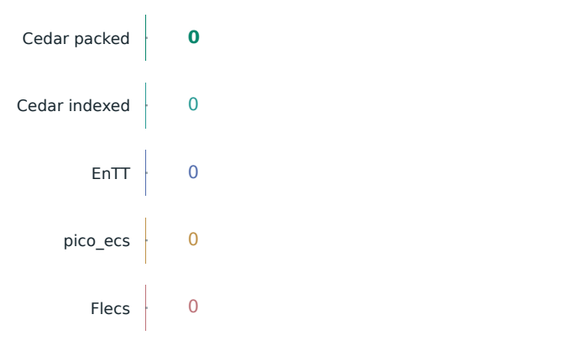
accounted_storage
world_cycles · N=262,144 · populated · step=2



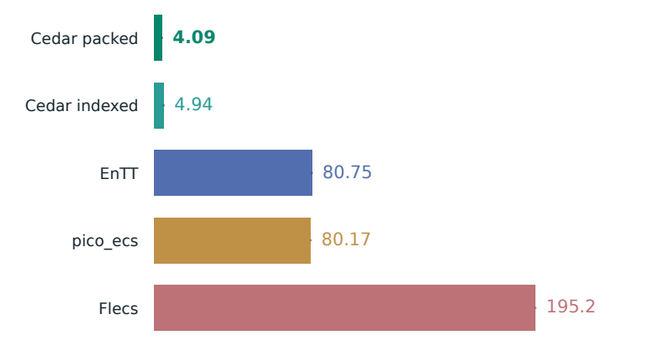
physical_footprint
world_cycles · N=262,144 · populated · step=2



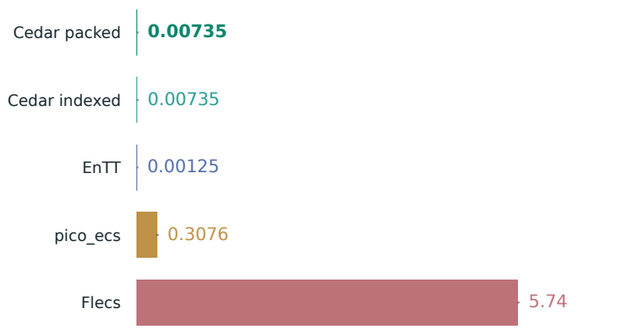
accounted_storage
world_cycles · N=262,144 · world_destroyed · step=3



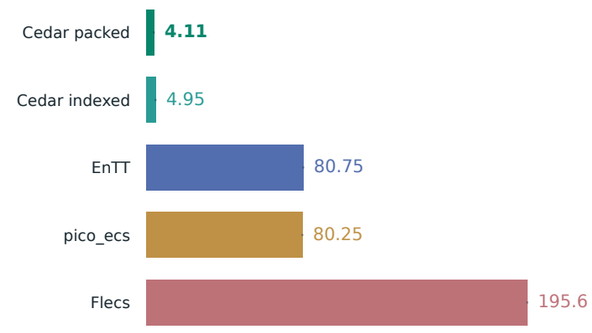
physical_footprint
world_cycles · N=262,144 · world_destroyed · step=3



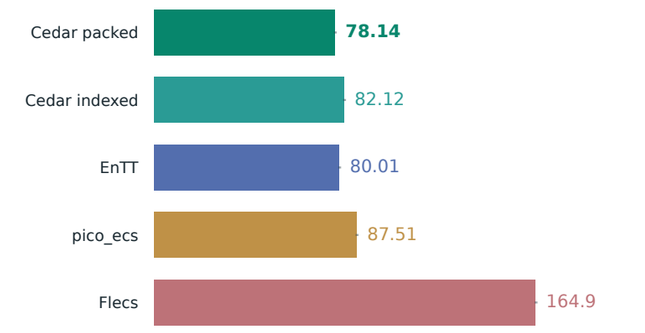
accounted_storage
world_cycles · N=262,144 · registered_again · step=4



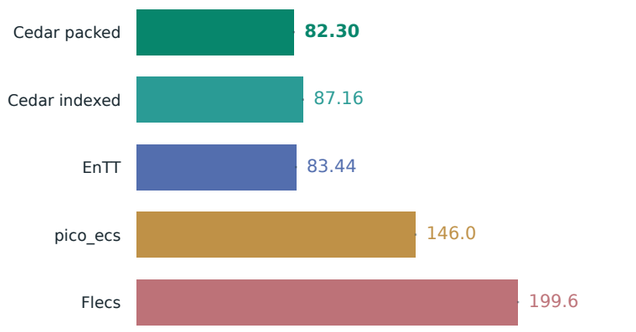
physical_footprint
world_cycles · N=262,144 · registered_again · step=4



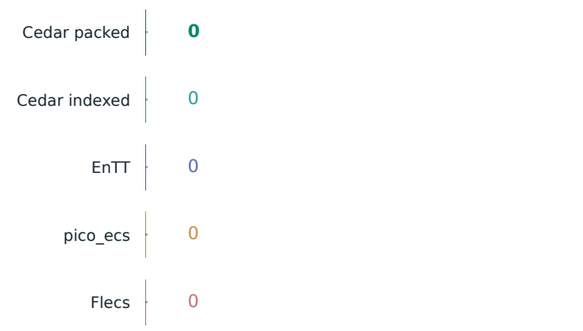
accounted_storage
world_cycles · N=262,144 · recreated · step=5



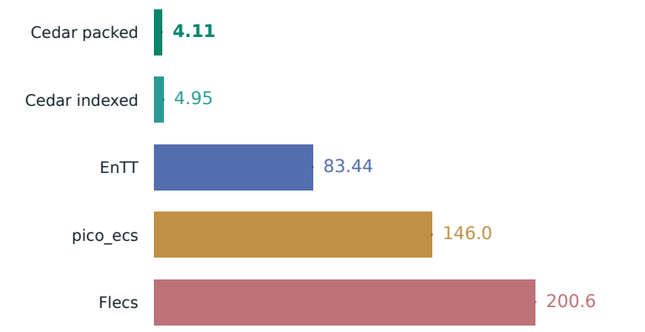
physical_footprint
world_cycles · N=262,144 · recreated · step=5



accounted_storage
world_cycles · N=262,144 · world_destroyed · step=6



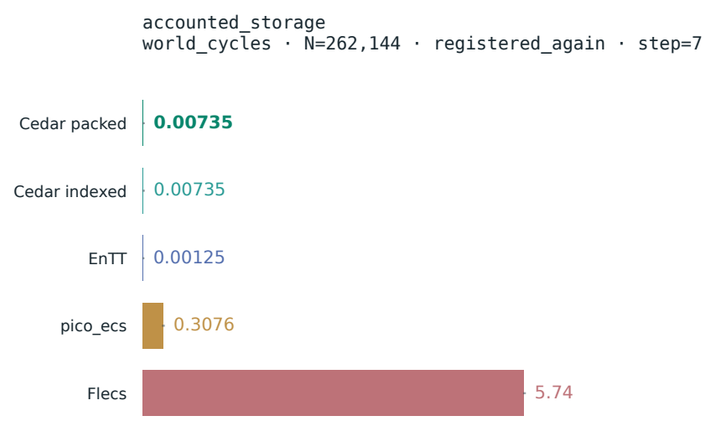
physical_footprint
world_cycles · N=262,144 · world_destroyed · step=6

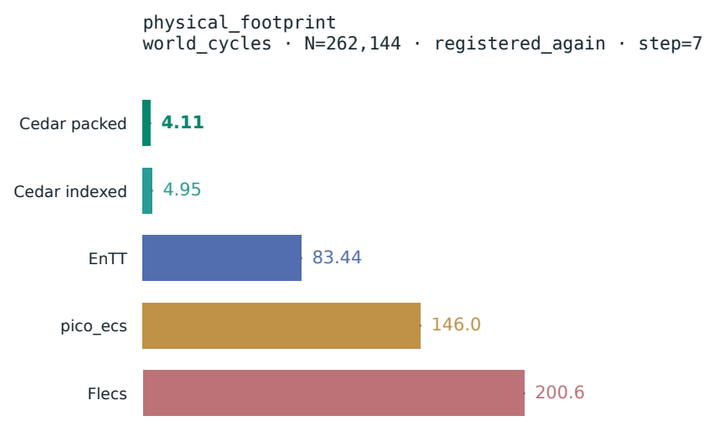


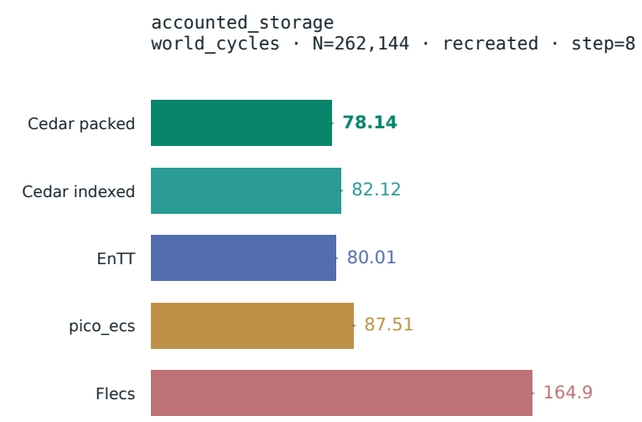
Memory comparison

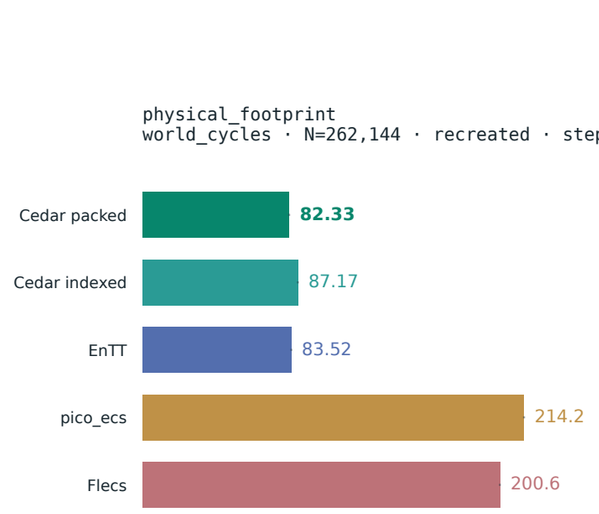
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

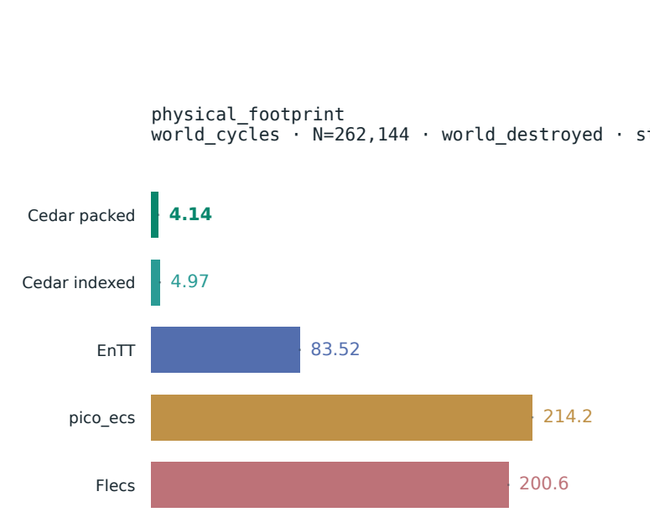


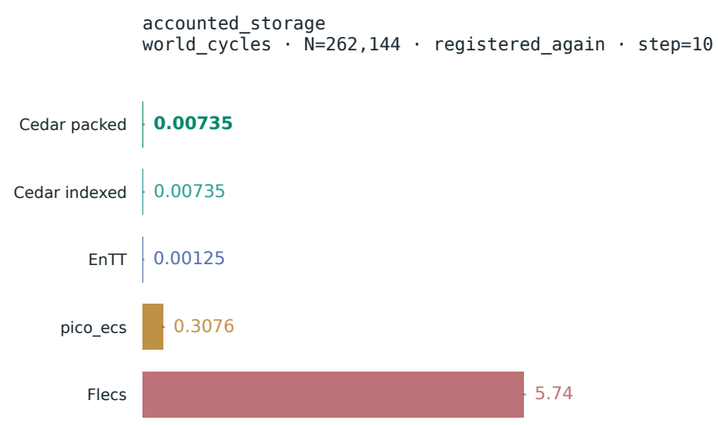


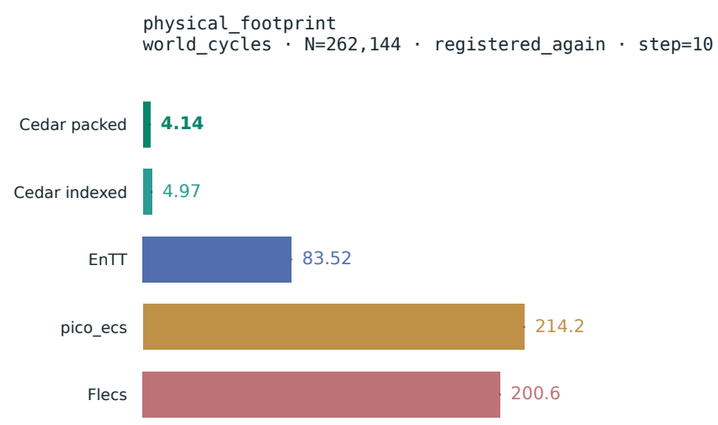


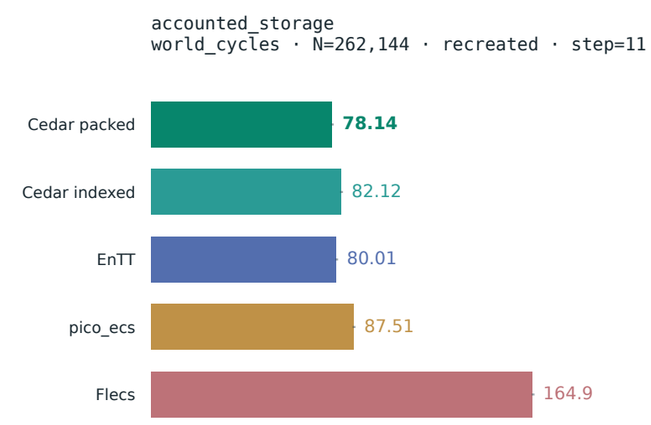








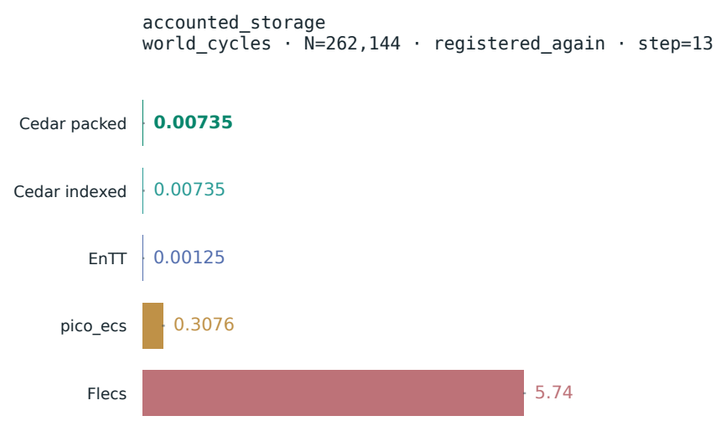


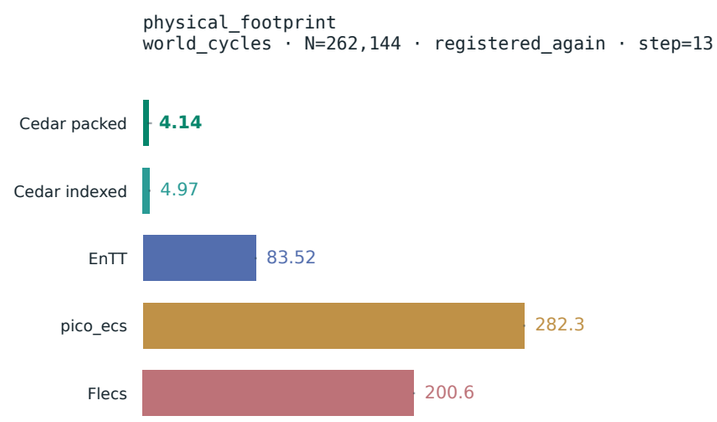


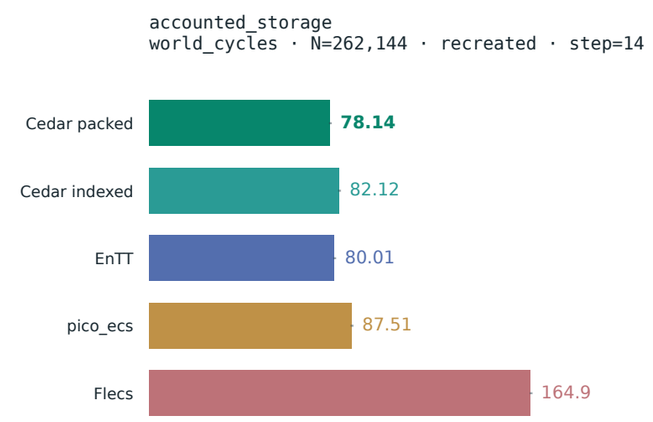
Memory comparison

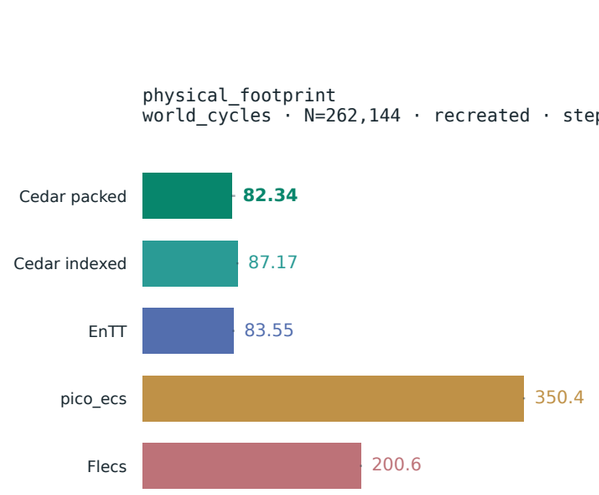
MiB ↓

Cedar packed · Cedar indexed · EnTT · pico_ecs · Flecs

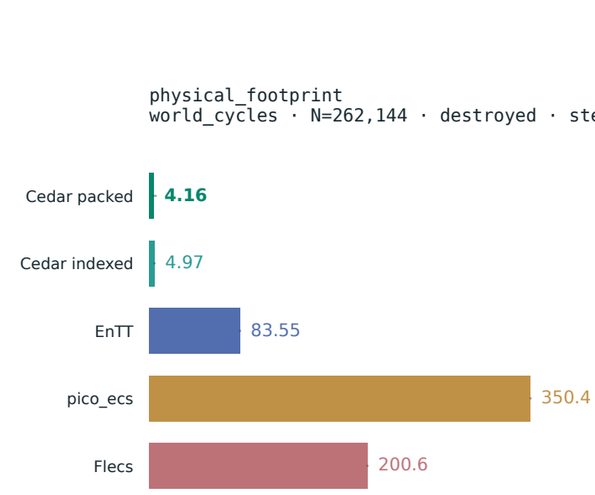






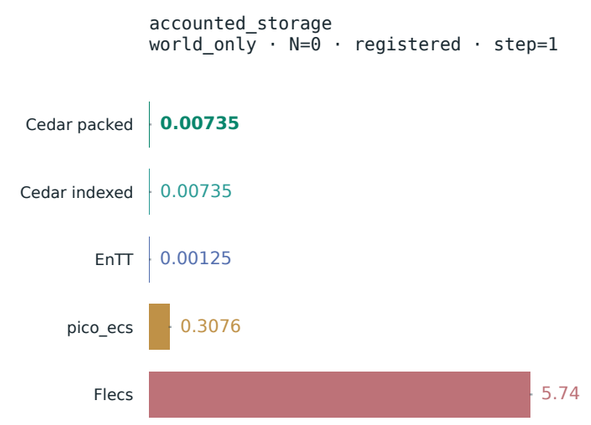


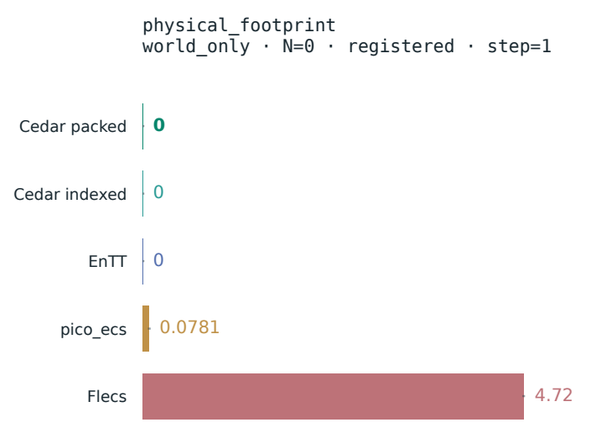


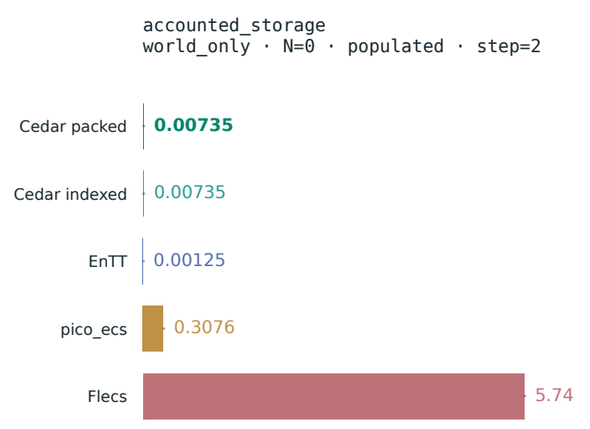


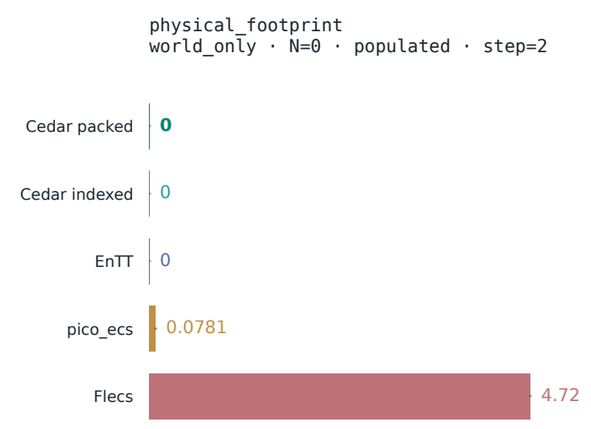












Memory comparison

MiB ↓

Cedar packed Cedar indexed EnTT pico_ecs Flecs



